

Research Methods for Managers



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Module - 1: Nature and Scope of Research Methodology

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Unit - 1.1: Research

Learning Objectives:

At the end of this unit, you will be able to:

- Understand definition and objectives of research
- Analyse characteristics of research
- Comprehend difference between research methods and research methodology

Introduction

We can understand the term 'research' to mean a collection of data that includes crucial information by taking research approaches into account. In other words, it is a collection of knowledge or data that has been studied by formulating a hypothesis and then coming up with significant findings in an organised manner. Research can be conducted on both an academic and a scientific level.

It can be broadly defined “as a systematic investigation towards understanding a multifaceted social phenomenon or a process”. It monitors the scientific approach to gain knowledge. The maximum important characteristic of this methodology is its thrust on objectivity.

To what extent is the research using the scientific approach advantageous in studying the problems of society? How can we attain reliable knowledge about the countless aspects of human experience? To be more exact how can the scientific approach be of value in consideration of social phenomena? In this Unit we will deliberate these questions. Our approach would be first, to recognize the meaning of the terms 'research' and 'scientific research', secondly, examine using the scientific method, its request in social sciences, its expectations and finally to take a close look at the methodology to find out how it can help rural advance professionals to understand the glitches of rural development.



1.1.1 Definition and Objectives of Research

According to Rajasekar et. al. (2006), research is a logical and systematic search for new and useful information on a particular topic. It is an investigation of finding

solutions to scientific and social problems through objective and systematic analysis. It is a search for knowledge, that is, a discovery of hidden truths. Here knowledge means information about matters. The information might be collected from different sources like experience, human beings, books, journals, nature, etc. Research can lead to new contributions to the existing knowledge. Only through research is it possible to make progress in a field. Research is done with the help of study, experiment, observation, analysis, comparison and reasoning. Research is in fact ubiquitous. More precisely, research seeks predictions of events and explanations, relationships and theories for them.

From the viewpoint of objectives, research can be classified as:

- descriptive
- correlational
- explanatory
- exploratory

Descriptive research attempts to describe systematically a situation, problem, phenomenon, service or programme, or provides information about, say, living condition of a community, or describes attitudes towards an issue.

Correlational research attempts to discover or establish the existence of a relationship/ interdependence between two or more aspects of a situation.

Explanatory research attempts to clarify why and how there is a relationship between two or more aspects of a situation or phenomenon.

Exploratory research is undertaken to explore an area where little is known or to investigate the possibilities of undertaking a particular research study (feasibility study/ pilot study).

In practice most studies are a combination of the first three categories.

Research Inquiry Mode: From the process adopted to find answer to research questions the two approaches are:

- Structured approach
- Unstructured approach

Structured approach: The structured approach to inquiry is usually classified as quantitative research. Here everything that forms the research process is objectives, design, sample and the questions that you plan to ask of respondents is predetermined. It is more appropriate to determine the extent of a problem, issue or phenomenon by quantifying the variation. e.g., how many people have a particular problem? How many people hold a particular attitude?

Unstructured approach: The unstructured approach to inquiry is usually classified as qualitative research. This approach allows flexibility in all aspects of the research process. It is more appropriate to explore the nature of a problem, issue or phenomenon without quantifying it. Main objective is to describe the variation in a phenomenon, situation or attitude. e.g., description of an observed situation, the historical enumeration of events, an account of different opinions different people have about an issue, description of working condition in a particular industry.

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Both approaches have their place in research. Both have their strengths and weaknesses. In many studies you have to combine both qualitative and quantitative approaches. For example, suppose you have to find the types of cuisine / accommodation available in a city and the extent of their popularity. Types of cuisine is the qualitative aspect of the study as finding out about them entails description of the culture and cuisine. The extent of their popularity is the quantitative aspect as it involves estimating the number of people who visit restaurant serving such cuisine and calculating the other indicators that reflect the extent of popularity. Different researchers have categorised research into what suits their discipline or field of research the best.

Rajsekar, et. al. (2006) proposes two types of research (basic research and applied research) as detailed below:

Basic Research

Basic research is an investigation on basic principles and reasons for occurrence of a particular event or process or phenomenon. It is also called theoretical research. Study or investigations of some natural phenomenon or relating to pure science are termed as basic research. Basic researches sometimes may not lead to immediate use or application. It is not concerned with solving any practical problems of immediate interest. But it is original or basic in character. It provides a systematic and deep insight into a problem and facilitates extraction of scientific and logical explanation and conclusion on it. It helps build new frontiers of knowledge. The outcomes of basic research form the basis for many applied research. Researchers working on applied research have to make use of the outcomes of basic research and explore the utility of them. Research on improving a theory or a method is also referred as fundamental research. For example, suppose a theory is applicable to a system provided the system satisfies certain specific conditions. Modifying the theory to apply it to a general situation is basic research.

Applied Research

In applied research one solves certain problems employing well known and accepted theories and principles. Most of the experimental research, case studies and interdisciplinary research are essentially applied research. Applied research is helpful for basic research. Research, the outcome of which has immediate application is also termed as applied research. Such research is of practical use to current activity. For example, researches on social problems have immediate use. Applied research is concerned with actual life research such as research on increasing efficiency of a machine, increasing gain factor of production of a material, pollution control, preparing vaccination for a disease, etc. Obviously, they have immediate potential applications.

Other Types of Research

Other types of research include action research (fact findings to improve the quality of action in the social world), explanatory research (searching explanations for events and phenomena, for example finding answer to the question why are the things like what they are?), exploratory research (getting more information on a topic) and comparative research (obtaining similarities and differences between events, methods, techniques, etc.). Within each research group, there are classifications of other research categories

The type of research challenge that an organisation faces will dictate the research strategy, not the other way around. The study's design phase defines which tools to employ and how to utilise them.

Impactful research reduces data bias and enhances trust in the accuracy of obtained data. In experimental research, the desired outcome is often believed to be a design that produces the smallest margin of error. The essential components are as follows:

- Accurate purpose statement
- Techniques to be implemented for collecting and analysing research
- The method applied for analysing collected details
- Type of research methodology
- Probable objections for research
- Settings for the research study
- Timeline
- Measurement of analysis

Proper research design sets your study up for accomplishment. Successful research studies provide perceptions that are accurate and unbiased.



1.1.2 Characteristics of Research

Neutrality: When planning your study, you may need to make assumptions about the data you hope to collect. The research findings should be free of bias and unbiased. Understand many people's perspectives on the final evaluated scores and conclusions and examine those who agree with the produced outcomes.

Reliability: When conducting research on a regular basis, the researcher expects consistent results. To assure the quality of the outcomes, your design should specify how to formulate research questions. Only a dependable design will allow you to get the desired goals.

Validity: A variety of measuring tools are available. However, the only correct measuring tools are those that assist a researcher in gauging results in accordance with the research purpose. The questionnaire created as a result of this design will thereafter be valid.

Generalization: The conclusion of your design should be applicable to the entire population, not just a subset of it. A generalised design indicates that your survey may be done with same accuracy on any segment of a population.

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Because the aforementioned criteria influence how respondents react to research questions, all of the aforementioned qualities should be balanced in a successful design.

To choose which model to use for a study, a researcher must first grasp the many sorts of research designs.

1.1.3 Difference between Research Methods and Research Methodology

The methodical and rigorous search for acceptable knowledge on a certain issue is referred to as research. It entails stating the problem, formulating a hypothesis, gathering and analysing data and reaching conclusions based on the facts and data gathered. And, in order to do so, the researcher employs research methods when conducting research.

Research techniques are frequently mistaken with research methodology, which entails a scientific examination of the research methods in order to identify a solution to the problem at hand. As a result, it seems appropriate to clarify the distinctions between research method and research methodology at this point. Take a look.

Method of Research Definition

The term “research method” refers to all of the approaches used by a researcher to do research in order to address a specific problem. The research method refers to the strategies and procedures used in the course of examining a research subject.

It includes both qualitative and quantitative methods of conducting research operations, such as surveys, case studies, interviews, questionnaires and observation, among others. These are the ways that aid in data collection and research in order to reach specific goals such as hypothesis testing or development. It includes all of the tools and behaviours employed at various levels of the research activity, such as making observations, collecting data, processing data, drawing inferences, making decisions and so on. There are three types of research methods:

- First Category: Data collection methods are described. When present data is insufficient to obtain a solution, such strategies are applied.
- Second Category: Incorporates data analysis techniques, such as identifying patterns and establishing a relationship between data and unknowns.
- Third Category: Consists of methods used to verify the accuracy of the results acquired.

Definition of Research Methodology

Research Methodology, as the name implies, is the study of ways for solving research problems. It is the science of learning how to do research in a systematic manner. It refers to the careful examination of the methods used in the field of study in order to verify that the conclusions reached are legitimate, reliable and credible. The researcher provides an overview of the many procedures he has taken to comprehend the subject at hand, as well as the logic behind the methods used by the researcher during the study. It also clarifies why a certain approach or methodology was chosen

above others, so that the results gained may be evaluated by the researcher or by a third party.

Differences Between Research Methods and Research Methodology

The distinctions between research methods and research methodology are easily discernible on the following grounds:

1. The procedure or methodology used by the researcher to conduct research is characterised as the research method. Research methodology, on the other hand, is a set of approaches employed scientifically to solve a research topic.
2. The research method is nothing more than the behaviour or instrument used in selecting and developing research techniques. In contrast, research methodology denotes the science of analysis, or the proper execution of research.
3. The research technique is concerned with the execution of experiments, tests, surveys, interviews and so on. In contrast, research methodology is concerned with learning many approaches that can be used in the execution of an experiment, test, or survey.
4. The research approach includes a variety of investigation procedures. In contrast, research methodology consists of a comprehensive approach aimed at achieving a certain goal.
5. The goal of the research approach is to find a solution to the situation at hand. In contrast, research methodology aims to apply proper techniques in order to find solutions.

Because the latter is a subset of the former, the scope of research methodology is broader than that of research method. The researcher should be familiar with the research technique as well as the methods in order to fully comprehend the research topic. In a nutshell, research method refers to the approach that can be used to investigate the nature of the world around us. On the contrary, research methodology serves as a foundation for understanding the factors that influence the success of the methodologies used.

Comparison Chart

BASIS OF COMPARISON	RESEARCH METHOD	RESEARCH METHODOLOGY
Meaning	Research Method implies the methods employed by the researcher to conduct research.	Research methodology signifies way to efficiently solving research problems.
What is it?	Behaviour and instrument used in the selection and construction of the research technique.	Science of understanding, how research is performed methodically.
Encompasses	Carrying out experiment, test, surveys and so on.	Study different techniques which can be utilized in the performance of experiment, test, surveys etc.

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Comprise of	Different investigation techniques.	Entire strategy towards achievement of objective.
Objective	To discover solution to research problem.	To apply correct procedures so as to determine solutions.

Check Your Understanding

Fill in the blanks:

1. In _____ one solves certain problems employing well known and accepted theories and principles.
2. The term _____ refers to all of the approaches used by a researcher to do research in order to address a specific problem.
3. Research is the study of _____ ways for solving research problems.
4. To assure the quality of the outcomes, a design should specify how to _____ research questions.
5. The methodical and rigorous search for acceptable knowledge on a certain issue is referred to as _____.
6. The goal of the research _____ is to find a solution to the situation at hand.
7. _____ is the science of learning how to do research in a systematic manner.

Summary

- Research is a method by which one acquires dependable and convenient information about a phenomenon or a process.
- The information might be collected from different sources like experience, human beings, books, journals, nature, etc. Research can lead to new contributions to the existing knowledge.
- Descriptive research attempts to describe systematically a situation, problem, phenomenon, service or programme, or provides information about, say, living condition of a community, or describes attitudes towards an issue.
- Correlational research attempts to discover or establish the existence of a relationship/ interdependence between two or more aspects of a situation.
- Explanatory research attempts to clarify why and how there is a relationship between two or more aspects of a situation or phenomenon.
- Exploratory research is undertaken to explore an area where little is known or to investigate the possibilities of undertaking a particular research study (feasibility study/pilot study).
- The term "research method" refers to all of the approaches used by a researcher to do research in order to address a specific problem.

- Research Methodology, as the name implies, is the study of ways for solving research problems. It is the science of learning how to do research in a systematic manner.
- Research method refers to the approach that can be used to investigate the nature of the world around us. On the contrary, research methodology serves as a foundation for understanding the factors that influence the success of the methodologies used.

Activity

1. How can the research methodology be utilized to give the utmost benefits?

Glossary

- Methodical: done according to a systematic or established form of procedure. subset
- Perspective: a mental view
- Interdisciplinary: relating to more than one branch of knowledge.
- Hypothesis: a supposition or proposed explanation made on the basis of limited evidence as a starting point for further investigation.
- Execution: the carrying out or putting into effect of a plan, order, or course of action.

Questions and Exercises

1. What do you mean by research? Enlighten its implication in modern times.
2. Differentiate between research methods and research methodology.
3. Define the different types of research, clearly pointing out the difference between an experiment and a survey.
4. What do you mean by Research Methodology?
5. Elucidate the uses of Social Research.

Further Readings

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Answers to Check your Understanding

1. applied research
2. "Research method"
3. Methodology
4. formulate
5. research
6. approach
7. Research Methodology

Unit - 1.2: Research Process

Notes

Learning Objectives:

At the end of this unit, you will be able to:

- Understand research criteria and research process
- Know different approaches of research: deductive, inductive, quantitative, qualitative, etc.
- Analyse types of research design

Introduction

The Research Process is a method of many scientific steps in directing the research work. Each step is interweaved with other steps. At first, the research problem is involved in the process. At that point, it improves in the following steps in a progressive manner. In general, an investigator follows a seven-step procedure when doing research. A research proposal is mostly used in research process. It will be there because the proposal accepts the research project regardless of whether you are capable of conducting research. Extant the particular goals and specific purposes of your research correctly while writing a research proposal.



Figure: Image showing processes in research

1.2.1 Research Criteria and Research Process

Good Research Criteria (Scientific Research) is expected to meet the following criteria:

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1. The research's objective should be clearly specified and common notions should be used.
2. The study technique should be sufficiently described. Detail that will allow another researcher to replicate the findings for additional progress, while maintaining the continuity of what has already been established been accomplished
3. The study procedure should be thoroughly designed. Aimed to produce as objective results as feasible.
4. The researcher should report shortcomings in the research with complete candour and procedure design and assess its impact on the findings.
5. Data analysis should be thorough enough to expose its significance. The relevance of the findings, as well as the methods of analysis employed, should be evaluated. The data's authenticity and dependability should be assessed and carefully examined.
6. Conclusions should be limited to those that are supported by data. The study and confined to those for which the data provide an answer adequate foundation.
7. Increased trust in research is warranted if the researcher is experienced, has an excellent research reputation and is a person of integrity.

In other words, we can list the characteristics of an excellent research as follows:

1. Good research is systematic: This indicates that research is conducted in a methodical manner. Structured with certain stages to be completed in a specific order according to a well-defined set of rules. The study design does not preclude creative thinking.
However, it categorically rejects the use of hunches and intuition in reaching conclusions.
2. Good research is logical: This means that the research is directed by logic. The laws of logical thinking and the inductive logical process in conducting research, deduction and reasoning are quite useful.
The process of reasoning from a component to the whole is known as induction. Deduction, on the other hand, is the process of reasoning from a given premise to a conclusion derived from that very premise. Indeed, in the context, logical thinking makes research more meaningful of making decisions.
3. Effective research is empirical: It signifies that the research is relevant. Relates mostly to one or more components of a real-world event and deals with tangible facts that serves as a foundation for research's external validity results.
4. Good research can be replicated: This feature enables study. The study's findings must be replicated in order for the findings to be validated a solid foundation for decision-making.

Research process can be followed in the following steps:**Step 1: Classify and Define the Research Problem**

The initial step in the process is to identify a problem or grow a research question. The research problem may somewhat hinder what the agency classifies as a problem, such as, more or less knowledge or information that is needed by the activity or the desire to identify a recreation tendency nationally. However, the research problem originates up with the continuing phenomenon or issues.

Frame Your Question:

- Your investigation may start as a general idea or an exact question, statement or thesis.
- Know what you want to pay attention to before you begin. A research problem is a statement about an area of concern, a problem that needs to be fixed, a problem that needs to be solved, or a perplexing question that exists in scholarly literature, theory, or practise and connects to the need for expressive, receptive and attentive exploration.

The research problem is typically posed in the form of a question in nearly all social science self-controls. A research problem does not state how to do something, it suggests a vague or broad proposition, or poses a value question.

Step 2: Review the Literature

Once the research problem is well-known and defined, the next step is to assess the existing research. The researcher must learn more about the topic through investigation. To do this, the researcher must review the literature related to the research problem. This step provides fundamental knowledge about the problem area. The review of literature also instructs the researcher about what studies have been accompanied in the past, how these studies were directed and the conclusions formulated by that research.

Get Background Material:

- Read almost all your topic using websites rather than encyclopaedias.
- It familiarises you with the subject, assists you in focusing on the most important aspects and can assist you in deciding whether to extend or narrow your emphasis.
- These bases often include bibliographies that you can “piggyback” to discover more sources on your topic.

Review of literature helps the researcher to determine the type of statistics that can be used in the research as well as the past and current trends in the research topic. The information exposed during this phase helps the programmer fully comprehend the magnitude of the problem, recognize the future aspects of the research and ascertain a strategy to combat the effects/impacts of the research topic.

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Step 3: Formulating a Hypothesis

In this step, the researcher types the problem precisely:

- The examination work is subject focused and refined.
- Then the researcher steps on to how the problem would be approached. The setting of the research problem can elect to formulate a fixed hypothesis.
- A hypothesis is tested. Effective research work formulates a hypothesis in such a way that the collected factual data determination provides evidence that either supports or refutes the hypothesis. The preparation of hypotheses in research will make you more knowledgeable.
- In the end, the hypothesis tries into a practical theory. The following approach should be used by researchers to develop working hypotheses:
 - a. Consultations with colleagues and experts about the problem, its origin and the goals of finding an answer;
 - b. Examination of data and records, if obtainable, concerning the tricky for possible trends, peculiarities and extra clues;
 - c. Evaluation of similar studies in the area or of the educations on similar problems; and
 - d. Exploratory personal research consists of original field meetings on a limited scale with attentive gatherings and additional individuals in order to risk greater insight into the practical aspects of the problem.

Step 4: Research Design

The researcher then needs to find out a research design. Research design agrees how the research materials will be composed. One or more research methods, intended for, experiment, survey, interview, etc. are chosen contingent on the research objectives.

- In some research circumstances, a survey may be suitable. In new facts, interviews or case studies or reflection might be more appropriate.
- Find the techniques to Choose a Research Design. Research Design truly provides insights into "how" to conduct study using a particular Research Methodology. Basically, every single researcher has a list of research queries that need to be assessed that can remain with research design.

In additional words, the function of research design is to arrange for the collection of relevant evidence with nominal expenditure of effort, time and money. Nonetheless how all these can be achieved be contingent mainly taking place on the research purpose.

Research purposes may be assembled into four categories:

- a. Exploration
- b. Description

- c. Diagnosis and
- d. Experimentation.

There are numerous research designs, for instance:

- Experimental and hypothesis challenging
 - Experimental designs can be moreover informal designs (such as before-and after deprived of control, after-only by means of control, before-and-after with control)
 - Formal designs (for example completely randomized plan, randomized block design, Latin square design, simple too complex factorial designs), out of which the researcher necessity select one for his individual project.
- Non-experimental assumption testing

Step 5: Move out the Research Procedure

Once the research design is decided, the assistant data records information. The researcher ensues with the research. Practical difficulties may come up in this stage. For example, the research technique may not suit properly. The interviewer might be disinclined to let carry out the research as planned. Moreover, a false understanding could potentially bias the outcome of the study. So, as soon as you collect data, you need to know the current techniques of data collection in order to collect necessary and relevant info with regard to research.

Step 6: Preparing Research Results

Now work out the insinuations of the data you gathered. Your tasks are not over yet. Rather problems might just begin! It is barely easy to clear out the inferences of the gathered materials. While it is possible to elucidate the research questions, approximately investigations are less conclusive. So, interpret your exploration results in order to report the findings. No matter what type of study you're doing, you'll reach a point where your head is filled of ideas that came from your analysis. You should write them down dejectedly as they come to you. Now you must combine all of those elements and ideas into a written narrative that makes sense to the reader and adequately represents your mission.

Step 7: Reporting Research Findings

The final step of the research process plan is to report the research findings. Define the significance of the research study. Work out how they relate to the previous research findings. Usually, the research report published as a journal, article, or book. This is the most recent stage in terms of the individual research project. Mostly, a research report converses questions that persisted unanswered and suggest auxiliary research in the forthcoming in general.

Notes

Notes

1.2.2 Different Approaches of Research: Deductive, Inductive, Quantitative, Qualitative, etc.

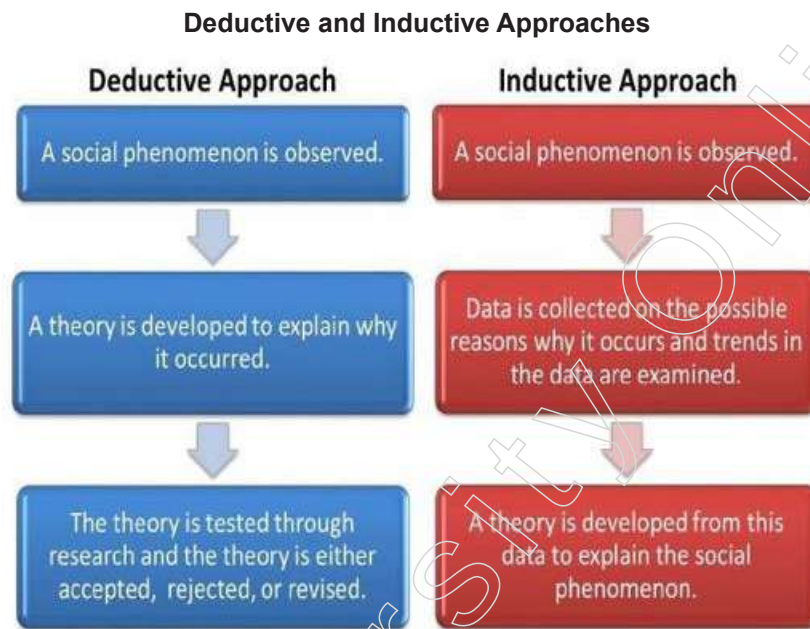


Figure: Image showing approaches made under deductive and inductive research approach

a) Inductive Research Approach

It is common to conduct inductive research when there are few or no existing works on a topic because there is no theory to test. The inductive method consists of three stages which can be explained with the given examples as following:

1. Observation
 - o A low-cost commercial airline flight is delayed
 - o Dogs A and B have fleas
 - o Elephants be contingent on water to exist
2. Detect a pattern
 - o Additional 20 flights from low-cost airlines are delayed
 - o All observed dogs have fleas
 - o All observed animals depend on water to exist
3. Grow a theory
 - o Low-cost airlines continuously have delays
 - o All dogs take fleas
 - o All biological life hinge on water to exist

Restrictions of the Inductive Approach

A conclusion drawn on the foundation of an inductive method cannot once be proven, then again it can be invalidated.

Example

You witness 1000 flights from low-cost airlines. Altogether of them experience a delay, which is in line with your theory. However, you can never demonstrate that flight 1001 will also be overdue. Still, the larger your dataset, the additional reliable the conclusion.

b) Deductive Research Approach

As soon as you conduct deductive research, you always begin with a theory (the outcome of inductive research). Reasoning deductively means challenging these theories. You cannot conduct deductive research if there is no nearby theory.

The deductive research approach consists of four stages which can again be explained with following examples:

1. Start with a present theory.
 - o Low-cost airlines always have delays
 - o All canines have fleas
 - o All biological life be subject to on water to exist
2. Formulate a hypothesis based on existing theory.
 - o If person along for the ride fly with a low-cost airline, then they will always experience postponements
 - o All pet dogs in my bed sitting room building have fleas
 - o All land-living mammals depend on water to exist
3. Bring together data to test the hypothesis.
 - o Collect flight numbers of low-cost airlines
 - o Examine all dogs in the building for fleas
 - o Examine all land mammal species to understand if they are dependent on water
4. Analyse the results: Does the data discard or support the hypothesis?
 - o 5 out of 100 flights of low-cost airlines are not hindered = reject hypothesis
 - o 10 out of 20 dogs didn't have fleas = cast-off hypothesis
 - o All land mammal species depend on water = upkeep hypothesis

Restrictions of a Deductive Approach

The conclusions of deductive reasoning can only be correct if all the premises set in the inductive training are true and the footings are clear.

Example

- All dogs abstain fleas (premise)
- Benno is a dog (premise)
- Benno takes fleas (conclusion)

Notes

Qualitative and Quantitative Approach

Any researcher would need an understanding of the research design types to see which is more appropriate for the study or which one brings to fruition the most accurate results. To do this, there are three broad ways to approach the design, as we will discuss below.

It is imperative to recognize that systematic opinion and testing can be accomplished using a wide variety of methods. Numerous people think of scientific inquiry sternly in terms of laboratory experimentation. Nevertheless, it is neither possible nor necessary to study all phenomena of interest below well-ordered laboratory conditions.

The strategy of any study begins with the selection of a theme and a research methodology. These initial decisions reflect conventions about the social world, how skill should be conducted and what constitutes legitimate problems, answers and criteria of “proof.” Poles apart approaches to research embrace both theory and method. Two general methods are widely recognized:

- Quantitative research
- Qualitative research

I. Quantitative Research

The first strategy is to design the study from a quantitative standpoint. This method is best suited for a research purpose in which actionable knowledge is linked to a statistical conclusion. The quantitative method, as the name implies, frames numbers as a representation of data. Because numbers are objective, a quantitative approach is required, for example, when making data-driven business decisions when profit margins are dependent on the smallest of details and/or figures.

It is sometimes referred to as a “top-down” strategy (Burney, 2008) and it entails removing pieces from the general to the specific. In this manner, the researchers reach a conclusion based on the premises or available facts. Because quantitative data can be quite large, statistical software and other services, the most popular of which are illustrated below, are used to analyse it.

In a quantitative research technique, objectivity is highly valued. As a result, researchers go to tremendous pains to ensure that their findings are not tainted by their own presence, behaviour, or expectations. One option is to conduct self-examination to ensure that their methodology or conclusions are devoid of unjustified biases or presumptions. A quantitative approach to research must imply that the design must account for and/or control for external variables; they can never be totally eliminated and must thus be acknowledged in the interpretation of the findings.

Researchers frequently begin with one or more hypotheses and the correlations between the variables they intend to study when approaching a design with quantitative aim. The design must also consider stronger forms of methodology and instruments used to measure and validate the obtained data, a clear plan of action, a statistical approach for data analysis and a reliable way to display the results.

It is an autopsy into an identified problem, founded on testing a theory, measured with numbers and examined using statistical techniques. The goal of quantitative

research is to define whether the predictive generalizations of a theory hold true. A study based on a qualitative procedure of inquiry, on the other hand, seeks to consider a social or human problem from multiple perspectives. Qualitative research is showed in a natural setting and includes a process of building a multifaceted and holistic depiction of the phenomenon of interest.

The choice of which research approach to use in a hypothetical study should be based on the topic of interest, the resources available, the researcher's abilities and training and the research's target audience. Even though some research may combine quantitative and qualitative procedures, there are major differences in the assumptions that underpin these approaches in their 'pure' manner, in addition to the data gathering and analysis actions performed.

Quantitative Research Methods

Quantitative research methods can be used for:

- Descriptive

In descriptive research, you simply want an overall summary of the variables in your study.

- Correlational

Correlational research looks into the relationships between the variables in your study.

- Experimental research

In experimental research, you examine whether there is a cause-and-effect relationship between variables in a systematic manner.

Correlational and experimental research can both be used to statistically test hypotheses or predictions. Based on the sampling method used, the results may be generalised to larger populations.



Figure: Image depicting methods in quantitative research

Notes

Assumptions for Original Quantitative Methods

- Realism is objective, “out there,” and self-governing of the researcher -- therefore reality is somewhat that can be studied objectively;
- The investigator ought to remain distant and self-governing of what did you say is being researched;
- The morals of the researcher do not inhibit with, or grow part of, the research - research is value-free;
- Research is generated primarily on deductive forms of logic in addition theories and hypotheses are proved in a cause-effect order; and
- The goal is to produce generalizations that donate to theory that empower the researcher to predict, elucidate and comprehend some occurrence.

Table: Methods of Quantitative Research

Methods of Research	How to Use	Illustrations
Experiment	Controlling or manipulating an independent variable in order to assess its impact on a dependent variable.	To see if a procrastination intervention can reduce procrastination in college students, randomly assign equal-sized groups to either a procrastination intervention or a comparable task. After the intervention, you compare the groups' self-ratings of procrastination behaviours.
Survey	In person, over the phone, or online, pose questions to a group of people.	To investigate first-year international college students' experiences with culture shock, you distribute questionnaires with rating scales.
Systematic Observation	Identify an interesting behaviour or occurrence and observe it in its natural environment.	To investigate college classroom participation, you sit in on classes and observe them, counting and recording the prevalence of active and passive behaviours among students from various backgrounds.
Secondary Research	Data gathered for other purposes, such as national surveys or historical records, should be collected.	You collect relevant questionnaire data from widely available longitudinal studies to assess whether attitudes toward climate change have changed since the 1980s.

Types of Quantitative Methods

1. Experiments

True testing is characterized by random task of topics to experimental conditions and the practice of investigational controls.

2. Quasi-Experiments

Quasi-experimental readings part almost all the features of experimental designs without that they include non-randomized assignment of themes to experimental conditions.

3. Surveys

Surveys comprehend cross-sectional and longitudinal lessons by means of questionnaires or talks for data collection through the intent of approximating the physiognomies of a large population of interest based on a smaller sample on or after that population.

Analysis of Quantitative Data

Once the data has been collected, it may need to be processed before it can be analysed. Survey and test data, for example, may need to be converted from words to numbers. Then, using statistical analysis, you can answer your research questions.

Descriptive statistics provide a summary of your data and include averages and variability measures. You can also use scatter plots, graphs and frequency tables to visualise your data and check for any outliers or trends. Based on your data, you can use inferential statistics to make predictions or generalisations. You can either test your hypothesis or estimate the population parameter using your sample data.

Illustration of Descriptive and Inferential Statistics

You believe that first-year college students procrastinate more than fourth-year students. Using 7-point self-rating scales, you collect data on the two groups' procrastination levels. To begin, you use descriptive statistics to create a data summary. You calculate the mean (average) and mode (most common rating) of procrastination for the two groups and plot the data to observe if there are any outliers.

Following that, you use inferential statistics to check your hypothesis. You discover a significant difference and support for your hypothesis using a t-test to compare the mean ratings of the two groups.

Benefits of Quantitative Research

Quantitative research is frequently employed in order to standardize data collection and generalise findings. This approach has the following advantages:

i) Replication

Because of standardised data collection protocols and tangible definitions of abstract concepts, the study can be repeated.

ii) Direct comparisons of outcomes

The study can be replicated in different cultural settings, at different times, or with different groups of participants. The outcomes can be statistically compared.

Notes

iii) Large numbers of samples

Through quantitative data analysis, data from large samples can be processed and analysed using dependable and consistent procedures.

iv) Validation of hypotheses

Using formalised and established hypothesis testing methods necessitates careful consideration and reporting of your research variables, data collection, predictions and testing methods prior to reaching a conclusion.

II. Qualitative Research

A qualitative research approach, on the other hand, seeks to establish a link between acquired data and observations. It generates a multitude of raw data, whether gathered by statistical means or otherwise, because it is about recording, analysing and uncovering the web of interconnection that underpins related issues. As a result, the nature of a qualitative technique lends itself nicely to exploratory research (Blaikie, 2009).

A qualitative approach, in contrast to quantitative approaches, is its polar opposite. It approaches the study's conclusion in an inductive manner. It is also known as a "bottom-up" method since it infers meaning or looks for patterns based on the data that they have gathered.

In the social sciences, qualitative research is widely used. It is concerned with studying and revealing the social constructs that human civilizations are framed in, as well as considering the significance of the human experience through the lens of beliefs, behaviours and emotions. A qualitative approach is so concerned in acquiring an understanding of what works as seen and felt by the participant (that is, subjectively).

While qualitative research employs some form of quantitative analysis, the method through which it collects data allows for greater flexibility. Unlike quantitative methods, which collect and aggregate data in computer-readable forms to be crunched later, qualitative approaches record data from observation and interaction with participants in textual format. Furthermore, the methodologies used vary greatly, with open-ended, exploratory and broad-ranging processes and few, if any, assumptions on the side of the researchers in order to make the data as pure and accurate as possible.

Finally, quantitative and qualitative methodologies should not be viewed as precise, polar opposites or categories. Rather, they should be viewed as distinct extremes of a continuum or vast system (Creswell, 2015). For example, a study paper may be more quantitative than qualitative, or vice versa (Creswell & Creswell, 2018).

Qualitative research entails gathering and analysing non-numerical data (e.g., text, video, or audio) in order to better understand concepts, opinions, or experiences. It can be used to gain detailed insights into a problem or to generate new research ideas. Qualitative research is the inverse of quantitative research, which involves the collection and analysis of numerical data for statistical analysis. Qualitative research is commonly used in the social sciences and humanities, such as sociology, anthropology, health sciences, education, history and so on.



Figure: Methods in qualitative research

Assumptions Underlying Qualitative Methods

- Various authenticities exist in any given situation - the researcher's, individuals who were actually investigated and the reader or audience who interpreted the results; these multiple perspectives, or voices, of informers (i.e., subjects) are contained within in the study;
- The researcher interacts with those studies while also working hard to reduce the distance between the researcher and the people being studied;
- The researcher is fully aware of and acknowledges the significance of the research; and research is context-bound;
- Exploration is based on inductive systems of logic; types of interest occur from informants (subjects), rather than being identified a priori by the researcher;
- The goal is to find and identify patterns or theories that can help explain a phenomenon of interest; and
- Determining truthfulness entails verifying information with informants or "triangulating" between disparate sources of evidence (e.g., collecting information from different sources).

Methodologies for Conducting Qualitative Research

To understand how people perceive the world, qualitative research is used. Although there are many methodologies to qualitative research, they all have one thing in common: they are flexible and focus on retaining rich meaning when interpreting data. Ethnography, grounded theory, action research, narrative research and phenomenological research are all examples of common approaches. They have some similarities, but their goals and perspectives are quite different.

Notes

Notes

Table: Methodologies for qualitative research

Methodology	What does it entail?
Ethnography	To understand the cultures of groups or organisations, researchers immerse themselves in them.
Grounded Theory	Inductive theory development is used by researchers to collect rich data on a topic of interest.
Action Research	To drive social change, researchers and participants work together to connect theory and practice.
Narrative Research	Researchers study how stories are told in order to better understand how participants perceive and interpret their experiences.
Phenomenological Research	A phenomenon or event is investigated by researchers by describing and interpreting the lived experiences of participants.

Three General Types of Qualitative Methods:

1. Case Studies

In an instance study, the researcher discovers a single entity otherwise phenomenon ('the case') bounded by time and activity (e.g., a program, event, institution, or social group) and collects detailed information through a diversity of data collection processes over a sustained period of time. The instance study is an evocative record of an individual's understandings and/or behaviours kept by an outside observer.

2. Ethnographic Studies

In ethnographic research, the researcher studies an intact cultural assembly in a natural setting over a specific period of time. A cultural cluster can be any group of individuals who segment a common social know-how, location, or other social representative of interest - this might range from an ethnographic learning of women in distress in crisis shelters, to children in foster conservation, to a study of a cultural gathering in Africa.

3. Phenomenological Studies

In a phenomenological education, human experiences are inspected over and done with the detailed explanation of the people existence studied - the objective is to comprehend the 'lived experience' of the individuals being studied. This approach involves researching a small group of people intensively over a longer period of time.



Figure: Difference between qualitative and quantitative research

Example of Qualitative Research

You decide to use an ethnographic approach to investigate the culture of a large tech company. You work for the company for several months and collect data using a variety of methods, including:

- You take field notes and reflect on your own experiences with the company culture.
- You send out open-ended surveys via email to employees in all the company's offices to see if the culture varies by location.
- You conduct in-depth interviews with employees in your office to learn more about their experiences and points of view.

Benefits of Qualitative Research

Qualitative research frequently attempts to preserve the perspective and voice of participants and it can be adjusted as new research questions emerge. Qualitative research is useful for the following purposes:

i) Flexibility

As new ideas or patterns emerge, the data collection and analysis process can be modified. They are not predetermined in advance.

ii) Environments that are natural

Data is collected in real-world settings or in a naturalistic manner.

iii) Valuable insights

Extensive descriptions of people's experiences, feelings and perceptions can be used to design, test and improve systems or products.

iv) New idea generation

Researchers can discover novel problems or opportunities as a result of open-ended responses that they would not have considered otherwise.

Check Your Understanding

Fill in the blanks:

1. The process of reasoning from a component to the whole is known as _____.
2. The study's findings must be _____ in order for the findings to be validated a solid foundation for decision-making.
3. The _____ is typically posed in the form of a question in nearly all social science self-controls.
4. One or more research methods, intended for, experiment, survey, interview, etc. are chosen contingent on the research _____.
5. The strategy of any study begins with the selection of a theme and research _____.

Notes

Notes

6. In a quantitative research technique, _____ is highly valued.
7. _____ and _____ can both be used to statistically test hypotheses or predictions.
8. _____ readings part almost all the features of experimental designs without that they include non-randomized assignment of themes to experimental conditions.
9. _____ research is commonly used in the social sciences and humanities, such as sociology, anthropology, health sciences, education, history and so on.
10. Ethnography, grounded theory, action research, narrative research and phenomenological research are all examples of common _____.

Summary

- The Research Process is a method of many scientific steps in directing the research work. Each step is interweaved with other steps.
- The study's findings must be replicated in order for the findings to be validated a solid foundation for decision-making.
- The research problem may somewhat hinder what the agency classifies as a problem, such as, more or less knowledge or information that is needed by the activity or the desire to identify a recreation tendency nationally.
- Review of literature helps the researcher to determine the type of statistics that can be used in the research as well as the past and current trends in the research topic.
- Effective research work formulates a hypothesis in such a way that the collected factual data determination provides evidence that either supports or refutes the hypothesis.
- One or more research methods, intended for, experiment, survey, interview, etc. are chosen contingent on the research objectives.
- The deductive research approach consists of four stages and the inductive method consists of three stages.
- Researchers frequently begin with one or more hypotheses and the correlations between the variables they intend to study when approaching a design with quantitative aim.
- Correlational and experimental research can both be used to statistically test hypotheses or predictions.
- Quasi-experimental readings part almost all the features of experimental designs without that they include non-randomized assignment of themes to experimental conditions.
- Survey and test data, for example, may need to be converted from words to numbers. Then, using statistical analysis, you can answer your research questions.

- Using formalised and established hypothesis testing methods necessitates careful consideration and reporting of your research variables, data collection, predictions and testing methods prior to reaching a conclusion.
- Ethnography, grounded theory, action research, narrative research and phenomenological research are all examples of common approaches.

Activity

1. How can we ensure the quality in research?

Glossary

- Naturalistic: derived from real life or nature, or imitating it very closely.
- Distress: extreme anxiety, sorrow, or pain.
- Informant: a person who gives information to another.
- Statistically: according to or by means of statistics.
- Physiognomy: a person's facial features or expression, especially when regarded as indicative of character or ethnic origin.
- Presumption: an idea that is taken to be true and often used as the basis for other ideas, although it is not known for certain.
- Magnitude: the great size or extent of something.
- Ascertain: find (something) out for certain; make sure of.

Questions and Exercises

1. "Empirical research in India in particular creates so many problems for the researchers". – Illuminate
2. State the problems that are usually faced by Social Researchers.
3. Transcribe short notes on: (1) Design of the research project; (2) Ex post facto research; (3) Motivation in research; (4) Objectives of research; (5) Criteria of good research; (6) Research and scientific method.
4. Describe its significance and describe compile the different types of research.
5. Clarify the steps in research process with the help of flow chart of the research process.
6. Elucidate the uses of Social Research.

Further Readings

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Answers to Check Your Understanding

1. induction
2. replicated
3. research problem
4. objectives
5. methodology
6. objectivity
7. Correlational, experimental research
8. Quasi-experimental
9. Qualitative
10. approaches

Unit - 1.3: Types of Research Design

Notes

Learning Objectives:

At the end of this unit, you will be able to:

- Understand pure & applied Research Design
- Know causal & correlational Research Design
- Examine cross-sectional & longitudinal Research Design
- Analyse experimental, semi-experimental & non-experimental Research Design
- Evaluate descriptive & exploratory Research Design

Introduction

A research design, also known as a research strategy, is an approach for answering a series of questions (McCombes, 2019). It is a framework that contains the methods and procedures for gathering, analysing and interpreting data. In other words, the study design outlines how the researcher will investigate the research's fundamental problem and is thus a component of the research proposal.

The design of this study determines the type of data collected and, as a result, the results. Research design, depending on the type, also determines the other constituent parts of a study, such as variables, hypotheses, experiments, technique and statistical analysis (Creswell et al., 2018).

However, many people mix up study design and research technique. The distinction is that the former offers an overview of how to tackle the problem, whilst the latter specifies how to implement the design. Both are essential in constructing a thesis statement. If you need assistance with this aspect of your research, you can refer to our advice on how to construct a thesis statement for a research paper.

Goals

The goal of excellent research design is to make the data address the research problem as clearly, accurately and unbiasedly as feasible. Arriving at outcomes entails correctly specifying the type of results needed to test a theory, evaluate, or characterise a phenomenon. Without doing so, interpreting data would appear weak and flimsy and it will most likely not address the problem that the researcher set out to solve.

Design is rudimentary in any study endeavour since everything finally emanates from the selected design and because this option is the most closely tied to the scholar's beliefs and research objectives (Vogt et al., 2012). When the conclusion is considered to have the least amount of bias, the research strategy has served its aim. One of its goals is to create research designs with the smallest margin of error possible.

To accomplish this, sound research design adheres to the following basic tenets:

- Identify the problem.
- Justify why it is a problem.
- Review literature surrounding the problem.

Notes

- Specify hypotheses regarding the problem.
- Describe the data to test the hypotheses and how it will be collected.
- Define how this data will be interpreted and analysed to find out if the hypotheses are true or false.

The sort of study design is one of the most important factors in determining the quality, relevance and accuracy of a result. As a result, before beginning to outline a proposal, it is usually a good idea to differentiate the type of research by incorporating it in the research design.

There are several approaches to a research design type, however the literature has not always been clear on these approaches (Abutabenjeh, 2018). In fact, previous publications have made distinguishing between types, methods and approaches nearly impossible, with some older references referring to fixed and flexible designs (Bouma, 1994).

There are two major types of research designs.

Taking its dynamics into account, the research design is divided into two perspectives: quantitative research design and qualitative research design. Furthermore, there are four main aspects of research design: reliability, neutrality, validity and generalisation. Furthermore, a researcher should have a firm grasp on how their project might be incorporated into the research design. Let us define Quantitative and Qualitative Research Designs:

Design of Quantitative Research

In Quantitative Research Design, a researcher explores the numerous variables in a project while including numbers and statistics to interpret its findings. The primary method of data gathering, measurement and meta-analysis is the use of visuals, figures and pie charts (it is information about the data by the data).

Design of Qualitative Research

This style of research is diametrically opposed to quantitative research design. It is explanatory in nature and it is continually looking for solutions to “What’s” and “How’s.” It primarily focuses on why a particular theory exists and what the respondent’s response to it would be. This enables a researcher to reach a conclusion based on accurate findings. Case studies are primarily employed in Qualitative Research Design to comprehend various social difficulties.

The 5 Different Types of Research Designs

- Now that we’ve established the basic categories of research, we can separate quantitative and qualitative research into the four major types of research designs: Descriptive Research Design
- Correlational Research Design
- Experimental Research Design
- Diagnostic Research Design
- Explanatory Research Design

These 5 types of Research Designs are measured the closest and exact to true experiments and are chosen in terms of accuracy, relevance as well as quality.

To make designing research as modest as possible, we have broken down the types of research design into major ones, as explained below.

1.3.1 Pure & Applied (with Examples)

Pure study seeks to solve issues like, “How do gases behave?” Applied research would be involved in the process of designing particular preparation for a gas in order for it to be generated and delivered in a cost-effective and efficient manner. This distinction appears to be simple, yet it is not always possible to distinguish between what is “pure” and what is “applied.”

Applied Research

Applied research is a research methodology that develops practical solutions to specific issues, whereas pure research is a research approach that tries to broaden knowledge in a field of study. This indicates that applied research is problem-focused, whereas pure research is knowledge-focused.

Applied research, is primarily concerned with finding solutions to issues through the collection and analysis of empirical data in order to produce valid study results. Pure research, on the other hand, tries to expand the boundaries of knowledge by developing new theories or revising old ones.

Applied research is defined primarily by its emphasis on delivering a practical solution to a specific problem, whereas pure research is defined primarily by its emphasis on advancing knowledge. In other words, pure research is theoretical, whereas applied research is practical.

Pure Research

Pure research leads to the acquisition of new knowledge as well as the expansion of current knowledge, whereas applied research does not result in the acquisition of new knowledge. Applied research is solution-specific since it only focuses on using knowledge to solve existing problems.

Discoveries from pure research have mostly been responsible for breakthroughs in various fields of study, but findings from applied research have primarily been effective in tackling specific research challenges. As a result, pure research is inherently universal, whereas applied research is inherently confined.

Pure research is explanatory and analytical in nature, whereas applied research is action-oriented and synthetic in nature. Pure research is solution-focused and primarily concerned with knowledge development rather than the implementation of research findings, which is the focus of applied research.

Applied research assists companies and individuals in solving specific problems, as opposed to pure research, which focuses solely on advancing knowledge without providing answers to current problems. This does not, however, deny the reality that findings from Pure research can be used to propose solutions to issues.

Notes

Unlike pure research, which is somewhat subjective in its approach, applied research is a form of investigation that is objective. Applied research is typically unbiased since it produces results by exposing empirical evidence to standardised scientific techniques, making it a more valid research method.

Examples of “Pure” Research

Many “pure” studies are of the “what is this?” or “how does it work?” sort. Many instances can be found in the early history of chemistry. The composition of matter (earth? air? fire? water? all of the above?) was a point of contention among ancient Greek philosophers. They didn’t intend to do anything with their newfound knowledge; they simply wanted to know.



Ancient Greek philosophers. “A Greek Philosopher and His Disciples”, Antonio Zucchi

Studies on the elements were primarily “pure” scientific types of experiments (particularly after Mendeleev’s periodic table was released). Is this a real thing? What are its characteristics? The scientists had no practical application in mind, but they were intrigued by the environment around them.

Examples of “Applied” Research (Technology)



Notes

Today, there is a lot of “applied” research going on. In general, no new scientific principles are found; instead, existing information is applied to the development of a new product. The use of x-rays in medicine is a good example of this type of study. Wilhelm Röntgen discovered how to create x-rays using a cathode ray tube in the late 1800s. He noticed that this new light had the ability to pass through the body. Recognising this, Röntgen anticipated that x-rays would be extremely useful in identifying and determining disease and injury in the body.

Röntgen is regarded as the inventor of the X-ray. While dealing with prior scientists' tests and techniques, Röntgen discovered that even with secured chambers for light, an unknown energy was released that reacted with the phosphorescent substances he was employing in his research. In 1901, he was awarded the Nobel Prize in Physics for his discovery and subsequent articles. He tested the new beams' ability to travel through the body using his wife's hand.

Examples of “In-Between”

It might be difficult to distinguish between pure and practical research at times. What may begin as a simple question may lead to some really useful information. Scientists examining the biology of a disease-causing microbe may quickly discover information that suggests a method to create a drug that would neutralise the microorganism. The substance may be used to learn more about biochemistry, but it may also be utilised to treat the disease.

Haemoglobin is a protein found in red blood cells that distributes oxygen throughout the body. Haemoglobin was investigated by scientists primarily to learn how it functioned. This research resulted in a better comprehension of how the protein changes form when oxygen binds to it. This knowledge was then used to assist individuals suffering from sickle cell anaemia, a condition characterised by a defective haemoglobin structure that causes haemoglobin molecules to cluster together when oxygen exits the protein. The study of protein structure alone resulted in a better understanding of a common ailment and paved the way for the creation of remedies.

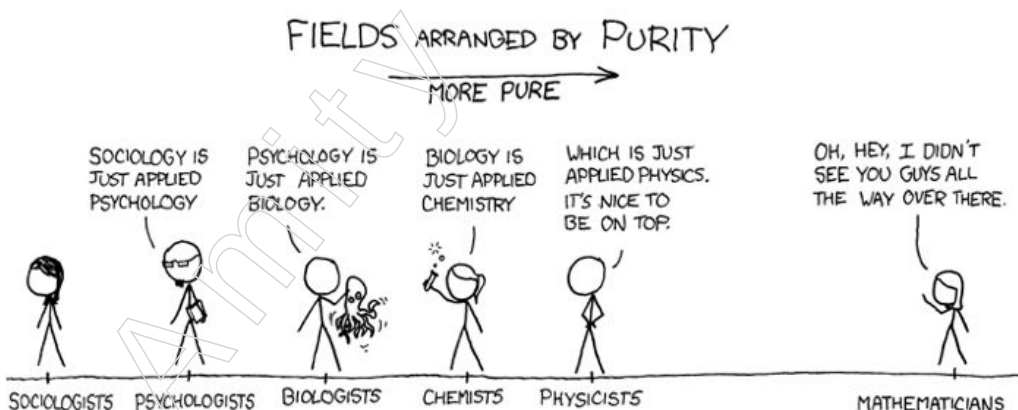


Figure: What is an applied science or research is relative perspective depending on the field?

- Examples in Education

Applied research in education is utilised to enhance educational processes by

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providing practical answers to pedagogical challenges. Pure research, on the other hand, is utilised in education to establish new pedagogic theories that explain various behaviours by students and teachers within the learning environment.

Examples of Pure research in education include:

1. How does the human memory function?
2. How do youngsters learn to speak different languages?

Among the many examples of applied education research are:

1. A research designed to pique pupils' interest in religious subjects.
 2. A research to increase teacher-student interaction in the classroom.
- Examples in Health

Applied research assists health and medical professionals in developing evidence-based solutions to major health issues. Pure research, on the other hand, assists medical practitioners in gaining insights into various health difficulties such as the genesis and symptoms of diseases and infections, which can be valuable in establishing a cure for such conditions.

Examples of pure health research include:

1. An inquiry of the Human Papilloma Virus's secondary symptoms (HPV).
2. An investigation into diarrhoea symptoms.

Examples of applied health research include:

1. An inquiry into the medicinal qualities of mushrooms.
 2. An examination of the negative effects of alcohol usage.
- Examples in Psychology

In industrial psychology, applied research is particularly valuable for developing answers to problems linked to workplace behaviour, organisational policy and personnel recruiting processes. Pure psychology study is utilised to obtain insights into various psychological circumstances in order to better comprehend these behaviours.

Pure research in psychology examples include:

1. What causes panic attacks?
2. What are the signs and symptoms of an anxiety disorder?

Applied research in psychology includes the following examples:

1. What therapy options are available for anxiety disorders?
2. What are some strategies for increasing employee productivity in the workplace?

Different Uses of Applied and Pure Research

Pure research is beneficial for acquiring unique information on an idea,

phenomena, or topic of study, whereas applied research is useful for discovering practical answers to identified issues. Pure research investigates the functions and characteristics of newly found phenomena in order to better comprehension of these notions, hence fuelling scientific and technological advances. Applied research, on the other hand, aids in the development of answers to specific problems or the development of new technology.

- Research Purpose

Curiosity and the desire to expand one's knowledge in various domains drive pure research. Applied research, on the other hand, is motivated by the desire to provide answers to various in order to solve a problem.

This implies that applied research is primarily concerned with reviewing empirical evidence for answers, whereas pure research investigates data samples to learn more about them. This type of information enhances one's understanding of the subject.

- Research Context

Pure research is undertaken in a controlled study environment, such as a laboratory, whereas conceptual research is conducted in a non-sterile or constrained real-life situation. The sterile study atmosphere in Pure research allows the researcher to rigorously monitor the research subjects' behaviour and traits.

However, in applied research, the researcher permits the variables of the study to freely interact with one another in an unlimited environment where other variables or third factors may intervene. This enables the researcher to have a more comprehensive understanding of the research problem and come at credible and effective solutions.

- Scope of the Research

In general, applied research has a narrower scope than pure research. This is due to the fact that, unlike pure research, which may be applied to a wide range of concepts, applied research concentrates on a single subject and its research outputs are primarily relevant to that subject.

Pure research can be considered a more universal comprehensive search than conceptual research because it deals with numerous notions across different subject matters. Fundamental research investigates knowledge in numerous dimensions in order to collect new data and improve an existing body of knowledge.

- Theory Development

Pure research attempts to develop theoretical approaches research findings and, as a result, improve the body of knowledge, whereas applied research aims to develop research findings that help solve practical issues. Pure research is concerned with principles and theories, whereas applied research is concerned with solutions.

Pure research typically tries to develop hypotheses and generalisations that explain an idea, subject, or phenomenon and are broadly applicable. Applied research, on the other hand, investigates empirical evidence in order to fit its findings with a specific issue.

- Research Outcomes

The researcher arrives at legitimate findings or conclusions that confirm or refute

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the research hypotheses after doing applied research by testing empirical evidence. These findings usually provide answers to the specific study questions, which are the rationale for the applied research.

Pure study, on the other hand, results in new theories, new dimensions to existing theories, or fresh information that improves on a body of knowledge. Pure research results do not directly serve as inventive answers to actual problems.

- **Research Approach**

Theoretical research is theoretical, whereas applied research is practical. Pure research, in this sense, produces hypotheses and builds on existing theories with the goal of contributing to an existing knowledge base.

Applied research, on the other hand, is more descriptive and practical. It is primarily concerned with the utility and worth of research outputs in terms of their end use, i.e., how they might be used to address existing issues and produce new ones.

Similarities between Applied and Pure Research

- **Data Collection Methods**

In order to acquire relevant data and arrive at the most objective study conclusions, pure and applied research use similar data gathering procedures. To obtain information and arrive at research conclusions, they often employ qualitative and quantitative data collection methods such as interviews, questionnaires, surveys and focus groups.

- **Intersectionality**

The findings of pure research are frequently used as the foundation for applied research.

- **Inductive and Deductive Reasoning**

To support research hypotheses, both pure and applied research methodologies employ both inductive and deductive reasoning. The researcher moves from the idea to the observation in deductive reasoning, while the researcher moves from the observation to the idea in inductive reasoning.

Researchers must comprehend the parallels and distinctions between applied and pure research approaches. As previously stated, the primary distinction between applied and pure research is the goal of the research, that is, what the research is intended to accomplish.

Aside from the objective of the research, applied and pure research differ in a variety of ways, including the research outputs, nature of the research and its contexts, to name a few. However, in order to arrive at objective results, both research approaches use similar data collection processes, such as observation and interviews.

1.3.2 Causal & Correlational (with Examples)

Correlation denotes a statistical relationship between variables.

It describes a relationship between variables: when one changes, so does the other. A correlation is a statistical measure of a variable's relationship with another one. These variables alter in tandem. However, this covariation is not always owing to a

direct or indirect causal relationship. Causation refers to the fact that changes in one variable induce changes in the other; there is a cause-and-effect relationship between the two variables. The two variables are linked and there is also a causal relationship between them.

Causation may not necessarily imply correlation, but correlation always suggest causation.

Correlational research methods can only show correlations between variables, whereas experimental designs can test causation.

Correlational Research

You collect data on your variables without changing them in a correlational study methodology.

Example 1: Collect survey data to see if there is a link between physical activity levels and self-esteem. You question participants about their current levels of exercise and use an inventory to assess their self-esteem.

Physical activity level is found to be positively correlated with self-esteem: lower levels of physical activity are associated with lower self-esteem, whereas greater levels of physical activity are associated with better self-esteem.

Correlational research has a high level of external validity, allowing you to generalise your findings to real-world situations. However, many studies have limited internal validity, making it difficult to link changes in one variable to changes in the other.

When performing controlled experiments is unethical, expensive, or difficult, these research strategies are widely used. They are also used to investigate relationships that are unlikely to be causal.

Example 2: To investigate if exposure to violent media is associated with aggression, you collect data on children's video game use and behavioural tendencies. You ask parents to report how many weekly hours their child spends playing violent video games and you interview parents and teachers on their children's behaviour.

There is a positive association between the variables: youngsters who spend a lot of time playing violent video games exhibit more aggressive conduct.

Causal Research

Controlled studies are the only way to truly demonstrate causal relationships between variables. Experiments are used to test formal predictions, known as hypotheses, in order to establish causation in one direction at a time.

Because the internal validity of the experiments is high, cause-and-effect relationships can be evidenced with reasonable confidence.

Because you modify an independent variable before measuring changes in a dependent variable, you can establish directivity in one direction.

Example 1: Testing directionality in an experimental design

You believe that physical activity level affects self-esteem, so you conduct an experiment to test this hypothesis. You implement a physical activity intervention and

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track self-esteem changes. To establish directionality, your physical activity intervention must occur prior to any observed change in self-esteem.

To see if this relationship is bidirectional, you'll need to create a new experiment to see if self-esteem affects physical activity level.

You can also eliminate the influence of third variables in a controlled experiment by using random assignment and control groups.

Random assignment aids in the even distribution of participant characteristics across groups, ensuring that they are similar and comparable. A control group allows you to compare an experimental modification to a similar treatment or no treatment at all.

Example 2: As an independent variable in an experimental design, you randomly assign each participant to either a control or an experimental group. Random assignment eliminates the effects of independent variables participant characteristics like as age or mental health status, which may have an impact on your results.

The experimental group receives the physical activity intervention, whereas the control group receives an unrelated, equivalent intervention. Any variations between groups can be attributed to your intervention by maintaining all variables constant across groups, except for your independent variable treatment.

Correlational Research

A correlational research approach investigates correlations between variables but preventing the researcher from controlling or manipulating any of them. Correlational investigations demonstrate the strength and/or direction of a relationship between two (or more) variables. Correlational studies or correlational study designs can have a positive, negative or zero correlation.

Correlational Studies Direction or Types	What Happens?	Example
Positive correlation	Both variables change in the same direction	As the prices of petrol increase, the fare of auto increases too.
Negative correlation	The variables change in opposite directions	As tea consumption increases, tiredness decreases
Zero correlation	There is no relationship between the variables	Tea consumption is not correlated with height

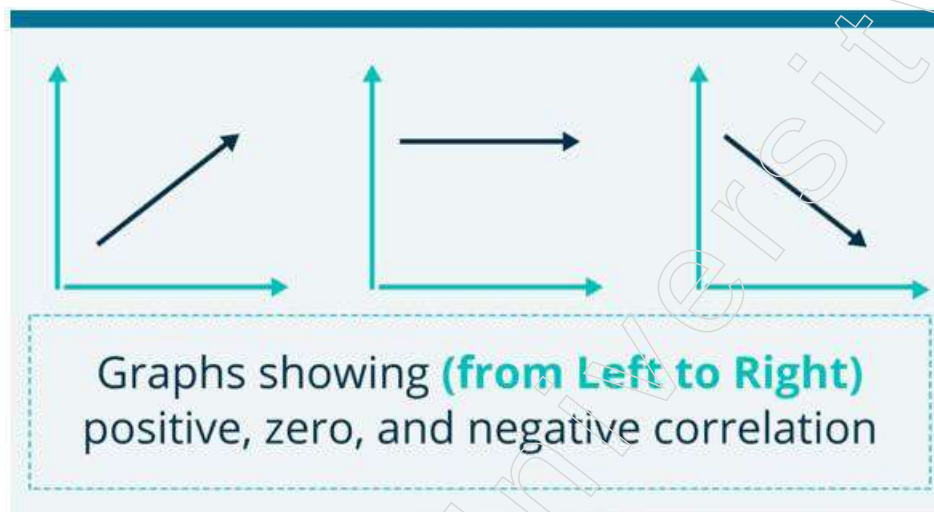
The correlational research design is ideal for quickly gathering data from natural settings. This enables you to apply your findings to real-world situations in an outwardly legitimate manner. Correlational studies research is a possible option in a few situations, such as:

- To investigate non-causal relationships

- To explore causal relationships between variables

To put new measurement instruments to the test like correlational research, like experimental research, seeks to establish a relationship between two variables. The distinction is that, whereas experimental research seeks to track changes in variables (causal), correlational research seeks to identify associations and similarities between them (Sassower, 2017).

As a non-experimental technique, it instead depends on statistical analysis to assess the link between these factors. Pearson's correlation coefficient (Mukaka, 2012) is a statistical approach used to calculate the amount of correlation between two variables. It has a value between -1 and +1. There is a relationship between the two the more it leans toward a positive value. A negative number indicates that the variables are related but only indirectly proportionally, whereas zero indicates that there is no relationship.



Correlational design example

You may use a correlational approach to see if the recent increase in online instruction coincides with any changes in test scores.

However, this approach cannot establish a causal relationship between the two variables. Many other factors, such as increasing stress and health difficulties among kids and teachers, could have influenced any decrease in test scores.

1.3.3 Cross-sectional & Longitudinal (With Examples)

Cross-sectional Study

A cross-sectional study is an observational study in which data is obtained as a whole to study a population at a single point in time in order to investigate the relationship between variables of interest.

1. In an investigation, a researcher records the information about participants without changing or manipulating their natural environment.
2. The most essential feature of a cross-sectional study is its ability to compare diverse samples at the same time. If a researcher wants to understand the

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relation between joggers and cholesterol levels, he or she might choose two age groups of daily joggers, one below 30 but more than 20 and the other above 30 but less than 40 and compare these to cholesterol levels among non-joggers in the same age groups.

3. At this point, the researcher can build gender subsets, but cannot consider past cholesterol levels because this would be outside the parameters for cross-sectional investigations.
4. Cross-sectional studies allow for the investigation of multiple factors at the same time. Researchers can look at age, gender, income and other factors in relation to jogging and cholesterol for very little or no extra expense.
5. However, there is one disadvantage to cross-sectional research: it cannot show a definitive relationship between cause and effect (a cause-and-effect relationship is one in which one action (cause) causes another occurrence (effect), for example, without an alarm, you may oversleep.)
6. This is primarily due to the fact that a cross-sectional study provides a picture of a particular point in time; this study does not assess what happened before or after. As a result, in the preceding example, it is difficult to determine whether the daily joggers had low cholesterol levels prior to starting jogging or if the activity assisted them in lowering previously high cholesterol levels.

Examples of cross-sectional research

A cross-sectional study collects data from subjects or participants who are similar in all respects except the one under consideration. Throughout the cross-sectional investigation, this variable remains constant. In contrast to a longitudinal study, where factors can alter over the research, this is a one-time event. Consider the following instances to gain a better understanding:

- **Retail:** Cross-sectional research on men and women in a certain age range can be undertaken in retail to identify parallels and variations in gender-related spending habits.

In business, researchers can undertake a cross-sectional study to learn how people from diverse socioeconomic backgrounds in one geographic segment react to a single modification in an offer.

- **Healthcare:** Researchers in the healthcare field may employ cross-sectional research to better understand how children aged 2 to 12 in the United States are prone to calcium insufficiency.
- **Education:** A cross-sectional research in school is especially useful in determining how kids who scored in a specific grade range in the same basic courses function with a new curriculum.
- **Psychology:** A cross-sectional study is research that involves diverse groups of people who do not share the same variable of interest (such as the variable you're focusing on), but do share other relevant variables. Age range, gender identity, socioeconomic level and so on are examples of these.

Cross-sectional research enables scholars and strategists to quickly gather actionable data that aids in decision-making and the provision of goods or services.

Example 1: Gender and phone sales

To promote sales, phone firms rely on complex and creative features. The predicted adoption rate and prospective phone sales are validated by research conducted by a phone manufacturer across the target demographic market. In a cross-sectional study, researchers recruit men and women from various areas and age groups.

If the results suggest that Asian women will not buy the phone because it is too bulky, the phone manufacturer can change the design to make it less bulky. They can also create and promote a smaller phone to appeal to a broader range of women.

Example 2: Men and cancer

A medical study assessing the prevalence of cancer in a given population is another example of a cross-sectional study. People of all ages, races, geographical places and social backgrounds can be evaluated by the researcher. If a considerable number of males in a certain age range are more likely to develop the condition, the researcher can conduct additional research to determine why. In this instance, a longitudinal study with the same subjects across time is preferable.

Different Types of Cross-Sectional Studies

When conducting a cross-sectional research project, you will engage in either descriptive or analytical research. Examine their descriptions to see if they apply to your task.

- **Descriptive research:** A cross-sectional study could be completely descriptive. A cross-sectional descriptive survey determines how frequently, broadly, or severely a variable of interest occurs across a certain demography.

Consider the above-mentioned retail case. In that cross-sectional study, researchers make specific observations in order to uncover spending trends. They might utilise the insights to develop new products and services, as well as advertise existing ones. They aren't necessarily investigating why these gendered patterns emerge in the first place.

- **Analytical study:** Analytical cross-sectional research looks into the relationship between two related or unrelated variables. This methodology isn't completely infallible, though, because outside variables and results, as well as associated investigations, are all occurring at the same time.

For example, to determine whether coal workers are at risk of developing bronchitis, just the factors in the mine are considered. What it does not take into account is that a susceptibility to bronchitis may be inherited, or that this health condition may have existed in coal miners prior to their employment in the mine.

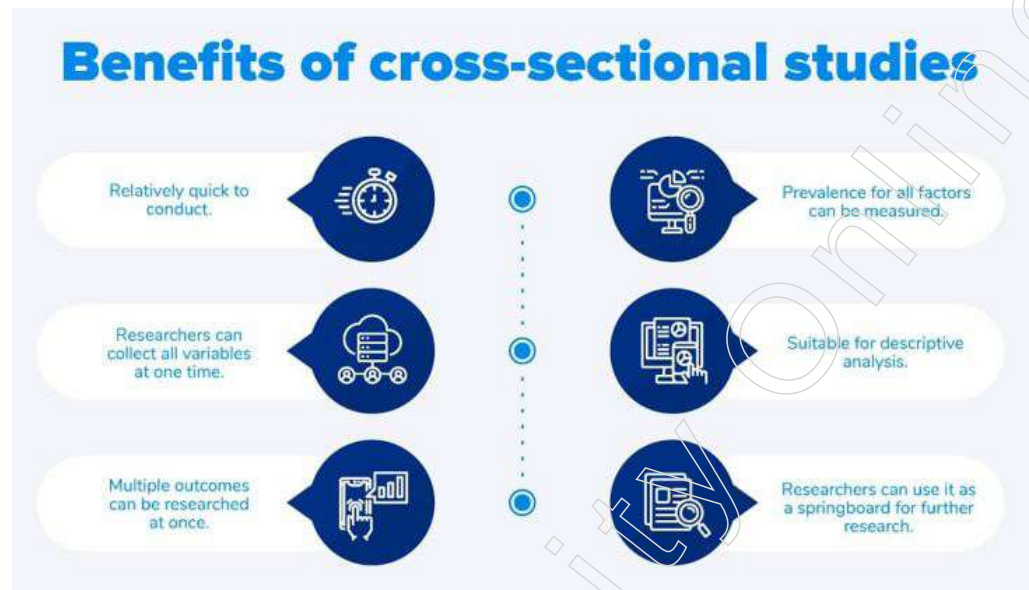
Other medical studies have demonstrated that coal mining is harmful to the lungs, but you don't want such assumptions to influence your current study.

Researchers typically employ both descriptive and analytical research approaches in a real-world cross-sectional study.

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Benefits of cross-sectional studies



- Relatively quick to carry out.
- Researchers can collect all variables at once and several outcomes can be researched at the same time.
- The prevalence of all factors can be determined.
- Appropriate for descriptive analysis.
- It can be used as a springboard for further research by researchers.

Cross-Sectional Versus Longitudinal Research

Longitudinal research may be preferable if you are seeking for an approach that studies people and factors over time. You might also follow up your cross-sectional research with a longitudinal study. It's easy to mix up the two study approaches, therefore we've separated them here:

- Although both are quantitative research approaches, there are certain variations when comparing cross-sectional studies vs. longitudinal studies.
- In cross-sectional investigations, variables are collected at a certain point in time. Longitudinal studies are conducted over numerous sessions and the factors may vary.
- Cross-sectional studies are preferred by researchers when looking for common points between variables. Nonetheless, because of their nature, longitudinal studies are used to further dissect the data from the cross-sectional study.

Longitudinal Study

A longitudinal study, like a cross-sectional study, is an observational study in which data is collected from the same sample over a long period of time. Longitudinal research can run from a few years to decades, depending on the type of information sought.

1. The advantage of performing a longitudinal study is that researchers can take notes, make observations and discover any changes in the features of their subjects. One of the most essential points to note here is that the longitudinal study goes beyond a particular point in time. As a result, they are able to define a proper chronology of events that transpired.
2. Continuing with the example, in a longitudinal study, a researcher would like to look at changes in cholesterol levels among women over the age of 30 but under the age of 40 who have jogged frequently over the last ten years. Because it would be able to control for cholesterol levels at the start of the jogging practise in a longitudinal study setup, longitudinal studies are more likely to reveal a cause-and-effect link.
3. Overall, research should drive design; but, as research improves, it can help establish which design is more appropriate. Cross-sectional studies can be completed more quickly than long-term investigations. As a result, a researcher may begin with a cross-sectional study and then move on to longitudinal investigations if necessary.

Implementing Longitudinal Study

If you wish to do a longitudinal study, you have two options: collect your own data or use data that has already been collected by someone else.

- Using data from other sources

Many governments or academic institutions conduct longitudinal studies and make the data available to the general public for free. For example, through the UK Data Service website, anybody can access data from the 1970 British Cohort Study, which has followed the lives of 17,000 British people from their births in a single week in 1970.

These statistics are often quite reliable and allow you to study changes over time. They are, however, more limited than data that you collect yourself. To protect the participants' identity, the data collected is frequently aggregated so that it can only be evaluated at the regional level. You will also be limited to the variables that the original researchers chose to study.

If you pursue this route, you should carefully analyse the dataset's source as well as the data that is given to you.

- Collecting your own data

If you opt to collect your own data, the method you use will be dictated by the sort of longitudinal study you conduct. You have the option of conducting a retrospective or prospective study.

1. In a retrospective study, you collect information about past events.
2. In prospective research, you select a group of individuals and track them throughout time while collecting data in real time.

Retrospective investigations are less expensive and take less time than prospective research, but they are more susceptible to measurement error.

Examples of retrospective vs. prospective

In retrospective research, you may look at patients' past medical records to discover if those who had this malignancy had previously smoked. In prospective

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research, you may watch a group of smokers and non-smokers over time to discover if they acquire cancer.

Advantages And Disadvantages of Longitudinal Studies

Longitudinal studies, like any other study design, involve trade-offs. They offer a distinct set of advantages, such as:

- Longitudinal studies enable researchers to track people in real time. This allows you to better establish the true sequence of events, providing insight into cause-and-effect linkages.

Example: A cross-sectional study on the impact of police on crime may discover that more police are related with higher crime, leading to the incorrect conclusion that police cause crime when it is the other way around. Longitudinal research, on the other hand, would be able to examine the rise or decline in crime after raising the number of police in an area.

- Longitudinal studies also allow for many observations of the same person throughout time. This means that any changes in the outcome variable cannot be attributed to individual variations.

Example: Assume you intend to investigate how a specific weight-training regimen influences sports performance. If you choose a longitudinal study, the effect of inherent skill on performance should be avoided because it will not vary throughout the course of the investigation.

- Prospective longitudinal studies eliminate the possibility of recollection bias, or the inability to recall prior events correctly.

Example: Suppose you're researching the effect of low-carb diets on weight loss. If you asked your participants to remember how many carbs they ate or how much they weighed in the past, they might struggle. These characteristics can be tracked in real time in longitudinal research.

- Longitudinal studies have various drawbacks, including Longitudinal studies are time-consuming and often more expensive than other types of studies, requiring significant commitment and resources to be effective. Because longitudinal studies repeatedly watch participants over time, any potential insights from the study may take some time to emerge.

Example: In a study looking at the associations between smoking and stomach cancer, you have to wait several years to observe any findings since the harmful effects of smoking accrue over time.

- Attrition, which occurs when study participants drop out, is widespread in longitudinal studies and can lead to erroneous conclusions.

Example: In your study on the impact of low-carb diets on weight reduction, individuals who are not seeing much progress may become disappointed and, as a result, are more likely to leave out. As a result, the diet may appear to be more successful than it is!

Cross-Sectional Vs Longitudinal Example

To investigate the link between smoking and stomach cancer You start with a cross-sectional study to investigate if there is a correlation between smoking and oesophageal cancer and you find that there is in males but not in women.

You then decide to conduct a long-term study to better investigate this association in guys. You would not have managed to focus on guys in particular without first doing a cross-sectional survey.

1.3.4 Experimental, Semi-experimental & Non-experimental (With Example)

Experimental study is scientific inquiry that employs two sets of variables. The first set serves as a constant against which the differences in the second set are measured. Experimentation is used in quantitative research methodologies, for example.

If you lack sufficient data to back your conclusions, you must first establish the facts. Experimental research collects data to assist you in making better judgments.

Experimentation is used in any research undertaken under scientifically appropriate settings. The effectiveness of experimental investigations is dependent on researchers confirming that a variable change is due only to modification of the constant variable. The study should identify a noticeable cause and effect.

Experimentation can be carried out in the following circumstances:

- Time is an important aspect in establishing a cause-and-effect relationship.
- Consistent behaviour between cause and effect.
- You want to comprehend the significance of cause and effect.

Types of Experimental Research Design

The traditional definition of experimental design is “the strategies employed to collect data in experimental investigations.”

There are three sorts of experimental designs:

- Pre-Experimental Research Design
- True Experimental Research Design
- Quasi-Experimental Research Design

The type of design is determined by how you categorise research individuals based on conditions or groups.

1. Pre-Experimental Research Design: After adopting cause and effect elements, a group, or groups, are kept under watch. You will undertake this research to determine if further investigation is required for these specific groups.

Pre-experimental research can be further classified into three types:

- One-shot Case Study Research Design
- One-group Pre-test-post-test Research Design
- Static-group Comparison

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2. **True Experimental Research Design:** True experimental study is the most accurate type of research because it relies on statistical analysis to prove or disprove a theory. Only real design can demonstrate a cause-effect connection within a group of experimental designs.

In a real experiment, three elements must be met:

- There is a Control Group that will not be subjected to changes and an Experimental Group that will be subjected to the modified variables.
- A variable that the researcher can manipulate
- Distribution at random

This type of experimental inquiry is common in the physical sciences.

3. **Quasi-Experimental Research Design:** The term “quasi” denotes resemblance. A quasi-experimental design is comparable to but not the same as an experimental design. The designation of a control group distinguishes the two. In this study, an independent variable is modified, but group members are not assigned at random. In field settings, where random allocation is either irrelevant or not required, quasi-research is used.

Advantages of Experimental Research

Before taking your idea to market, you can test it in a controlled environment through experimental research. It also provides the best approach for testing your theory because of the following benefits:



- Researchers have a firmer grip on variables to get desired outcomes.
- The subject or industry has no bearing on the success of experimental research. It can be used for research purposes by any industry. The outcomes are precise.
- After reviewing the results, you can apply your conclusions to similar ideas or circumstances.
- You can determine the cause and effect of a hypothesis. Researchers can dig deeper into this association to come up with more in-depth ideas.
- Experimentation is an excellent beginning place. The information you gather will serve as a foundation for developing new ideas and conducting additional study.

Exemplification of Experimental Business Research

Market researchers can use experimental research to a wide range of testing requirements. Experimentation can be applied to almost every situation that demands proof, confirmation or is obscured by uncertainty.

The following is an example of how a firm could benefit from this research:

A product manager must persuade higher-ups at a denim firm to introduce a new product line at a specific department store. The goal of this launch is to improve sales, grow the company's floor presence and broaden the offers.

The manager must demonstrate that this phrase is required in order for the company to present the idea to the department store. The product manager can then do experimental research to establish a compelling case for their theory, that a new line can increase sales.

The product manager conducts experimental research by conducting a test in a few places where the new denim line is sold. The locations of these stores differ to represent the target market sales before and after the debut. The test will last a month to see if the hypothesis (the new line resulting in higher attention and sales) can be proven.

This is a field experiment. The product manager must monitor the new product line's sales and foot traffic, paying close attention to spikes in revenue and overall sales in order to justify the new line.

Example of Experimental Research for Survey

Survey research is diametrically opposed to experimental research, as opposed to the other major types of research such as exploratory, descriptive and correlational research. This is due to the observational nature of surveys, whereas experimental research, as the name implies, relies on experiments, that is, testing out changes and evaluating the reactions to the changes.

Despite the fact that survey research and experimental research are diametrically opposed, they are not mutually exclusive. In reality, surveys are an effective way to acquire additional insight into an existing experiment or to identify variables before performing an experiment in the first place.

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As a result, firms can use a wide range of surveys to supplement their experimental study. Here are some of the most common survey methods that work in tandem with experimentation:

1. The quantitative survey
 - Discovers statistical significance within variables.
 - Useful because causal research is primarily quantitative.
2. The retrospective survey
 - Examines previous events, occurrences and attitudes about the variables.
 - Indicates whether and how the variables changed.
3. The prospective survey
 - Capable of detecting causal elements between variables throughout time.
 - Helpful in developing theories.
4. The customer experience survey
 - Assists organisations in identifying variables that contribute to or result from specific types of consumer interactions.
 - Enables firms to measure CX in relation to survey replies.
5. The pulse survey
 - Measures several essential aspects of a firm or organisation; conducts employee surveys.
 - Deployed more regularly, allowing variables to be tracked in real time.
6. The qualitative survey
 - Uses open-ended inquiries to help answer the what, why and how.
 - Extensively extracts key high-level information.

Semi-experimental Research

A semi-experimental or quasi-experimental study, like a true experiment, seeks to establish a cause-and-effect link between a variable.

A quasi-experiment, unlike a true experiment, does not rely on random allocation. Subjects are instead assigned to groups based on the non-criteria.

In instances where genuine experiments cannot be employed for ethical or practical reasons, quasi-experimental research is a useful tool.

Example of a True Experiment vs a Quasi-experiment

Assume you're curious about the impact of a new psychological therapy on people suffering from depression.

Example: True experimental design

In order to conduct a real trial, randomly assign half of the patients in a mental health clinic to undergo the new treatment. The other half, known as the control group, receives normal depression treatment.

Patients fill out a document explaining their symptoms every few months to check if the new medication is considerably better (or worse) than the standard one.

However, the administrators of the mental health clinic may refuse to give you permission to randomly assign their patients to therapy for ethical concerns. In this circumstance, a proper experiment cannot be carried out.

A quasi-experimental design can be used instead.

Example: Quasi-experimental design

You discover that a handful of the clinic's psychotherapists have elected to attempt the new therapy, whilst others who serve comparable patients have chosen to continue with the standard regimen.

These pre-existing groups can be used to compare the symptom development of patients getting the novel therapy to those receiving the traditional course of treatment.

Although the groups were not assigned at random, if any systematic differences between them are properly accounted for, you can be pretty confident that any changes are due to the therapy and no other confounding variable.

Various Types of Quasi-Experimental Designs

There are numerous sorts of quasi-experimental designs. In this section, we will go through three of the most popular types: non-equivalent groups design, regression discontinuity and natural experiments.

- Non-equivalent groups design

The researcher chooses existing groups that appear similar, but only one of the groups receives the therapy in non-equivalent group design.

In a real experiment with random assignment, the control and treatment groups are deemed equal in all ways except one: treatment. However, in a quasi-experiment with non-random groups, they may differ in other ways they are non-equivalent groups.

Researchers use this type of design to adjust for any confounding variables by controlling for them in their analysis or by selecting groups that are as similar as possible.

The most prevalent sort of quasi-experimental design is this one.

Example: You believe that a new after-school programme will result in improved grades. You select two groups of children from separate schools, one of which implements the new programme and the other does not.

You can determine whether the programme has an effect on grades by comparing pupils who participate to those who do not.

- Discontinuity in regression

Many prospective treatments that researchers want to investigate are built around an essentially arbitrary cut-off, with those above the threshold receiving the treatment and those below it not.

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Near this threshold, the differences between the two groups are frequently so minor as to be negligible. As a result, researchers can utilise those who are just below the threshold as a control group and those who are just beyond it as a treatment group.

Example: Some high schools in the United States are reserved for high-achieving pupils who must achieve a particular score on a test in order to be admitted. Those who pass this test will most likely vary from those who do not in a methodical way.

However, because the exact cut-off score is arbitrary, students near the threshold those who barely pass the exam and those who fail by a razor-thin margin tend to be extremely similar, with the slight differences in their scores owing mostly to random chance. As a result, any disparities in outcomes must be attributed to the school they attended.

You can investigate the long-term outcomes of these two groups of children to test the influence of attending a selective school (those who barely passed and those who barely failed).

- **Natural experiments**

Normally, researchers control which group the individuals are allocated to in both laboratory and field investigations. An external event or scenario ("nature") result in the random or random-like assignment of individuals to the treatment group in a natural experiment.

Natural experiments, despite the fact that some use random assignments, are not regarded legitimate experiments because they are observational in nature. Despite the fact that the researchers have no control over the independent variable, they can use this incident after the fact to evaluate the treatment's effect. However, because they could not afford to cover everyone who qualified for the programme, they instead assigned spaces in the programme by a random lottery.

Researchers were able to analyse the program's impact by randomly assigning the enrolled persons as a treatment group and the others who were eligible but did not win the lottery as a control group.

Non-experimental Research

Non-experimental research is study that does not include an independent variable. Instead, the researcher observes the perspective in which the phenomenon takes place and analyses it to obtain information.

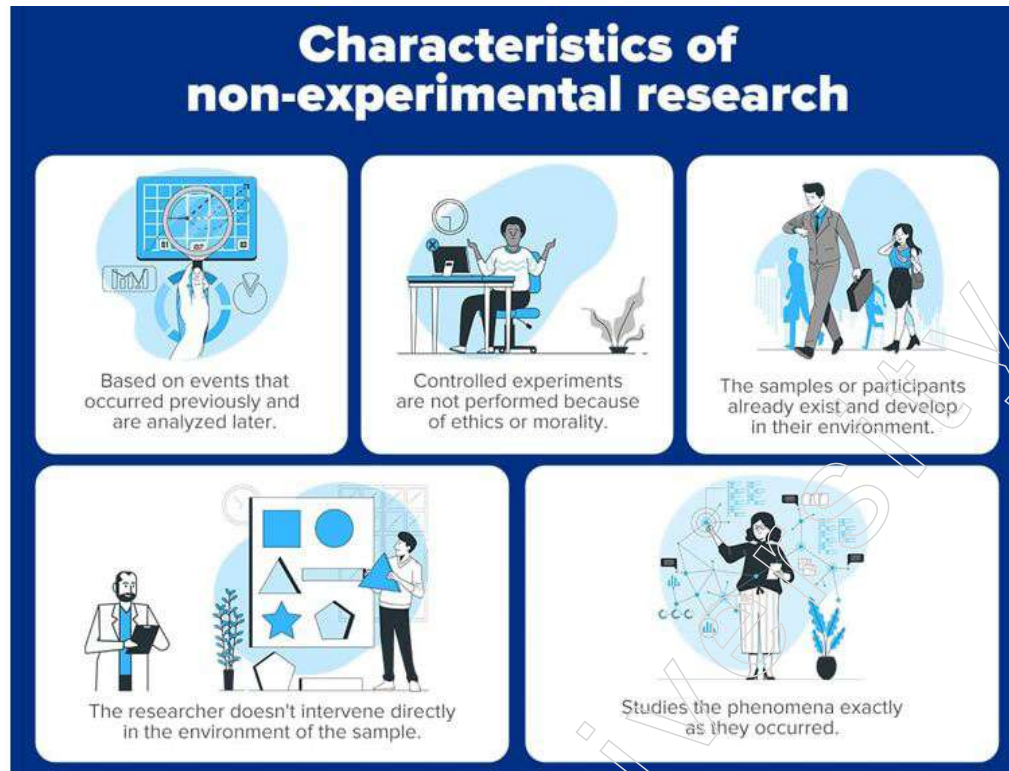
Unlike experimental research, in which the variables are kept constant, non-experimental research is undertaken when the researcher cannot control, manipulate, or alter the subjects during the study and must rely on interpretation or observations to reach a conclusion. This indicates that the method cannot establish a true cause and effect relationship if it is based on correlations, surveys, or case studies.

Characteristics of Non-experimental Research

- Most studies are based on previously occurring events that are later studied.
- Controlled experiments are not undertaken in this method for reasons such as ethics or morality.

- There is no creation of study samples; rather, the samples or investigators already exist and evolve in their environment.
- The researcher does not try to intervene in the sample's environment.

This method investigates things exactly as they occur.



Types of Non-experimental Research

Non-experimental research can take the following systems:

Cross-Sectional Research:

Cross-sectional research is used to observe and analyse the exact time of the research in order to encompass different study groups or samples. This type of research is classified as follows:

- Descriptive: When statistics are observed in the presence of one or more factors.
- Causal: It is in charge of describing the reasons and relationships that exist between variables at a specific point in time.

Longitudinal Research:

A longitudinal study seeks to examine the evolution and changes in the relationships between variables through time. Longitudinal studies are classified as follows:

- Trend: When they look at the changes that the research group has experienced in general.
- Group evolution: When the study group is comprised of a smaller sample size.

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- Panel: It is in charge of assessing individual and group changes in order to determine what causes them.

Uses of Non-experimental Research

Non-experimental research can be used in the following situations:

- When the research issue is about one variable instead of a statistical relationship between two variables.
- The variables in the study question have a non-causal statistical relationship.
- There is a causal relationship between the research question and the independent variable, but the independent variable cannot always be changed.
- In exploratory or broad study, a specific experience is confronted.

Advantages and Disadvantages

Non-experimental research has the following advantages:

- It is extremely adaptable during the study process.
- The cause of the phenomena is understood and its impact is being studied.
- The researcher can define the study group's characteristics.

The following are some of the drawbacks of nonexperimental research:

- The groups do not reflect the full population.
- Errors in approach may arise, resulting in biases.

Distinction Between Experimental and Non-Experimental Research

Changing variables and randomly assigning conditions to individuals is what experimental research entails. Experimental research designs are employed in health, biology and social science because they can determine the reason. Control and validity are strictly enforced in experimental study designs. Despite the fact that they need a lot of resources, they can provide some really fascinating results.

Non-experimental research, on the other hand, is typically descriptive or correlational, with the researcher making no specific adjustments. Simply said, you describe the situation as it is or a relationship between variables. It is difficult to determine causal effects when there is no control. In this type of research, validity is still an issue. However, it is more about the measurements than the effects.

1.3.5 Descriptive & Exploratory (with Examples)

1. Descriptive Research:

Descriptive research refers to research that designates a phenomenon or a cluster under study and it is simple to conduct it in social sciences due to manipulation of variables. It is mostly useful in gathering data on a certain population, circumstances and events. Descriptive research is used to collect data and then use statistical analysis to gain more or less insight from that data.

Example of descriptive research include population census, product marketing surveys etc.

Features of Descriptive research

The term descriptive research refers to research questions, plan of the study/study design and data analysis related to that topic. We call it an observational research method because none of the research study variables are influenced in any way.

The distinctive characteristics of descriptive research are discussed as following:

1. Quantitative research: Descriptive research remains a quantitative research method that attempts to gather quantifiable information for statistical analysis of the population sample. The aforementioned is a popular market research instrument that allows us to collect and define the demographic section's nature.
2. Uncontrolled variables: In descriptive research, none of the variables are biased in any way. This uses observational methods to conduct the research. Hence, the environment of the variables or their performance is not in the hands of the investigator.
3. Cross-sectional studies: Descriptive research is typically conducted as a cross-sectional study, with different sections assigned to the matching group.
4. The basis for further research: Researchers conducts additional research on the data gathered and analyzed from descriptive research using various research techniques. The numbers can also help point on the way to the types of research methods rummage-sale for future research.

Requests of Descriptive Research through Examples

A descriptive research method can be used in a variety of ways and for a variety of reasons. However, before enrolling in any survey, the review goalmouths and survey design are vital. Despite following these ladders, there is no means to know if one will encounter the research outcome. To what extent should descriptive research be funded? To comprehend the ultimate objective of research goalmouths, the following practices are currently practiced by organizations conducting descriptive research today:

- Describe respondent characteristics: The determination of by means of close-ended investigations is to draw concrete expectations about the respondents. This perhaps will be the need to create patterns, traits and actions of the respondents. It may also be to recognize from a respondent, their attitude, otherwise opinion around the phenomenon. Let's say, Millennials, for example, devote 60 minutes per week to internet browsing. Overall, this information assists the cluster researchers in making informed commercial decisions.
- Amount of data trends: Researchers extend data trends over time with the statistical abilities of an expressive research design. Consider whether a clothing company conducts research on transformed demographics similar age groups from 24-35 and 36-45 on a new range introduction of autumn wear. If one and only of individuals groups does not respond well to the fresh launch, it provides insight into what the new clothes are like and what is not.

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- **Conduct comparisons:** Establishments likewise use a descriptive research design to recognize how diverse groups answer to an exact product or service. For instance, an apparel brand makes a survey asking general enquiries that measure the brand's image. The equivalent study moreover asks demographic questions like to age, income, gender, geographical location, etc. This consumer research aids the organization appreciate what aspects of the variety appeal to the population and which aspects do not. It also aids in the development of type product or marketing fixes or the creation of a new product line to cater to in height growth potential groups.
- **Validate obtainable conditions:** Investigators broadly customize descriptive research to help ascertain the research object's prevailing conditions and underlying patterns. Researchers detect each variable and conduct an in-depth analysis in accordance with the non-invasive research method and the use of quantitative observation and approximately aspects of qualitative observation. Researchers are similarly accustomed to validating somewhat existing circumstances that may or may not be prevalent in a population.
- **Conduct research at diverse times:** The education can be accompanied at different eras to determine any similarities or differences. This also allows for the evaluation of a number of variables. Studies on current situations, while intended for verification, can also be repeated to draw trends.

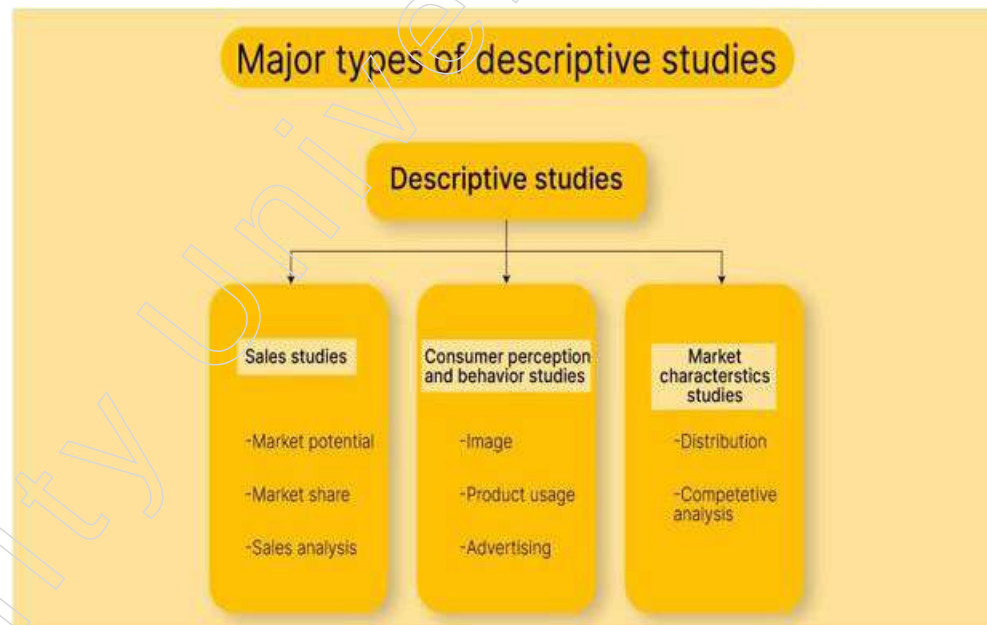


Figure: Types of descriptive studies

Benefits of Descriptive Research

Notable advantages of descriptive research are as following:

- **Data collection:** A detective can conduct unkind descriptive research using the same methods as the observational method, case study method and survey method. Amongst these three, all primary information collection approaches

are covered, which offers a lot of information. This can be used for future research or even emerging a hypothesis of your research subject.

- Varied: In the meantime, the data composed is qualitative and quantitative, it contributes a holistic indulgent of a research topic. The information is varied, diverse and in addition thorough.
- Natural environment: Descriptive research permits the research to be displayed in the respondent's natural surroundings, ensuring that first-class and honest data is compiled.
- Rapid to perform and cheap: As the sample size is usually large in descriptive research, the data collection remains quick to mien and is low-cost.

Descriptive Research Approaches

In general, there are three distinctive methods to determine descriptive research. They are:

- Observational method

The observational method is the utmost effective method to conduct this research, in addition researchers make use of together quantitative and qualitative observations.

A quantitative opinion is the gathering of data objectively, which is primarily engrossed on numbers and values. It recommends "associated with, of or depicted in footings of a quantity." Consequences of quantitative observation are derivative using statistical and numerical examination methods. It implies observation of any object associated through a numeric value such as age, shape, weight, capacity, scale, etc. For example, the researcher can track if present customers will refer the brand by means of a simple Net Promoter Score question.

Qualitative observation doesn't involve measurements or figures but instead just monitoring characteristics. In this instance, the researcher perceives the defendants from a distance. Since the objects of study are in a relaxed environment, the features observed are natural and effective. In a descriptive research design, the researcher can pick to be either a complete spectator, an observer as a participant, a member as an observer, or a full participant. For instance, in a supermarket, a researcher can monitor from afar and note down the consumers' selection and purchasing trends. This proposes a more in-depth look into the purchasing familiarity of the customer.

- Case study method

Case studies consist of in-depth research in addition study of individuals or groups. Case studies start with a hypothesis and widen at further possibilities of studying a phenomenon. Case studies, therefore, should not be dismissed as a means of controlling cause and effect because they are unable to provide accurate predictions and may contain a researcher's bias. The fact that there may be an atypical defendant in the survey is another reason why case studies are not an unwavering technique of supporting descriptive research. Labeling those primes as faulty generalisations and detracting from external validity.

- Survey research

In survey research, people's response is collected with surveys or questionnaires or polls. They will use a common market research tool to get input from respondents.

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The correct survey questions should be included in a learning to collect relevant data. It has a responsibility to be a balanced combination of open-ended and closed-ended inquiries. The inspection approach may be demonstrated both online and offline, making it the best choice for descriptive studies with a large sample size.

Illustrations of Descriptive Research

More or less, the examples of descriptive research are:

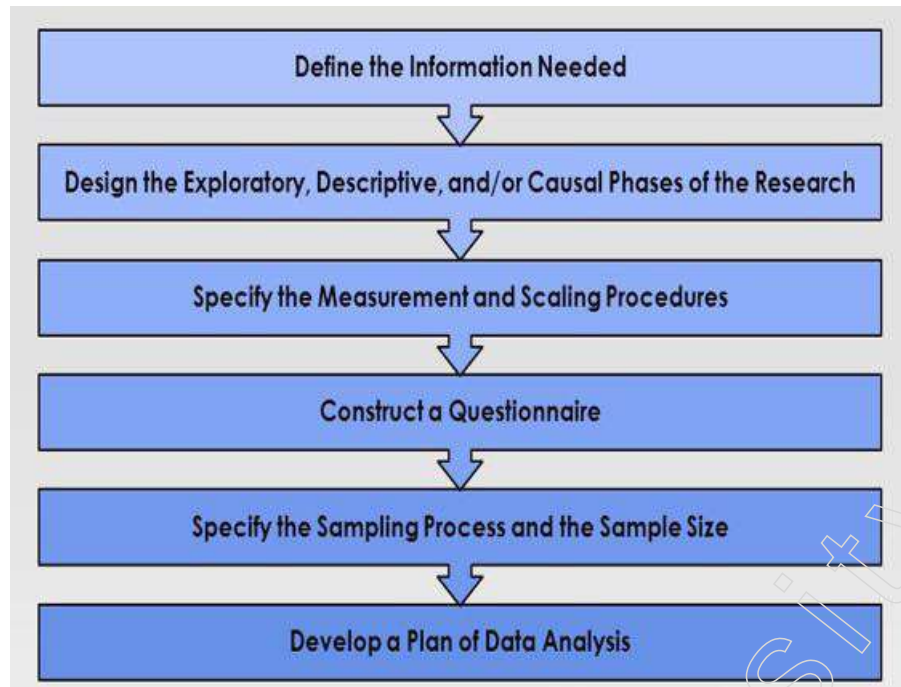
1. A department food group is introducing a new line of barbecue ribs and wants to know which flavours of rubs people favour. They do this type of research study using numerous methods and tools, such as observational approaches at supermarkets, to understand the number one flavour palette. By graphing while gathering in-depth demographic data, it is possible to gain insights into the preferences of different markets. This can also help personalise the rubs and spreads to the demographic's various favoured essences. This type of study supports the firm in tightening their commercial model and boosting marketing in key marketplaces.
2. Another scenario in which this research could be useful is if a school district wants to assess teachers' views on using technology in the classroom. The researcher can evaluate what they can help apprehend whether a full-fledged putting into practice can encounter a difficulty by conducting surveys and seeing their familiarity with technology over and done with observational ways. If the pupils are squeezed in any manner as a result of the change, this also contributes in indulgence.

Exploratory Research

Exploratory research is described as research conducted to study an undefined problem. It is carried out to gain a better knowledge of the current problem, but it will not produce definite conclusions. A researcher begins with a broad concept and uses this study as a vehicle to find difficulties that can be the focus of future research. A key consideration here is that the researcher be willing to shift course in response to the discovery of fresh data or insight. Typically, such research is conducted when the problem is in its early stages. It is also known as the methodology or interpretive research since it is used to answer questions such as what, why and how.

Example 1: Consider the following scenario. A juice bar owner believes that increasing the variety of juices will result in an increase in customers, but he is unsure and requires additional information. The owner aims to do an exploratory study to find out and thus decides to conduct exploratory research to determine whether expanding their juice choices will enable him to gain more clients or if there is a better option.

Example 2: A podcast survey template can be used to obtain feedback on podcast consumption metrics from both existing listeners and other podcast listeners who are not already subscribed to this channel. This enables the podcast's author to develop curated content that will reach a broader audience.



Exploratory & Conclusive Research Differences

	Exploratory	Conclusive
Objective:	To provide insights and understanding.	To test specific hypotheses and examine relationships.
Characteristics:	Information needed is defined only loosely. Research process is flexible and unstructured. Sample is small and non-representative. Analysis of primary data is qualitative.	Information needed is clearly defined. Research process is formal and structured. Sample is large and representative. Data analysis is quantitative.
Findings /Results:	Tentative.	Conclusive.
Outcome:	Generally followed by further exploratory or conclusive research.	Findings used as input into decision making.



Figure: Steps in research designing

Styles and Methodologies of Exploratory Research

Even though it may sound a little problematic to research something that has very little evidence about it, there are numerous methods which can help a scholar figure out the best research strategy, data collection methods and choice of subjects. There are two types of research that can be conducted, namely primary and secondary

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research. Under these two types there are numerous methods that can be cast-off by a researcher. The information gathered from this research can be either qualitative or quantitative. The following are some of the most commonly used investigation designs:

i) Primary Research Approaches

Primary research consists of information gathered directly from the subject. It can be through a collection of people or even an individual. Such research can be approved and made available without reservation by the researcher himself, or they can hire a third party to carry it out on their behalf. Primary research is precisely carried out to see the sights of a specific problem that necessitates an in-depth investigation.

- **Surveys/polls:** Surveys/polls are reused to collect data from a predefined group of respondents. It is one of the most crucial quantitative techniques. Various types of surveys or polls can be used to discover opinions, trends, etc. Through the advancement in technology, surveys can now be sent online and are just as simple to access. For example, using a survey app on tablets, laptops, or smooth mobile phones. This information is also available to the researcher in factual time as well. In order to achieve higher response rates, most organisations now offer short-length reviews and rewards to respondents.

For instance: A survey is sent to an assumed set of audience to understand their sentiments about the size of mobile phones as soon as they purchase one. Based on such information, the association can delve deeper into the topic and make business-related decisions.

- **Interviews:** Even though you may get a lot of information on or from public sources, however sometimes an individual interview can give in-depth information on the subject being studied. Such an investigation is a qualitative research process. A meeting through a subject substance expert can give you meaningful insights that a generalized public source would not be able to provide. Interviews continue to be available in person or over the phone and they use open-ended questions to get meaningful information about the topic.

For instance: An interview by means of an employee can give you added insights to find out the degree of job satisfaction, or an interview with a subject matter expert of quantum model can offer you in-depth information on that topic.

- **Focus groups:** Focus group is yet another widely used method in exploratory research. In such a technique, an assembly of individuals is chosen and are allowed to express their insights on the subject that is being studied. Nonetheless, it is critical to state unequivocally that when selecting individuals in a focus cluster, they should share a common background and have comparable involvements.

For instance: A focus group helps a study recognise customers' opinions if they were to buy a phone. This type of research can assist the researcher in understanding what are the consumer values when purchasing a phone. The aforementioned could be screen size, trademark value or even the dimensions. On the basis of which the organisation can understand, what are customer purchasing attitudes, consumer opinions and so on.

- **Observations:** Opinion research can include both qualitative and quantitative observations. Such a study is designed to identify an individual and entice the outcome based on or after their reaction to certain parameters. There is no direct interaction with the subject in such a study.

For instance: An FMCG company wants to be acquainted with how its consumer respond to the new shape of their product. The canvasser observes the customers initial reaction and collects the data, which is at that time used to draw inferences from the collective information.

ii) Secondary Research Methods

Secondary research is the collection of information from previously published primary research. In such research, you collect information from a variety of sources, including case studies, publications, newspapers, books, etc.

- **Online research:** In today's world, this is one of the most unusual ways to collect information on any topic. A large amount of data is available on the internet and the researcher can download it whenever he needs it. The candidness and authenticity of the cause websites from which the researcher is gathering information is an important consideration for such a study.

Perhaps: A researcher wants to know what one hundredth of people prefer a particular brand of phone. The researcher unprejudiced enters the information he needs in a search engine and change to multiple links by means of related material and statistics.

- **Literature research:** Nonfiction research is one of the most cost-effective methods for testing a hypothesis. There is tremendous quantity of information obtainable in libraries, online sources, or even moneymaking databases. Information can be obtained from newspapers, magazines, records from library, documents as of government agencies, specific topic connected articles, literature, annual reports, statistical books, from research organizations in addition so on.

Nevertheless, a few things have to be reserved in mind while researching on or after these sources. Government activities have authentic data but is available from time to time. Correspondingly, research from educational institutions is usually overlooked, despite the fact that educational institutions conduct more research than any other entity. Furthermore, marketable bases are responsible for information on major topics like political agendas, demographics, financial information, market trends and information, etc.

For example: A company has low sales. It can be easily explored from available statistics and market literature if the problem is market related or organization related or if the topic being studied is regarding financial situation of the country, then research data can be accessed through government documents or commercial sources.

- **Case study research:** Case study research can help a researcher in finding more information through carefully analyzing existing cases which have gone through a similar problem. Such scrutinies are very important and serious

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especially in today's commercial world. The researcher's goal is to ensure that he carefully analyses the situation in relation to all of the variables present in the previous case in contrast to his own case. It is commonly used by corporate organisations, the social sciences sector, or unevenly in the health-care sector.

For instance: A specific orthopedic surgeon has the uppermost success rate for performing knee surgeries. A lot of extra hospitals or doctors have taken up this case in order to recognize and benchmark the process in which this surgeon looks after the procedure in order to improve their success rate.

Ladders to Behaviour in Exploratory Research

- Classify the problem: A researcher recognizes the subject of research and the misbehaving is addressed by resounding out multiple methods to respond to the questions.
- Create the hypothesis: When the researcher discovers that there are no prior studies and that the problem is not precisely resolved, the researcher will create a hypothesis based on the questions gathered while classifying the problem.
- Further research: As soon as the data is obtained, the researcher will decide to continue his research and complete the descriptive investigation. Qualitative approaches are used to further investigate the topic in part and find out whether the data is correct or not.

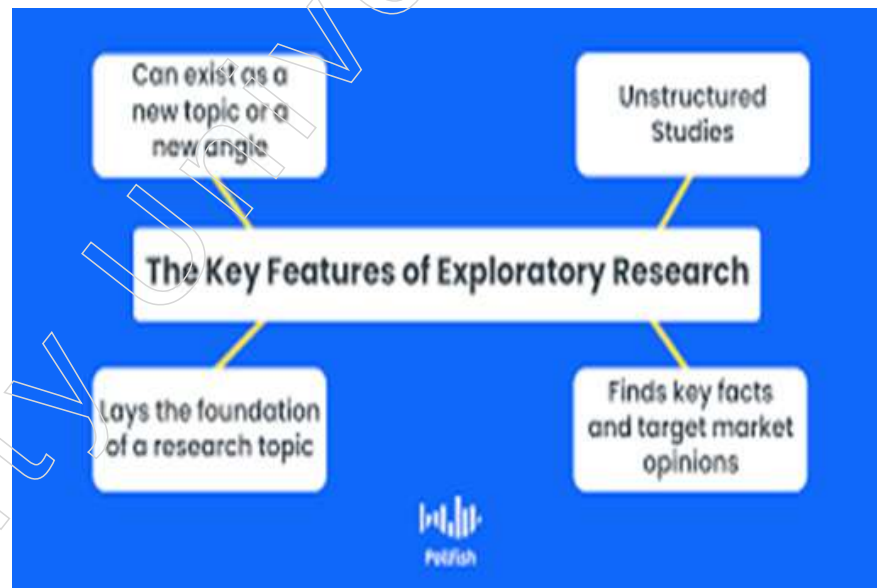


Figure: Key features of exploratory research

Main Features of Exploratory Research

- They continue to be informal trainings.
- It is typically low-cost, collaborative and open-ended.
- It will help a researcher answer questions that are close to what the problem is. What exactly is the purpose of the study? In addition, what topics possibly will be studied?

- To move forward with exploratory research, there is usually no prior research completed or the remaining ones do not provide a precise enough answer to the problem.
- It is time-consuming research and needs patience and has risks related with it.
- The researcher's willpower must go complete all the information available for the specific study he is conducting.
- There are no set of rules to transmit the research per se, as they are supply, broad and scattered.
- The research must have importance or value. If the delinquent is not essential in the industry, the research is ineffective.
- The research should also include a few theories which can explain its results/ findings, as this will make it easier for the researcher to evaluate it and move forward in his research
- Such research typically produces qualitative data, however in certain cases, quantitative data can be universal/generalised for a larger sample through the use of surveys as well as experiments.

Benefits of Exploratory Research

- The researcher maintains a high level of adaptability and is able to adapt to changes as the research progresses.
- It is usually inexpensive.
- It aids in the establishment of a research foundation, which can lead to additional research.
- It enables the researcher to determine whether or not the topic is worth investing time and money in and whether or not it is worth pursuing.
- It can assist additional researchers in identifying potential causes of the problem, which can then be investigated further in the study to determine which of them is the most likely cause of the problem.

Difficulties of Exploratory Research

- Though it can point you in the right direction in terms of what the answer is, the aforementioned is often inconclusive.
- The main disadvantage of exploratory research is that they provide qualitative data. Clearing up of such information can be judgmental as well as biased.
- Since in most cases, exploratory exploration encompasses a smaller sample, the results cannot be precisely interpreted for a universal population.
- Many times, when the data is gathered through secondary research, there is an unintended consequence of that data being obsolete and outdated.

Importance of Exploratory Research

Exploratory research is approved out when a topic needs to be thoroughly understood, especially if it has never been done before. The goal of such research is

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to observe the problem and its surroundings rather than to derive a conclusion from it. Such generous research will provide a researcher with a strong foundation for reconnoitering his ideas, selecting the appropriate research design and determining the outcome variables that are actually important for the analysis. Most importantly, such research can assist organisations or researchers without taking up a lot of time and resources, as it will power enable the researcher to determine if it worth pursuing.

Descriptive Research

Descriptive research is referred to as an observational research method because none of the variables in the study are modified throughout the research process. Descriptive research is a subset of quantitative market research or social research that entails performing survey research with quantitative variables on a commercial research tool or a social research tool.

Example: Assume a UK-based brand is attempting to establish itself through New York and wants to identify the demographics of purchasers who typically purchase from companies comparable to its own. The information received from the survey will be limited to the demographics of the population in this case. It will reveal information about the purchasing habits of different age groups in New York. Because the brand is attempting to establish itself in New York, it will not investigate why such tendencies exist. All they seek to know is the population's purchasing habits, not why such relationships exist.

Characteristics Of Descriptive Research

Descriptive study is quantitative in nature as it strives to collect information and statistically analyse it.

- Quantitative Analysis:

Descriptive research is a useful research tool that allows a researcher to collect data and explain its demographics using statistical analysis. As a result, it is a quantitative research technique.

- Nature of variables:

The factors in descriptive research are not controlled. They are not coerced in any way. Descriptive research is largely observational, which means that the researcher has no control over the nature and behaviour of the variables under study.

- Cross-sectional research:

Different portions of the same group are researched in descriptive research. For example, in order to analyse New York fashion preferences, the researcher can study both Gen Z and Millennials from the same demographic in New York.

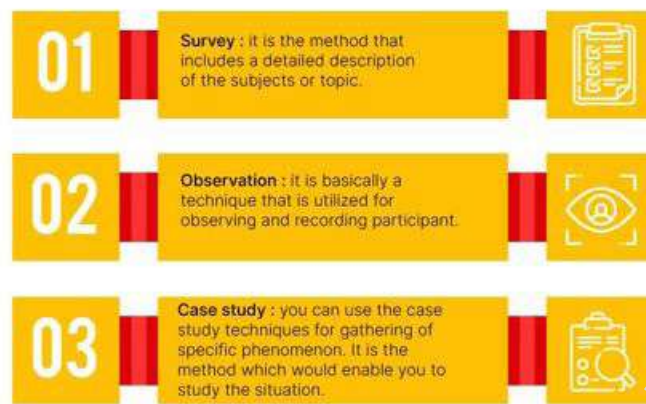
- Directs future research:

Researchers can further investigate the data acquired here because descriptive research identifies and describes patterns between variables. It directs academics to investigate why such patterns have been discovered and the relationship between them. As a result, it directs researchers in the direction of intelligent market research.

Methods Of Descriptive Research

Descriptive research can be directed through three diverse methods:

Methods of descriptive research



- **Observational Method:**

In behavioural sciences, observation is a common method of data collecting. W.L. Prosser summarised the power of observation as follows:

“There is still no individual who would accept dog tracks in the mud against with the certified testimony of a hundred eyewitnesses that no dog had passed by.”

Observation is the systematic monitoring and recording of behavioural and non-behavioural behaviours and situations in order to gather knowledge about the phenomena of interest ‘Behavioural Observation’ is:

- Nonverbal analysis, such as body movement. eye movement
- Linguistic analysis, which includes listening for sounds like ohs! and abs!
- Extralinguistic analysis, which examines pitch, timbre and pace of speech, among other things.
- Spatial examination of how people interact with one another.

Non-behavioural observation includes record analysis, such as newspaper archives, physical condition analysis, such as examining the quality of grains in gunny bags and process analysis, which involves observing any process. Observations can be divided into several categories.

Type of Observation

1. **Structured vs. Unstructured Observation:** In structured observation, the problem is explicitly stated, thus the behaviour to be observed and the technique by which it will be measured are specified in detail ahead of time. This decreases the possibility of observer bias in research, for example, a study on plant safety compliance can be studied in a structured manner. Unstructured analysis is performed when the problem has not been well described and it is not possible to specify what is to be observed. As a result, a researcher watches all relevant phenomena and has a great degree of leeway in terms of what they note and record, for example, watching a student’s behaviour in class would include monitoring their overall behaviour in

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the classroom setting. The data gathered through unstructured analysis should be thoroughly examined to ensure that no bias is introduced.

2. **Disguised vs. Undisguised Observation:** This classification is based on whether or not the subjects should be aware that they are being observed. The subjects are unaware that they are being observed in disguised observation. Hidden cameras, one-way mirrors and other gadgets are used to observe their activities. Because the subjects are unaware that they are being observed, they behave naturally. The disadvantage is that it may take several hours of observation before the subjects exhibit the phenomenon of interest. A disguised observation could be:
 - o Direct observation occurs when the researcher personally observes the behaviour.
 - o Indirect observation, which is the effect or result of the observed behaviour.

The subjects are conscious that they are being observed in undisguised observation. There is a concern that the subject will engage in a usual activity during this form of observation. The presence of an observer may disrupt the subject, but it is impossible to predict how long this disturbance will last. According to studies, such descriptions are fleeting and the patients quickly return to regular behaviour.

3. **Participant vs. Non-Participant Observation:** Participant observation occurs when the observer contributes in the situation while observing. g. a researcher investigating slum residents' lifestyles through participant observation will live in slums. His function as an observer could be hidden or revealed. He is able to watch in an intelligent manner since he has become a part of the situation. The observer may get sympathetic to the subjects as a result of this strategy, making it difficult to view his research objectively.

Non-participant observation occurs when the observer maintains outside the context and does not become involved or engage in the circumstance.

4. **Natural vs. Contrived Observation:** In natural observation, behaviour is noticed as it occurs in the actual setting, such as consumer preferences observed directly at Pizza Hut when customers purchase pizza. The benefit of this procedure is that real results are produced; yet, it is an expensive and time-consuming method.

In manufactured observation, the phenomenon is observed in an artificial or simulated context, for example, consumers are induced to order in a setting that looks like a restaurant but is not an actual one. This form of observation has the advantage of being brief and documenting behaviour is simple. However, because the consumers are aware of their surroundings, they may not exhibit genuine behaviour.

5. **Classification on the Basis of Mode of Administration:** This includes the monitors and records the behaviour as it occurs. The recording is done in accordance with an observation schedule. Personal observations not only document what has been defined, but also identify and document unanticipated behaviours that defy pre-established responses.

o **Mechanical Observation:** Instead of humans, mechanical equipment are used to record behaviour. The gadgets record the behaviour as it happens and the data is later sorted and evaluated. Aside from cameras, other devices include a galvanometer, which measures the emotional arousal induced by a specific stimulus, an audiometer

and a people metre, which record which channel on TV is being viewed, with the latter also recording who is observing the channel, a coulometer, which records eye movement and so on.

- o **Audit:** The process of getting information through the physical examination of data. The audit, which is a physical object count, is usually performed by the investigator himself. A store audit or a pantry audit are both types of audits. Store audits are conducted by distributors or manufacturers in order to analyse market share, purchase patterns and so on. For example, the researcher may verify the store records or do an inventory analysis to record the data. The pantry audit entails the researcher creating an inventory of product brands, quantities and package sizes at a consumer's home, usually during a personal interview. An audit of this type is used to complement or test the accuracy of information supplied in the direct questionnaire.
- o **Content Analysis:** Content analysis is the objective, systematic and proactive description of communication's manifest content. This method is made up of two parts: observation and analysis. It entails analysing the contents of a spoken or written communication. A quantitative examination of the existence of a specific property in a document can be performed using content analysis by recognising and counting the presence of a specific characteristic, such as to determine which politicians regularly employ secularism in their speeches, key terms that are associated with secularism are identified. Following that, the speeches of other politicians are evaluated to see how many times the terms appear. The politician whose speech contains the most key words is the one who regularly employs secularism to attract voters, nonetheless, content analysis is now utilised for qualitative analysis, in which the basic message of the content is assessed.

Conducting an Observation Study

When carrying out an observation study, care should be taken to ensure that it is free of errors and bias: The fundamental steps are as follows.

1. **Specify the type of research.** If the research is exploratory in character, basic observation will suffice. Another research has been more systematic. Depending on the study environment, the observation could be unstructured or structured.
2. **Describe the contents of the observation.** The researcher should define the variables of interest that will be observed, as well as additional variables that will influence the study. For each identified variable, the definition and measurement words to be used to record the results should be stated.
3. **Observer instruction.** An observer must be trained in order to record and notice the correct things. He should be able to remember information, view occurrences objectively and concentrate well. The more unstructured an observation, the more reliant on the observer to secure the research outcomes. As a result, the experience of an observer in such a situation is required.
4. **Create an observation form.** The observation form is a plan for observation that addresses the following:

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- o Who, i.e., the subjects to be studied, as well as the mediators who assist in studying.
- o What, i.e., the aspects to be observed. The characteristics to be observed must be provided.
- o Who, i.e., the subjects to be studied, as well as the mediators who assist in studying.
- o When, i.e., the observation time. It must be chosen if the observation should take place at a specific time or if any time is fine.
- o Who, i.e., the subjects to be studied, as well as the mediators who assist in studying.
- o Where, i.e., the location of the observation. The methods utilised to pick the location for observation must also be described.
- o Who, i.e., the subjects to be studied, as well as the mediators who assist in studying.
- o How, that is, the method of observation. Details about how the data will be observed, such as whether it will be observed directly or indirectly, by a single or several observers, or by a personal or mechanical approach, should be provided.
- o Who, i.e., the subjects to be studied, as well as the mediators who assist in studying.
- o Once the plan has been developed, the observer is ready to measure and record the phenomena of interest.
- o Who, i.e., the subjects to be studied, as well as the mediators who assist in studying.
- o Who, i.e., the subjects to be studied, as well as the mediators who assist in studying.

All research includes some form of observation, which might be quantitative or qualitative. A quantitative observation entails collecting data objectively, generally in numerical form. The information gathered should be related to or comprehended in terms of a number.

Statistical survey data analysis is used to assess quantitative observations. Observations of any variable associated to a numerical value, such as age, shape, weight, height, scale and so on, are examples of quantitative observations.

A researcher, for example, can learn about a customer's contentment with recent purchases by asking them to score their satisfaction on a Likert scale ranging from 1 (very dissatisfied) to 7 (extremely delighted).

Qualitative observations, which do not include numerical measurements, monitor the qualities of a phenomenon.

From a safe distance, researchers observe respondents in a naturalistic setting. Because the respondents are in their natural surroundings, the traits seen by the researchers are interesting and provide additional information. A researcher, for

example, can observe and record the observations of customers in a supermarket by observing their selection and purchase behaviours. This provides a thorough understanding of the customer.

It is critical to ensure high survey response rates in any type of research to increase the quality of insights. See our webinar recording for four methods to increasing survey response rates and collecting high-quality insights.

- **Survey Methodology:**

The survey approach entails capturing respondents' responses using surveys or questionnaires. Polls can also be included in surveys. They are the most commonly used method for gathering market research data. Surveys are commonly used to elicit feedback from respondents. A good descriptive study should include a survey that includes both open-ended and closed-ended questions.

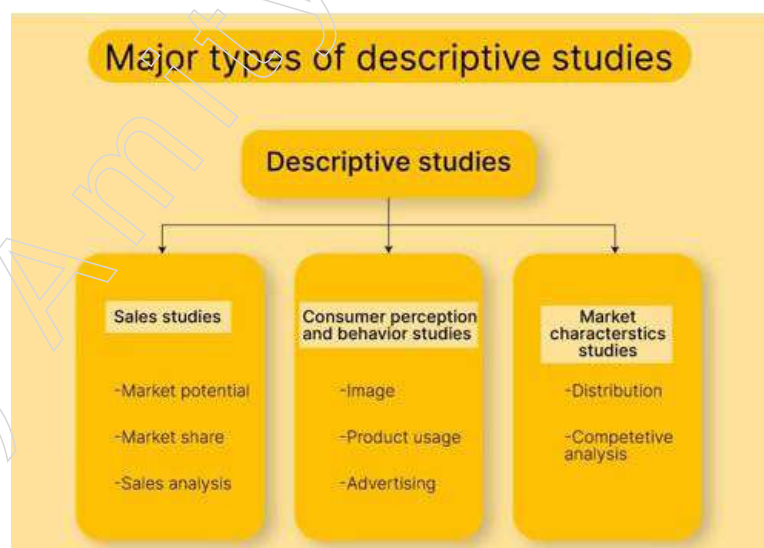
The survey approach has the greatest advantage of being able to be conducted both online and offline. One of the reasons that the survey method is the preferred method for descriptive research is that it allows for the collecting of vast volumes of data in a short period of time.

With Voice's Omnichannel survey software, you can conduct online surveys, offline surveys, phone surveys and self-completion surveys.

- **Case Study Technique:**

A case study is an in-depth examination of an individual or a group. Case studies typically result in the formation of a hypothesis to further investigate a phenomenon. Case studies are limited in scope since they do not allow the researcher to draw cause-and-effect findings or make precise predictions. This is because these relationships may represent a bias on the part of the researchers rather than a naturally occurring phenomena. Another reason case studies are limited in scope is that they may simply reflect an outlier respondent in the survey. An atypical responder is someone who is not like the average consumer and if researchers draw judgements about the entire target group based on one customer, the external validity of the study may suffer.

Some Examples of Conducting Descriptive Studies Under Market Research



Notes

- Sales Studies:

Researchers can utilise descriptive research to examine the market's potential; what is trending in the market and which products have a good possibility of selling well. They can also investigate what factors influence market shares and when they are anticipated to increase or decrease. They can also investigate the demographics of those purchasing things, such as their age group, gender and socioeconomic background.

- Consumer Perception and Behaviour Research:

Researchers can use descriptive research methodologies to examine what customers believe about the brand, what their attitudes (understand them using our brand perception survey form) are about the items sold by a specific brand and how they utilise other rival products. They can also examine which advertising methods have been successful in increasing positive brand perceptions. They can also examine consumer consumption behaviour (using a customer insights survey template to collect insightful data) and how it is influenced by product pricing.

- Market Characteristics Research:

Another way that academics utilise descriptive research methods is to examine the market distribution of products; which nations have higher sales, which countries have less products but sell out rapidly. Researchers can also examine competitors' brand management strategies, determining which strategies are effective and which are not.

Applications of Descriptive Research

A descriptive research method can be utilised for a variety of purposes. Even after setting survey goals, designing questions and gathering information via surveys, there is no way of knowing whether or not the research you are performing will achieve your predictions. Here are some examples of how organisations use descriptive research to grasp the objectives of research goals:

Defining the characteristics of respondents:

Because most descriptive research methods collect data using closed-ended questions, it aids in reaching objective judgments about the respondents. Descriptive research aids in the identification of trends, traits and behaviours among respondents. It also tries to understand respondents' attitudes and opinions regarding specific topics. For example, researchers can learn how many hours young adults spend on the internet, what they think about social media platforms and how significant they think these platforms are. All of this data will assist the organisation in making informed decisions about their products and brands

- Analysing trends in data:

Researchers can analyse data trends over time with the use of statistical analysis provided by descriptive research methodologies. Consider an apparel firm that releases a new line of clothing; they may conduct research to see how Gen Z and Millennials react to the new launch. If they discover that the new clothing line has been successful for one group (Gen Z) but not for the other, the corporation may discontinue production of clothing for the latter.

- Comparing various groups:

Comparing distinct groups of clients depending on their demographics is likewise related to the prior issue. A descriptive analysis could be performed to investigate how various groups of individuals react to distinct services provided by a corporation. For example, how does income, age, gender, income and so on effect consumer spending behaviour? This type of descriptive research assists businesses in determining what they should do to boost brand appeal in various segments of the public.

- Validating established responder patterns:

Because descriptive research methods are non-invasive and rely on quantitative data (for the most part), researchers can draw conclusions about why customers' existing purchase patterns exist. They can also utilise the information to launch a more in-depth investigation in the future.

- Conducting research at various periods

Descriptive analysis can be performed at several points in time to determine whether the trends are similar or dissimilar at different points in time. Studies can also be repeated in order to validate the original study's findings and derive proper conclusions.

- Discovering correlations between variables:

Descriptive research methods are also used to determine correlations and the degree of relationship between variables. For example, suppose a study is focusing on men's ages and spending habits. There is a chance of detecting a negative connection between the two variables, indicating that as men get older, they spend less money on sports equipment.

Advantages Of Descriptive Research

Descriptive research offers numerous advantages, including:

- Primary Data Collection

Data for a descriptive study is gathered through primary data gathering methods such as case studies, observational methods and surveys. This type of data collection provides us with valuable information that may be used for future research. It can also be used to generate hypotheses or to define your study purpose.

- Multiple Data Collection

Descriptive research can also be carried out through the collection of qualitative or quantitative data. As a result, it is more diversified, versatile and diverse, as well as more extensive and elaborate.

- Behavioural Observation

The descriptive research observational method allows researchers to watch respondent behaviour in natural situations. This also assures that the data gathered is of high quality and truthful.

- Cost-Effective

Descriptive research is inexpensive and data gathering for this study can be completed rapidly.

Notes

Disadvantages of Descriptive Research

Descriptive research has various drawbacks, which are as follows:

- **Deceptive Information**

Respondents may give deceptive or erroneous answers if they believe the queries are probing into personal topics. Respondents may also be influenced by the presence of the observer and participate in deception. This is referred to as the spectator effect.

- **Biases in Research**

The researchers' own prejudices may have an impact on the study's outcomes. This is referred to as the experimenter effect.

- **Representative Problem**

There is also the issue of representativeness, which means that a case study or data from a small sample does not effectively represent the entire population.

- **Limited Points**

Descriptive research has a limited reach because it simply analyses the "what" of research and does not evaluate the "why" or "how" of research.

Check Your Understanding

Fill in the blanks:

1. Before beginning to outline a proposal, it is usually a good idea to differentiate the type of research by incorporating it in the _____.
2. There are four main aspects of _____ are reliability, neutrality, validity and generalisation.
3. _____ are primarily employed in Qualitative Research Design to comprehend various social difficulties.
4. _____ research aids in the development of answers to specific problems or the development of new technology.
5. Theoretical research is _____, whereas applied research is _____.
6. The effectiveness of experimental _____ is dependent on researchers confirming that a variable change is due only to modification of the constant variable.
7. _____ research is study that does not include an independent variable.
8. There are two types of research that can be conducted, namely _____ and _____ research.

Summary

- A research design is a framework that contains the methods and procedures for gathering, analysing and interpreting data.

- Before beginning to outline a proposal, it is usually a good idea to differentiate the type of research by incorporating it in the research design.
- There are four main aspects of research design: reliability, neutrality, validity and generalisation.
- Case studies are primarily employed in Qualitative Research Design to comprehend various social difficulties.
- Pure research is explanatory and analytical in nature, whereas applied research is action-oriented and synthetic in nature.
- Applied research in education is utilised to enhance educational processes by providing practical answers to pedagogical challenges.
- Theoretical research is theoretical, whereas applied research is practical.
- In order to arrive at objective results, both research approaches use similar data collection processes, such as observation and interviews.
- Causation may not necessarily imply correlation, but correlation always suggests causation.
- Controlled studies are the only way to truly demonstrate causal relationships between variables.
- A cross-sectional study is an observational study in which data is obtained as a whole to study a population at a single point in time in order to investigate the relationship between variables of interest.
- Longitudinal research may be preferable if you are seeking for an approach that studies people and factors over time.
- The effectiveness of experimental investigations is dependent on researchers confirming that a variable change is due only to modification of the constant variable.
- Market researchers can use experimental research to a wide range of testing requirements. Experimentation can be applied to almost every situation that demands proof, confirmation, or is obscured by uncertainty.
- Non-experimental research is study that does not include an independent variable.
- Descriptive research is used to collect data and then use statistical analysis to gain more or less insight from that data.
- There are two types of research that can be conducted, namely primary and secondary research.
- Descriptive research aids in the identification of trends, traits and behaviours among respondents.
- Descriptive research is a subset of quantitative market research or social research that entails performing survey research with quantitative variables on a commercial research tool or a social research tool.

Activity

1. What measures do you suggest for analysing trends in data?

Notes

Glossary

- Traits: a distinguishing quality or characteristic, typically one belonging to a person.
- Objectively: in a way that is not influenced by personal feelings or opinions.
- Suffice: be enough or adequate.
- Extralinguistic: not involving or beyond the bounds of language.
- Obsolete: no longer produced or used; out of date.
- Adaptability: the quality of being able to adjust to new conditions.
- Delinquent: showing or characterized by a tendency to commit crime, particularly minor crime.
- Depict: show or represent by a drawing, painting, or other art form.

Questions and Exercises

1. Explain the Descriptive Research Approaches. Give proper illustrations of Descriptive Research.
2. Give a detailed account of Styles and Methodologies of Exploratory Research.
3. How can the Types of Observation be illustrated?
4. What are the best ways of conducting an Observation Study?
5. Illustrate the Case Study Technique.

Further Readings

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Answers to Check Your Understanding

1. research design
2. research design
3. Case studies
4. Applied
5. theoretical, practical
6. imply
7. Non-experimental
8. primary, secondary

Notes

Notes**Unit-1.4: Planning a Research Project****Learning Objectives:**

At the end of this unit, you will be able to:

- Understand identifying and defining a problem
- Evaluate research problem aspects & considerations
- Describe formulation of a research problem

Introduction

'When curiosity turns to serious matters, it's called research.'

– From Aphorisms (1880-1905) by Marie von Ebner-Eschenbach

Planning research projects is a time-honoured intellectual exercise that demands both inventiveness and acute analytical abilities. The goal of this Guide is to make the procedure systematic and simple to follow. While there is a lot of freedom and discovery involved – from the topics you choose to the data and methods you use - there are basic conventions and limits that apply regardless of your academic level or field of study. From high school to doctorate students and from art history to archaeology, research planning entails broadly comparable procedures, such as creating a question, developing an argument or making predictions based on past research and then selecting the resources needed to answer your issue.

At its most basic, research planning entails four distinct steps: orienting oneself to knowledge creation; identifying your research question; examining prior research on your question; and then selecting relevant facts to develop your own responses. Because the focus of this Guide is on planning a research project rather than executing a research project, this section will not go into depth on data collecting or analysis; those processes come after you plan the project. Furthermore, the subject is broad: year-long doctoral courses are devoted to data and analysis. Instead, the fourth and final section of this section will detail some fundamental tactics you could use to develop a data-selection and analysis procedure that is relevant to your research issue.

Key points to plan a research project

- While there is no 'best' way to design research, there are four general steps to planning a project: orienting yourself to knowledge creation; defining your research question; reviewing previous research on your question; and then selecting and analysing relevant data to formulate your own answers.
- Research planning is always an iterative process: as you progress through the four processes, it is common to go back and update prior aspects. Expect your research question, in particular, to go through several rounds of revision as you learn more about your topic, past research on that topic and the opportunities for you to conduct your own study.
- Good research questions tend to generate new research questions. This can be aggravating for individuals who want to get right to work. Make room for the unexpected: this is how knowledge normally evolves. Many of history's most

important discoveries have been made by people who were seeking for something quite different. There are ways to arrange your research planning process without overburdening yourself; the two exercises below are a good place to start and other approaches can be found in the Links and Books section.

1.4.1 Identifying and Defining a Problem

a) Research Delinquent

A research problem relic is an affirmation about a specific area of concern, a situation to be improved, a trouble to be eliminated, or a troubling question that happens in scholarly literature, in theory, or in practice that opinions to the need for expressive understanding and deliberate investigation. The research problem is typically posed in the form of a question in most of the social science corrections.

An exploration problem category does not specify how to do something, proposes a vague or broad proposition, or poses a value question. The determination of a problem declaration is to:

1. Familiarize the reader to the importance of the theme being studied. The booklover is preoccupied with to the inference of the education and the research queries or hypotheses to be monitored.
2. Dwells the problem into a particular setting that defines the parameters of what is to be investigated.
3. Make available the framework for reporting the results and signposts what is probably essential to conduct the study and explain how the discoveries will present this information.

Considerations in Selecting a Research Problem



- Interest
- Magnitude
- Measurement of concepts
- Level of expertise
- Relevance
- Availability of data
- Ethical issues

(i) Classify the Broad Study Area

This is an intense idea to consider about the topic of your interest. You ought to identify the field in which you would like to work for a long time since your theoretical study or graduation. It will aid you tremendously to get a stimulating research topic. For example, if you do advancement in sociology, you must decide your research study area in sociology. You might pick social problems corresponding unemployment, road accident, communal health, etc. For instance, if you do post-graduation in Computer Science and specialisation in Cyber Security, you must adopt your research study

Notes

area in Cyber security. You have sway over problems associated with cyber threats, cybercrime, cyber trends and so on.

ii) Divide the Broad Study Area hooked on Subareas

In this stage, you want to divide and stipulate your broad area of research into some sub-areas. You would refer with your supervisor in this regard. Write down sub-areas. Perhaps, if you select unemployment as your wide-ranging study area, then divide it under unemployment and social stability, joblessness & crime, unemployment & individual frustration, etc. Now, your research title could be something like, "How does unemployment lead to criminal activity?" or "How does unemployment lead to mental frustration among unemployed people?"

Let's say, if you select Cyber security as your extensive study area, formerly divide it into network safekeeping, web security, database security connected with cybercrime, etc.

iii) Mark-up your Concentration

It is nearly impossible to study all sub-areas. That's why you must recognize your area of interest. You should choose issues that are important to you. Your interest must be the maximum important element of your research study. Once you have selected your research study of interest, you would delete other sub-areas in which you do not feel interested. Keep in mind that if you lose interest in your investigation study, you will not get any results.

(iv) Learning Research Questions

In this period of formulating a research problem, you would point out your research queries under the area of attention as you decided in the previous stage. If you select unemployment as your research topic, your enquiries might be "how does unemployment affect individual social status?" "How it affects social stability?" "How it creates frustration among individuals?" Describe the research problem or question you intend to investigate. The more you revise the research problem it will be just as pertinent and fruitful towards solving the problem indeed.

(v) Set Obtainable Objective

Make your investigation's root objectives and sub-objectives clear. The primary purpose of research is to answer research questions. If you choose to investigate the "Impression of unemployment on individual social status" as your research problem or research question, set out what you want to discover and address at that point. Perhaps your primary goal influence will be to examine the unemployment rate in a specific society or state thus, the sub-objectives would be its effects on people's social lives. Setting attainable specific main and sub-objectives is critical.

(vi) Assess your Aims

You should evaluate your objectives immediately to ensure that you have a good chance of achieving them when you are finished with your research study. Assess your purposes in terms of time, budget, resources and technical proficiency at your

hand. You ought to also assess your research queries in light of reality. Regulate what outcome will bring your study. If you can measure accurately the persistence of the research study it will fetch significant results in the long run. In detail, research purposes determine the assessment of the study you are successful to work out.

(vii) Check Back

Before beginning research, you will go over all of the steps in expressing a research problem as well as everything you have completed thus far for the purpose of your investigation study. Then, ask yourself where your enthusiasm is. Do you have adequate resources to step up? If you are somewhat satisfied, then you frontward to undertake your research work. You can alter any of your plans in the light of genuineness if it necessitates.

1.4.2 Research Problem Aspects & Considerations

The following list summarises the 10 most crucial features of an excellent research problem for a thesis. The list allows you to assess any research problem and determine how well it measures up. Obviously, few issues will meet all 10 criteria, but good problems should meet the majority of them. A few remarks about each of them are in order.

Characteristics of a good thesis research problem are:

1. The problem can be stated clearly and concisely.
2. The problem generates research questions.
3. It is grounded in theory.
4. It relates to one or more academic fields of study.
5. It has a base in the research literature.
6. It has potential significance/importance.
7. It is do-able within the time frame and budget.
8. Sufficient data are available or can be obtained.
9. The researcher's methodological strengths can be applied to the problem.
10. The problem is new and it is not already answered sufficiently.

Fundamentals of Educational Research

To understand these points in detail:

1. The problem can be stated clearly and concisely.

Unless a problem can be expressed simply and concisely, it is most likely a bad or non-problem. The easiest method to put the problem statement to the test is to write it down in a brief sentence or paragraph and share it with others. If the problem cannot be articulated clearly in a single paragraph, it has challenges and will not survive as a suitable problem. Of course, it is difficult to convey complex topics in simple terms and it may take several weeks and numerous draughts until the statement is satisfactory. A good critic is vital. It's probably flaky if your spouse or mother can't understand it.

Notes

2. The problem generates research questions.

A number of more particular research questions should be generated as a result of the problem. These frame the problem as a question and indicate different features or components of the problem. The research questions help to address the more general statement and give a foundation for the investigation. Formulating these questions can be difficult, especially when it comes to expressing them at the appropriate degree of abstraction.

3. It is grounded in theory.

Theoretical and/or conceptual frameworks are used to analyse good problems. They connect the particular of what is being examined to a broader theoretical framework, which aids in the interpretation of the results and their relevance to the field.

4. It relates to one or more academic fields of study.

Good problems are those that relate to academic subjects that have followers and bounds. They usually have journals to which their followers subscribe. Research challenges that lack apparent linkages to one or two such disciplines of study are often doomed. Without such a field, determining where the problem lies in the universe of knowledge becomes impossible.

5. It has a base in the research literature.

A well-stated problem will be related to research literature, which is related to the previous principles. Tough problems frequently have a well-defined body of literature created by a small group of scholars and published in a small number of publications. With some issues, it may be tough to develop linkages and a literary foundation at first, but there should be one somewhere.

6. It has potential significance/importance.

This is the crucial 'so what?' question: Who cares once the problem is solved? Assume you've solved the problem and answered all of the questions and then consider whether you're any further along. The problem must be important to the researcher at the very least, but it should also be important to others.

7. It is doable within the time frame and budget.

There are logistical considerations in terms of your ability to conduct the research. It is pointless to pursue an issue that cannot be researched. Do not conduct an education study in India unless you have the funds to travel there and collect data, which may take years. This issue explains why so few theses are related to longitudinal data. The only exceptions are research firms with a long history of collecting and studying data on a specific community. Terman's (1954) study of genius, in which a defined sample was followed over 30 years, is a good example.

8. Sufficient data are available or can be obtained.

In other circumstances, there is not enough data to address the issue. Historical figures may have perished, archival records may have been lost, or access to specific

surroundings may be restricted. As previously said, conducting research on a foreign country is tough unless you can travel there and acquire local data. Using an existing database is one underutilised approach. Some data banks have been established over time and include several potentials for further investigation of new problems and issues.

Intricate Aspects of Research Problem

For your research problem to be effective, make sure that it has these elementary characteristics:

- Highlights important issues or needs;
- Based on truthful evidence (its non-hypothetical);
- Being untroublesome and relevant; and
- Signifies a testable and expressive hypothesis (avoiding useless answers).

1.4.3 Formulation of a Research Problem

Conveying your research problem allows you to make a purpose of your study clear to yourself and target readers. Emphasize your paper on providing pertinent data to address it. A problem statement is an operative and essential tool to keep you on pathway with research and evaluate it.

There are five ways to put the research problem into words:

1. Postulate research objectives

A lively account that defines all goals can help you maintain your composure and develop effective and eloquent research. They should be uncomplicated in order to bring you success. An imperfect goal will help you stay focused on your studies. This notice also assists professors in estimating the queries your research project answers and not the same methods that you use to state them.

2. Review of situation or environment for evaluating framework of research problem

The aforementioned is necessary to work hard to define and test all kinds of environmental variables to make your project successful. Why do you need to do that? This phase can help you describe if the important answers of your study will deliver sufficient data to be worth considering. Classify specific environmental variables that may have an impact on your research and begin expressing effective methods to control all of them.

3. Examine the nature of research problem

Research problems may be an assortment from simple to complex and the whole thing depends on a range of variables and their relationships. More or less of them can be unswervingly relevant to specific research questions, even though others are completely unimportant for your development. Why should you understand their nature? This knowledge aids you to develop effective clarifications. Consider nearby focus groups and other relevant details to provide the essential insight hooked on a specific question to gain a deep sense of understanding of all dimensions.

Notes

4. Determine adaptable relationships

Scientific, social and extra studies often focus on creating a convinced sequence of repeating behaviors over time. Finishing the entire process involves:

- Recognizing the variables that affect conceivable solutions to your research problem;
- Decisive on the degree to which you can use and rheostat all of them for study purposes;
- Shaping functional relationships amongst existing variables;
- Indicate the most critical variables for an explanation of your research problem. All through the formulation stage, it's necessary to reflect and generate as numerous potential approaches and variable relations as you can.

5. Alternative approaches

There are various consequences that each course of action or approach can bring, which is why you must anticipate them. Why connect possible outcomes? It is the primary goal of any research process.

Exploratory research is approved out when a topic needs to be thoroughly understood, especially if it has never been done before. The goal of such research is to observe the problem and its surroundings rather than to derive a conclusion from it. Such generous research will provide a researcher with a strong foundation for reconnoitering his ideas, selecting the appropriate research design and determining the outcome variables that are actually important for the analysis. Most importantly, such research can assist organisations or researchers without taking up a lot of time and resources, as it will power enable the researcher to determine if it worth pursuing.

Check Your Understanding

Fill in the blanks:

1. Good research questions tend to _____ new research questions.
2. The research _____ is typically posed in the form of a question in most of the social science corrections.
3. Theoretical or conceptual frameworks are used to analyse _____.
4. Conveying your _____ allows you to make a purpose of your study clear to yourself and target readers.
5. Research problems may be an assortment from simple to complex and the whole thing depends on a range of _____ and their relationships.

Summary

- Planning research projects is a time-honoured intellectual exercise that demands both inventiveness and acute analytical abilities.
- Good research questions tend to generate new research questions.
- The research problem is typically posed in the form of a question in most of the social science corrections.

- Your primary goal influence will be to examine the unemployment rate in a specific society or state thus, the sub-objectives would be its effects on people's social lives. Setting attainable specific main and sub-objectives is critical.
- The easiest method to put the problem statement to the test is to write it down in a brief sentence or paragraph and share it with others.
- Theoretical and/or conceptual frameworks are used to analyse good problems.
- Conveying your research problem allows you to make a purpose of your study clear to yourself and target readers. Emphasize your paper on providing pertinent data to address it.
- Research problems may be an assortment from simple to complex and the whole thing depends on a range of variables and their relationships.
- Scientific, social and extra studies often focus on creating a convinced sequence of repeating behaviours over time.
- Exploratory research is approved out when a topic needs to be thoroughly understood, especially if it has never been done before.

Activity

1. What would be the most appropriate steps according to you to examine the nature of research problem?

Glossary

- Formulation: the action of devising or creating something.
- Variable: not consistent or having a fixed pattern; liable to change.
- Unswervingly: not changing or becoming weaker; steady or constant.
- Pertinent: relevant or applicable to a particular matter; apposite.
- Extensive: covering or affecting a large area.
- Interpretation: the action of explaining the meaning of something.
- Relevance: the quality or state of being closely connected or appropriate.
- Concisely: in a way that is brief but comprehensive.

Questions and Exercises

1. What do you understand by identifying and defining a problem?
2. How will you evaluate research problem aspects and its considerations?
3. Describe formulation of a research problem in detail.
4. Illustrate the basic steps for expressing a research problem?
5. Why are the fundamentals of Educational Research required to be kept in mind for research?

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Notes

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Answers to Check Your Understanding

1. generate
2. problem
3. good problems
4. research problem
5. variables

Unit - 1.5: Applications of Research

Notes

Learning Objectives:

At the end of this unit, you will be able to:

- Understand application of research in marketing
- Evaluate application of research in finance
- Analyse application of research in human resource management
- Describe application of research in production
- Know application of research in entrepreneurship

Introduction

Research is a creative process and a systematic effort that is conducted in order to enhance knowledge that can be but is not limited to society culture human or any specialised topic. Research is a procedure that is used to establish facts or confirm them, as well as the outcomes of earlier work or to address current problems.

It is sometimes done solely to gain knowledge about a specific issue in a specific domain or sector. There are various forms of research that are employed in various businesses, ranging from discovering a new fact to performing an old fact with the use of modern technology.

Depending on the nature of the research and the methodology adopted to address the research, various methodologies can be employed to perform the research. Humanities, scientific, social, marketing, creative, practitioner, business, technological and other forms of research because research has a wide range of applications in every domain, it can be found in practically every industry.

The following are a few examples of research applications in various fields.

Applications of Research explained with Examples

1. Medicine Industry



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Research is frequently utilised in the medical business and pharmaceuticals to conduct tests and discover new drugs to cure various disorders. Pharmaceuticals are able to create new molecules and treat diseases such as Mumps, Measles, Polio and others thanks to research.

The medicinal sector has numerous uses for research and it would not be an exaggeration to say that the entire medical industry is based solely on research.

2. Medicinal Tech



Medicines are one component of the pharmaceutical sector; medical technology, on the other hand, represent a significant financial leap. The sickness can be easily treated with medicines, but in order to locate or diagnose it, high-end technological pharmaceutical items are required.

These products are technological in nature and have evolved over time. Detecting diseases with obsolete devices can be difficult and it is critical that diseases be discovered early in order to begin treatment.

3. Business research

In the last few decades, business has found vast applications for research. Different types of organisations have discovered a wide range of research applications in order to attract customers, establish better brands and make better products.

4. Environment research

Almost every administration in every country has rediscovered the importance of environmental research applications. Various non-governmental organisations (NGOs) and environmental rehabilitation organisations assist in understanding the current state of the environment and forecasting potential calamities to the public, allowing them to make appropriate changes.

Climate change and its consequences on our environment necessitate extensive environmental research. It is equally critical to educate people about potential natural disasters caused by the use of hazardous materials and to avoid using them before it is too late.

1.5.1 Application of Research in Marketing

Marketing research can be applied to various aspects of the marketing mix. Marketing research has a wide range of applications, including product, distribution, price, advertising and promotion, sales and market and so on.

What are the uses of marketing research?

Marketing research is the methodical examination of problems, the development of models and the gathering of data for the purpose of improving decision making and control in the marketing of goods and services.

Different Types of Marketing Research Applications

Marketing research applications can be considered under the following headings:

1. Product Research

Marketing research can be used in product planning and development to evaluate new product ideas, assess the need to change existing product mix, test new product acceptance, test product positioning, test package aesthetic appeal, product protection and ability to withstand transportation and stocking.

It entails researching product design, features, quality, brand tracking, brand preferences, product lines and product line rationalisation.

2. Advertising Research

Marketing research can be applied in a variety of ways in the advertising industry. It could be used to perform copy testing.

Marketing research can be used to investigate crucial elements of advertising copy such as the core subject, ideas, appeals, headlines and an evaluation of its attention value, communication clarity, memory value, conviction value and so on.

Marketing research can be used to determine the most cost-effective media plan for determining advertising effectiveness and measuring audience size.

It can also be used for media research, motivation research and determining the success of an ad campaign.

3. Distribution Research

Identification of existing and potential distribution channels, selection of appropriate intermediaries, determination of channel expectations, reduction of distribution cost, motivation for channels of distribution, measuring and evaluating the performance of the channels and different intermediaries, measuring relative effectiveness of different channels, assessment of dealer support and reasons for conflict are all broad areas of distribution research.

4. Pricing Research

The goal of this study is to determine the price expectations of consumers, as well as their reactions and responses to different price levels of items, in order to determine demand elasticity.

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It also includes determining demand elasticity.

It also includes determining consumer price expectations in various market segments, testing alternative pricing strategies, the evolution of competitor pricing strategies, the study of the factors influencing pricing decisions, understanding consumer pricing psychology and the evolution of firms' pricing policies and strategies.

5. Consumer Research

Customer research investigates a wide range of issues concerning consumer attributes and behaviour.

The primary goal of conducting study is for the organisation to gain a better knowledge of its customers.

The company may conduct consumer research to determine the composition of the product's customers, to locate customers, to determine demographic characteristics of current and potential customers, to determine psychographic characteristics of consumers, to identify buying motives, to understand consumer consumption patterns, reasons for consumer dissatisfaction, shifting consumption patterns, brand preferences and to prepare consumer profiles.

6. Market Research

The goal of this study is to acquire information about the company's market and to identify the market dynamics at work.

The scope of market research includes an assessment of market trends, determining the size of the company's current and potential markets, the evolution of the impact of government legislation, policies and schemes on the company's marketing operations and determining the strengths and weaknesses of competitors' marketing programmes, policies and strategies.

Market research can also be used to determine the Company's market share, forecast sales for the future period and effectively use market segmentation for product differentiation strategy.

7. Sales Research

The primary goal of sales research is to determine the sales potential of a company's products as well as the evolution of the company's sales performance.

Many subjects in sales research can be thoroughly investigated.

1.5.2 Application of Research in Finance

"If you can't count it, you can't manage it."

Accounting and finance are professions that are profoundly ingrained in practically every aspect of management, business concerns and day-to-day activities of any individual or organisation.

Accounting study investigates the implications of economic events on the accounting process, as well as the effects of reported information on economic events.

It covers a wide range of academic topics such as financial accounting, managerial accounting, auditing and taxation. Academic accounting research use the scientific approach to “address all areas of the accounting profession,” whereas research conducted by practising accountants focuses on solving problems for a client or group of clients. Although changes in accounting education and accounting academics in recent decades have resulted in a schism between academia and accounting practise, academic accounting research can make a substantial contribution to accounting practise. Whereas research in finance is not new to us, research in finance assists financial specialists and other individuals involved in studying how market developments and trends affect the financial business.

Definition

Webster’s Third International Dictionary of the English Language defines research as “studious inquiry or examination, especially critical and exhaustive investigation or experimentation, having for its aim the discovery of new facts and their correct interpretation, the revision of accepted conclusions, theories, or laws in the light of newly discovered facts, or practical applications of new or revised conclusions, theories, or laws. In the Encyclopaedia of Social Sciences, Research is defined as “the manipulation of things, concepts or symbols for the purpose of generalization to extend, correct or verify knowledge whether that knowledge aids in the construction of a theory or in practice of an art.”

Busha in his publication Research Methods in Librarianship says that Research is “a systematic quest for knowledge that is characterized by disciplined enquiry. Efficient and effective approach to expand knowledge is the conduct of special, planned and structured investigations.”

“Research is the process for looking for a specific answer to a specific question in an organized, objective, reliable way”, Payton, 1979 In order to understand research in deepest form it can be from what research is not about. That it is not mere information gathering:

- mere transformation of facts from one location to another;
- mere searching for information;
- a catch-word used to get information; and
- playing with words and presenting already published or searched data.

‘Research’ as an acronym, each letter of the word, standing for a particular aspect as given below:

R = Rational way of thinking

E = Expert and exhaustive treatment

S = Search and solution

E = Exactness

A = Analysis

R = Relationship of facts

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C = Critical observation, careful planning, constructive attitude and condensed generalization

H = Honesty and hard-work

Objectives of Research

The goal of the research is to find solutions to issues through the use of scientific procedures. To be specific, the research must be identified by taking into account the system requirements. During the process of determining the study objectives, the researcher must finalise the following:

- To formulate, that is, to create research in order to comprehend the subject from all perspectives.
- To be descriptive, that is, to accurately depict all of the qualities of a specific process, individual, group, situation, or institution
- To diagnose, that is, to examine the frequency with which something occurs
- To put a hypothesis of a causal relationship between variables to the test, i.e., hypothesis-based research studies.

In a social setting, research is carried out with the assistance of questions for research, hypotheses and boundary of the study.

Research Questions

Problems that have not yet been resolved or handled by anyone else, such as the objective of the research, the location of the research, the current state of the research and possible methods and means of resolving the research problem under investigation.

Hypothesis

A hypothesis is a presumption or assumption made by the researcher about the population or condition under study.

Boundary of the Study

It is a limit set by the researcher himself to restrict his work or to concentrate on a certain area within the given/available set-up and resources. The study's objectives can be characterised as qualitative or quantitative. The quantitative objectives seek to optimise specific measures of the system of study's performance, such as a study to estimate the demand for a product with high precision. The qualitative objectives seek to determine the relevance of hypotheses in a study of importance, such as a study of the impact of a company's sales force's level of education on its sales revenue.

Features

The following features dominated finance and financial management research:

- Estimating financial requirements: it is difficult to carry out any activity in real time as well as in the near future without finance. With thorough study, financial requirements can be accurately anticipated.
- Financial Management: The management of financial matters with the goal of

maximising profits. Choosing Capital Structure: research aids in understanding the capital structure of the organisation or department under consideration.

- Selecting a Source of Finance: Selecting source of finance is an ever going and ongoing process and through research activities it can always under scanner
- Selecting an Investment Pattern: Investment is always important, whether from an individual or institutional standpoint and understanding prior data and patterns can help.
- Proper Cash Management: Various cash management approaches can be employed and discovered through research and practise.
- Implementing Financial Controls: Putting suitable spending controls in place and implementing finance management research can be highly beneficial.
- Proper Use of Surpluses: Undertaking research activities simply determines how, where and in what manner surpluses should be employed and assigned.

Importance Financial and accounting research plays a vital role in guiding and influencing the path forward through the current fog of uncertainty that afflicts this industry. Financial research can help guide prudent investment selections in the asset management arena. To enable asset managers to ride through the current volatility, a greater emphasis on risk management approaches and asset allocation is required.

Finance and accounting are two dynamic fields in which ongoing research is required since data evolves and changes faster than in any other field. These two interconnected sectors have taken over every element of business and personal life. As the financial services business continues to deliver homogenous products and services, research, product development and technology will be required to generate distinct distinction in product offerings. Disintermediation is forcing financial services firms to cut costs by implementing more efficient technologies and to expand into new markets and product and service categories.

How will it be able to serve a fast rising population with differentiated product offers that tap into, align with and create value for the consumers' needs? To do this, it will be necessary to establish a unit whose exclusive focus will be on aligning and matching products with consumer wants, as well as leveraging research and product development to offer these items and services to them. Organizations that excel at R&D are constantly striving to identify improved processes and product management approaches in order to maintain their market leadership. Different research topics studied under financial management are:

- Ratio analysis
- Portfolio management
- Inflation
- Deflation
- Economic growth model
- Balance of payment
- Primary market study
- Secondary market study

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- Capital budgeting
- Study of financial parameters of organizations
- Economic evaluation of alternatives

1.5.3 Application of Research in Human Resource Management

Everything you need to know about HR research. HR research is, “the task of searching for and analysing of facts to the end that HR problems may be solved or principles and laws governing their solutions derived.”

HR research entails conducting searches, re-examinations, re-assessments and revaluations. In other words, research is an intentional and methodical examination meant to evaluate hypotheses using structured questions.

Meaning and Definition

The study of human resource practises and activities reveals the extent to which policies and procedures succeed or fail. HRM research gives an insight of what works, what does not work, what needs to change, the nature and extent of change. HR research is, “the task of searching for and analysing of facts to the end that HR problems may be solved or principles and laws governing their solutions derived.”

According to Michael J. Jucius, “HR research is the task of searching for and analysing of facts to the end that HR problems may be solved or principle and laws governing their solution derived”.

According to Dale Yoder, “HR research is a shortcut to knowledge and understanding which can replace the slower, more precarious road of trial and error in experience. It implies searching investigations, re-examinations, reassessments and revaluation. It is a purposive and systematic investigation designed to test carefully considered hypotheses or thoroughly framed questions”. “According to another expert, HR research is “the task of searching for and analysing facts, to understand how personnel problems may be solved or principles governing their solutions derived.”

Human Research Management research entails doing extensive investigations, re-examinations, re-assessments and revaluations. In other words, research is an intentional and methodical examination meant to evaluate hypotheses using structured questions. It is not just a collection of unstructured observations and it aims to answer specific issues. It is objective in the sense that it recognises and limits bias and prejudice at all stages of the process.

- It is systematic in the sense that it begins with a thorough design or plan and then conducts the research in accordance with that design.
- It is frugal; that is, it identifies methods and procedures for solving issues at the lowest possible cost.
- It is reproducible; that is, several researchers can utilise it independently at the same time.
- It is a well-planned and executed research, analysis and fact-finding procedure. It is carried out to confirm or disprove specific hypotheses or hunches. It adds to existing knowledge.

Accountability and Research Procedure

A researcher must adhere to a specific study technique. First and foremost, he must characterise the problem; that is, he must determine the problem that will be answered by developing a research plan. He must then describe his purpose; that is, he must determine the goal that will be achieved as a consequence of his study, such as formulating and testing a hypothesis and collecting data as a result of his observation of individuals, the distribution of a questionnaire and the conduct of interviews.

After gathering data, he classifies, analyses and interprets it, drawing conclusions, making generalisations, or developing new hypotheses. Finally, he writes a formal report that includes statements, tables, charts and other explanatory or illustrative material and he submits the findings of his research to people who need them.

The findings, plans and experiences of research projects are generally documented in a number of publications produced by an organisation, as well as in a number of other journals, technical or business periodicals; they are also covered in seminar reports, conference proceedings and monographs. Yoder classifies these into three categories – (a) those professing a major interest in the field of personnel and labour relations; (b) those having a specialised focus on one or more of these; and (c) journals covering wider interests, which include reports on research in the manpower management area.

Research is not solely the duty of any one division or department within an organisation. The personnel department has the initial responsibility, but it should be assisted by line supervisors and executives at all levels of management. The support that trade unions and other organisations such as educational institutions, private research groups and government agencies may provide should not be overlooked, but should be appropriately utilised.

In terms of research, psychologists, sociologists, economists, mathematicians and specialists in business administration, political science and other fields should all be included. According to Jucious, "The field of research requires the resources of several types of researchers and different kinds of tools. To seek answers through the methodology and principles of a single specialty is to build upon a weak foundation. Rather, research calls for a cosmopolitan attitude and interdisciplinary cooperation. The specialists who try to build a fence around all Aspects of research do themselves and industry a serious disservice."

Need of Research in Human Resource Management

Human resource research is frequently critical in resolving HR issues because it is difficult to make successful decisions without reliable information.

The following factors necessitate human resource research:

- (i) Expanding on current knowledge: Knowledge expansion is a constant and never-ending activity. Human resource knowledge is expanding at a quicker rate. HR research makes a substantial contribution to the expansion of existing HRM knowledge.
- (ii) Evaluation of proposed programmes, practices and activities: Before implementation, proposed programmes, practices and activities must be

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extensively evaluated. Human resource research gives critical information for evaluation.

- (iii) Evaluation of current and new policies and practises: Action research calls for the adoption of policies and practises based on research findings.
- (iv) Prediction of Personnel Issues: Employee unhappiness with a variety of issues has resulted in HR concerns. These issues will result in labour disputes. Human resource research can foresee potential difficulties and provide preventative measures.

Human Resource Management research is needed for several reasons, including:

- The economy's knowledge growth is huge and ongoing. HR expertise is expanding at a quicker rate. HR research makes a substantial contribution to the creation of new knowledge and the consolidation of current information.
- Human resource research provides the inputs required for a full evaluation of proposed programmes, practises and activities.
- Current HR policies, programmes and practises must be revised in light of HR research.
- HR research on many elements of human behaviour assists the organisation in identifying areas of employee happiness and dissatisfaction. While it may take initiatives to increase happiness, it may also take proactive actions to reduce employee dissatisfaction and the organization's oncoming trouble.

Some of the most essential aspects of Human Resource Management research are:

- (i) Human resource research is conducted with a specific goal in mind. In other words, it strives to provide particular answers.
- (ii) Human resource research is objective. It recognises and limits bias and prejudice throughout the process.
- (iii) It is methodical. It includes a detailed plan or design. Based on such design, an investigation is carried out.
- (iv) HR research is frugal. It identifies ways and approaches for resolving problems at the lowest possible cost.
- (v) Human resource research is reproducible. It can be utilised concurrently by multiple researchers.
- (vi) It is a well-planned and well-executed study and analysis.
- (vii) It is carried out in a systematic manner in order to check, verify, or reject hints, assumptions, or hunches.
- (viii) It adds to knowledge and pushes the boundaries of comprehension.

Manpower and human resource research encompasses all of the various disciplines that fall within the purview of personnel administration. The extent of such research can range from very simple to highly sophisticated, or from short and affordable to long and expensive. The four most dominating areas of research, according to most studies, are selection, training and development, attitudes and leadership and assessment devices.

The human resource researcher seeks to uncover the fundamental relationships that can lead to better personnel decisions in areas such as turnover, absenteeism, compensation levels and structure, job satisfaction, employee morale, assessment of managerial potential, training effectiveness, grievance handling, labour relations and collective bargaining.

Selection, opinion measurement, training and development, assessment, motivation, organisational effectiveness, management obsolescence, counselling and retirement are frequently highlighted as areas of high or low appearance in human resource research. Managerial selection and development, as well as general employee motivation, have been regarded as the two major human resource sectors in need of greater research.

The following are the primary goals of Human Resource Management research:

- i. Promoting and assessing current circumstances
- ii. Making predictions about future conditions, events and behavioural patterns.
- iii. Assessing the impact and outcomes of present policies, programmes, practises and activities.
- iv. Providing a logical foundation for revising current policies, programmes, procedures and activities.
- v. Evaluating proposed human resource policies, programmes and activities.
- vi. Evaluating and reviewing the relationship between organisational and HRM strategies.

The precise goals are as follows:

- (i) To assess and measure current conditions.
- (ii) Forecasting future conditions, occurrences and behavioural patterns.
- (iii) To assess the impact and outcomes of present policies, programmes, practises and activities.
- (iv) To provide a logical foundation for revising current policies, programmes, procedures and activities.
- (v) Assessing proposed policies, programmes and activities.
- (vi) Keeping management ahead of rivals by replacing old products with new products, old procedures with new techniques and old organisational practises with new organisational practises.
- (vii) To find ways and means of continuously improving employees' abilities and attitudes at a good or high level.
- (viii) To assess and revise the relationship and intertwining of organisational and human resource management methods.

Human Resource Management Research Methodology

The HR research process begins with an employee's symptom of dissatisfaction. To be proactive, management should recognise potential future employee problems and

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propose ways to prevent them. If problem avoidance is not practicable, management should do research to solve HRM issues.

The processes of Human Resource Management research are as follows:

1. Resolving the Research Issue

A researcher must first solve a research challenge before proceeding with an investigation. He can identify a research problem from the industrial environment, a review of the literature, observation, discussions with specialists and so on. The researcher must be crystal clear about what he aims to explore and why.

2. Establishing Goals

Once the research problem has been identified, he must develop relevant objectives and hypotheses. Otherwise, study would be a waste of time.

3. Improving the Research Method

- i. It refers to the data collection procedure. There are two kinds of data. The first is primary data and the second is secondary data. Primary data is data acquired directly from respondents, whereas secondary data is data collected from documents, libraries, records, periodicals, journals, books and so on.
- ii. The researcher's next step is to develop the research instruments, such as a questionnaire or timetable in the case of survey research.
- iii. In the event of an experimental study, the researcher may collect data even by observation.
- iv. The researcher must fix the sample in areas where the universe is vast. There are numerous sampling techniques. In order to obtain reliable data, he or she must select the appropriate sampling method and sample size. He/she can contact everyone and cover the entire population where the universe is small.
- v. The study period must be determined.

4. Data Collection

The researcher must collect information from the respondents.

5. Interpretation and analysis

After processing, the acquired data must be analysed. In analysis, tables, charts and diagrams are used. Statistical packages are also utilised to validate the results of the analysis. The analysed data is interpreted and then conclusions are drawn. The final section of the analysis focuses on the ramifications of the findings.

6. Report Writing

All of the findings are included in the report part and lastly, policymakers are given recommendations.

Human Resource Management Research Methods

Personnel research can be conducted using a variety of approaches and technologies. A research design must be chosen from among the different options accessible. The basic rule is to select the technique that promises to produce quality with the least amount of complexity, effort and money.

Typically, the research methodologies or instruments that are available are:

1. Historical Research

Past records and documentation are thoroughly investigated and interviews with former personnel are done. Almost all large organisations keep track of numerous personnel issues such as absenteeism, turnover, accident rates, wage structures and so on. The key element of this method is "its methodical research, which employs a long time span or longitudinal dimension."

2. Case Studies

These entail conducting an analytical investigation of the relationships that are significant in a specific situation or collection of circumstances. Although the precise meaning of a case study's conclusions is restricted to its specific previous scenario, thorough analysis and intelligent generalisation can be gained from it, endowing it with broader importance and application.

Individual case studies may result in the formation of broad hypotheses that can be used to establish the groundwork for additional or more intensive future study. The fundamental advantage of this method is that it allows the researcher to conduct a full, in-depth analysis of key episodes or situations; nevertheless, its main disadvantage is that it is historical in character and does not always represent general conditions.

3. Market Research

The focus of a survey research is on gathering original data by delivering a questionnaire or performing a structured interview. Certain study hypotheses are developed and survey questions are created to collect data. The relationship between observable phenomena, potential causes and related efforts is then computed and conclusions are drawn.

This method is time-consuming and expensive and it has been criticised for emphasising the necessity of data collection rather than the importance of analysing these facts and developing a theory on their basis.

4. Statistical Research

These disciplines are concerned with the gathering, analysis, classification and interpretation of mathematical and quantitative data. They place a premium on qualification, mathematical manipulation and statistical inference. They may employ averages, means, medians, modes, dispersion measures, trends, regressions and correlations. Because of the advancement of high-speed modern electronic data-processing technology, their use is becoming more common.

5. Mathematical Models

In most cases, mathematical models are employed in research to describe the

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relationships between the variables being studied. They aid in the development and testing of designs and sequences of equations that, in terms of mathematical notation, define the behaviour of interacting factors. These mathematical models also assist us in investigating both relatively simple and extremely complicated relationships and developing decision rules with broad applicability.

6. Simulation

Computers have popularised simulation-based designs. The procedure starts with the formulation of a hypothesis. It is used to research production and inventory management issues, as well as marketing, purchasing, employing and training workers and collective bargaining.

7. Action or field research

This strategy has proven to be the most effective in studying group behaviour in communities and workplaces. It entails complex design problems, because the observer himself becomes a variable in the observation process. This self-involvement on the side of the researcher provides him with new ideas, which he obtains through an active interaction that would not have been attainable through passive observation.

Following are the research approaches in Human Resource Management:

1. Experimental Study Approach:

Experimental studies attempt to determine how one variable effect another. In other words, it looks at how one variable behaves when the other is constant. The casual relationship has been formed. For example, in order to establish a relationship between incentive and output, other variables such as employee personality, ambient effect and so on are maintained under control.

Employees can be led to work for a number of days under identical conditions in this setting by adjusting incentives. If it is discovered that output varies with a given incentive, it can be concluded that, all else being equal, specific incentives influence output.

2. Exploratory Study Approach:

Exploratory studies place a premium on the discovery of new ideas and insights in order to gain a better understanding of a phenomenon or gain new insight into a research challenge in order to generate hypotheses. For example, a software corporation wants to figure out why there is such a high rate of attrition among programmers. It may develop specific hypotheses to be empirically tested. As a result, the reasons for attrition in this specific category of personnel are discovered in order to take corrective action. This type of research could pave the way for larger studies. It does not require a lot of money or time. It is believed to be adaptable in terms of data collection.

3. Historical Research Approach:

Historical research explores the origins and evolution of human resource problems in order to isolate and comprehend underlying elements. It is founded on historical

records and papers. Historical research allows you to see current events through the lens of your past experiences. It conducts systematic research on a time-span or longitudinal dimension.

Historical study sheds light on the evolution of a problem across time and then proposes solutions to it. This type of research is expensive because it takes a long time. Furthermore, the time lag between the occurrence of an event and its inquiry is a key issue. It calls into doubt the veracity of data until it can be empirically verified.

4. Case Studies Approach:

It aims to examine how an individual or an institution addresses a certain situation. This style of research allows a researcher to develop broad theories that will aid future research on similar issues. Example studies, on the other hand, are not comparable because each case gives a distinct contextual perspective.

5. Survey Research Approach:

The researcher obtains data from a representative sample population or the entire universe using a standardised questionnaire or interview schedule in this sort of research. He or she develops hypotheses. The obtained data is tallied, analysed and conclusions are drawn. This form of study is time intensive and expensive because the investigation must contact each and every sample responder. Examples include investigations into work satisfaction, absenteeism, attrition, morale and so on.

Planning and Conducting Research Programmes

In light of this, the processes required to organise and carry out a research programme are as follows:

1. Identify the research issue.

This should be one that piques the researcher's curiosity and has a clear link to a recognised theory or an important topic worth investigating. The investigation should yield new insights into the subject. At this point, background reading in the form of a preliminary study of the literature, mainly academic journals, but also books, particularly those based on research, is required to determine what has previously been accomplished in this field and any gaps. The setting in which the research will be conducted must be stated and supported.

2. Create an introductory research question.

This serves as the research's justification. It is, in effect, a declaration that answers the questions, 'What is the purpose of this research endeavour and what is its potential contribution to growing knowledge?' At this stage, it is based on the results of the first work done in Step 1, but it will be refined and reformed later when more information about the research is available.

3. Go over the literature.

A literature review will mostly look at academic journals. The goal is to determine what is already known about the topic, identify current theoretical frameworks and learn about other related studies.

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4. Create a theoretical framework.

The research must be carried out within a defined theoretical framework. This will outline the models, concepts and theories that can be used and developed to address the study topic. If a suitable framework does not exist, a grounded theory approach may be necessary, in which the researcher directly uses empirical evidence to establish the concepts and relationships that will be contained in the theory accepted as the study framework. It is critical to understand the assumptions, conditions and constraints underpinning the investigation.

5. Finish developing the research question.

The initial research question must be finalised in light of the results of the preceding procedures. The final research question will identify the topics to be studied and the issues to be explored. It will include a statement of intent outlining the goals of the research. This statement leads to the development of hypotheses or propositions that will be investigated by survey or experimentation during the research programme.

6. Come up with hypotheses or assertions.

A hypothesis is a tentative declaration of a relationship between two concepts that allows the ramifications of the statement being true to be tested. Hypotheses (which may be multiple) suggest the shape that the research project Human Resource Management will take in terms of gathering and analysing the evidence needed to test them. Hypotheses can be appended to the research question statement. A proposition is a proposal made as an explanation for an occurrence, a potential situation, or a type of behaviour that will be investigated by the investigation.

7. Design the research.

This entails planning ahead of time what research philosophy will be used. Is it to be positivist, phenomenological, or a combination of the two? The approach must then be established. A decision must be taken as to whether the research will be quantitative, qualitative, or a combination of the two. Methods for gathering and analysing evidence, as well as testing hypotheses or claims, will be discussed. The sources of evidence will be identified, as well as how they will be accessed. This will include analysing primary and secondary source documents, conducting additional literature studies, conducting surveys and conducting fieldwork. The design must clearly show how it will address the research topic while being compatible with the theoretical framework. If it is later discovered that this is not the case, the design will have to be changed.

8. Create a research plan.

This will address how the research will be carried out, the timeframe and the resources (money, personnel, software and so on) that will be required. Careful project planning is required.

9. Write and submit a proposal.

This will justify the research by outlining the research question as well as the recommended approach. It will also define the programme and the resources that will be necessary.

10. Carry out the research study.

This entails gathering and analysing evidence from a variety of sources in order to answer the research question and prove or disprove ideas. The relevance of the findings in respect to the study question and hypotheses will be explored and references to pertinent information from the literature will be made. This entails conducting an extensive literature study, gathering data, employing critical evaluation methods and employing statistical analysis where appropriate.

11. Create conclusions.

These compile all of the evidence. They answer the research question and explain why ideas were accepted or rejected. The findings' relevance will also be evaluated in terms of how they contribute to the advancement of existing knowledge and understanding. Any study limitations should also be highlighted.

12. Make suggestions.

These outline the managerial guidelines that emerged from the research. If the research was conducted within an organisation, they may additionally specify any necessary follow-up measures.

1.5.4 Application of Research in Production

In the last few decades, production has found vast applications for research. Different types of organisations have discovered a wide range of research applications in order to attract customers, establish better brands and make better products.

Here are a few examples of applications of research in Production:

1. Product research

Creating a new product necessitates extensive market research. Companies should research existing products and markets, as well as customer requests and wants.

In order to design a superior product and offer exceptional results in the market, the producer should also research what is missing in the market and what customers demand.

Companies hire third parties to perform research on existing products so that more products can be launched in a similar but more appealing way to buyers. Disrupting an existing base is one method for launching a product and research is undertaken appropriately. However, a second strategy would involve establishing and entering a new base rather than competing in the present market.

This would necessitate more research in order to identify the missing connection that may be monetized by the organisations. One option is to type the existing client base, while another is to establish a totally new customer base and market and both require equally solid research. For example, when the new iPhone was released, Apple was able to successfully generate a new client base for smartphones from an existing customer base by refusing to compete in the existing market.

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2. Marketing research

It is a type of marketing research that is very specialised and small in scope and it is carried out in order to improve advertising efficiency and reach customers more effectively. Pre-testing is a concept in advertising research in which the commercial is reviewed by a selected audience and their feedback is taken into account and the ads are modified or changed accordingly.

The goal of this study is to comprehend the brand influence, as well as the product impact and advertising effect, on the minds of the audience. Their attention and emotions are taken into account and feedback is provided as needed to improve the commercial.

3. Case studies

A few firms do case studies in order to gain a full understanding and detailed perspective on a customer, a company or a product or service for a specific instance that interests him or her.

Case studies assist organisations by offering a detailed assessment of a certain product. They also help to grasp the customer's attitudes and opinions about a company's goods in a far more detailed and comprehensive manner.

4. Questionnaires and surveys

Survey questions are a representation of study findings. The service assists the researcher in understanding and comparing the opinions of diverse clients for survey participants. The analysis can be used to do a much more in-depth investigation of the customer's topic.

It also enables the researcher to deliver insights into fresh concepts for client opinions that he would otherwise be ignorant of. If something works well for a product, it is the service that assists in determining what specifically works for the product and what exactly works against it.

5. Focus groups

Focus groups necessitate the investigation of diverse methodologies. The major aim for conducting focus group research is to gather ideas and honest feedback from customers about a certain brand's product or service.

The purpose of a focus group is to provide a sample group and sample feedback from a limited number of people so that the results can be extra favourable and applied to a larger audience and actual market. The feedback from the focus group is taken into account and appropriate adjustments to the product are made as a result.

6. Competition evaluation

In production, research is conducted to examine the competition. This enables producers to uncover flaws in their own products and redesign them in accordance with the market or client.

Studying the competition also assists producers in evolving and bringing themselves up to speed with the competition. They can compare themselves to

competing items and adjust their offerings accordingly. Competition analysis also assists producers in understanding each company's market share in relation to the competition.

1.5.5 Application of Research in Entrepreneurship



Research in Entrepreneurship is the process of gathering thorough information about all aspects of business and applying that information to maximise the company's sales and profits. A study like these assists businesses in determining which product or service is the most profitable or in high demand. In simple terms, it is the gathering of information or knowledge for professional or commercial purposes in order to assess prospects and goals for a firm.

Research in Entrepreneurship can be done for anything and everything. In general, when individuals talk about Research in Entrepreneurship, they imply asking research questions to determine where money may be spent to boost sales, profitability, or market share. Such study is essential for making intelligent and informed decisions.

For example, suppose a mobile phone business want to introduce a new model to the market. They are, however, unaware of the most popular mobile phone dimensions. As a result, the company conducts a Research in Entrepreneurship using various methods to gather information, which is then evaluated and conclusions drawn as to what dimensions are most in-demand. This will enable the researcher to make wise decisions to position his phone at the right price in the market and thus acquire a larger market share.

Research in Entrepreneurship: Types and Methodologies

Research in Entrepreneurship is a component of the business intelligence process. It is frequently carried out to establish whether a firm can flourish in a new territory, to study their competition, or to simply choose a marketing strategy for a product. This study can be carried out utilising either qualitative or quantitative research approaches.

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1. Quantitative research methods

Quantitative research methods are those that deal with numbers. It is methodical empirical research that employs statistical, mathematical, or computational approaches. Such procedures often begin with data gathering and then progress to statistical analysis utilising various methodologies. The following are some of the research methods utilised in Research in Entrepreneurship.

Survey research

Survey research is one of the most extensively utilised methods of data collection, particularly for performing Research in Entrepreneurship. Surveys involve posing numerous survey questions to a group of people using various methods such as online polls, online surveys, questionnaires and so on. Nowadays, most large organisations employ this strategy to collect data and use it to analyse the market and make suitable business decisions. Survey research employs various types of surveys, such as cross-sectional surveys, which are used to collect data from a specific group of people at a specific point in time and longitudinal surveys, which are used to collect data from a specific group of people over time in order to understand changes in the respondents' behaviour. Because of technological advancements, surveys can now be sent online via email or social media.

For example, suppose a corporation wishes to know the NPS score for their website, i.e., how pleased individuals who visit their website are. An increase in traffic to their website or the audience spending more time on their website might result in higher ranks on search engines, allowing the company to generate more leads and boost its visibility. As a result, the corporation can ask people who visit their website a few questions via an online survey to understand their ideas or obtain feedback and then make relevant modifications to the website to boost satisfaction.

Correlational Research

Correlational research is used to better understand the link between two entities and the impact each has on the other. Correlational research allows the researcher to correlate two or more variables using mathematical analysis methodologies. Such study can aid in the understanding of patterns, correlations and trends, among other things. It is also feasible to achieve the desired results by manipulating a single variable. In general, a conclusion cannot be formed solely on the basis of correlational study.

For example, a study might be undertaken to better understand the relationship between colours and gender-based audiences. Using such research and determining the target audience, a corporation can select the manufacture of specific colour products to be released in the market. This allows the organisation to better understand the supply and demand for its products.

Comparative-Causal Studies

Causal-Comparative research is an approach based on comparison. It is used to determine the cause-and-effect relationship between variables. It is also known as quasi-experimental research since it entails setting an independent variable and examining the impact on the dependent variable. Manipulation is not used in this type of research; rather, changes in variables or groups under the impact of the same

changes are noted. Drawing conclusions from such study is a little problematic because independent and dependent factors will always present in a group, thus all other characteristics must be considered before drawing any conclusions from the research.

For example, a study may be done to examine the impact of superior educational facilities in rural areas. Such a study can be conducted to examine the changes in a group of people from rural areas when they are supplied with decent educational facilities and before that.

Another example is to consider the impact of dams on farmers and crop output in that area.

Experimental Research

The goal of experimental research is to test a theory. Such study may be valuable in Research in Entrepreneurship since it can inform the product company about some behavioural characteristics of its customers, which can lead to increased revenue. In this method, an experiment is performed on a group of people to watch and then study their behaviour when subjected to specified criteria.

For example, recently, experimental research was undertaken to determine whether specific colours have an effect on the hunger of its consumers. A subset of the audience was then exposed to those specific colours while eating and the subjects were observed. Certain hues, such as red and yellow, have been shown to enhance hunger. As a result, such study was a windfall to the hospitality business. Many fast food restaurants, such as McDonald's and KFC, use these colours in their interiors, branding and packaging.

Another example of inferences taken from experimental research, which is commonly employed by most bars/pubs around the world, is that loud music causes people to drink more in less time. This was demonstrated through experimental research and was a significant discovery for many business owners throughout the world.

Online research / Literature Research

One of the oldest strategies is literature research. It is incredibly cost effective and it can yield a lot of information. Online research or literature research entails obtaining information from existing documents and studies, which can be found in libraries, annual reports and so on. With the growth of technology, such study has become even more straightforward and accessible to everyone. An individual can conduct direct research online for any information that is required, which will provide him with in-depth knowledge about the issue or the company. Such research is primarily employed by marketing and salespeople in the business sector to better understand the market or their customers. Such research is carried out using current information obtained from multiple sources, albeit care must be taken to validate the sources from which the information is to be gathered.

For example, a salesperson learns that a particular organisation is looking for a solution that their company supplies. As a result, the salesperson will first hunt for a decision maker within the firm, examine what department he is from and learn about what the target organisation is looking for and what they are into. Using this knowledge,

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he can tailor his solution to be exactly right when he pitches it to this client. He can even contact the customer directly by seeking a way to engage with him online.'

2. Qualitative research methods

Qualitative research is a vital tool in Research in Entrepreneurship. Qualitative research entails gathering information through open-ended conversational ways of communication. Such study allows the researcher to not only grasp what the audience thinks, but also why he thinks it. In such studies, detailed information can be acquired from subjects based on their responses. There are several different types of qualitative research methodologies that are commonly utilised, including interviews, focus groups, ethnographic research, content analysis and case study research. Such methodologies are critical in Research in Entrepreneurship because they allow the researcher to understand the consumer. What inspires consumers to buy and what does not is what will lead to better sales, which is the primary goal of every firm.

The following are a few ways that are frequently employed in today's world by most firms.

Interviews

Interviews are comparable to surveys in that they may contain the same questions. The distinction is that the respondent can answer these open-ended questions at length and the direction of the conversation or the questions being asked can be adjusted based on the subject's response. Such a method typically provides the researcher with thorough information about the subject's perspective or opinions. Conducting interviews with subject matter experts can also provide vital information to some businesses.

For example, a telecom manufacturer conducted an interview with a group of women to determine why they have fewer female clients. After interviewing them, the researcher discovered that some of the models had less feminine hues, therefore ladies opted not to purchase them. Such information might be crucial to a firm, such as a telecom manufacturer and can thus be used to gain market share by targeting female clients by introducing some feminine colours into the market.

Another example would be to interview a subject matter expert in social media marketing. An interview of this type can help a researcher understand why certain types of social media advertising methods work for a firm and why others don't.

Focus Groups

A focus group is a group of people who have been specially chosen to understand their perspectives and actions. It is typically a small group of people chosen with the parameters for their target market audience in mind to discuss a specific product or service. This strategy provides a researcher with a bigger sample size than an interview or a case study while utilising conversational communication.

Nowadays, focus groups can be sent online surveys to collect data and answer why, what and how questions. Such a strategy is critical for testing new concepts or goods before they are released to the public.

For example, a focus group study is undertaken to determine which dimension of screen size is most preferred by the present target market. A strategy like this allows

a researcher to delve further if the target market is more concerned with the phone's screen size, features, or colours. Using this information, a company may make informed judgments about its product range and increase its market share.

Ethnographic Research

Ethnographic research is one of the most difficult to conduct, yet the results can be incredibly exact. Such study is employed infrequently since it is time-consuming and can be costly. It requires the researcher to adapt to the natural environment and watch its target audience in order to collect data. Such an approach is typically utilised to comprehend cultures, obstacles, or other events that may occur in that specific area.

For example, the internationally recognised show "Undercover Boss" is an excellent example of how ethnographic research may be used in business. In this show, the top management of a huge business works as an ordinary employee in his own company to understand what improvements may be made, what the culture is in the organisation and to discover and reward hard-working workers. It can be seen that the researcher had to spend a significant amount of time in the natural setting of the employees in order to adapt to their ways and processes. While observing in this scenario, the researcher could obtain the knowledge he need first-hand without losing any information or prejudice and he could improve specific aspects of his business.

Case Study Research

Case study research is one of the most essential types of Research in Entrepreneurship. Most firms use it as marketing collateral to attract new customers. Case study research is used to measure client happiness, document difficulties and provide solutions. These inferences are used to highlight the benefits that the consumer received for choosing their specific firm. Such study is frequently applied in other sectors such as education, social sciences and so on. Case studies are offered by firms to new clients to demonstrate their capabilities and so such research is critical in the business sector.

For example, a service provider may have offered a testing solution to one of its clients. Case study research is undertaken to determine what obstacles were encountered during the project, what the scope of their work was, what the goal was to be attained and what solutions were provided to address the challenges. The study can conclude with the benefits that the company supplied through their solutions, such as shorter time to test batches, easier system deployment or integration, or even cost reduction. Such a study demonstrates the company's capacity and can thus be provided as empirical evidence to the new possibility.

Website Visitor Profiling/research

Website intercept surveys or website visitor profiling/research is a new concept that has emerged and is highly useful in the commercial industry. Using surveys to obtain direct feedback from your website visitors is an interesting method. In today's world, a lot of business is generated online, so it is critical to understand your website visitors because they are your future clients. Collecting feedback is crucial for any business since no business can be successful unless it understands its customers. To stay on

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top, a corporation must keep its customers delighted and aim to turn them into loyal customers.

A website intercept survey is an online survey that allows you to target visitors to understand their intent and collect feedback to evaluate the consumers' online experience. This can be used to capture information such as visitor intention, behaviour path and overall website satisfaction.

Depending on the information sought, various types of internet intercept surveys can be used to collect replies. Pop-ups, also known as Modal boxes and on-page surveys are two of the most common.

For example, suppose a prospective consumer is seeking for a specific product that a company is selling. Once he is routed to the page, an intercept survey will begin tracking his intent and path. After the transaction, the customer is presented with a pop-up or an on-page survey to rate the website. Such research allows the researcher to put this data to good use and therefore understand the customers' goal, his path and modify any areas of the website based on the responses, which leads to satisfied consumers and so higher revenues and market share.

Advantages of Research in Entrepreneurship

- Research in Entrepreneurship aids in identifying opportunities and dangers.
- It aids in the identification of problems and with this information, intelligent decisions may be taken to address the issue correctly.
- It aids in better understanding of customers and, as a result, can be used to improve communication with customers or stakeholders.
- By completing Research in Entrepreneurship ahead of time, risks and uncertainties can be reduced.
- Financial consequences and investments that will be required can be efficiently planned through Research in Entrepreneurship.
- Such research can aid in the tracking of competitiveness in the business sector.
- Research in Entrepreneurship can help a corporation decide where and how much to spend.
- Research in Entrepreneurship can help a company stay current with the market and its trends and relevant innovations can be made to stay ahead of the competition.
- Research in Entrepreneurship aids in determining a company's reputation.

Disadvantages of Research in Entrepreneurship

- Research in Entrepreneurship can be expensive.
- The majority of the time, Research in Entrepreneurship is based on assumptions.
- Research in Entrepreneurship can be time-consuming.
- Because of a biased demographic or a tiny focus group, Research in Entrepreneurship might occasionally provide erroneous findings.

- Because of the fast-changing nature of the markets, Research in Entrepreneurship findings can quickly become obsolete.

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The Importance of Research in Entrepreneurship

Research in , is one of the most effective methods for understanding customers, the market and competitors. This type of research assists businesses in understanding market demand and supply. Using such research can assist firms in lowering costs and developing solutions or goods that are targeted to market demand and the appropriate audience.

In-house Research in Entrepreneurship can help senior management establish an effective team and train or coach employees as needed. Research in Entrepreneurship allows a company to watch its competitors and, as a result, provide you an advantage in staying ahead of them. Failures can be prevented by conducting such research because it can inform the researcher as to whether the moment is proper to launch its product/solution and also whether the audience is appropriate. It will aid in understanding the brand's value and measuring client happiness, both of which are critical in order to consistently innovate and meet customer requests. This will aid the company's revenue and market share growth. Research in Entrepreneurship also aids in the recruitment of qualified candidates for various positions within the organisation. By undertaking such research, a corporation can perform a SWOT analysis, i.e., identify its strengths, weaknesses, opportunities and threats. With the use of this knowledge, intelligent decisions may be made to ensure corporate success.

Research in Entrepreneurship is the first step that any business owner must take in order to set up his business, survive, or thrive in the market. The primary reason for the relevance of such research is that it assists firms in growing in terms of revenue, market share and brand value.

In the last few decades, business has found vast applications for research. Different types of organisations have discovered a wide range of research applications in order to attract customers, establish better brands and make better products.

Check Your Understanding

Fill in the blanks:

1. _____ research is a vital tool in Research in Entrepreneurship.
2. _____ and _____ are two dynamic fields in which ongoing research is required since data evolves and changes faster than in any other field.
3. Marketing research can be applied in a variety of ways in the advertising industry and could be used to perform _____.
4. Almost every administration in every country has rediscovered the importance of _____ research applications.
5. _____ research is used to better understand the link between two entities and the impact each has on the other.
6. Research in _____ is the first step that any business owner must take in order to set up his business, survive, or thrive in the market.

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Summary

- Research is a creative process and a systematic effort that is conducted in order to enhance knowledge that can be but is not limited to society culture human or any specialised topic.
- Almost every administration in every country has rediscovered the importance of environmental research applications.
- Marketing research is the methodical examination of problems, the development of models and the gathering of data for the purpose of improving decision making and control in the marketing of goods and services.
- Marketing research can be applied in a variety of ways in the advertising industry. It could be used to perform copy testing.
- Market research can also be used to determine the Company's market share, forecast sales for the future period and effectively use market segmentation for product differentiation strategy.
- Accounting study investigates the implications of economic events on the accounting process, as well as the effects of reported information on economic events. It covers a wide range of academic topics such as financial accounting, managerial accounting, auditing and taxation.
- Problems that have not yet been resolved or handled by anyone else, such as the objective of the research, the location of the research, the current state of the research and possible methods and means of resolving the research problem under investigation.
- The qualitative objectives seek to determine the relevance of hypotheses in a study of importance, such as a study of the impact of a company's sales force's level of education on its sales revenue.
- Finance and accounting are two dynamic fields in which ongoing research is required since data evolves and changes faster than in any other field.
- Organizations that excel at R&D are constantly striving to identify improved processes and product management approaches in order to maintain their market leadership.
- In production, research is conducted to examine the competition. This enables producers to uncover flaws in their own products and redesign them in accordance with the market or client.
- Research in Entrepreneurship can be done for anything and everything.
- Research in Entrepreneurship implies asking research questions to determine where money may be spent to boost sales, profitability, or market share. Such study is essential for making intelligent and informed decisions.
- Correlational research is used to better understand the link between two entities and the impact each has on the other.
- Qualitative research is a vital tool in Research in Entrepreneurship.
- Ethnographic research is one of the most difficult to conduct, yet the results can be incredibly exact.

- A website intercept survey is an online survey that allows you to target visitors to understand their intent and collect feedback to evaluate the consumers' online experience.
- Research in Entrepreneurship is the first step that any business owner must take in order to set up his business, survive, or thrive in the market.

Activity

1. What applications of research do you recommend for finance? Explain with examples.

Glossary

- Deployment: the movement of troops or equipment to a place or position for military action.
- Prejudice: preconceived opinion that is not based on reason or actual experience.
- Validate: check or prove the validity or accuracy of (something).
- Relevant: closely connected or appropriate to what is being done or considered.
- Entail: involve (something) as a necessary or inevitable part or consequence.
- Justification: the action of showing something to be right or reasonable.
- Journals: a newspaper or magazine that deals with a particular subject or professional activity.
- Incentive: a thing that motivates or encourages one to do something.
- Obsolescence: the process of becoming obsolete or outdated and no longer used.
- Adhere: stick fast to (a surface or substance).
- Prudent: acting with or showing care and thought for the future.

Questions and Exercises

1. State the problems that are usually faced in application of research in marketing.
2. Evaluate the disadvantages of Research in Entrepreneurship.
3. Mention the types and methodologies Research in Entrepreneurship.
4. Analyse application of research in human resource management.
5. Give an overview of application of research in production.
6. Analyse the planning and conducting research programmes.
7. Explain the research approaches in Human Resource Management.

Further Readings

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Answers to Check Your Understanding

1. Qualitative
2. Finance, accounting
3. copy testing
4. environmental
5. Correlational
6. Entrepreneurship

Module - 2: Research Methods and Data Collection Technique

Notes

Structure:

- 2.1 Research Modelling
 - 2.1.1 Types
 - 2.1.2 Stages
- 2.2 Data Collection
 - 2.2.1 Introduction to Data Collection
 - 2.2.2 Data Collection Methods
 - 2.2.3 Survey
 - 2.2.4 Observation
 - 2.2.5 Questionnaire
 - 2.2.6 Tabulating & Validating Data
- 2.3 Questionnaire Designing
 - 2.3.1 Steps in Constructing a Questionnaire
 - 2.3.2 Types of Questions
 - 2.3.3 Format of Questionnaire
- 2.4 Measurement & Scaling
 - 2.4.1 Introduction to Measurement & Scaling
 - 2.4.2 Scaling Techniques
 - 2.4.3 Attitude Measurement Scales
- 2.5 Sampling
 - 2.5.1 Sampling Plan
 - 2.5.2 Sampling Frame
 - 2.5.3 Sample Selection Methods
 - 2.5.4 Probability Sampling Techniques
 - 2.5.5 Non-probability Sampling Techniques
 - 2.5.6 Sampling and Non-sampling Errors

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Unit - 2.1: Research Modelling

Learning Objectives:

At the end of this unit, you will be able to:

- Understand the types and stages of Research Modelling

Introduction

Research method by which one acquires dependable and convenient information about a phenomenon or a process can be broadly defined “as a systematic investigation towards understanding a multifaceted social phenomenon or a process”. It monitors the scientific approach to gain knowledge. The maximum important characteristic of this methodology is its thrust on objectivity.

To what extent is the research using the scientific approach advantageous in studying the problems of society? How can we attain reliable knowledge about the countless aspects of human experience? To be more exact how can the scientific approach be of value in consideration of social phenomena? In this Unit we will deliberate these questions. Our approach would be first, to recognize the meaning of the terms ‘research’ and ‘scientific research’, secondly, examine using the scientific method, its request in social sciences, its expectations and finally to take a close look at the methodology to find out how it can help rural advance professionals to understand the glitches of rural development.



2.1.1 Types

A research design is a framework for conducting research. There are numerous research designs, each serving a distinct purpose and achieving a distinct goal. In this section, we will look at important types of research design:

1. Historic research

answers the question, "How did belongings use to be?" When examining documents, historical researchers have encountered two key issues: primary vs. secondary sources in addition external versus internal blame.

Someone who was a member of or a first-hand witness to an event provided a primary source. Someone who got his or her information about an event from someone else prepared a secondary source.

External blame is a reference to the authenticity of the text. Once a document has been strongminded to be genuine (external criticism), investigators need to determine if the content is correct (internal criticism).

We perform historical research for a number of reasons:

- to evade the mistakes of the historical
- to put on lessons from the past to present problems
- to use the previous to make predictions near the present
- to know present practices and procedures in light of the past
- to observe trends athwart time

2. Action Research

- Action research stays a three-step spiral course of
 - (1) Planning which encompasses fact-finding,
 - (2) Captivating action and
 - (3) Fact-finding around the consequences of the action. (Lewin, 1947)
- Action research is a route by which practitioners endeavour to study their problems methodically in order to guide, correct and assess their decisions and action. (Corey, 1953).
- Act research in education is a study showed by colleagues in a school setting of the outcomes of their activities to recover instruction. (Glickman, 1990)
- Accomplishment research is a fancy way of saying let's study what is happening at our school and choose how to make it a healthier place. (Calhoun, 1994)

Circumstances that Provision Action Research

- An ability where a majority of teachers wish to progress some aspect (s) of edification in their school.
- Mutual agreement about how collective results will be made and implemented.
- A squad that is eager to lead the initiative.

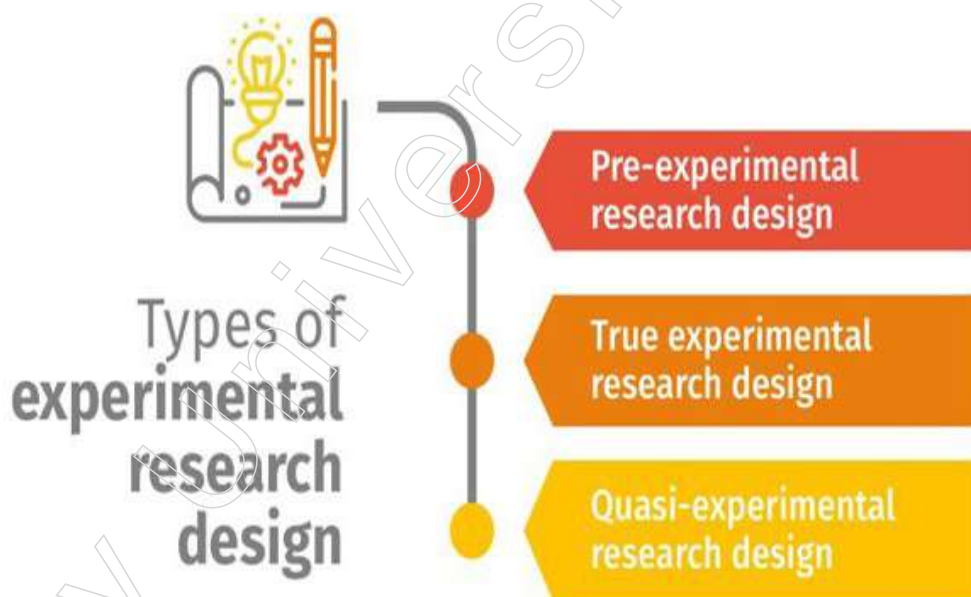
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- Education groups that meet commonly.
- A straightforward knowledge of the action research cycle and the logic for its use.
- Somebody to provide technical help and/or support.

The Action Research Series

- Classify a part of interest/problem.
- Recognize data to be collected, the arrangement for the results and a timeline.
- Gather and bring together the data.
- Scrutinize and understand the data.
- Choose upon the action in the direction of be taken.
- Estimate the accomplishment of the action.

3. Experimental Research



Experimental research refers to studies in which the researcher manipulates a variable to arrive at a conclusion or judgement, which is difficult to achieve in social sciences due to the presence of operating factors. It's helpful for determining the cause and impact of an underlying link and correlation. Experimental research does the same type of analysis, but it always looks at where the data from an experiment comes from.

Specimen of experimental research consist of laboratory experiments.

Forms of Experimental Research Project

The common definition for experimental research is, "The approaches used to collect data in experimental educations."

There are three primary forms of experimental research design:

- Pre-experimental investigation design

- True experimental research scheme
- Quasi-investigational research design

The manner you classify research subjects, grounded on conditions or groups, regulates the type of design.

1. Pre-experimental research design: Following the implementation of cause-and-effect elements, a collection of, or numerous groups, are kept under watch. You'll assess this research to see if more investigation is required for these specific groups.

You be able to breakdown pre-experimental research into three kinds:

- One-shot Case Learning Research Plan
 - One-set Pre-test-post-test Research Strategy
 - Static-group Link
2. Exact experimental research plan: True experimental investigation relies on statistical scrutiny to prove or refute a hypothesis, making it the most accurate system of research. Of the categories of experimental design, only true policy can find a cause-effect relationship inside a group. In a true experiment, three features need to be gratified:
 - In attendance is a Control Group, which won't remain subject to changes, in addition an Experimental Group, which determination experience the reformed variables.
 - A mutable which can be wrought by the researcher
 - Random spreading

This investigational research method commonly arises in the physical sciences.

3. Quasi-experimental research design: The term "Quasi" indicates similarity. A quasi-experimental plan is similar to experimental, nevertheless it is not the same. The variance between the two is the task of a control group. In this research, a self-governing variable is manipulated, nonetheless the participants of a collection are not randomly allocated. Quasi-research is used in ground settings where random task is either irrelevant or not required.

Benefits of Experimental Investigation

It's energetic to test new ideas or theories. Why put time, exertion and subsidy into some degree that may not work?

Experimental research permits you to test your impression in a controlled environment beforehand taking it to market. It also makes available the best method to test your theory, thank you to the subsequent advantages:

- Researchers partake a stronger hold over variables to achieve desired results.
- The topic or industry does not impact the efficiency of experimental research. At all industry can instrument it for research purposes.
- The consequences are exact.

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- Afterward analysing the results, you can smear your findings to comparable ideas or situations.
- You can classify the cause and outcome of a hypothesis. Researchers can further examine this relationship to regulate more in-depth ideas.
- Experimental research styles an ideal starting point. The data you assemble is a substance on which to build more thoughts and comports more research.

The foremost feature that distinguishes experimental research on or after other types of research is that the investigator deploys the independent variable. A number of experimental cluster designs in experimental research are in attendance. A majority of these fit the criteria for experimental research, while others take care of it.

- In accurate experimental research, the researcher not only operates the independent variable, he or she also randomly allotted individuals to the numerous treatment groupings (i.e., control and treatment).
- Now quasi experimental research, the researcher makes sure to randomly assign topics to treatment and control groups. In contrast to popular belief, the treatment is not distributed randomly among participants. In some circumstances, a researcher may assign one complete group to action and one whole group to control at random. In this scenario, quasi-experimental research entails using whole assemblies in an experiment rather of randomly assigning people to research conditions. (Some researchers express this latter situation differently. For our sequence, we will permit this explanation).
- Trendy causal comparative (ex post facto) research; the cluster groups are already formed. It does not encounter the standards of an experiment for the reason that the independent moveable is not manipulated.

4. Correlation

A connection has direction and can be one or the other, positive or negative (note exceptions listed later). By means of a positive correlation, individuals who score overhead (or below) the average (mean) on one amount tend to score in the same way above (or below) the average on the other measure. The scatterplot of a progressive correlation grows (from left to right). With negative relationships, an individual who scores above your head average on one and only measure tends to score lower than average on the other (or vice versa). The scatterplot of a negative relationship falls (on or after left to right).

A correlation can be different in the degree or strength of the connection (with the Pearson product-moment correlation coefficient that connection is linear). Zero designates no relationship flanked by the two measures and $r = 1.00$ or $r = -1.00$ designates a perfect relationship. The strength can be somewhere between 0 and + 1.00. Note: The sign r is used to signify the Pearson product-moment correlation coefficient aimed at a sample. The Greek letter rho is cast-off for a population. The

sturdier the correlation—the closer the value of r (correlation coefficient) come from to + 1.00—the more the scatterplot will plan beside a line.

5. T test

The T test is one kind of inferential statistics. It is used to regulate whether in attendance is a significant transformation between the means of two groups. We assume responsibility for the dependent variable fitting a normal distribution using all inferential statistics. We may classify the likelihood of a given result as soon as we assume a normal distribution occurs. We stipulate the equal of probability (alpha level, level of significance, p) we are willing to agree to take before we collect data ($p < .05$ is a mutual value that is used). After we collect data, we estimate a test statistic with a formula. We link our test statistic with a serious value start on a table to see if our results fall surrounded by the acceptable level of probability. Modern computer programs estimate the test statistic for us and also make available the exact likelihood of procurement that test statistic by means of the quantity of topics we have.

2.1.2 Stages

There are numerous techniques to research in any topic of study, regardless of whether it is applied or basic research. Each research study will be unique in some respects according to the time, setting, atmosphere and location in which it is conducted.

Nonetheless, all research initiatives share the same goal of expanding our understanding of the problem and so all pass through certain fundamental stages, constituting a process known as the research process.

Understanding the research process is required to carry out research properly and to sequence the steps inherent in the process.

These 8 stages in the research process are:

1. Identifying the problem.
2. Reviewing literature.
3. Setting research questions, objectives and hypotheses.
4. Choosing the study design.
5. Deciding on the sample design.
6. Collecting data.
7. Processing and analysing data.
8. Writing the report.

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The above-mentioned research process is, in essence, a component of a research proposal. It is an outline of the steps you propose to take in carrying out a research project.

A detailed inspection of the preceding stages indicates that, for the most part, each of these stages is dependent on the others.

One cannot examine data (step 7) unless data has been acquired (step 6). It is also true that a report (step 8) cannot be written before data has been collected and analysed (step 7).

The system of interdependent linked stages that is research is thus a system of interdependent related stages. Violation of this procedure may result in irreversible damage to the study.

It is also true that the researcher has various options available to him or her at each of the above-mentioned stages. A research method can be likened to a road map.

The map analogy is important for the researcher because there are various possibilities to follow at each stage of the research process.

Our key goal in our study decision is to select the optimal alternative in terms of time limits, money and human resources.

Before going over the steps of the research process, we'll go over the phrase 'iterative,' which appears in the oval-shaped diagram in the centre of the schematic

diagram. Iteration is ultimately the key to a successful research project: the process of returning to the identification of research problems, technique, data gathering and so on, which leads to new ideas, revisions and improvements.

When reviewing the research project with advisers and peers, it is common to discover that new research questions must be introduced, variables must be omitted, added, or redefined and other adjustments must be made. As a planned study is studied and re-examined from various angles, it may begin to evolve and take on a new shape.

This is to be expected and it is a necessary component of a solid research project.

Furthermore, it is critical to assess study methodologies and data to be collected from many perspectives to ensure a holistic approach to the research issue.

There is rarely a single approach or formula for producing a good research study, but it is critical to recognise that the research process is cyclical and iterative.

The Research Process Model 1

Step 1: Decide on a topic and asking queries

Outline your research problem, look for potential topics, do some preliminary background research and create questions to keep track of your findings.

Step 2: Recognizing resources

Number out what resources you will need to Grade A, answer your questions and crack your research problem. These resources include people, books, magazines, papers, videos, websites and any extra basis of information.

Step 3: Preparation your search

Narrow or broaden your topic, create topic and keyword lists to search, prioritize your queries, create interview questions, agenda interviews and organize your hunt time.

Step 4: Hunting and assembly

Utilize all your localizing skills, including skimming in addition scanning, using indexes and tables of matters, look for gaps in information, as well as recorded bibliographic evidence.

Step 5: Scrutinizing and Organizing

Agree what to keep, what to leave out, how to determine the most relevant information, how to organize your proceedings into useful parts. List out what the best sources obtainable are. Decide which sources are unswerving and credible. Verify evidences and sources.

Step 6: Creating it our own

Even when you've gathered a lot of knowledge about your issue, you'll still want to personalize it by adding your own thoughts to it. You have the ability to impact how

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facts are analysed, interpreted, compared, or viewed. You can develop an altogether new argument or solve an issue in a novel way by combining evidence from a variety of sources.

Step 7: Connecting your findings

This stage consists of all the issues in allocation what you've learned with your audience. This takes into account determining a format (paper, web page, PowerPoint exhibition, video, etc.); recognizing the audience; choosing the tone and style of communication; proofreading, editing and reviewing; citing your causes; and formulating a bibliography.

Step 8: Observing back on what you educated and how you learned

This stage entails reflecting on what you've learnt and the steps you've taken to complete your research. Your teacher might ask you to do some replicating throughout a project. You may be requested to evaluate yourself in contradiction of a rubric, stick to a log or journal about your progress, label the search process you use, in addition to honestly evaluate your own strong point and weaknesses as a researcher.

Research Method Model 2 (Stripling & Pitts)

This model breaks down the process into extra steps:

Step 1: Select a broad topic.

Step 2: Get an impression of the topic.

Step 3: Thin the topic.

Reflection Point: Is my theme a good one?

Step 4: Advance a thesis or announcement of purpose.

Replication Point: Does my thesis of statement of determination represent an operative, overall thought for my research?

Step 5: Communicate questions to direct research.

Replication Point: Do the questions deliver a footing for my research?

Step 6: Tactic for research and formation.

Reflection Argument: Be situated the research / production strategy practicable?

Step 7: Detection / Analyze / Assess foundations.

Reflection Point: Are my cradles usable and passable?

Step 8: Assess indication / Take notes / Bring collected bibliography.

Reproduction Point: Is my investigation comprehensive?

Step 9: Launch conclusions / Organize data into an outline.

Reflection Point: Are my finishes based on researched proof?

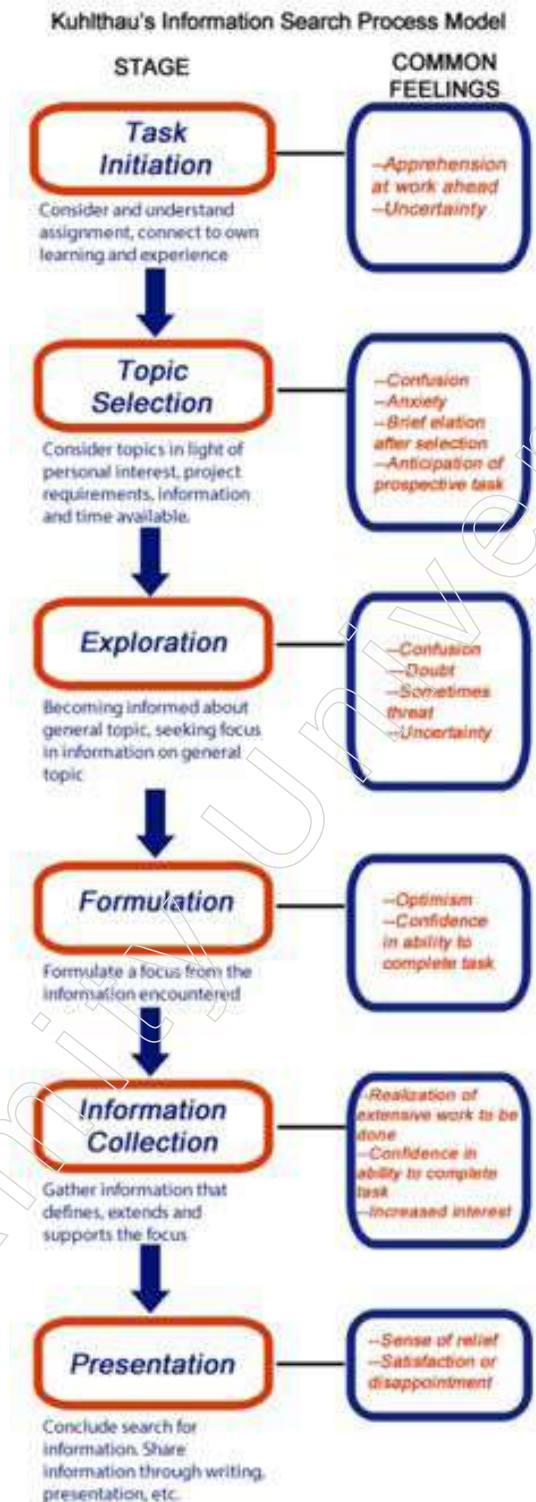
Fixes my outline logically form conclusions and evidence?

Step 10: Make and present ending product.

Mirror image Point: Is my paper / project adequate?

Research Process Model 3 (C. Kuhlthau)

This model is distinct in that it incorporates the expansion of thoughts surrounding a study topic as well as the emotions associated with the search process.



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Check Your Understanding

Fill in the blanks:

1. The maximum important characteristic of _____ this methodology is its thrust on objectivity.
2. Research _____ is a framework for conducting research.
3. The term "Quasi" indicates _____.
4. _____ research permits you to test your impression in a controlled environment beforehand taking it to market.
5. A number of experimental cluster designs in experimental research are in _____.
6. Understanding the _____ is required to carry out research properly and to sequence the steps inherent in the process.
7. The map analogy is important for the researcher because there are various possibilities to follow at each stage of the _____.
8. There is rarely a single approach or formula for producing a good research study, but it is critical to recognise that the research process is _____ and _____.

Summary

- A research design is a framework for conducting research. There are numerous research designs, each serving a distinct purpose and achieving a distinct goal.
- External blame is a reference to the authenticity of the text. Once a document has been strongminded to be genuine (external criticism), investigators need to determine if the content is correct (internal criticism).
- Experimental research refers to studies in which the researcher manipulates a variable to arrive at a conclusion or judgement, which is difficult to achieve in social sciences due to the presence of operating factors.
- There are three primary forms of experimental research design i.e., Pre-experimental investigation design, True experimental research scheme and Quasi-investigational research design.
- True experimental investigation relies on statistical scrutiny to prove or refute a hypothesis, making it the most accurate system of research.
- Experimental research permits you to test your impression in a controlled environment beforehand taking it to market.
- A number of experimental cluster designs in experimental research are in attendance.
- In accurate experimental research, the researcher not only operates the independent variable, he or she also randomly allotted individuals to the numerous treatment groupings (i.e., control and treatment).
- The T test is one kind of inferential statistics. It is used to regulate whether in attendance is a significant transformation between the means of two groups.

- There are numerous techniques to research in any topic of study, regardless of whether it is applied or basic research.
- Understanding the research process is required to carry out research properly and to sequence the steps inherent in the process.
- The system of interdependent linked stages that is research is thus a system of interdependent related stages. Violation of this procedure may result in irreversible damage to the study.
- There is rarely a single approach or formula for producing a good research study, but it is critical to recognise that the research process is cyclical and iterative.

Activity

1. Prepare a comparative hypothesis for all the three Research Processes.

Glossary

- Violation: the action of violating someone or something.
- Cluster: a group of similar things or people positioned or occurring closely together.
- Incorporates: take in or contain (something) as part of a whole; include.
- Cyclical: occurring in cycles; recurrent.
- Iterative: relating to or involving the repetition of a process or utterance, especially of a mathematical or computational process.

Questions and Exercises

1. What are the different types of Research Processes? Give an overview.
2. How would the different stages of research be explained. Illustrate with examples.
3. What are the circumstances that provision Action Research?
4. What are the forms of Experimental Research Project?

Further Readings

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Answers to Check Your Understanding

1. research
2. design
3. similarity
4. Experimental
5. attendance
6. research process
7. research process
8. cyclical, iterative

Unit - 2.2: Data Collection

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Learning Objectives:

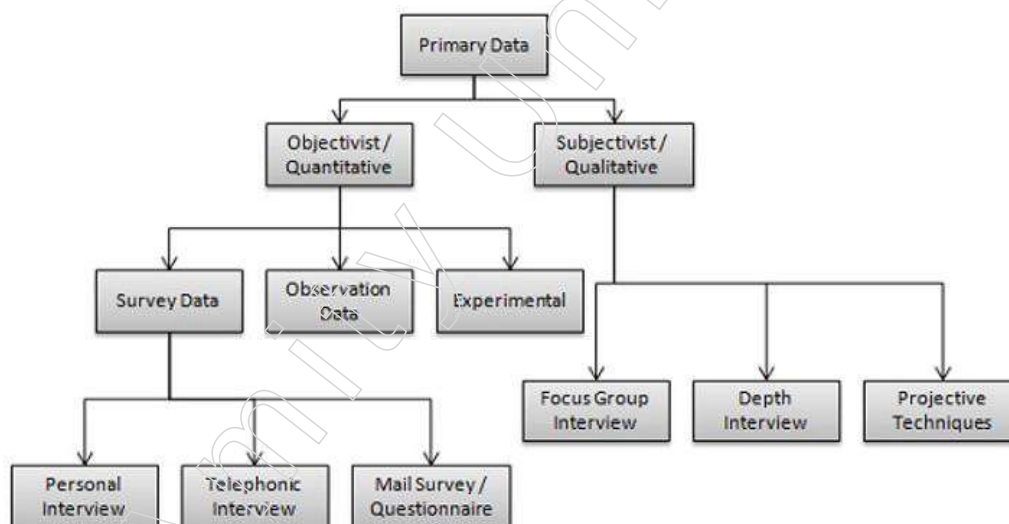
At the end of this unit, you will be able to:

- Understand data collection
- Know data collection methods
- Analyse survey
- Comprehend observation
- Examine questionnaire
- Learn tabulating & validating data

Introduction

Data collection is described as the process of collecting, measuring and evaluating correct insights for study using recognised and accepted methodologies. A researcher's hypothesis may be evaluated based on the evidence acquired. In most cases, regardless of the subject of study, the first and most important stage is data collection. The approach to data collection for different fields of study varies depending on the information needed.

The most important purpose of data collection is to guarantee that information-rich and reliable data is collected for statistical analysis so that data-driven research findings can be drawn.



2.2.1 Introduction to Data Collection

A study's data can be either primary or secondary in nature. By definition, primary data is information obtained first-hand by the researcher particularly for the goal of addressing the community at hand. An objectivist or subjectivist survey research is possible. An objectivist approach is more rigid and scientific. In this scenario, the hypothesis is developed utilising a widely available standard method. There is little

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or no leeway from the defined methods or queries. A hypothesis test is required by a subjectivist technique, but its processes are less stringent. At his or her option, the researcher may record data utilising unstructured ways.

2.2.2 Data Collection Methods

Information collection is a process of gathering information from all the pertinent sources to find responses to the research problem, test the hypothesis and assess the outcomes. Data collection methods can be alienated into two categories: secondary approaches of data collection and primary ways and means of data collection.

1. Primary Data Collection Methods

Prime data collection methods can be alienated into two groups: quantitative and qualitative.

Quantitative data collection methods are grounded in mathematical calculations in numerous formats. Techniques of quantitative data collection and analysis comprise questionnaires with closed-ended questions, systems of correlation and regression, mean, mode and median in addition others.

Quantitative methods are cheaper in the direction of application and they can be used within shorter duration of time likened to qualitative methods. Moreover, owing to a great level of standardization of quantitative methods, it is easy to make judgements of findings.

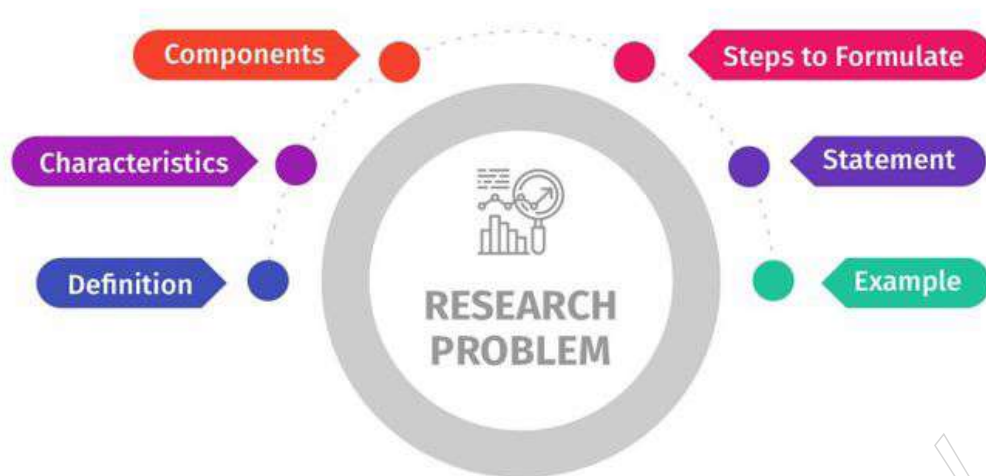
Qualitative research methods, on the contrary, do not encompass information or mathematical calculations. Qualitative research is meticulously associated with arguments, sounds, feeling, emotions, colors and other foundations that are non-quantifiable.

Qualitative studies goal to ensure greater level of complexity of understanding and qualitative data collection methods consist of interviews, questionnaires by means of open-ended questions, focus groups, reflection, game or game-playing, case studies etc.

2. Secondary Data Collection Methods

Secondary data is information that has already been published in books, newspapers, magazines, journals and other online resources. In these springs, there is a great amount of data regarding your research subject in business studies, almost regardless of the natural surroundings of the research region. Consequently, application of suitable set of criteria to select secondary data to be used in the study plays a vital role in terms of increasing the stages of research validity and reliability.

These measures include, but not limited to date of periodical, credential of the author, reliability of the foundation, quality of discussions, depth of analyses, the range of contribution of the text to the improvement of the research area etc.



Data is a set of facts, statistics, objects, symbols as well as events acquired from various sources. Organisations gather data in order to make better judgments. Organisations would struggle to make effective decisions without data, thus data is collected at various moments in time from various audiences.

For example, before releasing a new product, a company must collect data on product demand, customer preferences, rivals and so on. If data is not collected ahead of time, the organisation's newly launched product may fail for a variety of reasons, including a lack of demand and an inability to meet client expectations.

Although data is a significant asset for any company, it serves no value until it is analysed or processed to get the intended results.

Data collection methods can be divided into two categories: primary data collection methods and secondary data collection methods.

Methods for Collecting Primary Data

Primary data is gathered through first-hand experience and has never been utilised before. Primary data gathering procedures yield extremely accurate data that is specific to the research's purpose.

There are two types of primary data collection methods: quantitative methods and qualitative approaches.

1. Quantitative Methods:

Statistical methods are commonly used in quantitative strategies for market research and demand forecasting. Demand is forecasted using previous data in these ways. These major data collection methods are typically utilised to develop long-term forecasts. Statistical procedures are highly dependable because the element of subjectivity in these methods is minimal.

- Analysis of Time Series

A time series is a sequential order of values of a variable, often known as a trend, at equal time intervals. Using patterns, a firm can forecast demand for its products and services over a given period of time.

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- Techniques for Smoothing

Smoothing techniques can be utilised when the time series lacks notable trends. They eliminate a random variance in past demand. It aids in recognising patterns and demand levels in order to forecast future demand. The simple moving average approach and the weighted moving average method are the two most used methods for smoothing demand forecasting tools.

- Barometric Method

This method, also known as the leading indicators approach, is used by researchers to forecast future trends based on present events. When previous occurrences are used to forecast future events, they are referred to as leading indicators.

2. Qualitative Methods:

Qualitative approaches are very beneficial when no historical data is available. Alternatively, there is no need for numbers or mathematical calculations. Qualitative research is intimately related with non-quantifiable components such as words, sounds, feelings, emotions, colours and so on. These methods are based on personal experience, judgement, intuition, hypothesis, emotion and so forth.

Quantitative methods do not give the motivation for participants' responses, frequently do not reach disadvantaged populations and take a lengthy time to collect data. As a result, it is advisable to mix quantitative and qualitative methodologies.

- Surveys

Surveys are used to collect data from the target population in order to gain insights into their preferences, opinions, choices and comments on their products and services. Most survey software provides a variety of question types from which to choose.

You can also save time and work by using a pre-made survey template. Online surveys can be tailored to match the brand of the company by changing the theme, logo and so on. They can be disseminated via a variety of means, including email, website, offline app, QR code, social media and so forth. You can choose the channel based on the type and source of your audience.

Once the data has been gathered, survey software can generate a variety of reports and run analytics algorithms to uncover hidden insights. A survey dashboard can include statistics such as response rate, completion rate, demographic filters, export and sharing options and so forth. By integrating survey builder with third-party apps, you may reduce the amount of time spent on online data collecting.

- Polls

Polls are made up of a single or multiple-choice question. Polls can be used when a fast pulse of the audience's sentiments is desired. Because they are brief, it is simpler to elicit comments from the public.

Online polls, like surveys, can be embedded in a variety of platforms. Once respondents have responded to the question, they might be told where they stand in relation to other respondents' responses.

- Interviews

In this strategy, the interviewer asks the respondents questions either in person or over the phone. In face-to-face interviews, the interviewer asks the interviewee a series of questions in person and takes notes on the interviewee's responses. If meeting the subject is not possible, the interviewer can conduct a telephone interview. When there are only a few respondents, this kind of data collecting is appropriate. If there are many participants, it is extremely time-consuming and tedious to repeat the process.

- Delphi Technique

Market specialists are presented with the estimates and assumptions of forecasts generated by other industry experts using this strategy. Based on the information offered by other experts, experts may review and update their estimations and assumptions. The ultimate demand projection is based on the consensus of all experts on demand forecasts.

- Focus Groups

A focus group is a small group of people, usually 8-10 persons, who discuss the common areas of the problem. Each person shares his or her thoughts on the subject at hand. A moderator oversees the group members' discussions. The group comes to an agreement at the end of the conversation.

- Questionnaire

A questionnaire is a printed set of open-ended or closed-ended questions. Respondents are expected to provide answers based on their knowledge and experience with the problem at hand. The questionnaire is a component of the survey, although the end-goal of the questionnaire may or may not be a survey.

Secondary Data Collection Methods

Secondary data is information that has previously been used. The researcher can acquire data from both internal and external sources within the firm.

1. Internal sources of secondary data:

- Organisation's health and safety records
- Mission and vision statements
- Financial Statements
- Magazines
- Sales Report
- CRM Software
- Executive summaries

2. External sources of secondary data:

- Government reports
- Press releases
- Business journals
- Libraries

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- Internet

Secondary data collecting methods, like primary data collection methods, can use both quantitative and qualitative methodologies. Secondary data is more freely accessible and thus less time-consuming and expensive than primary data. However, the veracity of the data acquired cannot be validated using secondary data gathering methods.



2.2.3 Survey

Surveys are used to collect data from the target population in order to gain insights into their preferences, opinions, choices and comments on their products and services. Most survey software provides a variety of question types from which to choose.

You can also save time and work by using a pre-made survey template. Online surveys can be tailored to match the brand of the company by changing the theme, logo and so on. They can be disseminated via a variety of means, including email, website, offline app, QR code, social media and so forth. You can choose the channel based on the type and source of your audience.

Once the data has been gathered, survey software can generate a variety of reports and run analytics algorithms to uncover hidden insights. A survey dashboard can include statistics such as response rate, completion rate, demographic filters, export and sharing options and so forth. By integrating survey builder with third-party apps, you may reduce the amount of time spent on online data collecting.

Do those who go out at the gym live better, happier lives? How secure is New York City? How effective is your HIV/AIDS education programme? These are the kinds of questions that statistics cannot solve. Surveys necessitate careful planning and execution, as well as considerations for survey format, bias and weariness and so on. Survey data must be carefully examined by statistical analysis professionals, who must also apply their own discretion and experience to extract the most valuable information from a survey. Governments can use surveys to measure the effectiveness of a project, corporations can use surveys to analyse the response to a specific product and social scientists can use surveys to do quantitative research.



"I can prove it or disprove it! What do you want me to do?"

2.2.4 Observation

Opinion research can include both qualitative and quantitative observations. Such a study is designed to identify an individual and entice the outcome based on or after their reaction to certain parameters. There is no direct interaction with the subject in such a study.

For instance: An FMCG company wants to be acquainted with how its consumer respond to the new shape of their product. The canvasser observes the customers initial reaction and collects the data, which is at that time used to draw inferences from the collective information.

All research includes some form of observation, which might be quantitative or qualitative. A quantitative observation entails collecting data objectively, generally in numerical form. The information gathered should be related to or comprehended in terms of a number.

Statistical survey data analysis is used to assess quantitative observations. Observations of any variable associated to a numerical value, such as age, shape, weight, height, scale and so on, are examples of quantitative observations.

A researcher, for example, can learn about a customer's contentment with recent purchases by asking them to score their satisfaction on a Likert scale ranging from 1 (very dissatisfied) to 7 (extremely delighted).

Qualitative observations, which do not include numerical measurements, monitor the qualities of a phenomenon.

From a safe distance, researchers observe respondents in a naturalistic setting. Because the respondents are in their natural surroundings, the traits seen by the researchers are interesting and provide additional information. A researcher, for example, can observe and record the observations of customers in a supermarket by observing their selection and purchase behaviours. This provides a thorough understanding of the customer.

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It is critical to ensure high survey response rates in any type of research to increase the quality of insights. See our webinar recording for four methods to increasing survey response rates and collecting high-quality insights.

2.2.5 Questionnaire

A questionnaire is a procedure containing a set of questions, which are filled by the respondents.

According to Goode Hatt, "In general, the questionnaire refers to a device for securing answers to questions by using a form which the respondent fills in himself."

The goal of a questionnaire is twofold:

- to gather information from respondents spread throughout a large area
- to succeed in obtaining trustworthy and reliable information in a short period of time.



- Opinion poll is as an instrument for research, which can be made of a list of questions, alongside with the optimal of answers, printed or typed in a system on a form cast-off for acquiring specific info from the respondents.
- In general, inquiry form is delivered to the individuals concerned either by post or mail, demanding them to answer the questions and mail it back.
- Informers are expected to read and comprehend the questions and reply in the planetary provided in the survey itself.
- The questionnaire is organized in such a way that it interprets the required data into a series of questions, which informers can and will response.

Characteristics of a Good Questionnaire

The list below are features of good questionnaires:

- The aforementioned ought to consist of a well-written gradient of questions.

- The survey should deal with an important or important topic to create interest among defendants.
- It must seek only that data which cannot be attained from other sources.
- It has a duty to be as short as conceivable but would be comprehensive.
- The situation should be attractive.
- Guidelines should be vibrant and complete.
- It had better be represented in good psychological demand proceeding from over-all to more exact responses.
- Twice over negatives in questions must be avoided.
- Hitting two questions in one question also must be dodged. Every question would seek to get hold of only one definite information.
- It should be planned to collect information which can be hand-me-down afterward as information for analysis.

Design of Questions Voguish Surveys

The questions enquired can take two methods:

- Restricted questions, also named closed-ended, request the respondent to make varieties — yes or no, check matters on a list, or first-class from multiple choice responses.
- Restricted queries are easy to tabularize and compile.
- Unhindered questions are open-ended and permit respondents to segment feelings and opinions that are vital to them around the matter at hand.
- Unhindered questions are not easy to formulate and compile, but they allow defendants to reveal the deepness of their emotions.
- If the goal is to collect data from all respondents, then probing with limited questions that are easily quantifiable is preferable.
- If ranges of emotions or complexity of sentiment is to be studied, then create a scale to put a figure on those feelings.

Usages of Questionnaires

- Questionnaires are a mutual and inexpensive investigation tool used by isolated companies, government sections, individuals, groups, NGOs etch to get response, research, collect data as of consumer, clienteles or from general public liable on the want.
- Surveys are the maximum important part of primary surveys.
- One and only of the greatest benefits of questionnaires deceits in their uniformity — all defendants see faithfully the same questions.
- The aforementioned is an inexpensive method, notwithstanding of the size of the universe.
- Unrestricted from the bias of the interviewer, as the plaintiffs answer the queries in his own words.

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- Respondents partake enough time to reflect and answer.
- In arrears to its large coverage, respondents alive in distant areas can likewise be reached appropriately.

Limits of Questionnaire

- In the questionnaire, there is a high risk of collecting inaccurate and incomplete information, as people may not be able to comprehend the question effectively.
- The degree of non-reply is high.



2.2.6 Tabulating and Validating Data

Tabulation is the process of summarizing raw data in addition demonstrating it in compressed form for further analysis. Therefore, making tables is a very important step. Investigators can tabulate data manually or digitally. The best choice is based on the size and type of study, alternative prices, time constraints and the availability of computers and computer programmes at the outset. Hand tabulation is enough when the number of feedback forms is minimal and their length is short. The counting the number of observations (cases) that are off the record into certain categories:

1. Solitary-way tabulation: Categorization of single variables current in a study
2. Cross-tabulation: Simultaneously giving two or more variables in the learning
3. Classifying the number of respondents who have responded two or more questions consecutively

Table may be separated into:

- (i) Incidence tables,
- (ii) Reply tables,
- (iii) Contingency slabs,
- (iv) Uni-variate tables,
- (v) Bi-variate
- (vi) counters,

(vii) Statistical table as well

(viii) Time sequence tables.

In general, a research table has the subsequent parts:

- (a) Table figure,
- (b) Title of the table,
- (c) Caption
- (d) Stub (row heading),
- (e) Physique,
- (f) Head note,
- (g) Foot note.

As per a general rule the following stages are necessary in the preparation of table:

Title of table: The table ought to be first given a brief, simple and vibrant title which may express the basis of classification.

Columns and rows: Each bench should be prepared in objective adequate number of columns and rows.

Captions and stubs: The pillars and rows must be given simple and indistinct captions and stubs.

Ruling: Columns and rows must be divided by means of thin or thick rulings.

Arrangement of items: Similar figures must be arranged side by side.

Deviations: These should be organized in the column near the inventive data so that their presence may easily be eminent.

Size of columns: This would be conferring to the requirement.

Arrangements of items: This should be according to the problematic.

Special emphasis: This can be completed by writing important data in bold or exceptional letters.

Unit of measurement: The part should be noted underneath the lines.

Approximation: This must also be noted lower than the title.

Foot – notes: These may be set below the table.

Total: Totals of each column and magnificent total ought to be in one line.

Source: Foundation of data must be given. For most important data, write primary data

Data validation talks about to the process of safeguarding the accuracy and excellence of data. It is instigated by building several checks into a system or report to guarantee the logical constancy of input and stored data.

In computerized systems, data is entered by means of minimal or no human

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supervision. As a result, it is necessary to certify that the data that enters the scheme is correct and meets the looked-for quality standards. The data will be of little use if it is not entered properly in addition can make bigger downstream recording issues. Pointless data, even if entered correctly, will incur linked costs for scrubbing, transforming and storage.

Categories of Data Validation

There are many types of data validation. Maximum data validation measures will perform one or more of these authorizations to ensure that the data is correct beforehand storing it in the database. Mutual types of data validation checks consist of:

1. Data Type Check

A data type check settles that the data that comes in has the correct data type. For instance, a field might only accept numeric data. If this is the case, at that time any data covering other typescripts such as letters or exceptional symbols should be disallowed by the system.

2. Code Check

A code check makes certain that a field is nominated from a valid list of values or follows assured formatting rules. It is, for example, more convenient to verify the validity of a postal code by comparing it to a list of valid codes. The identical concept can be applied to extra items such as country codes in addition NAICS engineering codes.

3. Range Check

A range check will verify whether input data falls inside a predefined range. Latitude and longitude are usually used in geographic data. A latitude value would be between -90 and 90, despite the fact that a longitude value must be between -180 and 180. In the least values out of this range be situated invalid.

4. Format Check

Numerous data types follow a certain predefined set-up. A mutual use case is date columns that stay stored in a fixed format similar "YYYY-MM-DD" or "DD-MM-YYYY." A data validation technique that ensures dates be situated in the appropriate format helps uphold consistency across data and through time.

5. Consistency Check

A consistency check is a type of logical check that confirms that the data has been entered in a logically consistent way. A case in point is examination if the delivery date is afterward the shipping day for a parcel.

6. Uniqueness Check

Approximately data like IDs or e-mail addresses be situated unique by nature. A database must likely have unique entries on these fields. A uniqueness check ensures that an item is not arrived multiple times hooked on a database.

Practical Example

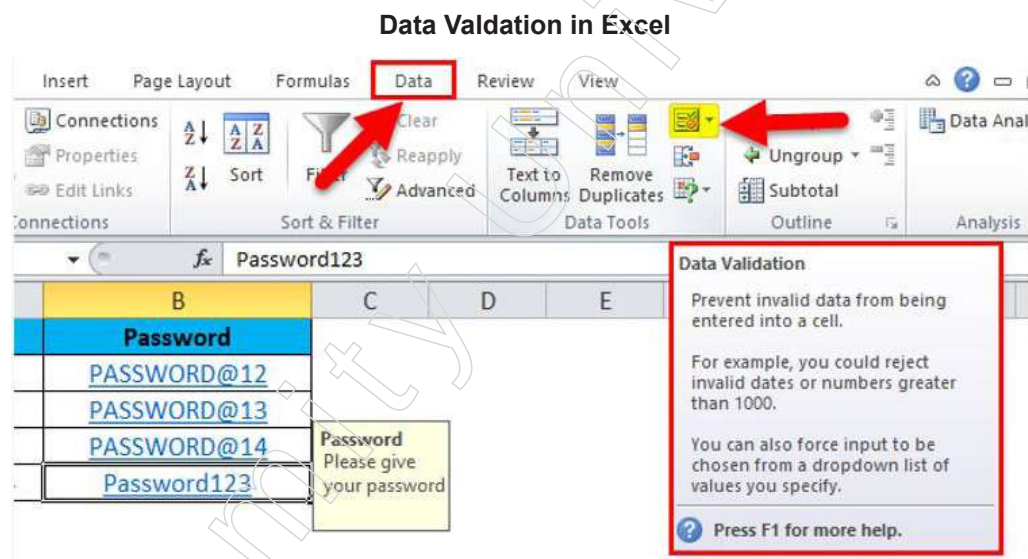
Contemplate the example of a retailer that collects data on its equipment but fails to form a proper check on the postal code. The oversight possibly will make it problematic to leverage the data for data and business intelligence. More than a few problems can occur if the postal code is not cross the threshold or entered indecorously.

It can be problematic to define the location of the store in round about mapping software. A stock postal code will also help make insights about the neighborhood anywhere the store is located. Shorn of a data check on the postal code, it is extra likely to lose the worth of data. It will result in additional costs if the data requirements to be recollected or the postal code needs to be physically entered.

A modest solution to the problem would be to place a check in place that guarantees a valid postal code is entered. The explanation could be a dropdown blackboard or an auto-complete procedure that allows the user to choose the postal code as of a list of valid codes. Such a kind of data validation is named a code validation otherwise code check.

Data Validation in Excel

The subsequent example is an outline to data validation in Excel. The data validation button further down the data tab arrange for the user with different types of data validation forms based on the data type in the cell. It likewise allows the user to describe custom validation checks via Excel formulas. The data validation can be brought into being in the Data Tools segment of the Data tab now the ribbon of Excel:



Data errors are not commonplace. They also frequently obstruct many data users' ability to conduct quick and timely data analysis. What if there was a way to quickly identify problems in your data and speed up the process of fixing them before sharing or analysing it?

Types of Errors Identified in the Validation Process

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1. Structural Errors

- Blank-header: There is a blank header name. All cells in the header row must have a value.
- Duplicate-header: There are multiple columns with the same name. All column names must be unique.
- Blank-row: Rows must have at least one non-blank cell.
- Duplicate-row: Rows can't be duplicated.
- Extra-value: A row has more columns than the header.
- Missing-value: A row has less columns than the header.

2. Content Errors

- Schema-error: Schema is not valid.
- Non-matching-header: The header's name in the schema is different from what's in the data.
- Extra-header: The data contains a header not defined in the schema.
- Missing-header: The data doesn't contain a header defined in the schema.
- Type-or-format-error: The value can't be cast based on the schema type and format for this field.

When you want to gain a sense of what all those thousands of rows of data in your table really imply, a tabulation is a wonderful place to start. Tabulations enable you to calculate metrics like as the total, average, or highest value of a certain column while grouping those estimates based on common values in one or more other columns.

Data validation is an important step in ensuring the accuracy of a study or report. For example, erroneous or missing information in the input data could arise due to poor data source quality, or errors could occur during the analysis stage where data sources are merged/joined or edited inappropriately. As a result, data validation can or should be undertaken at the data cleansing step prior to analysis, as well as during the reporting stage. Manual validation of tabular reports with hundreds of records is a time-consuming and error-prone procedure and errors in high-stakes reports are unacceptable and embarrassing.

Data validations can be performed in a variety of ways, including checking for data types, formats, uniqueness, the presence of missing values when they are not accepted, cardinality checks, data integrity validation and business rules/logic validation, among others. In most data base systems, data validation is an automated process, however the level of validation varies from system to system. Data bases, on the other hand, are not the sole data sources for analytical tasks. As a result, the quality of a data collection is never guaranteed and validation is critical in the analytical work environment. The automation of data validation contributes significantly to the efficient development of high-quality reports.

Check Your Understanding

Fill in the blanks:

1. A _____ is a printed set of open-ended or closed-ended questions.

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2. All _____ includes some form of observation, which might be quantitative or qualitative.
3. A quantitative observation entails collecting data objectively, generally in _____ form.
4. There are two types of _____ methods: quantitative methods and qualitative approaches.
5. A _____ test is required by a subjectivist technique, but its processes are less stringent.
6. Secondary data is information that has _____ been used.
7. A _____ makes certain that a field is nominated from a valid list of values or follows assured formatting rules.
8. _____ is the process of summarizing raw data in addition demonstrating it in compressed form for further analysis.

Summary

- Data collection is described as the process of collecting, measuring and evaluating correct insights for study using recognised and accepted methodologies.
- A hypothesis test is required by a subjectivist technique, but its processes are less stringent.
- Data collection methods can be alienated into two categories: secondary approaches of data collection and primary ways and means of data collection.
- Primary data gathering procedures yield extremely accurate data that is specific to the research's purpose.
- There are two types of primary data collection methods: quantitative methods and qualitative approaches.
- Qualitative approaches are very beneficial when no historical data is available.
- Quantitative methods do not give the motivation for participants' responses, frequently do not reach disadvantaged populations and take a lengthy time to collect data.
- Tabulation is the process of summarizing raw data in addition demonstrating it in compressed form for further analysis.
- Market specialists are presented with the estimates and assumptions of forecasts generated by other industry experts using this strategy.
- A questionnaire is a printed set of open-ended or closed-ended questions.
- Secondary data is information that has previously been used.
- Surveys are used to collect data from the target population in order to gain insights into their preferences, opinions, choices and comments on their products and services.

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- Governments can use surveys to measure the effectiveness of a project, corporations can use surveys to analyse the response to a specific product and social scientists can use surveys to do quantitative research.
- All research includes some form of observation, which might be quantitative or qualitative.
- Qualitative observations, which do not include numerical measurements, monitor the qualities of a phenomenon.
- A quantitative observation entails collecting data objectively, generally in numerical form.
- A questionnaire is a procedure containing a set of questions, which are filled by the respondents.
- Restricted questions, also named closed-ended, request the respondent to make varieties — yes or no, check matters on a list, or first-class from multiple choice responses.
- Surveys are the maximum important part of primary surveys.
- Data validation talks about to the process of safeguarding the accuracy and excellence of data.
- Maximum data validation measures will perform one or more of these authorizations to ensure that the data is correct beforehand storing it in the database.

Activity

1. How do the observers schedule their observations (pick the class, day, time)? Can you contemplate of a better technique to do this?

Glossary

- Automated: operated by largely automatic equipment.
- Cardinality: the number of elements in a set or other grouping, as a property of that grouping.
- Indecorously: not in keeping with good taste and propriety; improper.
- Consistency: conformity in the application of something, typically that which is necessary for the sake of logic, accuracy, or fairness.
- Gradient: an inclined part of a road or railway; a slope.

Questions and Exercises

1. What do you understand by data collection?
2. Stretch three advantages of observation as a data collection technique.
3. Elaborate ten criteria of good questionnaire.
4. What are the diverse ways of recording interview responses?
5. What is the necessity to learn tabulating?
6. What techniques are applied for validating data?

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Answers to Check Your Understanding

1. questionnaire
2. research
3. numerical
4. primary data collection
5. hypothesis
6. hypothesis
7. code check
8. Tabulation

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Unit - 2.3: Questionnaire Designing

Learning Objectives:

At the end of this unit, you will be able to:

- Understand steps in constructing a questionnaire
- Analyse types of questions
- Know format of questionnaire

Introduction

Questionnaire design is more of an art than a science. It is an endeavour to compile a list of questions by constantly checking for faults, learning what to eliminate and what to retain. However, a fundamental design strategy can be devised. A questionnaire is typically designed in three stages:

1. Developing a design strategy
2. Constructing the questionnaire
3. Drafting and refining the questionnaire.

2.3.1 Steps in Constructing a Questionnaire

Phase I: Developing a Design and a Strategy

1. **Specify the Information Required:** The researcher must be able to specify a list of information requirements. In most cases, this activity was completed when the research project or research design was created. The hypothesis given previously serves as a guideline in establishing the information requirement. The hypothesis specifies the relationship between the variables and the researcher may ideally create the data needed to prove or disprove the hypothesis.
2. **Determine the Communication Approach:** This refers to the method used to perform the survey, such as a personal interview, in-depth interview, telephone, mail, computer and so on. This approach choice will have an impact on the type of questionnaire to develop. The type of communication technique used is impacted by criteria such as the location of the respondents, the time and budget available, the nature of the study and so on. The communication strategy chosen results in various introductions, instructions, layout and so on. Once the communication strategy has been decided upon, a selection is made on the type of questionnaire that will be created.
3. **Type of Questionnaire:** In this step the researcher stipulates how the information will be collected by stating the type of questionnaire requisite. There are four sorts of questionnaires.
 - o **Structured-Undisguised Questionnaire:** This is the most common variety and it entails employing questions with plain, direct wording in a logical arrangement. For all replies, the wording and order stay the same. They are quite simple to implement and tabulate.

- o Unstructured-Disguised Questionnaire: The polar opposite of the previous form, this questionnaire conceals the objective of the research and displays no obvious order or patterns. To collect data, such a questionnaire typically use projective methods. The respondent is given a concealed or hidden stimulus and the response is unstructured.
- o Unstructured-Undisguised Questionnaire: The objective of the study is evident in this sort of questionnaire, although the questions are often open-ended. "How do you feel about prohibiting student union elections?" Respondents may react in an unstructured style. In most cases, these questionnaires are utilised in in-depth interviews.
- o Structured-Disguised Questionnaire: The goal of this questionnaire is to conceal the purpose of the study while allowing for easy coding and analysis. This method is founded on the concept that direct queries might impact or prejudice responses; but, if the questions are disguised, we can ask the respondents. What they know, not what they feel, for example, the previous question will be worded as:

What impact does the election of a student union have?

- (a) It raises awareness.
- (b) It interferes with research.
- (c)
- (d)

Although such questionnaires allow for easy tabulation and analysis, they are rarely widely used because to the time and effort required to create disguised questions.

Phase II: Constructing the Questionnaire

It takes many phases to construct a questionnaire:

1. Determine Question Content: This phase begins the effort of developing particular questions that will generate the data needed for the study. Certain considerations should be made when framing the questions:
 - o Is the question required? Every query should serve some purpose in terms of delivering more and real information.
 - o Is the question finished? The breadth of the query should be broad enough to expose all of the facts that a researcher requires.
 - o Is it necessary to ask a single question or a series of questions? There should be no 'double barrelled questions,' which combine two questions into one, such as 'Are this year's elections transparent and in accordance with electoral commission guidelines?' This is an erroneous approach. Instead, the following two questions should have been asked to obtain the needed information:

Are this year's elections transparent?

Where the election commission's recommendations were strictly followed.

- o Is the respondent verbal? Because of his incapacity to order his ideas, the respondent may be unable to provide an effective response.

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- o Is the respondent aware? The respondent's knowledge level should be considered, which means that the content of the query should correspond to the respondent's knowledge level.
- o Can the respondent recall? The questions should not be too difficult for the respondent to remember. He should not make any assumptions about his recollection. Take a little test and answer the following questions:

What was the most recent film you saw?

When was the last time you went out to eat?

When was the last time you went to a temple?

These questions, while simple, test your ability to recall information.

- o Is the respondent willing to provide an answer? This is important in situations where the inquiries are sensitive, such as those about a person's faith, finances, family life and so on.
2. Decide on a Response Strategy: After deciding on the content of the questions, the following step is to decide on the structured answer technique. (Close answer with set alternative questions) or an unstructured response method (open response with open ended questions). Some of the response strategies are as follows:
- o Dichotomous questions
Do you have a digital camera?
☐ Yes ☐ No
 - o Multichotomous questions
Which brand do you prefer for buy a digital camera?
☐ Sony
☐ Cannon
☐ Nikon
☐ Kodak
 - o Checklist questions
What features do you look for in your digital camera?
☐ Picture lucidity
☐ Screen dimensions
☐ Video recording capacity
☐ Cost-effective
☐ Smart physical looks
☐ Free service for 1 year
☐ Large memory size
 - o Scale questions may be of the rating or ranking type

Ranking

Which of the following media has been instrumental in influencing your choice of digital camera? Rank them giving 1 to the most effective media, 2 to the next most and so on.

- Television
- Company Brochures
- Newspapers
- Net Advertising
- FM Radio

Rating

Rate each of the following factors in terms of 'most desired, "somewhat desired,' and 'least desired' that a digital camera should have.

	Most Desired	Some What Desired	Least Desired
High Picture Clarity	☐	☐	☐
Large Memory	☐	☐	☐
Big Screen	☐	☐	☐
Zooming Function	☐	☐	☐
Picture Editing Function	☐	☐	☐
Small Size	☐	☐	☐
Video Recording Facility	☐	☐	☐

3. Determine the Wording of the Inquiry

This state is concerned with the wording of each question. The researcher must exercise extreme caution while structuring the question, as a badly phrased inquiry will either result in an incorrect response or in a refusal to answer. When posing a question, keep the following points in mind:

- o Use simple words: Because the questionnaire is not a test of the respondent's vocabulary, the words should be basic, such as "where, in your opinion, does this organisation's forte lie?" Because some people may not understand the definition of the word 'forte,' it is better to frame it simply as 'Where, in your opinion, does the ability of this organisationally?'
- o Avoid technical jargon: Using technical terms may leave even the most educated respondents unable to answer the questions.
- o Avoid using vague queries: terms like 'occasionally,' 'often, "sometimes,' 'you,' and so on are all problematic.

How often do you go to see a movie at a theatre?

- ☐ Never
- ☐ Occasionally

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☐ Sometimes

☐ Often

This question is insignificant since everybody has a different clarification of occasionally and sometimes.

o Avoid biased wording: Biased or leading questions are those that guide the respondent towards an answer or offer him a hint to the questions. Such inquiries should be ignored since they obscure the question's intent, for example.

- 'Do you think the government is doing the right thing by permitting FDI in the retail sector?'
- 'Do you think it's appropriate to utilise low-cost car accessories?'

These questions compel respondents to think in a specific way.

The level of personalisation should be limited, for example, 'what must our government do today?'

- Increase defence spending, even if it means raising taxes. It is also possible to write the alternative as
- Increase defence spending even if it means paying more taxes.

The second option is more individualised and both options will generate distinct results. There is no hard and fast rule dictating which method should be employed. However, the amount of personalisation should be chosen so that the issue is presented in a more realistic manner.

Phase III: Drafting and Refining the Questionnaire

1. Decide On Question Sequence:

We now reach the stage of designing the questionnaire and the sequencing of questions is critical. The following points should be kept in mind:

- o Begin with a simple and intriguing question. It makes the recipient feel at ease. An alternative is to employ a 'funnel strategy,' in which broad questions are asked first, followed by particular inquiries.
- o The questions must be arranged logically. Jumping between topic to topic would disrupt the respondent's flow and cause him to lose interest in completing the inquiry.
- o Classification questions should come later. The personal questions are classified in a questionnaire. The reason for asking classification questions before target questions is to avoid antagonising the respondent prior to actually getting to the heart of the study. For example, respondents who readily offer their viewpoint on preferred car types are likely to balk at responding with the same freedom if their income is asked first.
- o Difficult and challenging questions should be asked early in the questionnaire to avoid frightening the respondent.
- o Question branching must be done with caution. Branching is the process of instructing the responder where to go next in the questionnaire based

on his answers to previous questions. Branching is easier in the event of a telephone or personal interview schedule, but it should be avoided when the questionnaire is to be mailed because it may confuse the respondent.

2. Determine the Physical Characteristics

The physical appearance of the respondents influences how they reply to the questionnaire. As a result, the following issues should be kept in mind:

- o Print on high-quality paper with high-definition ink so that it may be easily read. The questionnaire should appear professional and be simple to complete.
- o The size of the questionnaire is critical. A short questionnaire is preferable than a long one, provided that the shortness does not come at the sacrifice of appearance. If a tiny size is attained by cramming the questionnaire, errors will occur, resulting in less informative answers.
- o An opening letter should accompany a questionnaire. This must introduce the study and make respondent recognise the value of his abilities. The importance of study, as well as the significance of respondent responses, should be highlighted in the letter.
- o The directions should be written in a clear and respectful manner. The manner of reporting the responses, such as check mark, cross, or circle, should be clearly defined. If the respondent is to skip particular questions, use 'go to' instructions and if an entire segment is to be skipped, use different colours for each area.

Pre-Testing the Questionnaires

Pre-testing is the process of employing a questionnaire on a temporary basis on a small set of respondents to assess how the questionnaire performs. Pre-testing aids in identifying faults in individual questions as well as the sequence of questions. Pre-testing can be used to examine several characteristics such as the ability of the questionnaire to create respondent interest, the perception of the meaning of questions, the continuity of the questions and the time required to complete the questionnaire. Pretesting can be divided into three categories:

- o Researcher Pre-testing: Entails a small group of fellow experts testing the instrument in its early stages of development. These researchers can offer crucial feedback on how to improve the questionnaire.
- o Respondent Pre-testing: In this case, a small sample of the desired respondent population is used for pre-testing. The instruments are administered to the sample in their completed form and then inputs can be taken from them on the survey.
- o Collaborative Pre-testing: When respondents are informed about their involvement in pre-testing, the test becomes collaborative. A deep questioning of each and every question is done in such a pre-testing, which is normally a time demanding process.
- o Non-Collaborative Pre-test: A non-collaborative pre-test occurs when

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respondents are not informed of their role in pre-testing. Because they are not being told, their participation will be rather limited. However, this method has the advantage of being carried out in a context that is identical to the real world.

2.3.2 Types of Questions

There are two kinds of questionnaire questions: open-ended questions and closed-ended questions. Take a look at the following explanation to better understand the format of the questionnaire:

Open Format Questions

Open format questions, often known as open-ended questions, allow your audience to express themselves freely. These questions have no preconceived set of answers and the respondent is allowed to answer whatever he or she feels is appropriate. You can acquire honest, insightful and even unexpected suggestions by incorporating open format questions in your questionnaire. This category includes qualitative questions.

An ideal questionnaire would include an open-ended question at the end that solicits comments and/or suggestions for improvement from responders.

Closed Format Questions

Closed format or closed-ended questions are multiple choice questions in which respondents are limited to selecting one of the offered multiple choice responses. There is no set number of multiple choice options that should be provided; the number can be even or odd.

One of the primary benefits of incorporating closed format items in your questionnaire design is the ease with which preliminary analysis can be performed. Because the answer set is known, these questions are great for generating statistical data and percentages. Closed-ended questions can also be posed at different intervals to different groups to effectively track their opinion about a product/service/company over time. There are seven different sorts of closed-ended inquiries.

Example of an Open Format Question



State your opinion about the quality of ABC Corporation's products and services ?

.....

.....

.....

Example of a Closed Format Question

Which are the gadgets that you cannot live without?

1. Cell Phone	2. Laptop	3. PDA
4. iPOD	5. Digital Camera	6. XBOX 360

Classification of Closed Format Questions for Questionnaire Design

Investigators can develop polling or survey questions in seven different methods in order to acquire correct statistical data from their respondents. The following are seven sorts of closed-ended questions that can be included in your questionnaire design:

1. Leading Questions

Leading questions are those that force your audience to give you a specific type of answer. All of the answers to a leading question would be equally plausible. A leading question might be one with options such as fair, good, great, poor, superb, excellent and so on. These questions are designed to elicit an opinion from the audience in a small number of words.

Example of a Closed-Ended Leading Question

How would you rate the products of XYZ Corporation?

1 Fair	2 Good	3 Excellent	4 Superb
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2. Importance Questions

Respondents in significance questions are typically asked to rate the importance of a specific topic on a scale of 1 to 5. These questions can help you understand what matters to your respondents and assist you to make business-critical decisions.

Example of a Closed-Ended Importance Question

Cost-effective services are:

1 Extremely Important	2 Very Important	3 Somewhat Important	4 Not Very Important	5 Not at all Important
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3.. Likert Questions

Likert scale questions can help you determine how strongly your respondents agree with a specific proposition. Such questions can also assist you determine how your customers feel about a certain issue, product, or service.

Example of a Closed-Ended Likert Question

ABC Corporation's Products have to improve on quality?

1 Strongly Agreed 2 Agreed 3 Neither Agree or Disagree 4 Disagree 5 Strongly Disagree

4. Dichotomous Questions

These are simple questions to which responders must respond with a yes or no. One significant disadvantage of dichotomous questions is that they cannot examine replies other than yes or no; there is no room for a medium perspective.

Example of a Closed-Ended Dichotomous Question

Do you like the Products of ABC Corporation?

Yes No

5. Bipolar Questions

Bipolar questions are the ones having two exciting answers written at the opposite ends of the scale. The respondents are asked to mark their responses between those two.

Example of a Closed-Ended Bipolar Question

How would you describe the services of XYZ Corporation?

Efficient X Inefficient

Fast X Slow

Reliable X Unreliable

6. Rating Scale Questions

In rating scale questions, respondents are asked to grade a specific issue on a scale ranging from poor to good. Rating scale questions normally feature an even number of options, thus respondents do not have the option of selecting a medium option.

Example of a Closed-Ended Rating Scale Question

How would you rate the services of XYZ Corporation?

1 Good	2 Fair	3 Poor	4 Very Poor
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7. Buying Propensity Questions

Buying propensity inquiries attempt to analyse customers' future intentions and determine the respondent's purchasing intention. These questions ask respondents whether they want to buy a specific product, what requirements they want met and whether they would buy such a thing in the future.

Example of a Closed-Ended Buying propensity Question

If mobile phones had an in-built mp3 player, would you prefer to buy it ?

1 Definitely	2 Probably	3 Probably Not	4 Not Sure	5 Definitely Not
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Knowing the questionnaire format and the kind of questions to ask in a questionnaire are crucial, but there is something more important. It is knowing which questions should be avoided in a survey or poll.

Take care to avoid following type of questions when preparing a questionnaire:

Embarrassing Questions

Questions that elicit information about respondents' personal and private lives are considered embarrassing. Such queries should be avoided since they risk eroding your respondents' trust.

Positive/ Negative Connotation Questions

Because the majority of verbs, adjectives and nouns in the English language have either positive or negative connotations, inquiries are inevitably interpreted as either positive or negative. Strong negative or positive overtones must be avoided while defining a question.

Hypothetical Questions

Hypothetical inquiries are based on fantasy and assumption. These questions compel respondents to express their thoughts on a certain topic and the data gathered as a result is inconsistent and ambiguous. In questionnaires, hypothetical questions should be avoided.

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2.3.3 Format of Questionnaire

A questionnaire is a collection of questions designed to be used in survey research for academic and business objectives. The questions were created by the researchers in order to collect data for quantitative analysis. The survey questionnaire is designed to collect information from respondents.

The questionnaire is divided into three sections: the questionnaire introduction, demographic information or respondent personal information and research questions. Quantitative research questionnaires are often referred to as survey questions or surveys by researchers. The questionnaire, on the other hand, differs from the quantitative survey and qualitative interview questions.

The following parameters are to be kept in mind:

Size:

- It should be smaller in size than that of the schedule.
- The extent in length and breadth should be appropriate.
- It should not be more than two or three pages as to the nature of the research.

Appearance:

- It should be constructed on a good quality paper and printing.
- It should have an attractive layout.

Clarity:

- The questions should be short, clear in terms, tenure and expression.

Sequence:

- The question should be arranged according to the importance and preference.

Communicability:

- The questions of the questionnaire should be able to keep the interest of the respondents

Span:

- The length of the questions of the questionnaire should be as short as possible.
- The questionnaire should not be long in length.

Check Your Understanding

Fill in the blanks:

1. There are two kinds of questionnaire questions namely _____ and _____.
2. The _____ differs from the quantitative survey and qualitative interview questions.

3. _____ questions can help you determine how strongly your respondents agree with a specific proposition.
4. Hypothetical inquiries are based on _____ and _____.
5. _____ questions normally feature an even number of options, thus respondents do not have the option of selecting a medium option.
6. There is no set number of _____ options that should be provided as the number can be even or odd.

Summary

- A questionnaire is typically designed in three stages viz. Developing a design strategy, constructing the questionnaire and drafting and refining the questionnaire.
- The questions must be arranged logically.
- Jumping between topic to topic would disrupt the respondent's flow and cause him to lose interest in completing the inquiry.
- Pre-testing is the process of employing a questionnaire on a temporary basis on a small set of respondents to assess how the questionnaire performs.
- There are two kinds of questionnaire questions: open-ended questions and closed-ended questions.
- Open format questions, often known as open-ended questions, allow your audience to express themselves freely.
- An ideal questionnaire would include an open-ended question at the end that solicits comments and/or suggestions for improvement from responders.
- Closed format or closed-ended questions are multiple choice questions in which respondents are limited to selecting one of the offered multiple choice responses.
- Pollsters can develop polling or survey questions in seven different methods in order to acquire correct statistical data from their respondents.
- Likert scale questions can help you determine how strongly your respondents agree with a specific proposition.
- Bipolar questions are the ones having two exciting answers written at the opposite ends of the scale.
- In rating scale questions, respondents are asked to grade a specific issue on a scale ranging from poor to good.
- Hypothetical inquiries are based on fantasy and assumption.
- A questionnaire is a collection of questions designed to be used in survey research for academic and business objectives.
- The questionnaire is divided into three sections: the questionnaire introduction, demographic information or respondent personal information and research questions.

Notes

Activity

1. Assume that you are an entrepreneur planning to launch a new product. How would you prepare a questionnaire to research about the requirement of the product as well as the future prospects?

Glossary

- Inconsistent: not staying the same throughout.
- Ambiguous: (of language) open to more than one interpretation; having a double meaning.
- Eroding: gradually wear away
- Proposition: a statement or assertion that expresses a judgment or opinion.
- Preliminary: denoting an action or event preceding or done in preparation for something fuller or more important.

Questions and Exercises

1. A questionnaire is typically designed in three stages. What are those stages and how are they applied?
2. Give an account of all the types of questionnaires.
3. What does it take to construct a questionnaire?
4. How useful is the Pre-Testing the Questionnaires?
5. Classify the closed format and the open format questions for a questionnaire.
6. What care must be taken to avoid ineffective type of questions when preparing a questionnaire?

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Answers to Check Your Understanding

1. open-ended questions, closed-ended questions.
2. questionnaire
3. Likert scale
4. fantasy, assumption
5. Rating scale
6. multiple choice

Notes

Notes**Unit - 2.4: Measurement & Scaling****Learning Objectives:**

At the end of this unit, you will be able to:

- Understand measurement & scaling
- Analyse scaling techniques
- Comprehend attitude measurement scales

Introduction

In Research, measurement is when we use a yardstick to assess the weight, height, or another property of a physical thing, we say we are measuring. We also use measurement to determine how much we like a music, a painting, or the personalities of our friends. As a result, we can measure both physical objects and abstract concepts.

2.4.1 Introduction to Measurement & Scaling

Measurement is a difficult and time-consuming endeavour, especially when dealing with qualitative or abstract phenomena. By measurement, we mean the process of assigning numbers to things or observations, with the level of measurement determined by the rules that govern the assignment of the numbers. It is simple to assign numbers to certain attributes of objects, but it is more difficult to do so for others. Evaluating social conformity, IQ, or marital adjustment, for example, is far less evident and requires much more attention than measuring physical weight, biological age, or a person's financial holdings. In other words, whereas attributes like weight, height and so on may be measured directly using some standard unit of measurement, it is more difficult to assess properties like motivation to succeed, stress tolerance and so on. When measuring the length of a pipe using a yard stick, we can expect excellent precision; however, if the notion is abstract and the measurement equipment are not standardised, we are less confidence in the accuracy of the measurement results.

Technically, measurement is the process of translating aspects of a domain onto other aspects of a range based on some correspondence rule. We invent some type of scale in the range (in terms of set theory, range may refer to some set) and then translate or map the qualities of objects from the domain (in terms of set theory, domain may refer to some other set) onto this scale while measuring. For example, if we want to find the male to female attendance ratio when conducting a survey of people who go to a concert, we can categorise those who go by sex. According to set theory, this procedure involves mapping the observed physical features of persons attending the show (the domain) onto a sex classification (the range). The correspondence rule is as follows: If the object in the domain appears to be male, assign it to "0," and if it appears to be female, assign it to "1."

Similarly, a person's marital status can be recorded as 1, 2, 3, or 4, depending on whether the person is single, married, widowed, or divorced. We can also record "Yes" or "No" responses to a question as "0" and "1." (Or as 1 and 2 or perhaps as 59 and 60). Categorical data (qualitative or descriptive) can be converted into numerical

data in this artificial or nominal manner and if we code the various categories in this manner, we refer to the numbers we record as nominal data. Nominal data are just numerical in name because they share none of the attributes of the numbers with which we deal in conventional arithmetic. For example, if we record marital status as 1, 2, 3, or 4, we cannot write $4 > 2$ or $3 < 4$, nor can we write $3 - 1 = 4 - 2$, $1 + 3 = 4$ or $4, 2 = 2$. When there is nothing, we can do but put up inequalities, we refer to the data as ordinal data. For example, if one mineral can scratch another, it is assigned a greater hardness number and talc, gypsum, calcite, fluorite, apatite, feldspar, quartz, topaz, sapphire and diamond are assigned hardness values ranging from 1 to 10. We can write $5 > 2$ or $6 > 9$ with these numbers since apatite is harder than gypsum and feldspar is softer than sapphire, but we cannot write $10 - 9 = 5 - 4$ because the difference in hardness between diamond and sapphire is considerably bigger than the difference between apatite and fluorite. It would likewise be meaningless to claim that topaz is twice as hard as fluorite just because their hardness numbers on the Mohs scale are 8 and 4, respectively. In the context of ordinal data, the larger than symbol (i.e., $>$) can be used to denote "happier than," "preferred to," and so on. When we can generate differences in addition to inequalities, we refer to the data as interval data. Assume the following temperature values (in degrees Fahrenheit) are given to us: 58° , 63° , 70° , 95° , 110° , 126° and 135° . In this example, we can simply write $100^\circ > 70^\circ$ or $95^\circ < 135^\circ$, which signifies that 110° is warmer than 70° and 95° is cooler than 135° . We can also write $95^\circ - 70^\circ = 135^\circ - 110^\circ$ because equal temperature disparities require the same amount of heat to raise the temperature of an object from 70° to 95° or from 110° to 135° . On the other side, even if $126^\circ - 63^\circ = 2$, saying that 126° is twice as hot as 63° would be meaningless. To see why, switch to a centigrade scale, where the first temperature becomes $5/9 (126 - 32) = 52^\circ$ and the second temperature becomes $5/9 (63 - 32) = 17^\circ$ and the first value is now more than three times the second. This dilemma stems from the fact that both the Fahrenheit and Centigrade scales have artificial origins (zeroes), i.e., neither scale's number 0 indicates the lack of whatever amount we are attempting to measure. We refer to data as ratio data when, in addition to setting up inequalities and making differences, we can also create quotients (i.e., when we can execute all of the standard operations of arithmetic). In this respect, ratio data encompasses all of the standard measurements (or determinations) of length, height, money amounts, weight, volume, area, pressures and so on. The distinction between nominal, ordinal, interval and ratio data mentioned above is essential because the type of a collection of data may recommend the employment of specific statistical procedures. When assessing the qualities of objects or abstract notions, a researcher must be very aware of this issue.

Scaling refers to the processes of assigning numbers to different degrees of opinion, attitude and other notions. This can be accomplished in two ways:

- (i) making a judgement about some characteristic of an individual and then directly placing him on a scale defined in terms of that characteristic
- (ii) designing questionnaires in such a way that the score of an individual's responses assigns him a place on a scale. A scale is a continuum that consists of the highest point in terms of some attribute, such as preference, favourableness, etc. and the lowest point, as well as various intermediate places between these two extreme points.

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These scale point positions are so closely associated that when the first point is the highest, the second point represents a higher degree in terms of a particular characteristic than the third point and the third point indicates a higher degree than the fourth and so on. Individuals are thus assigned numbers for gauging degree distinctions in attitudes/opinions based on their scale-positions. All of this becomes clearer when we discuss scaling techniques (s). As a result, the term “scaling” refers to the processes used to attempt to obtain quantitative measures of subjective abstract notions. Scaling has been defined as a “procedure for the assignment of numbers (or other symbols) to a property of objects in order to impart some of the characteristics of numbers to the properties in question.”

2.4.2 Scaling Techniques

Scaling techniques for measuring data gathered from respondents.

The phrase scaling refers to attempts to objectively quantify one's attitude. Attitude is influenced by a variety of environmental and internal influences. Appropriate scales are established based on the attitude to be measured. Scaling is a technique for measuring qualitative responses of respondents, such as sentiments, perceptions, likes, dislikes, interests and preferences.

Types of Scales

Most frequently used Scales are:

1. Nominal Scale
2. Ordinal Scale
3. Interval Scale
4. Ratio Scale

Self-Rating Scales

1. Graphic Rating Scale
2. Itemized Rating Scales
 - a. Likert Scale
 - b. Semantic Differential Scale
 - c. Stapel's Scale
 - d. Multi-Dimensional Scaling
 - e. Thurston Scales
 - f. Guttman Scales/Scalogram Analysis
 - g. The Q Sort technique

Four types of scales are generally used for Marketing Research.

1. Nominal Scale

This is a very basic scale. It entails categorising facts/options into various

alternative categories, which are usually exhaustive as well as mutually exclusive. These are the least limiting of all the scales because they are only numerical. Credit card numbers, bank account numbers, employee identification numbers and so on are examples of Nominal Scale. It is simple and extensively used for studying the relationship between two variables. Numbers are nothing more than labels in a Nominal Scale and they are employed expressly to define different kinds of responses. The following example exemplifies -What is your gender?

☐ Male

☐ Female

Another example is a survey of retail stores done on two dimensions way of maintaining stocks and daily turnover.

How do you stock items at present?

☐ By product category

☐ At a centralized store

☐ Department wise

☐ Single warehouse

Daily turnover of consumer is?

☐ Between 100 – 200

☐ Between 200 – 300

☐ Above 300

A two way classification can be made as follows

Daily/Stock Turnover Method	Product Category	Department wise	Centralized Store	Single Warehouse
100 – 200				
200 – 300				
Above 300				

'Mode' is frequently used for response category.

2. Ordinal Scale

Ordinal scales are the most basic scales for evaluating attitudes in marketing research. Because the numbers have the virtue of rank order, it is more powerful than a nominal scale. The scale is used to determine the importance of certain product attributes/benefits as perceived by respondents.

Example 1: Rank the following attributes (1 - 5), on their importance in a microwave oven.

1. Company Name
2. Functions
3. Price

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4. Comfort

5. Design

The respondents selected the most vital attribute as number one and the least important as number five. In addition to numbers, letters and symbols can be used to rate on an ordinal scale. Such a scale makes no attempt to quantify the degree of favourability of certain ranks.

Example 2 - If there are 4 different types of fertilizers and if they are ordered on the basis of quality as Grade A, Grade B, Grade C, Grade D is again an Ordinal Scale.

Example 3 - If there are 5 different brands of Talcom Powder and if a respondent ranks them based on say, "Freshness" into Rank 1 having maximum Freshness Rank 2 the second maximum Freshness and so on, an Ordinal Scale results.

'Median' and 'mode' are meaningful for ordinal scale.

3. Interval Scale

In the case of Interval Scales, the distance between the various categories, unlike in Nominal, or numbers, unlike in Ordinal, are equal. Rating Scales are another name for Interval Scales. An Interval Scale has an arbitrary Zero point and subsequent integers at equal intervals. A thermometer is an excellent example of an interval scale. Illustration 1 - How do you rate your present refrigerator for the following qualities.

Company Name	Less Known	1	2	3	4	5	Well Known
Functions	Few	1	2	3	4	5	Many
Price	Low	1	2	3	4	5	High
Design	Poor	1	2	3	4	5	Good
Overall Satisfaction	Very Dis-Satisfied	1	2	3	4	5	Very Satisfied

With such a scale, the researcher can state that position 5 on the scale is higher than position 4 and the distance from 5 to 4 is the same as the distance from 4 to 3. However, because no zero position has been defined, such a scale does not allow the conclusion that position 4 is twice as strong as position 2. The data from the Interval Scale can be utilised to get the Mean scores of each attribute across all respondents. It is also possible to compute the Standard Deviation (a measure of dispersion).

4. The Ratio Scale

Unless a baseline item is provided for comparison, ratio scales are rarely employed in marketing research. In the Interval scale example above, a score of 4 in one quality does not necessarily imply that the respondent is twice as satisfied as the responder who scores 2 on the scale. A ratio scale has a natural zero point and subsequent numbers are put at equally spaced intervals. Scales, for example, are used to measure physical quantities such as length, weight and so on.

In physical contexts, ratio scales are fairly common. The most adaptable are quantitative replies that construct a ratio scale mathematically. Ratio scales have all of the properties of interval scales and the ratios of the numbers on these scales have

meaningful interpretations. Data on some demographic or descriptive parameters will have ratio-scale qualities if gathered through open-ended questions. Consider the following inquiries: Q 1) What is your annual income before taxes? _____ \$

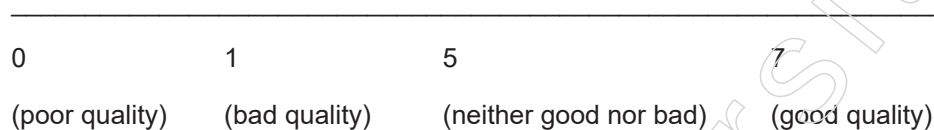
2) How far is the Theatre from your home ? _____ miles

These issues have a natural, unambiguous beginning position, which is zero. Calculating and understanding ratio makes sense because the beginning point is not picked arbitrarily. For example, a respondent earning \$40,000 per year earns twice as much as one earning \$20,000 per year.

Self-Rating Scales

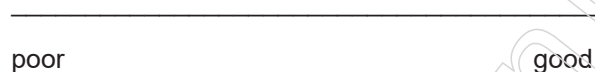
1. Graphic Rating Scale

The respondents rate the objects by placing a mark at the appropriate position on a line that runs from one extreme of the criterion variable to another. Example



Brand 1

This is also called continuous rating scale. The customer can inhabit any position. Here one attribute is taken ex-quality of any brand of ice-cream.



Brand 2

This line might be vertical or horizontal, with scale points. On the continuous scale, there is no additional signal. There is a range specified. We evaluate the physical distance between the left extreme position and the response point on the line to quantify the responses to the question “express your overall impression about ice-cream Brand 2 by inserting a tick mark at appropriate position on the line.” The greater the distance, the better the response or attitude toward the brand.

Its restriction is that coding and analysis will take a long time because we must first measure the physical distances on the scale for each responder.

2. Itemized Rating Scales

These scales are not the same as continuous rating scales. Each category is related with a number of brief descriptions. They are frequently employed in marketing research. They basically take the shape of multiple-choice questions. Likert, Semantic, Staple and Multiple Dimension are the most prevalent. Thurston and Guttman are two others.

a. Likert Scale

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Rensis Likert created it. Respondents are asked to indicate their level of agreement or disagreement with a series of statements. Each scale item contains five response options ranging from strongly agree to strongly disagree.

5	4	3	2	1
Strongly agree	Agree	Indifferent	Disagree	Strongly disagree

Each statement is assigned a numerical score ranging from 1 to 5. It can also be scaled as -2 to +2.

-2	-1	0	1	2
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For example, quality of Mother Dairy ice-cream is poor then Not Good is a negative statement and Strongly Agree with this means the quality is not good.

Each level of agreement is assigned a numerical value and the total score of the responders is calculated by adding these numbers together. The total score of a response shows a person's specific perspective.

Likert Scales are ordinal in nature, allowing one to rank attitudes but not assess the difference between them. They require around the same amount of effort to construct as the Thurston scale and are thought to be more discriminating and trustworthy due to the greater range of responses normally provided in Likert scale.

A normal Likert scale contains between 20 and 30 statements. To create a good Likert Scale, first generate a huge pool of statements related to attitude measurement and then exclude the statements that are vague and non-discriminatory from the pool.

As a result, the Likert scale is a five-point scale spanning from 'strongly agree' to 'strongly disagree.' This method does not involve a judgement gap.

b. Semantic Differential Scale

This is a seven point scale and the end points of the scale are associated with bipolar labels.

1						7
Unpleasant	2	3	4	5	6	
Submissive						Pleasant Dominant

Suppose we want to know personality of a particular person. We have options:

1. Unpleasant/Submissive
2. Pleasant/Dominant

Bi-polar refers to two opposing currents. Individuals can receive a score ranging from 1 to 7 or -3 to 3. Profiles are created based on these responses. We can analyse two or three products and acquire profile analysis by connecting these profiles. Depending on the number of factors, it might take any shape.

Profile Analysis

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'Mean' and 'median' are used for comparison. This scale aids in determining the overall similarities and contrasts between objects.

When using Semantic Differential Scale to create an image profile, it gives a solid foundation for comparing photos of two or more items. The main advantage of this scale is its simplicity in obtaining results when compared to more complex scaling approaches. The method is simple and quick to implement, yet it is extremely sensitive to slight variations in attitude, very adaptable, dependable and generally valid.

c. Stapel's Scale

Jan Stapel came up with the idea. This scale has a few distinguishing characteristics: -

1. Each item has only one word or phrase that describes the dimension it represents.
2. There are ten response categories for each item.
3. There are an odd number of categories for each item.
4. The response categories are labelled numerically but not verbally.

For example, assume we ask respondents to assess the quality of ice cream on a scale of +5 to -5 on the following categories. Choose a plus sign for terms that best describe the ice cream. Choose a negative number for adjectives that you believe do not appropriately represent the ice cream's quality. As a result, we can choose any value between +5, for terms we believe are very accurate and -5, for words we believe are very inaccurate. This scale is typically displayed vertically.

+5

+4

+3

+2

+1

High Quality

-1

-2

-3

-4

-5

This is a unipolar rating scale.

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d. Multi-Dimensional Scaling

It is a collection of analytical approaches used to analyse customer attitudes regarding perceptions and preferences. It is used to investigate:

1. The key features of a specific class of products perceived by customers when considering the product and by which they compare the various ranks.
2. Determine which brands compete most directly with one another.
3. Determine whether consumers would be interested in a new brand with a unique combination of features not existing in the market.
4. What is the optimum combination of product features for the consumer?
5. What sales and advertising messaging are compatible with consumers' opinions of brand?

It is a computer-based method. Respondents are asked to categorise the various brands as similar, extremely similar, not similar and so on. A high degree of quality of fit is obtained at the expense of a variety of other characteristics. The computer algorithm then computes a lack of fit index. The goal is to select a small number of dimensions that will alleviate the majority of the tension. After developing the configuration for the consumer's choice, the following stage is to determine the preference in relation to the product under consideration. These strategies seek to identify and quantify the product features that are essential to consumers.

This scaling is based on the erroneous premise that a buyer comparing different brands will perceive the differences based on only one quality. For example, what are the prerequisites for enrolling in an M.Com course? The responses could be to pursue a postgraduate degree, enter the teaching profession, gain knowledge, or appear in the NET. You cannot make a judgement based just on one of the traits. As a result, when consumers are deciding between brands, they base their selection on a variety of factors. In practise, customer perceptions involve various traits and each consumer perceives each brand as a composite of a number of different attributes. This is a flaw on this scale.

Whenever we have a choice between several options, we should go with multidimensional scaling. There are numerous applications for such scaling, including market segmentation, product life cycle management, vendor evaluations and advertising medium selection.

The disadvantage of this scale is that the concept of similarities and preferences is difficult to define. Furthermore, the distances between the things are perceived to be varied.

e. Thurston Scales

These are also called as scales with equal looking intervals. They are used to assess people's attitudes regarding a specific notion or construct. A significant number of statements relating to the notion or construct being measured are gathered for this purpose. The judges assess these remarks on an 11-category scale, with each category expressing a varying level of support for the concept. The items are then rated based on the mean or median ratings supplied by the judges and used to build a questionnaire

comprising twenty to thirty items chosen more or less evenly across the range of evaluations.

The assertions are written in a way that allows people to agree or disagree with them. The scale is then presented to a group of respondents and their scores are computed by calculating the mean or median value of the items agreed upon. A person who disagrees with every item receives a score of zero. So, the advantage of this scale is that it is a measurement scale with intervals. However, it is a time-consuming and labour-intensive process. They are frequently employed in psychology and education research.

f. Guttman Scales/Scalogram Analysis

It is based on the premise that items can be organised along a continuum so that a person who agrees with or finds acceptable one item will also agree with or find acceptable all other items expressing a less extreme position. For example, children should not be permitted to watch obscene programmes, or the government should prohibit certain programmes from airing on television. They are all tied to one thing.

In this scale, each score indicates a distinct set of replies and so the overall score of each individual is calculated. This magnitude requires a significant amount of time and work to build.

They are widely utilised in political science, ethnography, public opinion polling, research and psychology.

g. The Q Sort technique

It is used to quickly distinguish between a huge number of items. It employs a rank order technique and the objects are sorted into heaps based on their resemblance to some criteria. The number of objects to be sorted should be between 60 and 140. For example, in this case, we're looking at nine different brands. We categorise the brands based on flavour into tasty, moderate and non-tasty.

We can also categorise based on price—low, medium and expensive. Then we can determine whether individuals prefer low-cost, high-cost, or moderate-cost brands. We can categorise sixty brands or divide them into three piles. As a result, the number of things is to be divided into three piles: low, medium and high.

2.4.3 Attitude Measurement Scales

Attitudes are constructions that cannot be directly observed. As a result, the most typical method of evaluating attitudes is to analyse some aspect of behaviour and then draw conclusions about the attitudes that may be responsible for that behaviour.

Types of Attitude Measuring Scales

Many various procedures for measuring attitudes have been devised, while self-report surveys are the most commonly used. In general, an attitude scale or questionnaire will present the subject with a series of items to which she or he must react and the investigator will draw inferences about that individual's attitudes based on the responses.

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Attitude surveys, physiological and behavioural observation measures are among the tools used.

1. Attitude Surveys

This is the most prevalent method for assessing attitudes. An attitude survey involves the investigators distributing a questionnaire or asking a series of questions over the phone. A respondent expresses his or her feelings by answering a series of questions. These queries can be open-ended or closed-ended. The open-ended questions allow the respondent to respond in his or her own terms.

For example, a respondent could be asked, 'What qualifications do you believe are required for the Prime Minister of India?' Although this style of question offers detailed information, the responses can be difficult to interpret. As a result, the majority of questions on an attitude survey are closed-ended or restricted, such as 'Are women qualified to be President of India?'

Respondents would check a box to indicate their response, such as yes, no, or don't know. This style of question asks responders to choose from a limited set of options.

2. Rating Scales

Another type of survey question is the rating scale, in which respondents circle a number on a scale to indicate how much they agree or disagree with a statement. The Thurstone and Likert Measures, named after their founders, are the two most prevalent forms of attitudes scales.

a. Thurstone Scales

Thurstone (1929) is credited with developing the first attitude-measurement approach. He is known as the "Father" of Attitude Scaling. He created one of the first tools for evaluating people's views about religion. It is composed of assertions concerning a specific problem, with each statement assigned a number value indicating how positive or unfavourable it is deemed to be (See Exhibit I).

Exhibit I: Thurstone Scale Items Assessing Attitudes Toward Euthanasia (Tordella and Neutens, 1979)

- Each item's scale values (range = 1 to 5) is given in parentheses. The respondents do not know the values.
- A person with a terminal illness has the right to decide to die. (4.15)
- Inducing death for merciful reasons is wrong. (1.65)
- A person should not be kept alive by machines. (2.44)
- Euthanasia gives a person a chance to die with dignity. (4.29)
- The taking of human life is wrong, no matter what the circumstances. (1.36)

People tick each of the items with which they agree and a mean score is calculated to reflect their attitude. Thurstone scales are highly complex and time demanding because the scale's developer must ensure that a wide variety of conceivable perspectives is reflected in the statements utilised. This is especially crucial given that

the only replies permitted are agreement and disagreement. As a result, their principal disadvantage is that they are time-consuming and thus expensive to make.

b. Likert Type Scales

Likert (1932) created a technique that simplifies this process by allowing participants to choose from a range of possible replies, usually in the form of a five-point scale ranging from strongly agree, agree, uncertain, disagree and strongly disagree (See Exhibit II).

Exhibit II: Sample Liker Scale

For each of the statements below, please indicate the extent of your agreement or disagreement by placing a tick (✓) in the appropriate box.

1. People who commit murder should be hanged				
Strongly agree	Agree	Neither agree nor disagree	Disagree	Strongly disagree
2. Trial jury should be abolished				
Strongly agree	Agree	Neither agree nor disagree	Disagree	Strongly disagree

There is no need for the judges to categorise each statement while using this method because the categorization is embedded into the scale. The Likert scales are scored by assigning a score from 1 to 5 to each of the responses. The score is then totalled to provide a final assessment of the individual's attitude.

c. Guttman Scales

Louis Guttman (1947) created a scale similar to Likert and Thurstone's, but he put great strain on the instrument by evaluating only one attribute (a property called one-dimensionality, a single dimension underlies the responses to the scale).

Guttman realised that on one-dimensional scales, people who agreed with a more extreme test item would also agree with the fewer severe items that came before it. The items are arranged in a hierarchical order, so that a good reaction to one implies a favourable response to each of the things preceding it.

The subject's level of the attribute under examination is represented by the sum of the affirmative responses. This scale's attitude construct is more narrowly focused than the other scaling approaches.

The following is an example of a Guttman scale measuring the discriminative attitude:

- Would you marry a refugee?
- Would you accept a refugee as a close friend?
- Do you find refugees living in your neighbourhood acceptable?
- Should refugees be allowed to live in the same neighbourhood, as others?

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d. Bogardus' Social Distance Scale

This was developed by Bogardus (1926) as a means of measuring racial or ethnic prejudice, a prejudice being seen as an extreme form of an attitude.

The respondent is given a series of statements concerning the possible relationships and is asked to say to what extent he would accept the relationship depicted.

The scale asks the people, whether they would be willing to accept each group:

- As close relatives by marriage (1.00)
- As my close personal friends (2.00)
- As neighbours on the same street (3.00)
- As co-workers in the same occupation (4.00)
- As citizens in my country (5.00)
- As only visitors in my country (6.00)
- Would exclude from my country (7.00)

A score of 1.00 for a group indicates no social distance and therefore no prejudice.

Respondents would be asked to check the boxes next to the relationships they think acceptable and the scale would offer an overall assessment of racial bias.

e. Semantic Differential Scale

According to Osgood et al. (1957), people evaluate attitudes along three dimensions: evaluative (good-bad), potency (strong-weak) and activity (active-passive).

To quantify each of these three dimensions, they created the semantic differential, a scale comprised of several different dimensions against which the attitude object could be rated. Typically, the dimensions would be:

1. Pleasant Unpleasant
2. Strong Weak
3. Fast Slow
4. Active Passive
5. Hard Soft
6. Good Bad

The attitude object is assessed along each of these dimensions and the results can then be used to discern between the three dimensions of the attitude, namely evaluative (1,6), potency (2,5) and activity (3,4). (3,4).

Using this technique, it is feasible to determine not only whether the attitude objects are judged to be favourable or unfavourable, but also why this is the case in terms of the three dimensions.

3. Physiological Measurements

Physiological reactions reflect emotional reactions (skin response, pulse rate,

dilation of the pupils etc.). As a result, it makes sense to examine physiological reactions in order to identify more objective attitude indicators.

a. Galvanic Skin Response

The Galvanic Skin Response has been the most important objective indication of attitudes (GSR). The GSR measures the skin's electrical resistance, which changes when people are emotionally stimulated. As a result, the GSR enables one to assess a subject's emotional response to an attitude object.

Porier and Lott (1967), for example, conducted an experiment in which black and white experimenters accidentally touched their white participants with their hands. The deeper the subject's racial preconceptions (as determined by a questionnaire beforehand), the more the subject's galvanic skin response altered when touched by the black experimenter, compared to being handled by the white experimenter.

As seen by this example, the primary issue with psycho-physiological indicators of attitudes like as the GSR is that they assess the intensity of emotional responses but not their direction. Many other aspects of attitude, such as novelty and unexpectedness, have an impact on the metrics.

b. Facial Electromyogram

The face Electromyogram (EMG) is an objective attitude indicator that can be used to assess the quality or direction of one's attitude (its positivity/negativity). The facial EMG enables the detection of differences between positive and negative sentiments. When people hear a message that they agree with rather than disagree with, there is a relative rise in EMG activity in one set of muscles but a relative drop in another. The majority of these alterations are invisible to the human eye.

In practise, physiological markers of attitudes are rarely used. One factor could be the previously indicated insensitivity of most instruments to the quality of an attitudinal response. Another key reason is the requirement of using technological devices, which are difficult to use in field sittings.

4. Behavioural Measures

These approaches extract attitude metrics from open, observable behaviour patterns. The subjects are aware that they are being observed in the majority of behaviour observations. In the non-reactive measurement, however, the individuals are monitored without their knowledge, or, even more indirectly, some of their behavioural patterns are examined.

Attitudinal measurements of this type are known as behaviour indicators, observation approaches and unobtrusive measures. In one early study, for example, researchers counted the number of bumper stickers supporting a specific candidate on cars in a parking lot to determine voting preferences (Wrightsman, 1969).

By looking through garbage cans, the other researchers assessed views toward competing cola brands. Others sought to establish which exhibit was the most popular at a museum by analysing the amount of wear and tear on various portions of the carpet (Webb, Campbell, Schwartz, Sechrist & Grove, 1981).

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Unobtrusive assessment approaches are less susceptible to conscious distortions than self-report methods; nevertheless, this benefit comes at the expense of substantial ambiguities of interpretation (such as the uncertain validity of the obtained results) and ethical issues.

It is frequently difficult to understand exactly what these objective signs indicate in relation to an attitude and they are frequently determined by reasons or situational restrictions other than attitudes.

The lost-letter method is an excellent example of inconspicuous measurement. A researcher may not get honest answers on a questionnaire if she tries to gauge a community's attitude toward, say, outsiders. She can, however, use the lost-letter strategy provided she has some stamps and envelopes.

The researcher conducts the following: she addresses an envelope to someone with a foreign-sounding name at a local address. She stamps the envelope and then drops it on a busy street near the post office, where it will be readily found and mailed.

She drops a stamped envelope addressed to someone whose name does not seem foreign as a baseline control. To gather a sample, the operation is repeated. The envelopes that turn up are numbered and compared, with names that sound foreign and those that don't. This shows the general attitude toward outsiders.

5. Cognitive Measures

A new test has been developed in recent years to tap our implicit attitudes, self-concepts and other crucial parts of our cognitive system. The term implicit refers to mental associations that are largely automatic. The Implicit Association Test (IAT), established by Greenwald, McGhee and Schwarz, is the most well-known implicit measurements test (1998).

Implicit attitudes are attitudes that we carry but are unaware of, therefore we are unable to directly report them. These implicit attitudes can only be measured in a roundabout way. The IAT seeks to assess the strength of the link between the two concepts.

For example, the IAT instructed test-takers to place a stimulus, which could be a word or a picture, to a pair of targets as rapidly as possible. The IAT aims to delve into our unconscious associations. It has been used to investigate the unconscious roots of various biased attitudes.

Check Your Understanding

Fill in the blanks:

1. When using _____ Scale to create an image profile, it gives a solid foundation for comparing photos of two or more items.
2. _____ Scaling is a collection of analytical approaches used to analyse customer attitudes regarding perceptions and preferences.
3. _____ Scales are also called as scales with equal looking intervals.
4. Scaling techniques for measuring data gathered from _____.
5. 'Median' and 'mode' are meaningful for _____ scale.

Summary

- Measurement is a difficult and time-consuming endeavour, especially when dealing with qualitative or abstract phenomena.
- Scaling refers to the processes of assigning numbers to different degrees of opinion, attitude and other notions.
- Attitude is influenced by a variety of environmental and internal influences. Scaling is a technique for measuring qualitative responses of respondents, such as sentiments, perceptions, likes, dislikes, interests and preferences.
- It entails categorising facts/options into various alternative categories, which are usually exhaustive as well as mutually exclusive.
- 'Mode' is frequently used for response category.
- A ratio scale has a natural zero point and subsequent numbers are put at equally spaced intervals. Scales, for example, are used to measure physical quantities such as length, weight and so on.
- Respondents are asked to indicate their level of agreement or disagreement with a series of statements in Likert Scale.
- Attitudes are constructions that cannot be directly observed. As a result, the most typical method of evaluating attitudes is to analyse some aspect of behaviour and then draw conclusions about the attitudes that may be responsible for that behaviour.
- Physiological reactions reflect emotional reactions like skin response, pulse rate, dilation of the pupils etc.
- The face Electromyogram (EMG) is an objective attitude indicator that can be used to assess the quality or direction of one's attitude and its positivity or negativity.
- A new test has been developed in recent years to tap our implicit attitudes, self-concepts and other crucial parts of our cognitive system.

Activity

1. Prepare a PPT to establish the various measurement and scaling techniques used in research.

Glossary

- Biased: unfairly prejudiced for or against someone or something.
- Substantial: of considerable importance, size, or worth.
- Feasible: possible to do easily or conveniently.
- Prejudice: preconceived opinion that is not based on reason or actual experience.
- Preconceptions: a preconceived idea or prejudice.
- Unobtrusive: not conspicuous or attracting attention.
- Implicit: implied though not plainly expressed.

Notes

Notes

Questions and Exercises

1. What is Multi-Dimensional Scaling?
2. Explain the most frequently used Scales?
3. How can the Self-Rating Scales be defined and explained?
4. Elucidate the types of Attitude Measuring Scales.
5. How are physiological reactions used in order to identify more objective attitude indicators?
6. What cognitive measures to carry out scaling?

Further Readings

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Answers to Check Your Understanding

1. Semantic Differential
2. Multi-Dimensional
3. Thurston
4. respondents
5. ordinal

Unit - 2.5: Sampling

Notes

Learning Objectives:

At the end of this unit, you will be able to:

- Understand sampling plan
- Know about sampling frame
- Analyse sample selection methods
- Describe probability sampling techniques
- Comprehend non-probability sampling techniques
- Examine sampling and non-sampling errors

Introduction

When full census data cannot be collected, statisticians collect sample data by developing specific experiment designs and survey samples. Statistics itself also provides tools for prediction and forecasting through statistical models.

To use a sample as a guide to an entire population, it is important that it truly represents the overall population. Representative sampling assures that inferences and conclusions can safely extend from the sample to the population as a whole. A major problem lies in determining the extent that the sample chosen is actually representative. Statistics offers methods to estimate and correct for any bias within the sample and data collection procedures. There are also methods of experimental design for experiments that can lessen these issues at the outset of a study, strengthening its capability to discern truths about the population.

Sampling theory is part of the mathematical discipline of probability theory. Probability is used in mathematical statistics to study the sampling distributions of sample statistics and, more generally, the properties of statistical procedures. The use of any statistical method is valid when the system or population under consideration satisfies the assumptions of the method. The difference in point of view between classic probability theory and sampling theory is, roughly, that probability theory starts from the given parameters of a total population to deduce probabilities that pertain to samples. Statistical inference, however, moves in the opposite direction inductively inferring from samples to the parameters of a larger or total population.

2.5.1 Sampling Plan

A sampling plan is a detailed outline of which capacities will be occupied at what times, on which material, in whatsoever manner and by whom. Sampling plans must be designed in such a way that the following data will cover an illustrative sample of the parameters of attention and allow for all questions, as stated in the goals, to be answered.

The steps complex in evolving a sampling plan are:

1. be familiar with the parameters to be measured, the diversity of possible values and the mandatory resolution

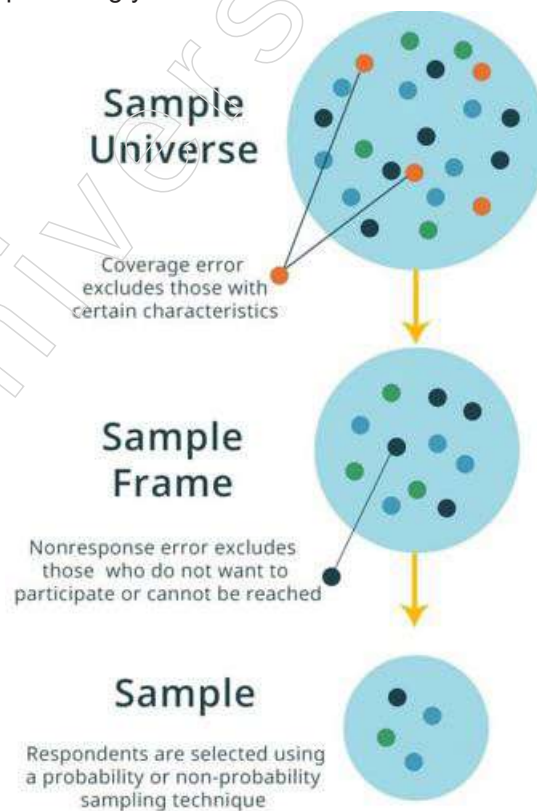
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2. innovativeness a sampling scheme that essentials how and when samples will be taken
3. select illustration sizes
4. scheme data storage set-ups
5. allocate parts and responsibilities

It's uncommon that you can acquire data from every single person in a group of people, no matter how quickly you perform study into them. You select a sample in its position. The sample is a group of people that will essentially take part in the study.

To draw valid inferences from your results, you partake to prudently decide how you will select a sample that is demonstrative of the group as a whole. In attendance are two kinds of sampling procedures:

- Probability sampling comprises random selection, letting you to mark strong arithmetical inferences just about the whole group.
- Non-probability sampling includes non-random choice based on convenience or extra criteria, permitting you to with no trouble collect data.



2.5.2 Sampling Frame

A sample frame is a list of everything in your population. It's a comprehensive list of everyone or anything you wish to learn about. The distinction between a population and a sampling frame is that the population is broad, whereas the frame is narrow. For example, the population could be "People who live in Jacksonville, Florida." The frame would name all of those people, from Adrian Abba to Felicity Zappa. A couple more examples:

Notes

2. Exclude all persons who are not part of the target population.
3. Contains correct information that can be utilised to contact specific individuals.

Other broad considerations you should make:

- A one-of-a-kind identifier for each member. This might be a straightforward numerical identifier (e.g., from 1 to 1000). Make certain that there are no duplicates in the frame.
- A sensible arrangement of the list. Sort them alphabetically, for example.
- Current information. This may need to be examined on a regular basis (e.g., for address changes).

In rare instances, obtaining a sampling frame may be impossible or extremely complex. Obtaining a list of prostitutes in your city, for example, is unlikely (mainly due to the fact that most prostitutes will not want to be found). To compensate for the lack of a sample frame, procedures such as snowball sampling must be used at times. Snowball sampling is the process of finding one (or a few) persons for a survey or experiment. You then request that they find someone else who is willing to participate. Then that person finds someone else and so on, until you have a sufficient number of people.

Sample Space vs. Sampling Frame

A sampling frame is a list of things from which you draw a sample. A sample space is a catalogue of every conceivable outcome of an experiment. For example, you might have a sampling frame of people's names in a specific town for a survey on family size that you're intending to perform. The sample area includes all conceivable outcomes from your survey: one person, two people, three people, ten people, or more.

2.5.3 Sample Selection Methods

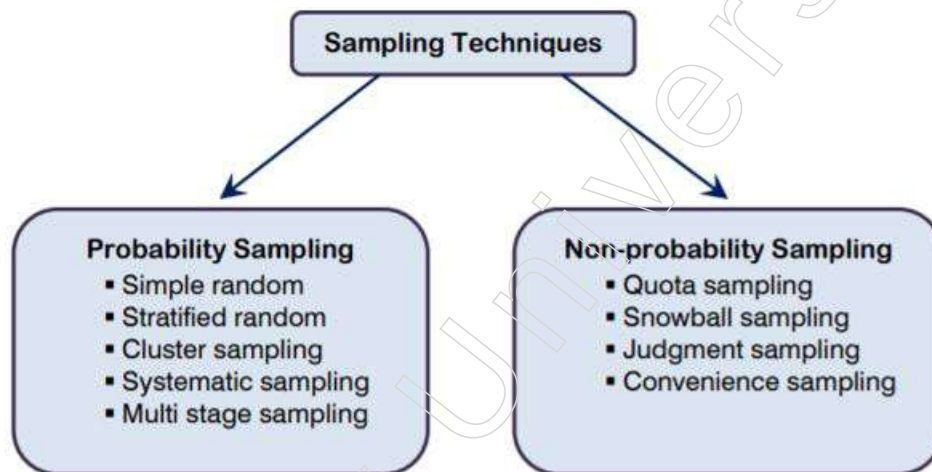
It is unlikely that the researcher will be able to collect data from all situations in order to answer the research questions. As a result, a sample must be chosen. The population refers to the complete set of instances from which the researcher sample is drawn. Because researchers do not have the time or resources to analyse the entire population, they use sampling techniques to decrease the number of cases.

The stages that are likely to go through when conducting sampling.

1. Clearly Define Target Population
2. Select Sampling Frame
3. Choose Sampling
4. Technique
5. Determine
6. Sample Size
7. Collect Data
8. Assess
9. Response Rate

- A. Stage 1: Define the Target Population Clearly: The first step in the sampling procedure is to define the target population precisely. The number of people residing in a country is generally referred to as its population.
- B. Stage 2: Choose a Sampling Frame: A sampling frame is a list of real-world cases from which a sample will be selected. The sampling frame must be population-representative.
- C. Stage 3: Select a Sampling Technique: Prior to delving into the many forms of sampling methods, it is important to define sampling and the reasons why researchers are likely to select a sample. Sampling is the process of selecting a subset from a predetermined sampling frame or the complete population. Sampling can be used to draw conclusions about a population or to make generalisations on current theory. In essence, this is determined by the sample technique used. In general, sampling techniques can be divided into two types:
- Probability or random sampling
 - Non- probability or non- random sampling

Before choosing specific type of sampling technique, it is needed to decide broad sampling technique.



- D. Stage 4: Calculate Sample Size: A random sample must be of sufficient size in order to generalise from it and prevent sampling errors or biases. What is adequate is determined by a number of factors, which frequently perplex people doing surveys for the first time. This is because what matters here is not the fraction of the study community sampled, but the absolute size of the sample chosen in relation to the population's complexity, the researcher's goals and the types of statistical manipulation that will be utilised in data analysis. While it is true that the larger the sample, the less likely the findings would be biased, diminishing returns can soon set in when samples exceed a certain size, which must be balanced against the researcher's resources (Gill et al., 2010). To put it clearly, higher sample numbers reduce sampling error, but only at a slower rate. There are several statistical formulas available for calculating sample size. There are various techniques for computing the sample size for categorical data, each employing a distinct formula. $n \times p (100-p)$ The needed sample size is $z^2 / E^2 n$. P denotes the frequency of occurrence of a situation or condition. E denotes the maximum allowable error as a percentage.

Notes

Z is the value that corresponds to the required level of confidence. This formula is comprised of two major components (Bartlett et al., 2001).

First, there are factors to consider when estimating the degrees of precision and risk that the researcher is willing to accept. E denotes the margin of error (the level of precision) or the risk that the researcher is willing to take (for example, the plus or minus figure reported in newspaper poll results). In social research, a margin of error of 5% is considered acceptable. So, for example, if 40% of respondents in a work satisfaction survey said they were unsatisfied, the range would be between 35% and 45%. Because sampling error is inversely related to the square root of n, the smaller the value of E, the larger the sample size necessary; yet, a big sample cannot guarantee precision (Bryman and Bell, 2003). Z refers to the degree of certainty that the survey findings show correct results. This refers to the degree to which we may be certain that the sample survey accurately estimated the characteristics of the population. Z is the statistical value that corresponds to the desired level of confidence. The basic notion here is that if a population is sampled frequently, the average value of a variable or question received will be equal to the genuine population value. Typical levels of confidence in management research are 95 percent (0.05: a Z score of 1.96) or 99 percent (0.01: Z=2.57). A 95 percent degree of confidence means that 95 out of 100 samples will contain the true population value within the margin of error (E). The second critical component of a sample size formula is the estimation of population variation or heterogeneity (P). Management scholars are frequently concerned with selecting sample size for challenges involving population percentages or proportions estimation (Zikmund, 2002).

The variance of a proportion or the percentage incidence of how a specific question, for example, will be answered is P in the calculation (100-P). Where P is the percentage of a sample that has a characteristic, such as the 40% of respondents who were dissatisfied with their pay and (100-P) is the percentage (60%) that does not have the feature or belief. The main challenge is determining how to estimate the value of P before to conducting the poll. According to Bartlett et al. (2001), researchers should use 50% as an estimate of P because this will maximise variance and provide the largest sample size (Bartlett et al., 2001). The formula for estimating sample size of the population has almost little effect on how well the sample is likely to depict the population and as Fowler (2002) points out, it is unusual for it (the population fraction) to be an essential consideration when deciding on sample size (Fowler, 2002).

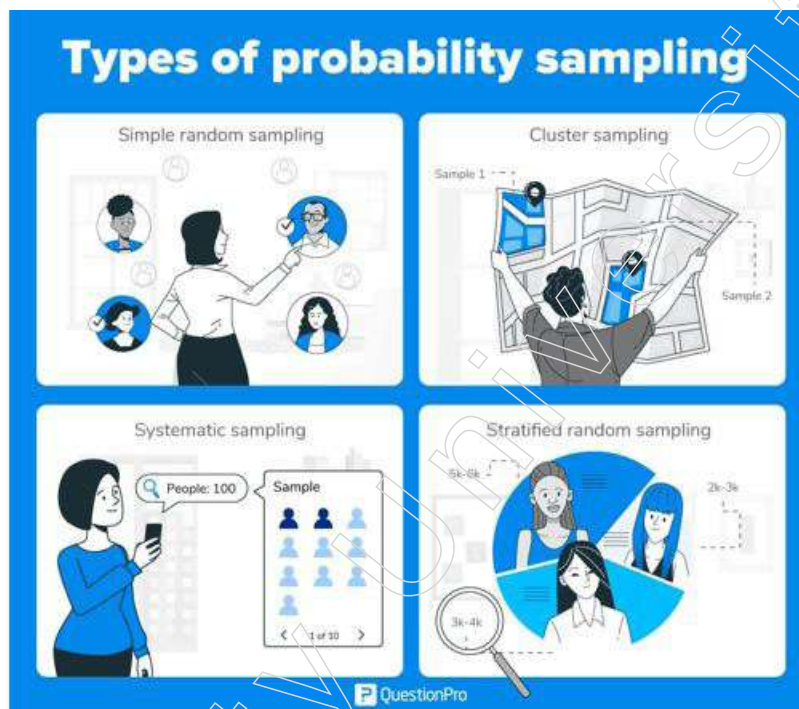
The sample sizes represent the number of replies obtained rather than the number of surveys issued (this number is often increased to compensate for non-response). However, response rates for postal and e-mailed surveys are rarely 100 percent in most social and management surveys. Increasing the sample size by up to 50% in the initial distribution of the survey is probably the most popular and time efficient technique to ensuring minimum sample size is reached (Bartlett et al., 2001).

- E. Stage 5: Collect Data: After determining the target population, sampling frame, sampling procedure and sample size, the following step is to collect data.
- F. Stage 6: Evaluate Response Rate: The number of cases who agreed to participate in the study is referred to as the response rate. These examples are drawn from the original sample. In practise, most researchers never attain a response rate of 100%. Reasons for this could include the respondent's rejection to answer, ineligibility to

respond, incapacity to react, or the respondent's whereabouts but researchers' inability to contact them. To summarise, response rate is crucial since each non-response has the potential to skew the final sample. In some ways, clearly defining the sample, using the appropriate sampling technique and obtaining a big sample can help to lessen the chance of sample bias.

The various sorts of sampling methods/techniques were explained in this work. In addition, the six processes for conducting sample were discussed. As previously stated, there are two kinds of sampling methods: probability sampling and non-probability sampling. Each of these strategies use a variety of sampling methodologies. Quota sampling, Snowball sampling, Judgment sampling and Convenience sampling are examples of non-probability sampling. Probability sampling comprises Simple random, Stratified random, Cluster sampling, Systematic sampling and Multi stage sampling.

2.5.4 Probability Sampling Techniques



Probability sampling means that each single member of the group has a chance of being selected. It is regularly used in quantitative research. Probability sampling methods are the most valid ones if you need to obtain results that are representative of the entire population.

In attendance are four main kinds of probability sample.

1. Simple random sampling

In a simple random sample, every single member of the population partakes an equal chance of being selected. Your sample frame should consist of the whole population.

To decide this type of sampling, you can use tools like random number producers or additional techniques that are based completely on chance.

Notes

Example

You are given the task of selecting a basic random sample of 100 employees from Company X. You assign a number from 1 to 1000 to each employee in the company database and you use a random number generator to generate the first 100 numbers.

2. Systematic sampling

Systematic sampling be situated similar to simple random sampling, but then again it is usually to some extent easier to conduct. Every single member of the population is registered with a number, but then again instead of randomly generating numbers, individuals are selected at regular intervals.

Example

Altogether employees of the company are registered in alphabetical order. From the first 10 numbers, you arbitrarily select a starting point: number 6. On or after number 6 onwards, each 10th person on the list is selected (6, 16, 26, 36 and so on), in addition you end up by means of a sample of 100 people.

If you practice this technique, it is important to make sure that there is no hidden pattern in the list that can skew the sample. For example, if the HR database cluster's employees via team and team members are registered in order of seniority, there is a risk that your study might skip over individuals in junior roles, resulting in a sample that is skewed in the direction of senior employees.

3. Stratified sampling

Stratified sampling includes dividing the population into subpopulations that may differ in important ways. It permits you draw more detailed conclusions by ensuring that every single subgroup is appropriately represented in the sample.

To customize this sampling method, you split the population into subgroups (called strata) based on the germane characteristic (e.g., gender, age range, income bracket, job role).

Grounded on the overall proportions of the population, you compute how many people should be sampled from respectively subgroup. At that time, you use random or orderly sampling to choose a sample from each subgroup.

Example

The business has 800 female employees and 200 male employees. You need to ensure that the example reflects the gender balance of the corporation, so you sort the population into two strata created on gender. Then your run-through random sampling on each set, choosing 80 women and 20 men, which provides you a representative sample of 100 people.

4. Cluster sampling

Cluster sampling also includes dividing the population into subgroups, but each subgroup ought to have similar characteristics to the whole sample. Instead of sampling individuals from the beginning of each subgroup, you interview entire groupings at random.

If it's feasible, you could count every single person in each sampled cluster. If the bunches are large, you can use one of the strategies above to sample individuals from the areas surrounding each cluster. This is named multistage sampling.

This method is used for dealing with large and detached populations, but there is extra risk of error in the sample, as there will possibly be substantial differences surrounding clusters. It's difficult to promise that the sampled clusters are really representative of the complete population.

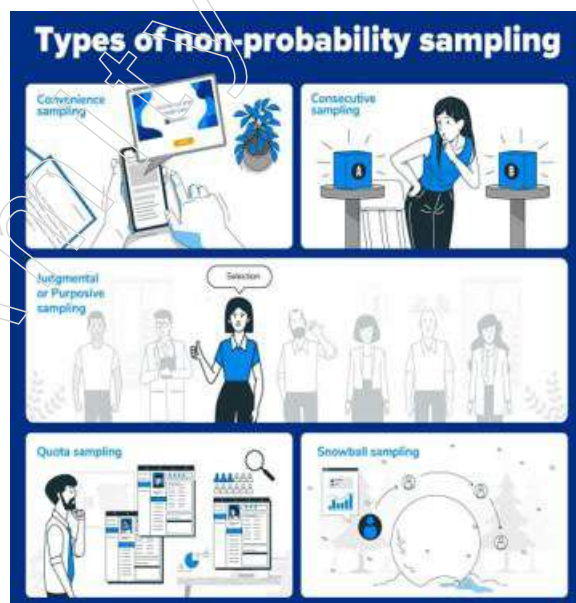
Example

The corporation has offices in 10 cities from corner to corner the country (all with roughly the similar number of employees in similar roles). You don't partake the capacity to transportable to every office to bring together your data, so you use random sampling to hand-picked 3 offices – these are your clusters.

5. Multi-stage sampling

Multi-stage sampling is a method of gradually transitioning from a broad to a narrow sample (Ackoff, 1953). If, for example, a Malaysian publisher of an International Journal of Academic Research in Management Volume 5, Issue 2, 2016, ISSN: 2296-1747 publishes an International Journal of Academic Research in Management Volume 5, Issue 2, 2016, ISSN: 2296-1747 Copyright© Helvetic Editions LTD - All Rights Reserved - 2016 www.elvedit.com If 22 automobile publications wanted to conduct a survey, they could simply select a random sample of Malaysian car owners from the whole population. This is obviously both costly and time demanding. A less expensive option would be to use multi-stage sampling. In essence, this would entail separating Malaysia into geographical regions. Following that, some of these zones are chosen at random and then subdivided, maybe based on local authority areas. Following that, some of them are chosen at random and separated into smaller sections, such as towns or cities. The primary goal of multi-stage sampling is to pick samples that are concentrated in a few geographical areas. This, once again, saves time and money.

2.5.5 Non-probability Sampling Techniques



Notes

In a non-probability sample, persons are selected founded on non-random criteria and not each individual has a reason of being included.

This form of sample is easier and inexpensive to access, but it has a higher risk of sampling bias. That means the implications you can type about the population are feebler than with probability samples and your conclusions may be more limited. If you use a non-probability sample, you ought to still aim to mark it as representative of the population as possible.

In exploratory and qualitative research, non-probability sampling approaches are frequently browsed through. The goal of this type of research is not to test a hypothesis about a large group, but to gain an early understanding of a small or understudied population.

1. Convenience sampling

A convenience sample simply comprises the individuals who occur to be most easy to get to the researcher.

This is a laidback and inexpensive way to gather initial data, nevertheless there is no way in the direction of tell if the sample is demonstrative of the population, so it can't yield generalizable results.

Example

You are researching sentiments about student support facilities in your university, so after collectively of your classes, you ask your fellow students to wide-ranging a survey on the topic. This is an appropriate way to gather data, nevertheless as you only surveyed students enchanting the same classes by means of you at the same level, the example is not illustrative of all the students at your university.

2. Voluntary response sampling

Comparable to a convenience sample, a voluntary response sample is chiefly based on ease of access. As an alternative of the researcher indicating participants and directly contacting them, individuals volunteer themselves (e.g., by re-joining to a public online survey).

Voluntary response samples are continuous, at least to some extent biased, as some people determination inherently be more probable to volunteer than others.

Example

You direct out the survey to all students at your campus and a lot of students decide to complete it. This can definitely give you some insight hooked on the topic, but the people who answer back are more likely to be individuals who have strong opinions approximately the student support services, so you can't be sure that their views are demonstrative of all students.

3. Purposive sampling

This form of sampling, also identified as judgement sampling, involves the researcher by means of their expertise to select an example that is most useful to the purposes of the research.

It is common practice in qualitative research, wherever the researcher wants to gain detailed knowledge around a specific phenomenon somewhat than make statistical inferences, or anywhere the population is very small and specific. An actual purposive sample obligation has clear criteria in addition rationale for inclusion.

Example

You want to distinguish more about the sentiments and experiences of disabled students at your university, so you decisively select a number of students with different sustenance needs in order to gather a wide-ranging range of data on their experiences by means of student services.

4. Snowball sampling

If the inhabitants are hard to access, snowball sampling can be utilized to recruit participants through existing participants. The quantity of people you have access to “snowballs” as you change to in contact by way of more people.

Example

You are investigating experiences of homelessness in your city. In the meantime, there is no list of all homeless individuals in the city, probability control group isn't possible. You meet one individual who agrees to play a part in the research and she puts you in contact by means of other homeless people that she be acquainted with in the area.

Sample size is the quantity of completed answers your survey receives. It's called a sample because it only characterizes part of the cluster of people (or target population) whose feelings or behavior you care about. For instance, one way of sampling is to use a “random sample,” somewhere respondents are preferred entirely by chance from the populace at large.

At this juncture are three key footings you'll need to recognize to calculate your sample size and provide it context:

Population size: The entire number of people in the group you are demanding to study. If you were pleasing a random sample of people transversely the U.S., then your population size would be around 317 million. Similarly, if you remain surveying your company, the dimension of the population is the total amount of employees.

Margin of error: A one hundredth that tells you how much you can think your survey consequences to reproduce the views of the complete population. The smaller the margin of error, the sooner you are to having the exact response at a given self-confidence level.

Sampling confidence level: A fraction that discloses how self-assured you can be that the inhabitants would first-class an answer within a convinced range. For example, a 95% confidence equivalent means that you can be 95% certain the outcomes lie flanked by x and y numbers.

2.5.6 Sampling and Non-sampling Errors

A sampling error happens when the sample hand-me-down in the study is not characteristic of the whole population. Sampling errors frequently come about and

Notes

therefore, researchers continuously compute a margin of error during final results as a statistical practice. The margin of error is the quantity of error allowed for a miscalculation to signify the difference amongst the sample and the real population.

At this time are the top four market research mistakes while sampling:

- **Population description error:** A population description error happens when researchers don't distinguish precisely who to review. For example, envisage a research study around kid's apparel. Who is the correct person to survey? It can be together parents, only the mother, in addition the child. The paternities type of purchase decisions, nonetheless the kids may influence their high-quality.
- **Sample surround mistake:** Sampling surround mistakes arise as rapidly as researchers target the sub-population incorrectly while choosing the sample. For example, pick a sampling frame on or afterward the telephone white pages book may take erroneous inclusions for the aim that people shift their cities. Erroneous exclusions take place when people wish to un-list their numbers. Wealthy households may have extra than one connection, in consequence leading to manifold inclusions.
- **Selection error:** A selection error come about when respondents self-select themselves to contribute in the study. Individual the interested ones respond. You can regulator selection errors by successful the extra step to request replies from the entire sample. Pre-survey preparation, follow-ups and a neat in addition clean survey design determination boost respondents' partaking rate. Also, try procedures like CATI surveys in addition in-person interviews to take full advantage of responses.
- **Sampling errors:** Sampling errors take place due to a gap in the representativeness of the respondents. It majorly come about when the researcher looks after his plan sample carefully. These example errors can be controlled and detached by creating a cautious sample design, having a large enough example to reflect the complete population, or by means of an online sample or review audiences to assemble responses.

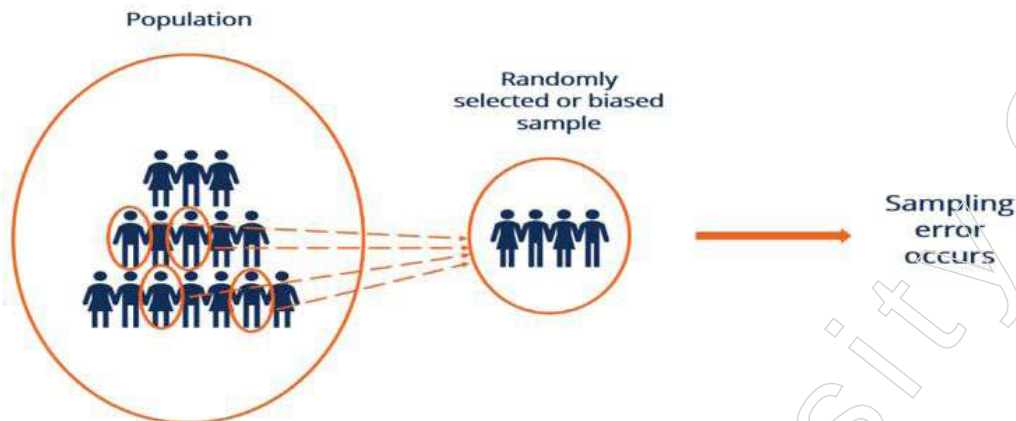
Regulating your Sampling Error

Statistical theories assist researchers' extent the probability of sampling mistakes in sample size and population. The size of the sample measured from the population primarily regulates the extent of the sampling error. Larger sample sizes tend to meet a lower rate of errors. Researcher's use a metric known as the boundary of error to comprehend and evaluate the margin of error. Usually, a self-assurance level of 95% is well-thought-out to be the desired Sampling errors are relaxed to identify. At this time are a few simple stages to diminish sampling error:

1. **Upsurge sample size:** A larger sample size outcomes in a more accurate result for the study gets closer to the definite population size.
2. **Division of the population hooked on groups:** Test groups giving to their size in the population as an alternative of a random sample. For example, if people of

a certain demographic make up 20% of the population, you can be certain that your education is entirely made up of this variable.

3. Distinguish your population: Study your population and comprehend its demographic mix. Be acquainted with what demographic routine your product and service make and make sure you solitary target the sample that substances confidence level.



Non-sampling inaccuracy refers to an error that arises from the data gathering process that causes the data to differ from the true values. It differs from sampling error, which is any divergence between the sample values and the universal values that may occur as a result of an infinite sampling size.

Non-sampling errors can originate in innumerable forms, including non-response error, dimension error, interviewer error, modification error and processing error.

Mechanics of Non-Sampling Error

Non-sampling error can happen over a sample or an entire people (census) is taken. It falls further down two types:

1. Random errors

Random errors can be situated errors that cannot be accounted for and impartially happen. In statistical investigations, it is assumed that one unintentional error cancels out the others, therefore they are of little to no concern.

2. Systematic errors

Systematic errors affect the sample of the education and, as a result, often creates useless data. A systematical error is constant and repeatable, so the study's makers must take great care to alleviate such an error.

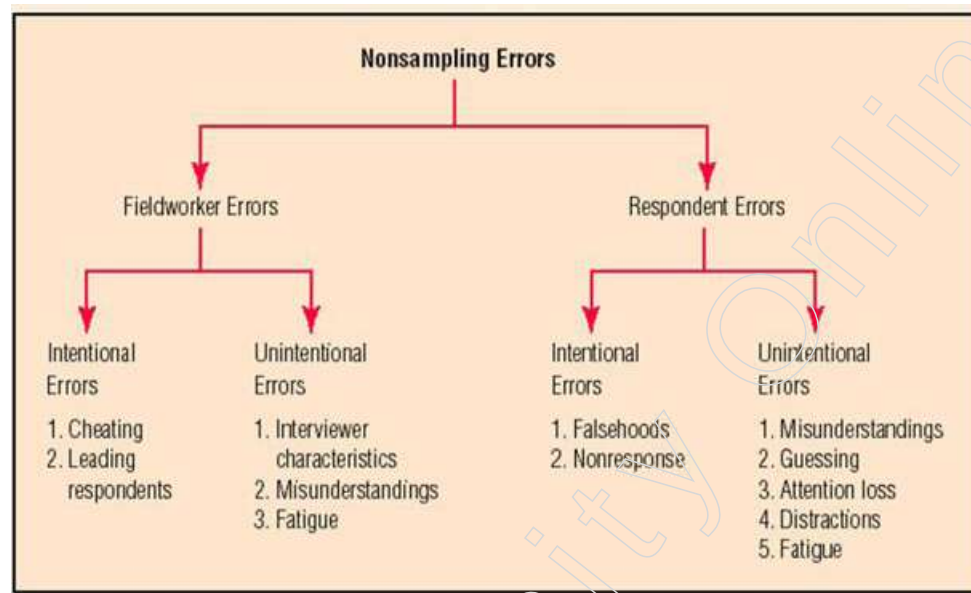
Non-sampling errors can take place from several characteristics of a study. The most common non-sampling errors consist of errors in data entry, biased queries and decision-making, non-responses, untrue information and inappropriate examination.

Kinds of Non-Sampling Errors

There are numerous types of non-sampling errors, including:

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Nonsampling Errors Associated with Fieldwork



1. Non-response mistake

A non-response error occurs when the number of persons who choose to participate in a survey differs from the number of people who do not. In new words, it exists when people have assumed the option to take part but choose not to; consequently, their survey consequences are not assimilated into the data.

2. Measurement error

A measurement error refers to all faults relating to the measurement of each sampling unit, as opposed to errors relating to how they were chosen. The fault often arises when there are confusing questions, low-quality data in arrears to sampling fatigue (i.e., somebody is tired of enchanting a survey) and low-quality dimension tools.

3. Interviewer error

Interviewer error come about when the interviewer (or administrator) types an error when transcribing a response. In qualitative research, an examiner may lead a interviewee to answer in a definite way. In quantitative research, an interviewer will possibly ask the question differently, which leads to an altered result.

4. Adjustment error

An adjustment error designates a situation where the analysis of the data is adjusted so that it is not completely accurate. Methods of adjustment error consist of errors with weighting the data, data scrubbing and imputation.

5. Processing error

A processing error occurs when there is a problem through processing the data that causes an error of some kind. An instance will be if the data remained entered incorrectly or if the data file is unethical.

Sampling Error vs. Non-Sampling Error

Although sampling error and non-sampling error are frequently used in the same context, there are significant differences between the two ideas. They comprise:

1. In contrast to non-sampling error, which gets up when a mistake occurs, sampling error might alter to your feet even when no apparent fault has occurred.
2. Sampling error come about as soon as the sample is not characteristic of the worldwide truth, even though non-sampling error is meticulous to a certain study design.
3. Sampling error can be abridged greatly as sampling size upsurges, but non-sampling error needs more methodical procedures to reduce.
4. Sampling error is often caused by internal factors, whereas non-sampling error is shaped by external factors not totally connected to a survey, study, or census.

How to Reduce Errors

Reducing non-sampling error is not as effortlessly attained as reducing sampling error. With sampling error, you can decrease the risk of mistake by simply growing the sample size. It will not work for non-sampling error, which is every so often very hard to detect and eliminate (unless exact methodical consideration is assumed to the basis of the error).

To efficiently reduce non-sampling error, cautious consideration must be taken by individuals designing the study to guarantee the validity of the results. By way of means, a researcher may plan a mechanism hooked on the study to decrease the error even though subsequently not presenting another error.

For instance, a researcher may pay the individual a bonus contingent on the exactness of their data entry, or they may film all interviews to guarantee that the interviewer halts on topic in addition on script.

Check Your Understanding

Fill in the blanks:

1. _____ sampling is the process of finding one or a few persons for a survey or experiment.
2. _____ sampling includes dividing the population into subpopulations that may differ in important ways.
3. In exploratory and qualitative research, _____ sampling approaches are frequently browsed through.
4. The margin of error is the quantity of error allowed for a _____ to signify the difference amongst the sample and the real population.
5. In _____ research, an interviewer will possibly ask the question differently, which leads to an altered result.

Summary

- Sampling theory is part of the mathematical discipline of probability theory.

Notes

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- To draw valid inferences from your results, you partake to prudently decide how you will select a sample that is demonstrative of the group as a whole.
- A sample frame is a list of everything in your population. It's a comprehensive list of everyone or anything you wish to learn about.
- The distinction between a population and a sampling frame is that the population is broad, whereas the frame is narrow.
- A sampling frame is a list of things from which you draw a sample. A sample space is a catalogue of every conceivable outcome of an experiment.
- Before choosing specific type of sampling technique, it is needed to decide broad sampling technique.
- Probability sampling means that each single member of the group has a chance of being selected. It is regularly used in quantitative research.
- Cluster sampling includes dividing the population into subgroups, but each subgroup ought to have similar characteristics to the whole sample.
- The primary goal of multi-stage sampling is to pick samples that are concentrated in a few geographical areas.
- Comparable to a convenience sample, a voluntary response sample is chiefly based on ease of access.
- Statistical theories assist researchers' extent the probability of sampling mistakes in sample size and population.
- The size of the sample measured from the population primarily regulates the extent of the sampling error.
- Researcher's use a metric known as the boundary of error to comprehend and evaluate the margin of error.
- Non-sampling inaccuracy refers to an error that arises from the data gathering process that causes the data to differ from the true values.
- Random errors can be situated errors that cannot be accounted for and impartially happen.
- A non-response error occurs when the number of persons who choose to participate in a survey differs from the number of people who do not.
- A processing error occurs when there is a problem through processing the data that causes an error of some kind.
- Although sampling error and non-sampling error are frequently used in the same context, there are significant differences between the two ideas.
- To efficiently reduce non-sampling error, cautious consideration must be taken by individuals designing the study to guarantee the validity of the results.

Activity

1. What best suited techniques do you suggest to reduce errors in sampling?

Glossary

- Contingent: subject to chance.
- Upsurge: an upward surge in the strength or quantity of something; an increase.
- Unethical: not morally correct.
- Assimilated: take in (information, ideas, or culture) and understand fully.
- Alleviate: make (suffering, deficiency, or a problem) less severe.
- Divergence: separate from another route, especially a main one and go in a different direction.
- Acquainted: make someone aware of or familiar with.

Questions and Exercises

1. What concepts are required to understand sampling plan?
2. What do you know about sampling frame? Elaborate.
3. Analyse the sample selection methods.
4. Describe the probability sampling techniques in detail.
5. What non-probability sampling techniques are implemented to achieve proper research?
6. Distinguish between sampling and non-sampling errors.

Further Readings

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Answers to Check Your Understanding

1. Snowball
2. Stratified
3. non-probability
4. miscalculation
5. quantitative

Notes

Module - 3: Data Analyses Techniques

Structure:

- 3.1 Descriptive Statistics
 - 3.1.1 Definition & Characteristics of Descriptive Research
 - 3.1.2 Applications of Descriptive Research(with Examples)
 - 3.1.3 Descriptive Research Methods
- 3.2 Hypothesis Testing
 - 3.2.1 Introduction to Hypothesis
 - 3.2.2 Types of Hypothesis
 - 3.2.3 Testing Hypothesis and Significance Levels
 - 3.2.4 Type I & Type II Error(with Example)
 - 3.2.5 One-tailed and Two-tailed Test
 - 3.2.6 Confidence Intervals
 - 3.2.7 Bayesian Statistics
- 3.3 Parametric & Non-Parametric Tests
 - 3.3.1 Z-test
 - 3.3.1 T-test
 - 3.3.3 F-test
 - 3.3.4 Correlation & Regression
 - 3.3.5 Chi-square Test
 - 3.3.6 Factor Analysis
- 3.4 Data Analysis
 - 3.4.1 Introduction to SPSS
 - 3.4.2 Creating Data File
 - 3.4.3 Run Example on Parametric Test
 - 3.4.4 Run Example on Non-parametric Test

Unit - 3.1: Descriptive Statistics

Notes

Objectives

At the end of this unit, you will be able to:

- Understand the definition and characteristics of descriptive research.
- Learn about the applications of descriptive research (with examples).
- Comprehend descriptive research methods.

Introduction



Descriptive statistics are used to characterize the fundamental characteristics of a study's data. They give quick summaries of the sample and the metrics. They are the foundation of practically every quantitative data analysis, along with simple graphical analysis.

Descriptive analyses are sometimes distinguished from descriptive statistics. When you use descriptive statistics, you're merely describing what the data is or indicates. When using inferential statistics, you're attempting to draw inferences that go beyond the data at hand. For example, we use inferential statistics to try to infer what the population would think based on the sample data. Alternatively, we can use inferential statistics to determine if an observed difference between groups is reliable or one that occurred by chance in this study. As a result, we use inferential statistics to extrapolate general conditions from our data, while descriptive statistics are used to merely describe what's going on in our data.

Descriptive statistics are used to display quantitative data in a logical and understandable manner. We may have a number of measures in a research investigation. Alternatively, we can assess a huge number of people using any method. Descriptive statistics assist us in making sense of enormous amounts of data. Each descriptive statistic condenses a large amount of information into a concise summary.

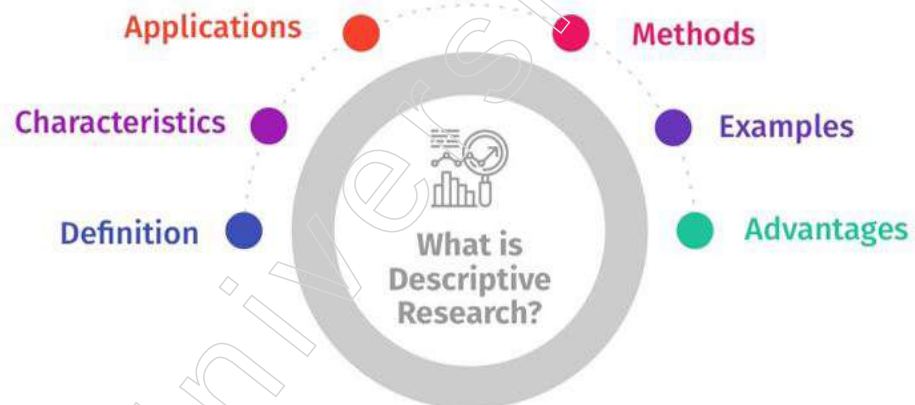
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When you try to explain a large number of observations with a single indicator, you risk distorting the data or losing critical information. The batter's batting average does not indicate whether he is hitting home runs or singles. It's impossible to determine if she's been in a rut or on a roll. The GPA does not indicate whether the student took challenging or easy courses, or if the courses were in their primary subject or from other disciplines. Despite these drawbacks, descriptive statistics provide a useful summary that can be used to compare persons or other units.

The properties of a data set are summarized and organized using descriptive statistics. A data set is a collection of responses or observations from a sample or population as a whole.

The initial step of statistical analysis in quantitative research is to define properties of the replies, such as the average of one variable (e.g., age) or the relationship between two variables (e.g., age and creativity).

3.1.1 Definition & Characteristics of Descriptive Research



When a researcher wants to find out about different aspects including traits, frequency, trends, correlations, and classifications, they utilize a descriptive study approach. When the researcher has little information on the study problem, they can use it. The methods used in descriptive research are those that describe the features of the variables being studied. This approach focuses on addressing questions about the “what” rather than the “why” of the study topic. Instead of focusing on the “why,” descriptive research focuses on merely describing the characteristics of the demographics under investigation.

Since none of the variables in the study are altered throughout the research process, descriptive research is referred to as an observational research method.

Consider the case of a UK-based firm attempting to establish itself in New York and wishing to learn more about the demographics of purchasers who often purchase from brands comparable to its own. The demographics of the population will be the exclusive focus of the information acquired from the survey. It will reveal information about the purchasing habits of different age groups in New York. Because the brand is attempting to establish itself in New York, it will not investigate why such tendencies exist. They're just interested in learning about the general public's purchasing habits, not why such linkages exist.

Descriptive research is a type of quantitative market research or social research that entails conducting surveys using quantitative variables on a market research or social research tool.

Descriptive research is a form of study that focuses on describing a population, condition, or phenomenon. It focuses on answering the questions of how, what, when, and where but not why.

This is due to the fact that it is important to have a thorough understanding of a research issue before looking at why it exists in the first place.

Before asking why, an investor considering investing in the ever-changing Amsterdam housing market should first understand the current state of the market, how it changes (increasing or decreasing), and when it changes (time of year). This is where descriptive research comes into the picture.

Characteristics of Descriptive Research



The word descriptive research refers to the research questions, study design, and data analysis that were undertaken on that topic. It's called an observational research approach because none of the variables in the research study are changed in any way.

The following are some distinguishing aspects of descriptive research:

1. **Quantitative research:** Descriptive research is a type of quantitative study that aims to collect measurable data so that a population sample may be statistically analyzed. It is a widely used market research instrument that allows us to collect and define the characteristics of a demographic category.
2. **Uncontrolled variables:** None of the variables in descriptive research are impacted in any way. The research is carried out using observational approaches. As a result, the researcher has no control over the variables' nature or behavior.
3. **Cross-sectional studies:** Descriptive research is often a cross-sectional study in which different portions of the same group are investigated.
4. **The basis for further research:** The following provides the foundation for future research: Researchers use a variety of research approaches to dig

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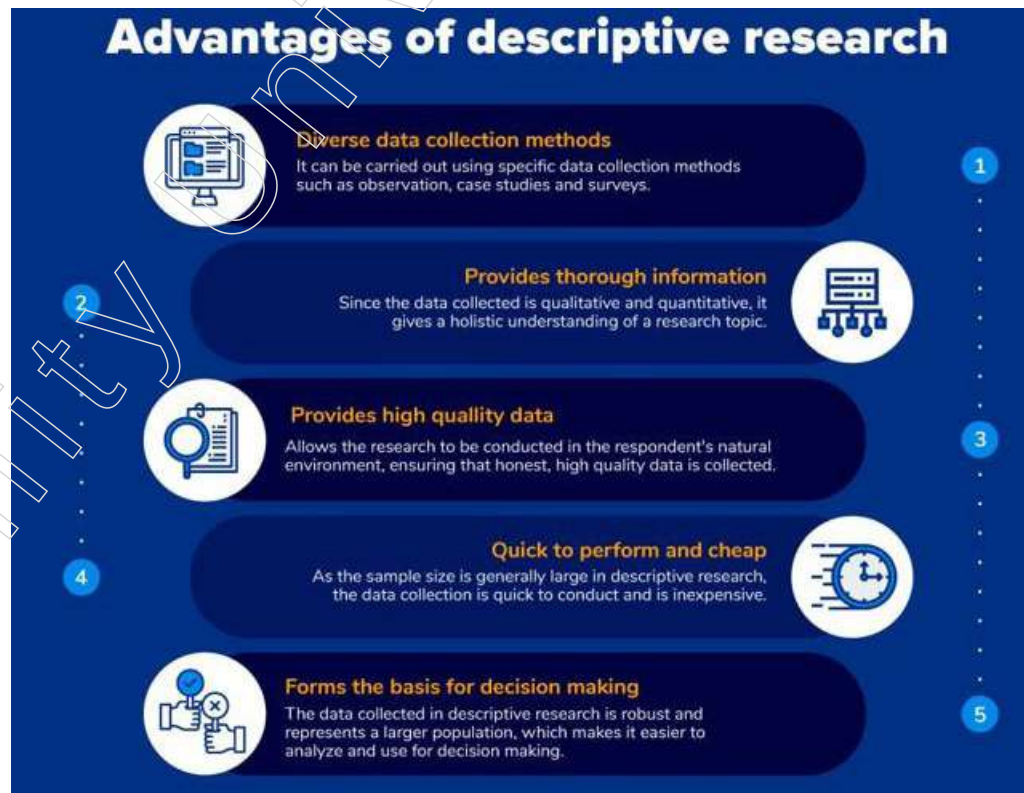
deeper into the data collected and processed during descriptive research. The information can also point to the types of research methodologies that will be used in the future.

5. Natural setting: In most cases, descriptive studies are carried out in natural settings. For example, you can distribute survey questionnaires to random people or use an observational method to watch people's behavior in a certain area. For instance, if you want to learn about people's purchasing habits, you can go to a supermarket and observe them.

Advantages of Descriptive Research

The following are some of the major advantages of descriptive research:

- Data collection: Descriptive research can be done in a variety of ways, including observational, case study, and survey methods. All of the key data collection methods are covered by these three, which provides a wealth of information. This information can be useful for future study or even to generate a hypothesis about your research topic.
- Variable: The qualitative and quantitative data collected provides full knowledge of a research issue. The data is rich, diverse, and comprehensive.
- Natural setting: Descriptive research allows the researcher to conduct the study in the respondent's natural setting, resulting in high-quality and honest data.
- Data gathering is quick and inexpensive in descriptive research because the sample size is usually big.



Source: <https://www.questionpro.com/blog/descriptive-research/>

Figure: Advantages of Descriptive Research

Disadvantages of Descriptive Research

Descriptive research has a few drawbacks, which are as follows:

- Inaccurate Information

If respondents believe the questions are evaluating personal matters, they may give deceptive or erroneous responses. The presence of the observer can also affect respondents, causing them to engage in deception. The observer effect is the name for this phenomenon.

- Biases in research

The study's results may be influenced by the researchers' personal prejudices. The experimenter effect is the name for this phenomenon.

- Representative Issues

There's also the issue of representativeness, which means that a case study or data from a tiny sample doesn't adequately represent the entire community.

- Score is limited.

Descriptive research has a limited reach, as it merely analyses the "what" of research rather than evaluating the "why" or "how" questions.

3.1.2 Applications of Descriptive Research (With Examples)

For a variety of reasons, descriptive research is used. There is no way of knowing whether or not the study you are performing will satisfy the predictions you have made, even after setting survey goals, survey designs, and data collection through surveys. Here are some examples of how organizations use descriptive research to grasp the objectives of research goals:

1. Define respondent characteristics: Close-ended questions are used to define respondent characteristics so that definite conclusions can be drawn about the respondents. This could be because of the necessity to determine the respondents' patterns, traits, and habits. It could also be to ascertain a respondent's attitude or viewpoint on the issue. Understanding how many hours per week millennials spend on the internet, for example. All of this data aids the research company in making well-informed business judgments.
2. Analyze data patterns: With the statistical capabilities of a descriptive research design, researchers may track data patterns across time. Consider how an apparel firm might conduct research on distinct demographics, such as age groups 24-35 and 36-45 while launching a new autumn wear line. If one of those groups does not take to the new launch well, it provides information about what outfits are popular and which are not. Customers' unfavorable clothing and apparel are discontinued by the brand.
3. Conduct comparisons: Organizations also utilize a descriptive research design to learn how different groups react to a certain product or service. For example, an apparel company might construct a survey that asks generic questions about the company's image. Demographic questions such as age, income, gender, and geographic region are also asked in the same study. This consumer research aids the company in determining which components of the

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brand appeal to the general public and which do not. It also aids in product or marketing improvements, as well as the creation of a new product line to cater to groups with strong growth potential.

4. **Validate existing conditions:** Descriptive research is commonly used by researchers to determine the current conditions and underlying patterns of the research object. Researchers watch each variable and conduct an in-depth study thanks to the non-invasive research method and the use of quantitative observation and some features of qualitative observation. It's also used by researchers to confirm any existing conditions that may be prevalent in a group.
5. **Carry out research at various times:** To see if there are any parallels or differences, the analysis might be done at different times. This also enables the evaluation of any number of variables. Studies on current situations can also be repeated to identify trends for verification.

3.1.3 Descriptive Research Methods



Descriptive research can be prepared in three ways.

1. Case study method

The case study approach entails an in-depth and extensive examination of the subject, person, or case under investigation. To conduct the research, a case study uses a formal research process. Using the findings of the case study, a research hypothesis may be developed that can be utilized to broaden the scope of research.

Case study research, on the other hand, is not appropriate for determining the relationship between cause and effect because it does not yield precise data. Furthermore, the case study method's outcome is specific to that instance and related cases and cannot be generalized. Case studies are focused on unique and interesting problems that are complex and difficult to solve, and they provide further information about a specific case.

In a medical case study, for example, researchers investigate a rare medical situation in order to learn more about it. Scientists employ case study approaches to

learn about rare phenomena in the same way.

2. Observational Method

Non-experimental research includes observational research. Observational research is described as a sort of research in which the researcher observes the subject's continuing behaviors. Observational research is commonly utilized in the domains of marketing and social science. The researcher observes the actions of the participants under investigation in their natural environment in observational research.

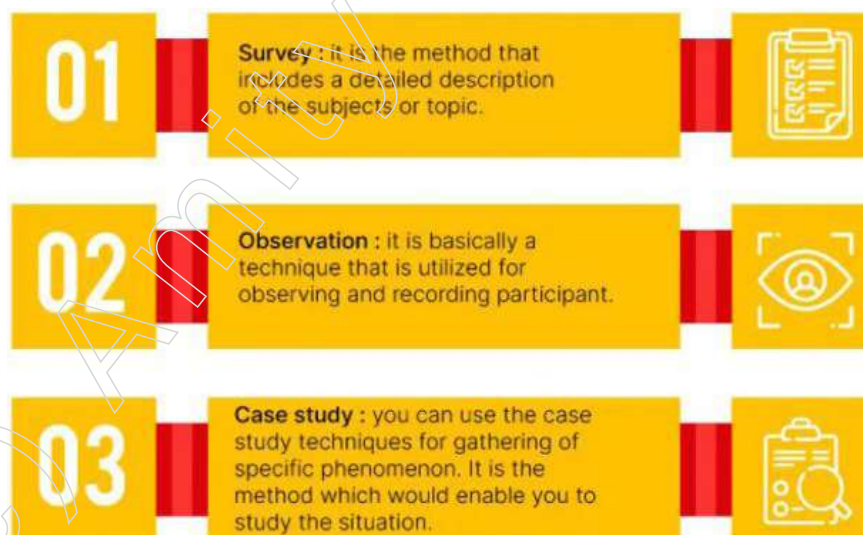
Observational methods differ from experimental research methods in that an artificial environment is constructed for the subjects under study in experimental research methods. Naturalistic observation and participant observation are two forms of observational studies. A naturalistic observation study is one in which subjects are observed in their natural state. People being observed in a research project are aware of the observation in participant observation. They are invited to participate in an observational study. Observational research methods are appropriate for studying the behavior of research subjects. However, this study is unable to provide information about the underlying causes of the subjects' behaviors.

3. Survey Research

Survey research is one of the most popular and straightforward methods for gathering information and data. On paper or in any digital format, a questionnaire is created with questions linked to the study problem. These questionnaires are delivered to people at random in the hopes of achieving an accurate response. In university and commercial research, the survey research approach is commonly utilized. Survey research, also known as primary research, can be used with other research methodologies to produce precise results.

Furthermore, survey research data can be used by other researchers as secondary research data.

Methods of Descriptive research

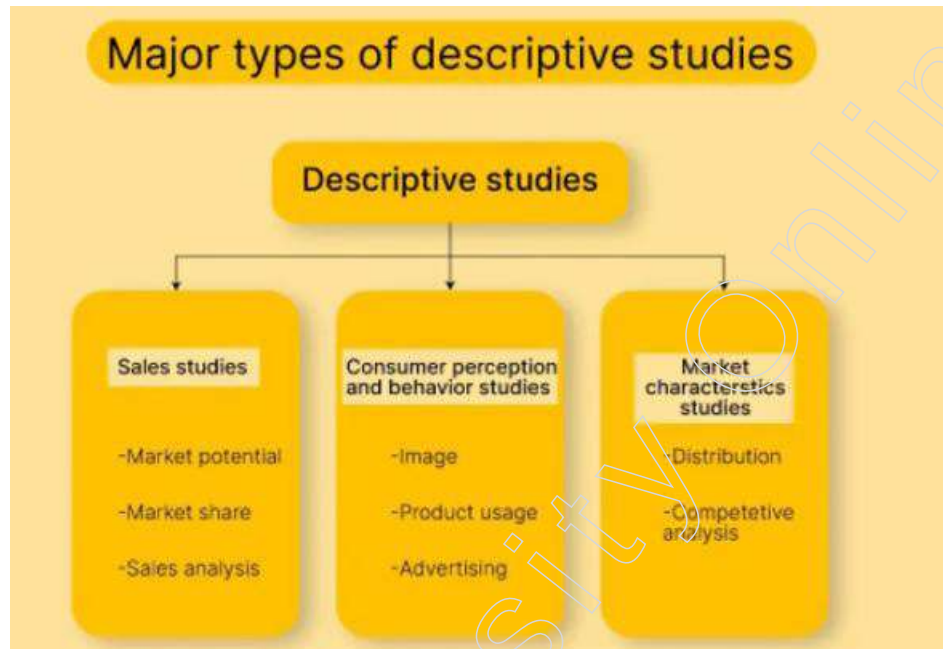


Source: <https://www.voxco.com/wp-content/uploads/2021/03/MicrosoftTeams-image-9-2.jpg>.web

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Examples of Descriptive Research



The following are some examples of descriptive research:

- A specialized food company that is releasing a new line of barbecue rubs wants to know what types of rubs consumers prefer. They do this type of research study utilizing numerous ways, such as observational methods at supermarkets, to understand the desired flavor palette. By polling as well as gathering in-depth demographic data, it is possible to gain insight into market preferences. This can also assist adapt the rubs and spreads to the various meats that that group like. This form of research aids the firm in fine-tuning their business model and boosting marketing in key markets.
- Another application for this research is if a school district wants to assess teachers' views on using technology in the classroom. The researcher can measure what they can help comprehend if a full-fledged implementation can confront an issue by conducting surveys and evaluating their comfort with technology through observational approaches. This also aids in determining whether the pupils are affected in any manner by the change.

Other difficulties and research questions that can lead to descriptive research include the following:

- Market researchers are interested in observing consumer behavior.
- A corporation wishes to assess its employees' morale.
- Students will access online lessons rather than textbooks, according to a school district.
- To see if the company's wellness programs improve employees' overall health.

Check Your Understanding

1. The word descriptive research refers to the research questions, study design, and data analysis that were undertaken on that topic. It's called an _____ approach because none of the variables in the research study are changed in any way.

2. The _____ approach entails an in-depth and extensive examination of the subject, person, or case under investigation.
3. _____ research is described as a sort of research in which the researcher observes the subject's continuing behaviors.
4. _____ is one of the most popular and straightforward methods for gathering information and data. On paper or in any digital format, a questionnaire is created with questions linked to the study problem. These questionnaires are delivered to people at random in the hopes of achieving an accurate response.
5. Descriptive statistics are used to characterize the fundamental characteristics of a study's data. True / False
6. Descriptive statistics are used to display quantitative data in a logical and understandable manner. True / False
7. The properties of a data set are summarized and organized using descriptive statistics. A data set is a collection of responses or observations from a sample or population as a whole. True / False
8. When a researcher wants to find out about different aspects including traits, frequency, trends, correlations, and classifications, they utilize a descriptive study approach. When the researcher has little information on the study problem, they can use it. True / False
9. Descriptive research is a type of quantitative market research or social research that entails conducting surveys using quantitative variables on market research or social research tool. True / False

Summary

- Descriptive statistics are used to characterize the fundamental characteristics of a study's data. They give quick summaries of the sample and the metrics. They are the foundation of practically every quantitative data analysis, along with simple graphical analysis.
- Descriptive analyses are sometimes distinguished from descriptive statistics. When you use descriptive statistics, you're merely describing what the data is or indicates. When using inferential statistics, you're attempting to draw inferences that go beyond the data at hand.
- Descriptive statistics are used to display quantitative data in a logical and understandable manner. We may have a number of measures in a research investigation. Alternatively, we can assess a huge number of people using any method. Descriptive statistics assist us in making sense of enormous amounts of data. Each descriptive statistic condenses a large amount of information into a concise summary.
- The properties of a data set are summarized and organized using descriptive statistics. A data set is a collection of responses or observations from a sample or population as a whole.
- The initial step of statistical analysis in quantitative research is to define properties of the replies, such as the average of one variable (e.g., age) or the relationship between two variables (e.g., age and creativity).

Notes

- The methods used in descriptive research are those that describe the features of the variables being studied. This approach focuses on addressing questions about the “what” rather than the “why” of the study topic. Instead of focusing on the “why,” descriptive research focuses on merely describing the characteristics of the demographics under investigation.
- Descriptive research is a type of quantitative market research or social research that entails conducting surveys using quantitative variables on market research or social research tool.
- The word descriptive research refers to the research questions, study design, and data analysis that were undertaken on that topic. It's called an observational research approach because none of the variables in the research study are changed in any way.
- For a variety of reasons, descriptive research is used. There is no way of knowing whether or not the study you are performing will satisfy the predictions you have made, even after setting survey goals, survey designs, and data collection through surveys.
- Descriptive research can be prepared in three ways.
- The case study approach entails an in-depth and extensive examination of the subject, person, or case under investigation. To conduct the research, a case study uses a formal research process. Using the findings of the case study, a research hypothesis may be developed that can be utilized to broaden the scope of research.
- Non-experimental research includes observational research. Observational research is described as a sort of research in which the researcher observes the subject's continuing behaviors. Observational research is commonly utilized in the domains of marketing and social science. The researcher observes the actions of the participants under investigation in their natural environment in observational research.
- Observational methods differ from experimental research methods in that an artificial environment is constructed for the subjects under study in experimental research methods. Naturalistic observation and participant observation are two forms of observational studies. A naturalistic observation study is one in which subjects are observed in their natural state.
- Survey research is one of the most popular and straightforward methods for gathering information and data. On paper or in any digital format, a questionnaire is created with questions linked to the study problem. These questionnaires are delivered to people at random in the hopes of achieving an accurate response.

Activity

1. Prepare a presentation on the topic “Descriptive Research”.
2. Formulate a case study describing the methods of descriptive research.

Questions & Exercises

1. What is descriptive research?

2. What are the characteristics of descriptive research?
3. What are the methods of descriptive research?
4. What are the applications of descriptive research?
5. State the advantages of descriptive research.
6. Briefly explain the disadvantages of descriptive research.

Glossary

- Descriptive Statistics: The properties of a data set are summarized and organized using descriptive statistics. A data set is a collection of responses or observations from a sample or population as a whole
- Descriptive Research: Descriptive research is a type of quantitative market research or social research that entails conducting surveys using quantitative variables on a market research or social research tool. The methods used in descriptive research are those that describe the features of the variables being studied. This approach focuses on addressing questions about the “what” rather than the “why” of the study topic. Instead of focusing on the “why,” descriptive research focuses on merely describing the characteristics of the demographics under investigation.

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Check your Understanding – Answers

1. observational research
2. case study
3. Observational
4. Survey research
5. True
6. True
7. True
8. True
9. True

Unit - 3.2: Hypothesis Testing

Notes

Objectives:

At the end of this unit, you will be able to:

- Understand the introduction to the hypothesis.
- Learn about the types of hypothesis.
- Comprehend the testing hypothesis and significance levels.
- Learn about type I and type II errors (with example).
- Understand one-tailed and two-tailed test.
- Evaluate confidence intervals.
- Comprehend Bayesian statistics.

Introduction

Data investigation is the process of capturing valuable data by reviewing, cleansing, transforming, and displaying it using one of several methods, such as descriptive analysis, reversion analysis, dispersion analysis, factor analysis (self-governing variable to find the pattern), and time series, which are all based on mathematical and statistical approaches, or decision trees (tree-like structure for choosing various paths), machine learning, and artificial intelligence.

In most cases, the hypothesis is thought to be the most important tool in research. Its major purpose is to generate fresh ideas for experiments and observations. Indeed, many experiments are conducted with the explicit goal of testing theories. Decision-makers frequently encounter scenarios in which they want to test hypotheses based on available data and then make decisions based on the results. In social science, where direct knowledge of population parameter(s) is uncommon, hypothesis testing is a common approach for determining if sample data support a theory sufficiently to allow generalization. As a result of hypothesis testing, we can make probability claims about population parameters (s). The theory may not be proved beyond a reasonable doubt, but it is accepted in practice if it has withstood critical scrutiny. Before we go into how hypotheses are tested using various tests designed for the purpose, it's important to first define what a hypothesis is and the concepts that go along with it so that we can better comprehend the hypothesis testing methodologies.

3.2.1 Introduction to Hypothesis

An assumption (plural hypotheses) is a precise, testable declaration of what the researcher(s) predict will be the outcome of the study. It is stated at the start of the study. This customarily includes proposing a conceivable correlation between two variables: the independent variable (pardon the researcher changes) and the dependent variable (what the research actions). In research, in attendance is a convention that the hypothesis is on paper in two forms, the null hypothesis, in addition the alternate hypothesis (called the experimental hypothesis as soon as the method of examination is an experiment)?

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A hypothesis is a scientific guess assumption of any phenomena occurring over the earth's surface. A hypothesis is a conjectural statement of the relation between two or more variables. We may define a hypothesis as a testable statement of the potential relationship between two or more variables. It is an assumption that serves as a tentative explanation for a researcher; it is a question put to nature to be observed. Hypothesis is, therefore, any statement, proposition or assumption that serves as a tentative explanation of certain facts.

Example: Students attending regular classes perform better than the students not attending classes.

George A Lumberg has defined

"The hypothesis is a tentative generalization, the validity of which remains to be tested. In its most elementary stage, the hypothesis may be a guess, imaginative ideas, which become the basis for action or investigation."

According to Goode and Halt

"The formulation of a deduction however constitutes a hypothesis. If verified it becomes a part of theoretical construction."

According to Webster

"A hypothesis is a proposition, condition or principle which is assumed, perhaps without belief, in order to draw out its logical consequences and by this method to test its accord with facts which are known or may be determined."

When people talk about a hypothesis, they usually mean a simple assumption or a guess that has to be proven or denied. A hypothesis, on the other hand, is a formal inquiry that a researcher wants to answer. As a result, a hypothesis can be described as a proposition or a series of propositions put out as an explanation for the occurrence of a certain group of phenomena, either as a provisional guess to lead some inquiry or as a highly probable explanation in light of known facts. A research hypothesis is frequently a predicted statement that connects an independent variable to a dependent variable and may be tested using scientific procedures.

Consider the following statements: "Students who undergo counseling will demonstrate a bigger improvement in inventiveness than students who do not receive counseling." "Car A is performing as well as automobile B," for example. These are hypotheses that can be verified and tested objectively. As a result, we can conclude that a hypothesis expresses our search criteria and is a notion that can be tested to assess its validity.

Hypothesis characteristics: A hypothesis must have the following characteristics:

- (i) The hypothesis should be explicit and clear. The inferences drawn on the basis of a theory that is not clear and explicit cannot be trusted.
- (ii) The hypothesis must be able to be tested. Many times, research programmes have become bogged down in a swamp of untestable theories. The researcher may conduct some preliminary research in order to make the idea testable. A hypothesis is testable if it can lead to additional deductions, which may then be validated or refuted by observation.

- (iii) If the hypothesis is a relational hypothesis, it should state the relationship between variables.
- (iv) Hypotheses should be specific and limited in scope. A researcher must keep in mind that narrower hypotheses are more testable, and he should develop them.
- (v) A hypothesis should be stated in as basic a manner as feasible so that it is easily understood by all parties involved. However, it is important to realize that the simplicity of a hypothesis has nothing to do with its importance.
- (vi) The hypothesis must be consistent with the majority of known facts, i.e., with a large body of established facts. To put it another way, it should be the one that judges believe is the most plausible.
- (vii) The hypothesis should be testable in a reasonable amount of time. Even the best hypothesis should not be used if it cannot be tested in a reasonable amount of time, because one cannot spend a lifetime collecting evidence to test it.
- (viii) The hypothesis must account for the facts that led to the necessity for an explanation. This means that by combining the hypothesis with other well-known and accepted generalizations, the original problem condition should be deduced. As a result, a hypothesis must truly explain what it claims to explain, and it must be supported by empirical evidence.

Characteristics of Hypothesis

Hypothesis obligation possess the subsequent characteristics:

1. Hypothesis ought to be a conjectural statement.
2. It must be in the format of a clear in addition precise statement.
3. It must be specific and related to the glitches of research.
4. The aforementioned must be logical and must explain the affiliation between the variables.
5. It should be measurable and proficient of being tested.
6. Hypothesis have a duty to be consistent with most identified facts i.e., it must be consistent with a substantial body of well-known facts.
7. Hypothesis must elucidate the facts that gave rise to the need for description. Thus, hypothesis must essentially explain what it claims to clarify (Predictable)

Sources of Hypothesis

Subsequent are the sources of hypothesis:

- The resemblance sandwiched between the phenomena.
- Annotations from past studies, present-day practices and on or after the competitors.
- Scientific philosophies.
- Over-all patterns that effect the thoughtful process of people.

Instances of Hypothesis

Subsequent are the examples of hypothesis grounded on their types:

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- Consumption of sugary drinks every day leads to obesity is an illustration of a simple hypothesis.
- Altogether lilies have the same quantity of petals is an example of a null hypothesis.
- If a person sleeps for 7 hours, he will feel less tired than if he sleeps for less.
- Purposes of Hypothesis
- Subsequent are the functions accomplished by the hypothesis:
- Hypothesis contributes to making a comment and experiments possible.
- It is the starting opinion for the investigation.
- Hypothesis assists in confirming the observations.
- It helps in navigating the investigations in the right directions.

Basic Concepts Concerning Testing of Hypotheses

It is necessary to discuss basic ideas in the context of hypothesis testing.

- (a) Alternative hypothesis and null hypothesis: The terms null hypothesis and alternative hypothesis are frequently used in statistical analysis. The null hypothesis is what we call the idea that if we compare technique A to method B in terms of superiority and proceed on the assumption that both ways are equally good. In contrast, we may believe that method A is superior or method B is inferior, in which case we are proposing an alternate hypothesis.

“The null hypothesis is generally symbolized as H_0 and the alternative hypothesis as H_a . Suppose we want to test the hypothesis that the population means (μ) is equal to the hypothesized mean (μ_{H_0}) = 100. Then we would say that the null hypothesis is that the population mean is equal to the hypothesized mean 100 and symbolically we can express as:”

$$H_0 : \mu = \mu_{H_0} = 100$$

We should conclude that something else is true if our sample data do not support this null hypothesis. The alternative hypothesis is what we come up with after rejecting the null hypothesis. In other terms, the alternative hypothesis refers to the set of alternatives to the null hypothesis. If we accept H_0 , then we are rejecting H_a and if we reject H_0 , then we are accepting H_a . For $H_0: \mu = \mu_{H_0} = 100$, we may consider three possible alternative hypotheses as follows:

Alternative hypothesis	To be read as follows
$H_a : \mu \neq \mu_{H_0}$	(The alternative hypothesis is that the population mean is not equal to 100 i.e., it may be more or less than 100)
$H_a : \mu > \mu_{H_0}$	(The alternative hypothesis is that the population mean is greater than 100)
$H_a : \mu < \mu_{H_0}$	(The alternative hypothesis is that the population mean is less than 100)

Before the sample is drawn, the null hypothesis and alternative hypothesis are chosen (the researcher must avoid the error of hypothesizing from the data he gathers

and then testing the hypotheses with the same data). The following considerations are frequently kept in mind while choosing a null hypothesis:

- (a) The alternative hypothesis is usually the one that is sought to be proven, whereas the null hypothesis is the one that is sought to be disproved. As a result, a null hypothesis is a hypothesis we're seeking to rule out, while an alternative hypothesis is all the other possibilities.
- (b) If rejecting a hypothesis when it is truly true entails a high level of risk, it is considered the null hypothesis since the chance of rejecting it when it is true is (the level of significance), which is chosen to be very small.
- (c) The null hypothesis should always be a particular hypothesis, meaning that it should not declare or approximate a value.

In general, we proceed with hypothesis testing on the basis of the null hypothesis, while keeping the alternative hypothesis in mind. Why is this the case? The explanation is that if the null hypothesis is correct, one can assign probabilities to various possible sample findings, but this is not possible if the alternative hypothesis is used. As a result, the null hypothesis (also known as statistical hypothesis) is frequently used.

- (b) The level of significance: In the context of hypothesis testing, this is a crucial term. It is always a percentage (typically 5%) that should be determined with considerable care, consideration, and logic. If we set the significance level at 5%, H_0 will be rejected if the sampling result (i.e., seen evidence) has a probability of less than 0.05 if H_0 is true. In other words, a 5% level of significance indicates that the researcher is ready to take a 5% chance of rejecting the null hypothesis (H_0) if it turns out to be true. As a result, the significance level is the maximum value of the probability of rejecting H_0 if it is true, and it is normally calculated before the hypothesis is tested.
- (c) "Decision rule or test of hypothesis: Given a hypothesis H_0 and an alternative hypothesis H_a , we make a rule which is known as decision rule according to which we accept H_0 (i.e., reject H_a) or reject H_0 (i.e., accept H_a). For instance, if (H_0 is that a certain lot is good (there are very few defective items in it) against H_a) that the lot is not good (there are too many defective items in it), then we must decide the number of items to be tested and the criterion for accepting or rejecting the hypothesis. We might test 10 items in the lot and plan our decision saying that if there are none or only 1 defective item among the 10, we will accept H_0 otherwise we will reject H_0 (or accept H_a). This sort of basis is known as the decision rule."
- (d) "Type I and Type II errors: in the context of testing of hypotheses, there are basically two types of errors we can make."
- (e) "Two-tailed and One-tailed tests: In the context of hypothesis testing, these two terms are quite important and must be clearly understood."

3.2.2 Types of Hypothesis

The act of putting a hypothesis or a supposition to the test in regard to a statistical parameter is known as hypothesis testing. Hypothesis testing is used by analysts to determine whether or not a hypothesis is reasonable.

Hypothesis testing is an important stage in data science and statistics since it involves the verification of an assumption that could aid in the development of a

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statistical parameter. A researcher might, for example, form a hypothesis based on the assumption that the average of all odd integers is an even number.

The researcher will have to evaluate the hypothesis utilizing hypothesis testing procedures in order to determine its plausibility. In contrast to a hypothesis that is 'supposed' to be true based on little or no evidence, hypothesis testing necessitates the presence of convincing evidence in order to prove that a statistical hypothesis is correct.

Perhaps this is where statistics come into play. This procedure involves a lot of elements. However, before we can grasp the process of hypothesis testing in research technique, we must first understand the many sorts of hypotheses that are involved.

There are two main kinds of hypotheses:

1. Worldwide Hypothesis – which may be scientifically showed and universally accepted- in physical sciences- Newton's Law etc.
2. Inadequate Hypothesis – for limited area in social sciences – demonstrable, based on deductions.

According to Kerlinger, there are two kinds of hypothesis.

1. Simple Hypothesis – expressions relationship amongst two or more variables.
2. Change Hypothesis – shows change or comparison of two variables e.g., non-tribal's are more developed than the tribal's.

Alternative Hypothesis

The alternative hypothesis proposes that there is a link between the two variables under consideration (one variable has a result on the other).

When the independent variable is used, an experimental hypothesis predicts what change(s) will occur in the dependent constant.

It states that the results are important in terms of confirming the theory of nothingness explored and that they are not attributable to chance.

The alternative hypothesis takes the responsibility that the variances are not due to fluctuations of sampling, they are actual. H_0 : mean = 120 (Null hypothesis) H_1 : mean $\neq$ 120 (Alternative hypothesis) e.g., Mean is greater or lesser than 120.

Two more forms of hypothesis:

1. Directional Hypothesis – one-tailed hypothesis. It indicates the direction of a single variable. For example, in comparison to rural areas, urban areas are more prone to sickness.
2. Non-directional Hypothesis – two-tailed hypothesis. This hypothesis written has a fixed direction. In this hypothesis the researcher does not make any declaration about the direction of any variable e.g. There is a change between the growth level of rural and urban areas.

Significance:

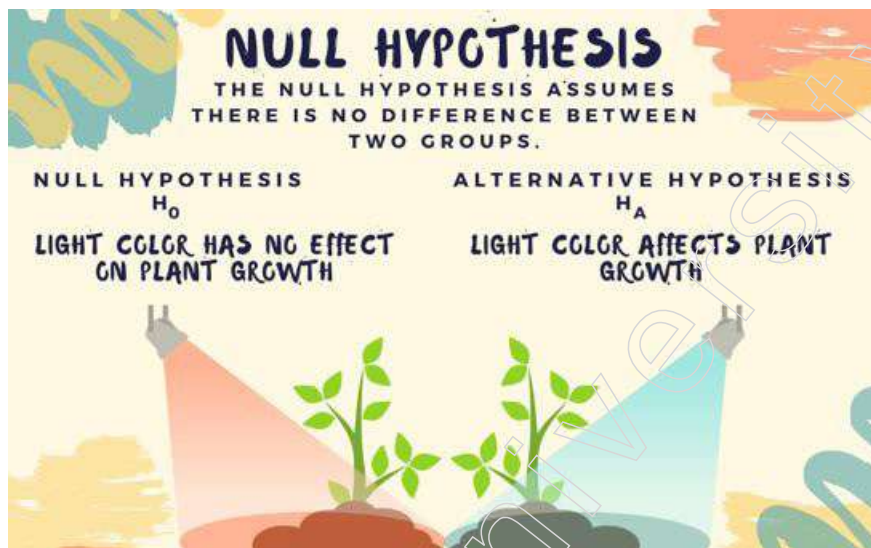
1. Hypothesis stretches point to enquiry.

2. It supports in deciding the direction to continue with
3. It benefits in selecting pertinent variables
4. It helps in portrayal specific ends

Null Hypothesis

A null hypothesis is a hypothesis that states that there is no association between the two variables studied (one variable does not affect the additional). There will be no vicissitudes in the dependent variable owing to the manipulation of the independent mutable.

It molds results that are attributable to chance and are unimportant in terms of the presence of the idea under investigation. The symbol is denoted by " H_0 ".



Non Directional Hypothesis

A non-directional (two-tailed) hypothesis envisages that the independent variant will have an outcome on the dependent variable, nonetheless the direction of the result is not specified. It just states that there will be a transformation.

E.g., there will be a variance in how numerous numbers are properly recalled by children and adults.

Directional Hypothesis

A directional (one-tailed) hypothesis predicts the direction of the dependent variable's effect on the independent variable. It envisages in which direction the variation will yield place. (I.e. greater, smaller, less, more)

E.g., adults will appropriately recall more lyrics than children.

Statistical Hypothesis

A statistical hypothesis is one that can be tested statistically to see if it is plausible.

One can determine the plausibility of a statistical hypothesis and determine if it is true or not by employing data sampling and statistical knowledge.

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Ladders in Preparation of Hypothesis

A hypothesis is a hesitant assumption drawn on or after practical knowledge or theory. A hypothesis is used as a guide in the search for further facts or ideas that a researcher is unaware of. However, the preparation of the hypothesis is one of the most difficult steps in the complete scientific research process.

For that reason, in this regard, we propose to point out the basic steps in the construction of a hypothesis. We are positive that these guidelines will be supportive in your research study.

1. Define Variables

First and foremost, you must specify your variables in order to formulate a hypothesis. What do you want to test? Determination test that plunders increase appraisal attainment. Or rewards weak reading achievements? What are your goals, they need to be clearly defined, calculable, and measurable. This will provide you by means of a clear idea of what to follow to attain results.

2. Training In-Depth the Variables

If we assume that your variables are Rewards and Achievements, then you'll want to investigate how rewards affect reading achievement. An in-depth study, rigorous queries, and data of recompenses increase reading achievements will make you able to settle your hypothesis. Specify reliant on and independent variables.

3. Stipulate the Nature of Relationship

Then, identify what relationship there happens to be between the variables. What variable effects the other? What is the dependent variable and what is the independent variable? In what way does Rewards impact achievements? If rewards acts as a key role in reading successes, then rewards is the independent variable.

4. Recognize Study Population

In research, the populace refers to the entire group of people who will be studied. If you want to test how rewards upsurge reading achievements in the United Kingdom, you would then want to study the entire population of the United Kingdom. Because the majority of the public avoids participating in reading achievements. As a result, the researcher must classify the participants in the study.

5. Brand Sure Variables be there as testable

Variables in your hypothesis need to be testable. Otherwise, the hypothesis would be worthless. For that reason, your research study needs to accept or reject a variable. Therefore, all the variables need to be tested. Only probable or rejected variables can be tested. Moreover, the sole goal of a research hypothesis is to test variables in the long run.

Bases rummage-sale in hypothesis formulation:

1. Individual experience- of area, process, items
2. Preceding researches

3. Investigation abstracts, journals, records, thesis, dissertations etc.
4. Theories/ models
5. Analogues/ Equivalents

Difficulties in the Formulation of hypothesis:

1. Absence of clean theoretical background
2. Lack of rational background
3. Nonexistence of knowledge of scientific method
4. Practical theory
5. Equivalences
6. Personal practice

Performing Hypothesis Testing

Let's move on to a better understanding of the process now that we've learned about the different sorts of hypotheses and their roles in hypothesis testing.

To begin the practice of hypothesis testing, a researcher must first establish two hypotheses: an alternative hypothesis and a null hypothesis.

To establish these two hypotheses, researchers must examine data samples, identify a reasonable pattern among the samples, and formulate a statistical hypothesis to test.

To begin hypothesis testing, a random sample population can be recruited. Only one of the two hypotheses, alternative and null, can be proven to be correct. It's possible that the presence of both hypotheses is essential for the process to succeed.

Either of the hypotheses will be rejected at the conclusion of the hypothesis testing procedure, while the other will be supported. Even if one of the two hypotheses is proven to be correct, no hypothesis can ever be completely validated.

As a result, only statistical samples and confirmed data can be used to support a theory. A step-by-step tutorial to hypothesis testing can be found here.

1. Establish the hypotheses

First and foremost, two hypotheses must be established: alternative and null, which will serve as the foundation for hypothesis testing.

These hypotheses kick off the testing procedure, which entails the researcher working with data samples to support either the alternative hypothesis or the null hypothesis.

2. Generate a testing plan

It is now time to create a testing plan based on the hypotheses that have been developed. A testing strategy, also known as an analysis plan, entails gathering data samples, deciding which statistic to use, and calculating the sample size.

All of these considerations are critical when conducting hypothesis testing.

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3. Analyze data samples

It's time to move on to the analysis phase once you've completed your testing strategy. Configuring statistical values of samples, pulling them together, and establishing a pattern from these samples are all part of data sample analysis.

A researcher must determine a set of items when evaluating data samples. -

- The degree of significance in hypothesis testing shows whether a statistical result may be considered significant if the null hypothesis is correct.
- A form of sampling-distribution and a test statistic are used in the testing procedure, which leads to hypothesis testing. A variety of testing procedures are available to aid in the examination of data samples.
- A test statistic is a numerical summary of a data collection that can be used to test hypotheses.
- P-value - The P-value interpretation is the likelihood of finding a sample statistic that is as extreme as the test statistic, showing the null hypothesis' plausibility.

4. Infer the results

The inference of outcomes based on the study of data samples determines whether the alternative hypothesis is true or not. The null hypothesis is rejected when the P-value is smaller than the significance level, and the alternative hypothesis is found to be reasonable.

Methods of Hypothesis Testing

We'll look at several ways of hypothesis testing now that we've looked at different parts of hypothesis testing. Overall, there are two sorts of hypothesis testing techniques. The following is a list of them:

Frequentist Hypothesis Testing

The frequentist hypothesis, also known as the classical approach to hypothesis testing, is a type of hypothesis testing that focuses on making assumptions based on existing evidence. The ostensible truths and assumptions are based on current data, and two hypotheses are proposed. Null Hypothesis Significance Testing is a common subtype of the frequentist approach (NHST). Since its debut in the mid-1950s, the NHST approach (including the null and alternative hypothesis) has been one of the most popular methods of hypothesis testing in the field of statistics.

Bayesian Hypothesis Testing

The Bayesian Hypothesis Testing, a far more unorthodox and modern form of hypothesis testing, claims to test a particular hypothesis in accordance with past data samples, known as prior probability, and current data that lead to the plausibility of a hypothesis. The result produced represents the hypothesis's posterior probability. The researcher uses 'prior probability and posterior probability' to conduct hypothesis testing on the spot in this method.

The Bayesian technique determines whether a hypothesis is true or untrue based on this prior probability. The Bayes factor, which is a key component of this approach,

represents the likelihood ratio between the null and alternative hypotheses. The Bayes factor is a measure of the likelihood of either of the two hypotheses that are used to test hypothesis.

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3.2.3 Testing Hypothesis and Significance Levels

Procedure for Hypothesis Testing

To test a hypothesis, the researcher must determine whether the hypothesis appears to be valid based on the evidence gathered. The basic question in hypothesis testing is whether the null hypothesis should be accepted or rejected. The processes we take to choose between rejecting and accepting a null hypothesis are referred to as the procedure for hypothesis testing.

The following are the steps involved in hypothesis testing:

- (i) Making a formal declaration: Making a formal declaration of the null hypothesis (H_0) as well as the alternative hypothesis (H_a) is the first step. This means that, given the nature of the study challenge, hypotheses should be articulated explicitly. For example, if Mr. Mohan of the Civil Engineering Department wants to test an ancient bridge's load bearing capacity, which must be greater than 10 tonnes, he can formulate his hypothesis as follows:

Null hypothesis $H_0 : \mu = 10 \text{ tons}$

Alternative Hypothesis $H_a : \mu > 10 \text{ tons}$

Let me give you another example. An aptitude test taken at the national level yielded an average score of 80. To assess a state's educational system, the average score of 100 randomly selected students from the state was 75. The state is curious if there is a major gap between the local and national results. In this case, the assumptions could be phrased as follows:

Null hypothesis $H_0 : \mu = 80$

Alternative Hypothesis $H_a : \mu \neq 80$

The formation of hypotheses is a vital stage that must be carried out with attention, taking into account the purpose and nature of the situation at hand. It also determines whether a one-tailed or two-tailed test should be used. We use a one-tailed test if H_a is of the type greater than (or lesser than), but we use a two-tailed test if H_a is of the type "whether greater or smaller."

- (ii) Selecting a significance level: The hypotheses are tested at a predetermined level of significance, which should be mentioned as well. In practise, either a 5% or a 1% threshold is used for this reason. The magnitude of the difference between sample means, the size of the samples, the variability of measurements within samples, and whether the hypothesis is directional or non-directional (A directional hypothesis predicts the direction of the difference between, say, means) are all factors that influence the level of significance. In summary, the level of significance must be appropriate for the goal and type of the investigation.
- (iii) Deciding the distribution to use: Once the level of significance has been determined, the following stage in hypothesis testing is to select the proper

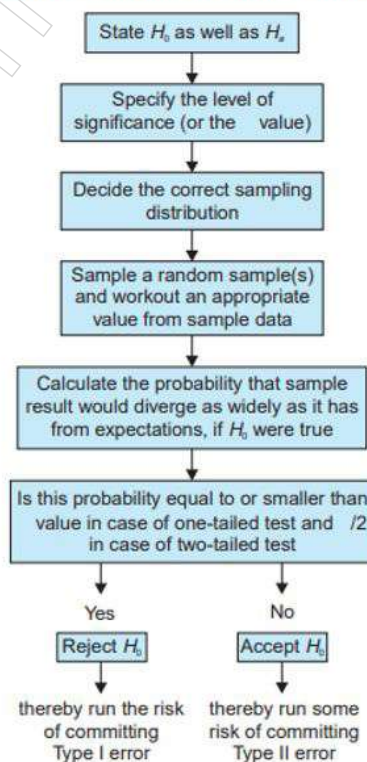
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- sampling distribution. The normal distribution and the t-distribution are the most common options. The rules for choosing the correct distribution are similar to those we discussed earlier in relation to estimate.
- (iv) Selecting a random sample and computing an appropriate value: The final step is to choose a random sample(s) and compute an appropriate value for the test statistic using the relevant distribution from the sample data. To put it another way, draw a sample to provide empirical data.
 - (v) Calculation of the probability: If the null hypothesis were true, one must compute the probability that the sample result would diverge as significantly from expectations as it has.
 - (vi) Comparing the probability: Another step is to compare the probability that has been calculated with the value for the significance level that has been specified. If the estimated probability is equal to or less than the value in a one-tailed test (and $/2$ in a two-tailed test), reject the null hypothesis (accept the alternative hypothesis), but accept the null hypothesis if the calculated probability is larger. If we reject H_0 , we run the danger of committing a Type I error (at most the level of significance), but if we accept H_0 , we run the risk of committing a Type II error (the amount of which cannot be determined as long as H_0 is ambiguous rather than precise).

Flow Diagram for Hypothesis Testing

For better understanding, the above-mentioned general approach for hypothesis testing can also be depicted in the form of a flowchart, as shown in Fig. below.

FLOW DIAGRAM FOR HYPOTHESIS TESTING



Source: Based on the flow diagram in William A. Chance's Statistical Methods for Decision Making, Richard D. Irwin INC., Illinois, 1969, p.48.

Tests of Hypotheses

As previously stated, hypothesis testing is used to assess the validity of an assumption (officially known as the null hypothesis) in order to choose between two competing hypotheses concerning the value of a population parameter. Hypothesis testing is a technique for determining whether a population hypothesis is likely to be true or false based on a sample of facts. For the goal of testing hypotheses, statisticians have devised numerous tests of hypotheses (also known as tests of significance), which can be classed as: (a) parametric tests or standard tests of hypotheses; and (b) non-parametric tests or distribution-free tests of hypotheses.

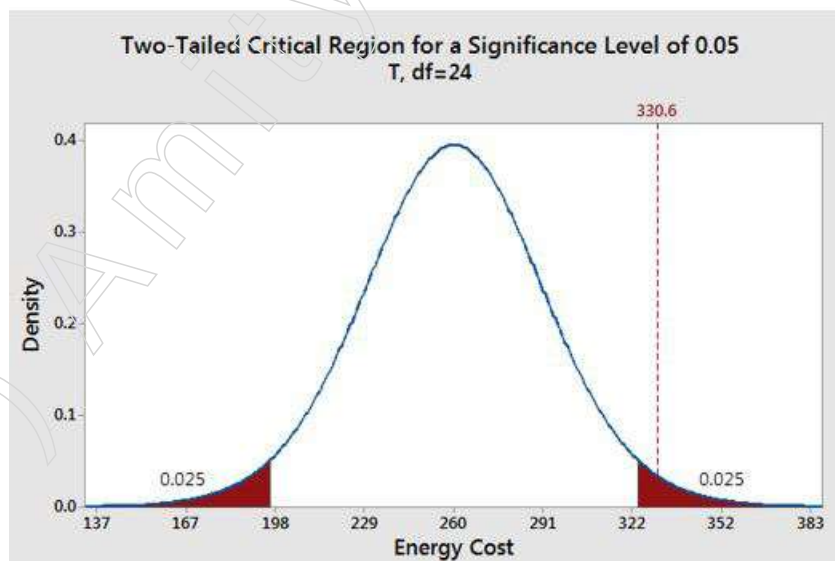
The parent population from which we collect samples is frequently assumed to have specific traits using parametric testing. Before parametric tests can be performed, assumptions such as observations come from a normal population, sample size is high, and assumptions about population parameters like mean, variance, and so on, must hold true. However, there are times when the researcher is unable or unwilling to make such assumptions. In such cases, we employ non-parametric tests, which are statistical procedures for evaluating hypotheses that do not rely on any assumptions about the parameters of the parent population. Furthermore, most non-parametric tests assume just nominal or ordinal data, whereas parametric tests require at least an interval scale of measurement.

As a result, in order to attain the same amount of Type I and Type II errors, non-parametric tests require more observations than parametric tests.

What is the significance level (alpha)?

The chance of rejecting the null hypothesis when it is true is known as the significance level, often known as alpha or α . A significance level of 0.05, for example, represents a 5% chance of finding that a difference exists when there is none.

Because of their technical character, these definitions can be difficult to comprehend. The topics are much easier to understand with a picture! The significance level defines how far the line on the graph will be drawn from the null hypothesis value. We need to shade the 5% of the distribution that is furthest away from the null hypothesis to graph a significance level of 0.05.



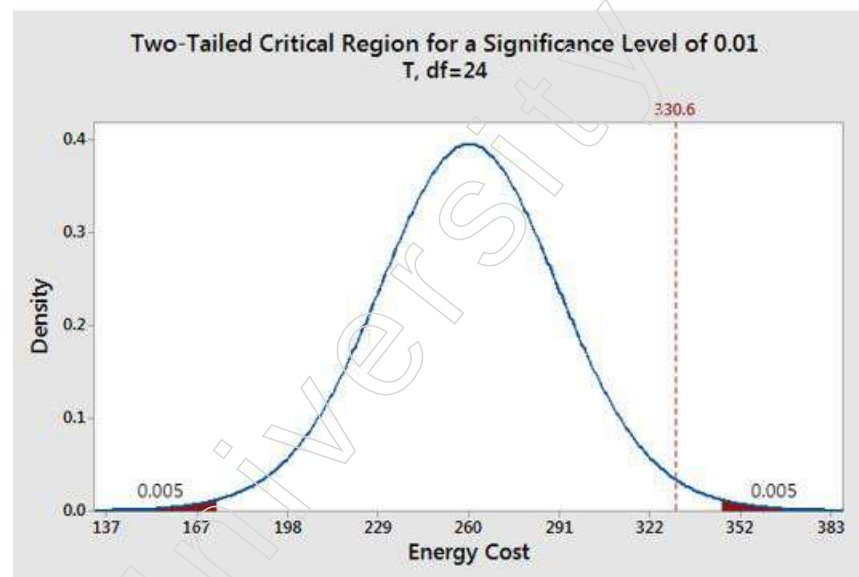
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The two shaded areas in the graph above are equidistant from the null hypothesis value, each with a probability of 0.025, for a total of 0.05. These shaded areas are referred to as the critical zone for a two-tailed test in statistics. If the population mean is 260, we should have a sample mean that is 5 percent of the time in the critical region. The crucial area specifies the distance between our sample statistic and the null hypothesis value before we can declare it is exceptional enough to reject the null hypothesis.

Our sample mean (330.6) is statistically significant at the 0.05 level since it falls within the critical zone.

We may also use the other usual significance level of 0.01 to evaluate if it's statistically significant.



Each of the two shaded areas has a probability of 0.005, for a total probability of 0.01. Our sample mean does not fall within the critical region this time, hence the null hypothesis is not rejected. This contrast demonstrates why you should determine your significance level before starting your research. It spares you from having to choose a significance level because it provides you with significant results quickly!

We were able to conclude that our results are statistically significant at the 0.05 level without utilising a P value thanks to the graph. When using statistical software's numeric output, though, you'll need to compare the P value to your significance level to make this judgment.

3.2.4 Type I & Type II Error (with Example)

A Type I mistake in statistics is a false positive result, whereas a Type II error is a false negative result.

In hypothesis testing, the chances of making these errors are inherent because making a statistical conclusion always contains uncertainty.

The significance level, or alpha (α), determines the likelihood of making a Type I error, whereas beta (β) determines the likelihood of making a Type II error. These dangers can be reduced by carefully planning your study design.

Type I vs. Type II Error as an Example

Based on your modest symptoms, you decide to get tested for COVID-19. There are two types of errors that could occur:

- Type I error (false positive): the test result indicates that you have coronavirus when you don't.
- Type II error (false negative): the test result indicates that you are free of coronavirus, while you are in fact infected.

Error in Statistical Decision-Making

You can use hypothesis testing to see whether your data supports or contradicts your study predictions.

The null hypothesis assumes that there is no difference between groups or no association between variables in the population while doing hypothesis testing. It's always used in conjunction with an alternative hypothesis, which is your study prediction of a true difference between groups or a true link between variables.

Example: Null and alternative hypotheses

You want to see if a new medicine can help with the symptoms of an autoimmune condition.

In this instance:

- H_0 is the null hypothesis, which states that the new medicine has no effect on disease symptoms.
- The alternative hypothesis (H_1) is that the medicine is beneficial at reducing the disease's symptoms.

Then, using your data and the findings of a statistical test, you decide whether the null hypothesis can be rejected. Because these decisions are dependent on probabilities, there is always the possibility of reaching an incorrect conclusion.

- If your results are statistically significant, it signifies that if the null hypothesis is true, they are extremely improbable to occur. You would reject your null hypothesis in this scenario. However, it's possible that this is a Type I error.
- If the null hypothesis is correct, your findings have a high chance of occurring even if they do not demonstrate statistical significance. As a result, your null hypothesis is not rejected. However, this might be a Type II error.

Example: Type I and Type II errors

When you receive false positive results, you conclude that the pharmacological intervention improved symptoms when it didn't. This is known as a Type I error. Other random causes or measurement errors could have caused these improvements.

A Type II error occurs when false negative results are obtained, leading to the conclusion that the medication intervention did not relieve symptoms when, in fact, it did. Your research may have overlooked vital markers of progress or attributed any gains to other variables.

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Type I and Type II Error

Null hypothesis is...	True	False
Rejected	Type I error False positive Probability = α	Correct decision True positive Probability = $1 - \beta$
Not rejected	Correct decision True negative Probability = $1 - \alpha$	Type II error False negative Probability = β

Type I Error

A Type I error occurs when the null hypothesis is rejected when it is actually true. It entails claiming that results are statistically significant when they were obtained only by chance or due to unrelated variables.

The significance level (alpha or α) you choose determines the risk of making this mistake. That's a number you select at the start of your research to determine the statistical likelihood of getting your results (p value).

In most cases, the significance level is chosen at 0.05 or 5%. If the null hypothesis is correct, your results have a 5% or less chance of occurring.

If your test's p value is less than the significance level, your findings are statistically significant and support the alternative hypothesis. Your results are statistically non-significant if your p value is greater than the significance level.

Example: Statistical significance and Type I error

You compare the symptoms of patients who received the novel pharmacological intervention to a control treatment in your clinical study. You get a p value of 0.035 using a t test. Because this p value is less than your alpha of 0.05, you believe your findings are statistically significant and reject the null hypothesis.

However, if the null hypothesis is correct, the p value indicates that your results have a 3.5 percent chance of occurring. As a result, there is still a chance of making a Type I mistake.

You can simply lower the significance level to lessen the Type I error likelihood.

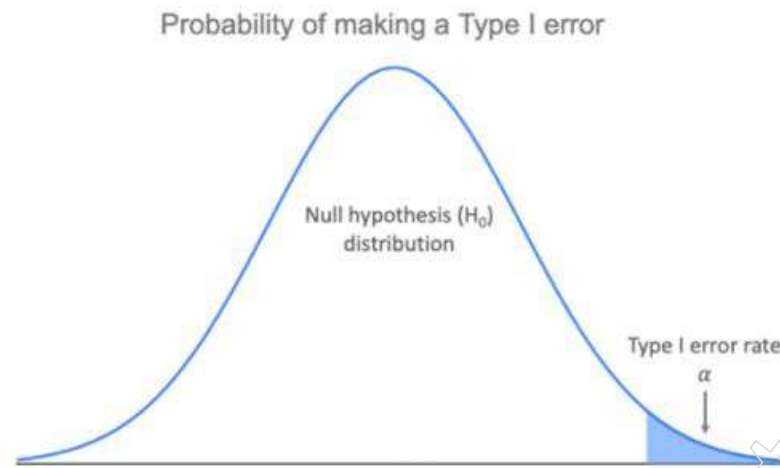
Type I error rate

If the study were repeated with new samples and the null hypothesis were true in the population, the null hypothesis distribution curve below illustrates the chances of getting all conceivable findings.

The shaded region towards the end shows alpha. In statistics, it's also known as a key zone.

If your data are statistically significant and fall within the critical region of this curve,

the null hypothesis is rejected. However, because the null hypothesis is true in this situation, this is a false positive result.



Type II Error

A Type II error occurs when the null hypothesis is not rejected when it is wrong. This is not the same as “accepting” the null hypothesis, because hypothesis testing can only tell you if the null hypothesis should be rejected.

A Type II error, on the other hand, is when you don’t realise there was an effect when there was. In actuality, your study may not have had sufficient statistical power to identify a significant effect.

When there is a true effect, the power of a test determines how well it can identify it. A power level of 80% or above is generally regarded as satisfactory.

A study’s statistical power is inversely related to the risk of a Type II mistake. The lesser the probability of committing a Type II error, the better the statistical power.

Example: Statistical power and Type II error

When planning your clinical trial, you do a power analysis and decide that your sample size will give you an 80% chance of detecting an effect size of 20% or larger. A 20% effect size indicates that the medication intervention decreases symptoms 20% more than the control treatment.

A Type II effect, on the other hand, may occur if the influence is smaller than this size. Due to insufficient statistical power, a smaller effect size is unlikely to be observed in your investigation.

Statistical power is determined by:

- The size of the effect: Larger effects are easier to notice.
- Measurement error: Errors in recorded data, both systematic and random, degrade power.
- Larger samples reduce sampling error while also increasing power.
- Significance level: As the significance level rises, so does the power.

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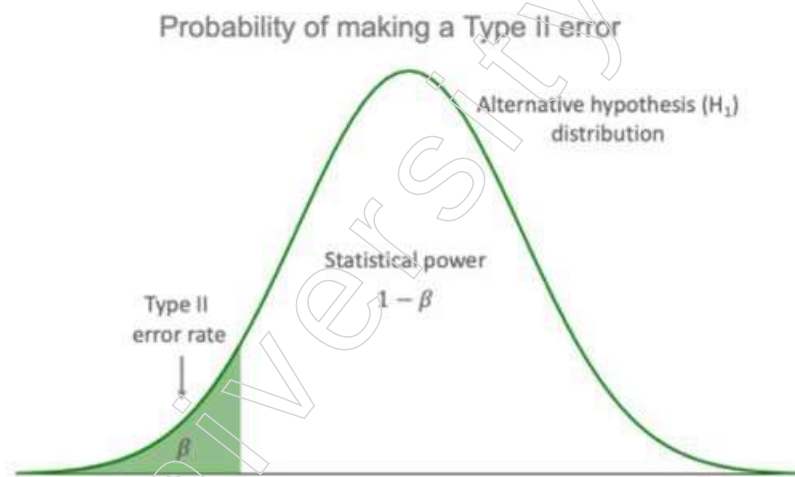
- Increase the sample size or the significance level to (indirectly) lower the likelihood of a Type II mistake.

Type II error Rate

If the study were repeated with new samples and the alternative hypothesis were true in the population, the alternative hypothesis distribution curve below illustrates the chances of getting all conceivable outcomes.

The shaded area on the left side represents the Type II error rate, which equals beta. The statistical power, which is $1 - \beta$, is represented by the remaining area under the curve.

Increasing your test's statistical power reduces your chances of producing a Type II error.



Trade-off between Type I and Type II errors

Type I and Type II error rates have an impact on each other. This is because statistical power is inversely connected to the Type II error rate and is affected by the significance level (the Type I error rate).

This implies that Type I and Type II mistakes have a significant tradeoff:

- Setting a lower significance level reduces the probability of Type I errors while increasing the chance of Type II errors.
- Increasing the test's power reduces the chance of Type II errors while increasing the risk of Type I errors.

The graph below illustrates this trade-off. There are two curves shown:

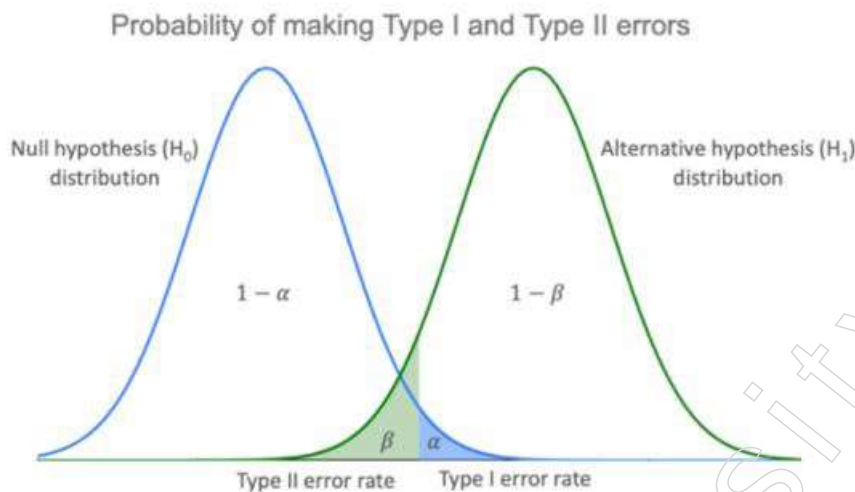
If the null hypothesis is true, the null hypothesis distribution gives all possible outcomes. The right conclusion for any point on this distribution is that the null hypothesis is not rejected.

If the alternative hypothesis is correct, the alternative hypothesis distribution indicates all conceivable outcomes. For each point on this distribution, the right conclusion is to reject the null hypothesis.

Where these two distributions intersect, Type I and Type II errors arise. The Type

I error rate is shown by the blue shaded region, whereas the Type II error rate is represented by the green shaded area.

You can indirectly alter the amount of the Type II mistake rate by adjusting the Type I error rate.



It's critical to achieve a balance between the dangers of Type I and Type II mistakes. Increasing beta always comes at the expense of lowering alpha, and vice versa.

Is a Type I or Type II error worse?

A Type I error is usually worse for statisticians. In practice, though, depending on your research environment, either form of inaccuracy could be worse.

A Type I error occurs when the primary statistical assumption of a null hypothesis is violated. This could result in new rules, practices, or treatments that are ineffective or wasteful.

Consequences of a Type I Error, for example

Despite the risks of severe side effects and little study on the outcomes, over a million people have been supplied the prescription based on the mistaken conclusion that the new medicinal intervention is successful. Because of the repercussions of this Type I mistake, other therapeutic choices have been ruled out in favor of this intervention.

A Type II error, on the other hand, occurs when a null hypothesis is not rejected. It may just result in missed possibilities for innovation, but this can have significant practical implications.

Example: Consequences of a Type II error

When a Type II error occurs, the medication intervention is deemed useless, even if it can actually relieve disease symptoms. This suggests that a medicine with significant clinical value isn't reaching a huge number of individuals who could benefit from it.

Notes

3.2.5 One-tailed and Two-tailed Test

These two words are extremely significant in the context of hypothesis testing and must be understood completely. If, for example, the sample mean is significantly higher or lower than the population mean, a two-tailed test rejects the null hypothesis.

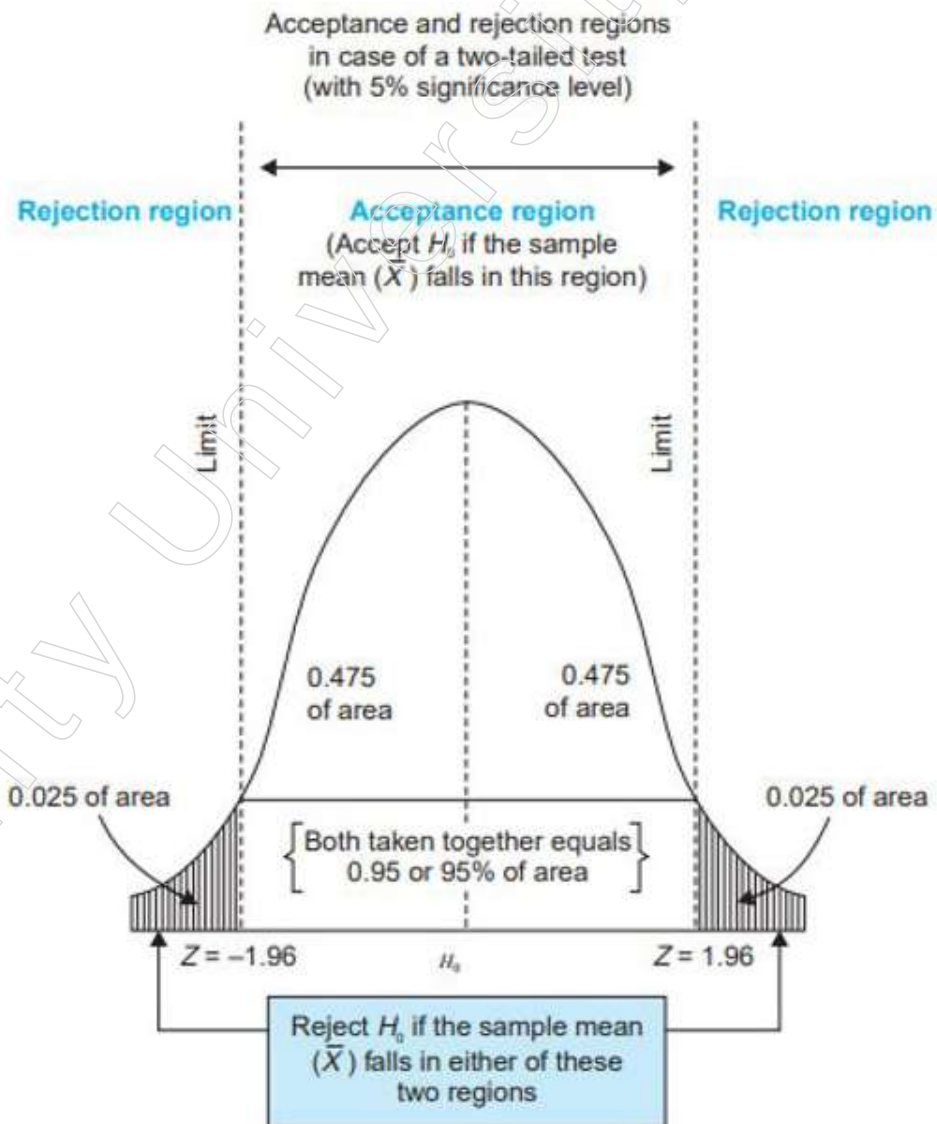
When the null hypothesis is a defined value and the alternative hypothesis is a value not equal to the null hypothesis's specified value, this test is appropriate. When we have the following equation, the two-tailed test is appropriate symbolically.

$$H_0: \mu = \mu_{H_0} \text{ and } H_a: \mu \neq \mu_{H_0}$$

which may mean,

$$\mu > \mu_{H_0} \text{ Or } \mu < \mu_{H_0}$$

As a result, in a two-tailed test, there are two rejection regions*, one on each tail of the curve, as shown below:



Arithmetically we can state:

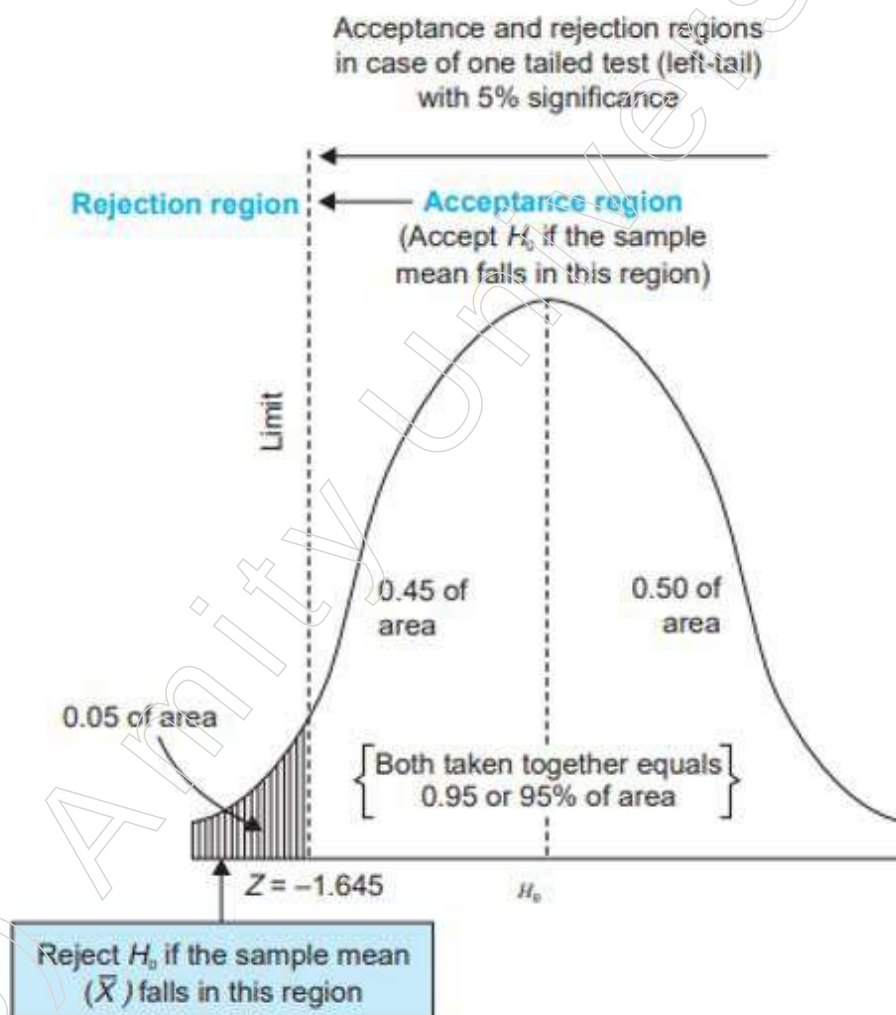
Acceptance Region $A: |Z| \leq 1.96$

Rejection Region $R: |Z| > 1.96$

The probability of the rejection area will be 0.05 (equally split on both tails of the curve as 0.025) and that of the acceptance region will be 0.95, as illustrated in the above curve, if the significance level is 5% and the two-tailed test is used. If we use $\mu = 100$ and our sample mean deviates significantly from 100 in any direction, we reject the null hypothesis; but, if our sample mean does not deviate significantly from, we accept the null hypothesis.

However, there are times when only a single-tailed test is suitable. When determining whether the population mean is lower than or higher than a hypothesised value, a one-tailed test would be utilized.

“For instance, if our $H_0: \mu = \mu_{H_0}$ and $H_a: \mu < \mu_{H_0}$, then we are interested in what is known as left-tailed test (wherein there is one rejection region only on the left tail) which can be illustrated as below”:



Arithmetically we can state:

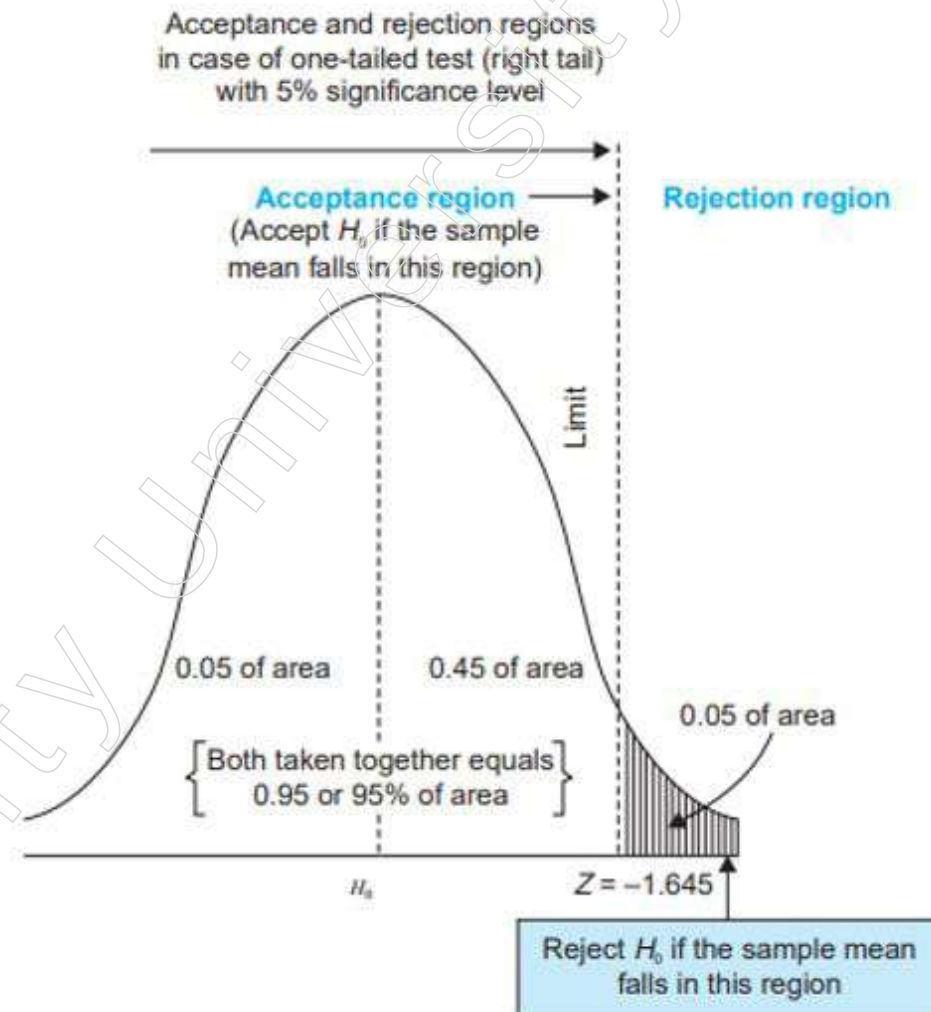
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Acceptance Region $A: Z > -1.645$

Rejection Region $R: Z \leq -1.645$

We will reject H_0 if our $\mu = 100$ and our sample mean deviates significantly from 100 in the lower direction, but we will accept H_0 at a particular level of significance if our $\mu = 100$ and our sample mean deviates significantly from 100 in the lower direction. If the significance threshold is kept at 5% in this example, the rejection zone will be equal to 0.05 of the area in the left tail, as seen in the graph above.

"In case our $H_0: \mu = \mu_{H_0}$ and $H_a: \mu > \mu_{H_0}$, we are then interested in what is known as one tailed test (right tail) and the rejection region will be on the right tail of the curve as shown below":



Arithmetically we can state:

Acceptance Region $A: Z \leq 1.645$

Rejection Region $A: Z > 1.645$

If our μ is equal to 100 and our sample mean is considerably different from 100 in the upward direction, we will reject H_0 ; otherwise, we will accept it. If the significance threshold in the aforementioned scenario is kept at 5%, the rejection zone will be equal to 0.05 of the area in the right-tail, as illustrated in the above curve.

It is important to recall that adopting H_0 on the basis of sample data does not imply that H_0 is correct. We just mean that there isn't any statistical evidence to refute it; we're not claiming that H_0 is true (though we act as if it is).

3.2.6 Confidence Intervals

A confidence interval is the margin of error that a researcher would encounter if they could ask a certain research question of every member of the target population and receive the same response that the sample members gave in the survey. If the researcher used a confidence interval between 4 and 60 percent of the survey participants said "Would recommend to friends," he could be confident that between 56 percent and 64 percent of the whole target group would reply "Would recommend to friends" when asked the same question.

In this example, the confidence interval is ± 4 .

3.2.7 Bayesian Statistics

For a long time, Bayesian statistics has been thought of as a branch of statistics; nonetheless, its contribution to the development of statistical inference is far more significant. Its conceptual foundation can be traced back to a very early and subjective interpretation of probability and its use in everyday human actions. It is linked to Carneades' (214–129/8 BC) concept of 'pithanon,' which means persuasion or persuasion, which permits humans to make decisions and act without necessarily having complete awareness of the situation and situations they are confronted with. The reader is directed to Bett for a thorough discussion of Carneades' philosophical perspective (1989).

Statistical inference, according to the Bayesian approach to statistics, involves more than just gathering, analysing, and interpreting data with endogenous variability, as the frequentist view of statistics suggests. It is about providing a realistic means of understanding uncertain events through the concept of probability, to paraphrase de Finetti (1974, 1975). When new information becomes available, it also gives a consistent approach for inductive mathematical reasoning. In a nutshell, statistics is the study of risk.

According to Savage (1954) and Lindley (1965, 2000), the only self-consistent (coherent) strategy to deal with uncertainty is to use the concept of probability. Uncertainty refers to data, parameters, and models while dealing with statistical problems. After expressing uncertainty about these, the next step is to consider the information provided by the data and update the uncertainty using probability calculus methods. The Bayes' theorem provides the essential mathematical instrument for doing so. As a result, the Bayesian approach offers a unified and consistent inference procedure.

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The rules of probability calculus, namely (a) the addition rule (finite or countable), (b) the product rule, and (c) the convexity rule (that the probability lies in the convex unit interval taking the value 1 for the certain event), must be demonstrated on the basis of axioms about logical and subjective human behaviour in an uncertain environment. The above three probability laws are the outcome of several axiomatic systems. See Savage (1954), DeGroot (1970), and Fishburn (1973) for a treatment of such axiomatic systems (1986).

The Bayesian approach, while straightforward in theory, can be difficult to apply in practise, especially when dealing with complex and highly organised statistical models. There are two key tasks to complete: (a) modelling uncertainty about the probabilistic behaviour of the statistical data at hand and the parameters involved, including models, and (b) using extremely efficient computer methods to estimate the parameters and confirm the models. Much of the statistical work in the first task is based on the concepts of coherence and exchangeability (both due to de Finetti (1937)), and it can be substantially simplified by employing so-called objective priors (diffuse, flat, reference or non informative priors). In terms of the second, the introduction of current and very efficient computer tools like Markov-chain Monte Carlo (MCMC) has made this a relatively simple approach. During the previous two decades, their versatility has had a huge impact on the Bayesian method and its growth across numerous disciplines.

Limitations of the Tests of Hypotheses

There are some limits to the aforementioned tests that a researcher should constantly be aware of. The following are significant limitations:

- (i) The tests should not be administered mechanically. It's important to remember that testing isn't a substitute for decision-making; tests are merely useful tools for decision-making. As a result, "correct statistical evidence interpretation is critical to making educated decisions." 6
- (ii) Tests do not explain why there is a difference between the means of two samples, for example. They just indicate whether the difference is due to sample variations or other factors, but they don't tell us what the other factor(s) are that are producing the difference.
- (iii) Because significance test results are dependent on probabilities, they cannot be conveyed with absolute certainty. When a test reveals a statistically significant difference, it simply means that the difference is unlikely to be attributable to chance.
- (iv) Statistical judgments based on significance tests cannot be regarded to be completely accurate evidence of the hypotheses' veracity. This is especially true in the case of small samples, where the likelihood of making erroneous assumptions is higher.

The size of the samples should be suitably increased for improved reliability.

All of these drawbacks suggest that in the case of statistical significance, inference techniques (or tests) must be accompanied with enough subject-matter knowledge and the capacity to make sound decisions.

Check your Understanding

1. A _____ is a scientific guess assumption of any phenomena occurring over the earth's surface.
2. A _____ hypothesis is frequently a predicted statement that connects an independent variable to a dependent variable and may be tested using scientific procedures.
3. The _____ hypothesis is what we call the idea that if we compare technique A to method B in terms of superiority and proceed on the assumption that both ways are equally good.
4. The _____ hypothesis proposes that there is a link between the two variables under consideration (one variable has a result on the other).
5. A _____ - _____ (two-tailed) hypothesis envisages that the independent variant will have an outcome on the dependent variable, nonetheless the direction of the result is not specified.
6. A _____ (one-tailed) hypothesis predicts the direction of the dependent variable's effect on the independent variable. It envisages in which direction the variation will yield place.
7. The _____ hypothesis, also known as the classical approach to hypothesis testing, is a type of hypothesis testing that focuses on making assumptions based on existing evidence.
8. The Bayesian Hypothesis Testing, a far more unorthodox and modern form of hypothesis testing, claims to test a particular hypothesis in accordance with past data samples, known as prior probability, and current data that lead to the plausibility of a hypothesis. True / False
9. A Type I error occurs when the null hypothesis is rejected when it is actually false. True / False
10. A _____ error, on the other hand, is when you don't realise there was an effect when there was.
11. A _____ interval is the margin of error that a researcher would encounter if they could ask a certain research question of every member of the target population and receive the same response that the sample members gave in the survey.

Summary

- Data investigation is the process of capturing valuable data by reviewing, cleansing, transforming, and displaying it using one of several methods, such as descriptive analysis, reversion analysis, dispersion analysis, factor analysis (self-governing variable to find the pattern), and time series, which are all based on mathematical and statistical approaches, or decision trees (tree-like structure for choosing various paths), machine learning, and artificial intelligence.
- A hypothesis is a scientific guess assumption of any phenomena occurring over the earth's surface. A hypothesis is a conjectural statement of the relation between two or more variables. We may define a hypothesis as a testable statement of

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the potential relationship between two or more variables. It is an assumption that serves as a tentative explanation for a researcher; it is a question put to nature to be observed. Hypothesis is, therefore, any statement, proposition or assumption that serves as a tentative explanation of certain facts.

- Hypothesis testing is an important stage in data science and statistics since it involves the verification of an assumption that could aid in the development of a statistical parameter. A researcher might, for example, form a hypothesis based on the assumption that the average of all odd integers is an even number. The researcher will have to evaluate the hypothesis utilizing hypothesis testing procedures in order to determine its plausibility. In contrast to a hypothesis that is 'supposed' to be true based on little or no evidence, hypothesis testing necessitates the presence of convincing evidence in order to prove that a statistical hypothesis is correct.
- A non-directional (two-tailed) hypothesis envisages that the independent variant will have an outcome on the dependent variable, nonetheless the direction of the result is not specified. A directional (one-tailed) hypothesis predicts the direction of the dependent variable's effect on the independent variable. It envisages in which direction the variation will yield place. A statistical hypothesis is one that can be tested statistically to see if it is plausible.
- The Bayesian Hypothesis Testing, a far more unorthodox and modern form of hypothesis testing, claims to test a particular hypothesis in accordance with past data samples, known as prior probability, and current data that lead to the plausibility of a hypothesis. The Bayesian technique determines whether a hypothesis is true or untrue based on this prior probability. The Bayes factor, which is a key component of this approach, represents the likelihood ratio between the null and alternative hypotheses.
- As previously stated, hypothesis testing is used to assess the validity of an assumption (officially known as the null hypothesis) in order to choose between two competing hypotheses concerning the value of a population parameter. Hypothesis testing is a technique for determining whether a population hypothesis is likely to be true or false based on a sample of facts. For the goal of testing hypotheses, statisticians have devised numerous tests of hypotheses (also known as tests of significance), which can be classed as: (a) parametric tests or standard tests of hypotheses; and (b) non-parametric tests or distribution-free tests of hypotheses.
- A Type I mistake in statistics is a false positive result, whereas a Type II error is a false negative result. In hypothesis testing, the chances of making these errors are inherent because making a statistical conclusion always contains uncertainty.
- A confidence interval is the margin of error that a researcher would encounter if they could ask a certain research question of every member of the target population and receive the same response that the sample members gave in the survey.
- Statistical inference, according to the Bayesian approach to statistics, involves more than just gathering, analysing, and interpreting data with endogenous variability, as the frequentist view of statistics suggests. It is about providing

a realistic means of understanding uncertain events through the concept of probability, to paraphrase de Finetti (1974, 1975). When new information becomes available, it also gives a consistent approach for inductive mathematical reasoning. In a nutshell, statistics is the study of risk.

Activity

1. Using the steps in the unit, write five different hypotheses based on specific ideas you are interested in researching. Make a table with each step as a heading and then fill out the pertinent information relating to that step. Conclude with writing a complete hypothesis.

Questions & Exercises

1. What are Type I and Type II errors?
2. How do you reduce the risk of making a Type I error?
3. How do you reduce the risk of making a Type II error?
4. What is statistical significance?
5. What is statistical power?

Glossary

- Change Hypothesis – shows change or comparison of two variables e.g., non-tribal's are more developed than the tribal's.
- Inadequate Hypothesis – for limited area in social sciences – demonstrable, based on deductions.
- Simple Hypothesis – expressions relationship amongst two or more variables.
- Worldwide Hypothesis – which may be scientifically showed and universally accepted- in physical sciences- Newton's Law etc.

Further Readings

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Check your Understanding – Answers

1. Hypothesis
2. Research
3. Null
4. Alternative
5. Non-directional
6. Directional
7. Frequentist
8. True
9. False
10. Type II
11. Confidence

Unit - 3.3: Parametric & Non-Parametric Tests

Notes

Objectives

At the end of this unit, you will be able to:

- Understand Z-test.
- Learn about T-test.
- Comprehend F-test.
- Understand correlation & regression.
- Evaluate Chi-square test.
- Analyze factor analysis.

Introduction

Tests that make assumptions about the parameters of the population distribution from which the sample is derived are known as parametric tests. The assumption that the population data is regularly distributed is frequently made. Because non-parametric tests are “distribution-free,” they can be applied with non-Normal variables. The non-parametric equivalents of a variety of parametric tests are shown in the table below.

Parametric test	Non-Parametric equivalent
Paired t-test	Wilcoxon Rank sum Test
Unpaired t-test	Mann-Whitney U test
Pearson correlation	Spearman correlation
One way Analysis of variance	Kruskal Wallis Test

“Non-parametric tests are valid for both non-Normally distributed data and normally distributed data, so why not use them all the time?”

In all circumstances, it would appear reasonable to utilize non-parametric tests, which would eliminate the need to test for Normality. Parametric testing, on the other hand, are favored for the following reasons:

1. We are rarely interested in a significance test on its own; we want to know something about the population from which the samples were drawn, and this is best accomplished using parameter estimates and confidence intervals.
2. Flexible modelling with non-parametric tests, such as controlling for confounding effects using multiple regression, is difficult.

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3. Non-parametric tests typically have less statistical power than their parametric counterparts. In other words, when significant differences do exist, one is more likely to notice them.

Parametric Tests

The main idea behind parametric tests is that we have a set of defined parameters that are utilized to determine a probabilistic model that may be used in Machine Learning.

Parametric tests are those for which we have prior knowledge of the population distribution (i.e., normal), or for which we can readily approximate the distribution to a normal distribution using the Central Limit Theorem.

The following are the parameters for adopting the normal distribution: Mean Standard Deviation

Finally, whether or not a test is parametric is entirely reliant on the population assumptions. There are numerous parametric tests available, including the following:

- With the use of a known standard deviation, determine the confidence interval for the population means.
- The confidence interval for population means and the unknown standard deviation must be determined.
- To determine the population variance's confidence interval.
- To calculate the confidence interval for the difference between two means when the standard deviation is unknown.

Non-parametric Tests

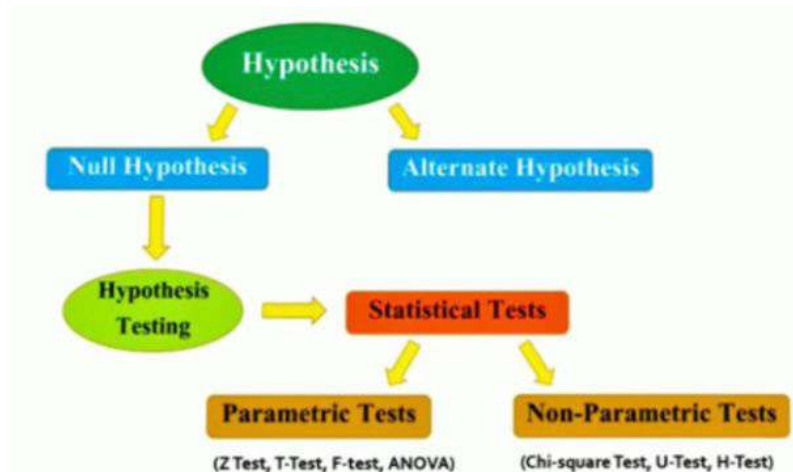
We make no assumptions about the parameters for the given population or the population we are researching in non-parametric tests. In fact, the population has no bearing on these exams.

As a result, no predefined set of parameters is accessible, and no distribution of any type (normal distribution, etc.) is available for usage.

Nonparametric tests are also referred to as distribution-free tests for this reason.

Non-parametric tests are becoming more common in today's world, and they have a lot of influence. Some of the reasons for this are:

- The key argument is that when employing parametric testing, there is no need to be polite.
- The second argument is that we don't have to make any assumptions about the population on which we're performing the research.
- The majority of nonparametric tests are simple to use and comprehend, implying that their complexity is minimal.



The z-test, t-test, and F-test are the most important parametric tests. All of these tests are predicated on the assumption of normalcy, which assumes that the data source is normally distributed.

Although the population may not be normally distributed in some circumstances, the tests will still be applicable because we usually deal with samples, and sampling distributions closely resemble normal distributions.

3.3.1 Z-test

The z-test is a statistical test that is based on the normal probability distribution and is used to determine the significance of a variety of statistical variables, most notably the mean. For determining the significance of the measure in question, the appropriate test statistic*, z, is calculated and compared to its probable value (read from table showing area under normal curve) at a set level of significance. In research investigations, this is the most commonly utilized test. This test is employed even when a binomial or t-distribution is appropriate, on the assumption that such a distribution tends to resemble a normal distribution as 'n' increases. In the event of a large sample or when population variance is known, the z-test is commonly employed to compare the mean of a sample to some hypothesised mean for the population. In the case of large samples or when population variance is known, the z-test is also used to determine the significance of differences between means of two independent samples. When n is big, the z-test is also used to compare the sample proportion to a theoretical value of population proportion or to judge the difference in proportions between two independent samples. This test can also be used to determine the significance of metrics like median, mode, and coefficient of correlation, among others.

Z-test is a statistical technique to determine whether the delivery of the test statistics can be approached by a normal distribution. It is the method to determine whether two sample means are just about the same or different as soon as their variance is recognized and the sample size is large (should be ≥ 30).

The minute to Use Z-test:

- The example size should be superior to 30. Otherwise, we should use the t-test.
- Samples ought to be drawn at random on or after the population.

Notes

- The standard deviance of the population must be known.
- Samples that are haggard from the population must be independent of each other.
- The information should be generally distributed, however for large sample size, it is supposed to partake a normal distribution.

Ladders to execute Z-test:

- First, recognize the null and alternate hypotheses.
- Control the level of significance (α).
- Discovery the critical value of z in the z-test by means of,
- Compute the z-test statistics. Further down is the formula for calculating the z-test statistics.

Z-Test

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma^2}{\sqrt{n}}}$$

$$z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Z-Test vs T-Test

wherever,

$\bar{X}$: mean of the sample.

μ : mean of the population.

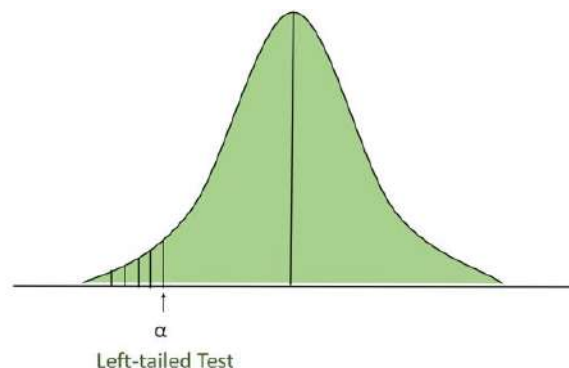
Sd: Standard deviation of the population.

n: sample size.

Nowadays, compare with the hypothesis and adopt whether to reject or not to throwaway the null hypothesis

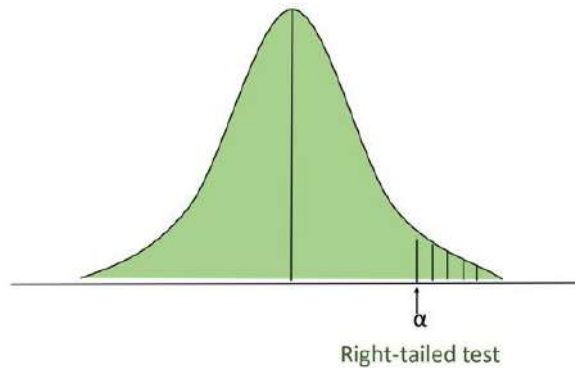
Kind of Z-test

Left-tailed Test: Our area of rejection is now placed to the extreme left of the delivery in this test. Our null hypothesis is that the call for worth is less than or equal to the mean population value at this moment.

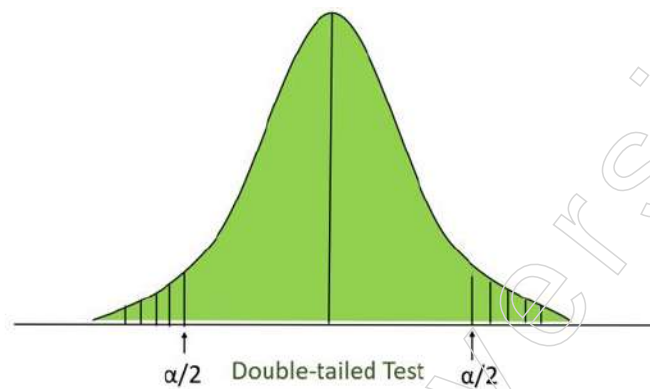


Right-tailed Test: Now, in this test, our rejection region is at the very edge of the distribution. Our null hypothesis at this moment is that the required value is less than or equal to the population mean value.

Notes



Two-tailed test: Our rejection zone, which is popular in this test, is positioned between the distribution's two extremes. Our null hypothesis at this moment is that the claimed worth is equal to the mean population value.



Underneath is the instance of performing the z-test:

If a sample of 400 male workers has a mean height of 67.47 inches, is it reasonable to regard the sample as a sample from a large population with a mean height of 67.39 inches and a standard deviation of 1.30 inches at a 5% level of significance?

Enchanting the null hypothesis that the mean height of the populace is equal to 67.39 inches, we tin can write:

$$H_0: \mu = 67.39"$$

$$H_a: \mu \neq 67.39"$$

$$\bar{x} = 67.47", \sigma = 1.30", n = 400$$

High and mighty the population to be normal, we can effort out the test statistic z as under:

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$Z = \frac{67.47 - 67.39}{\frac{1.30}{\sqrt{400}}}$$

As H_a is two-sided in the given question, we intend to be smearing a two-tailed test

Notes

for decisive the rejection regions at a 5% level of implication which comes to as under, by means of normal curve area table:

$$R : |z| > 1.96$$

The experimental value of t is 1.231 which is in the reception region since $R : |z| > 1.96$, and thus, H_0 is acknowledged.

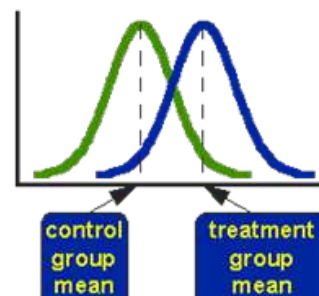
3.3.2 T-test

The t-test is based on the t-distribution and is considered an appropriate test for judging the significance of a sample mean or a difference in the means of two samples in the case of small sample(s) when population variance is unknown (in which case we use sample variance as an estimate of population variance). When two samples are related, we utilize the paired t-test (also known as the difference test) to determine the significance of the mean difference between them. It can also be used to determine the significance of simple and partial correlation coefficients. The relevant test statistic, t , is calculated from the sample data and then compared to its probable value based on the t-distribution (to be read from the table that gives probable values of t for different levels of significance for different degrees of freedom) at a specified level of significance for concerning degrees of freedom for accepting or rejecting the null hypothesis at a specified level of significance for concerning degrees of freedom for accepting or rejecting the null hypothesis. It should be highlighted that the t-test is only applicable when the population variance is unknown and the sample size is small.

A t-test is a statistical test for comparing the means of two groups. In hypothesis testing, t-test is frequently used to determine whether a method or treatment has an effect on the population of interest, or whether two groups differ from one another.

Additional example: Student's T-tests can be cast-off in real life to associate averages. For example, a drug company may famine to test a new cancer drug to find out if it advances life expectancy. In an experiment, there's permanently a control group (a group who are agreed a placebo, or "sugar pill"). The control group might show an average life expectancy of +5 years, even though the group taking the new drug strength have a life expectancy of +6 years. It would appear that the drug might work. Nonetheless it could be due to a fluke. To test this, investigators would use a Student's t-test to bargain out if the results are repeatable for a complete population.

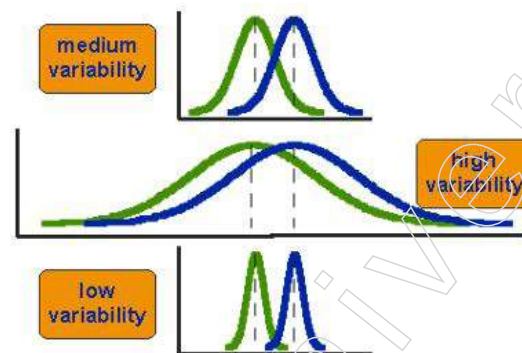
The t-test assesses whether the means of two collections are statistically changed from each other. This analysis is suitable whenever you want to associate the means of two groups, and especially suitable as the examination for the posttest-only two-group randomized experimental design.



Notes

Number shows the distributions for the treated (blue) and controller (green) groups in a study. In fact, the figure shows the idealized spreading – the actual distribution would regularly be depicted with a histogram or bar graph. The figure designates where the control and treatment group means are located. The question the t-test addresses is whether the means are statistically diverse.

What does it mean to say that the averages for two sets are statistically different? Consider the three situations shown in Figure. The primary thing to notice approximately the three situations is that the difference flanked by the means is the same in all three. Also, you must also notice that the three circumstances don't look the same – they tell very different stories. The top example shows a case through moderate variability of scores inside each group. The second situation shows the high variability case. The third shows the circumstance with low variability. Undoubtedly, we would accomplish that the two groups appear most dissimilar or distinct in the bottom or low-variability case. Why? For the reason that there is relatively little overlay between the two bell-shaped curves. In the extraordinary variability case, the group difference gives the impression least striking because the two bell-shaped scatterings overlap so much.



This brings us to a crucial conclusion:

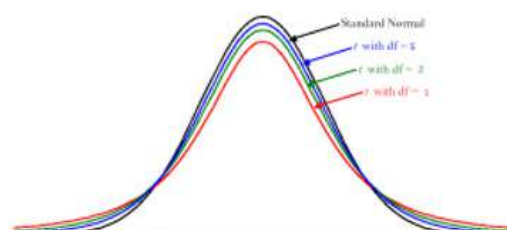
When comparing the changes in two groups' scores, we must consider the difference in the midpoint of their means in comparison to the feast or variability of their values. The t-test fixes just this.

T-test formula

The formulation for the two-sample t-test (a.k.a. the Student's t-test) is written below.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2(\frac{1}{n_1} + \frac{1}{n_2})}}$$

Student's t-distribution



Notes

This formula is made up of the following; t is the t -value, x_1 and x_2 are the means of the two clusters being compared, s_2 is the mutual standard error of the two groups, and n_1 and n_2 are the sum of comments in each of the groups.

A larger t -value indicates that the difference flanked by group means is bigger than the pooled standard error, indicating that the difference between the groups is more significant.

You can check whether your estimated t -value is bigger than what would be expected by chance by putting it in opposition to the numbers in a critical value chart. If this is the case, you can discard the null hypothesis and prove that the two groups are significantly different.

Whatever type of t -test have to use?

As soon as choosing a t -test, you will need to contemplate two things: whether the group's nothingness compared come as of a single population or two different populations, in addition whether you want to test the change in a specific direction.

One-sample, two-sample, or paired t -test?

- Unknown quantity: the groups come on or after a single population (e.g. measuring before and later an experimental treatment), accomplish a paired t -test.
- Uncertainty: the groups come from two changed populations (e.g. two different species, or people on or after two separate cities), accomplish a two-sample t -test (a.k.a. independent t -test).
- If at hand is one group being likened against a standard value (e.g. comparing the acidity of a watery to a neutral pH of 7), achieve a one-sample t -test.

One-tailed or two-tailed t -test?

- Accept you only care whether the two populations are diverse from one another, achieve a two-tailed t -test.
- Accept you want to recognize whether one population mean is greater than or less than the extra, accomplish a one-tailed t -test.

Say,

Understand a researcher wishes to assessment the hypothesis that a sample of size $n = 25$ with mean $\bar{x} = 79$ and standard deviation $s = 10$ was haggard at random from a population with mean $\mu = 75$ and unknown standard deviation. By means of the formula for the t -statistic, $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

The premeditated t equals 2. For a two-sided test at a mutual level of significance $\alpha = 0.05$, the serious values as of the t distribution on 24 degrees of liberty are -2.064 and 2.064 . The calculated t does not exceed these values, henceforth the null hypothesis cannot be situated rejected with 95 percent self-assurance. (The confidence level is $1 - \alpha$.)

3.3.3 F-test

The F -test compares the variance of two independent samples and is based on the F -distribution. This test is also used in conjunction with analysis of variance (ANOVA)

to determine the significance of several sample means at the same time. It's also used to determine whether multiple correlation coefficients are significant. For accepting or rejecting the null hypothesis, the test statistic, F , is calculated and compared to its likely value (as shown in the F -ratio tables for different degrees of freedom for higher and smaller variances at specified level of significance).

The term "F Test" is a catch-all word for tests that use the F -distribution. Frequently, when people talk about the F -Test, they are in fact talking about the F -Test to Associate Two Variances. On the other hand, the f -statistic is rummage-sale fashionable change of tests for example regression study, the Chow test, and the Scheffe test. Depending on the expected quantity we are running an F -Test, we might use many types of technology to run the test. For the reason that doing F -test by hand, together with variances, is a complex in addition time-consuming task.

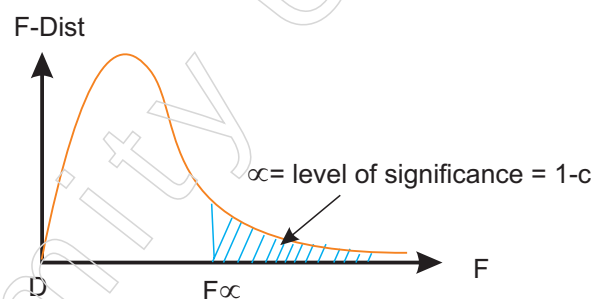
F Test Formula

Definition of F-test:

An exam statistic with an F -distribution under the null hypothesis is known as a F test in cutting-edge statistics. The aforementioned is used to associate the statistical models as per the data set available. George W. Snedecor, in honor of Sir Ronald A. Fisher, has given name to this formula as F Test Formula

By means of an F Test using technology, the following steps are listed below:

- State-owned the null hypothesis through the alternate hypothesis.
- Compute the F -value, by means of the formula.
- Treasure trove the F Statistic which is the perilous value for this test. This F -statistic formulation is the ratio of the modification of the collection means alienated by the mean of the within-group variances.
- Finally, sustenance or throwaway the Null Hypothesis.



Notes: F -Dist Skewed to right

F Distribution

Formula for F-Test to Associate Two Variances:

A Statistical F Test practices an F Statistic to association two variances, σ_1 and σ_2 , by intermediate them. As a result, the result will always be a positive amount because variances are always positive. Consequently, the reckoning for equaling two variances by means of the F -test is:

Notes

$F\text{-value} = \frac{\text{variance}_1}{\text{variance}_2}$

i.e., $F\text{-value} = \frac{\sigma_1^2}{\sigma_2^2}$

To re-count the variances of two transformed sets of values, the F-test formula is used. Working with the F distribution under the null hypothesis, we first compute the mean of two detailed observations before manipulating their variance.

F-tests

The test for joint significance has its own formula which takes the following form:

$$F = \frac{RSS_R - RSS_U / m}{RSS_U / n - k}$$

m – number of restrictions

k – parameters in unrestricted model

RSS_U – unrestricted RSS

RSS_R – restricted RSS

Explained Examples for F Test Formula

1: Perform an F-Test on the subsequent samples:

Sample-1 partaking variance = 109.63, sample size = 41.

Sample-2 taking Variance = 65.99, sample size = 21.

Solution:

Step-1: First write the hypothesis statements as:

H_0 : No difference in variances.

H_a : Difference in variances.

Step-2: Calculate the F-critical value. At this time take the highest variance as the numerator in addition the lowest variance as the denominator:

$F\text{ Value} = \frac{\sigma_1^2}{\sigma_2^2}$

$F\text{ Value} = \frac{109.63}{65.99}$

$F\text{ Value} = 1.66$

Step-3: Compute the degrees of freedom as:

The degrees of freedom in the table should be the sample size - 1, so for sample-1 it is 40 and for sample-2 it is 20.

Step-4: Select the alpha level. As, no alpha level is situated given in the question, consequently we may routine the standard level of 0.05. This wants to be halved intended for the test, so use 0.025.

Step-5: We will bargain the critical F-Value via the F-Table. We will usage the table with 0.025. Critical-F for (40, 20) at alpha (0.025) is 2.287.

Step-6: Relate the calculated value to the customary table value. If our calculated

value is advanced than the table value, at that time we may throwaway the null hypothesis. At this time, $1.66 < 2.287$. So, we cannot discard the null hypothesis.

3.3.4 Correlation & Regression

Correlation and linear regression are the most often used approaches for examining the relationship between two quantitative variables. The strength of a linear relationship between two variables is measured by correlation, whereas regression expresses the relationship as an equation. For example, in patients presenting to an emergency department (ED), we may utilize correlation and regression to see if there is a link between age and urea level, and if urea level can be predicted for a given age.

3.3.5 Chi-square Test

Among the many tests of significance devised by statisticians, the chi-square test is an important one. Chi-square is a statistical measure used in sample analysis to compare a variance to a theoretical variance. It is symbolically expressed as χ^2 (pronounced Ki-square).

It “may be used to evaluate if categorical data exhibits dependency or the two classifications are independent” as a non-parametric* test. When categories are included, it can also be used to compare theoretical populations to actual data.”

As a result, the chi-square test can be used to solve a wide range of problems. In fact, the test is a tool that allows all researchers to (i) evaluate the goodness of fit; (ii) test the significance of association between two attributes; and (iii) test the homogeneity or importance of population variance.

Testing of Hypothesis Procedures: chi square test

To regulate whether the association sandwiched between two qualitative variables is statistically significant, researchers need to conduct a test of implication called the Chi-Square Test. There are five steps to compartment this test.

Step 1:

Express the hypotheses

- Null Hypothesis:

H₀: In attendance is no significant association amongst students' educational level and their preference aimed at online or face-to-face instruction.

or H₀: In attending is no difference in the circulation of instructional preferences between undergraduate also graduate students.

- If there is no link between the two variables, the people will be evenly distributed over the table's cells.
- The alternative hypothesis aimed at a chi-square test is continuously two-sided. (It is technically multi-sided for the differences may happen in both guidelines in each cell of the table).
- Alternative Hypothesis: H_a:

Notes

In attendance is a significant connotation between students' educational level and their partiality for online or face-to-face instruction.

Or H_a : In attendance is a significant change in the distribution of instructional preferences amongst undergraduate and graduate students.

Step 2: Stipulate the expected values for apiece cell of the table (when the null hypothesis is true)

- The predictable values specify whatever the values of each cell of the table would be if there was no suggestion between the two variables.
- The formula for computing the expected values needs the sample size, the row totals, and the pillar totals

Step 3: Assuming H_0 is true, associate the observed counts by way of the sample by way of the expected counts to check if the data provide undeniable evidence against the null hypothesis.

- The practical values are the actual values computed from the sample.
- When doing a chi-square test, statistical software determines how to combine the anticipated and observed counts for each cell.

Step 4: Calculate the test statistic

- The chi-square statistic links the observed values to the predictable values.
- This examination statistic is used to determine whether the variance between the observed and expected morals is statistically noteworthy

Chi square Test in Excel

J1										
=CHISQ.TEST(B2:E4,B8:E10)										
	A	B	C	D	E	F	G	H	I	J
1	Observe	Hall A	Hall B	Hall C	Hall D	Total		Chi Square Value		0.03137
2	Chairs	100	92	89	81	362				
3	Tables	100	77	78	85	340				
4	Stools	50	81	75	76	282				
5	Total	250	250	242	242	984				
6										
7	Expected	Hall A	Hall B	Hall C	Hall D	Total				
8	Chairs	92.0	92.0	89.0	89.0	362				
9	Tables	86.4	86.4	83.6	83.6	340				
10	Stools	71.6	71.6	69.4	69.4	282				
11	Total	250	250	242	242	984				

Select a function:

- CHITEST
- CHISQ.TEST
- MOD
- ABS
- COT
- ENCODEURL
- HLOOKUP

CHISQ.TEST(actual_range,expected_range)
Returns the test for independence: the value from distribution for the statistic and the appropriate

3.3.6 Factor Analysis

Factor analysis is a data reduction approach that allows academics to look at topics that are difficult to quantify directly. Factor analysis produces easy-to-understand, actionable data by distilling a huge number of variables into a few comprehensible underlying elements.

When you use this strategy in your study, you'll be able to discover trends and see

themes throughout your datasets faster, allowing you to figure out what the data points have in common.

Factor analysis, unlike statistical procedures such as regression analysis, does not require defined variables.

The most common application of factor analysis is to determine the relationship between all of the variables in a dataset.

The Objectives of Factor Analysis

Consider factor analysis to be shrink wrap. When applied to a big amount of data, it reduces the size of the set to make it more manageable and understandable.

The main goal of factor analysis can be subdivided into four smaller goals:

- To determine the exact number of factors required to explain common themes among a group of variables.
- To find out how closely each variable in the dataset is linked to a common theme or component.
- To provide an understanding of the dataset's common factors.
- To figure out how much each observed data point represents each theme or component.

When to Use Factor Analysis

It might be difficult to know when to utilise which statistical approaches to extract the most information from your data.

Keep your aim in mind when contemplating factor analysis.

Factor analysis can be divided into three types. If your goal fits into any of these categories, factor analysis should be your statistical approach of choice:

- When you need to build a hypothesis about a link between variables, you should use exploratory factor analysis.
- To test a hypothesis concerning the link between variables, confirmatory factor analysis should be employed.
- Construct validity should be used to determine whether your survey is measuring what it claims to be measuring.

How To Ensure Your Survey is Optimized for Factor Analysis

If you know you'll be conducting a factor analysis on survey answer data, there are a few things you can do ahead of time to guarantee that your analysis is simple, informative, and actionable.

Identify and target a sufficient number of respondents

The lifeblood of factor analysis is large datasets. For factor analysis to generate significant results, you'll need a big group of survey respondents, which you can commonly find through panel services.

While factors like population size and your topic of interest will determine how

Notes

many respondents you need, it's advisable to keep an open mind and think "the more the better."

The more inquiries you have, the better.

As many specific questions as feasible should be included in your survey design. If your survey simply contains a few broad items, factor analysis will fall flat.

The ultimate purpose of factor analysis is to reduce a broad notion by taking into account more specific, contextual information, therefore this method will give you the results you want.

Pursue quantitative data.

You should avoid using open-ended survey questions if you intend to undertake a factor analysis.

You'll save a lot of time while conducting your factor analysis if you provide answer possibilities in the form of scales (whether they're Likert Scales, numerical scales, or even 'yes/no' scales). Simply be sure to use the same scaled answer options as much as feasible.

Factor analysis is a method to condense the data of countless variables into a just a few variables. For this reason, it is also sometimes named "dimension reduction." You can decrease the "dimensions" of your data and attach to one or more "super-variables." The most common method is identified as Principal Component Analysis (PCA).

In what way Factor Analysis Can Aid You?

Issue analysis is useful in:

1. Shrinking variables
2. Uncovering groups of responses

Say you ask numerous questions all driving at different, nonetheless closely related, features of customer satisfaction:

1. How pleased are you with our product?
2. Would you indorse our product to a friend or family member?
3. How possible are to you purchase our manufactured goods in the future

On the other hand, you only want one variable to characterize a customer satisfaction score. One choice would be to average the three question responses. One more option would be to generate a factor dependent variable. This can be completed by running PCA and possession the first Principal Component (also known as a factor). The benefit of PCA over an average is that it routinely weights individually all the variables in the calculation.

Say you partake in a list of questions and you don't know accurately which responses will move together in addition which will move differently; for instance, purchase barriers of potential customers. The succeeding are conceivable barriers to purchase:

1. Fee is prohibitive
2. Complete implementation costs
3. We can't touch a consensus vogueish our organisation
4. Manufactured goods are not consistent by means of our business strategy
5. I want to develop an ROI, but then again cannot or have not
6. We are protected into a contract through another product
7. The product benefits don't outweigh the cost
8. We partake no motive to switch
9. Our IT section cannot sustenance your product
10. We do not have adequate technical resources
11. Your creation does not have a feature we need
12. Extra (please specify)

At this time are some of the benefits of smearing dimensionality reduction to a dataset:

- Space prerequisite to store the data is abridged as the number of dimensions emanates down
- Less dimensions central to less computation/training time
- Approximately algorithms do not achieve well when we have a large scope. So, plummeting these dimensions needs to occur for the algorithm to be useful
- It receipts care of multicollinearity by eliminating redundant features. For example, you partake two variables – 'time used up on treadmill in minutes' and 'calories burnt'. These variables are extremely correlated as the more time you apply running on a treadmill, the added calories you will burn. Hereafter, there is no point in storing both as just one of them does what you require
- It benefits in visualizing data. As debated earlier, it is very difficult to visualize data in advanced dimensions so plummeting our space to 2D or 3D may allow us towards plot and detect patterns more clearly

Instances of Factor Analysis Educations

Factor analysis, counting PCA, is often tied in tandem with segmentation studies. Customers or client's strength be segmented by means of PCA itself or its' strength be an intermediary step to decrease variables beforehand using KMeans to make the segments.

Factor analysis offers simplicity after reducing variables. For long trainings with large blocks of Matrix Likert scale questions, the figure of variables can turn out to be unwieldy. Making the data easier to comprehend with factor analysis helps analyst to emphasis and clarify the results.

Notes

Sample Questions

Choosing the right questions for factor analysis is both an art and a science. Choosing which variables to decrease takes some experimentation, endurance and creativity. Factor analysis the whole thing well on Likert scale inquiries and Sum to 100 question types.

Factor analysis the whole thing well on matrix blocks of the succeeding question genres:

Psychographics (Agree/Disagree):

- I worth family
- I have faith in brand represents value

Behavioral (Agree/Disagree):

- I am obtaining the cheapest option
- I am a bargain customer

Attitudinal (Agree/Disagree):

- The budget is not improving
- I am satisfied with the product

Action-Based (Agree/Disagree):

- I love sporting
- I occasionally shop online throughout work hours

Behavior and psychographic questions are particularly suited for factor analysis.

Check your Understanding

1. Tests that make assumptions about the parameters of the population distribution from which the sample is derived are known as _____ tests.
2. In all circumstances, it would appear reasonable to utilize non-parametric tests, which would eliminate the need to test for _____.
3. _____ tests are also referred to as distribution-free tests for this reason.
4. The _____ is a statistical test that is based on the normal probability distribution and is used to determine the significance of a variety of statistical variables, most notably the mean.
5. When two samples are related, we utilize the paired t-test (also known as the difference test) to determine the significance of the mean difference between them. True / False
6. The z-test compares the variance of two independent samples and is based on the F-distribution. True / False
7. Correlation and linear regression are the most often used approaches for examining the relationship between two quantitative _____.
8. _____ is a statistical measure used in sample analysis to compare a variance to a theoretical variance.

9. _____ analysis produces easy-to-understand, actionable data by distilling a huge number of variables into a few comprehensible underlying elements.
10. You should avoid using open-ended survey questions if you intend to undertake a factor analysis. True / False

Summary

- Tests that make assumptions about the parameters of the population distribution from which the sample is derived are known as parametric tests. The assumption that the population data is regularly distributed is frequently made. Because non-parametric tests are “distribution-free,” they can be applied with non-Normal variables. The non-parametric equivalents of a variety of parametric tests are shown in the table below.
- Parametric tests are those for which we have prior knowledge of the population distribution (i.e., normal), or for which we can readily approximate the distribution to a normal distribution using the Central Limit Theorem.
- We make no assumptions about the parameters for the given population or the population we are researching in non-parametric tests. In fact, the population has no bearing on these exams.
- As a result, no predefined set of parameters is accessible, and no distribution of any type (normal distribution, etc.) is available for usage. Nonparametric tests are also referred to as distribution-free tests for this reason. Non-parametric tests are becoming more common in today’s world, and they have a lot of influence. Some of the reasons for this are:
- The z-test is a statistical test that is based on the normal probability distribution and is used to determine the significance of a variety of statistical variables, most notably the mean. For determining the significance of the measure in question, the appropriate test statistic*, z, is calculated and compared to its probable value (read from table showing area under normal curve) at a set level of significance.
- The t-test is based on the t-distribution and is considered an appropriate test for judging the significance of a sample mean or a difference in the means of two samples in the case of small sample(s) when population variance is unknown (in which case we use sample variance as an estimate of population variance). When two samples are related, we utilize the paired t-test (also known as the difference test) to determine the significance of the mean difference between them. It can also be used to determine the significance of simple and partial correlation coefficients.
- The F-test compares the variance of two independent samples and is based on the F-distribution. This test is also used in conjunction with analysis of variance (ANOVA) to determine the significance of several sample means at the same time. It’s also used to determine whether multiple correlation coefficients are significant.
- Correlation and linear regression are the most often used approaches for examining the relationship between two quantitative variables. The strength of a linear relationship between two variables is measured by correlation, whereas regression expresses the relationship as an equation.

Notes

- Among the many tests of significance devised by statisticians, the chi-square test is an important one. Chi-square is a statistical measure used in sample analysis to compare a variance to a theoretical variance. It is symbolically expressed as χ^2 (pronounced Ki-square).
- Factor analysis is a data reduction approach that allows academics to look at topics that are difficult to quantify directly. Factor analysis produces easy-to-understand, actionable data by distilling a huge number of variables into a few comprehensible underlying elements.

Activity

1. Write a report on factor analysis. What is it? How is it performed? What conclusions can be drawn from it? Etc. Write a minimum of 1000 words.

Questions & Exercises

1. Non-parametric tests are valid for both non-Normally distributed data and Normally distributed data, so why not use them all the time?
2. Briefly describe the important parametric tests used in context of testing hypotheses. How such tests differ from non-parametric tests? Explain.
3. What is a t-test? When it is used and for what purpose(s)? Explain by means of examples.
4. What is the z test?
5. What is the t test formula?
6. What are the five steps to run a Chi-Square test?

Glossary

- Comportment: behavior; bearing

Further Readings

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Check your Understanding – Answers

1. Parametric
2. Normality
3. Nonparametric
4. Z-test
5. True
6. False
7. Variables
8. Chi-square
9. Factor
10. True

Notes

Unit - 3.4: Data Analysis

Objectives

At the end of this unit, you will be able to:

- Learn about SPSS.
- Understand creating data file.
- Analyze run example on the parametric test.
- Comprehend the run example on the non-parametric test.

Introduction



Data analysis is the most important aspect of any study. The data analysis process summarizes the information gathered. It entails the use of analytical and logical reasoning to data in order to identify patterns, correlations, and trends.

Data analysis is the systematic use of statistical and logical techniques to describe the scope of the data, modularize the data structure, condense the data representation, illustrate via images, tables, and graphs, and evaluate statistical inclinations, probability data, and derive meaningful conclusions. By removing the superfluous noise caused by the remainder of the data, these analytical approaches allow us to infer the underlying inference. Because data generation is a constant activity, data analysis is also a continuous, iterative process that includes both data collecting and data analysis. One of the most important aspects of data analysis is ensuring data integrity.

Transportation, risk and fraud detection, consumer interaction, city planning, healthcare, web search, digital advertising, and more are all examples of how data analysis is applied.

In the case of healthcare, as we just saw with the pandemic outbreak, Coronavirus hospitals are facing the difficulty of coping with the demand of treating as many patients as possible. Using data analysis to monitor machine and data utilization in such instances provides for efficiency gains.

Make the following pre-requisites for good Data Analysis before delving any deeper:

- Ensure that the essential analytical skills are available.
- Ensure that data collecting and analysis processes are implemented properly.
- Calculate statistical significance.
- Check for erroneous analysis.
- Ascertain the presence of valid and unbiased inferences.
- Ensure that the data, data sources, data analysis methods, and inferences drawn are accurate and reliable.
- Consider the scope of the investigation.

3.4.1 Introduction to SPSS

While Alchemer includes powerful built-in reporting options that are simple to use and present for most internet surveys, NPS surveys, and employee satisfaction surveys, most researchers consider SPSS to be the best-in-class solution for in-depth statistical analysis.

SPSS is for Statistical Package for the Social Sciences, and it's a statistical data analysis programme utilized by a variety of scholars. SPSS is a software suite designed for data management and statistical analysis in the social sciences. It was first released in 1968 by SPSS Inc., and was later purchased by IBM in 2009.

Although IBM SPSS Statistics is the official name, most users still refer to it as SPSS. SPSS is widely regarded as the gold standard for social-science data analysis, thanks to its simple and English-like command language and comprehensive user manual.

Market researchers, health researchers, survey firms, government bodies, education researchers, marketing organizations, data miners, and others utilize SPSS to handle and analyze survey data collected with an online survey platform like Alchemer.

To get the most out of their research and survey projects, most leading research agencies use SPSS to analyze survey data and extract text data.

SPSS Statistics is a statistical analysis software package that allows you to perform interactive or batch statistical analysis. It was long manufactured by SPSS Inc. before being purchased by IBM in 2009. IBM SPSS Statistics is the brand name for current versions (after 2015).

The name of the software was changed from Statistical Package for the Social Sciences (SPSS) to Statistical Product and Service Solutions to reflect the original market.

Overview

SPSS is a commonly used statistical analysis application in the social sciences. Market researchers, health researchers, survey firms, the government, education researchers, marketing groups, data miners, and other professionals use it. The original SPSS manual (Nie, Bent, & Hull, 1970) has been called one of "sociology's most influential publications" because it allows ordinary researchers to perform their own statistical analysis. Data management (case selection, file reshaping, creating derived

Notes

data) and data documentation (a metadata dictionary is saved in the datafile) are aspects of the base software in addition to statistical analysis.

SPSS Statistics has a lot of capabilities that may be accessed by pull-down menus or by programming with a proprietary 4GL command syntax language. The advantages of command syntax programming include reproducible output, the simplification of repetitive operations, and the handling of sophisticated data manipulations and analysis. Furthermore, certain complicated apps can only be created in syntax and cannot be accessed using the menu system. The pull-down menu interface also creates command syntax, which can be presented in the output if the default parameters are altered. The “paste” button in each menu can also be used to paste them into a syntax file. Using the provided Production Job Facility, programmes can be run interactively or unattended.

In addition, command language subroutines can be written in a “macro” language. A Python programmability extension can dynamically generate command syntax programmes using information from the data dictionary and data. Although SaxBasic is still accessible, the Python programmability extension, introduced in SPSS 14, has largely supplanted the less effective SAX Basic “scripts” for most uses. Furthermore, the Python extension enables SPSS to execute any of the statistics included in the free R software package. SPSS can now be driven externally by a Python or VB.NET programme using supplied “plug-ins” starting with version 14. (These two scripting features, as well as many other scripts, are supplied on the installation disc and are generally installed by default starting with Version 20.)

Internal file structure, data types, data processing, and matching files are all constrained by SPSS Statistics, which makes programming much easier. SPSS datasets are two-dimensional tables with rows representing cases (such as individuals or households) and columns representing measurements (such as age, sex, or household income). There are just two forms of data defined: numeric and text (or “string”). All data processing is done in a case-by-case manner through the file (dataset). One-to-one and one-to-many matches are possible, but many-to-many matches are not. There is also a separate Matrix session where data can be processed as matrices utilizing matrix and linear algebra operations, in addition to the cases-by-variables structure and processing.

The graphical user interface offers two perspectives, which can be switched between by clicking one of the two tabs in the SPSS Statistics window’s bottom left corner. The ‘Data View’ displays the cases (rows) and variables in a spreadsheet format (columns). The data cells, unlike spreadsheets, can only contain numbers or text, and formulas cannot be placed in them. The ‘Variable View’ shows the metadata dictionary in a table format, with each row representing a variable and displaying the variable name, variable label, value label(s), print width, measurement type, and a number of additional details. Manually editing cells in both views defines the file structure and allows data entry without requiring command syntax. For tiny datasets, this may be adequate. Larger datasets, such as statistical surveys, are more frequently prepared with data entry software, entered during computer-assisted personal interviewing, scanned and analyzed with optical character recognition and optical mark recognition software, or directly captured through online questionnaires. The data is then imported into SPSS.

Data can be read and written from ASCII text files (including hierarchical files), other statistics packages, spreadsheets, and databases using SPSS Statistics. ODBC and SQL allow SPSS Statistics to read and write to external relational database tables.

Statistical output is to a proprietary file format (*.spv file, supporting pivot tables) for which a standalone reader can be obtained in addition to the in-package viewer. Text, Microsoft Word, PDF, Excel, and more formats can be exported from the proprietary output. Alternatively, output can be saved as text, tab-delimited text, PDF, XLS, HTML, XML, SPSS dataset, or a variety of graphic image formats (with the OMS command) (JPEG, PNG, BMP and EMF).

SPSS Statistics comes in a variety of variants. SPSS Statistics Gradpacks are heavily discounted editions of SPSS Statistics that are solely available to students. SPSS Statistics Server is a client/server architecture version of SPSS Statistics. Additional functionality can be added to the main software through add-on packages (examples include complex samples which can adjust for clustered and stratified samples, and custom tables which can create publication-ready tables). SPSS Statistics can be purchased on an annual or monthly subscription basis.

On August 8, 2017, SPSS Statistics released version 25. Within the Advanced Statistics and Custom Tables add-on, SPSS v25 offers new and advanced statistics such as random effects solution findings (GENLINMIXED), robust standard errors (GLM/UNIANOVA), and profile plots with error bars. V25 also contains new Bayesian Statistics capabilities, such as robust new charting tools, including new default templates and the ability to share with Microsoft Office products, as well as publication-ready charts.

The SPSS logo used prior to the renaming in January 2010.



Functionalities of SPSS

- Data Examination.
- General Linear Model.
- Correlation.
- ANOVA.
- Regressions.
- Cluster analysis.
- Time series.
- Graphics and graphical interface.
- Data Transformations.
- Descriptive statistics.
- Reliability tests.
- T-tests.
- MANOVA.

Notes

SPSS's Primary Functions

SPSS has four programmes to help researchers with their sophisticated data analysis requirements.

Statistics Program

The Statistics module in SPSS offers a variety of basic statistical functions, including frequencies, cross-tabulation, and bivariate statistics, to name a few.

Modeling Software

Researchers can use the Modeler software in SPSS to create and validate predictive models using advanced statistical processes.

Program for Text Analytics in Surveys

The Text Analytics for Surveys tool from SPSS assists survey administrators in gaining valuable insights from open-ended survey responses.

Designer of Visualization

With SPSS's Visualization Designer application, researchers may easily build a range of graphics from their survey data, such as density charts and radial boxplots.

SPSS offers data management options in addition to the four programmes mentioned above, allowing researchers to do case selection, construct derived data, and reshape files.

SPSS also includes data documentation, which allows researchers to keep track of their metadata. This metadata dictionary serves as a centralised store for data-related information such meaning, relationships with other data, provenance, usage, and format.

In SPSS, you can use a number of different statistical approaches, including:

- Frequencies, cross-tabulation, and descriptive ratio statistics are examples of descriptive statistics.
- Analysis of variance (ANOVA), means, correlation, and nonparametric tests are examples of bivariate statistics.
- Linear regression is a type of numerical outcome prediction.
- Prediction techniques such as cluster analysis and factor analysis are used to identify groups.

SPSS is groundbreaking software mostly used by research experts which aid them process critical data in modest steps. Working with data is a diverse and time-consuming process, but with the help of specific strategies, this programme can easily handle and function data. These procedures are used to examine, transform, and yield a characteristic pattern between altered data variables. In addition, the output can be received through a graphical demonstration, allowing the user to quickly recognise the result. Recite the following to understand the aspects that have a role in data management and completion.

1. **Data Transformation:** This method is used to change the arrangement of the data. After changing the data type, it integrates identical type of data in one place in addition it becomes easy to manage it. You will be able to insert the changed data into SPSS, and in addition, it will transform its structure as per the scheme specification and requirement. It means that no matter what operating system you use, SPSS can still operate with old data.
2. **Regression Analysis:** It is used to comprehend the relation amongst dependent and interdependent variables that are stored in a data file. It also clarifies how a change in the value of a mutually dependent variable can affect the dependent data. The most important need of regression analysis is to realize the type of relationship formed by different variables.
3. **ANOVA (Analysis of variance):** The aforementioned is a statistical method to associate events, collections or processes, and find out the change between them. It can assist you in determining which strategy is best for completing a task. By observing at the result, you can invent the feasibility and efficiency of the particular means.
4. **MANOVA (Multivariate analysis of variance):** This technique is used to associate data of random variables whose value is unidentified. MANOVA system can also be used to analyze different forms of population and whatever factors can distress their choices.
5. **T-tests:** It is used to understand the variance between two sample types, in addition researchers apply this method to find out the variance in the interest of two categories of groups. This test can also comprehend if the produced output is worthless or useful.

This software was industrialized in 1960, but far along in 2009, IBM acquired it. They have completed some momentous vicissitudes in the programming of SPSS and nowadays it can perform various types of research task in a number of fields. Due to this, the routine of this software is protracted to many industries in addition organizations, for example marketing, health care, schooling, surveys, etc.

3.4.2 Creating Data File

Data collection is the process of gathering, collecting, extracting, and storing large amounts of data in organized or unstructured formats such as text, video, audio, XML files, records, or other picture files that will be utilized in data analysis later on.

“Data collection” is the first phase in the big data analysis process before beginning to examine the patterns or usable information in the data. The data to be studied must be gathered from a variety of reliable sources.

Notes



The information received is known as “knowledge,” and the data gathered is known as “raw data,” which is not helpful right now. However, after cleaning the impure data and employing it for further analysis, the information obtained is known as “knowledge.” Knowledge can take numerous forms, including business knowledge, enterprise product sales, sickness treatment, and so on. The primary purpose of data collecting is to gather data that is rich in information.

Set up a data file to keep track of everything, preferably before you start collecting data. Your data file can be a distinct file or part of the same database as the one you’re using to track answers. Some key data security considerations can be found in the section on designing data collection procedures and tools. Also, keep the following pointers in mind as you prepare for data collecting. Your data file will eventually contain all of the replies that parents supplied on their surveys, regardless of how you structure the two files. Take some time to think about it right now.

- How will the data file be organized?
- Which software package will be used to manage quantitative and qualitative data?

If you’re using a spreadsheet programme like Microsoft Excel or a statistical software programme like SPSS or SAS to enter data from paper surveys by hand or scanning them, you can use a spreadsheet programme like Excel or a statistical software programme like SPSS or SAS. Here are some pointers on how to set up the data file:

Tip 1: Each column should correspond to a different analytical variable, and each row should comprise all of the data for one parent (each with a unique ID number). Survey items, respondent demographic information, district or school information, and so on are examples of variables. Take a look at Table below for an example.

Example of organizing data in rows and columns in a data file

ID	AGE	SEX	RACE	DATE	Q1	Q2	Q3
001							
002							
003							
004							

Q1 = survey item 1, Q2 = survey item 2, etc.

Notes

Tip 2: Make a table that includes information about each variable in the data file.

The name of the variable (which may be the survey item number or an abbreviated label that describes the variable)

A description of each variable, as well as the unit of measurement and data type (e.g., text, numeric, date)

The entry's maximum length (i.e., how many characters can be typed into the file)

The potential values, as well as any constraints on values

If you're using a programme like SPSS, the application may prompt you to do so; if you're using Microsoft Excel, you may wish to create this table in a separate worksheet or file. For an example of a variable information table that corresponds to the example in Table above, see Table below.

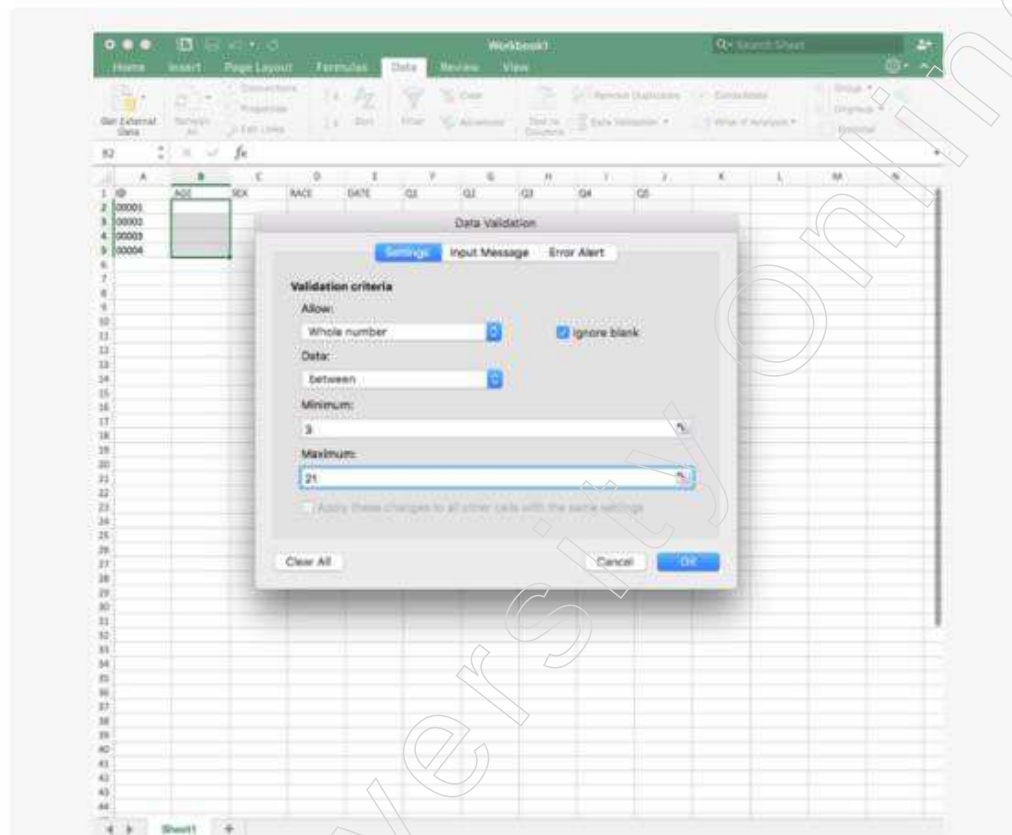
Example of table with information about each variable included in the data file

Variable name	Description	Units	Data type	Max Size	Values and value restrictions (Blank=none)
Age	Age of Child	Years	Numeric	2	3-21
Sex	Gender	-	Numeric	1	0= Female 1= Male
Race	Race/ethnicity	-	Numeric	1	1= American Indian or Alaska Native 2= Asian 3= Black or African American 4= Hispanic/Latino 5= Native Hawaiian or Other Pacific Islander 6= White 7= Two or more races
Date	Date survey completed	-	date	8	dd/mm/yy
Q1	Survey Item1: (Description of Item)	-	numeric	1	1=Very strongly disagree 2=Strongly disagree 3=Disagree 4=Agree 5=Strongly agree 6=Very strongly agree

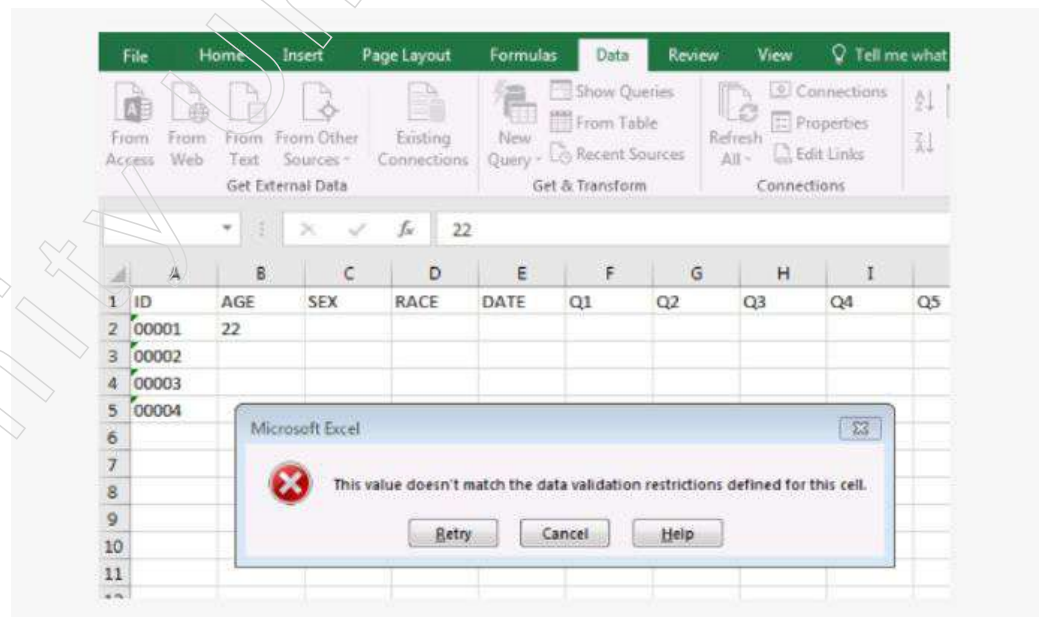
To reduce data entry errors, set constraints on cells in the data file. Cells can be set up to accept only text, numbers, or dates, for example. You can also limit the number of characters that can be entered in a cell and the range that can be used (for example, only allow ages 3 to 21 for a student's age). For an example of how to do this in Excel, see Figure below.

Notes

Example of setting age restrictions on cells in Excel data files

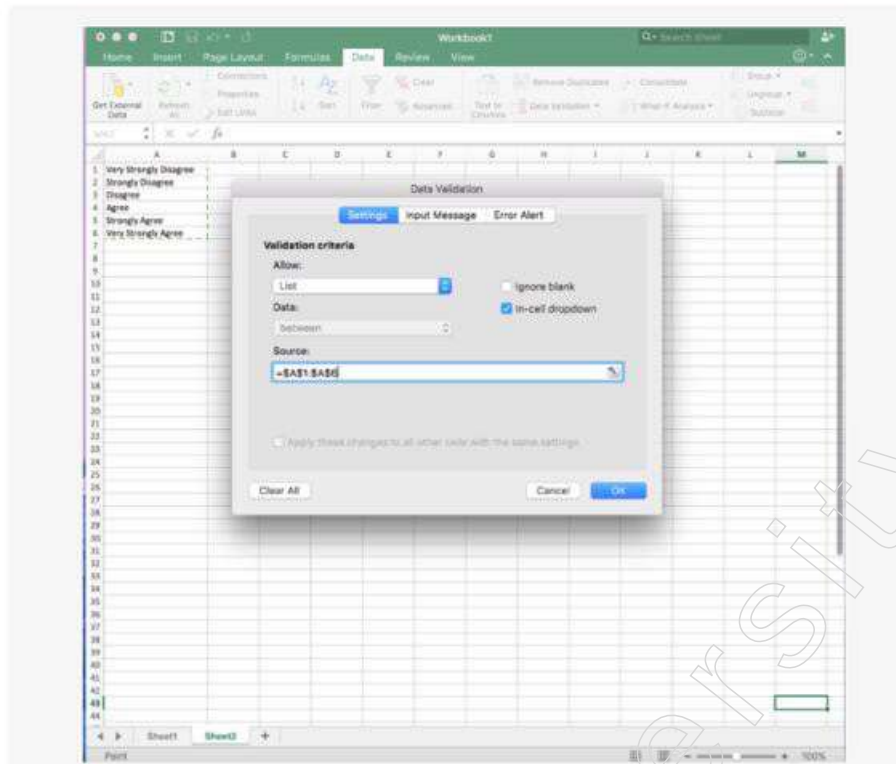


The error notice shown below is an example of what you'll get if you try to input an age that isn't within the predefined range.

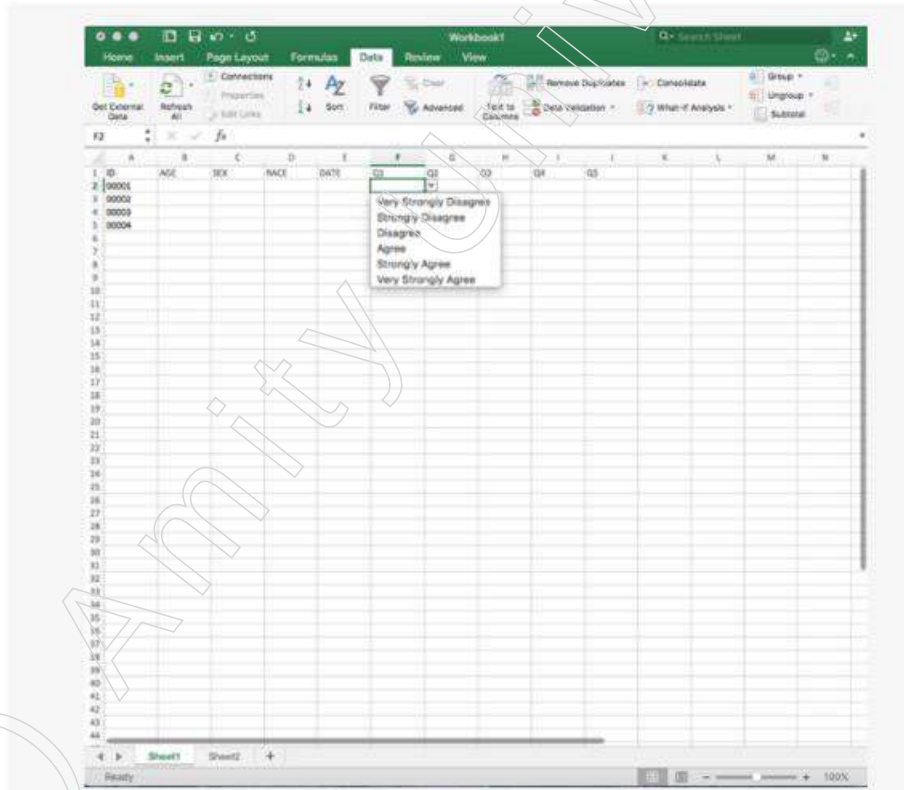


Tip 4: Create drop-down menus whenever possible to assist assure data entry consistency. Users will only be able to select predefined responses from the drop-down menu. This is illustrated through graphs. Step-by-step instructions for creating drop-down menus in Excel can be found by searching on Google.

Part 1 of creating drop-down menus in Excel



Part 2 of creating drop-down menus in Excel



Tip 5: To make qualitative data analysis easier, create a separate data file for responses to open-ended items. You can use a spreadsheet, such as Excel, or a software application, such as NVivo, ATLAS.ti, or Dedoose, that is created expressly

Notes

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for analysing qualitative data. For further information on how to analyse open-ended responses, see the section on interpreting qualitative data.

Many online survey tools allow you to automatically export survey data to your data file. The processes for exporting the data are normally described in the online survey program's instructions, and you will often have the choice of exporting the data to Excel, SPSS, or SAS. When generating your web survey, you can take some measures to guarantee that your data file is set up appropriately when you retrieve the results.

As an example,

Consider using checkboxes or radio buttons in your web survey to limit how participants enter responses and ensure that your data is clean and consistent when you download it, as shown in Figure below.

2. What is your child's gender?

☒ Female

☐ Male

Check that all items are formatted correctly, so that a parent, for example, can check more than one box if a question asks respondents to “choose all that apply,” as illustrated in Figure below.

2. Which race/ethnicity best describes your child? (Check all that apply.)

☒ American Indian or Alaska Native

☐ Asian

☐ Black or African American

☐ Native Hawaiian or Other Pacific Islander

☐ White

☐ Two or more races

Ok

3.4.3 Run Example on Parametric Test

Nonparametric tests exist in a parallel universe to parametric testing.

Parametric tests of means	Nonparametric tests of medians
1-sample t-test	1-sample Sign, 1-sample Wilcoxon
2-sample t-test	Mann-Whitney test
One-Way ANOVA	Kruskal-Wallis, Mood's median test
Factorial DOE with a factor and a blocking variable	Friedman test

Tests that make assumptions about the parameters of the population distribution from which the sample is derived are known as parametric tests. The assumption that the population data is regularly distributed is frequently made.

Advantages

Advantage 1: With skewed and nonnormal distributions, parametric testing can produce reliable results.

Many people aren't aware of this, but parametric studies can yield valid results even if your continuous data isn't distributed regularly. All you have to do now is make sure your sample size matches the requirements for each of the analyses listed below. These needs have been discovered through simulation research. For additional information on these studies, go here.

Parametric analyses	Sample size requirements for nonnormal data
1-sample t-test	Greater than 20
2-sample t-test	Each group should have more than 15 observations
One-Way ANOVA	- For 2-9 groups, each group should have more than 15 observations - For 10-12 groups, each group should have more than 20 observations

The central limit theorem allows you to utilize these parametric tests with non-normally distributed data.

Advantage 2: When groups have varied levels of variability, parametric testing can produce reliable results.

Nonparametric tests don't require normally distributed data, which is true. Nonparametric tests, on the other hand, contain additional criteria that can be difficult to meet. In a nonparametric analysis, all of the groups must have the same variability (dispersion). When group variability differs, nonparametric analyses may not produce correct results.

Parametric studies, such as the 2-sample t-test or one-way ANOVA, on the other hand, allow you to examine groups with uneven variances. It's as simple as checking the appropriate box in most statistical tools! When you utilize a parametric analysis, you don't have to worry about groups having varying levels of variability.

Advantage 3: Parametric tests offer more statistical power than nonparametric tests.

Parametric tests have a higher power in most circumstances. A parametric analysis is more likely to find an effect if one exists.

Run Example on Parametric Test [ANOVA (One-way analysis of variance) is an example of parametric test]

Notes

The one-way analysis of variance (ANOVA) is performed to see if there are any statistically significant differences in the means of two or more unrelated groups (although you tend to only see it used when there are a minimum of three, rather than two groups). For example, you could divide students into three independent groups and conduct a one-way ANOVA to see if exam performance varies based on test anxiety levels (e.g., low, medium and high-stressed students). It's also vital to remember that the one-way ANOVA is an omnibus test statistic that can't tell you which specific groups were statistically significantly different from each other; it can only say that at least two of them were.

This "fast start" instruction will show you how to use SPSS Statistics to do a one-way ANOVA, as well as how to analyze and report the findings. We also demonstrate how to conduct a post hoc test using SPSS Statistics because the one-way ANOVA is frequently followed by a post hoc test. Before we go over the technique, you should be aware of the several assumptions that your data must meet in order for a one-way ANOVA to produce a valid result. Following that, we'll talk about these assumptions.

Assumptions

When you decide to use a one-way ANOVA to analyze your data, you must first ensure that the data you wish to analyze can really be analyzed using a one-way ANOVA. You must do this because a one-way ANOVA is only appropriate if your data "passes" the six assumptions that a one-way ANOVA must meet in order to provide a valid result. In practice, checking for these six assumptions takes a little longer because it requires you to click a few more boxes in SPSS Statistics and think a little harder about your data, but it's not a tough effort. Do not be startled if one or more of these assumptions is violated when analyzing your own data using SPSS Statistics before we expose you to these six assumptions (i.e., is not met). This is frequent when working with real-world data rather than textbook examples, which often only demonstrate how to perform a one-way ANOVA when everything goes smoothly. Don't worry, though. Even if your data contradicts certain assumptions, there is usually a way to fix it.

Assumption 1: The interval or ratio level should be used to measure your dependent variable (i.e., they are continuous). Revision time (measured in hours), intelligence (measured using an IQ score), exam performance (measured from 0 to 100), weight (measured in kg), and so on are examples of variables that fit this condition. In our article *Types of Variables*, you can learn more about interval and ratio variables.

Assumption 2: Your independent variable should be made up of at least two categorical, independent groupings. A one-way ANOVA is typically used when there are three or more categorical, independent groups, although it can also be employed when there are just two groups (but an independent-samples t-test is more commonly used for two groups). Ethnicity (e.g., 3 groups: Caucasian, African American, and Hispanic), physical activity level (e.g., 4 groups: sedentary, low, moderate, and high), occupation (e.g., 5 groups: surgeon, doctor, nurse, dentist, therapist), and so on are examples of independent variables that fit this condition.

Assumption 3: You should have observation independence, which indicates that there should be no association between the observations in each group or between the groups. Each group, for example, must have diverse participants, with no person

belonging to more than one group. This is more of a research design issue than a testable assumption, yet it is a key assumption of the one-way ANOVA. If your study fails to meet this criterion, you'll need to use a different statistical test for the one-way ANOVA (e.g., a repeated measures design).

Assumption 4: No substantial outliers should exist. Outliers are single data points in your data that do not follow the expected pattern (for example, in a study of 100 students' IQ scores, where the mean score was 108 with only a small variation between students, one student had a score of 156, which is extremely unusual and may even put her in the top 1% of IQ scores globally). Outliers might have a negative impact on the one-way ANOVA, lowering the validity of your results. Fortunately, you can readily discover any outliers when using SPSS Statistics to execute a one-way ANOVA on your data. We (a) show you how to discover outliers using SPSS Statistics; and (b) outline some of the alternatives you have for dealing with outliers in our updated one-way ANOVA guide.

Assumption 5: For each category of the independent variable, your dependent variable should be approximately regularly distributed. We talk about the one-way ANOVA only requiring approximately normal data because it is "robust" to violations of normality, which means that the assumption can be slightly violated and still yield good findings. The Shapiro-Wilk test of normalcy, which is easily tested using SPSS Statistics, can be used to check for normality. In our improved one-way ANOVA guide, we not only show you how to conduct it, but we also explain what to do if your data contradicts this assumption (i.e., if it fails it more than a little bit).

Assumption 6: The variances must be homogeneous. The Levene's test for homogeneity of variances in SPSS Statistics can be used to evaluate this assumption. If your data contradicts this assumption, you'll need to perform a Welch ANOVA rather than a one-way ANOVA, which you can do with SPSS Statistics, as well as a different post hoc test. We (a) show you how to perform Levene's test for homogeneity of variances in SPSS Statistics, (b) explain some of the things you'll need to consider when interpreting your data, and (c) present possible ways to continue with your analysis if your data fails to meet this assumption, such as using a Welch ANOVA instead of a one-way ANOVA and a Games-Howell test instead of a Tukey post hoc test.

SPSS Statistics can be used to test assumptions 4, 5, and 6. You should check that your data fulfils assumptions 1, 2, and 3 before proceeding, though you don't need SPSS Statistics to do so. Remember that the findings of a one-way ANOVA may not be legitimate if the statistical tests on these assumptions are not performed appropriately. This is why we devote a piece of our expanded one-way ANOVA tutorial to assisting you with this.

Example

A manager wants to boost his company's production by boosting the speed with which his staff can use a specific spreadsheet tool. He hires an outside agency to train him in this spreadsheet programme because he lacks the necessary abilities in-house. There are three levels of classes available: beginning, intermediate, and advanced. He is unsure which course is required for the type of job they do at his company, so he enrolls ten employees in the beginner, intermediate, and advanced courses. He

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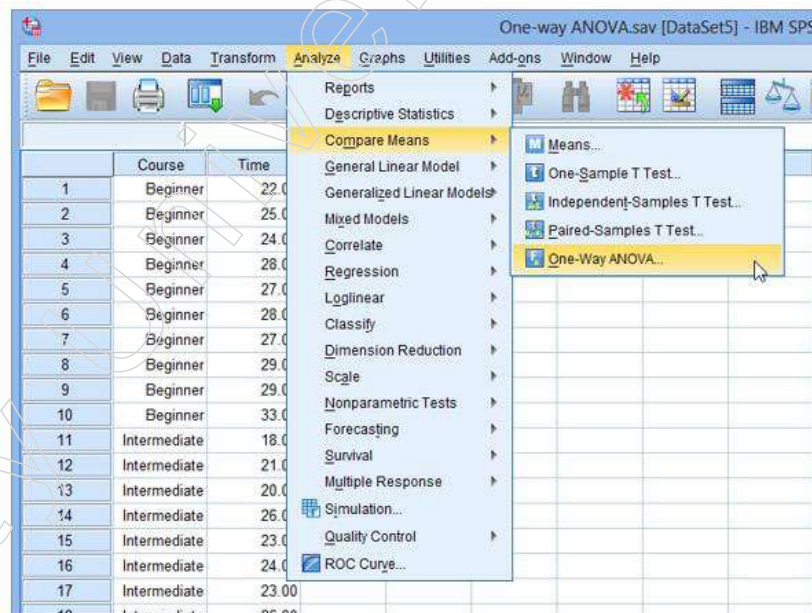
offers them a problem to tackle using the spreadsheet programme when they return from training, and he times how long it takes them to complete the problem. He then analyses the three courses (beginning, intermediate, and advanced) to determine if the average time it took to accomplish the task differs.

We split the groups for analysis in SPSS Statistics by defining a grouping variable named Course (i.e., the independent variable) and assigning a value of “1” to the beginners’ course, “2” to the intermediate course, and “3” to the advanced course. The variable Time was used to record the amount of time it took to finish the set issue (i.e., the dependent variable).

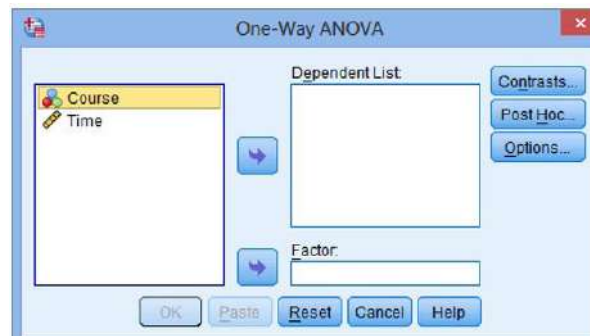
Test Procedure in SPSS Statistics

When the six assumptions in the preceding section, Assumptions, have not been violated, the eight steps below explain to you how to analyze your data using a one-way ANOVA in SPSS Statistics. We explain to you how to interpret the test results at the end of these eight phases. If you need assistance ensuring that your data meets assumptions 4, 5, and 6, which are required when doing a one-way ANOVA and can be tested using SPSS Statistics, please contact us.

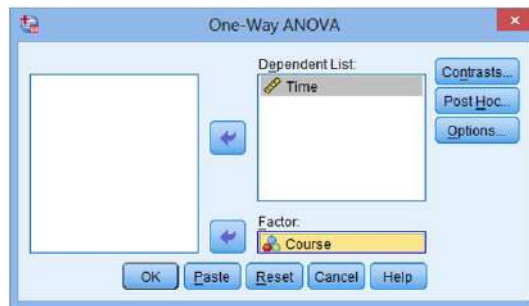
Step 1: Click Analyze > Compare Means > One-Way ANOVA... on the top menu, as shown below.



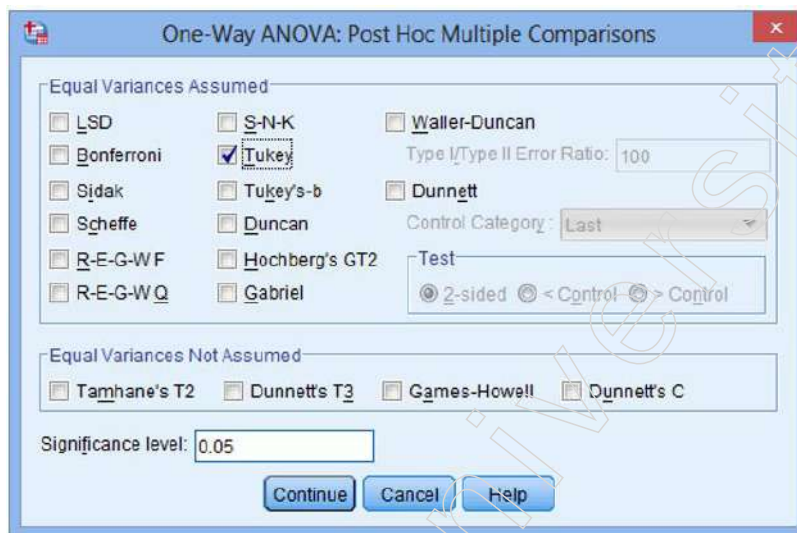
Step 2: You will be presented with the One-Way ANOVA dialogue box:



Step 3: Transfer the dependent variable, Time, into the Dependent List: box and the independent variable, Course, into the Factor: box using the appropriate Right arrow buttons (or drag-and-drop the variables into the boxes), as shown below:

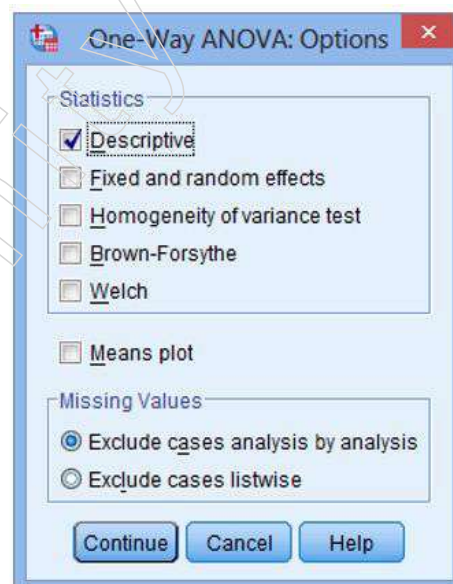


Step 4: Click on the Post hoc button. Tick the Tukey checkbox as shown below:



Step 5: Click on the Continue button.

Step 6: Click on the Options button. Tick the Descriptive checkbox in the – Statistics– area, as shown below:



Notes

“NOTE: When testing for some of the assumptions of the one-way ANOVA, you will need to tick more of these checkboxes.”

Step 7: Click on the Continue button.

Step 8: Click on the OK button.

3.4.4 Run Example on Non-parametric Test

Since non-parametric tests are “distribution-free,” they can be applied with non-Normal variables.

Advantages

Advantage 1: Nonparametric tests look at the median, which can be more useful in some cases.

Nonparametric analyses have an advantage in some datasets because they evaluate the median rather than the mean. The mean isn't necessarily the best indicator of a sample's central tendency. Even if you can run a valid parametric analysis on skewed data, that doesn't always mean it's the preferable method. Let me illustrate using the salary distribution.

Salaries are typically biased to the right. The majority of earnings are clustered around the median, which is the point where half of the workers are paid more than the other half. There is, however, a lengthy tail that extends into the higher pay ranges. The extended tail of the distribution moves the mean away from the central median value. Both of these distributions are common in salary distributions.

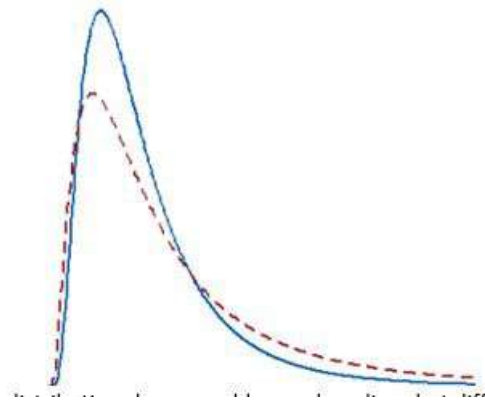
When several very high-income individuals join the sample, the mean increases by a large amount, despite the fact that the majority of people's salaries remain the same. They're still grouped together around the median.

Parametric and nonparametric test results can give you different outcomes in this case, and both can be correct.

If you take a big random sample from each population, the difference between the means is statistically significant for both distributions. Despite this, there is no statistically significant difference in the medians. This is how it works.

Changes in the tail have a significant impact on the mean in skewed distributions. This mean change can be detected using parametric tests. The median, on the other hand, is unaltered, and a nonparametric analysis can legitimately show that the median has not moved appreciably.

You must select whether the mean or median is the best choice for your study, as well as whatever type of difference to look for.



These two distributions have roughly equal medians but different means.

Advantage 2: When our sample size is limited and our data is potentially nonnormal, nonparametric tests are appropriate.

When your sample size isn't large enough to meet the conditions in the table above and you're not confident your data follows the normal distribution, use a nonparametric test. Be aware that normality tests with small sample numbers may lack the capacity to yield useful results.

This is a challenging scenario. Nonparametric studies have lower power to begin with, and a small sample size further adds to the difficulty.

Advantage 3: Ordinal data, ranking data, and outliers can all be analyzed using nonparametric tests.

Outliers can skew the results of parametric tests, which can only assess continuous data. Nonparametric tests, on the other hand, may assess ordinal and ranked data without being tripped up by outliers.

If outliers represent exceptional conditions, you may be able to lawfully exclude them from your dataset. However, outliers are sometimes a legitimate element of a research area's distribution and should not be removed.

Since different tests can analyze different types of data and have varying abilities to manage outliers, you should double-check the assumptions for nonparametric analysis.

Mann-Whitney U Test using SPSS Statistics

Introduction

When the dependent variable is ordinal or continuous but not normally distributed, the Mann-Whitney U test is used to compare differences between two independent groups. For example, you could use the Mann-Whitney U test to see if views against wage discrimination differ by gender when attitudes are measured on an ordinal scale (i.e., your dependent variable would be "attitudes towards pay discrimination" and your independent variable would be "gender", which has two groups: "male" and "female"). Alternatively, you might use the Mann-Whitney U test to see if salaries changed based on educational level when measured on a continuous scale (i.e., your dependent variable would be "salary" and your independent variable would be "educational level",

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which has two groups: “high school” and “university”). Although this is not always the case, the Mann-Whitney U test is typically thought of as a nonparametric alternative to the independent t-test.

The Mann-Whitney U test, unlike the independent-samples t-test, allows you to draw different conclusions about your data based on the assumptions you make about the distribution of your data. These results can range from simply declaring whether the two populations are different to establishing whether the medians of the two groups differ. These various conclusions are based on the shape of your data's distributions, which we'll go over in more detail later.

We walk you through all of the steps necessary to understand when and how to apply the Mann-Whitney U test in our improved Mann-Whitney U test guide, including how to interpret and report your results using SPSS Statistics. Subscribing to Laerd Statistics gives you access to this upgraded Mann-Whitney U test tutorial. When the key assumption of the Mann-Whitney U test is broken, we show you the foundations of the test using one of SPSS Statistics' methods in this “fast start” guide. Before we show you how to conduct it, we'll go over the various assumptions that must be met by your data for a Mann-Whitney U test to be valid. Following that, we'll talk about these assumptions.

Assumptions

When you decide to use a Mann-Whitney U test to analyse your data, you must first ensure that the data you wish to analyse can really be analysed using a Mann-Whitney U test. You must do this because a Mann-Whitney U test is only applicable if your data “meets” four assumptions that must be met for a Mann-Whitney U test to produce a valid result. In practise, verifying for these four assumptions takes a little longer because it requires you to click a few more boxes in SPSS Statistics and think a little harder about your data, but it's not a tough effort.

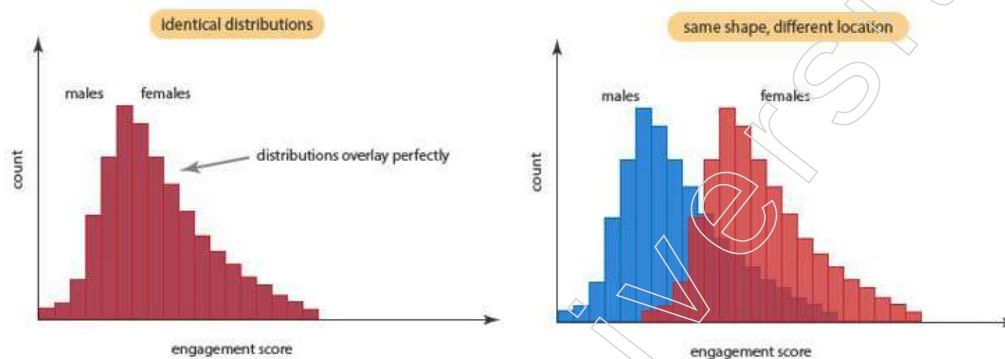
Do not be startled if one or more of these assumptions is violated when analysing your own data using SPSS Statistics before we expose you to these four assumptions (i.e., is not met). This is frequent when working with real-world data rather than textbook examples, which often only demonstrate how to perform a Mann-Whitney U test when everything goes smoothly! Don't worry, though. Even if your data contradicts certain assumptions, there is usually a way to fix it. First, consider the following four assumptions:

Assumption 1: The ordinal or continuous level should be used to measure your dependent variable. Likert items (e.g., a 7-point scale from “strongly agree” to “strongly disagree”), among other ways of rating categories, are examples of ordinal variables (e.g., a 5-point scale explaining how much a customer liked a product, ranging from “Not very much” to “Yes, a lot”). Continuous variables include things like revision time (measured in hours), intelligence (measured by IQ score), exam performance (measured from 0 to 100), and weight (measured in kg).

Assumption 2: Your independent variable should be divided into two distinct category groups. Gender (2 groups: male or female), employment status (2 groups: employed or unemployed), smoker (2 groups: yes or no), and so on are examples of independent variables that fit this condition.

Assumption 3: Observation independence indicates that there should be no association between the observations in each group or between the groups themselves. Each group, for example, must have diverse participants, with no person belonging to more than one group. Although this is more of a research design issue than a testable assumption, it is a key assumption of the Mann-Whitney U test. If your study fails to meet this criterion, you'll need to use another statistical test for the Mann-Whitney U test (e.g., a Wilcoxon signed-rank test). You can utilize our Statistical Test Selector, which is part of our upgraded material, if you're not sure if your study fits this assumption.

Assumption 4: When your two variables are not normally distributed, a Mann-Whitney U test might be utilized. To understand the results of a Mann-Whitney U test, you must first decide if your two distributions (i.e., the distribution of scores for both groups of the independent variable; for example, 'males' and 'females' for the independent variable, 'gender') are the same shape. Take a look at the diagram below to see what this means:



The distribution of scores for 'males' and 'females' in the two diagrams above has the same shape. Because the two distributions are identical (i.e., both distributions are identical, so they are 'on top of each other' in the diagram, with the blue-colored male distribution underneath the red-colored female distribution), the distribution of scores for 'males' (illustrated in blue on the diagram on the right) cannot be seen in the diagram on the left. Even though both distributions have the same shape in the diagram on the right, they are in different places (i.e., the distribution of one of the groups of the independent variable has higher or lower values than the second distribution – in our example, females have 'higher' values than males overall).

It is exceedingly improbable that your two distributions will be identical when you analyze your own data, although they may have the same (or a 'similar') shape. If they indeed have the same shape, you can compare the medians of your dependent variable (e.g., engagement score) for the two groups (e.g., males and females) of the independent variable (e.g., gender) you're interested in using SPSS Statistics. If your two distributions have different shapes, you can only compare mean ranks using the Mann-Whitney U test.

As a result, while doing a Mann-Whitney U test, you must additionally utilize SPSS Statistics to assess whether your two distributions are of the same shape or not. This necessitates a few extra SPSS Statistics procedures, but it's a simple step-by-step approach that we demonstrate in our expanded Mann-Whitney U test guide. We show you how to perform a Mann-Whitney U test in this "fast start" instruction, assuming that

Notes

your two distributions do not have a comparable shape and that you can only compare mean ranks, not medians.

SPSS Statistics can be used to test assumption #4. You should check that your data fulfills assumptions 1, 2, and 3 before proceeding, albeit you don't need SPSS Statistics to do so. Remember that if you don't check assumption 4, you won't know whether you're comparing mean ranks or medians correctly, and your Mann-Whitney U test findings could not be legitimate. This is why a section of our expanded Mann-Whitney U test tutorial is dedicated to assisting you with this.

Example

Cholesterol (a form of fat) concentration in the blood is linked to the chance of getting heart disease, with greater cholesterol concentrations indicating a higher risk and lower concentrations indicating a lower risk. Your chance of having heart disease can be reduced by lowering cholesterol levels in the blood. The concentration of cholesterol in your blood rises when you are overweight and/or physically sedentary. Exercise and weight loss can both help to lower cholesterol levels. However, it is unclear whether exercise or weight loss is the most effective way to lower cholesterol levels. As a result, a researcher chose to see if a weight loss or exercise intervention would be more helpful in lowering cholesterol levels. To this goal, the researcher gathered a random sample of overweight males who were inactive. The participants were then divided into two groups: Group 1 followed a calorie-controlled diet (the 'diet' group) and Group 2 followed an activity-training programme (the 'exercise' group). Cholesterol concentrations were compared between the two groups at the end of the treatment programmes to determine whether treatment programme was more effective.

Setup in SPSS Statistics

The scores for cholesterol concentration, our dependent variable, were input into SPSS Statistics under the variable name Cholesterol. Then, to represent our independent variable, we built a grouping variable called Group. We assigned a value of "1" to the diet group and a value of "2" to the exercise group because our independent variable comprised two groups: "diet" and "exercise." If you don't identify your two groups, SPSS Statistics won't be able to tell them apart, and the Mann-Whitney U test won't be performed.

We show you how to appropriately enter data into SPSS Statistics to perform a Mann-Whitney U test for both the Nonparametric Tests > Independent Samples and Legacy Dialogs > 2 Independent Samples procedures indicated in the point above in our updated Mann-Whitney U test tutorial.

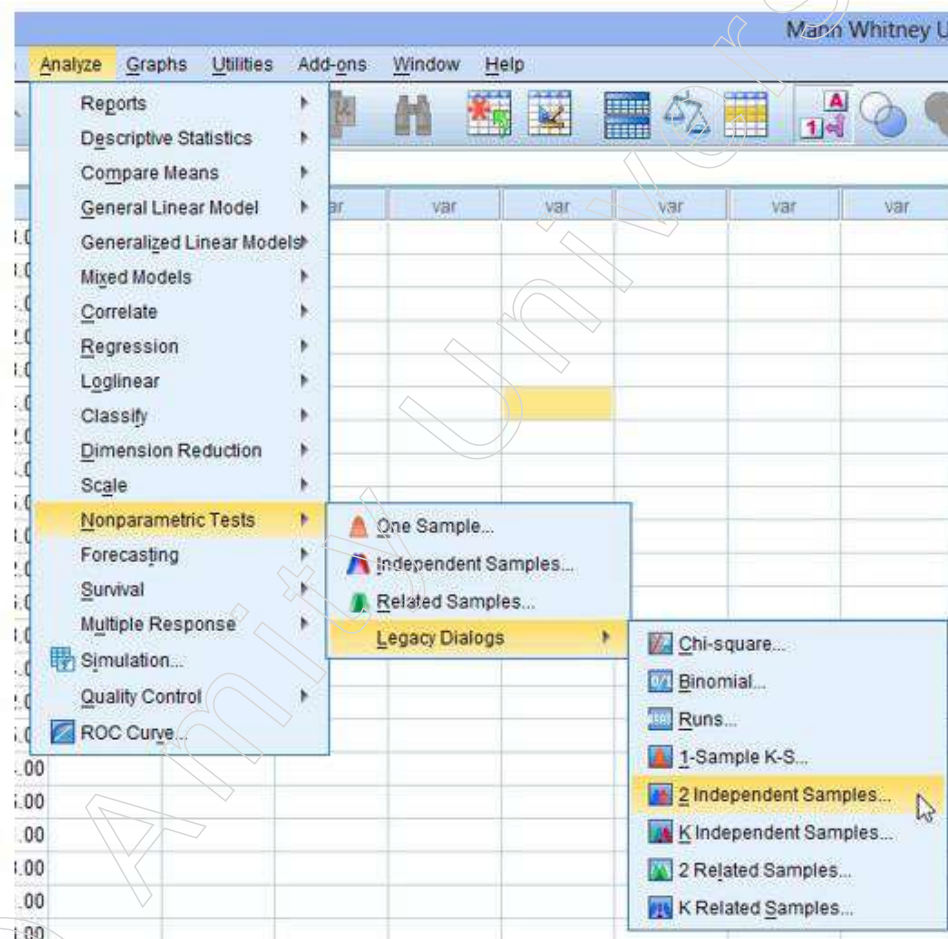
Test Procedure in SPSS Statistics

If you remember from assumption #4, the SPSS Statistics technique for analysing your data using a Mann-Whitney U test varies depending on the shape of the two independent variable distributions. We are referring to the two distributions of the independent variable, Group (i.e., the distribution of scores for Group 1 – the 'diet' group – and Group 2 – the 'exercise' group) in our example, where our dependent variable is cholesterol concentration, Cholesterol. We'll show you how to use a Mann-Whitney U test in SPSS Statistics to analyse your data when the two distributions have

a different shape and you need to compare the mean ranks of your dependent variable rather than medians.

You'll need to access the Procedures section of our enhanced Mann-Whitney U test guide to determine whether your two distributions have the same or different shapes, or if you want to know how to use SPSS Statistics to perform a Mann-Whitney U test when your two distributions have the same shape and you need to compare medians rather than mean ranks. In addition, the 10 steps below teach you how to use the Legacy Dialogs > 2 Independent Samples procedure in SPSS Statistics to perform a Mann-Whitney U test. As previously stated, there are two ways to execute a Mann-Whitney U test in SPSS Statistics: the Nonparametric Tests > Independent Samples process and the Legacy Dialogs > 2 Independent Samples procedure. If your two distributions have the same shape, we recommend using the Nonparametric Tests > Independent Samples technique, but the Legacy Dialogs > 2 Independent Samples procedure is OK if your two distributions have different shapes. In our updated Mann-Whitney U test guide, we show you both approaches.

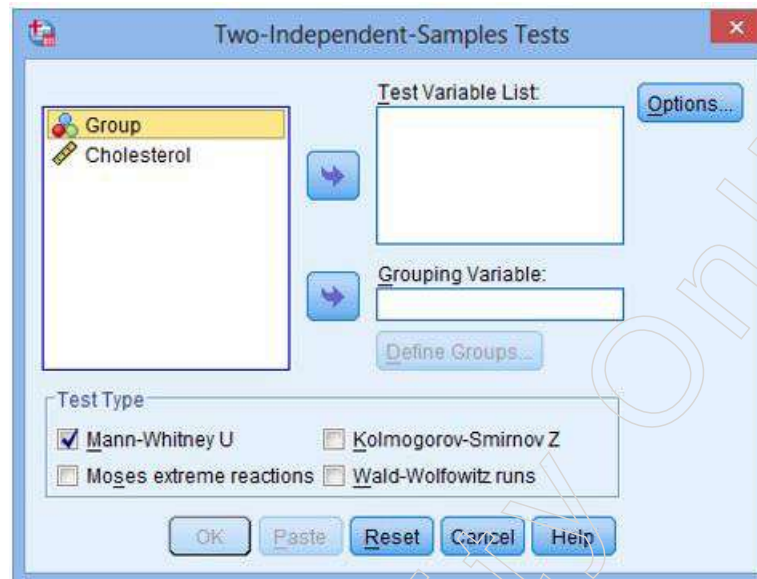
Step 1: Click Analyze > Nonparametric Tests > Legacy Dialogs > 2 Independent Samples... on the top menu, as shown below:



Step 2: You will be presented with the Two-Independent-Samples Tests dialogue box, as shown below:

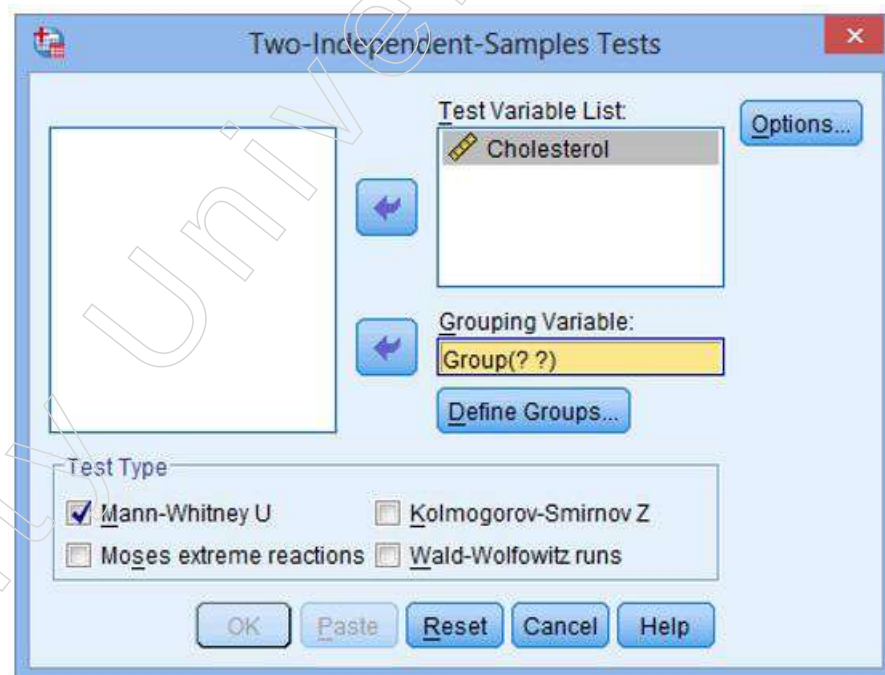
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Step 3: Transfer the dependent variable, Cholesterol, into the Test Variable List: box and the independent variable, Group, into the

Grouping Variable: box by using the Right arrow button or by dragging-and-dropping the variables into the boxes.



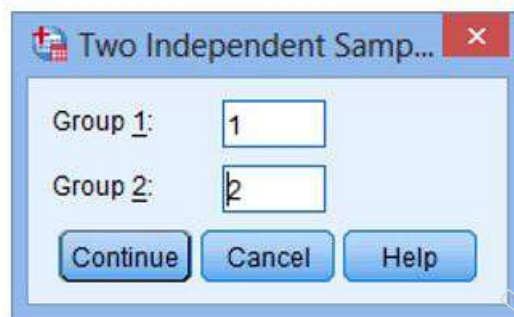
“Note: Make sure that the Mann-Whitney U checkbox is ticked in the –Test Type– area and the Grouping Variable: box is highlighted in yellow (as seen above). If it is not highlighted in yellow, simply click your cursor in the Grouping Variable: box to highlight it.”

Step 4: Click on the Define groups button. The button will not be clickable if you have not highlighted the Grouping Variable: box as instructed in Step 4. You will be presented with the following screen:

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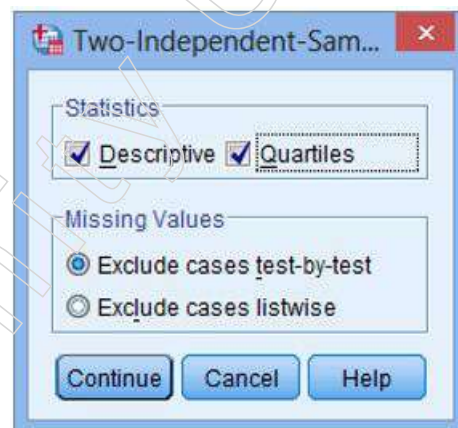
Step 5: Enter 1 into the Group 1: box and enter 2 into the Group 2: box. Remember that we labelled the Diet group as 1 and the Exercise group as 2.



“Note: If you have more than two groups in your study (e.g., three groups: diet, exercise and drug groups), but only wanted to compare two (e.g., the diet and drug groups), you could type 1 into the Group 1: box and 3 into the Group 2: box (i.e., if you wished to compare the diet with drug group).”

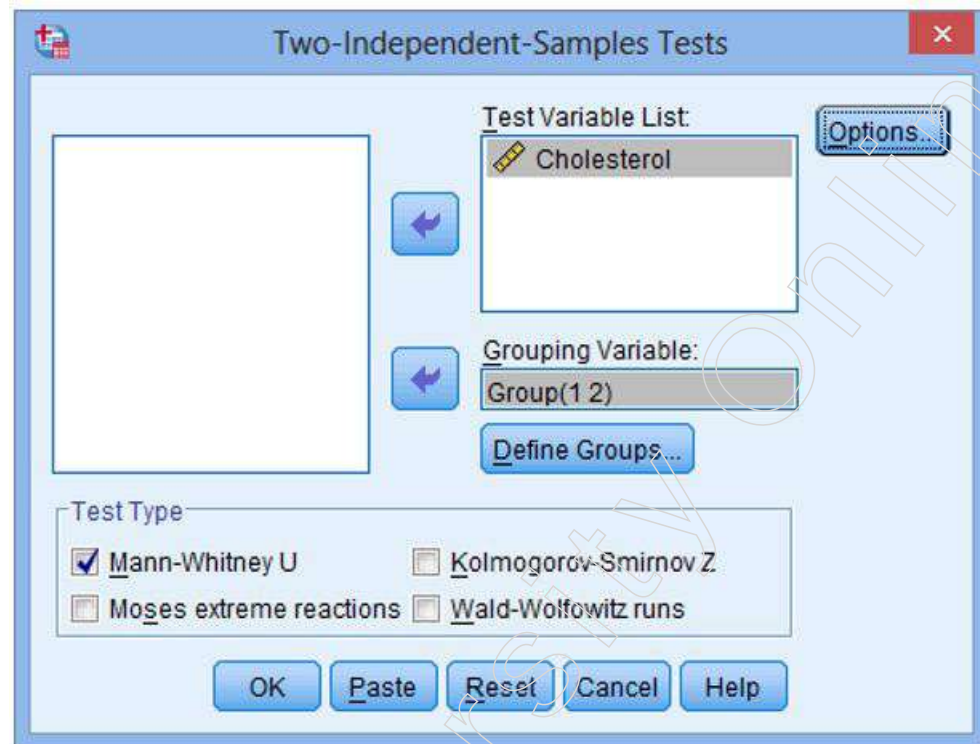
Step 6: Click on the Continue button.

Step 7: If you wish to use this procedure to generate some descriptive statistics, click on the Options button and then tick Descriptive and Quartiles within the – Statistics– area. You will be presented with the dialogue box below:



Step 8: Click on the Continue button, which will bring you back to the main dialogue box with the Grouping Variable: box now completed, as shown below:

Notes



Step 9: Click on the OK button. This will generate the output for the Mann-Whitney U test.

Check your Understanding

- _____ is the systematic use of statistical and logical techniques to describe the scope of the data, modularize the data structure, condense the data representation, illustrate via images, tables, and graphs, and evaluate statistical inclinations, probability data, and derive meaningful conclusions.
- One of the most important aspects of data analysis is ensuring data _____.
- _____ is a software suite designed for data management and statistical analysis in the social sciences.
- Manually editing cells in both views defines the file structure and allows data entry without requiring command syntax. True / False
- _____ system can also be used to analyze different forms of population and whatever factors can distress their choices.
- Data _____ is the process of gathering, collecting, extracting, and storing large amounts of data in organized or unstructured formats such as text, video, audio, XML files, records, or other picture files that will be utilized in data analysis later on.
- _____ can take numerous forms, including business knowledge, enterprise product sales, sickness treatment, and so on.
- The central limit theorem allows you to utilize these parametric tests with non-normally distributed data. True / False
- This "_____" instruction will show you how to use SPSS Statistics to do a one-way ANOVA, as well as how to analyze and report the findings.

10. Since non-parametric tests are “_____ - _____,” they can be applied with non-Normal variables.

Notes

Summary

- Data analysis is the systematic use of statistical and logical techniques to describe the scope of the data, modularize the data structure, condense the data representation, illustrate via images, tables, and graphs, and evaluate statistical inclinations, probability data, and derive meaningful conclusions. By removing the superfluous noise caused by the remainder of the data, these analytical approaches allow us to infer the underlying inference. Because data generation is a constant activity, data analysis is also a continuous, iterative process that includes both data collecting and data analysis. One of the most important aspects of data analysis is ensuring data integrity.
- SPSS is for Statistical Package for the Social Sciences, and it's a statistical data analysis programme utilized by a variety of scholars. SPSS is a software suite designed for data management and statistical analysis in the social sciences. It was first released in 1968 by SPSS Inc., and was later purchased by IBM in 2009.
- Manually editing cells in both views defines the file structure and allows data entry without requiring command syntax. For tiny datasets, this may be adequate. Larger datasets, such as statistical surveys, are more frequently prepared with data entry software, entered during computer-assisted personal interviewing, scanned and analyzed with optical character recognition and optical mark recognition software, or directly captured through online questionnaires. The data is then imported into SPSS.
- Data collection is the process of gathering, collecting, extracting, and storing large amounts of data in organized or unstructured formats such as text, video, audio, XML files, records, or other picture files that will be utilized in data analysis later on.
- Many online survey tools allow you to automatically export survey data to your data file. The processes for exporting the data are normally described in the online survey program's instructions, and you will often have the choice of exporting the data to Excel, SPSS, or SAS. When generating your web survey, you can take some measures to guarantee that your data file is set up appropriately when you retrieve the results.
- When you decide to use a one-way ANOVA to analyze your data, you must first ensure that the data you wish to analyze can really be analyzed using a one-way ANOVA. You must do this because a one-way ANOVA is only appropriate if your data “passes” the six assumptions that a one-way ANOVA must meet in order to provide a valid result. In practice, checking for these six assumptions takes a little longer because it requires you to click a few more boxes in SPSS Statistics and think a little harder about your data, but it's not a tough effort.
- Nonparametric analyses have an advantage in some datasets because they evaluate the median rather than the mean. The mean isn't necessarily the best indicator of a sample's central tendency. Even if you can run a valid parametric analysis on skewed data, that doesn't always mean it's the preferable method. Let me illustrate using the salary distribution.

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- The Mann-Whitney U test, unlike the independent-samples t-test, allows you to draw different conclusions about your data based on the assumptions you make about the distribution of your data. These results can range from simply declaring whether the two populations are different to establishing whether the medians of the two groups differ. These various conclusions are based on the shape of your data's distributions, which we'll go over in more detail later.

Activity

1. Find a raw set of data online and upload it to excel. Then following the steps in the unit, run the data through a one-way ANOVA.

Questions & Exercises

1. What are the prerequisites for good data analysis?
2. What are the primary functions of SPSS?
3. What are the six assumptions of one-way ANOVA?
4. What is the Mann-Whitney U test?

Glossary

- Superfluous: unnecessary, especially through being more than enough
- Syntax: the structure of words and phrases in computers or language
- Vicissitudes: a alteration between opposite or contrasting things

Further Readings

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10. Miles, Mathew B. and Huberman, A. Michael (1994): Qualitative Data Analysis. 2nd edn., New Delhi: Sage Publications.
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Check your Understanding – Answers

1. Data analysis
2. Integrity
3. SPSS
4. True
5. MANOVA
6. Collection
7. Knowledge
8. True
9. Fast start
10. Distribution-free

Notes

Module - 4: Associative and Predictive Techniques

Structure:

- 4.1 Correlation
 - 4.1.1 Correlation Coefficient
 - 4.1.2 Pearson Product-moment Correlation (Karl Pearson's Coefficient of Correlation - Introduction)
 - 4.1.3 Spearman Rank Correlation
 - 4.1.4 Linear-Nonlinear-Relationship
 - 4.1.5 Use Cases of Correlation in Research (Case Study Buisness Problem)
- 4.2 Regression
 - 4.2.1 Introduction to Regression
 - 4.2.2 Linear Regression
 - 4.2.3 Two Variable Linear Regression
 - 4.2.4 Regression Lines
 - 4.2.5 Regression Coefficients
 - 4.2.6 Method Least Square Understanding
 - 4.2.7 Maths Behind Least Square
 - 4.2.8 Polynomial Regression
 - 4.2.9 Understanding Model parameters
 - 4.2.10 Matrix Approach Multiple Regression
 - 4.2.11 Model Performance Measure R-sqaure
 - 4.2.12 Interpretation R-Square
 - 4.2.13 Model Evaluation MAPR RMSE
 - 4.2.14 Model Evaluation Multicollineraity Effect VIF
 - 4.2.15 Steps to Good Model
 - 4.2.16 Summary Correlation Regression
 - 4.2.17 Curve Fitting Excel
 - 4.2.18 Correlation Regression Analysis Excel
 - 4.2.19 CaseStudy BuisnessProblem and Data
 - 4.2.20 Casestudy CorrMatrix RegressionAnalysis
 - 4.2.21 CaseStudy Conclusion
 - 4.2.22 Use Cases of Regression in Research (Case study Corr Matrix Regression Analysis)

Unit - 4.1: Correlation

Notes

Objectives:

In this Unit, you will be able to:

- Understand Correlation Coefficient
- Learn Pearson Product-moment Correlation (Karl Pearson's Coefficient of Correlation - Introduction)
- Identify Spearman Rank Correlation
- Measure Linear-Nonlinear-Relationship
- Understand Use Cases of Correlation in Research (Case Study Business Problem)

Introduction

Research is a comprehensive and extensive investigation of a specific topic, concern, or issue using the scientific method. It's the adult version of elementary school science fair projects, in which you try to learn something by doing an experiment. This is best performed by transforming the problem into a question, with the goal of doing research to answer the queries.

Research may be about anything, and we hear about numerous kinds of research in the news. Cancer research has produced 'Breakthrough Cancer-Killing Treatment Has No Side Effects in Mice,' and 'Baby Born with HIV Cured.' Each of them started with an issue or a problem (such as cancer or HIV) and a query, such as, 'Does medicine X diminish malignant tissue or HIV infections?'

But all I've mentioned so far is what research has shown (kind of like stating baking leads to apple pie; it doesn't actually tell you anything other than the two are related). You must have a problem, worry, or issue that has evolved into a question before you can begin investigating it. These can be gleaned via watching the world, past studies, professional literature, or peer recommendations. Because your query must be answered, research truly begins with the appropriate question. Research cannot actually address questions such, 'How can I cure cancer?' It's too hazy and untestable.

The presence of a question induces an internal feeling of 'I need to know something.' To continue with the baking analogy, this internal feeling of desiring something is analogous to having a need for apple pie. Because you're reading this in the psychology area, we'll take a psychological approach to this, and perhaps eliminate some of the baked parallels.

Characteristics of Research

— All research initiatives should have the following eight key qualities. These are:

- Empirical – Based on tried-and-true scientific procedures developed from real-world observations and experiments.
- Logical – Adheres to a set of sequential operations based on sound ideas.

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- Cyclic – Research should start with a question and conclude with a question, i.e. research should lead to new lines of inquiry.
- Controlled – Strenuous efforts are made to maintain all factors constant, save those under inquiry.
- Hypothesis-based – The research design provides data that fits the study objectives and can confirm or reject the hypothesis. It makes the research investigation reproducible and gives the results credibility.
- Analytical – Data is created, collected, and analysed using tried-and-true processes that provide high accuracy and reproducibility while minimising possible mistakes and abnormalities.
- Objective – sound judgement is used by the researcher to ensure that the research findings are valid.
- Statistical treatment – The researcher employs good judgement to assure the validity of the study findings.
- Types of research



Exploratory Research

Exploratory research is defined as study that seeks to answer a variety of questions. Researchers perform exploratory studies to investigate a set of questions, as the term implies. Exploratory research does not give a solution to the subject under investigation. It is carried out to investigate previously unstudied issue areas. It establishes the groundwork for future study. It's a type of framework for future, more in-depth data collecting and analysis.

Descriptive Research

Descriptive research is data-driven and seeks to broaden understanding of contemporary challenges. Descriptive research is concerned with characterising the behaviour of the people being investigated. It simply requires one variable to be studied. Descriptive research focuses on describing the phenomena, understanding it, and confirming the study's conclusions. As an example, its primary goal is to increase information about current situations through data collecting. Descriptive research examines the behaviour of a sample population. To carry out the investigation, just one variable is necessary. The three main goals of descriptive research are to describe, explain, and validate the findings. Descriptive research is referred to as an

observational research approach since none of the variables in the study are modified throughout the research process.

Explanatory Research

Explanatory research design employs a researcher's views and opinions on one issue as a leading point for subsequent investigations; it is also used to explore hypotheses further. The research focuses on describing previously unknown patterns of a phenomenon and delving into the specifics of research issues such as what, why, and how.

Correlational Research

The study of the link between two variables is known as correlational research. This form of study examines only two variables in order to identify and render the link between variables suspected of being related in some way.

This study attempts to make sense of the factors discovered in earlier phases of research. Although correlational research is insufficient to draw conclusions about cause and effect relationships, it is required to perform it in order to determine whether a link between variables exists in the first place.

It is a non-experimental observational research method; there is no controlling or modification of the variables involved. The connection between the variables might be positive, negative, or 0 percent (non-existent).

Causal Research

Causal research is based on the task of establishing cause and effect correlations. As such, it entails performing experiments and market testing in a controlled environment. It is more scientific than any of the other study categories.

This type of study employs data from correlational and explanatory research to try to uncover causal linkages. Because correlation does not imply causation, causal research investigates whether variables having a negative or positive correlation influence the other variable(s) in the study.

Causal research has two goals: determining which factors constitute the cause and which constitute the effect, and understanding the link between the causal variables after the effect has occurred.

Experimental Research

A scientific research plan is rigorously followed in experimental study. It is essentially scientific, more so than causal research, in that it virtually, if not completely, applies the scientific approach to discovering a solution.

This is the last step of the research process, in which all of the information from the preceding phases is used to execute an experiment to test a hypothesis. It might also come after causal research, which is a type of experimental study.

Researchers can run additional tests on the variables for which they discovered causal links, in which they can investigate ways to reverse or lessen an unfavourable

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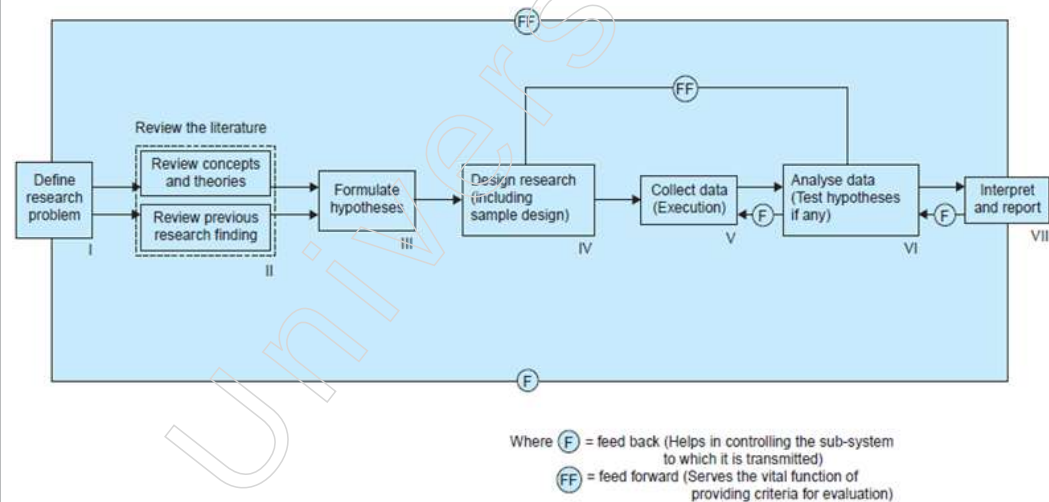
correlation. Alternatively, more tests might inform a company how to maximise the advantages of a desired association.

• The Research Process

The Research Process is a series of scientific stages that must be followed in order to do research. Each step is related to the ones before it. The procedure begins with the research problem. Then it proceeds sequentially to the following phases. In general, a researcher conducts research in seven phases. A Research Proposal is largely required in research activity. It is because the proposal accepts the research project whether or not you are able to undertake research. So, when writing a research proposal, be sure to include the exact strategies and explicit objectives of your research.

Steps of the research process

The research process consists of a sequence of activities or stages required to carry out research efficiently, as well as the intended sequencing of these steps. The diagram in Figure depicts a research process very effectively. According to the diagram, the research process consists of a number of closely connected tasks.



Step 1: Identify and defining the Research Problem

The first stage is to identify an issue or formulate a research topic. The research challenge might be anything that the agency perceives as a problem, some knowledge or information that the agency requires, or the desire to detect a national recreation trend. However, the research challenge raises the topic of a continuous phenomenon or difficulties. Create Your Question:

- Your research may begin with a broad concept or a precise question, assertion, or thesis.
- Before you begin, decide what you want to focus on.

A research problem is a declaration about an area of concern, a situation that needs to be better, a challenge that needs to be solved, or a problematic question that exists in academic literature, theory, or practise and indicates the need for meaningful understanding and purposeful exploration. The research challenge is frequently stated in the form of a question in various social science subjects. A research topic does not tell you how to achieve anything, it does not make a vague or wide claim, and it does not pose a value question.

Step 2: Review the Literature

After identifying and defining the research problem, the following stage is to examine previous research. The researcher must gain a better understanding of the subject under inquiry. To do this, the researcher needs to conduct a literature study on the research problem. This stage offers basic information regarding the situation. The examination of literature also informs the researcher about previous investigations, how these studies were done, and the findings in the issue area.

Obtain Background Information

- Use websites or encyclopaedias to learn more about your issue.
- It exposes you to the issue, assists you in focusing on its important parts, and can assist you in deciding whether to extend or limit your emphasis.
- These sites frequently provide bibliographies from which you may “piggyback” to identify further information on your topic.

In the obesity study, the review of literature enables the programmer to discover horrifying statistics related to the long-term effects of childhood obesity in terms of health issues, death rates, and projected medical costs. In addition, the programmer finds several articles and information from the Centres for Disease Control and Prevention that describe the benefits of walking 10,000 steps a day. The information discovered during this step helps the programmer fully understand the magnitude of the problem, recognize the future consequences of obesity, and identify a strategy to combat obesity (i.e., walking).

Step 3: Formulating a Hypothesis

The researcher refines the problem in this stage.

- The research is topic-specific and polished.
- The researcher next moves on to how the problem would be tackled. The nature of the study challenge may influence the formulation of a specific hypothesis.
- A hypothesis is validated. Effective research work formulates a hypothesis in such a way that acquired factual data either supports or disproves it. The formulation of hypotheses in research will make you more knowledgeable.
- Eventually, the hypothesis becomes a workable theory.

The following technique should be used by researchers to establish working hypotheses.

- a) Discussions with co-workers and specialists regarding the problem, its origins, and the goals of finding a solution;
- b) Examine data and records pertaining to the problem, if available, for possible patterns, anomalies, and other indications;
- c) A review of comparable research in the region or studies on similar difficulties; and
- d) Exploratory personal inquiry, which entails limited-scale original field interviews with relevant parties and persons in order to gain a better understanding of the practical elements of the situation.

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Step 4: Research Design

The researcher must then devise a study plan. The research design determines how research materials will be gathered. Depending on the study objectives, one or more research methods, such as an experiment, survey, or interview, are used.

- A survey may be appropriate in some research circumstances. In other cases, interviews, case studies, or observation may be preferable.
- Research Design elucidates “how” to do research utilising a certain Research Methodology. Essentially, every researcher has a list of research questions that must be evaluated, which may be accomplished through study design.

In other words, the purpose of study design is to allow for the collection of meaningful evidence with the least amount of effort, time, and money. However, how all of this is accomplished is primarily determined by the study objective. There are four types of research objectives.

- a) Exploration,
- b) Description,
- c) Diagnosis, and
- d) Experimentation.

There are numerous study designs, for example,

- Experimental and hypothesis testing

Experimental designs might be formal or casual (such as before-and after without control, after-only with control, before-and-after with control)

Formal designs (such as entirely randomised designs, randomised block designs, Latin square designs, basic and sophisticated factorial designs) must be chosen by the researcher for his or her particular study.

- Non-experimental hypothesis testing.

Step 5: Carry out the Research Process

The researcher collects data and records information while the research strategy is being chosen. The researcher continues his or her investigation. During this point, practical challenges may develop. For instance, the research approach may not be appropriate. The interviewer may be hesitant to allow the research to go as intended. Furthermore, a faulty interpretation might possibly skew the study's findings. So, when collecting data, you must be aware of the most successful data gathering strategies in order to get essential and relevant information for study.

Step 6: Preparing Research Results

Now, consider the ramifications of the information you've obtained. Your difficulties are not yet over. Problems may only be getting started! It is not easy to sort through the implications of the items obtained. While it is feasible to narrow down the research issues, certain studies are less definitive. So, in order to convey the findings, interpret your study findings. Whatever type of study you are conducting, there will come a time when your brain will be filled with ideas generated by your analysis. Ideally, you should jot them down as they occur to you. Now you must condense all of those materials and

concepts into a written narrative that is understandable to the reader and does honour to your mission.

Step 7: Reporting Research Findings

The research findings are reported in the final phase of the research process overview. Explain the importance of the research project. Determine how they connect to past study findings. The research report is usually published as a journal article or book. This is the final stage of the particular research study. A research report mostly covers outstanding questions and suggests additional study in general. This also represents how you compose your research paper. You must properly document your study findings.

- Approaches of Research: Deductive, Inductive

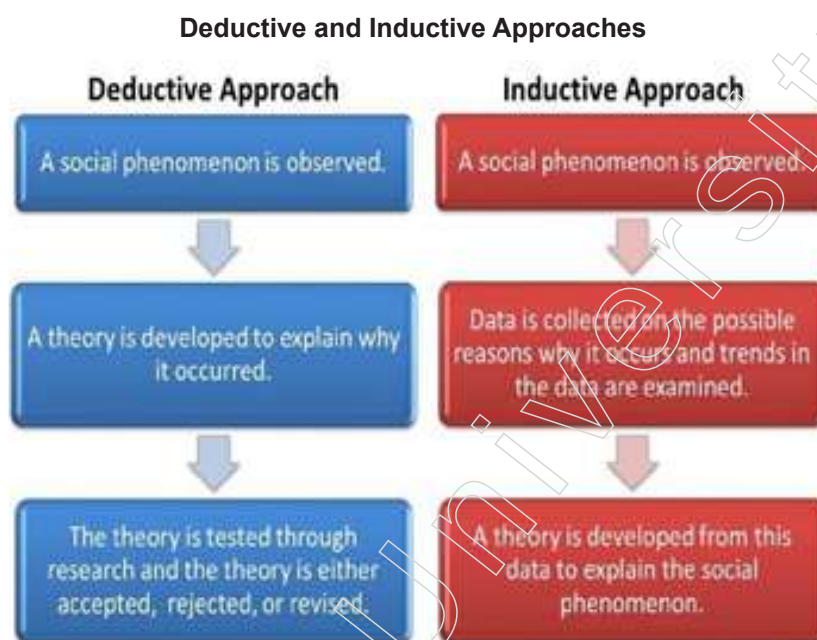


Figure: Image showing approaches made under deductive and inductive research approach

a) Inductive Research Approach

Since there is no theory to test, it is customary to do inductive research when there are few or no current works on a topic. The inductive technique consists of three steps, which may be described using the following examples:

1. Observation

- o A low-cost commercial airline flight has been rescheduled.
- o Fleas have infested dogs A and B.
- o Elephants be contingent on water to survive.

2. Detect a pattern

- o An additional 20 low-cost airline flights have been rescheduled.
- o Fleas have been found on all of the dogs observed.

Notes

- o All observed creatures rely on water to survive.

3. Grow a theory

- o Low-cost planes are always delayed.
- o All dogs have fleas.
- o All biological life depends on water to live.

Restrictions of the Inductive Approach

A conclusion formed on the basis of an inductive technique cannot be proven once and then refuted.

Example

You witness 1000 low-cost airline flights. They all perceive a delay, which is consistent with your idea. However, you can never prove that flight 1001 will be late as well. Nonetheless, the broader your dataset, the more trustworthy the result.

b) Deductive Research Approach

When conducting deductive research, you always start with a theory (the outcome of inductive research). Deductive reasoning entails criticising these theories. There can be no deductive inquiry if there is no adjacent theory.

The deductive research technique is divided into four stages, which may be shown with the following examples:

4. Start with a present theory

- o There are usually delays with low-cost carriers.
- o All canines have fleas
- o All biological life must be dependent on water in order to exist.

5. Formulate a hypothesis based on existing theory

- o If the passengers accompanying the trip fly with a low-cost airline, they will constantly face delays.
- o Fleas have infested all of the pet pets in my bedsitting room building.
- o All land-dwelling animals rely on water to survive.

6. Bring together data to test the hypothesis

- o Gather low-cost airline flight numbers.
- o Flea-check all pets in the building.
- o Examine all terrestrial mammal species to see if they are water reliant.

7. Analyse the results: Does the data discard or support the hypothesis?

- o 5 out of 100 low-cost airline flights are not hampered = reject hypothesis
- o 10 of 20 dogs were flea-free = cast-off theory
- o All land mammal species need on water for survival = maintenance theory

Restrictions of a deductive approach

Deductive reasoning conclusions can only be right if all of the premises established in inductive training are true and the footings are obvious.

Example

- All dogs are flea-free (premise)
- Benno is a canine (premise)
- Benno collects fleas (conclusion)

Correlation is a statistical concept that describes how two variables move in tandem with one another. When two variables move in the same direction, they are said to be positively correlated. They have a negative correlation if they travel in opposite directions.

Correlation expresses the degree of a link between two variables numerically as the correlation coefficient. The correlation coefficient has values ranging from -1.0 to 1.0.

A perfect positive correlation is one with a correlation coefficient of precisely one. This means that as one security goes up or down, the other security follows in lockstep in the same direction. A complete negative correlation indicates that two assets move in opposing directions, whereas a zero correlation indicates that there is no linear link at all.

Correlations are significant in finance because they are used to anticipate future trends and manage portfolio risks. Correlations between assets may now be easily calculated utilising a variety of software applications and internet platforms. Correlations, along with other statistical ideas, are critical in the development and pricing of derivatives and other complicated financial products.

In modern finance, correlation is an extensively utilised concept. A trader, for example, may use previous correlations to forecast whether a company's stock would rise or decrease in reaction to a change in interest rates or commodity prices. Similarly, a portfolio manager may seek to limit risk by ensuring that the assets in their portfolio are not unduly connected with one another.

Example of Correlation

It is critical for investment managers, traders, and analysts to assess correlation since the risk-reduction advantages of diversification rely on this statistic. Financial spreadsheets and software can easily compute the value of correlation.

Suppose that an analyst has to calculate the correlation between the following two data sets:

X: (41, 19, 23, 40, 55, 57, 33)

Y: (94, 60, 74, 71, 82, 76, 61)

The process of discovering the connection consists of three parts. The first step is to add up all the X values to get SUM(X), then add up all the Y values to get SUM(Y), and multiply each X value by its corresponding Y value to get SUM(X,Y):

Notes

$$\text{SUM}(X) = (41 + 19 + 23 + 40 + 55 + 57 + 33) = 268$$

$$\text{SUM}(Y) = (94 + 60 + 74 + 71 + 82 + 76 + 61) = 518$$

$$\text{SUM}(X,Y) = (41 \times 94) + (19 \times 60) + (23 \times 74) + \dots (33 \times 61) = 20,391$$

In the next step take each X value, square it, and sum up all these values to find $\text{SUM}(x^2)$. The same must be done for the Y values:

$$\text{SUM}(X^2) = (41^2) + (19^2) + (23^2) + \dots (33^2) = 11,534$$

$$\text{SUM}(Y^2) = (94^2) + (60^2) + (74^2) + \dots (61^2) = 39,174$$

Noting that there are seven observations, n, the following formula can be used to find the correlation coefficient, r:

$$r = \frac{n \times (\sum(X,Y) - (\sum(X) \times \sum(Y)))}{\sqrt{(n \times \sum(X^2) - \sum(X)^2) \times (n \times \sum(Y^2) - \sum(Y)^2)}}$$

where: r=Correlation coefficient n=Number of observations

The correlation, in this example would be:

$$r = (7 \times 20,391 - (268 \times 518)) / \text{Square Root}((7 \times 11,534 - 268^2) \times (7 \times 39,174 - 518^2)) = 3,913 / 7,248.4 = 0.54$$

4.1.1 Correlation Coefficient

Correlation coefficients are used to assess the strength of a link between two variables.

A correlation coefficient, represented by r, measures the degree of relationship. It is a measure of linear connection and is commonly referred to as Pearson's correlation coefficient after its creator. If a curved line is required to express the connection, different and more difficult correlation measures must be utilised.

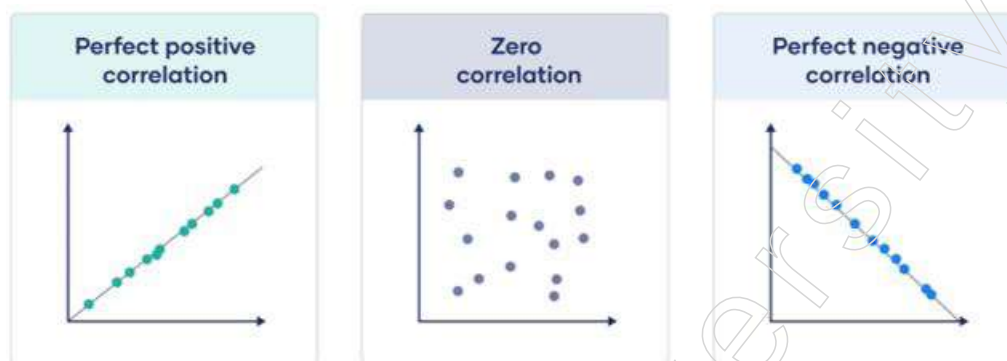
The correlation coefficient is evaluated on a scale ranging from + 1 through 0 to – 1. The degree of correlation between two variables is denoted by + 1 or -1. When one variable rises as the other rises, the correlation is positive; when one falls as the other rises, the correlation is negative. The complete lack of correlation is indicated by the value 0.

The correlation coefficient (ρ) is a metric that measures how closely two variables' movements are related. The Pearson product-moment correlation, which generates the most frequent correlation coefficient, is used to quantify the linear connection between two variables. This correlation coefficient, however, may not always be an acceptable measure of reliance in a non-linear connection.

A correlation coefficient is a number ranging from -1 to 1 that indicates the degree and direction of a link between variables. In other words, it indicates how comparable two or more variables' measurements are across a dataset.

Notes

Correlation coefficient value	Correlation type	Meaning
1	Perfect positive correlation	When there is a change in one variable, the other variables also change in the same direction.
0	Zero correlation	There is no relationship between the variables.
-1	Perfect negative correlation	When there is change in one variable, the other variables change in the opposite direction.



What Does Correlation Measure?

Correlation in statistics investigates and evaluates the direction and intensity of association between variables, therefore correlation measures co-variation rather than causality. As a result, correlation should never be interpreted as establishing a cause-and-effect relationship. For example, there is a correlation between two variables X and Y, which indicates that when the value of one variable changes in one way, the value of the other variable changes in either the same direction (i.e., positive change) or the opposite direction (i.e., negative change). Furthermore, if the correlation exists, it is linear, which means that the relative movement of the two variables may be represented by drawing a straight line on graph paper.

Scatter Diagram

A scatter diagram is a diagram that depicts the values of two variables, X and Y, as well as how these two variables relate to one another. Variable X values are supplied along the horizontal axis, whereas variable Y values are given on the vertical axis.

When the regression model is employed later, one of the variables is specified as an independent variable and the other as a dependent variable. The independent variable X is thought to have some effect or influence on the dependent variable Y in regression. Correlation approaches are symmetric in terms of the two variables, with no indication of causality or direction of impact included in the statistical analysis. In the following example, a scatter diagram is shown. The identical example is used subsequently to calculate the correlation coefficient.

Notes

Types Of Correlation Coefficients

All correlation coefficients have a highly distinct description based on their application areas and characteristics. Covariance, Pearson's Spearman's, and Polychoric Correlation Coefficient will be used throughout this study as the four primary correlation coefficients. The sort of correlation coefficient to choose is determined on the variables being researched.

1) Covariance Correlation Coefficient

In general, covariance correlation can be used when the characteristics have a regularly distributed relationship. It is simple to use to calculate the total of the variances of two specified attributes.

The orientation of the characteristics may be deduced using this coefficient. However, if it is necessary to measure the dependencies of the variables, an extra metric must be used because the covariance coefficient cannot respond to this connection.

The formulation of the covariance correlation is given below in simple terms.

$$\text{Covariance (X, Y)} = (\text{sum (x - mean(X)) * (y - mean(Y))}) * 1/(n-1)$$

2) Pearson's correlation

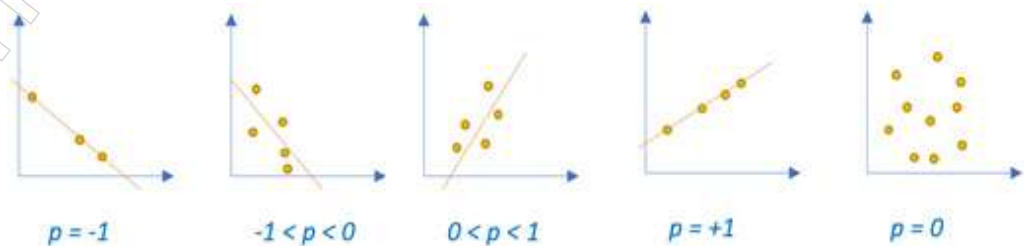
A linear relationship between two variables is also required for Pearson's correlation. It is known as the "Pearson product-moment correlation" in the literature.

The covariance correlation mathematical formula can be used to construct Pearson's correlation formulation.

$$\text{Pearson (X, Y)} = \text{Covariance (X, Y)} / (\text{stdDev(X)} * \text{stdDev(Y)})$$

Pearson correlation coefficient parameters can be viewed in five various ranges based on the variables' present placement on the x and y axes; however, the correlation's range is susceptible to change.

When the data points show a downward trend, this is referred to as a Negative Correlation. When, on the other hand, an upward trend is found, it may be labelled as a Positive Correlation between the two examined variables. p can be regarded as the correlation coefficient in the illustration below. You may look at the correlation alignments based on the distributions of the data points.



Pearson's product moment correlation coefficient is abbreviated as ρ for a population parameter and r for a sample statistic. It is employed when the variables under consideration are regularly distributed. This coefficient is influenced by extreme values, which can exaggerate or diminish the strength of the link, and is thus incorrect

when one or both variables are not normally distributed. The formula for computing the sample Pearson's correlation coefficient for a correlation between variables x and y is given by

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\left[\sum_{i=1}^n (x_i - \bar{x})^2 \right] \left[\sum_{i=1}^n (y_i - \bar{y})^2 \right]}}$$

where x_i and y_i are the values of x and y for the i th individual.

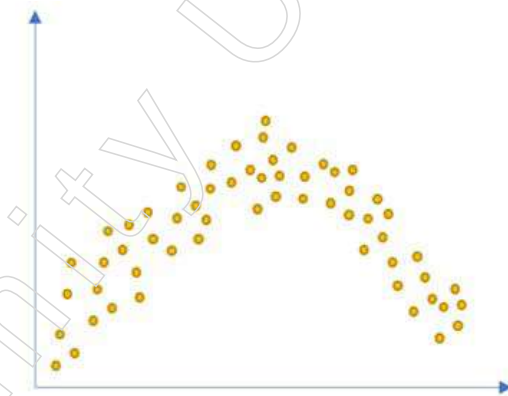
3) Spearman's rank correlation coefficient

Spearman's rank correlation coefficient is abbreviated s for a population parameter and r_s for a sample statistic. When one or both variables are skewed or ordinal, it is suitable, and it is resilient when extreme values are present. The formula for computing the sample Spearman's correlation coefficient for a correlation between variables x and y is provided by

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

Where d_i is the rank difference between x and y.

Spearman's correlation, also known as "Spearman's Rank-Order Correlation" in academia, is used when there is a lack of regularly distributed nature within the variable collection.



Spearman's correlation formulation may also be generated from the covariance correlation formula by including a variable ranking in the formulation. Ranking may be characterised as naming the arrangement of the variable labels in order to determine the level of correspondence among the observation couple, which can then be used to assess their influence rate on each other.

$$\text{Spearman } (X, Y) = \text{Covariance}(\text{rank}(X), \text{rank}(Y)) / (\text{stdDev}(\text{rank}(X)) * \text{stdDev}(\text{rank}(Y)))$$

Notes

Notes

4) Polychoric correlation

The polychoric correlation or tetrachoric correlation may be defined as the sort of coefficient used to analyse the connection of two Gaussian distributed continuous data acquired through surveys and personality tests and used to produce rating scales. As an example, consider the "Likert Scale," which is a frequently used scaling system in the domains of social sciences.

In the case of continuous data, this coefficient may be derived using Pearson's correlation coefficient.

4.1.2 Pearson Product-moment Correlation (Karl Pearson's Coefficient of Correlation - Introduction)

Karl Pearson's Coefficient of Correlation

Karl Pearson created this approach, commonly known as the Product Moment Correlation Coefficient. Along with the Scatter Diagram and Spearman's Rank Correlation, it is one of the three most powerful and widely used methods for measuring the amount of correlation.

The Karl Pearson correlation coefficient approach is quantitative and provides a numerical number to determine the strength of the linear relationship between X and Y. A coefficient correlation of this type is denoted by the symbol 'r.'

The Karl Pearson Coefficient of Correlation formula is denoted as –

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

In this formula,

r = Pearson's correlation coefficient

n = number of paired scores

X = score of the first variable

Y = score of the second variable

XY = the product of the two paired scores

Karl Pearson (1948) created Pearson's product moment correlation coefficient, or Pearson's r, from a related theory provided by Sir Francis Galton in the late 1800s. It is not only the first correlational measure to be devised, but it is also the most widely used measure of association. All subsequent correlation measures are modifications of Pearson's equation designed to correct for breaches of the assumptions that must be satisfied in order to utilise Pearson's equation (Burns & Grove, 2005; Polit & Beck, 2006). Pearson's r expresses the strength, direction, and likelihood of a linear relationship between two interval or ratio variables.

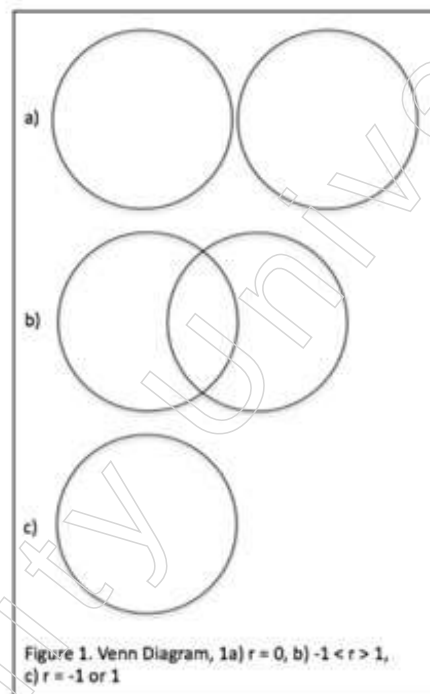
Pearson's Product Moment Correlation Coefficient - Pearson's r

Pearson's r, which can range between -1 and 1, is a measure of the linear

Notes

connection between two interval or ratio variables. It is the same as the point-biserial correlation; it assesses the link between a dichotomous (yes or no, male or female) variable and an interval/ratio variable (Cramer, 1998). Pearson's r has the benefit of being a straightforward way to analyse the relationship between two variables; whether they share variance (covary), whether the relationship is positive or negative, and the degree to which they correlate. The downside of using Pearson's r is that it cannot identify nonlinear interactions and may display a correlation of zero when the correlation has a relationship that is not linear. Furthermore, the sorts of variables that may be examined are restricted. In addition to Pearson's r , semi-partial and partial correlation may be used to quantify the connection between an outcome and a predictor variable after correcting for the impact of other predictor variables in the equation.

"Pearson's correlation is defined as the ratio of variance shared by two variables" (Cramer, 1998, p. 137). A Venn diagram can be used to demonstrate correlation (Figure 1). Each circle denotes the amount of variation in each variable. The circles in Figure 1a do not overlap, showing that there is no relationship between the two variables. In Figure 1b, the two circles overlap, and the degree of overlap reflects the extent of the connection. The circles would entirely cover each other in a perfect correlation (Figure 1c).



Pearson's r may be calculated using a variety of formulas. Pearson's r may be determined manually for small samples using:

$$r = \frac{n(\sum XY) - (\sum X)(\sum Y)}{\sqrt{[n(\sum X^2) - (\sum X)^2][n(\sum Y^2) - (\sum Y)^2]}}$$

Where:

r = Pearson's correlation coefficient

n = number of paired scores

Notes

X = score of the first variable

Y = score of the second variable

XY = the product of the two paired scores

Or similarly:

$$r = \frac{\text{covariance of variable A and B}}{\sqrt{(\text{variance of variable A}) \times (\text{variance of variable B})}}$$

Analogously:

$$r = \frac{\text{covariance of variable A and B}}{(\text{standard deviation of variable A}) \times (\text{standard deviation of variable B})}$$

"The greater the covariance, the more powerful the link." The covariance can never be more than the product of the two variables' standard deviations" (Cramer, 1998, p. 139). The denominator in the equations is always positive, while the numerator can be positive, negative, or zero, "...allowing r to be positive, negative, or zero accordingly" (Zar, 1999, p. 378).

Properties of Coefficient of Correlation

- The coefficient of correlation (r) value always lies between ± 1 . Such as:
 $r=+1$, perfect positive correlation
 $r=-1$, perfect negative correlation
 $r=0$, no correlation
- The correlation coefficient is unaffected by the origin or scale. By origin, it implies that when any non-zero constant is subtracted from the provided values of X and Y, the value of "r" stays constant. By scale, it implies that dividing or multiplying the values of X and Y by any constant has no influence on the value of "r."
- The correlation coefficient is the geometric mean of two regression coefficients. Symbolically, it is expressed as follows:

$$r = \sqrt{b_{xy} + b_{yx}}$$

- When the variables X and Y are independent, the coefficient of correlation is "zero." However, the opposite is not true.

Assumptions of Karl Pearson's Coefficient of Correlation

1. The connection between the variables is "Linear," which implies that when the two variables are plotted, the points plotted form a straight line.
2. A huge number of independent factors influence the variables under examination, resulting in a Normal Distribution. Variables such as price, demand, supply, and so on are impacted by such circumstances that the normal distribution is produced.
3. The variables are unrelated to one another.

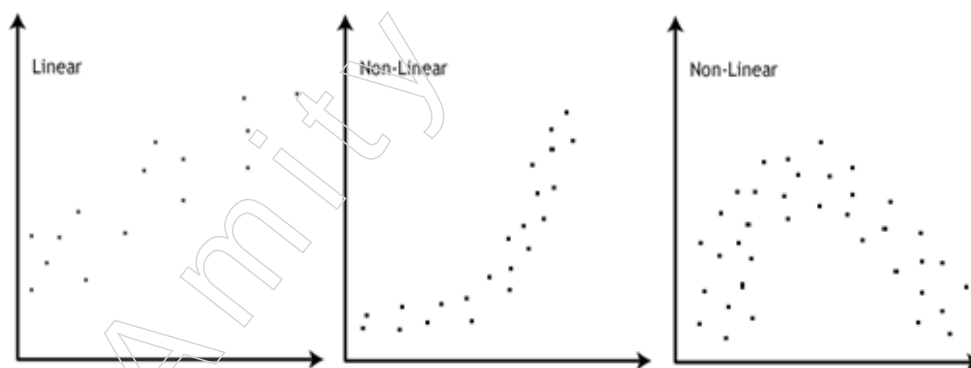
Assumptions

You must understand the many assumptions that your data must satisfy in order for a Pearson's correlation to produce a legitimate result. These assumptions are discussed more below.

When you want to analyse your data using Pearson's correlation, you must first ensure that the data you wish to analyse can be analysed using Pearson's correlation. You must do this because Pearson's correlation can only be used if your data "passes" four assumptions that Pearson's correlation requires in order to get a valid result. In practise, verifying for these four assumptions adds a little more time to your research, forcing you to click a few more buttons in SPSS Statistics when completing your analysis, as well as think about your data a little bit longer, but it is not a tough effort.

Before we go into these four assumptions, don't be shocked if one or more of them are broken when you analyse your own data with SPSS Statistics (i.e., is not met). This is not unusual when working with real-world data rather than textbook examples, which frequently only teach you how to use Pearson's correlation when everything goes smoothly! Don't worry, however. Even if your evidence contradicts certain assumptions, there is usually a way to overcome this. First, consider the following four assumptions:

- Assumption #1: Your two variables should be monitored on an interval or ratio basis (i.e., they are continuous). This condition is met by factors such as revision time (measured in hours), intellect (measured using an IQ score), exam performance (measured from 0 to 100), weight (measured in kg), and so on.
- Assumption #2: Your two variables have a linear connection. While there are other methods for determining if a linear relationship exists between your two variables, we recommend using SPSS Statistics to create a scatterplot in which you may plot one variable against the second variable and then visually evaluate the scatterplot for linearity. Your scatterplot may resemble one of the following:



If the connection in your scatterplot is not linear, you must either run a nonparametric counterpart of Pearson's correlation or modify the data, which you may accomplish with SPSS Statistics. In our enhanced guides, we show you how to: (a) create a scatterplot to check for linearity when using SPSS Statistics to perform Pearson's correlation; (b) interpret different scatterplot results; and (c) transform your data using SPSS Statistics if your two variables do not have a linear relationship.

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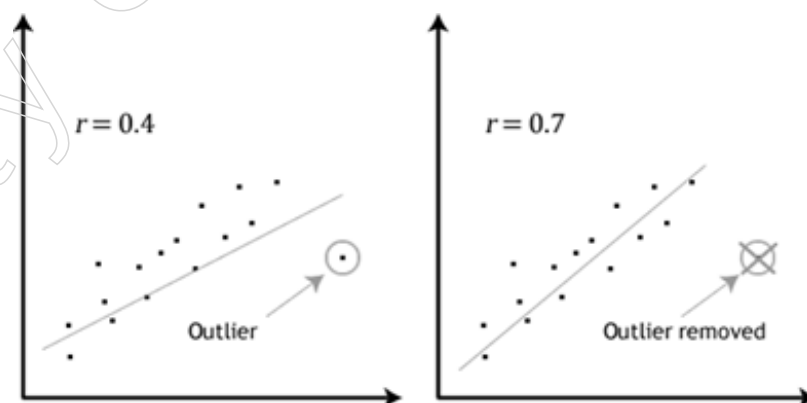
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Note: SPSS is an abbreviation for Statistical Package for the Social Sciences, and it is used by a variety of scholars for advanced statistical data processing. SPSS was designed for the administration and statistical analysis of social science data. SPSS Inc. established it in 1968, and it was subsequently bought by IBM in 2009.

Although it is officially known as IBM SPSS Statistics, most users still refer to it as SPSS. SPSS is widely regarded as the international standard for social-science data analysis because to its simple and English-like command language and amazingly extensive user manual. SPSS is used for processing and analysing survey data collected via an online survey platform like Alchemer by market researchers, health researchers, survey firms, government bodies, education researchers, marketing organisations, data miners, and many more. SPSS is commonly used to analyse survey data and mine text data in order to get the most out of research and survey initiatives.

Note: Pearson's correlation is used to determine the degree to which a connection is linear. In other words, it assesses if a linear component of relationship exists between two continuous variables. As a result, Pearson's correlation does not make any assumptions about linearity. However, you would not generally seek a Pearson's correlation to discover the strength and direction of a linear link if you already know that the relationship between your two variables is not linear. Instead, another statistical measure might better capture the relationship between your two variables. As a result, it is not uncommon to examine the relationship between your two variables in a scatterplot to determine whether running a Pearson's correlation is the best choice as a measure of association or whether another measure would be preferable.

- Assumption #3: No significant outliers should exist. Outliers are simply single data points within your data that do not follow the expected pattern (for example, in a study of 100 students' IQ scores, where the mean score was 108 with only a small variation between students, one student had a score of 156, which is very unusual and may even place her in the top 1% of IQ scores globally). The scatterplots below demonstrate the possible influence of outliers:



Pearson's correlation coefficient, r , is very sensitive to outliers, which can have a significant impact on the line of best fit and the Pearson correlation coefficient. As a result, adding outliers in your study might produce misleading findings in some circumstances. As a result, it is preferable if there are no outliers or if they are maintained to a minimum. Fortunately, when running Pearson's correlation on your data with SPSS Statistics, you can easily incorporate processes to filter for outliers. In our updated Pearson's correlation guide, we: (a) demonstrate how to discover outliers using

a scatterplot, which is a straightforward procedure when using SPSS Statistics; and (b) explore some of the alternatives open to you for dealing with outliers.

- **Assumption #4:** Your variables should have a roughly normal distribution. To estimate the statistical significance of the Pearson correlation, bivariate normality must be assumed; however, this assumption is difficult to assess, thus a simpler technique is more usually utilised. This less complicated technique entails determining the normalcy of each variable individually. The Shapiro-Wilk test of normalcy, which is readily checked for using SPSS Statistics, may be used to test for normality. In our improved Pearson's correlation tutorial, we not only teach you how to achieve this, but we also explain what you may do if your data violates this assumption.

4.1.3 Spearman Rank Correlation

It is named after Charles Spearman and is commonly indicated by the Greek letter ' ρ ' (rho). It is mostly used for data analysis.

It assesses the strength and directionality of the relationship between two ranking variables. But, before we get into the Spearman correlation coefficient, we must first grasp Pearson's correlation. A Pearson correlation is a statistical measurement of the strength of a linear relationship between two sets of data.

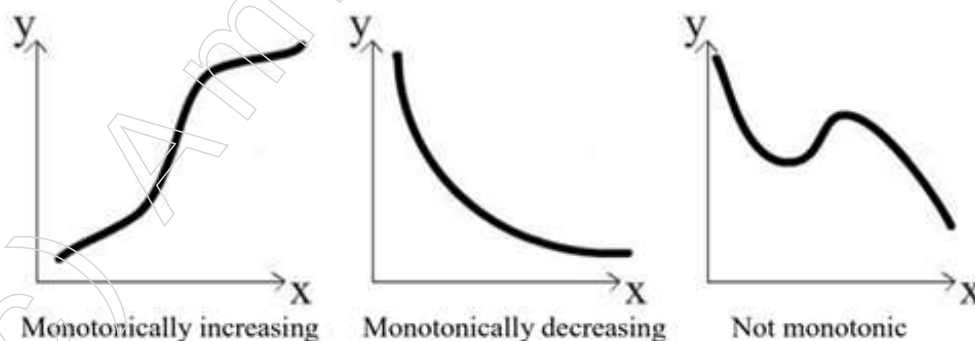
Before delving into Spearman's correlation, it is necessary to first grasp Pearson's correlation, which is a statistical measure of the strength of a linear relationship between paired data. Its computation and subsequent significance tests necessitate the following data assumptions:

- Interval or ratio level;
- Linearly related;
- Bivariate normally distributed.

If your data does not fit the above criteria, utilise Spearman's rank correlation!

Monotonic function

To grasp Spearman's correlation, you must first understand what a monotonic function is. As the independent variable rises, a monotonic function either never increases or never drops. The graphs below show monotonic functions:



- Monotonically increasing -as the x variable rises, the y variable will never increase;

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- Monotonically decreasing - as the x variable increases the y variable will never increase;
- Not monotonic - as the x variable increases the y variable sometimes decreases and sometimes rises.

Spearman's correlation coefficient

The Spearman's correlation coefficient is a statistical measure of the strength of a monotonic relationship between two sets of data. In a sample, it is indicated by r_s and is bound by design as follows:

$$-1 \leq r_s \leq 1$$

And its meaning is identical to Pearson's, for example, the closer r_s is to ± 1 , the greater the monotonic association. Correlation is an effect size; thus we can use the following advice for the absolute value of r_s to orally convey the strength of the correlation:

- .00-.19 "very weak"
- .20-.39 "weak"
- .40-.59 "moderate"
- .60-.79 "strong"
- .80-1.0 "very strong"

The following data assumptions must be met in order to calculate Spearman's correlation coefficient and assess its significance:

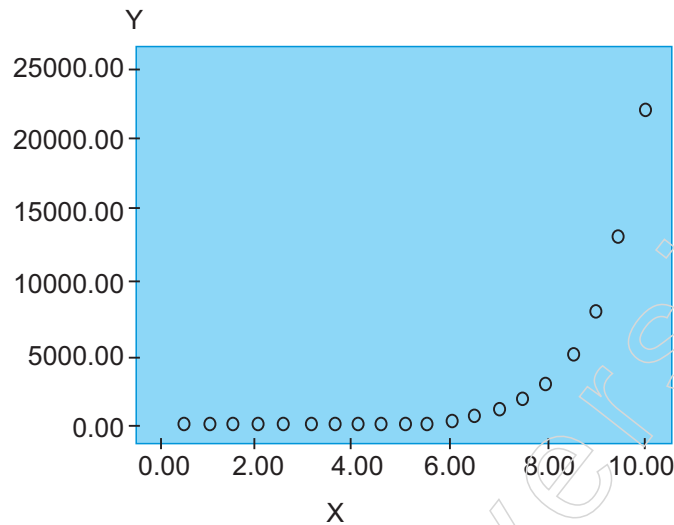
- Interval or ratio level or ordinal;
- Monotonically related.

It is worth noting that, unlike Pearson's correlation, there is no condition for normality, making it a nonparametric statistic. Let's look at some instances to see what I mean. The table below shows the x and y values for the relationship $y = \exp(x)$. We can observe from the graph that this is a properly growing monotonic connection.

	x	y
1	.5	1.6
2	1.0	2.7
3	1.5	4.5
4	2.0	7.4
5	2.5	12.2
6	3.0	20.1
7	3.5	33.1
8	4.0	54.6
9	4.5	90.0
10	5.0	148.4
11	5.5	244.7
12	6.0	403.4
13	6.5	665.1

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14	7.0	1096.6
15	7.5	1808.0
16	8.0	2981.0
17	8.5	4914.8
18	9.0	8103.1
19	9.5	13359.7
20	10.0	22026.5

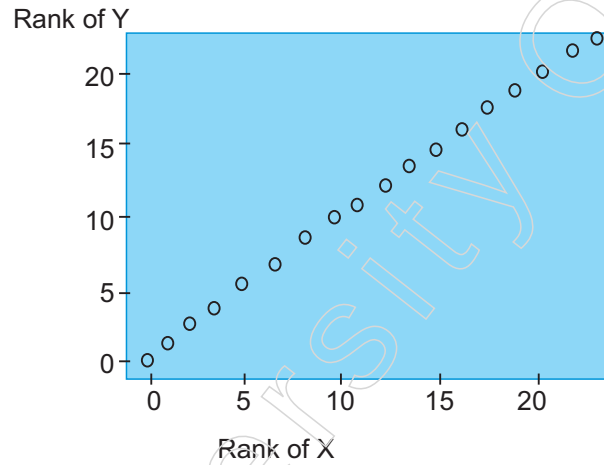


The Pearson's correlation calculation for this data yields a value of .699, which does not indicate that there is a perfect link between the variables. However, Spearman's correlation for this data is 1, indicating a perfect monotonic connection. Spearman's correlation is calculated by multiplying Pearson's correlation by the ranking values of this data. Ranking (from low to high) is calculated by assigning a rank of 1 to the lowest value, a rank of 2 to the next lowest, and so on. We can see from the plot of the ranking data that they are completely linearly connected.

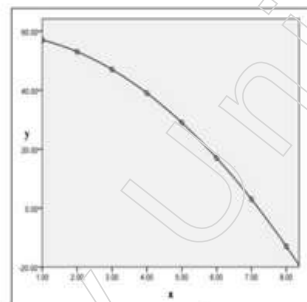
	x	Rank of x	y	Rank of Y
1	.5	1	1.6	1
2	1.0	2	2.7	2
3	1.5	3	4.5	3
4	2.0	4	7.4	4
5	2.5	5	12.2	5
6	3.0	6	20.1	6
7	3.5	7	33.1	7
8	4.0	8	54.6	8
9	4.5	9	90.0	9
10	5.0	10	148.4	10
11	5.5	11	244.7	11
12	6.0	12	403.4	12
13	6.5	13	665.1	13

Notes

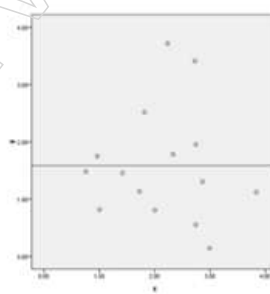
14	7.0	14	1096.6	14
15	7.5	15	1808.0	15
16	8.0	16	2981.0	16
17	8.5	17	4914.8	17
18	9.0	18	8103.1	18
19	9.5	19	13359.7	19
20	10.0	20	22026.5	20



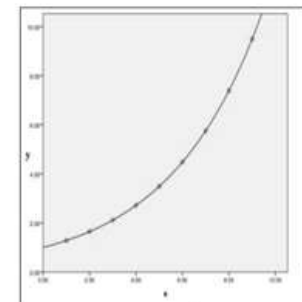
Various samples and their accompanying sample correlation coefficient values are shown in the figures below. The first three are monotonic correlation values of -1, 0 and 1:



$r_s = -1$
perfect -ve
monotonic correlation

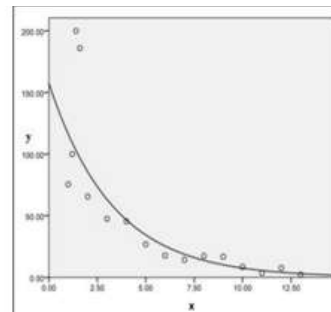


$r_s = 0$
no correlation

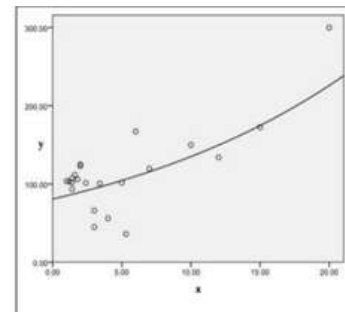


$r_s = 1$
perfect +ve
monotonic correlation

Invariably, we find the following values in a sample:

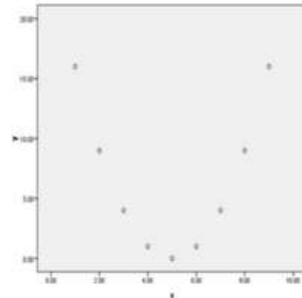


$r_s = -.941$
very strong -ve
monotonic correlation



$r_s = .372$
weak +ve
monotonic correlation

Note: Because Spearman's correlation coefficient is a measure of a monotonic connection, a result of $r_s = 0$ does not mean that the variables have no relationship. For example, in the scatterplot below, $r_s = 0$, implying that there is no (monotonic) correlation, yet there is a perfect quadratic relationship:



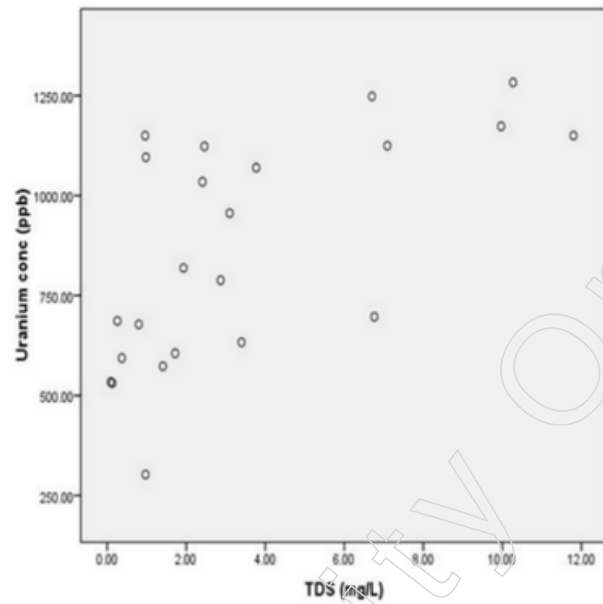
$r_s = 0$
perfect quadratic relationship

Example

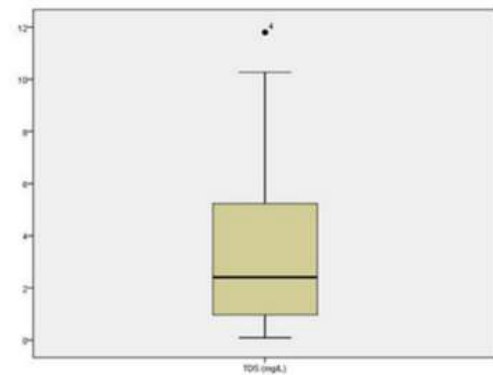
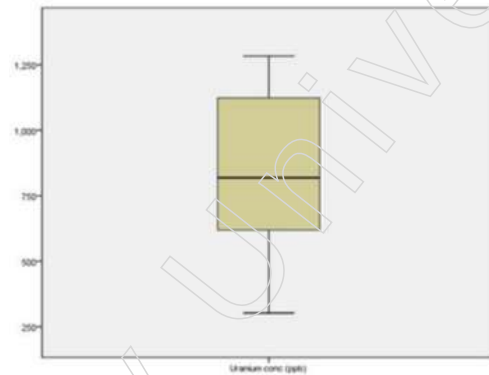
The following data set is made up of 23 groundwater samples that were collected and measured for Uranium content (ppb) and total dissolved solids (mg/L). Is it possible to find out if the two variables are related? If there are assumption breaches, we should first assess if Pearson's correlation is suitable or whether we should turn to Spearman's.

	Uranium conc (ppb)	TDS (mg/L)
1	678.10	.80
2	818.93	1.93
3	302.38	.97
4	1149.60	11.80
5	573.14	1.41
6	1034.55	2.41
7	633.25	3.40
8	1095.42	.98
9	1122.58	2.46
10	686.51	.26
11	1172.84	9.97
12	593.70	.37
13	1247.95	6.70
14	533.99	.09
15	605.51	1.72
16	696.96	6.76
17	1282.95	10.27
18	531.16	.13
19	788.36	2.87
20	956.06	3.10
21	1149.38	.96
22	1069.82	3.77
23	1124.17	7.09

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The scatterplot shows a clear positive relationship between Uranium and TDS. However, at TDS levels close to zero, there may be some evidence of non-linearity. However, this is debatable, so we'll go on to the other normalcy assumption. We must do some normality checks on the two variables. Examining data boxplots is a straightforward approach to accomplish this. These are provided below.



The boxplot for Uranium is pretty consistent with one from a normal distribution; the median is towards the centre of the box, and the whiskers are roughly equal in length. The boxplot for TDS is slightly troubling in that the median is near to the lower quartile and the lower whisker is shorter than the higher, implying positive skewness. There is also an outlier, and Pearson's correlation is sensitive to both these and skewness. Given our reservations about normality, we'll look at the skewness coefficients to see if there's any further evidence that one of the variables is skewed.

Descriptive Statistics			
	N	Skewness	
	Statistic	Statistic	Std. Error
Uranium conc (ppb)	23	-.148	.481
Valid N (listwise)	23		

Descriptive Statistics			
	N	Skewness	
	Statistic	Statistic	Std. Error
TDS (mg/L)	23	1.189	.481
Valid N (listwise)	23		

If the absolute values of the skewness coefficients are less than two times their standard errors, the skewness coefficients are not sufficiently big to justify worry. Using this technique, the skewness of the Uranium data is consistent with the data being

normal. However, the TDS skewness coefficient appears to be significant enough to raise the possibility of positive skewness ($1.189 > 2 \times .481$). As a result, we have worries about the normality of our data and should do a Spearman's correlation analysis. SPSS generates the following Spearman's correlation results

Correlations				
Spearman's rho	Uranium conc (ppb)	Correlation Coefficient	Uranium conc (ppb)	TDS (mg/L)
			1.000	.708
		Sig. (2-tailed)		.000
		N	23	23
	TDS (mg/L)	Correlation Coefficient	.708	1.000
		Sig. (2-tailed)	.000	
		N	23	23

** . Correlation is significant at the 0.01 level (2-tailed).

The considerable Spearman correlation coefficient value of 0.708, verifies the graph's findings: there appears to be a strong positive association between the two variables. As a result, high uranium levels are related with high TDS levels. However, we must do a significance test to see if there is any or no evidence to imply that linear connection exists in the population based on this sample. To do this, we compare the null hypothesis, H_0 , that there is no monotonic correlation in the population, to the alternative hypothesis, H_1 , that there is monotonic correlation; our data will show which of these opposing hypotheses is more likely to be true. If ρ_s is the Spearman's population correlation coefficient, we may phrase this test as follows:

$H_0: \rho_s \leq 0$

$H_1: \rho_s > 0$

Specifically, the null hypothesis of no monotonic correlation in the population against the alternative hypothesis of monotonic correlation in the population. Because SPSS provides a p-value of .000 for this test, we may conclude that we have very strong evidence to accept H_1 , i.e., we have some evidence that groundwater uranium and TDS levels are monotonically associated in the population.

This might be formalised as follows: "The association between 23 groundwater uranium and TDS levels was determined using Spearman's correlation. Uranium and TDS had a substantial, positive monotonic connection ($r_s = .71$, $n = 23$, $p < .001$)."

Spearman correlation coefficient: Formula and Calculation with Example

$$r_R = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Here,

n = number of data points of the two variables

d_i = difference in ranks of the "ith" element

The Spearman Coefficient, ρ , can have a value ranging from +1 to -1 where,

- A ρ value of +1 means a perfect association of rankings

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- A ρ value of 0 means no association of rankings
- A ρ value of -1 indicates a perfect negative association between rankings.

Closer the ρ value to 0, weaker is the association between the two rankings.

Before we can calculate the Spearman's Rank Coefficient of Correlation, we must first rank the data. It is critical to note that as one variable is increased, the other variable follows a monotonic relationship.

At each step, you must compare the values of the two variables. The calculations are as follows:

The results of nine students in History and Geography are shown in the table below.

Step 1- Make a table using the collected data.

Step 2- Begin by arranging the two data sets in descending order. To rank data, assign the rating "1" to the largest number in the column, "2" to the second largest number, and so on. Typically, the smallest value receives the lowest ranking. This should be done for both measurement sets.

Step 3- Insert a third column d into your data collection, where d represents the difference in rankings. For example, if the first student has a physics rank of 3 and a math rank of 5, the rank difference is 3. Square your d values in the fourth column.

History	Rank	Geography	Rank	D	d square
35	3	30	5	2	4
23	5	33	3	2	4
47	1	45	2	1	1
17	6	23	6	0	0
10	7	8	8	1	1
43	2	49	1	1	1
9	8	12	7	1	1
6	9	4	9	0	0
28	4	31	4	0	0
					12

Step 4- Add up all your values of d square, which is 12 ($\sum d$ square)

Step 5- Put these values in the formula

$$r_R = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)}$$

$$= 1 - (6 \cdot 12) / (9(81 - 1))$$

$$= 1 - 72 / 720$$

$$= 1 - 0.1$$

$$= 0.9$$

The Spearman's Rank Correlation for this data is 0.9 and as previously stated, if the p value is nearing +1 then they have a perfect rank relationship.

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4.1.4 Linear-Nonlinear-Relationship

A linear relationship implies that two quantities are proportionate to each other: doubling one causes the other to double as well.

Linear relationships are frequently used as an approximation to characterise any connection, despite the fact that there is no unique method to define a linear relationship in terms of the underlying nature of the numbers. A linear link between a person's height and weight, for example, differs from a linear relationship between a person's volume and weight. The second relationship makes more sense, but both are linear and, of course, incompatible with each other. Medications are frequently recommended in proportion to weight, particularly for youngsters. This is an illustration of a linear relationship.

Nonlinear connections, in general, are any nonlinear relationships. What is vital to remember while thinking about nonlinear interactions is that a broader range of potential dependencies is permitted. When there is minimal information available to assess the nature of the relationship, adopting a linear relationship is the simplest and hence, according to Occam's razor, a valid starting point. Additional information, on the other hand, typically exposes the necessity to apply a nonlinear connection.

Many of the nonlinear interactions that may exist are nonetheless monotone. This implies they either rise or drop but never both. Monotonic transitions can be smooth or abrupt. For example, a medicine may be useless until a specific threshold is reached, at which point it becomes effective. Nonlinear connections, on the other hand, can be non-monotonic. A medicine, for example, may grow gradually more beneficial over a specific range before becoming dangerous. As a result, the degree of assistance grows and declines, indicating a non-monotonic and nonlinear connection.

Even though a connection is monotonic and changes in one measure are smoothly connected to changes in the other, a linear relationship is not necessarily the best approximation. It is frequently beneficial to generalise to a power law connection. In a power law connection, when one quantity is doubled, the other is multiplied by a number that is not two, but it is always the same number. Many complex systems' quantity dependencies have been shown to be better represented by power laws than by linear connections. A power law is a more generic type of connection; hence it should be a better approximation. However, there are basic reasons for power law behaviour in complex systems in many circumstances.

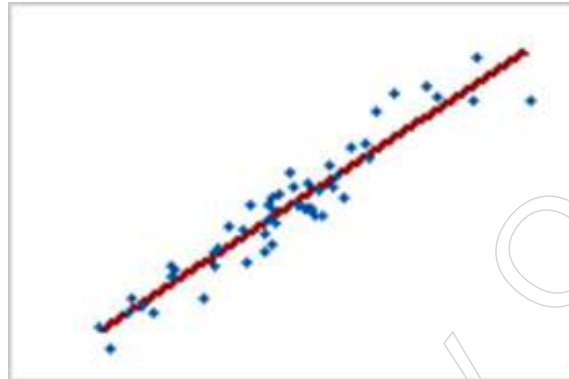
Linear, nonlinear, and monotonic relationships

It is critical to understand how two variables are connected while examining their connection. Although linear connections are the most prevalent, variables can also have nonlinear or monotonic relationships, as seen below. It is also conceivable that no link exists between the variables. To examine the association, you should first create a scatterplot of the variables.

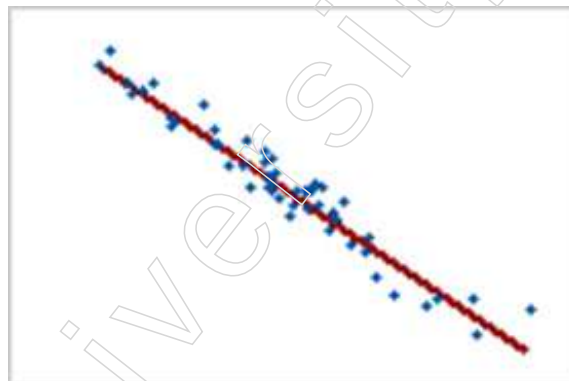
A linear connection is a data trend that may be represented by a straight line. Assume an airline wishes to determine the effect of fuel prices on flight expenses. They

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discover that every \$1 increase in the price of a gallon of jet fuel raises the cost of their LA-NYC travel by almost \$3500. This depicts a linear relationship between the cost of jet fuel and the cost of a flight.



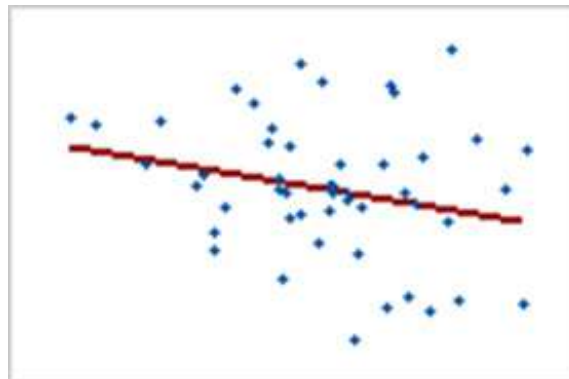
Plot 1: Strong positive linear relationship



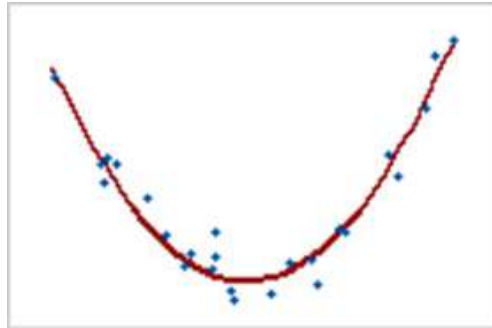
Plot 2: Strong negative linear relationship

A positive linear connection arises when both variables increase or decrease at the same pace. The points in Plot 1 closely follow the line, indicating a strong link between the variables. This association has a Pearson correlation value of $+0.921$.

A negative linear connection develops when one variable increases while the other variable declines. The points in Plot 2 closely follow the line, indicating a strong link between the variables. This connection has a Pearson correlation value of -0.968 .



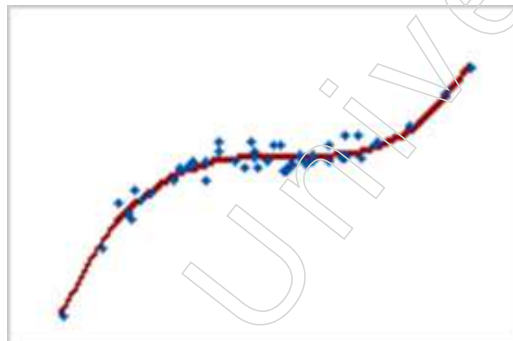
Plot 3: Weak linear relationship



Plot 4: Nonlinear relationship

If Plot 3's data points appear to be randomly dispersed. They are not near the line, indicating a very weak association, assuming one exists. This connection has a Pearson correlation value of -0.253 .

If the relationship between two variables is not linear, the rate of growth or decrease might vary when one variable changes, resulting in a "curved pattern" in the data. This curving trend might be better represented by a nonlinear function, such as a quadratic or cubic function, or it could be converted to become linear. Plot 4 depicts a significant link between two variables. The Pearson correlation coefficient, however, is only $+0.244$ since the connection is not linear. This relationship exemplifies why it is critical to plot the data in order to investigate any potential linkages.



Plot 5: Monotonic relationship

The variables in a monotonic connection tend to move in the same relative direction, but not always at the same pace. The variables of a linear connection move in the same direction at the same pace. Plot 5 demonstrates that both variables are rising at the same time, although not at the same rate. This is a monotonic but not linear connection. For this data, the Pearson correlation value is 0.843 , while the Spearman correlation is greater, at 0.948 .

Linear connections are monotone as well. Plot 1 shows a connection that is both monotonic and linear.

4.1.5 Use Cases of Correlation in Research (Case Study Business Problem)

Correlation Between Customer Relationship Management System Usage, Product Innovation, and Customer Satisfaction

Notes

Introduction

As organisations grew in size, the need to manage the interactions between business and client became more difficult (Akroush, et al., 2011). This difficulty prompted the advancement of technology to solve this need via a customer relationship management (CRM) system (Verrill, 2013). The CRM system began as a database to store customer contact information and evolved into a more complex management tool that increased corporate communication (Asghar, et al., 2013). The most recent CRM system evolution developed capabilities that expanded the flexibility of tracking consumers' expectations (Akroush, et al., 2011). The promise of highly precise customer information has resulted in increased CRM investment by US corporations. According to Mavuram (2013), 73 percent of US businesses have invested in CRM systems to update their operations. Nonetheless, some company executives continue to doubt that a CRM system can improve customer happiness and enhance revenue (Colman, et al., 2011). Broetzmann et al. (2015) and Mavuram (2013), for example, explored the reasons why most firms find it difficult to install a CRM system. The challenge is exacerbated by a paucity of resources in the scholarly literature supporting the usage of CRM systems for these two reasons. The purpose of this study was to address this issue by determining whether there is a link between the adoption of a CRM system and both customer happiness and product innovation.

This research gathered data on the advantages of adopting a CRM system, particularly in small to medium-sized businesses in the United States. The quantitative correlational design was used to examine if the selected organisations profited from the adoption of CRM systems in meeting the demands of their customers and enhancing their goods. In this connection, the predictor CRM system and the criterion variables of product innovation and customer happiness were analysed to see if they were related. The current CRM Behavior Theories, which defined the interconnected views of CRM linked with customer satisfaction management, were evaluated to identify the relationship between them (Labus and Stone, 2010).

Problem statement

Forming a relationship with customers was a very complicated process for businesses prior to the use of CRM systems (Mavuram, 2013). In theory, the CRM system would assist firms in better understanding their consumers because it involves activities such as recording customers' reactions to new items, which might lead to product development and innovation (Bovee and Thill, 2013; Darvish, et al., 2012; Ku, 2010). However, there is a paucity of empirical data suggesting that the usage of CRM systems is effective in improving small- to medium-sized businesses (Coltman, et al., 2011; Cuesta, 2014; Josiassen, et al., 2014). Doyle (2012) and Gummesson (2014) addressed this issue, claiming that determining the effective use of a CRM system for product innovation and consumer pleasure was challenging. Furthermore, Chen and Popovich (2013) asserted that researching the link between CRM system usage and customer satisfaction should not only focus on meeting consumer demands, but should also include providing continual healthy communication with the organisation. Doyle (2012) and Gummesson (2014) agreed on the need of demonstrating the value of CRM systems for product innovation and customer delight.

Research questions

The study looked at the connection between three variables: CRM system usage, product innovation, and customer satisfaction. The quantitative correlational design was used to address the problem by answering research questions and identifying variable interactions (Bettany-Saltikov and Whittaker, 2013). The hypotheses were explored in the study by finding statistical correlations between the variables. The first study question asked if the CRM system improves customer satisfaction with the company's products and services. The issue sought to ascertain whether the CRM system can give crucial information to the organisation in order to improve the delivery of quality products to consumers and thereby improve business outcomes.

RQ1: Is there a link, if any, between the adoption of a CRM system and customer satisfaction in small to medium-sized businesses in the United States? The second study question looked into whether adopting CRM systems helps with product innovation in order to meet consumer expectations. During the decline stage of a product's life cycle, innovation is required to meet customers' changing and evolving demands.

RQ2: Is there a link, if any, between the adoption of a CRM system and product innovation in small to medium-sized businesses in the United States?

The hypotheses

Correlation was utilised to determine whether the hypotheses were accepted or rejected.

H10: There is no substantial relationship between CRM system utilisation and customer satisfaction for small to medium-sized businesses in the United States.

H1a: For small- to medium-sized businesses in the United States, there is a substantial relationship between CRM system utilisation and customer happiness.

H20: For small to medium-sized businesses in the United States, there is no substantial relationship between CRM system utilisation and product innovation.

H2a: For small- to medium-sized businesses in the United States, there is a considerable relationship between CRM system utilisation and product innovation.

Significance of the study

The CRM system is often regarded as a great instrument for progress (Hassan and Parvez, 2013). However, many of the touted advantages have yet to be scientifically validated. This study on the adoption of CRM systems in small to medium-sized US enterprises might thus be useful on a number of levels. In the following business categories, valuable information may be gathered:

- Innovation: This quantitative correlational design can transfer knowledge that can lead to increased product innovation. The use of a CRM system provides mechanisms for transforming new ideas into valued goods. The advantage of

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using a CRM system is that it gives knowledge that is important for product innovation and the development of the company's financial gain.

- **Leadership:** The current study's findings may persuade business owners to implement CRM systems in their firms to aid in leadership. CRM systems enable executives to grow their firm by managing their relationships with customers in order to meet their demands.
- **Business organizations:** The current study can add to the literature of numerous schools of thought on various business strategies and customer service models, resulting in increased customer retention and profit. Business companies are still convinced to use CRM systems to develop and manage customer relationships.
- **Customers:** The current study's findings may aid customers by providing information that will assist them in determining their important responsibilities in the business. It contributes to the creation of care for consumers as stakeholders, prompting firms to build good communication between them. Businesses benefit from increased profitability as a result of satisfied customers.
- **Education:** The current study provided significant hypotheses that academics might utilise as beginning points in their discussions and arguments. This study gives a foundational understanding of the relationship between employing a CRM system and customer satisfaction. Further research is needed to establish a link between product innovation and the adoption of CRM systems.

Literature review

As the trend of corporate modernisation proceeded, suppliers and purchasers' tight connections deteriorated. Prior to the Industrial Revolution, traders knew their clients well, understood their needs, and even knew their names. When mass production became a reality, a barrier between merchants and buyers was built. As a result, finding buyers for standardised products became the primary necessity (Wang and Hui, 2012; Hailey, 2015).

The CRM system has evolved throughout the years from many business systems. Because of organisations' desire to increase their competitive advantages, the usage of CRM systems for medium-sized enterprises in the United States is becoming more common. CRM systems have begun to make information more accessible by launching updated software solutions that are flexible across sectors (Ku, 2010; Hailey, 2015). The CRM system is still used today by firms that rely on two factors: innovation and customer happiness (CRM Switch Staff, 2013). CRM systems were designed largely to meet the demands of customers (Bovée and Thill, 2013). CRM systems focus on one client at a time, creating a long-term relationship and then establishing an equally advantageous relationship (Gummesson, 2014). Product innovation and customer satisfaction were found to be directly related to the CRM system in the literature. CRM enables company representatives to focus on one consumer at a time, developing a durable and mutually beneficial connection first (Gummesson, 2014). Various firms are vying in the early twenty-first century to develop a fresh connection with existing clients in order to increase long-term customer loyalty and profitability for their company (Chen and Popovich, 2013; Hailey 2015). A CRM strategy may both reinforce and

develop new customer relationships (Wang and Hui, 2012). The CRM system enables a consistent approach that can maximise client interactions (Bovée and Thill, 2013). Knowing each consumer through a customer-centric data mining business strategy allows the firm to constantly and proactively provide and sell superior products or services for a longer length of time (Kindström and Kowalkowski, 2014). Aside from technology advancements, CRM system projects represent a significant shift in weight from managing product portfolios to managing customer portfolios, necessitating modifications to business processes.

Method

- Method and design

The quantitative technique with a correlational design sought to relate the adoption of a CRM system to product innovation and customer satisfaction in small- to medium-sized businesses in the United States. The quantitative correlational approach assisted in the discovery of a linear link between the predictor, CRM system utilisation, and criterion variables such as product innovation and customer happiness. The quantitative approach is used to calculate the study data (Neuman, 2011; Vogt, 2003). The nature of this study necessitated the acquisition of information from original sources. This work was completed by structuredly soliciting participants' perspectives in order to provide statistical maps and information to aid in describing the subject under consideration (Nenty, 2009; Neuman, 2011). The information was acquired via the SurveyMonkey electronic platform, where participants from small to medium-sized American businesses who utilise CRM systems were chosen to provide feedback on their CRM usage. This method resulted in a generalisation derived from a sample population as well as the assessment of various views and opinions within a certain cohort (Bettany-Saltikov and Whittaker, 2013).

- Setting, population, and sample

Individuals working in the categories of top managers, middle managers, and first-line managers of small- to medium-sized US businesses that use the CRM system for product innovation and customer satisfaction formed the research population. The survey questions were answered by 97 people in total. Participants who accepted to participate in the survey were routed to the survey form, while those who declined received a thank you message and their survey monkey session was automatically stopped. The statistical programme G*Power version 3.1.9.2 was used to compute the minimal sample size of the participants. Because it employs a type of random selection, the probability sampling approach was applied.

- Materials and instruments

The survey results are used to generate data for the instrument (Black, 1999; Vogt, 2003). The questions centred on the participants' impressions and expectations of using the CRM system in terms of product innovation and customer satisfaction. The study's findings were tallied based on survey responses from participants. All confidential information pointing to the participants' identities was kept private. There was no mention of any of the participants' names. The answer to a specific problem drives research quality (Davies and Dodd, 2002). Incorporating trustworthiness through consistent processes in this study showed that the acquired data was valid and reliable, and that data collection was correct and error-free.

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Results of analysis

The study findings articulated the participants' thoughts, attitudes, and behaviour to identify how the usage of CRM systems determines the link between customer happiness and product innovation. The 97 participants' data was divided into three separate layers of positions. These were the senior executives, middle managers, and front-line managers from the chosen organisations. The survey respondents served as the foundation for determining the relationship between CRM system utilisation, product innovation, and customer happiness. Data received from participants was utilised to examine how the usage of a CRM system affects product innovation and customer satisfaction. Because the variables in this study are handled as continuous, the researcher opted to analyse the data using ordinal analysis. Using the Likert-scale methodology, each variable is made up of six statements that the study participants scored from strongly agree to strongly disagree. The IBM SPSS version 23 programme aided in determining the dependability of the data in this research investigation. Cronbach's alpha is a measure of scale reliability's internal consistency, as illustrated in Table 1.

Table 1: Cronbach's alpha parameters (source: Authors' own research)

Cronbach's Alpha	Internal Consistency
$\alpha \geq 0.9$	Excellent
$0.9 \geq \alpha \geq 0.8$	Good
$0.8 \geq \alpha \geq 0.7$	Acceptable
$0.7 \geq \alpha \geq 0.6$	Questionable
$0.6 \geq \alpha \geq 0.5$	Poor

Using Pearson correlation, the first study question established a link between CRM system utilisation and customer happiness. Six statements from RQ1 served as the foundation for measuring the first two variables: CRM system utilisation and customer satisfaction. Using IBM SPSS version 23, two hypotheses were formed from RQ1 and tested for acceptance or rejection. The Pearson correlation coefficient r has a value range of +1 to -1. A value of 0 implies that the two variables have a negative correlation. The standard deviation (std) revealed that the data is clustered around the mean. Smaller standard deviations of 11 and 6, respectively, demonstrate this. The correlation coefficient (r) of 0.622 measures the degree and direction of a negative value less than 1.0 linear link between the CRM system and customer satisfaction. The value of r alternates between +1 and -1. It explains the relationship between the two variables. Customer satisfaction is exactly +1, indicating that there is a perfect linear link between CRM system utilisation and contentment. The findings resulted in the rejection of the null hypothesis.

The second study topic investigated the relationship between CRM systems and product innovation. RQ2 encompassed six assertions. These were about the relationship between the CRM system and product innovation. Using IBM SPSS version 23, two hypotheses were produced from RQ2 for evaluating acceptance or rejection. The standard deviation (std) revealed that the data is clustered around the mean. Smaller standard deviations of 113 and 129, respectively, demonstrate this. The correlation coefficient (r) of 0.371 quantifies the degree and direction of a linear link

between the CRM system and product innovation, which is 0.47. As a consequence, the Pearson correlation for the three variables was negative. The value of 0 indicates that there is a negative association between the two variables. However, when one variable's value declines, the value of the other variable rises. As a result, the null hypothesis is rejected as well. According to the conclusions of this study, CRM system utilisation has a substantial relationship with customer happiness. A complete positive linear connection indicates the perfect linear link between CRM system and customer satisfaction. The association between CRM system and product innovation, on the other hand, has a negative linear relationship. This indicates that the reduction is part of a linear connection.

Table 2: Analysis of the research question (Source: Authors' own research)

	N	Range	Mean	Std. Deviation	Variance
	Statistic	Statistic	Statistic	Statistic	Statistic
Awareness	97	4	4.00	1.146	1.313
Difficulty	97	4	4.03	1.159	1.343
Acclaims	97	4	3.95	1.014	1.029
Easy to use	97	4	4.06	1.019	1.038
Accurate result	97	4	3.96	1.030	1.061
Useful solution	97	4	3.74	1.013	1.027
Service satisfaction	97	4	4.14	0.890	0.791
Positive remarks	97	4	4.12	0.949	0.901
Saying good things	97	4	4.12	1.013	1.026
Customer loyalty	97	4	4.04	1.050	1.102
Feeling rewarded	97	4	4.01	1.015	1.031
Increased customers	97	4	3.97	1.015	1.030
Product innovation	97	4	4.01	1.046	1.094
New products	97	4	4.09	1.032	1.064
New ideas	97	4	3.72	1.125	1.265
Introduce more products	97	4	4.01	1.123	1.260
Competitive new products	97	4	4.01	1.015	1.031
Innovative products	97	4	3.92	1.038	1.076
Valid N (listwise)	97				

Discussion

The three variables in this quantitative correlation analysis are CRM system utilisation, product innovation, and customer happiness. The goal is to discover the relationship between these factors. The research questions were assigned at random to participants from small to medium-sized businesses in the United States. This procedure began when participants agreed to provide informed consent (located at the very first page). Those who agreed were instantly sent to the survey tool. To provide statistical calculation of the study outcomes, the study results were gathered and tabulated using IBM SPSS version 23. The goal is to investigate if the adoption of a CRM system helps the functioning of small- to medium-sized businesses in the United States.

The primary study question was to investigate the importance of CRM system utilisation to customer satisfaction and product innovation in the operations of small- to medium-sized businesses in the United States. Two sub questions supplemented the main research topic. The statements in each study question were scored by the participants using the Likertscale technique.

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RQ1: Is there a link, if any, between the adoption of a CRM system and customer satisfaction in small to medium-sized businesses in the United States?

According to the conclusions of this study, the use of a CRM system has a substantial relationship with customer happiness. The CRM system and customer satisfaction have a perfect linear connection. According to the findings, the implementation of a CRM system in a small- to medium-sized business in the United States increases the likelihood of providing amazing customer service, which leads to pleased consumers.

RQ2: Is there a link, if any, between the adoption of a CRM system and product innovation in small to medium-sized businesses in the United States?

According to the study's findings, the association between CRM systems and product innovation has a negative linear relationship with a modest influence. This merely demonstrated that the CRM system is primarily intended to improve customer service rather than provide business organisations with fresh product ideas to push to the market. The following were the study's hypotheses:

H₁₀: There is no substantial relationship between CRM system utilisation and customer satisfaction for small to medium-sized businesses in the United States.

H_{1a}: For small- to medium-sized businesses in the United States, there is a substantial relationship between CRM system utilisation and customer happiness. The null hypothesis is rejected, indicating that there is no meaningful relationship between CRM system use and customer satisfaction for small- to medium-sized businesses in the United States. Because a value less than 0 implies a negative relationship, as one variable's value falls, the value of the other variable increases.

H₂₀: For small to medium-sized businesses in the United States, there is no substantial relationship between CRM system utilisation and product innovation.

H_{2a}: For small- to medium-sized businesses in the United States, there is a considerable relationship between CRM system utilisation and product innovation.

The study's output is a collaboration between the researcher and the management of small- to medium-sized businesses in the United States. The participants agreed that using a CRM system had a considerable impact on customer satisfaction while having a modest impact on product organisation. The data gathered revealed that the adoption of a CRM system has a substantial association with customer satisfaction and, most likely, boosts the profitability of small and medium-sized businesses.

The CRM system, according to the participants, boosted their approachability and gave a complete view of the full customer life cycle. This quantitative correlational study may be significant and useful in the business process for the business community, customers, and academia, primarily because it gives information on rich company practises to increase customer happiness. Business organisations are examples of innovation because they include the deliberate use of information, imagination, and creativity in obtaining diverse benefits from resources.

According to the conclusions of this study, the business community, company executives, and customers may profit from efficient CRM system implementation:

- Management

The study's findings may be used to assess the amount to which the management of small- to medium-sized businesses in the United States is developing their business processes through the usage of a CRM system.

- Stakeholders

This study provides certain motives for corporate stakeholders to explore the adoption of a CRM system in their organisation to support any activities that might increase the firm's profitability.

- Leadership

The outcomes of this study suggest that company executives utilise a CRM system to gain a better understanding of their customers' interactions with their organisations' products and services. CRM systems provide tools that help business executives follow up with their customers on purpose by giving the information and procedures required to accomplish the company's mission.

Check your understanding

- 1) _____ is defined as study that seeks to answer a variety of questions. Researchers perform exploratory studies to investigate a set of questions, as the term implies.
 - a) Exploratory research
 - b) Corelative Research
 - c) Descriptive Research
 - d) Casual Research
 - e) None of the above
- 2) The study of the link between two variables is known as _____.
 - a) Exploratory Research
 - b) Explanatory Research
 - c) Correlational research
 - d) Scientific Research
 - e) None of the above
- 3) SPSS is an abbreviation for _____.
 - a) Statistical Package for the Social Sciences
 - b) Statistical Problem for the Social Sciences
 - c) Social Package for the Statistical Sciences
 - d) Spearman Problem for the Statistical Science
 - e) None of the above

Notes

- 4) CRM full form of _____.
a) Consumer Relational Management
b) Customer Relationship Management
c) Correlation Relationship Management
d) Coefficient Relationship Management
e) None of the above
- 5) The _____ is a statistical measure of the strength of a monotonic relationship between two sets of data.
a) Coefficient Correlation
b) Karl Pearson's Coefficient of Correlation
c) Spearman's correlation coefficient
d) Product Moment Correlation Coefficient.
e) None of the above
- 6) Which of the following is not a type of Correlation Coefficients?
a) Covariance Correlation Coefficient
b) Pearson's correlation
c) Spearman's rank correlation coefficient
d) Polychoric correlation
e) None of the above
- 7) _____ is referred to as an observational research approach since none of the variables in the study are modified throughout the research process.
a) Casual Research
b) Descriptive research
c) Experimental Research
d) Explanatory Research
e) None of the above
- 8) _____ is a comprehensive and extensive investigation of a specific topic, concern, or issue using the scientific method.
a) Research
b) D
c) D
d) D
e) None of the above

Summary

- Exploratory research does not give a solution to the subject under investigation. It is carried out to investigate previously unstudied issue areas.

- Causal research is based on the task of establishing cause and effect correlations. As such, it entails performing experiments and market testing in a controlled environment.
- Causal research has two goals: determining which factors constitute the cause and which constitute the effect, and understanding the link between the causal variables after the effect has occurred.
- Correlation is a statistical concept that describes how two variables move in tandem with one another.
- A correlation coefficient, represented by r , measures the degree of relationship. It is a measure of linear connection and is commonly referred to as Pearson's correlation coefficient after its creator.
- A scatter diagram is a diagram that depicts the values of two variables, X and Y , as well as how these two variables relate to one another.
- The Karl Pearson correlation coefficient approach is quantitative and provides a numerical number to determine the strength of the linear relationship between X and Y . A coefficient correlation of this type is denoted by the symbol ' r '.
- A linear relationship implies that two quantities are proportionate to each other: doubling one causes the other to double as well.
- A positive linear connection arises when both variables increase or decrease at the same pace.
- A negative linear connection develops when one variable increase while the other variable declines.

Activity

1. Write a Case study on correlation between owner brand and firm value.

Questions & exercise

1. Define Research. Write its qualities and types. Explain the research Process.
2. What is Correlation? Explain its Types.
3. Describe Pearson Product-moment Correlation and Spearman Rank Correlation.

Glossary

- Descriptive research is concerned with characterising the behaviour of the people being investigated.
- Explanatory research design employs a researcher's views and opinions on one issue as a leading point for subsequent investigations; it is also used to explore hypotheses further.
- Karl Pearson created this approach, commonly known as the Product Moment Correlation Coefficient.
- Karl Pearson (1948) created Pearson's product moment correlation coefficient, or Pearson's r , from a related theory provided by Sir Francis Galton in the late 1800s.

Notes

- A correlation coefficient is a number ranging from -1 to 1 that indicates the degree and direction of a link between variables.
- Correlation expresses the degree of a link between two variables numerically as the correlation coefficient. The correlation coefficient has values ranging from -1.0 to 1.0.
- The connection between the variables is "Linear," which implies that when the two variables are plotted, the points plotted form a straight line.
- The research process consists of a sequence of activities or stages required to carry out research efficiently, as well as the intended sequencing of these steps.
- Descriptive research is concerned with characterising the behaviour of the people being investigated. It simply requires one variable to be studied.
- Descriptive research examines the behaviour of a sample population. To carry out the investigation, just one variable is necessary. The three main goals of descriptive research are to describe, explain, and validate the findings

Further readings

1. <https://resources.pollfish.com/survey-guides/mastering-the-6-most-critical-types-of-research-for-any-research-endeavor/>
2. http://cuja.ac.in/E_Lecture/DCS/L4%20Basic%20steps%20of%20doing%20research.pdf
3. <https://towardsdatascience.com/types-of-correlation-coefficients-db5aa9ea8fd2>
4. https://www.researchgate.net/publication/262011045_Pearson's_Product_Moment_Correlation_Sample_Analysis/link/5c29791da6fdccfc70731d98/download
5. <https://statistics.laerd.com/spss-tutorials/pearsons-product-moment-correlation-using-spss-statistics.php>

Check your understanding - Answers

- 1) a)
- 2) c)
- 3) a)
- 4) b)
- 5) c)
- 6) e)
- 7) b)
- 8) a)

Unit - 4.2: Regression

Notes

Objectives:

In this Unit you will be able to:

- Understand Regression
- Learn Regression Coefficients
- Understand Linear Regression
- Explain Method Least Square Understanding
- Learn about Polynomial Regression
- Understand Model Evaluation MAPR RMSE
- Identify Model Evaluation Multicollinearity Effect VIF
- Learn Steps to Good Model
- Understand Curve Fitting Excel
- Learn about Correlation Regression Analysis Excel

Introduction

Regression is a statistical approach used in finance, investment, and other fields to identify the strength and type of a connection between one dependent variable (typically represented by Y) and a sequence of other variables (known as independent variables).

Regression assists investment and financial managers in valuing assets and understanding the links between factors such as commodity prices and the stocks of companies that trade in those commodities.

Regression analysis is a set of statistical processes used in statistical modelling to estimate the relationships between a dependent variable (often referred to as the 'outcome' or 'response' variable) and one or more independent variables (often referred to as 'predictors,' 'covariates,' 'explanatory variables,' or 'features'). Linear regression is the most popular type of regression analysis, in which the line (or a more sophisticated linear combination) that best fits the data according to a certain mathematical criterion is found. The ordinary least squares approach, for example, computes the unique line (or hyperplane) that minimises the sum of squared differences between the genuine data and that line (or hyperplane). This permits the researcher to estimate the conditional expectation (or population average value) of the dependent variable when the independent variables take on a specified set of values for precise mathematical reasons. Less prevalent types of regression employ somewhat different approaches to estimate alternative location parameters (e.g., quantile regression or Necessary Condition Analysis) or the conditional expectation over a larger range of non-linear models (e.g., nonparametric regression).

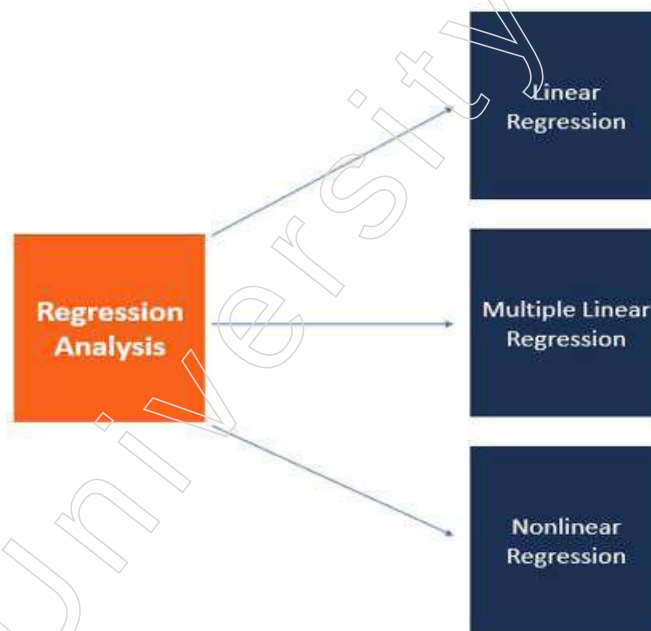
Regression analysis is typically utilised for two separate conceptual reasons. First, regression analysis is frequently used for prediction and forecasting, and its application overlaps significantly with the subject of machine learning. Second, regression analysis may be used to identify causal links between independent and dependent variables

Notes

in particular situations. Importantly, regressions show only correlations between a dependent variable and a set of independent variables in a given dataset. A researcher must carefully demonstrate why existing correlations have predictive power for a new context or why a link between two variables has a causal interpretation before using regressions for prediction or inferring causal relationships. The latter is especially crucial when utilising observational data to estimate causal links.

4.2.1 Introduction to Regression

Regression analysis is a collection of statistical methods for estimating connections between one or more independent variables and a dependent variable. It may be used to analyse the strength of the link between variables and to forecast their future relationship.



There are various types of regression analysis, including linear, multiple linear, and nonlinear. Simple linear and multiple linear models are the most frequent. Nonlinear regression analysis is typically employed for more difficult data sets with a nonlinear connection between the dependent and independent variables.

Regression analysis has extensive applications in many fields, including finance.

1) Regression Analysis – Linear Model Assumptions

Six key assumptions underpin linear regression analysis:

- The slope and intercept have a linear connection between the dependent and independent variables.
- The independent variable is not a random variable.
- The residual (error) value is zero.
- The residual (error) value remains constant across all observations.
- The residual (error) value is not associated across all data.
- The residual (error) values are normally distributed.

2) Regression Analysis – Simple Linear Regression

A model that examines the connection between a dependent variable and an independent variable is known as simple linear regression. The following equation expresses the simple linear model:

$$Y = a + bX + \epsilon$$

Where:

- Y – dependable Variable
- X – Independent (explanatory) variable
- a – Intercept
- b – Slope
- ϵ - Residual (error)

3) Regression Analysis – Multiple Linear Regression

Multiple linear regression analysis is identical to basic linear regression analysis, with the distinction that the model includes multiple independent variables. Multiple linear regression is mathematically represented as:

$$Y = a + bX_1 + cX_2 + dX_3 + \epsilon$$

Where,

- Y – Dependable Variable
- X_1, X_2, X_3 – Independent (explanatory) variables
- a – Intercept
- b,c,d – Slopes
- ϵ - Residual (error)

The requirements for multiple linear regression are the same as those for the basic linear model. However, because there are numerous independent variables in multiple linear analysis, there is additional essential condition for the model:

- Non - Collinearity: Independent variables should have a low correlation with one another. It will be difficult to analyse the genuine links between the dependent and independent variables if the independent variables are significantly associated with each other.

What is regression analysis and what does it mean to perform a regression?

Regression analysis is a reliable way for determining which factors have an effect on a certain topic of interest. The method of doing a regression allows you to accurately establish which elements are most important, which ones may be ignored, and how these factors interact with one another.

To completely understanding regression analysis, it is necessary to understand the following terms:

- Dependent Variable: This is the major aspect you're attempting to comprehend or forecast.

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- **Independent Variables:** These are the variables you believe have an effect on your dependent variable.

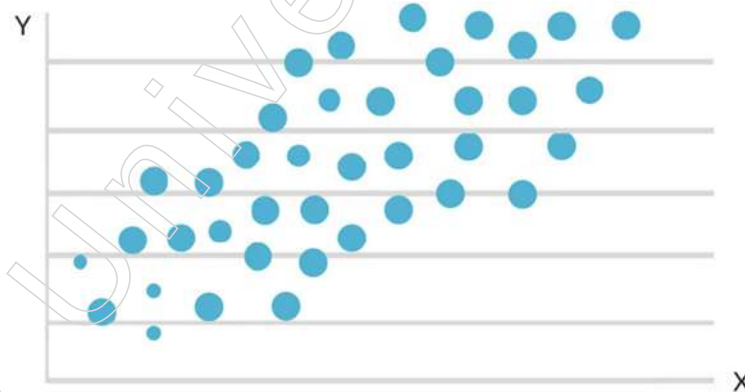
Attendee satisfaction with the event is our dependent variable in our application training example above. Our independent variables include the topics addressed, the length of sessions, the food supplied, and the cost of a ticket.

Ways to do Regression Analysis

To do a regression analysis, you must first establish a dependent variable that you believe is impacted by one or more independent factors. Then you'll need to create a complete dataset to work with. Administering surveys to your target consumers is an excellent approach to build this dataset.

Your survey should include questions about all of the independent variables in which you are interested. In this scenario, we'd like to know the historical levels of satisfaction with the events over the last three years (or whichever long you judge statistically significant), as well as any information about the independent factors. Perhaps we're especially interested in how the cost of a ticket to the event has influenced levels of enjoyment.

To investigate whether or whether there is a link between these two variables, we would first plot these data points on a chart, which would like the theoretical example below.

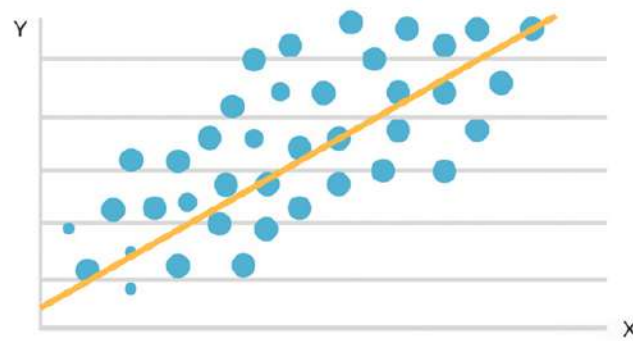


(The first step in determining if there is a link between your independent and dependent variables is to plot your data.)

The y-axis should be used to display our dependent variable (in this example, the degree of event satisfaction), while the x-axis should be used to depict our independent variable (the price of the event ticket). Once your data has been plotted, you may see correlations. If the hypothetical chart above accurately represented the influence of ticket pricing on event satisfaction, we might confidently state that the greater the ticket price, the higher the levels of event satisfaction.

But how can we tell the degree to which ticket price affects event satisfaction?

Draw a line across the centre of all of the data points on the chart to begin answering this question. This line is known as your regression line, and it may be computed exactly using a typical statistics application such as Excel. We'll use a theoretical chart to show what a regression line should look like once more.



The regression line depicts the connection between the independent and dependent variables. Excel will also supply a formula for the line's slope, which provides meaning to the relationship between your independent and dependent variables.

A regression line's formula can be something like $Y = 100 + 7X + \text{error term}$.

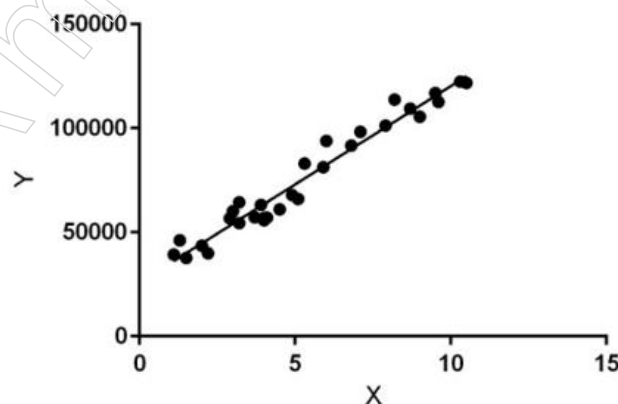
This means that if there is no "X," then Y equals 100. If X represents our increase in ticket price, we know that even if there is no increase in ticket price, event satisfaction will improve by 100 points.

Excel's slope formula includes an error term, as you may have seen. Because independent variables are never perfectly perfect predictors of dependent variables, regression lines always include an error term. This makes sense when considering the impact of ticket prices on event satisfaction — there are clearly other factors influencing event satisfaction aside from price.

Your regression line is only an estimate based on the data you have available. As a result, the higher your error term, the less definite your regression line is.

4.2.2 Linear Regression

Linear regression, also known as ordinary least squares (OLS) and linear least squares, is the workhorse of regression. It is a supervised learning-based machine learning algorithm. It carries out a regression task. Based on independent variables, regression models a target prediction value. It is mostly used to determine the link between variables and predicting. Different regression models differ in terms of the type of connection they evaluate between dependent and independent variables, as well as the number of independent variables utilised.



Notes

Notes

Linear regression is used to predict the value of a dependent variable (y) based on a given independent variable (x). As a result, this regression approach determines a linear connection between x (input) and y. (output). As a result, the name Linear Regression was chosen.

In the diagram above, X (input) represents job experience and Y (output) represents a person's wage. Our model's best fit line is the regression line.

Sir Francis Galton (1822–1911) was the first to suggest linear regression. Galton developed the word regression to express the finding that the majority of extremely tall dads had sons who were shorter than them, and the majority of very short fathers had sons who were taller than them. The tendency of this height growth was toward average (or mean) height. Regression to the mean was the name given to this occurrence. His investigation into this effect became known simply as regression. The primary goals of linear regression are to:

- Determine if a connection exists between one variable and another (or a group of variables);
- Explain the nature of this link, if it exists;
- Determine the precision of this connection.
- If many variables are employed, assess the relative contributions of each variable.

Linear regression makes all three of the preceding assumptions. However, as previously shown, you may use the central limit theorem to adjust for nonnormality. Engineering control charts rely on that fact, or the central limit theorem, by sampling a process flow and generating this group of means. The control charts are based on a collection of means rather than the underlying data from which they were generated. The fundamental procedure for doing linear regression on a data set is to estimate parameters (coefficients) for each possible predictor variable (X) to indicate the influence that variable has on the response variable (Y). These are expected to be linear and additive effects (i.e., you add them up to get the total effect). The investigation yielded the following general version of the regression equation:

$$Y = a + b_1 X_1 + b_2 X_2 + b_3 X_3 + \dots + b_n X_n \quad (1)$$

In Eq. (1), variable a is the Y value where $X=0$ (the intercept), variable b_1 is the coefficient for X factor #1 (X_1), variable b_2 is the coefficient for X factor #2 (X_2), and so on until the last X-variable is reached (X_n). The coefficients indicate the impacts of two sources of the X-link variables to the dependent variable (Y): the relative influence of X on Y and the effect of scale differences between X and Y. If the Y-variable and all of the X-variables have the same scale, then the coefficients indicate each X-relative variable's predictability. However, if the scales are different, the coefficients may not accurately reflect the relative predictabilities.

Variable scales are often extremely diverse. Standardization is one method for addressing the impacts of size. This is the process of converting all variables to a single scale. One popular way for standardisation (calculating the z-score) is to remove the mean from a number and divide by the standard deviation, resulting in a scale in standard deviation units. Normalization, which uses some function to adjust the scale

of the variable, is another approach for creating a common scale. The most often used function is to set the new value to $\text{Equal (old value - minimum value) / (maximum value - minimum value)}$.

Linear regression is a statistical method for modelling the connection between a scalar response and one or more explanatory factors (also known as dependent and independent variables). Simple linear regression is used when there is only one explanatory variable; multiple linear regression is used when there are more than one. This phrase differs from multivariate linear regression, which predicts several correlated dependent variables rather than a single scalar variable.

Relationships are modelled using linear predictor functions whose unknown model parameters are derived from data in linear regression. These kinds of models are known as linear models. The conditional mean of the answer given the values of the explanatory variables (or predictors) is most typically believed to be an affine function of those values; less frequently, the conditional median or another quantile is employed. Linear regression, like other kinds of regression analysis, concentrates on the conditional probability distribution of the answer given the predictor values, rather than the joint probability distribution of all of these variables, which is the realm of multivariate analysis.

Linear regression was the first form of regression analysis to be carefully explored and widely employed in practical applications. This is due to the fact that models that are linearly connected to their unknown parameters are easier to fit than models that are non-linearly related to their parameters, and the statistical features of the resulting estimators are easier to identify.

Linear regression offers a wide range of applications. The majority of applications fall into one of two general categories:

- If prediction, forecasting, or error reduction are the goals, linear regression can be used to fit a predictive model to an observed data set of response and explanatory variable values. If new values of the explanatory variables are obtained without an associated response value after creating such a model, the fitted model may be used to predict the response.
- If the goal is to explain variation in the response variable that can be attributed to variation in the explanatory variables, linear regression analysis can be used to quantify the strength of the relationship between the response and the explanatory variables, specifically to determine whether some explanatory variables have no linear relationship with the response at all, or to identify which subsets of explanatory variables may contain redundant information.

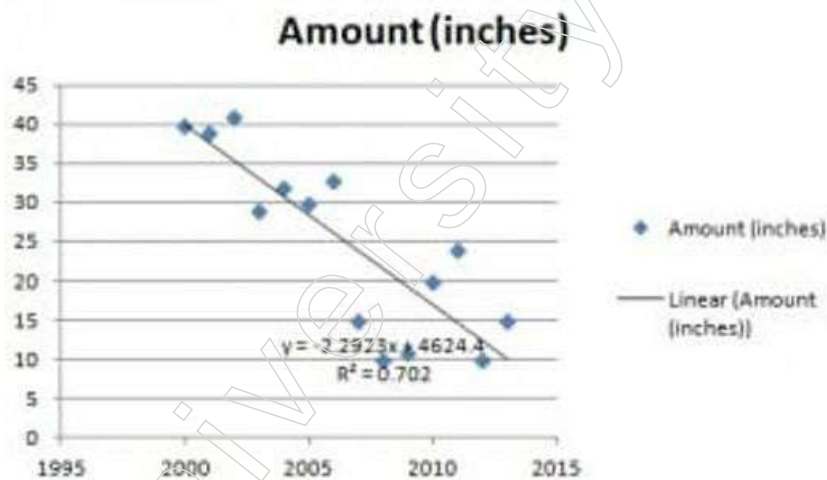
Linear regression models are frequently fitted using the least squares method, but they can also be fitted in other ways, such as by minimising the “lack of fit” in some other norm (as in least absolute deviations regression) or by minimising a penalised version of the least squares cost function (as in ridge regression (L2-norm penalty) and lasso regression) (L1-norm penalty). The least squares technique, on the other hand, may be used to fit models that are not linear. As a result, while the phrases “least squares” and “linear model” are sometimes used interchangeably, they are not equivalent.

Notes

Simple Linear Regression

You've undoubtedly worked with line graphs with one X axis and one Y axis. The X variable is sometimes referred to as the independent variable, whereas the Y variable is referred to as the dependent variable. One independent variable X is shown against one dependent variable Y in simple linear regression. In regression analysis, the independent variable is known as the predictor variable, while the dependent variable is known as the criterion variable. Many people, though, simply refer to them as the independent and dependent variables. Multiple independent variables are used in more complex regression techniques (such as multiple regression).

The results of regression analysis might be either linear or nonlinear graphs. The relationships between your variables may be described with a straight line in a linear regression. Non-linear regressions produce curved lines. (**)



Simple linear regression for the amount of rainfall per year

The Linear Regression Equation

Linear regression is a technique for simulating the connection between two variables. The equation is also known as the slope formula. The equation is written as $Y = a + bX$, where Y is the dependent variable (the variable plotted on the Y axis), X is the independent variable (the variable plotted on the X axis), b is the slope of the line, and an is the y-intercept.

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

The first step in determining a linear regression equation is determining whether or not the two variables have a connection. This is frequently a matter of judgement for the researcher. You'll also need an x-y formatted list of your data (two columns of data— independent and dependent variables).

Step 1: Create a chart of your data, filling in the columns the same way you would if you were looking for the Pearson's Correlation Coefficient.

Subject	Age X	Glucose Level Y	XY	X ²	Y ²
1	43	99	4257	1849	9801
2	21	65	1365	441	4225
3	25	79	1975	625	6241
4	42	75	3150	1764	5625
5	57	87	4959	3249	7569
6	59	81	4779	3481	6561
Σ	247	486	20485	11409	40022

From the above table, $\Sigma x = 247$, $\Sigma y = 486$, $\Sigma xy = 20485$, $\Sigma x^2 = 11409$, $\Sigma y^2 = 40022$. n is the sample size (6, in our case).

Step 2: Use the following equations to find a and b .

$$a = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$b = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$a = 65.1416$$

$$b = .385225$$

Find a :

- $((486 \times 11,409) - ((247 \times 20,485)) / 6 (11,409) - 2472)$
- $484979 / 7445$
- $= 65.14$

Find b :

- $(6(20,485) - (247 \times 486)) / (6 (11409) - 2472)$
- $(122,910 - 120,042) / 68,454 - 2472$
- $2,868 / 7,445$
- $= .385225$

Step 3: Put the values into the equation.

$$y' = a + bx$$

$$y' = 65.14 + .385225x$$

That's how to find a linear regression equation.

4.2.3 Two Variable Linear Regression

Multiple linear regression is a statistical approach for predicting the result of a variable based on the values of two or more variables. It is an extension of linear regression and is also referred to simply as multiple regression. The variable we aim to predict is referred to as the dependent variable, and the variables we use to forecast

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the value of the dependent variable are referred to as independent or explanatory variables.

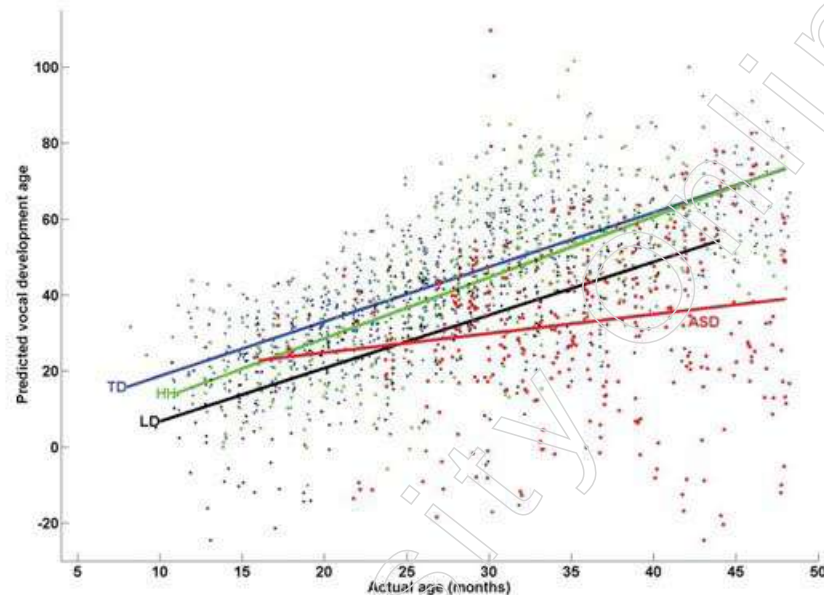


Figure: Multiple linear regression model predictions for individual observations

Multiple Linear Regression Formula

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon$$

Where:

- y_i is the variable that is reliant or forecasted.
- β_0 represents the y-intercept, or the value of y when both x_1 and x_2 are 0.
- β_1 and β_2 are the regression coefficients that describe the change in y as a function of a one-unit change in x_{i1} and x_{i2} , respectively.
- β_p is the slope coefficient for each independent variable.
- ϵ is the model's random error (residual) term.

Understanding Multiple Linear Regression

Simple linear regression allows statisticians to forecast the value of one variable based on knowledge about another one. Linear regression seeks to create a straight-line connection between two variables.

A variant of regression in which the dependent variable has a linear connection with two or more independent variables is known as multiple regression. Non-linearity occurs when the dependent and independent variables do not follow a straight line.

Both linear and non-linear regression use graphs to monitor a specific response using two or more variables. Non-linear regression, on the other hand, is typically difficult to implement since it is based on assumptions acquired by trial and error.

Assumptions of Multiple Linear Regression

The following assumptions underpin multiple linear regression:

1. A linear relationship between the dependent and independent variables

The basic assumption of multiple linear regression is that the dependent variable and each of the independent variables have a linear relationship. Creating scatterplots and visually inspecting the scatterplots for linearity is the best technique to check the linear connections. If the scatterplot shows a non-linear relationship, the analyst will need to conduct a non-linear regression or modify the data using statistical software such as SPSS.

2. The independent variables are not highly correlated with each other

Multicollinearity, which arises when the independent variables (explanatory variables) are highly interrelated, should not be present in the data. When independent variables exhibit multicollinearity, it is difficult to determine which variable contributes to the variance in the dependent variable. The Variance Inflation Factor approach is the most effective for testing the assumption.

3. The variance of the residuals is constant

Linear regression is based on the assumption that the amount of error in the residuals is comparable at each point in the linear model. This is referred to as homoscedasticity. When analysing the data, the analyst should plot the standardised residuals against the projected values to see if the points are evenly distributed over all independent variable values. To test the assumption, display the data on a scatterplot or use statistical software to generate a scatterplot that incorporates the full model.

4. Independence of observation

The model implies that the observations are unrelated to one another. Simply expressed, the model implies that residual values are independent. The Durbin Watson statistic is used to test for this assumption.

The test returns values ranging from 0 to 4, with values ranging from 0 to 2 indicating positive autocorrelation and values ranging from 2 to 4 indicating negative autocorrelation. The midpoint, a value of 2, indicates that there is no autocorrelation.

5. Multivariate normality

When residuals are regularly distributed, this is referred to as normalcy. Examine the distribution of residual values to put this assumption to the test. It may also be examined using one of two methods: a histogram with an overlaid normal curve or the Normal Probability Plot.

Importance of Multiple Linear Regression

Simple linear regression is a function that enables an analyst or statistician to predict one variable based on knowledge about another one. Linear regression can only be used when there are two continuous variables, one independent and one dependent. The parameter that is utilised to compute the dependent variable or result is known as the independent variable. A multiple regression model includes many explanatory variables.

The following assumptions underpin the multiple regression model:

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- The dependent variables and the independent variables have a linear relationship.
- The independent variables are not too connected with one another.
- y_i observations are selected at random and separately from the population.
- Residuals should be normally distributed, with a mean of zero and variance σ

The coefficient of determination (R-squared) is a statistical metric used to determine how much of the variation in outcome can be explained by changes in the independent variables. Even if the predictors are unrelated to the outcome variable, R² always rises when additional predictors are added to the MLR model.

As a result, R² cannot be used to determine which predictors should be included in a model and which should be discarded. R² can only be between 0 and 1, with 0 indicating that none of the independent variables can predict the result and 1 indicating that the independent variables can predict the outcome without mistake.

While maintaining all other variables constant, beta coefficients are appropriate for evaluating the results of multiple regression ("all else equal"). A multiple regression's outcome might be shown horizontally as an equation or vertically as a table.

4.2.4 Regression Lines

The Regression Line is the line that best fits the data, with the least overall distance between the line and the dots (variable values) depicted on a graph. In other terms, the regression line is a line that is utilised to minimise the squared deviations of predictions.

There are as many regression lines as there are variables. If we consider two variables, say X and Y, there will be two regression lines:

- Regression line of Y on X: This shows the most likely values of Y based on the values of X.
- Regression line of X on Y: This shows the most likely values of X based on the values of Y.

Regression Equations are the algebraic expressions of these regression lines. For each of the two regression lines, there will be two regression equations.

The degree of correlation between the variables is determined by the distance between these two regression lines, such that the closer the regression lines are to each other, the higher the degree of correlation, and the farther the regression lines are from each other, the lower the degree of correlation.

When the two regression lines coincide, i.e., only one line exists, the correlation is considered to be either perfect positive or perfect negative. If the variables are independent, the correlation will be 0, and the regression lines will be at right angles, i.e. parallel to the X and Y axes.

Note: Take note that the regression lines intersect at the point of average of X and Y. This indicates that if the perpendicular is drawn on the X axis from the place where the lines cross, we will receive the mean value of X. Similarly, if we draw a horizontal line on the Y axis, we get the mean value of Y.

The Least Square Regression

Linear regression, in general, finds the straight line. It is also known as the regression line of least squares. It is a representation in a bivariate dataset.

Assume that y is a dependent variable. The independent variable is X . This is the population regression line.

$$Y = a_0 + a_1x$$

"Where a_0 is the constant and a_1 is the regression coefficient and x is the value of the independent variable." If you are given a random sample of observations, you may estimate the population regression line by:

$$Y' = a_0 + a_1x$$

Where,

a_0 is the constant and

b_1 is the regression coefficient.

' x ' is the independent variable's value, and ' y ' is the anticipated value of the dependent variable.

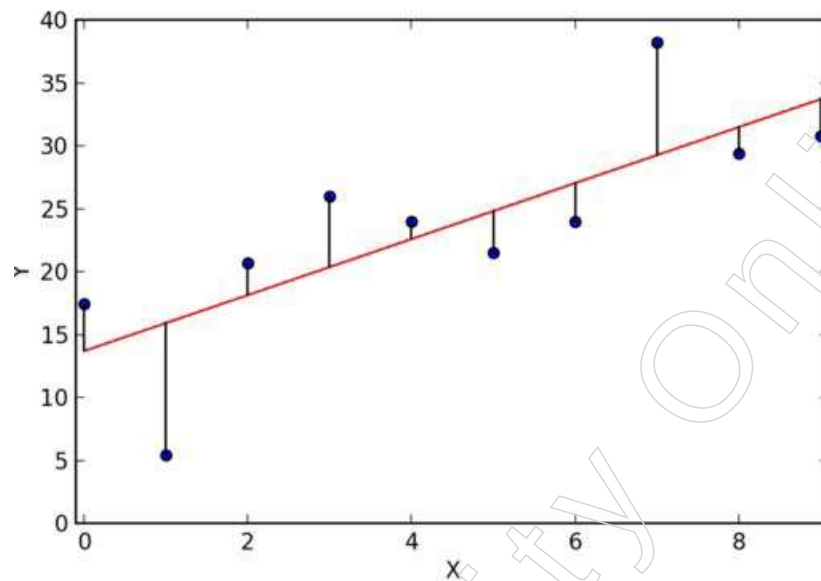
Important Properties of the Regression Lines

- The values of the regression coefficients stay constant. Because of the change in scale, there is a shift in origin. According to the property:
- If the variables x and y are replaced by u and v , respectively, $u = (x-a)/p$ and $v = (y-c)/q$, where p and q are constants. $B_{yx} = q/p \cdot b_{vu}$ $B_{xy} = p/q \cdot b_{uv}$.
- If there are two regression lines. Both of these lines cross at the same position $[x', y']$. Variables x and y are taken into account. The intersection of both regression lines, i.e., y on x and y , is $[x', y']$ according to the property. This is the answer to both equations with variables x and y .
- The geometric mean of the correlation coefficients between the two variables x and y will be found. In addition, the sign over the values of the correlation coefficients will equal the sign shared by both coefficients. So, if the regression of the coefficients are $b_{yx} = (b)$ and $b_{xy} = (b')$ then the correlation coefficient is $r = \pm \sqrt{b_{yx} \cdot b_{xy}}$, which means that in some circumstances, both coefficients produce a negative value and r is likewise negative. If both coefficient values are positive, r will be positive.
- The regression constant (a_0) equals to the y -intercept of the regression line. Where a_0 and a_1 are the regression coefficients.

Importance of Regression line

Forecasting techniques benefit from regression lines. Its goal is to describe the relationship between the dependent variable (y variable) and one or more independent variables (x variable). By inputting alternative values for the independent variables, the regression line equation serves as an analyst who can foresee future behaviours of the dependent variables.

Notes



Regression Line Formula: $y = a + bx + u$

Multiple Regression Line Formula: $y = a + b_1x_1 + b_2x_2 + b_3x_3 + \dots + b_t x_t + u$

Significance of linear regression

Regression lines are utilised in the financial and commercial sectors. Various financial analysts use linear regressions to anticipate stock prices, commodities prices, and to value a wide range of assets. Several businesses use linear regressions to anticipate sales, inventory, and a variety of other factors.

The least square assumptions are used to construct the regression lines. Regression analysis is often based on the sum of the squares of the observed values' departures from the lines of best fit. A regression line offers the best average value of one variable from any given value of the other variable.

In regression analysis, two regression lines are often used to represent the average connection between X and Y variables. It means that if there are two variables, X and Y, one line reflects the regression of Y on x and the other represents the regression of x on Y. (Fig. 1).

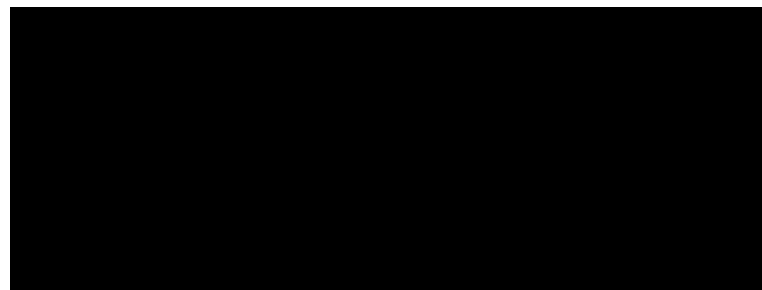


Figure 1: Regression lines of two variables

If the value of one variable is known, the equivalent value of variables on the other axis may be calculated along these lines. There is a significant degree of correlation between X and Y when the regression lines are closer to each other.

Another rationale for two regression lines is that regression lines are best-fit lines based on assumptions of least squares of observed value deviations. As a result, the best-fit lines are those that indicate the smallest values of the deviation squares of the observed values.

Deviations can be measured in two ways:

- (i) Horizontally, and
- (ii) Vertically.

Two lines are required to achieve a minimum of two deviations. The regression line of X is the line on which the sum of the squares of horizontal deviations is minimal, and the regression line of Y on X is the line on which the total of the squares of vertical deviations becomes smallest.

If $X = 0$, both variables are independent and will intersect at a straight angle. When the regression lines cross at the means of X and Y, a perpendicular line drawn from that point to the X axis will contact the axis on the mean value of X. Similarly, a perpendicular line drawn from that point to the Y axis will intersect the Y axis at the mean value of y. (Fig. 2).

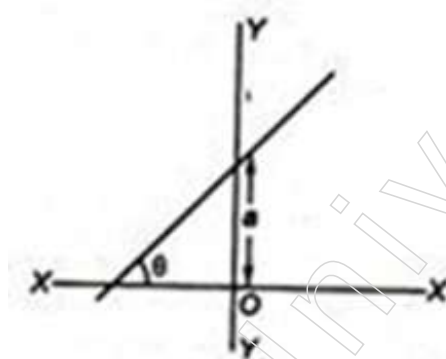


Figure: 2 (Equation of Regression line of Y on X)

$$Y = a + bx$$

$$b = \tan \theta \text{ Regression Coefficient of } y \text{ on } X$$

The bigger the angle divergence, the lower the correlation. The degree of association is shown by the steepness of the regression lines. The stronger the correlation, the steeper the regression lines X and Y or Y and X.

4.2.5 Regression Coefficients

The Regression Coefficient is the constant 'b' in the regression equation that represents the change in the value of the dependent variable as a function of the unit change in the independent variable.

There will be two regression coefficients if there are two regression equations:

- Regression Coefficient of X on Y: The regression coefficient of X on Y is denoted by the symbol b_{xy} , and it quantifies the change in X for each unit change in Y. It may be expressed symbolically as follows:

Notes

$$b_{xy} = r \frac{\sigma_x}{\sigma_y}$$

σ_x = standard deviation of X

σ_y = standard deviation of Y

When the deviations from the real means of X and Y are taken into account, the b_{xy} may be calculated using the following formula:

$$b_{xy} = \frac{\sum xy}{\sum y^2}$$

When calculating deviations from the assumed mean, the following formula is used:

$$b_{xy} = \frac{N \sum dxdy - \sum dx \sum dy}{N \sum dy^2 - (\sum dy)^2}$$

- Regression Coefficient of Y on X: The symbol b_{yx} is used to represent the change in Y that corresponds to a unit change in X. It may be expressed symbolically as follows:

$$b_{yx} = r \frac{\sigma_y}{\sigma_x}$$

In case, the deviations are taken from the actual means; the following formula is used:

$$b_{yx} = \frac{\sum xy}{\sum x^2}$$

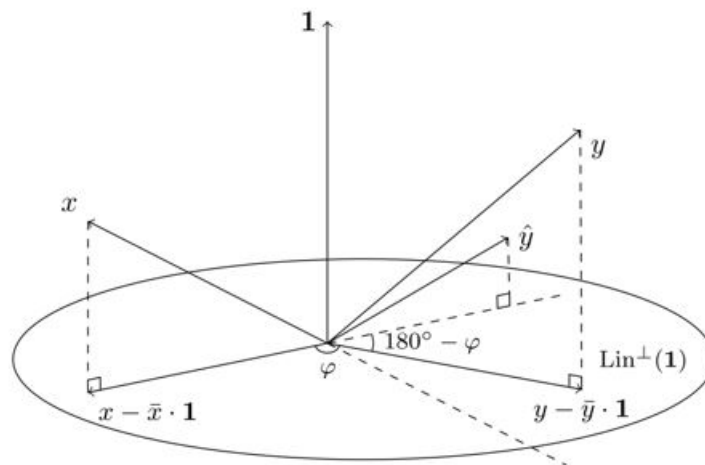
When the deviations from the expected means are taken into account, the b_{yx} may be determined using the following formula:

$$b_{yx} = \frac{N \sum dxdy - \sum dx \sum dy}{N \sum dx^2 - (\sum dx)^2}$$

The Regression Coefficient is also known as a slope coefficient since it defines the slope of the line, which is the change in the independent variable for each unit change in the independent variable.

The regression coefficients are a statistical metric used to determine the average functional connection between variables. One variable is reliant on the other in a regression analysis. It also assesses the degree to which one variable is dependent on another (s).

The regression coefficient was initially used to assess the association between fathers' and sons' heights. The slope coefficient is another name for regression coefficients. Because it influences the slope of the line, which is the change in the independent variable per unit change in the independent variable.



Classification of Regression Coefficient

- Simple partial and multiple
- Positive and negative
- Linear and non-linear

Some of the properties of regression coefficient:

- It is generally denoted by 'b'.
- It is expressed in the form of an original unit of data.
- If there are two variables, say x and y, the regression coefficient has two values. One will be acquired when x is independent and y is dependent, whereas the other will be obtained when y is independent and x is dependent. The regression coefficient of y on x is denoted by b_{yx} , while the regression coefficient of x on y is denoted by b_{xy} .
- The sign of both regression coefficients must be the same. If b_{yx} is positive, b_{xy} will be positive as well, and vice versa.
- If one regression coefficient is bigger than one, the others will be smaller.
- The correlation coefficient is equal to the geometric mean of the two regression coefficients.
 - o $R = \sqrt{b_{yx} \cdot b_{xy}}$
- Furthermore, the arithmetic means (am) of both regression coefficients are equal to or larger than the correlation coefficient.
 - o $(b_{yx} + b_{xy})/2 = \text{equal or greater than } r$
- The regression coefficients are unaffected by changes in the origin. They are, however, not independent of the scale shift. It indicates that subtracting any constant from the values of x and y has no influence on the regression coefficients. The regression coefficient will vary if x and y are multiplied by any constant.

Notes

4.2.6 Method Least Square Understanding

The least-squares approach is a type of mathematical regression analysis that is used to identify the best fit line for a set of data points, offering a visual representation of the connection between the data points. Each data point denotes the link between a known independent variable and an unknown dependent variable.

The regression analysis approach begins with a collection of data points that will be represented on an x- and y-axis graph. Using the least-squares approach, an analyst will construct a best-fit line that illustrates the probable link between independent and dependent variables.

The least-squares approach offers the overarching justification for where the line of greatest fit should be placed among the data points being analysed. The most common application of this method, which is also known as “linear” or “ordinary,” aims to generate a straight line that minimises the sum of the squares of the errors generated by the results of the associated equations, such as the squared residuals resulting from differences in the observed value and the value predicted based on that model.

The Line of Best Fit Equation

The equation for the line of best fit generated by the least squares approach provides the storey of the relationship between the data points. Computer software models that provide a summary of outputs for analysis, where the coefficients and summary outputs describe the dependency of the variables being evaluated, can be used to establish the line of best-fit equations.

Least Squares Regression Line

If the data indicates a less-than-linear relationship between two variables, the line that best matches this linear relationship is known as a least-squares regression line, which minimises the vertical distance between the data points and the regression line. The word “least squares” refers to the smallest sum of squares of mistakes, often known as the “variance.”

Regression analysis depicts dependent variables on the vertical y-axis and independent variables on the horizontal x-axis. These designations will be used to create the equation for the best-fit line, which will be found using the least-squares approach.

A non-linear least-squares issue, unlike a linear problem, has no closed solution and is often addressed by iteration. Carl Friedrich Gauss, who discovered the least-squares technique in 1795, is credited with its invention.

The least-squares approach is an important statistical method used to obtain a regression line or the best-fit line for a given pattern. An equation with specified parameters describes this procedure. The least squares approach is widely used in assessment and regression. This method is believed to be a typical strategy in regression analysis for approximating sets of equations with more equations than unknowns.

The technique of least squares actually describes the answer for minimising the sum of squares of deviations or mistakes in each equation's outcome. Find the formula

for the sum of squares of errors, which can assist you in determining the variance in observed data.

In data fitting, the least-squares approach is frequently used. The best fit result is supposed to lower the sum of squared errors or residuals, which are defined as the discrepancies between the observed or experimental value and the model's corresponding fitted value.

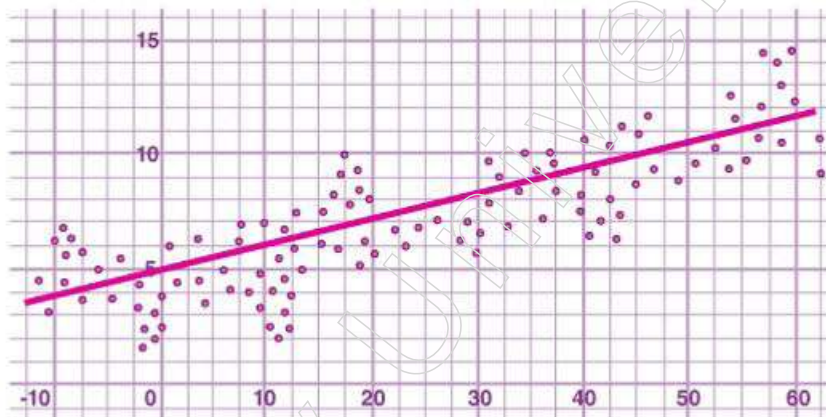
There are two basic categories of least-squares problems:

- Ordinary or linear least squares
- Nonlinear least squares

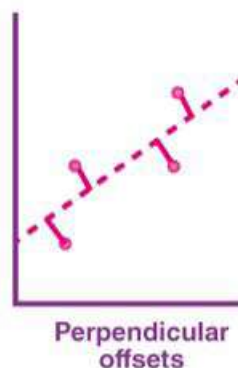
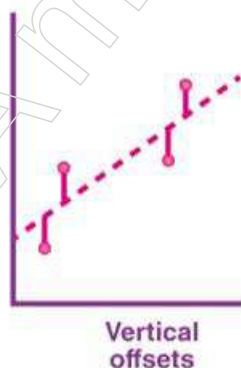
These are determined by the residuals' linearity or nonlinearity. In statistics, linear issues are frequently encountered in regression analysis. Non-linear issues, on the other hand, are commonly utilised in the iterative refinement approach, in which the model is approximated to the linear one with each iteration.

Least Square Method Graph

The line of greatest fit in linear regression is a straight line, as seen in the picture below:



The provided data points must be reduced by minimising the residuals or offsets of each point from the line. Vertical offsets are typically employed in surface, polynomial, and hyperplane issues, although perpendicular offsets are more commonly used in practise.



Notes

4.2.7 Math Behind Least Square

Method of Least Squares

Each fitted curve point reflects the link between a known independent variable and an unknown dependent variable.

In general, the least squares approach, also known as the method of linear or ordinary least squares, employs a straight line to fit through the provided locations. This line is known as the line of best fit because it minimises the sum of squares of the distances between the points.

In this strategy, the outcomes are often represented using equations with specific parameters. The technique of least squares actually describes the answer for minimising the sum of squares of deviations or mistakes in each equation's outcome.

The least squares approach is commonly used for data fitting. The best fit result minimises the sum of squared errors or residuals, which are the discrepancies between the observed or experimental value and the model's corresponding fitted value. Ordinary or linear least squares and nonlinear least squares are the two primary types of least squares techniques.

Mathematical Representation

It is a mathematical approach that produces a fitted trend line for a collection of data that meets the two requirements listed below.

1. The total of the variances between the actual and calculated values of Y is zero.
2. The sum of the squares of the discrepancies between the actual and calculated values is the smallest.

This function returns the line that is the best fit. This approach can produce findings that suit either a straight line or a parabolic trend.

The least squares approach, as explored in time series analysis, is used to discover the trend line that best fits a time series data.

Secular Trend Line

The following equation defines the secular trend line (Y):

$$Y = a + b X$$

Where,

Y = denotes the predicted value of the dependent variable.

a = Y-axis intercept, or the height of the line above the origin (when X = 0 and Y = a).

b = the line's slope (the rate of change in Y for a given change in X)

When b is positive, the slope is positive; when b is negative, the slope is negative.

X = independent variable (in this case it is time)

The following two equations must be solved concurrently in order to estimate the constants a and b:

$$\Sigma Y = na + b \Sigma X$$

$$\Sigma XY = a \Sigma X + b \Sigma X^2$$

If the midpoint of the time series is used as the origin, then the negative values in the first half of the series balance out the positive values in the second half, resulting in $\Sigma X = 0$. The above two normal equations will be as follows in this case:

$$\Sigma Y = na$$

$$\Sigma XY = b \Sigma X^2$$

In such a case the values of a and b can be calculated as under:

$$\text{Since } \Sigma Y = na$$

$$a = \frac{\Sigma Y}{n}$$

$$\text{Since, } \Sigma XY = b \Sigma X^2$$

Solved Example on Method of Least Squares

Fit a straight line trend on the following data using the Least Squares Method.

Period (year)	1996	1997	1998	1999	2000	2001	2002	2003	2004
Y	4	7	7	8	9	11	13	14	17

Solution:

Total of 9 observations are there. So, the origin is taken at the Year 2000 for which X is assumed to be 0.

PERIOD (YEAR)	Y	X	XY	X ²	REMARK
1996	4	-4	-16	16	Negative Region
1997	7	-3	-21	9	
1998	7	-2	-14	4	
1999	8	-1	-8	1	
2000	9	0	0	0	Origin
2001	11	1	11	1	Positive Region
2002	13	2	16	4	
2003	14	3	42	9	
2004	17	4	68	16	
Total (Σ)	ΣY = 90	ΣX = 0	ΣXY = 88	ΣX ² = 60	

The table shows that the value of n is 9, value of ΣY is 90, value of ΣX is 0, value of ΣXY is 88 and value of ΣX^2 is 60.

Substituting these values in the two given equations,

Notes

$$a = 909 \text{ or } a = 10$$

$$b = 8860 \text{ or } b = 1.47$$

$$\text{Trend equation is: } Y = 10 + 1.47 X$$

Least Square Method Formula

According to the least-square approach, the curve that best fits a given set of observations is one with the smallest sum of the squared residuals (or deviations or errors) from the provided data points. Assume that the provided data points are (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , ..., (x_n, y_n) , where all x 's are independent variables and all y 's are dependent variables. Assume that $f(x)$ is the fitting curve and that d reflects the error or departure from each supplied point.

Now, we can write:

$$d_1 = y_1 - f(x_1)$$

$$d_2 = y_2 - f(x_2)$$

$$d_3 = y_3 - f(x_3)$$

.....

$$d_n = y_n - f(x_n)$$

According to least-squares, the best-fitting curve is characterised by the property that the sum of squares of all deviations from supplied values must be smallest, i.e.,

$$S = \sum_{i=1}^n d_i^2$$

$$S = \sum_{i=1}^n [y_i - f_{x_i}]^2$$

$$S = d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2$$

Sum = Minimum Quantity

If we need to get the equation of the best fit line for the supplied data, we may apply the following formula.

The least square line equation is given by $Y = a + bX$

Normal equation for 'a':

$$\sum Y = na + b\sum X$$

Normal equation for 'b':

$$\sum XY = a\sum X + b\sum X^2$$

We may obtain the appropriate trend line equation by solving these two normal equations.

Thus, using the formula $y = ax + b$, we may obtain the line of best fit.

Solved Example

For a collection of data $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$, the Least Squares Model passes through the point $(\bar{x}, \bar{y})$, where $\bar{x}$ is the average of the x_i 's and $\bar{y}$ is the average of the y_i 's. The following example shows how to use the least square method to obtain the equation of a straight line or a least square line.

Question:

Consider the time series data given below:

x_i	8	3	2	10	11	3	6	5	6	8
y_i	4	12	1	12	9	4	9	6	1	14

Use the least square method to determine the equation of line of best fit for the data. Then plot the line.

Solution:

Mean of x_i values = $(8 + 3 + 2 + 10 + 11 + 3 + 6 + 5 + 6 + 8)/10 = 62/10 = 6.2$

Mean of y_i values = $(4 + 12 + 1 + 12 + 9 + 4 + 9 + 6 + 1 + 14)/10 = 72/10 = 7.2$

Straight line equation is $y = a + bx$.

The normal equations are

$$\sum y = an + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

X	Y	x^2	XY
8	4	64	32
3	12	9	36
2	1	4	2
10	12	100	120
11	9	121	99
3	4	9	12
6	9	36	54
5	6	25	30
6	1	36	6
8	14	64	112
$\sum x = 62$	$\sum y = 72$	$\sum x^2 = 468$	$\sum xy = 503$

Substituting these values in the normal equations,

$$10a + 62b = 72 \dots (1)$$

$$62a + 468b = 503 \dots (2)$$

$$(1) \times 62 - (2) \times 10,$$

$$620a + 3844b - (620a + 4680b) = 4464 - 5030$$

Notes

Notes

$$-836b = -566$$

$$b = 566/836$$

$$b = 283/418$$

$$b = 0.677$$

Substituting $b = 0.677$ in equation (1),

$$10a + 62(0.677) = 72$$

$$10a + 41.974 = 72$$

$$10a = 72 - 41.974$$

$$10a = 30.026$$

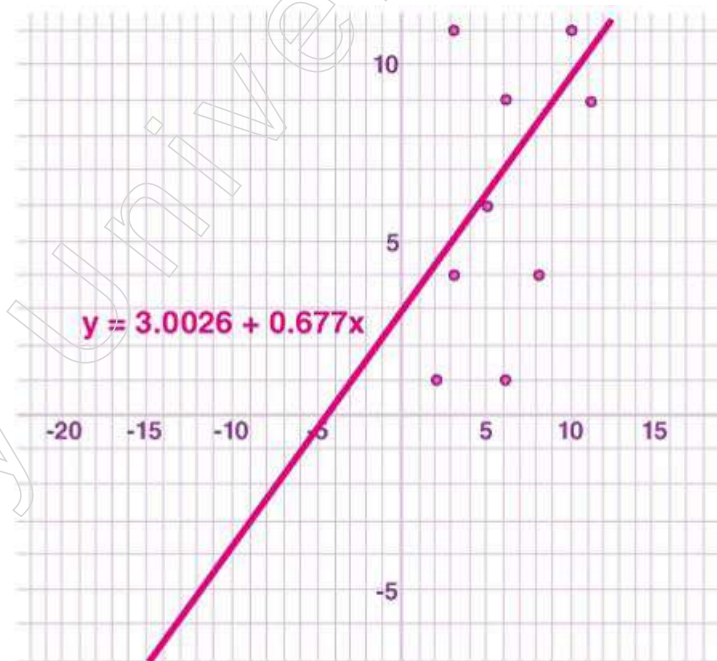
$$a = 30.026/10$$

$$a = 3.0026$$

Therefore, the equation becomes,

$$y = a + bx$$

$$y = 3.0026 + 0.677x$$



This is the required trend line equation.

Now, we can find the sum of squares of deviations from the obtained values as:

$$d1 = [4 - (3.0026 + 0.677 \cdot 8)] = (-4.4186)$$

$$d2 = [12 - (3.0026 + 0.677 \cdot 3)] = (6.9664)$$

$$d3 = [1 - (3.0026 + 0.677 \cdot 2)] = (-3.3566)$$

$$d4 = [12 - (3.0026 + 0.677 \cdot 10)] = (2.2274)$$

$$d5 = [9 - (3.0026 + 0.677*11)] = (-1.4496)$$

$$d6 = [4 - (3.0026 + 0.677*3)] = (-1.0336)$$

$$d7 = [9 - (3.0026 + 0.677*6)] = (1.9354)$$

$$d8 = [6 - (3.0026 + 0.677*5)] = (-0.3876)$$

$$d9 = [1 - (3.0026 + 0.677*6)] = (-6.0646)$$

$$d10 = [14 - (3.0026 + 0.677*8)] = (5.5814)$$

$$\sum d^2 = (-4.4186)^2 + (6.9664)^2 + (-3.3566)^2 + (2.2274)^2 + (-1.4496)^2 + (-1.0336)^2 + (1.9354)^2 + (-0.3876)^2 + (-6.0646)^2 + (5.5814)^2 = 159.27990$$

4.2.8 Polynomial Regression

Polynomial regression is a type of regression analysis in which the connection between the independent and dependent variables is represented by an n th degree polynomial. The least squares approach is commonly used to fit polynomial regression models. According to the Gauss Markov Theorem, the least square technique minimises the variance of the coefficients. Polynomial Regression is a subset of Linear Regression in which the polynomial equation is fitted to the data with a curvilinear connection between the dependent and independent variables.

Types of Polynomials

$$\text{Linear} \text{ ————— } ax + b = 0$$

$$\text{Quadratic} \text{ ————— } ax^2 + bx + c = 0$$

$$\text{Cubic} \text{ ————— } ax^3 + bx^2 + cx + d = 0$$

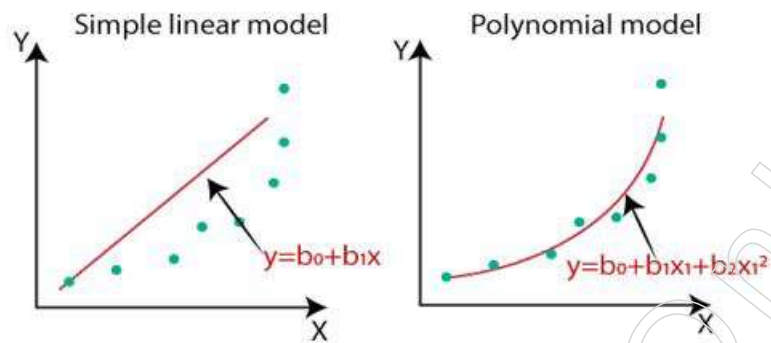
Assumptions of Polynomial Regression:

- A linear or curvilinear additive connection between the dependent variable and a collection of k independent variables (x_i , $i=1$ to k) can describe the behaviour of a dependent variable.
- The dependent variable's connection to any independent variable is linear or curved (specifically polynomial).
- The independent variables are unrelated to one another.
- The mistakes are independent, regularly distributed, with a mean of zero and a variance of one (OLS).

Need of Polynomial Regression

- We build our model and discover that it performs horribly.
- We detect a relationship between the actual value and the best fit line that we predicted, and it appears like the real value has some type of curve on the graph, although our line is nowhere close cutting the mean of the points.
- Polynomial Regression comes into action here; it predicts the best fit line that follows the pattern (curve) of the data, as seen in the figure below:

Notes



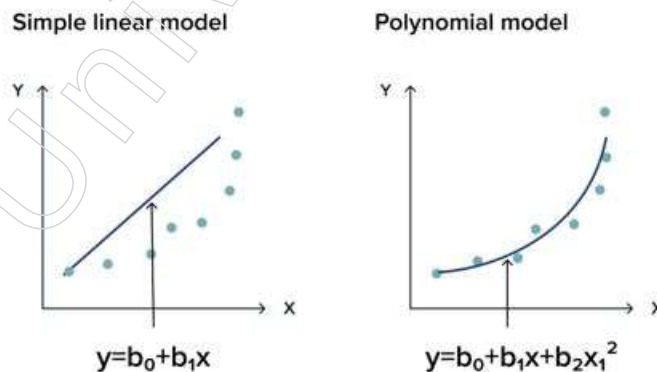
- One of the primary distinctions between Linear and Polynomial Regression is that Polynomial Regression does not need the relationship between the independent and dependent variables in the data set to be linear.
- Polynomial Regression is commonly employed when the Linear Regression Model fails to capture the points in the data and the Linear Regression Model fails to adequately describe the best outcome.

Math behind polynomial regression

The general equation for a polynomial regression model is:

$$y = b_0 + b_1x_1 + b_2x_1^2 + b_3x_1^3 + \dots b_nx_1^n$$

In this equation, y is the dependent variable, x is an independent variable, and b_0 – b_n are the parameters you can optimize.



Because the parameters of the regression are linear, you may fit the curve to your data using the same methods as for linear regressions – least squares and so on.

Actually, as the astute and mathematically astute may have recognised, this is only a specific example of multiple linear regression.

Let's go back to the weight-loss scenario.

In the instance of multiple linear regression, you want to know how different values, such as hours spent at the gym, sugar intake, and so on, affect weight loss. In the instance of polynomial regression, you want to know how different powers of one variable affect it. (x , x^2 , x^3 , and so on, where x is the sugar intake, for example.)

Despite the fact that the curve in the second example will be bent, the statistical estimation issue is the same in both circumstances.

Advantages of polynomial regression

Here are some advantages to use polynomial regression in your next machine learning model:

- Non-linear connections between variables can be modelled.
- You may employ a wide variety of different functions for fitting.
- Useful for exploration: you may check for the presence of curvature and its inflections.

Overall, it is a versatile technique that may be used to suit a wide range of data point distributions.

Disadvantages of polynomial regression

However, polynomial regression, like linear regression, is not a universal technique.

- Even a single outlier in the data plot might significantly skew the results.
- Public relations models are prone to overfitting. You can fit anything if you use enough parameters. According to John von Neumann, "with four parameters, I can fit an elephant; with five, I can make him shake his trunk."
- As a result of the previous, PR models may not generalise well outside of the data utilised.

Uses of polynomial regression in machine learning

Now let us have a look at some practical examples where polynomial regression is used.

Death rate prediction

When disasters occur, such as epidemics, fires, or tsunamis, disaster management teams must forecast the number of wounded or killed individuals in order to manage resources. The team must be prepared because it may take days, if not months, to lessen the impact of such incidents. Polynomial regression enables us to create versatile machine learning models that reflect the probable death rate by analysing a large number of dependant variables. In COVID-19 pandemics, for example, these characteristics can include if the patient has any chronic conditions, how frequently they are exposed to big groups of people, whether they have access to protective equipment, and so on.

Tissue growth rate prediction

Tissue growth rate prediction is employed in a variety of situations. To begin, polynomial regression is frequently employed to track oncology patients and the spread of malignant tumours. This form of regression aids in the development of a model that takes into account the non-linear nature of this spreading.

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Tissue growth rate prediction, on the other hand, is employed in monitoring ontogenetic growth; in other words, it allows doctors to track the development of the organism in the womb from an early stage.

Speed regulation software

Today, an increasing number of speed restriction software solutions powered by ML are geared towards avoiding risky behaviour rather than penalising violators of road rules. Predictive modelling with polynomial regression allows you to look for trends in driver behaviour and advise them about the importance of following the laws even before they exceed the speed limit. Preventative strategies have been shown to be more successful in lowering the number of traffic accidents.

4.2.9 Understanding Model Parameters

The most well-known regression model assessment measures are:

1. R-squared (R^2), which is the percentage of variance in the result explained by the predictor factors. In multiple regression models, R^2 is the squared correlation between the observed result values and the model's projected values. The better the model, the higher the R-squared.
2. Root Mean Squared Error (RMSE), which assesses the model's average error in predicting the result of an observation. The RMSE is defined mathematically as the square root of the mean squared error (MSE), which is the average squared difference between the observed actual result values and the projected values by the model. As a result, $MSE = \text{mean}((\text{observeds} - \text{predicted})^2)$ and $RMSE = \sqrt{MSE}$. The model performs better when the RMSE is low.
3. Residual Standard Error (RSE), also known as model sigma, is a version of RMSE that accounts for the number of predictors in the model. The model performs better when the RSE is low. In fact, the difference between RMSE and RSE is negligible, especially for large multivariate data sets.
4. Mean Absolute Error (MAE), The MAE, like the RMSE, gauges prediction error. It is the average absolute difference between observed and projected outcomes, calculated as $MAE = \text{mean}(\text{abs}(\text{observeds} - \text{predicted}))$. When compared to RMSE, MAE is less susceptible to outliers.

The issue with the aforementioned metrics is that they are sensitive to the introduction of other factors in the model, even if such variables do not contribute much to explaining the outcome. In other words, adding more variables to the model will always enhance the R^2 and decrease the RMSE. As a result, we require a more rigorous statistic to guide us.

In terms of R^2 , there is an adjusted version known as Adjusted R-squared, which corrects the R^2 for having too many variables in the model.

There are four more key metrics that are often used for model evaluation and selection: AIC, AICc, BIC, and Mallows Cp. These are an unbiased estimate of the MSE of the model prediction error. The model performs better when these indicators are lower.

4.2.10 Matrix Approach Multiple Regression

Because it allows for a concise, straightforward representation of regression analysis, matrix algebra is commonly employed for the derivation of multiple regression. In scalar concept, an estimated multiple regression model is given as: $Y = A + BX_1 + BX_2 + BX_3 + E$ or $Y = A + BX_1 + BX_2 + BX_3 + E$.

The same equation may be represented in a more compact and (believe it or not!) comprehensible form using matrix notation: $y = Xb + e$ or $y = Xb + e$.

Furthermore, matrix notation is adaptable, as it can accommodate any number of independent variables. Operations on the model matrix XX are done on all independent variables at the same time. Finally, you will see that matrix expression is often utilised in statistical presentations of OLS study findings. For all of these reasons, we begin by developing multiple regression in matrix form.

A matrix is a rectangular array of integers that is divided into rows and columns. As previously stated, operations on matrices are performed on all elements of a matrix at the same time. In this part, we will cover the fundamentals of matrix algebra, which are required to comprehend the representation of multiple regression in matrix form.

Multiple Linear Regression Model

$$y_i = \beta_0 + \beta_1 x_{i1} + e_i, \quad i = 1, 2, 3, \dots, n \quad (1)$$

Based on the regression equation, a basic linear regression depicts the relationship between the dependent variable y and the independent variable x .

The best fitting line may be identified using the least squares approach by reducing the sum of the squares of the vertical distance from each data point on the line. Gordon and Gordon (2004) and Scariano and Calzada (2004) provide more fascinating debate on this topic (2004). The dependent variable is associated to two or more independent variables in the multiple linear regression model. The generic model for k variables has the following form.

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + e_i, \quad i = 1, 2, \dots, n. \quad (2)$$

The basic linear regression model is used to find the straight line that best matches the data. The multiple linear regression model, on the other hand, is used to find the best fit plane for the data, such as with two independent variables. Models with more than two independent variables have a more complex structure, but they may still be analysed using multiple linear regression techniques. In multiple linear regression analysis, the least squares technique is utilised to estimate the regression coefficients in 2. The regression coefficients represent the unrelated contributions of the independent variables to predicting the dependent variable. Unlike simple linear regression, decisions about the degree of interaction or correlation between each of the independent variables must be made. The calculations required to compute the regression coefficients (β_i , $i = 1, \dots, k$), residual sum of squares (SSE), regression sum of squares (SSR), and so on are rather complex. In terms of data, the multiple regression model may be expressed using matrix notation to aid calculation.

A Matrix Approach to Multiple Linear Regression Analysis

Using matrices instead of vectors to describe observations, levels of regressor

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variables, regression coefficients, and random errors results in a more compact framework. The model is in the shape of $Y = X\beta + \epsilon$ (3)

and when written in matrix notation we have

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}. \quad (4)$$

It should be noted that Y is a $n \times 1$ dimensional random vector of observations, X is a $n \times (k + 1)$ matrix determined by the predictors, β is a $(k + 1) \times 1$ vector of unknown parameters, and ϵ is a $n \times 1$ vector of random errors. The first step in multiple linear regression analysis is to find the vector of least squares estimators, $\hat{\beta}$, which yields the linear combination $\hat{y}$ that minimises the length of the error vector. Essentially, the estimator $\hat{\beta}$ returns the smallest feasible sum of squares difference between $\hat{y}$ and y . Matrix notation can be used to represent $\hat{\beta}$ algebraically. A key need for multiple regression analysis is that the variables $x_1, x_2, \dots, x_n$ be linearly independent. This means that the correlation between each x_i is low.

Because the goal of multiple regression is to minimise the sum of squared errors, the regression coefficients that fit this requirement are obtained by solving the least squares normal equation.

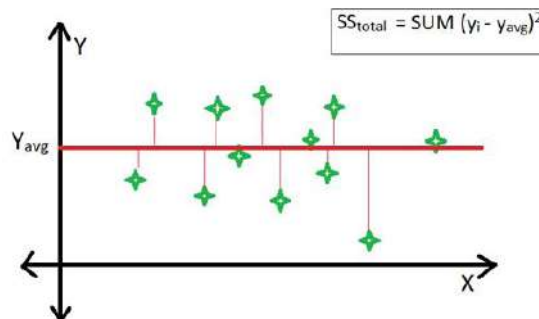
$$X^T X \hat{\beta} = X^T Y. \quad (5)$$

If the variables $x_1, x_2, \dots, x_n$ are linearly independent, then the inverse of $X^T X$ exists, namely $(X^T X)^{-1}$. By multiplying both the sides of the normal equation 5 by $(X^T X)^{-1}$, we get

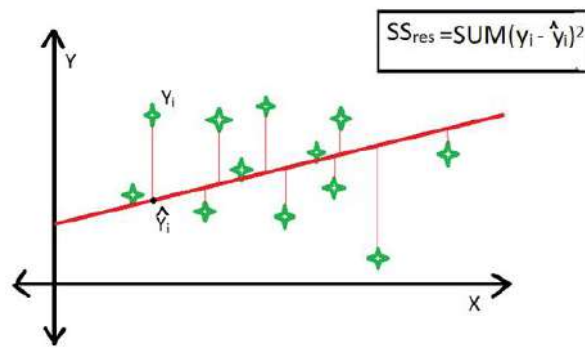
$$\hat{\beta} = (X^T X)^{-1} X^T Y. \quad (6)$$

4.2.11 Model Performance Measure R-square

R-squared is a statistical metric that measures a regression model's quality of fit. The optimal r-square value is 1. The closer the r-square number is to one, the better the model fits. R-square is a comparison of the residual sum of squares (SSres) and the total sum of squares (SStot). Total sum of squares is computed by adding the squares of the perpendicular distances between data points and the average line.



The sum of squares of the perpendicular distance between data points and the best fitted line is used to determine the residual sum of squares.



R square is calculated by using the following formula:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Where SS_{res} is the residual sum of squares and SS_{tot} is the total sum of squares.

The R-square approach may be used to assess the goodness of fit of regression models. The closer the r-square number is to 1, the better the model.

Note: It should be noted that the value of R-square might be negative if the model fitted is worse than the average fitting model.

Limitation of using R-square method –

- As additional variables are introduced to the model, the value of r-square always grows or remains constant, without recognising the relevance of the newly added variable (i.e., value of r-square never decreases on addition of new attributes to the model). As a result, with an increase in r-square value, non-significant features may also be included to the model.
- This is because SS_{tot} is always constant, and the regression model tries to reduce the value of SS_{res} by finding some connection with this new characteristic, causing the overall value of r-square to grow, resulting in a bad regression model.

R-square is a linear regression model goodness-of-fit measure. This statistic shows how much of the variance in the dependent variable the independent factors explain collectively. R-squared is a handy 0–100% scale that assesses the strength of the association between your model and the dependent variable.

Following the fitting of a linear regression model, you must decide how well the model fits the data, is it effective in explaining changes in the dependent variable? For regression analysis, there are numerous important goodness-of-fit statistics.

Assessing Goodness-of-Fit in a Regression Model

Linear regression finds the equation with the least difference between all of the observed and fitted values. To be more specific, linear regression determines the dataset's least sum of squared residuals.

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Residuals are the distance between the observed value and the fitted value.

A regression model is said to fit the data well if the discrepancies between the observed and projected values are modest and biased. In this application, unbiased indicates that the fitted values are neither systematically excessively high or too low over the observation space.

However, before analysing numerical metrics of goodness-of-fit, such as R-squared, the residual plots should be evaluated. By illustrating problematic patterns in the residuals, residual plots can expose a biased model significantly more effectively than numeric output.

R-squared and the Goodness-of-Fit

R-squared measures the spread of data points around a fitted regression line. It is also known as the coefficient of determination or, in the case of multiple regression, the coefficient of multiple determination. Higher R-squared values reflect less discrepancies between the observed data and the fitted values for the same data set.

R-squared is the percentage of the dependent variable variance explained by a linear model.

$$R^2 = \frac{\text{Variance explained by the model}}{\text{Total variance}}$$

R-squared is always between 0 and 100%:

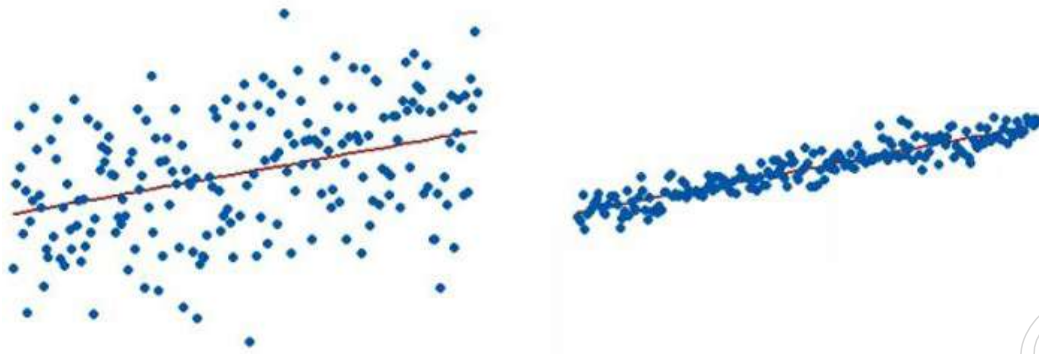
- 0% denotes a model that explains none of the fluctuations in the response variable around its mean. The dependent variable's mean predicts both the dependent variable and the regression model.
- 100% denotes a model that explains all fluctuations in the response variable around its mean.

Generally, the higher the R^2 , the better the regression model matches your data.

Visual Representation of R-square

We may display the fitted values against the observed values to see how R-squared values describe the dispersion around the regression line.

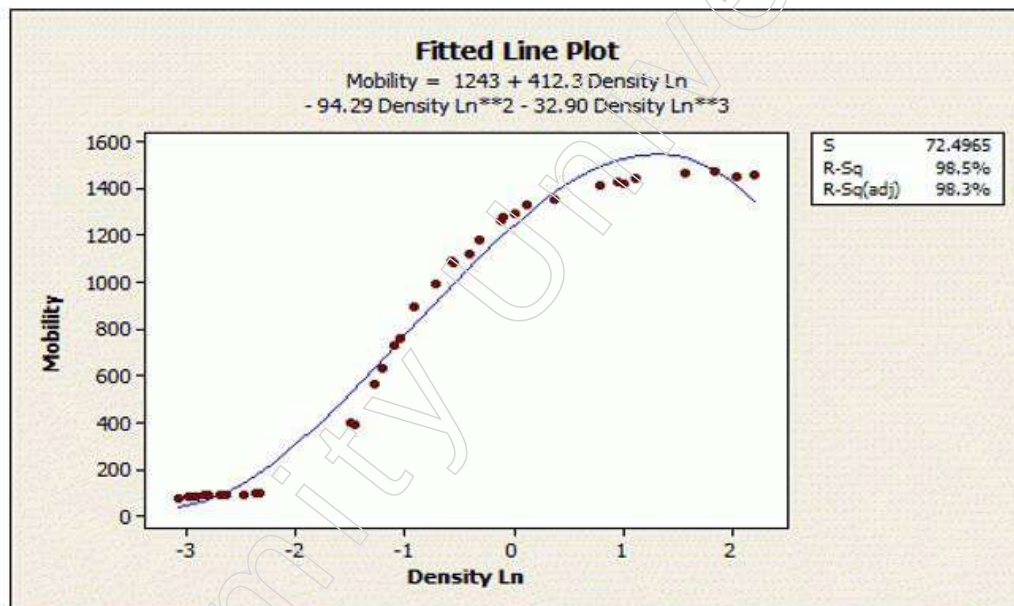
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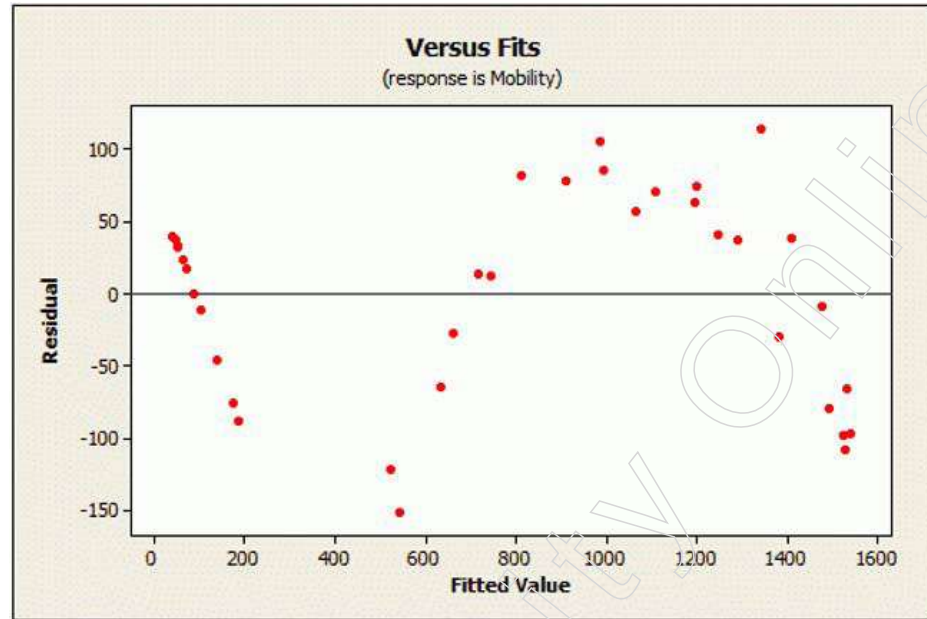
The R-squared for the regression model on the left is 15%, whereas it is 85% for the model on the right. The data points are closer to the regression line when a regression model accounts for more of the variation. In actuality, we'll never encounter a regression model with a perfect R². In such scenario, the fitted values equal the data values, and all of the observations fall on the regression line perfectly.

Are High R-squared Values Always Great?

A regression model with a high R-squared value might have a variety of issues. We may anticipate that a high R² implies a good model, but have a look at the graphs below. The fitted line plot represents the relationship between electron mobility and density.



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The data in the fitted line plot have a very low noise connection, and the R-squared is 98.5 percent, which appears to be excellent. The regression line, on the other hand, constantly under- and over-predicts the data along the curve, which is bias. The Residuals versus Fits figure highlights this undesirable tendency. The residuals in an unbiased model are randomly distributed around zero. Despite a high R2, non-random residual patterns imply a poor fit.

When our linear model is underspecified, we get this form of specification bias. In other words, major independent variables, polynomial terms, and interaction terms are lacking. Try adding terms to the model or fitting a nonlinear model to generate random residuals.

A lot of additional factors can cause our R2 to be falsely inflated. Overfitting the model and data mining are two of these factors. Either of these can yield a model that appears to offer a great fit to the data, but the results might be completely misleading.

An overfit model is one in which the model matches the sample's random idiosyncrasies. Chance correlations can be used in data mining. In either scenario, even with completely random data, we may construct a model with a high R2!

R Square Formula

R square metric is given by:

$$R^2 = 1 - \frac{\text{MSE}(\text{model})}{\text{MSE}(\text{baseline})}$$

$$\frac{\text{MSE}(\text{model})}{\text{MSE}(\text{baseline})} = \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (\bar{y} - \hat{y}_i)^2}$$

MSE (model): Mean Square Error of the predictions against actual.

MSE (baseline): Mean Square Error of mean prediction against actual.

Adjusted R Square:

The adjusted R-Square metric is a more sophisticated variant of the R-Square statistic. When additional variables are added to the model, the R-Square value either grows or remains constant. R-Square does not punish you for include variables that aren't useful to the model. However, adjusted R-Square increases only when a meaningful variable is included in the model. The adjusted R-square measure is provided by-

$$\bar{R}^2 = 1 - \left(1 - R^2\right) \left[\frac{n-1}{n-(k+1)} \right]$$

k: number of variables

n: number of observations

The number of variables is taken into consideration in adjusted R-Square. As we add more variables to the model, the denominator $n-(k+1)$ drops, while the equation as a whole rise.

If Adjusted R-Square does not rise, the variable included is not useful to the model. As a result, we deduct a bigger number from 1 and the Adjusted R-Square decreases.

Classification Model Performance Parameters

It is critical to determine the optimal model when developing categorization models such as churn or fraud prediction. This is possible with the assistance of model performance parameters. We need these parameters when our dependent variable is categorical; most of the time, we only deal with binary. Let's look at model performance parameters for categorization models.

1. Confusion Matrix:

A confusion matrix is a $n \times n$ matrix with N classes in the dependent variable or goal. The majority of the time, $n=2$ is used, resulting in a 2×2 matrix. Now, let's look at the statistics that may be derived from this confusion matrix.

Confusion Matrix	Target		
Positive	Negative		
Model	Positive	A	B
Negative	C	D	

1. Accuracy: $(A+D) / (A+B+C+D)$
2. Misclassification error: $(B+C) / (A+B+C+D)$
3. Positive Predictive Value or Precision: $A / (A+B)$
4. Negative Predictive Value: $D / (C+D)$
5. Sensitivity or recall or hit rate or true positive rate: $A / (A+C)$

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6. Specificity or true negative rate: $D / (B+D)$

In different business aims and areas, different criteria have been employed. If you are working on a high-risk project that requires high accuracy, such as fraud models, you will focus on recall, but if you are working on a churn model that requires as much coverage as possible, you would focus on precision.

There is one more metric that is affected by accuracy and recall. The harmonic mean of accuracy and recall values for a classification task is the F1-Score.

F1 Score formula is:

$$\frac{2 * (\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})}$$

2. Kolmogorov Smirnov chart:

The K-S chart, also known as the Kolmogorov-Smirnov chart, is used to assess the effectiveness of classification models. The degree of separation between the positive and negative distributions is measured by K-S. In most classification models, the K-S value will be between 0 and 100, with the higher the number indicating that the model is better at distinguishing between positive and negative situations.

This metric uses a percentile technique (discussed in the second paragraph) to determine a cut-off point to distinguish between predictive positive and predictive negative. Take the decile where you will obtain the most K-S as a cut off mark. After you've determined the cut off via K-S and assigned predictive positive and negative, you may use the confusion matrix to compute accuracy, recall, precision, and the remaining metrics.

See the table below for a more complete explanation of K-S:

Decile	Responders	Non-Responders	Total	Cumulative Responders	Cumulative Non-Responders	% Cumulative Responders	% Cumulative Non-Responders	K-S
0	0	0	0	0	0	0%	0%	0%
1	1,946	14,645	16,591	1,946	14,645	56%	9%	47%
2	721	15,958	16,679	2,667	30,603	77%	19%	58%
3	336	16,387	16,723	3,003	46,990	86%	29%	57%
4	204	16,440	16,644	3,207	63,430	92%	39%	53%
5	127	16,433	16,560	3,334	79,863	96%	49%	47%
6	96	16,577	16,673	3,430	96,440	99%	59%	39%
7	33	16,735	16,768	3,463	1,13,175	99%	69%	30%
8	10	16,730	16,740	3,473	1,29,905	100%	80%	20%
9	6	16,401	16,407	3,479	1,46,306	100%	90%	10%
10	3	16,739	16,742	3,482	1,63,045	100%	100%	0%
Total	3,482	1,63,045	1,66,527					

Depending on the company requirements, you may need to adjust your K-S cut off mark up and down. This is not statistical, but due to the nature of the business and domain, you may need to alter the cut off mark. For example, if you are constructing a fraud model and the business requires a high accuracy model, you will raise your cut off mark. Let's assume by K-S it was third, but due to business, you had to bump it up to second. You are now limiting yourself to 20% of the data.

3. AUC – ROC:

The most significant advantage of adopting a ROC curve is that it is unaffected by changes in the percentage of dependant variables. ROC is an abbreviation for Receiver Operating Characteristic Curve. Looking at the confusion matrix above, we can see that for a probabilistic model, the values for each measure vary. The ROC curve is a depiction of the sensitivity or True Positive rate vs the (1- specificity) or False Positive rate. An example of the ROC curve is shown below.

More area under the ROC curve indicates that the generated model is effective. Less area under the ROC curve indicates that the generated model has to be improved.

4. Concordance Discordance ratio:

This is a regularly used metric for calculating model performance. Let's use the example of two clients to better grasp the notion of concordance and discordance.

Your model predicts customers A and B. If you look at the table below, you will see that the ranking of this combination is the same in both real and projected. (A is churned, but B is not.) A has a higher probability than B). This is known as the concordance case.

Customer	Churn	Churn Probability
A	1	0.9
B	0	0.7

The inverse of this instance is termed discordance, and it occurs when the ranking of pairings does not match in real and expected. Customer L gets churned, although his churn probability is lower than M's. An example of discordance is shown below.

Customer	Churn	Churn Probability
L	1	0.6
M	0	0.8

When both the actual and anticipated values are equal, the situation is referred to as a tie.

Let us try to understand how concordance and discordance can assist us detect model performance now that we grasp the notion of concordance and discordance. In that data collection, n number of pairings have been formed. Assume the data collection contains three observations, A, B, and C. The initials of the couple will be AB, AC, and BC. How many of these pairs are concordant, and how many are discordant? We must

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subtract the concordance pair % from the overall pair. A higher concordance indicates that the generated model is good.

Aside from the above-mentioned model performance measures, we have other model performance parameters such as the gain and lift chart, the Gini coefficient, and so on, but we have covered all of the commonly used statistics.

- **Model Stability**

Creating a model and evaluating its performance is not the entire task. You must also determine whether the resulting model is consistent, overfitted, or underfitted. Model performance parameters verify this. There are a few checks you may do to determine the model's consistency. The following are some often used checks:

- **Train and test testing**

When building the model, divide the data set into train and test (70:30 or 60:40) so that you may construct a model on the training dataset and test it on the test dataset. The most essential thing to remember is that the variation in your model performance parameters for both train and test should be minimal. If they differ, it indicates that your model is not consistent, and you must make adjustments to the model to make it consistent.

- **Out of time testing**

This is a more extensive version of train and test testing. You have divided data into K pieces in K fold, let's suppose K is 10. Then, from the first through the ninth component, you construct a model and test it on the tenth part; accordingly, you will develop ten models and evaluate their performance each time.

- **K Fold Validation**

This is an extended form of train and test testing. In K fold, you have divided data into K parts, let's say K is 10. Then you develop a model on the 1st to 9th part and test on the 10th part, similarly, you will develop 10 models and check the model performance each time.

Model performance parameters will assist you in obtaining a consistent model. If the model performance parameters do not vary, the produced model is consistent. Model performance parameters are a critical component of model development. Understanding these factors and employing them wisely while constructing a model is essential. Model performance parameters aids in the creation of optimal and consistent models.

4.2.12 Interpretation R-Square

R-squared may be simply defined as how well the regression model fits the observed data values. Let's look at an example to see what I mean.

Consider a model with an R^2 value of 70%. This means that the model accounts for 70% of the fitted data in the regression model. When the R^2 value is high, it usually indicates a better fit for the model.

The validity of the statistical measure is not just determined by R^2 , but also by other criteria such as the type of the variables, the units in which the variables are

measured, and so on. As a result, a high R-squared value is not necessarily indicative of a good regression model and might sometimes suggest issues.

In general, a low R-squared value is a bad sign for a model. However, when additional factors are taken into account, a low R² value might result in a successful prediction model.

Calculation of R-squared

R-squared can be evaluated using the following formula:

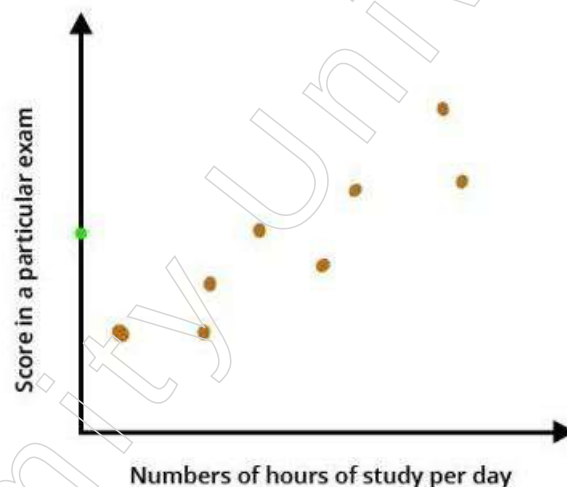
$$\text{R-squared} = \frac{SS_{\text{regression}}}{SS_{\text{total}}}$$

Where:

- $SS_{\text{regression}}$ – Explained sum of squares due to the regression model.
- SS_{total} – The total sum of squares.

The sum of squares due to regression evaluates how effectively the model represents the fitted data, whereas the total sum of squares analyses the variability in the data utilised in the regression model.

To understand the computation of R-squared more clearly, let us return to the prior circumstance where we have two factors: the number of hours of study each day and the score in a certain exam. In this case, the score represents the target variable, while the number of hours of study each day represents the independent variable.



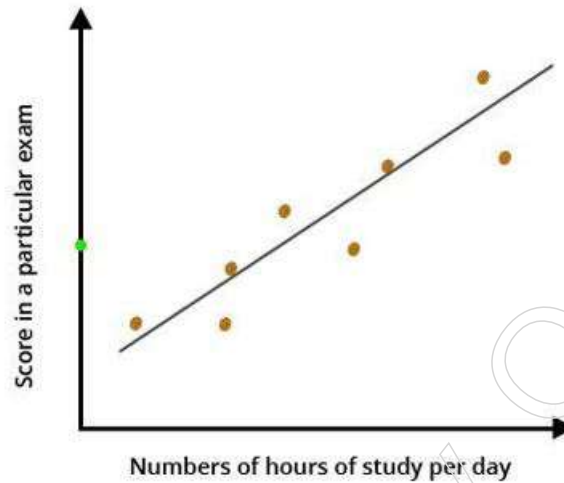
In this situation, we will require a basic linear regression model using the following equation:

$$\hat{y} = w_1x_1 + b$$

By minimizing the squared error over all data points, the parameters w_1 and b may be computed. The least square function is defined by the following equation:

$$\text{minimize } \sum (y_i - w_1x_{1i} - b)^2$$

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Now, to calculate the goodness-of-fit, we need to calculate the variance:

$$\text{var}(u) = 1/n \sum (u_i - \bar{u})^2$$

where, n represents the number of data points.

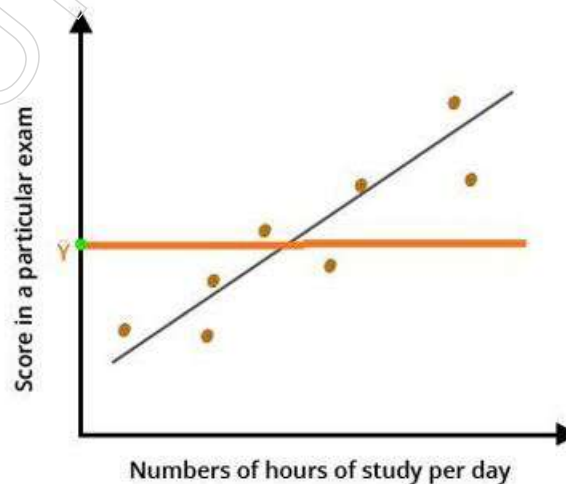
R-squared now computes the amount of variance explained by the model for the target variable, i.e. the function of the independent variable.

However, in order to do so, we must compute two things:

- Variance of the target variable:

$$\text{var}(\text{avg}) = \sum (y_i - \bar{Y})^2$$
- Variance of the target variable around the best-fit line:

$$\text{var}(\text{model}) = \sum (y_i - \hat{y})^2$$



Finally, R-squared equation can be calculated as follows:

$$R^2 = 1 - [\text{var}(\text{model})/\text{var}(\text{avg})] = 1 - [\sum (y_i - \hat{y})^2 / \sum (y_i - \bar{Y})^2]$$

Limitations of R-squared

Some of R-limitations squared are as follows:

- R-squared cannot be used to determine whether or not the coefficient estimations and projections are skewed.
- R-squared does not indicate whether or not the regression model is a good fit.

The residuals plots must be evaluated to identify the model's bias. A good model can have a low R-squared value, and a model with poor goodness-of-fit can have a high R-squared value.

Although R-squared is a simple way to estimate how well a regression model fits a dataset, it does not tell the whole picture. To understand the entire picture, you must be well-versed in R², as well as other statistical metrics and residual plots.

4.2.13 Model Evaluation MAPR RMSE

RMSE

The most commonly used assessment measure in regression situations is the root mean square error (RMSE). It is based on the premise that mistakes are impartial and have a normal distribution. The following are the essential aspects to consider with RMSE:

- The square root power enables the metric to display huge number variances.
- The squared structure of the measure contributes to more robust results by preventing the positive and negative error values from cancelling.
- It avoids the use of absolute error numbers in mathematical computations, which is extremely undesirable.
- Reconstructing the error distribution using RMSE is deemed more trustworthy when we have more samples.
- Outlier values have a large impact on RMSE. As a result, before utilising this measure, make sure you've eliminated any outliers from your data collection.
- When compared to mean absolute error, RMSE is more weighted and penalises significant mistakes.

RMSE metric is given by:

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (Predicted_i - Actual_i)^2}{N}}$$

where, N is Total Number of Observations.

Pros of the Evaluation Metric:

- It is simple to grasp;
- It may be used as a heuristic for training models.
- It is computationally simple and readily differentiable, which is desired by many optimization techniques.
- Because of the square root, RMSE penalises mistakes less than MSE.

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Cons of the evaluation metric:

- RMSE, like MSE, is affected by the size of the data. It grows in magnitude as the scope of the inaccuracy grows.
- One significant disadvantage of RMSE is its susceptibility to outliers, which must be eliminated for it to perform correctly.
- As the size of the test sample grows, so does the RMSE. This becomes a problem when we calculate the findings based on various test samples.

MAPE

Calculates the mean absolute percentage error (Deviation) function for both the forecast and the final results. MAPE is the average absolute percent error for each time period, which is calculated as actuals minus forecast divided by actuals.

The following is how the mean absolute percentage error (MAPE) is defined:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i} \cdot 100\%$$

Where:

y_i is the actual observations time series

$\hat{y}_i$ is the estimated or forecasted time series

n is the number of non-missing data points.

Pros of the Evaluation Metric:

- Because its error estimates are in percentages, MAPE is independent of the variable scale.
- All mistakes are standardised on a similar scale and are simple to comprehend.
- Because MAPE employs absolute percentage mistakes, the problem of positive and negative numbers cancelling out is eliminated.

Cons of the evaluation metric:

- One major disadvantage of MAPE is when the denominator value is zero. We are confronted with the "division by zero" difficulty since it is undefined.
- MAPE punishes negative mistakes more severely than positive errors. As a result, when we compare the accuracy of prediction techniques, it is biased since it will choose a method with numbers that are too low by default.
- Because the division operation is utilised, a change in the actual value for the same mistake will result in a different loss. Consider the following scenario: if the real value is 100 and the forecasted value is 75, the loss is 25%. When the actual value is 50 and the anticipated value is 75, the loss is 50%. However, the real fault would be the same in both circumstances. That is, 25.

4.2.14 Model Evaluation Multicollinearity Effect VIF

In regression analysis, a variance inflation factor (VIF) reveals multicollinearity.

When there is correlation between predictors (i.e. independent variables) in a model, it is called multicollinearity, and its existence might have a negative impact on your regression findings. The VIF calculates how much the variance of a regression coefficient is exaggerated owing to model multicollinearity.

VIFs are often computed using software as part of a regression study. The output will include a VIF column. VIFs are calculated by regressing a predictor against every other predictor in the model. This produces the R-squared values, which may subsequently be entered into the VIF calculation. The predictor you're looking at is denoted by "i" (e.g., x1 or x2):

$$\text{VIF} = \frac{1}{1 - R_i^2}$$

A variance inflation factor is a technique that may be used to determine the degree of multicollinearity. When testing the influence of numerous factors on a certain result, a multiple regression is utilised. The dependent variable is the outcome that is influenced by the independent variables—the model's inputs. When there is a linear connection, or correlation, between one or more of the independent variables or inputs, multicollinearity exists.

Because the inputs all influence each other, multicollinearity is a challenge in multiple regression. As a result, they are not truly independent, and it is difficult to determine how much the interaction of the independent variables influences the dependent variable, or outcome, inside the regression model.

In statistical terminology, a multiple regression model with significant multicollinearity makes estimating the connection between each independent variable and the dependent variable more challenging. Changes in the data or the structure of the model equation might cause substantial and unpredictable changes in the estimated coefficients on the independent variables.

There are tests for multicollinearity that may be done to confirm the model is properly stated and running effectively. One such measurement instrument is the variation inflation factor. The use of variance inflation factors aids in determining the severity of any multicollinearity concerns so that the model may be changed. The variance inflation factor quantifies how much an independent variable's behaviour (variance) is impacted or inflated by its interaction/correlation with other independent variables.

Variance inflation factors provide a fast estimate of how much a variable contributes to the regression's standard error. When there is high multicollinearity, the variance inflation factor for the variables involved is quite big. Following the identification of these variables, many ways may be utilised to reduce or combine collinear variables, hence addressing the multicollinearity problem.

Interpreting the Variance Inflation Factor

Variance inflation factors vary from one to ten. The numerical number for VIF indicates (in decimal form) how much the variance (i.e., the standard error squared) is inflated for each coefficient. A VIF of 1.9, for example, indicates that the variance of a specific coefficient is 90 percent more than what would be expected if there was no multicollinearity – if there was no association with other variables.

Notes

A general rule of thumb for understanding the variance inflation factor is as follows:

- 1 = not correlated.
- Between 1 and 5 = moderately correlated.
- Greater than 5 = highly correlated.

It is debatable how large a VIF must be before it presents problems. What is known is that the higher the VIF, the less trustworthy your regression results will be. A VIF greater than 10 shows a substantial correlation and is cause for worry. Some writers advocate for a more cautious level of 2.5 or higher.

A high VIF is not always a cause for worry. You may acquire a high VIF, for example, by using products or powers from other variables in your regression, such as x and x^2 . High VIFs for dummy variables representing nominal variables with three or more categories are often not a concern.

Multicollinearity

While multicollinearity does not lower a model's overall predictive power, it might result in statistically insignificant estimations of the regression coefficients. In some ways, it may be viewed as a type of double-counting in the model.

When two or more independent variables are closely connected or measure almost the same thing, the underlying impact is accounted for twice (or more) across the variables. It becomes difficult, if not impossible, to determine which variable truly influences the independent variable. This is a concern since many econometric models are designed to assess exactly this type of statistical relationship between the independent factors and the dependent variable.

Assume an economist wishes to see if there is a statistically significant link between the unemployment rate (independent variable) and the inflation rate (dependent variable). Incorporating additional independent variables relating to the unemployment rate, such as a fresh first jobless claim, would almost certainly introduce multicollinearity into the model.

The whole model may have high, statistically adequate explanatory power, but it cannot distinguish whether the effect is mostly attributable to the unemployment rate or to fresh initial jobless claims. This is what the VIF would discover, and it would recommend either deleting one of the variables from the model or finding a way to consolidate them to capture their combined influence, depending on which hypothesis the researcher is interested in testing.

What is Multicollinearity?

In a regression model, multicollinearity arises when two or more independent variables are significantly associated with one another.

In a regression model, this indicates that one independent variable may be predicted from another independent variable. Height and weight, for example, household income and water usage, automobile mileage and price, study time and leisure time, and so on.

To demonstrate this, let me use a simple example from our daily lives. Colin enjoys

watching television while eating chips. The more he watches television, the more chips he eats, and the happier he becomes!

Now, if we could quantify and measure Colin's pleasure while he's engaged in his favourite pastime, which do you believe would have a higher influence on his happiness? Should I eat chips or watch TV? That's tough to say since the moment we try to gauge Colin's contentment by eating chips, he begins watching television. And as soon as we try to gauge his contentment by watching television, he begins munching chips.

In Colin's instance, eating chips and watching television are highly connected, and we cannot discern the influence of the separate activities on his joy. This is the problem of multicollinearity!

Multicollinearity in Regression Analysis: Problems, Detection, and Solutions

When independent variables in a regression model are correlated, multicollinearity emerges. Because independent variables should be independent, this association is problematic. When fitting the model and interpreting the findings, a high degree of correlation between variables might pose issues.

Why is Multicollinearity a Potential Problem?

One of the primary goals of regression analysis is to determine the connection between each independent variable and the dependent variable. A regression coefficient is the mean change in the dependent variable for each 1 unit change in an independent variable while all other independent variables are held constant. That final section is critical for our understanding of multicollinearity.

The concept is that you may modify the value of one independent variable while leaving the rest alone. When independent variables are correlated, it means that changes in one variable are connected with changes in another. The stronger the relationship, the more difficult it is to modify one variable without affecting the others. Because the independent variables tend to change in unison, it becomes difficult for the model to estimate the link between each independent variable and the dependent variable individually.

There are two types of multicollinearities:

- o Structural multicollinearity: This arises when we construct a model term from other terms. In other words, rather than being present in the data itself, it is a byproduct of the model that we provide. For example, if you square term X to describe curvature, there is obviously a relationship between X and X^2 .
- o Data multicollinearity: This form of multicollinearity is inherent in the data rather than a result of our model. This type of multicollinearity is more common in observational investigations.

What Problems Do Multicollinearity Cause?

Multicollinearity leads to the following two sorts of problems:

- o The coefficient estimates might vary greatly depending on whatever other independent variables are included in the model. The coefficients become

Notes

extremely sensitive to even minor changes in the model.

- o Multicollinearity lowers the precision of the calculated coefficients, reducing the statistical strength of your regression model. You may not be able to rely on p-values to detect statistically significant independent variables.

Consider fitting a regression model in which the coefficient values, and even the signs, vary substantially depending on the variables included in the model. It's unsettling when slightly different models yield drastically different results. You don't feel like you understand the true impact of each variable.

Add to it the fact that you can't always rely on the p-values to choose which independent variables to include in the model. This difficulty makes it difficult to both identify and defend the proper model if many of your p-values are not statistically significant.

As the intensity of the multicollinearity rises, so do these undesirable consequences. However, these difficulties only influence the connected independent variables. Even if a model has high multicollinearity, other variables in the model may be entirely unaffected. The regression example with multicollinearity that follows displays these issues in action.

Do I Have to Fix Multicollinearity?

Multicollinearity makes it difficult to understand your coefficients and limits the capacity of your model to discover statistically significant independent variables. These are unquestionably severe issues. The good news is that you don't always have to figure out how to correct multicollinearity.

The requirement to eliminate multicollinearity is determined by the severity of the problem and the primary purpose of your regression model. Remember the following three points:

1. The severity of the difficulties grows in direct proportion to the degree of multicollinearity. As a result, if your multicollinearity is relatively modest, you may not need to remedy it.
2. Multicollinearity has an effect only on the correlated independent variables. As a result, if multicollinearity does not exist for the independent variables of special relevance, you may not need to address it. Assume your model includes the experimental variables of interest as well as some control variables. If considerable multicollinearity occurs for the control variables but not for the experimental variables, then the experimental variables may be interpreted without difficulty.
3. Multicollinearity influences the coefficients and p-values but not the forecasts, precision of predictions, or goodness-of-fit statistics. You don't need to decrease severe multicollinearity if your primary purpose is to generate predictions and you don't need to comprehend the function of each independent variable.

What causes Multicollinearity?

Multicollinearity may occur as a result of the following issues:

- Multicollinearity may exist as a result of faults in the dataset at the time of generation. These issues may be the result of poorly conceived studies, heavily observational data, or an inability to modify the data.
- When new variables are established that are reliant on existing variables, multicollinearity may emerge.
- Including variables that are same in the dataset.
- Incorrect usage of dummy variables might potentially lead to a multicollinearity issue. This is referred to as the Dummy variable trap.
- In some circumstances, a lack of data might lead to multicollinearity issues.
- In some circumstances, a lack of data might lead to multicollinearity issues.

Remedies for Multicollinearity

There is no one approach that will eradicate multicollinearity. Certain techniques might be beneficial:

1. Do Nothing

Live with what you have.

2. Drop a Redundant Variable

If a variable is redundant, it should not have been included in the model to begin with. So deleting it is simply resolving a specification error. Use economic theory to help you decide which variables to eliminate.

3. Transform the Multicollinear Variables

Multicollinearity may sometimes be reduced by re-specifying the model, for example, by creating a combination of the multicollinear variables. For example, instead of putting GDP and population in the model, include GDP/population (GDP per capita).

4. Increase the Sample Size

Increasing the sample size enhances an estimator's precision and lowers the negative impacts of multicollinearity. Adding data, on the other hand, is usually not possible.

Testing for Multicollinearity with Variance Inflation Factors (VIF)

You'll be well on your way to assessing if you need to remedy multicollinearity if you can determine which variables are affected by it and the degree of the association. Fortunately, there is a very easy test that can be used to determine multicollinearity in your regression model. The variance inflation factor (VIF) detects and quantifies the intensity of association between independent variables.

For each independent variable, statistical software computes a VIF. VIFs have no upper limit and begin at 1. A value of one implies that there is no relationship between this independent variable and any other variables. VIFs between 1 and 5 indicate a substantial association, although it is not severe enough to necessitate remedial action. VIFs larger than 5 indicate significant levels of multicollinearity in which the coefficients are poorly approximated and the p-values are suspect.

Notes

VIFs may be used to find correlations between variables and to assess the strength of the links. VIFs can be displayed by most statistical applications. Because observational studies are more prone to multicollinearity, assessing VIFs is especially crucial.

4.2.15 Steps to Good Model

The process of deciding which independent variables to include and exclude from a regression equation is known as model specification. How do you select the most appropriate regression model? The world is multifaceted, and attempting to describe it with a limited sample size is ineffective. In this essay, I'll teach you how to choose the best model for your needs. I'll go through statistical methodologies, potential problems, and practical advice for choosing your model. The variable selection process is frequently a combination of statistics, theory, and practical expertise.

When a researcher seeks to quantitatively characterise the link between independent factors and the dependent variable, the requirement for model selection typically arises. Typically, scientists measure a large number of variables but only include a subset of them in the model. Analysts aim to remove unrelated independent variables and include only those that have a real link with the dependent variable. During the specification phase, analysts generally experiment with alternative variable combinations and model types. For example, they can experiment with alternative terminology to explain interactions between variables and data curvature.

By incorporating the appropriate number of independent variables in the regression equation, the analysts must achieve a Goldilocks equilibrium.

- o Too few: Underspecified models tend to be biased.
- o Too many: Over specified models tend to be less precise.
- o Just right: Models with the correct terms are not biased and are the most precise.

To minimise biased findings, your regression equation should include any independent variables expressly tested as part of the study, as well as additional factors that impact the dependent variable.

Statistical Methods for Model Specification

Statistical evaluations can be used throughout the model specification process. A variety of measures and methods can assist you in determining which independent variables to include in your regression equation.

Adjusted R-squared and Predicted R-squared: Generally, you want to choose models with higher adjusted and forecast R-squared values. These statistics can assist you in avoiding the main problem with ordinary R-squared: it always rises when an independent variable is added. This attribute tempts you to provide an overly complicated model, which might lead to misleading results.

- o Adjusted R-squared rises only when a new variable improves the model by a greater margin than chance. Variables of poor quality might cause it to fall.

- o Predicted R-squared is a cross-validation approach that may also be used to reduce. Cross-validation divides your data to see if the model is generalizable beyond your dataset.

P-values for the independent variables: P-values smaller than the significance level in regression imply that the term is statistically significant. The procedure of “reducing the model” entails incorporating all candidate variables in the model and then continually deleting the one term with the highest non-significant p-value until your model includes only significant terms.

Stepwise regression and best subsets regression: These two automated model selection techniques are algorithms that determine which variables should be included in your regression equation. These automated approaches can be useful when you have a large number of independent variables and require assistance with the exploratory phases of the variable selection process. These processes can produce the Mallows' Cp statistic, which aids in balancing the precision-bias trade-off.

Real World Complications in the Model Specification Process

The good news is that statistical approaches can assist you with model specification. Unfortunately, there are a number of potential difficulties.

- o The quality of your best model is only as excellent as the data you collect. The accurate model's specification is dependent on your ability to measure the appropriate variables. In reality, when essential factors are left out of the model, the estimates for the variables that remain might be skewed. This is referred to as omitted variable bias. If a confounder cannot be included, try introducing a proxy variable to eliminate this bias.
- o The sample you obtain may be odd due to chance or technique. When working with samples, false positives and false negatives are unavoidable.
- o When independent variables in a regression equation are correlated, multicollinearity arises. When there is multicollinearity, tiny changes in the equation can result in huge changes in coefficients and p-values. It can also lower statistical significance in key variables. Multicollinearity makes model selection difficult for these reasons.
- o If you fit several models throughout the model selection process, you will discover variables that appear to be statistically significant but are simply associated by coincidence. This is due to the fact that all hypothesis tests have a false discovery rate. Even random data may be made to appear to have important associations with this form of data mining!
- o P-values, adjusted R-squared, projected R-squared, and Mallows' Cp can all be used to identify various regression equations. There isn't always an obvious answer.
- o Stepwise regression and best subsets regression can be useful in the early phases of model development. However, studies demonstrate that while these tools can get close to the correct answer, they seldom identify the proper model.

Practical Recommendations for Model Specification

Notes

The specification of regression models is both a science and an art. Statistical approach can be useful, but in the end, theory and other factors must take precedence.

Theory

Before you begin collecting data, it is essential to research the literature to establish a theoretical knowledge of the key independent variables, their interactions with the dependent variable, and the predicted coefficient signs and effect magnitudes. Building your expertise allows you to obtain the proper data in the first place and to define the optimum regression equation without having to resort to data mining.

Specification should not be only dependent on statistical measurements. In reality, the theoretical issues should serve as the cornerstone of your model selection procedure. Make sure your statistical results fit theory and, if required, make improvements. For example, if theory indicates that an independent variable is significant, you may include it in the regression equation even though its p-value is not significant. Investigate and either adjust the model or explain the discrepancy if a coefficient sign is the opposite of theory.

Simplicity

Analysts frequently believe that complex issues necessitate elaborate regression equations. However, research shows that simplifying models frequently results in more exact models*. When you have numerous models with equivalent predictive potential, select the simplest model since it is more likely to be the best.

Begin with simplicity and add complexity only when necessary. As a model grows more complicated, it is increasingly probable that you are adapting it to match the peculiarities in your own dataset rather than true population associations. Overfitting reduces generalizability and can produce untrustworthy results.

To avoid unnecessarily complicated models, don't aim for a high R-squared value haphazardly. Confirm that adding complexity is consistent with theory and results in reduced prediction ranges. Check for additional indicators of overfitting, such as projected R-squared.

Residual Plots

Check the residual plots during the specification process. Residual plots are a simple technique to prevent biased models and can assist you in making modifications. For example, when an underspecified regression equation is skewed, residual plots show patterns, indicating the need to represent curvature. The most basic model, which generates random residuals, is a strong candidate for being relatively exact and impartial. Statistical metrics, in the end, cannot tell you which regression equation is the best. They just do not comprehend the foundations of the issue. Your knowledge is always an important element of the model specification process.

4.2.16 Summary Correlation Regression

The two multivariate distribution-based analyses are correlation and regression. The term "multivariate distribution" refers to a distribution with many variables. Correlation is defined as the analysis that determines the existence or absence of a link

between two variables 'x' and 'y.' Regression analysis, on the other hand, predicts the value of the dependent variable based on the known value of the independent variable, assuming an average mathematical relationship exists between two or more variables.

Definition of Correlation -

Correlation is derived from the terms 'Co' (together) and the relationship between two things. Correlation occurs when it is noticed that a change in one variable's unit is reciprocated by an equivalent change in another variable, either directly or indirectly, during the study of two variables. Alternatively, the variables are said to be uncorrelated when there is no movement in one variable that corresponds to any movement in another variable in a specified direction. It is a statistical approach used to express the strength of the relationship between two variables.

Correlation may be both positive and negative. If the two variables move in the same direction, i.e., a rise in one variable causes an increase in the other, and vice versa, the variables are said to be positively linked. As an example, consider the terms investment and profit. A negative correlation occurs when the two variables move in opposite ways, so that a rise in one variable causes a decrease in the other, and vice versa. For instance, consider the price and demand for a product.

Definition of Regression

Regression is a statistical approach for estimating the change in the metric dependent variable owing to a change in one or more independent variables that is based on the average mathematical connection between two or more variables. It is a significant instrument in many human activities because it is a strong and versatile tool for forecasting past, present, or future occurrences based on past or current experiences. For example, a company's future earnings can be anticipated based on historical performance.

In a basic linear regression, there are two variables, x and y, where y depends on x or is impacted by x. In this case, y is referred to as a variable dependent or criteria, while x is referred to as a variable independent or predictor. The line of regression y on x is written as follows:

$$Y = a + bx$$

where, a = constant,

b = regression coefficient,

The a and b are the two regression parameters in this equation.

Difference between Correlation and Regression

Basis For Comparison	Correlation	Regression
Meaning	Correlation is a statistical measure that determines the association or co-relationship between two variables.	Regression describes how to numerically relate an independent variable to the dependent variable.

Notes

Usage	To represent a linear relationship between two variables.	To fit the best line and to estimate one variable based on another.
Dependent and Independent variables	No difference	Both variables are different.
Indicates	Correlation coefficient indicates the extent to which two variables move together.	Regression indicates the impact of a change of unit on the estimated variable (y) in the known variable (x).
Objective	To find a numerical value expressing the relationship between variables.	To estimate values of random variables on the basis of the values of a fixed variable.

4.2.17 Curve Fitting Excel

As we examine relevant mathematical models and formulae that we expect to find frequently in the corporate sector. This strategy works if we are given equations that model the processes, we are interested in. But what happens if we aren't provided equations? Many critical company functions are frequently defined by data. Past sales, material prices, and customer demand are some examples.

We can determine the best fitting curve given a data collection. A simple technique is to suppose that the data is the result of a beautiful formula. In real-world applications, we frequently notice that so-called noise may complicate matters. (For example, if I'm looking at fast food restaurant sales, our model will include noise from traffic and poor weather outdoors.) For the sake of this course, we will assume that the data is relatively clean, but with some noise. The task of developing the best fitting curve to data is divided into two parts:

1. We must first decide on the type of curve or model to be used.
2. We want to be able to adjust the model's parameters (constants) to get the greatest fit.

Coming up with a theoretical rationale why we wish to apply a certain model in a given scenario is covered in a big number of your business courses, both those you've previously taken and those you're about to take. Models that appear often in theoretical courses are given names and utilised without having to redo the model's theoretical base. (This is why we introduced the normal distribution and the logistic growth function, both of which do not appear to be simple equations.) In this course, we'll be satisfied with simple heuristic reasoning about which model to choose.

The second part of the challenge is determining how to set the parameters to produce the curve that best fits the data. A moment of contemplation reveals that determining the right meaning of "best fitting" is a difficult endeavour that is beyond the scope of this course. For the time being, we'll stick with the usual definition:

The best-fitting curve minimises the sum of the squares of the measured and anticipated values.

We'll return to that definition later in the course after we've learned more mathematics, but for now, just know that it's the standard definition and is utilised by Excel. Instead, we'll concentrate on using Excel to generate the best-fitting curve for the suitable model. Excel offers a built-in functionality that will determine the best fitting equation for a data set for a limited number of functions:

- Linear model
- Exponential model
- Polynomial model
- Logarithmic model
- Power model

We will demonstrate how to find an equation for a data collection, provided we know which model is best suited to describe the data.

Exercises: Using Excel to find best fit curves

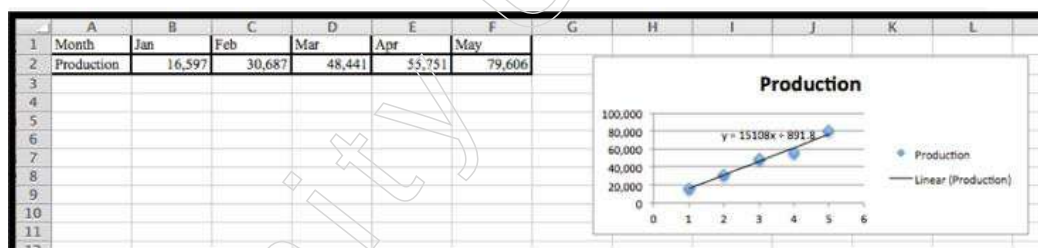
1. We have the following data on widget production:

Month	Jan	Feb	Mar	Apr	May
Production	16,597	30,687	48,441	55,751	79,606

- a. Find the best fitting linear function for the data.
- b. Give the production value that function predicts for May.
- c. Give the production value that function predicts for July.

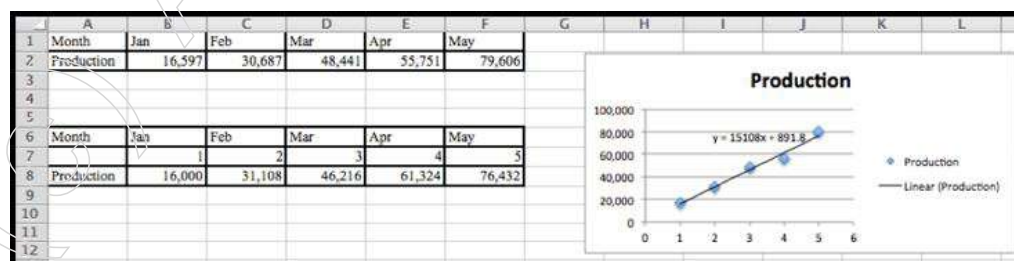
Solution.

- a. Produce a scatter plot of the data. Then, from the menu, select Chart – Chart layout – Trendlines. Find the linear approximation and choose “show the function” from the options menu.



Excel converts the months into numerical figures automatically. $y = 15108x + 891.8$ is the best linear function for the data.

- b. Make a new table and use the function to calculate the projected production levels.



Notes

Because Excel requires x to be a number, we must introduce a row and enter the proper numerical values: 1 for January, 2 for February, and so on.

May's output is expected to be 76,432.

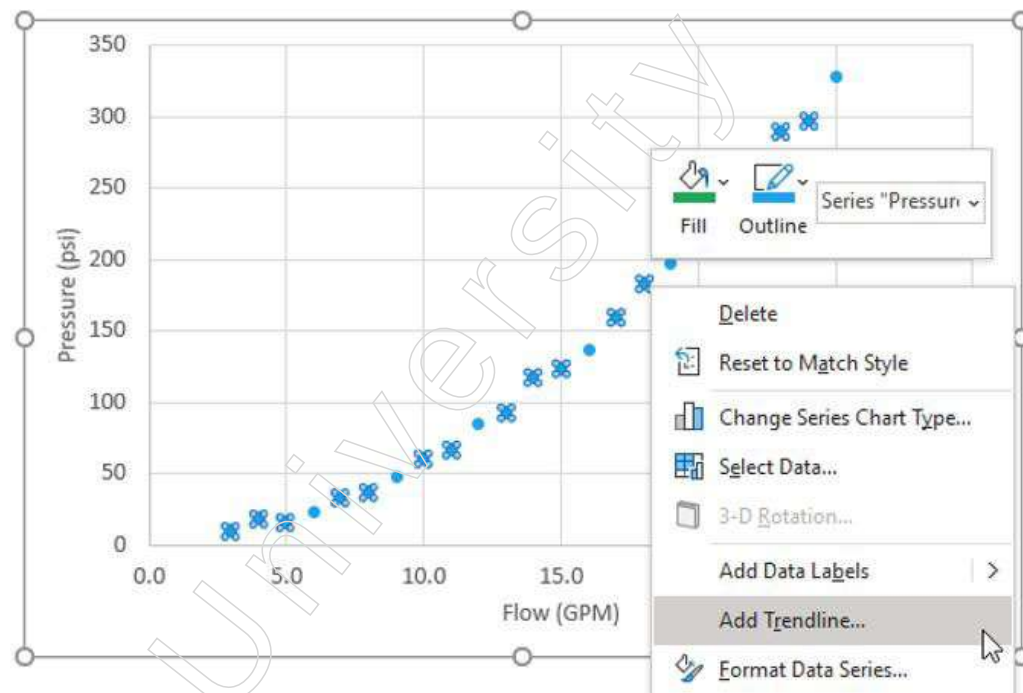
- c. To get the July production level, input $x = 7$ into the table and compute the output.

For the month of July, the linear model forecasts a total output of 106,648.

The convenient way to fit a curve to experimental data are Excel charts.

First, create a scatter chart.

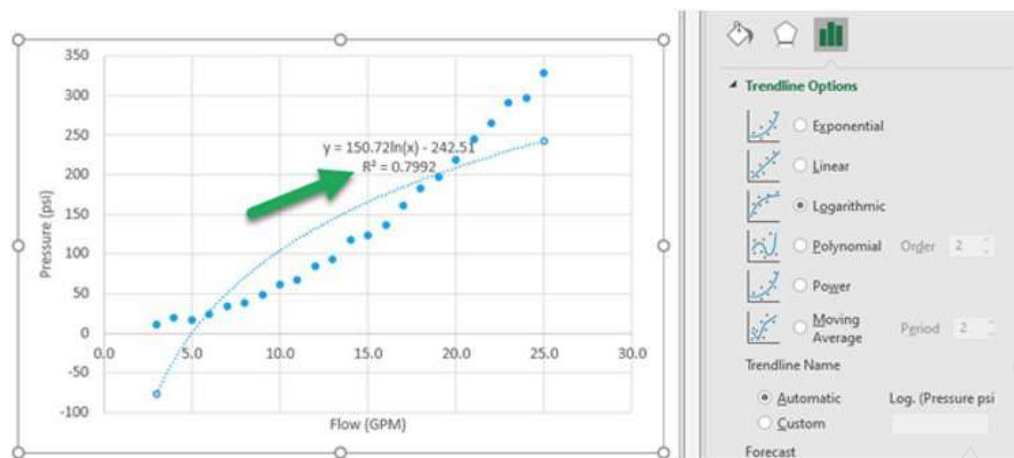
Then, right click on the data series and select "Add Trendline..."



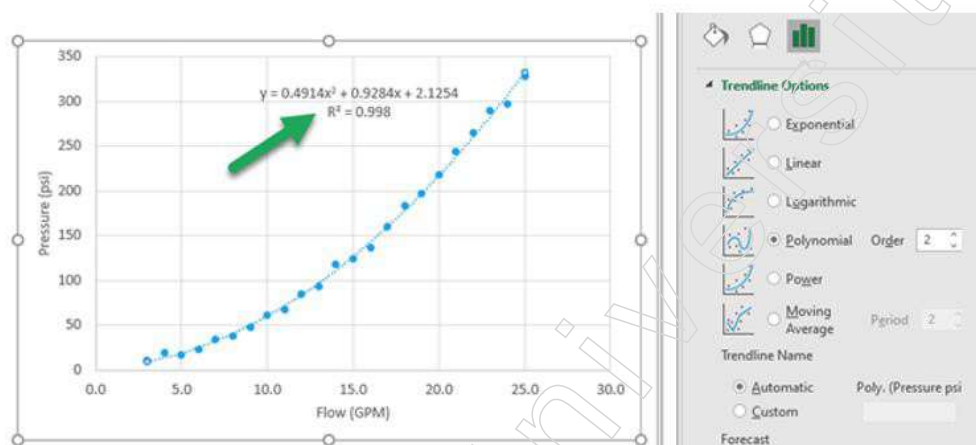
Select the Display Equation on chart and Display R-Squared value on chart choices in the Format Trendline pane.

Experiment with several types of curves to find which one optimises the value of R-squared.

A logarithmic equation with an R-squared value of 0.7992 fits the curve for this data set. We can also observe from the image below that it is not a good fit.



A second-order polynomial, on the other hand, fits the data with an R-squared value of 0.998. That is, the polynomial equation better matches the facts.



Although the trendline choice is easy, it is not necessarily the greatest option for determining the optimal equation to match the data.

It may only supply 1 or 2 significant digits for the equation coefficients in some circumstances. This limits the accuracy of any formula that employs these coefficients to the same number of significant digits.

Furthermore, there is no way to reference the equation's coefficients in the spreadsheet without manually putting them into cells. This might cause issues if the data is changed but the coefficients in the spreadsheets are not.

The LINEST function, on the other hand, provides a comprehensive technique to curve fit in Excel.

Curve Fitting in Excel

Although LINEST is an abbreviation for "linear estimate," we may use it for nonlinear data analysis with a few simple changes.

LINEST has the following syntax:

=LINEST(known_y's, [known_x's], [const], [stats])

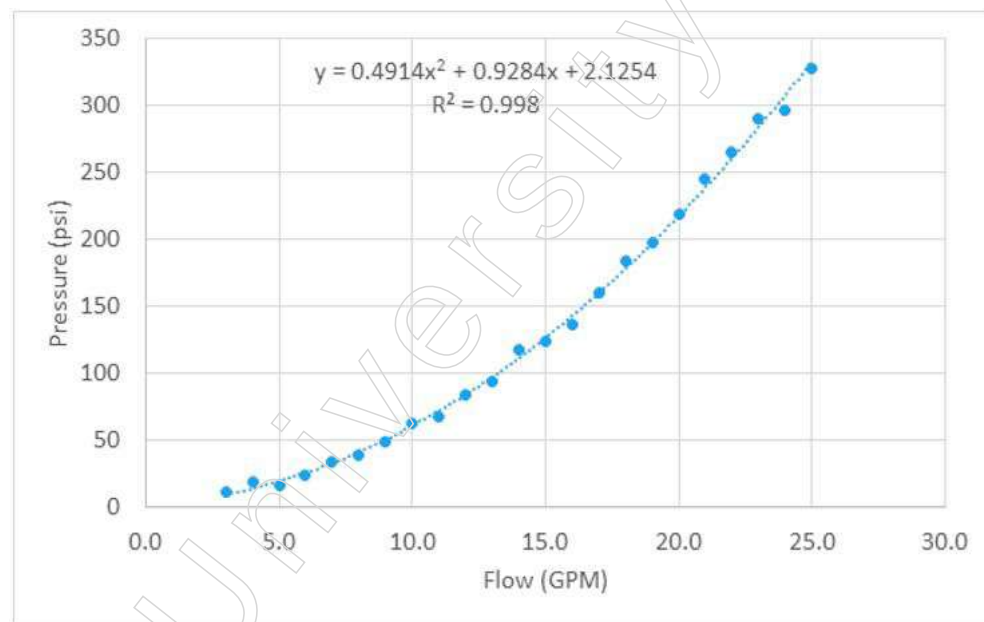
Where;

Notes

- known_y's are the y-values that correlate to the x-values that you are attempting to fit (or dependent variable)
- known_x's are the x-values that are known (or independent variable)
- const (optional) is set to TRUE if the y-intercept should be computed normally and FALSE if it should be set to ZERO.
- stats (optional) are an input that specifies if regression statistics should be returned. TRUE returns the statistics, whereas FALSE returns no statistics.

Polynomial Curve Fitting in Excel

Assume we have some experimental data of pressure drop vs. flow rate via a water valve, and after displaying the data on a chart and experimenting with different trendlines, we discover that the data is quadratic, as seen in the example above.



Despite the fact that this data is nonlinear, the LINEST function may be utilised to obtain the best fit curve for this data. We do this for polynomial equations by utilising array constants.

The fact that we may return the coefficients straight to cells is a benefit of utilising LINEST to obtain the coefficients that define the polynomial equation. We won't have to manually transfer them out of the chart this way.

Because the equation is quadratic, or a second order polynomial, it has three coefficients: one for x squared, one for x, and one for a constant. So we'll need to start by making a place to hold the equation's three coefficients.

	A	B	C	D	E	F	G	H
1								
2								
3								
4								
5								
6								
7								
8								
9								

Valve Pressure Drop	
Flow	Pressure
GPM	psi
3.0	10.93
4.0	18.95
5.0	16.71

Outputs
Coefficients

Create a space on the worksheet to store the coefficients.

Using LINEST for Nonlinear Regression in Excel

The LINEST function provides a set of coefficients as well as optional regression statistics. So, before entering the formula, we'll need to enter it as an array formula by selecting all three of the cells for the coefficients.

If the cells carrying the flow and pressure data are labelled "flow" and "pressure," respectively, the formula is as follows:

=LINEST(pressure, flow^{1,2}, TRUE, FALSE)

Data		Outputs
Flow GPM	Pressure psi	Coefficients
3.0	10.93	=LINEST(pressure, flow^{1,2}, TRUE, FALSE)
4.0	18.95	
6.71	6.71	
8.55	8.55	

In this situation, the known y's are the pressure readings, and the known x's are the flow rates.

Measurements that have been elevated to the first and second powers In Excel, the curly brackets "{" and "}" denote an array constant. Essentially, we are instructing Excel to construct two arrays: one of flow and another of flow-squared, and to fit the pressure to both of those arrays simultaneously.

Finally, the TRUE and FALSE parameters instruct the LINEST function to calculate the y-intercept naturally (rather than forcing it to zero) and to output no further regression data.

Because it is an array formula, we must input it by pressing Ctrl+Shift+Enter.

The function then returns the x^2 and x coefficients, as well as a constant (because we chose to allow LINEST to calculate the y-intercept).

Data		Outputs
Flow GPM	Pressure psi	Coefficients
3.0	10.93	x^2 0.491391
4.0	18.95	x 0.92836
		c 2.125445

The coefficients are the same as those produced by the chart trendline tool, but they are now in cells, making them much easier to utilise in further computations.

LINEST delivers the coefficient for the highest rank of the independent variable on the far left side of any polynomial equation, followed by the next highest, and so on, until the constant is returned.

Notes

Notes

In Excel, a similar approach may be used to fit Exponential, Logarithmic, and Power function curves.

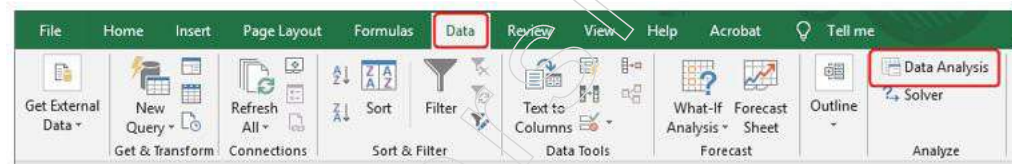
4.2.18 Correlation Regression Analysis Excel

- Using Excel to Calculate Correlation

Excel can compute correlation coefficients and perform a number of other statistical studies. Even if you don't use Excel on a daily basis, this piece is a great starting point for calculating and evaluating correlation.

In this post, I'll walk you through the process of having Excel compute Pearson's correlation coefficient and teach you how to evaluate the findings. In addition, I give links to important statistical materials that I've produced that provide straightforward explanations. We'll study and understand an example dataset together.

Check that Excel's Data Analysis ToolPak is installed before proceeding. Locate Data Analysis on the Data tab, as illustrated below.



If you don't see Data Analysis, you need to install that ToolPak.

Correlation Basics

When there is a correlation between two variables, it means that when one variable changes, the other variable tends to change in the same way. Direction and strength are the two most critical features to assess.

The direction of the correlation correlates to the sign of the coefficient, which might be positive or negative. Does the other variable likely to rise or decrease when one variable increases?

- o A positive correlation suggests that when one variable grows, so does the other.
- o A negative correlation means that while one variable rises, the other tends to fall.

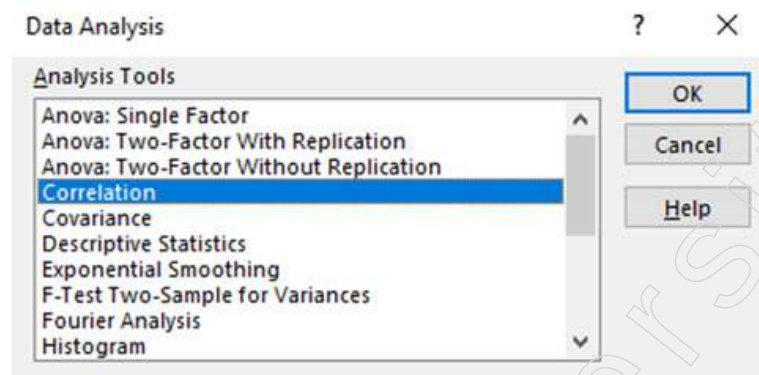
The correlation strength is shown by the values, which can vary from -1 to +1. The stronger the association, the closer the number is to -1 or 1. Values close to 0 indicate that there is no link.

Using Excel to Calculate Correlation

In this example, we'll look at two variables: adolescent girls' height and weight. These data were gathered during a real experiment. Arrange your data in columns or rows to use the correlation tool in Excel. My data is organised in columns, as illustrated in the example below.

	A	B
1	Height M	Weight kg
2	1.60	49.44
3	1.65	62.60
4	1.65	75.75
5	1.53	48.99
6	1.45	43.09
7	1.61	52.62
8	1.56	47.97
9	1.50	45.59
10	1.52	47.85

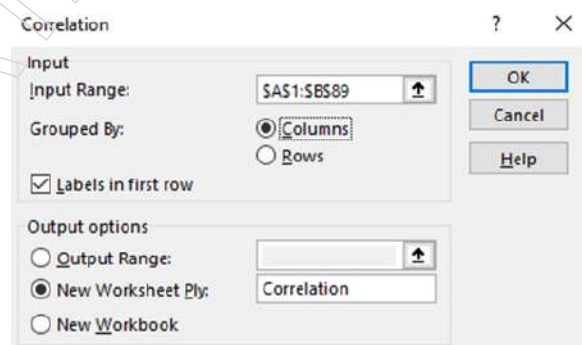
Click Data Analysis on the Data tab in Excel, as illustrated above. Choose Correlation from the Data Analysis window and then follow the instructions below.



Step-by-Step Instructions for Excel's Correlation Analysis

1. Select the range for the variables you wish to evaluate under Input Range. You must include at least two variables for correlation in your worksheet, and they must form a continuous block. For the variables you give, Excel computes all feasible pairwise correlations.
2. In Grouped By, specify how your variables should be grouped. Because this style is universal across statistical applications, I usually provide one variable per column. You may also include one variable per row.
3. Select the Labels in first row checkbox if the variable names in row 1 are meaningful. This option improves the readability of the output.
4. In the Output Options section, specify where you want Excel to display the findings. I prefer to have them on their own worksheet.
5. Click OK.

For our example, fill in the dialog box as shown below.



Notes

Notes

Interpreting Excel's Correlation Results

After Excel creates the statistical output, I autofit the columns for clarity.

	Height M	Weight Kg
Height M	1	
Weight Kg	0.705098543	1

We are curious about the association between Weight kg and Height M. Each variable is associated with itself in the two perfect values of 1!

The correlation between height and weight is 0.705. The positive coefficient corresponds to the upward slope shown on the scatterplot. The association's strength is 0.705, which is considered modest. It is undeniably real. Taller persons often weigh more. However, as we can see from the graph, there is not a perfect link. There is a trend, but there are exceptions to the pattern.

There is clearly a general trend, or inclination, for taller persons to weigh more in the height and weight statistics.

- Regression analysis in Excel - the basics

Regression analysis is used in statistical modelling to assess the associations between two or more variables:

The dependent variable (also known as the criterion variable) is the key factor you are attempting to comprehend and forecast.

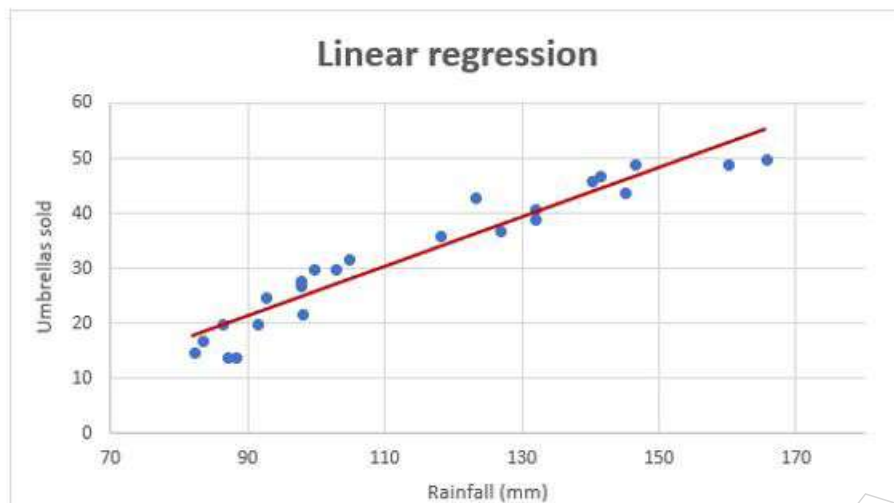
The elements that may impact the dependent variable are known as independent variables (also known as explanatory variables or predictors).

Regression analysis enables you to understand how the dependent variable changes when one of the independent factors changes, and it lets you to discover which of those variables has the most influence statistically.

A regression analysis model is technically based on the sum of squares, which is a mathematical method for determining the dispersion of data points. A model's purpose is to obtain the least feasible sum of squares and to draw a line that is closest to the data.

In statistics, there is a distinction between simple and multiple linear regression. A linear function is used to describe the connection between a dependent variable and one independent variable in simple linear regression. Multiple linear regression occurs when two or more explanatory factors are used to predict the dependent variable. Nonlinear regression should be used instead if the dependent variable is modelled as a nonlinear function since the data associations do not follow a straight line. This course will concentrate on a simple linear regression.

As an example, consider the sales of umbrellas over the previous 24 months and the average monthly rainfall during the same time period. Put this data on a graph, and the regression line will show the link between the independent variable (rainfall) and the dependent variable (umbrella sales):



Notes

Linear regression equation

Mathematically, a linear regression is defined by this equation:

$$y = bx + a + \varepsilon$$

Where:

- x is an independent variable.
- y is a dependent variable.
- a denotes the Y-intercept, or the predicted mean value of y when all x variables are equal to 0. It is the point on a regression graph where the line crosses the Y axis.
- b is the slope of a regression line, which represents the pace at which y changes as x varies.
- ε is the random error term, which is the difference between the actual and predicted values of a dependent variable.

Because predictors are never fully exact in real life, the linear regression equation always includes an error component. However, other systems, like Excel, calculate the error term behind the scenes. So, in Excel, you do linear regression using the least squares approach, looking for coefficients a and b that are such that:

$$y = bx + a$$

For our example, the linear regression equation takes the following shape:

$$\text{Umbrellas sold} = b * \text{rainfall} + a$$

There are a few different approaches to find a and b . The three primary ways for performing linear regression analysis in Excel are as follows:

- Regression tool included with Analysis ToolPak
- Scatter chart with a trendline
- Linear regression formula

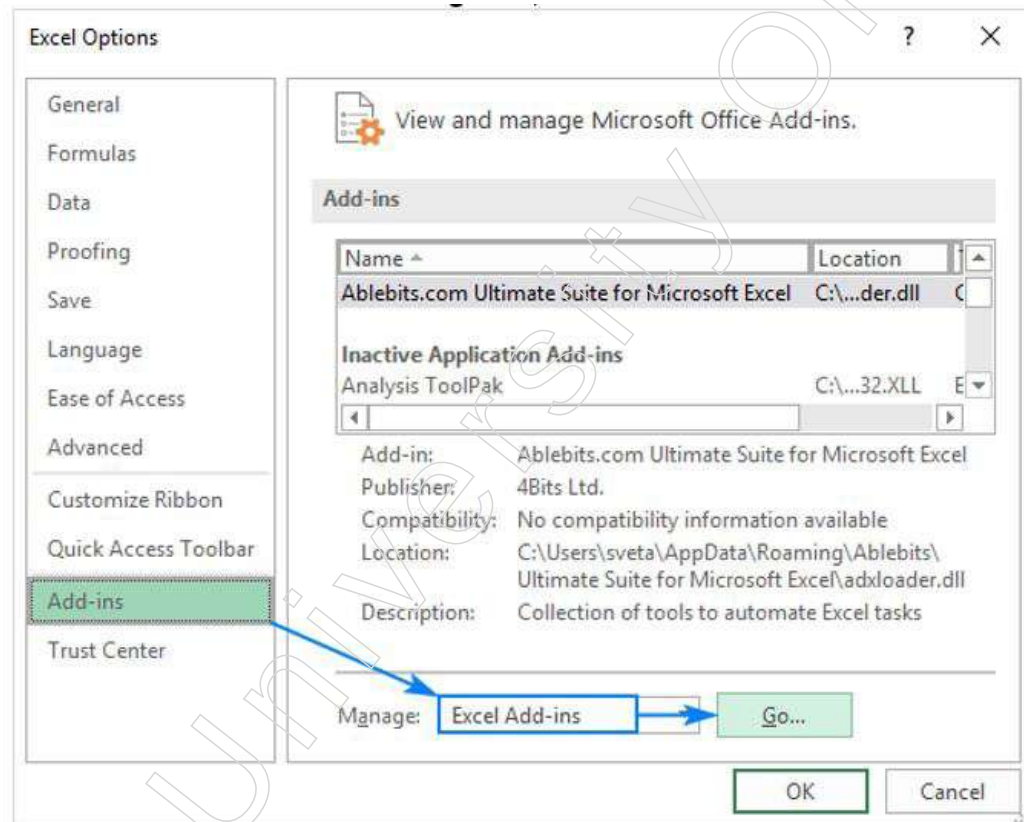
The exact instructions for each approach are provided below. How to Use Analysis ToolPak to Perform Linear Regression in Excel. This example demonstrates how to use a unique tool supplied with the Analysis ToolPak add-in to conduct regression in Excel.

Notes

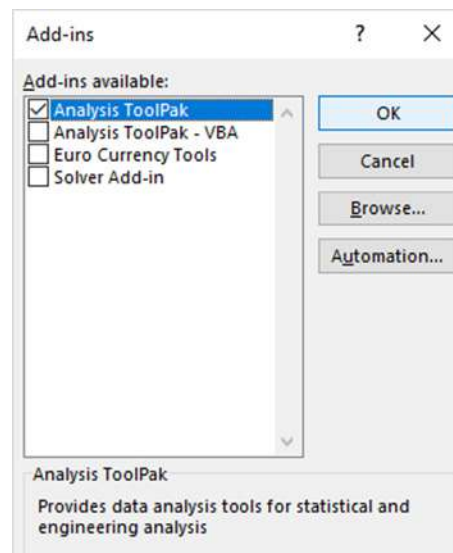
Enable the Analysis ToolPak add-in

Analysis ToolPak is available in all versions of Excel 2019, up to and including 2003, although it is not activated by default. As a result, you must manually switch it on. Here's how it works:

- 1) In Excel, go to File > Options.
- 2) In the Excel Options dialogue box, on the left sidebar, pick Add-ins, make sure Excel Add-ins is selected in the Manage box, and click Go.



1. In the Add-ins dialog box, tick off Analysis Toolpak, and click OK:



This will add the Data Analysis tools to the Data tab of your Excel ribbon.

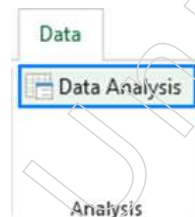
Run regression analysis

In this example, we'll use Excel to do a simple linear regression. In column B, we have the average monthly rainfall for the previous 24 months, which is our independent variable (predictor), and in column C, we have the number of umbrellas sold, which is the dependent variable. Of course, there are several more elements that might influence sales, but for the time being, we will concentrate just on these two variables:

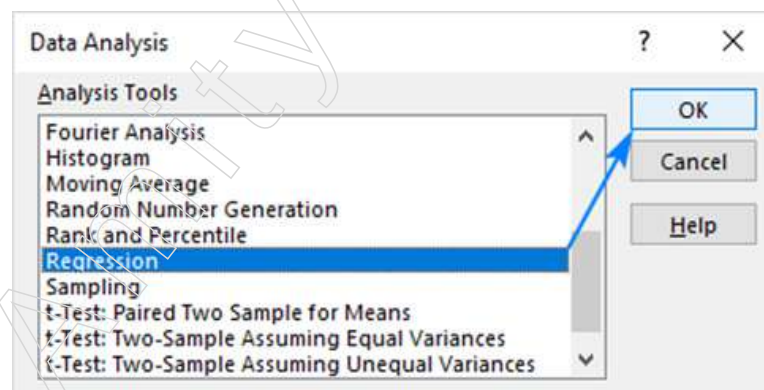
	A	B	C
1	Month	Rainfall (mm)	Umbrellas sold
2	Jan	82	15
3	Feb	92.5	25
4	Mar	83.2	17
5	Apr	97.7	28
6	May	131.9	41
7	Jun	141.3	47
8	Jul	165.4	50
9	Aug	140	46
10	Sep	126.7	37

Carry out the following procedures to execute regression analysis in Excel with Analysis Toolpak enabled:

1. Click the Data Analysis button in the Analysis group on the Data tab.



2. Click OK after selecting Regression.



3. In the Regression dialogue box, make the following changes:

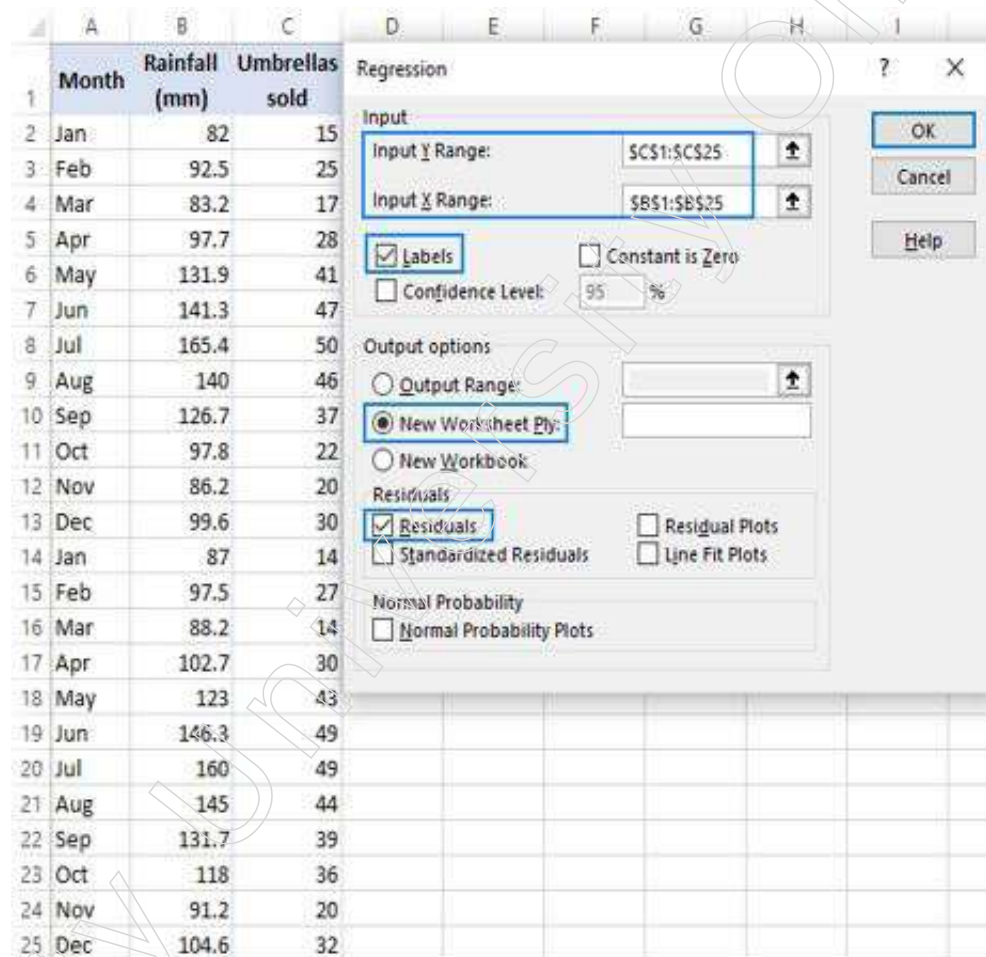
- o Choose the Input Y Range as your dependent variable. It's umbrella sales in our situation (C1:C25).
- o Choose the Input X Range, which is your independent variable. It's the average monthly rainfall in this case (B1:B25).

Notes

Notes

Select two or more adjacent columns with distinct independent variables if you are creating a multiple regression model.

- o Check the Labels box if your X and Y ranges have headers at the top.
- o Select your chosen Output option, in our example, a new worksheet.
- o Select the Residuals checkbox if you want to know the difference between the anticipated and actual numbers.



4. Click OK and observe the regression analysis output created by Excel.

Interpret regression analysis output

As previously demonstrated, doing regression with Excel is simple since all calculations are performed automatically. The interpretation of the findings is more difficult since you must understand what each number represents. The four key sections of the regression analysis output are broken down below

Regression analysis output: Summary Output

This part explains you how well the calculated linear regression equation fits your source data.

SUMMARY OUTPUT	
<i>Regression Statistics</i>	
Multiple R	0.957666798
R Square	0.917125697
Adjusted R Square	0.913358683
Standard Error	3.58141382
Observations	24

Here's what each piece of information means:

Multiple R. The Correlation Coefficient is used to calculate the strength of a linear relationship between two variables. The correlation coefficient can have any value between -1 and 1, and its absolute value represents the strength of the link. The greater the absolute value, the more powerful the relationship:

- 1 indicates a strong positive relationship
- -1 indicates a strong negative relationship
- 0 indicates no relationship at all

R Square. It is the Coefficient of Determination that is used to determine the quality of fit. It displays the number of points that fall on the regression line. The R² number is determined from the entire sum of squares, or more accurately, the sum of the original data's squared deviations from the mean.

R² in our case is 0.91 (rounded to two digits), which is rather good. This signifies that 91 percent of our values are consistent with the regression analysis model. In other words, the independent factors explain 91 percent of the dependent variables (y-values) (x-values). R Squared of 95 percent or above is considered a good fit.

Adjusted R Square. It is R squared multiplied by the number of independent variables in the model. For multiple regression analysis, this value should be used instead of R square.

Standard Error. Another goodness-of-fit statistic that reflects the precision of your regression analysis - the smaller the number, the more confident you may be in your regression equation. While R² reflects the amount of the dependent variable's variation that the model explains, Standard Error is an absolute statistic that displays the average distance the data points fall from the regression line.

Observations. It simply refers to the number of observations in your model.

Regression analysis output: ANOVA

The second component of the output is Analysis of Variance (ANOVA):

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	3122.775	3122.775	243.4623	2.21604E-13
Residual	22	282.1835	12.82652		
Total	23	3404.958			

Notes

It divides the sum of squares into discrete components that provide information about the degrees of variability in your regression model:

- df is the number of the degrees of freedom associated with the sources of variance.
- SS stands for the sum of squares. The lower your Residual SS is in comparison to your Total SS, the better your model matches the data.
- MS stands for mean square.
- F is the F statistic, often known as the F-test for the null hypothesis. It is used to assess the model's overall importance.
- Significance F is F's P-value.

The ANOVA component is rarely utilised for a simple linear regression study in Excel, but the last component is definitely worth a closer look. The Significance F value indicates how trustworthy (statistically significant) your results are. If Significance F is less than 0.05 (5%), your model is acceptable. If it is more than 0.05, you should probably pick another independent variable.

Regression analysis output: coefficients

This section contains detailed information on the elements of your analysis:

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	-19.07410899	3.372182168	-5.656310378	1.09E-05	-26.06758677	-12.08063122
Rainfall	0.45000132	0.02884018	15.6032773	2.22E-13	0.390190448	0.509812192

Coefficients is the most helpful component in this section. It allows you to construct a linear regression equation in Excel:

$$y = bx + a$$

Our linear regression formula for our data set, where y is the number of umbrellas sold and x is the average monthly rainfall, is as follows:

$$Y = \text{Rainfall Coefficient} * x + \text{Intercept}$$

With a and b values rounded to three decimal places, it becomes:

$$Y = 0.45 * x - 19.074$$

For example, with the average monthly rainfall equal to 82 mm, the umbrella sales would be approximately 17.8:

$$0.45 * 82 - 19.074 = 17.8$$

Similarly, you may calculate how many umbrellas will be sold with any other monthly rainfall (x variable) you provide.

Regression analysis output: residuals

When the estimated and actual number of sold umbrellas are compared to the monthly rainfall of 82 mm, the results are slightly different:

- Estimated: 17.8 (calculated above)

- Actual: 15 (row 2 of the source data)

What's the distinction? Because independent factors will never be ideal predictors of dependent ones. And the residuals can tell you how much the actual values differ from the expected ones:

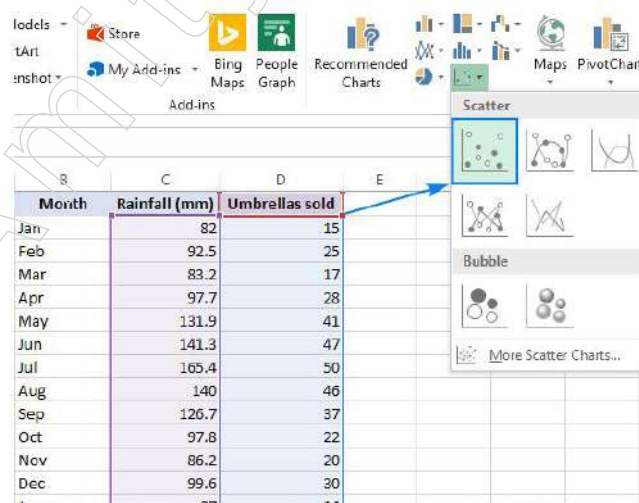
Residual output		
observation	Predicted umbrellas sold	Residuals
1	17.82599924	-2.825999237
2	22.55101131	2.448986904
3	18.56600082	-1.366000821
4	24.89101996	3.10898004
5	40.28106510.7189849	
6	44.51107751	2.488922493
7	55.35610932	5.356109317
8	43.92607579	2.073924208
9	37.94105824	-0.941058237
10	24.93602009	-2.936020092
11	19.71600478	0.283995219
12	25.74602247	4.252977532

The residual for the first data point (82 mm of rain) is roughly -2.8. So, we multiply this number by the expected value to get the real value: 17.8 minus 2.8 equals 15.

How to make a linear regression graph in Excel

Draw a linear regression chart if you need to rapidly visualise the relationship between the two variables. That's a piece of cake! Here's how it works:

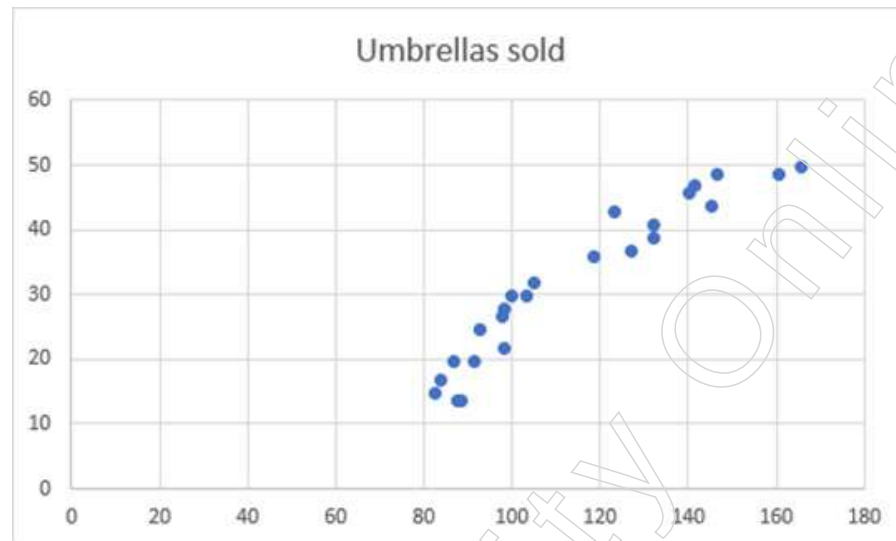
1. Choose the two columns containing your data, including headers.
2. On the Inset tab, under the Charts group, click the Scatter chart icon, and then pick the Scatter thumbnail (first one):



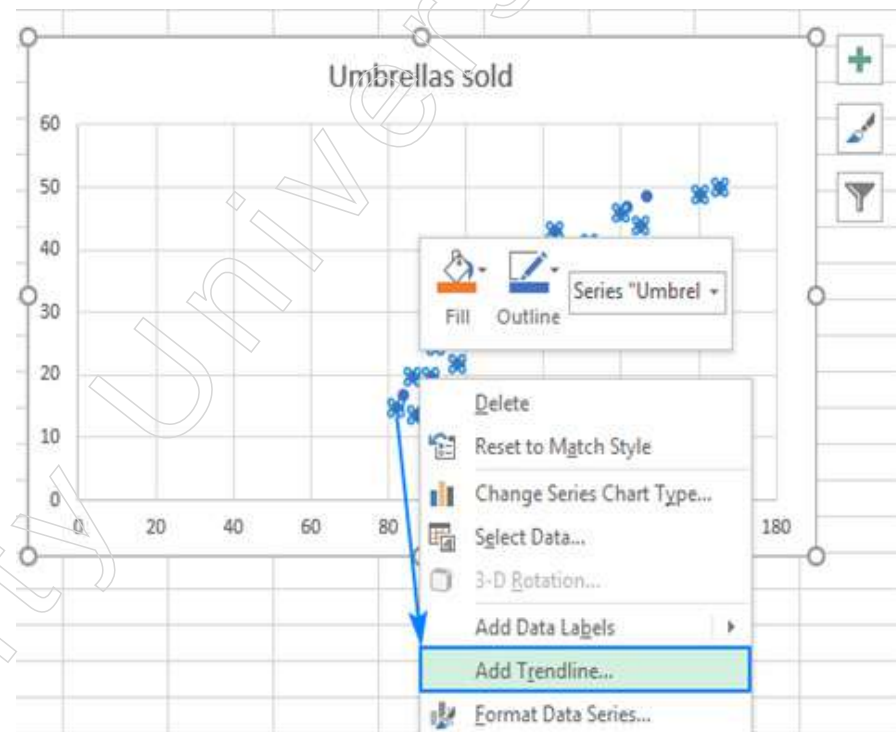
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This will insert a scatter plot in your worksheet, which will resemble this one:

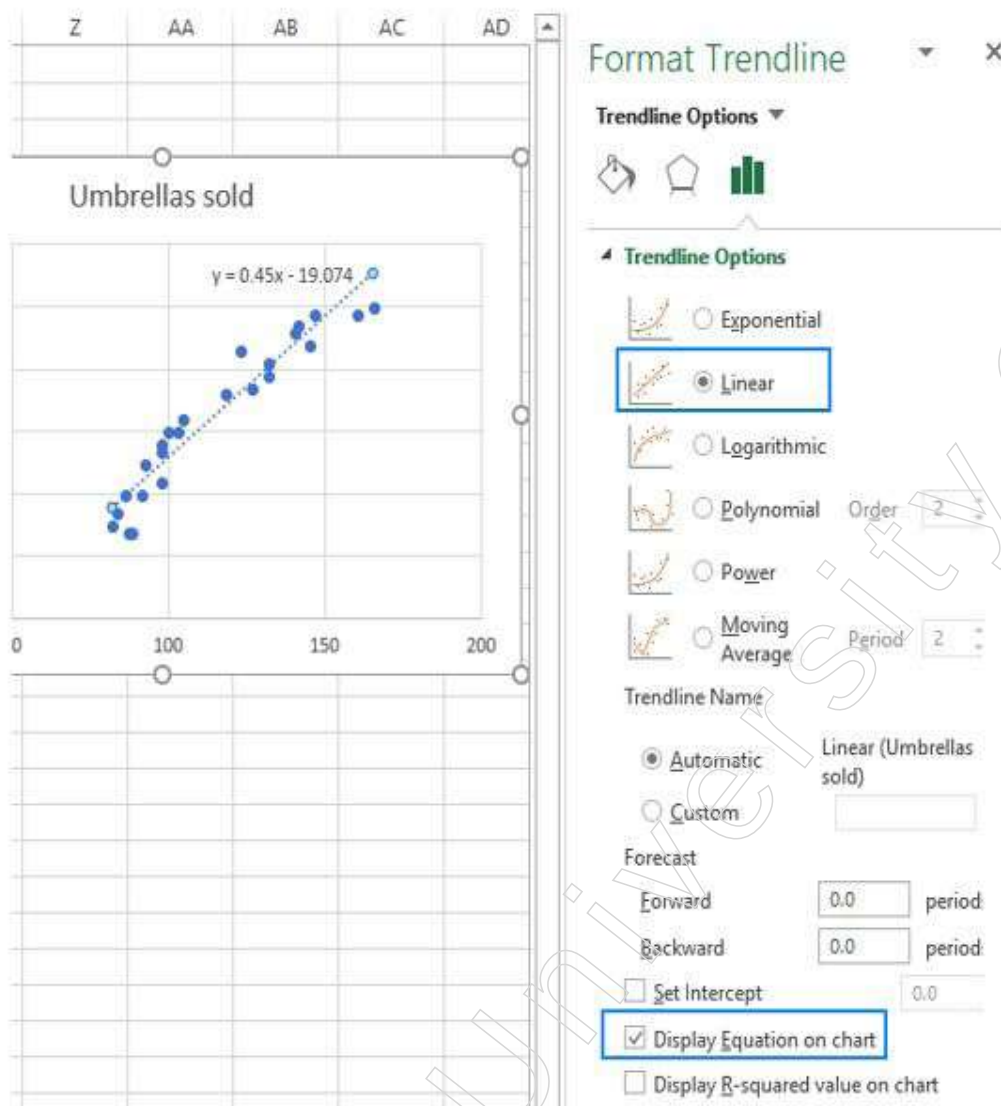


- Next, we'll draw the least squares regression line. To do so, right-click on any point and select Add Trendline... from the context menu.



- On the right pane, select the Linear trendline shape and, optionally, check Display Equation on Chart to get your regression formula:

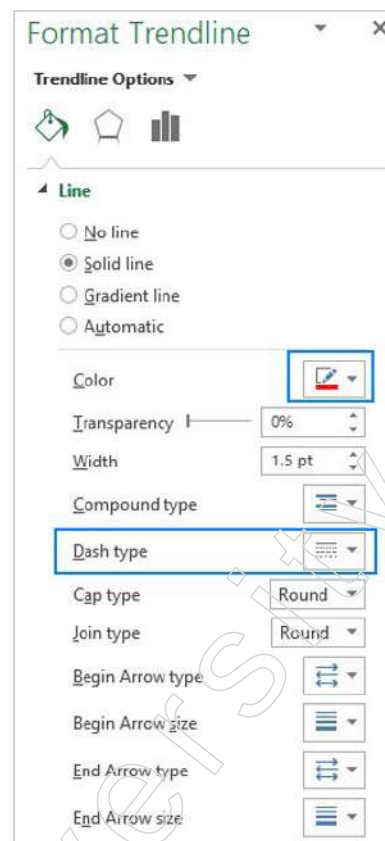
Notes



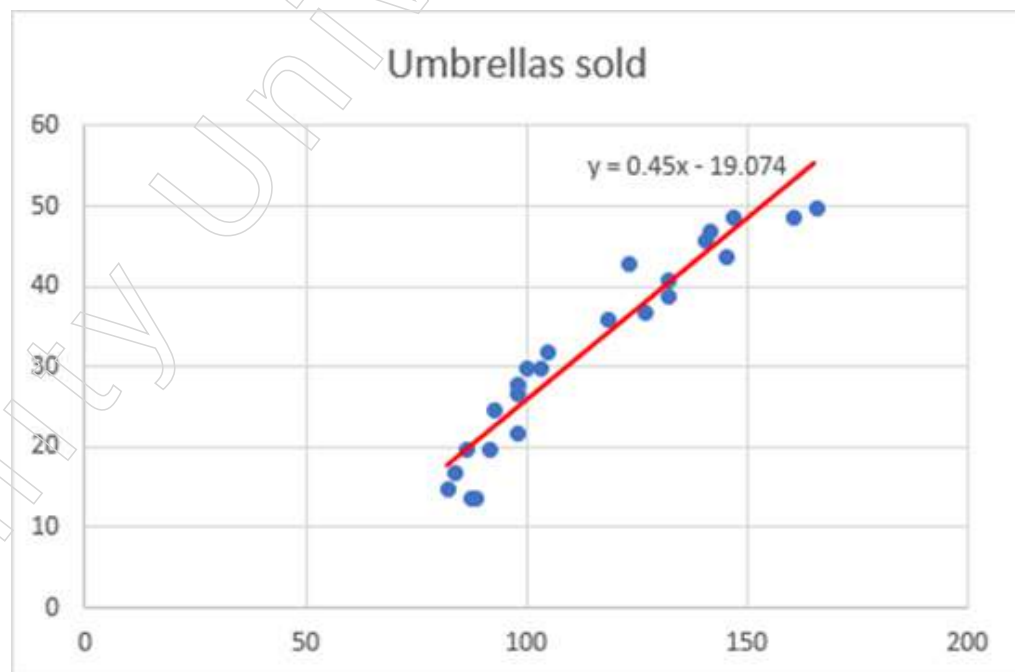
As you can see, the regression equation Excel generated for us is identical to the linear regression formula we developed using the Coefficients output.

5. Go to the Fill & Line menu and adjust the line to your satisfaction. You may, for example, change the line colour and use a solid line instead of a dashed line (choose Solid line in the Dash type box):

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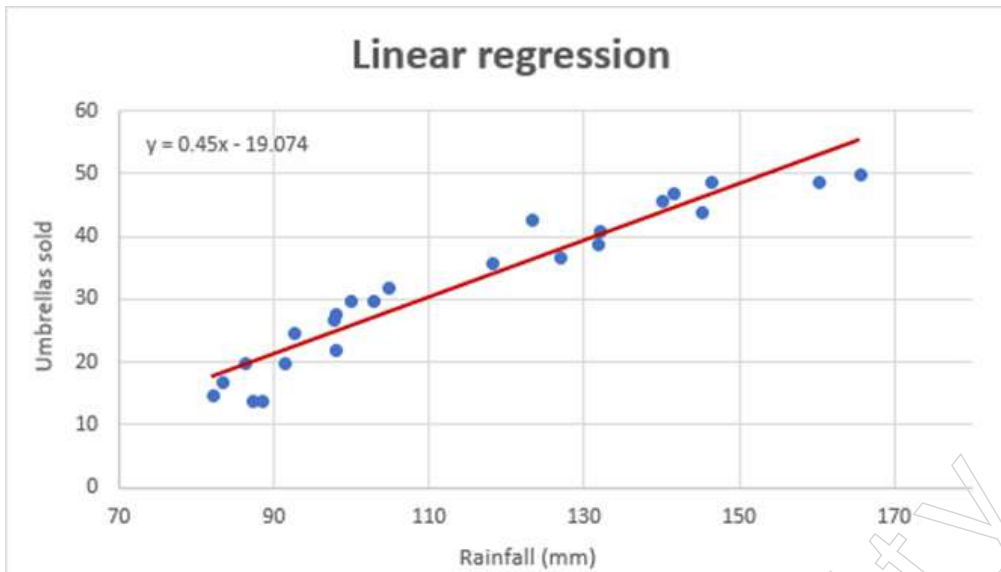


At this stage, your chart resembles a good regression graph:



Still, you might want to make a few additional changes:

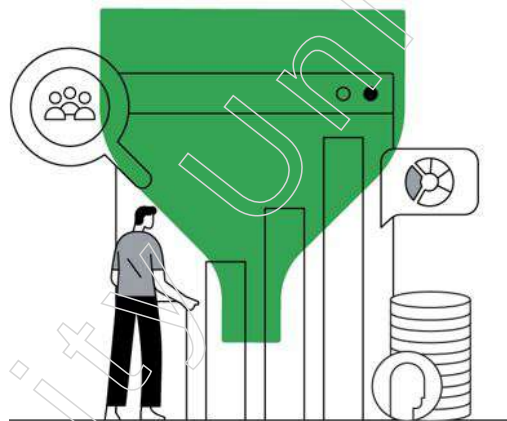
- Move the equation as you see appropriate.
- Create axis titles (Chart Elements > Axis Titles).
- If your data points begin in the centre of the horizontal and/or vertical axis, like in this example, you may want to remove the extra white space.



Please take note! The independent variable should always be on the X axis and the dependent variable on the Y axis in the regression graph. If your graph is drawn in the wrong order, switch the columns in your worksheet and redraw the chart. If you are not permitted to rearrange the source data, you may simply alter the X and Y axes in a chart.

4.2.19 Case Study Business Problem and Data

How 3 brands used data-smart ways to drive profitability



Knowing how to extract relevant insights from massive volumes of data is becoming a key competitive edge as advertising technology advances. Many businesses continue to utilise lead volume to maximise campaign budget. However, when consumer behaviour shifts and marketing budgets contract during the pandemic, it's critical to harness data and find marketing indicators that will assist produce the greatest profit.

Brands must understand the little activities consumers take throughout the customer journey in order to move beyond digital marketing ROI and enhance profitability. Knowing what motivates individuals to buy or lose interest may help companies make better investment decisions that result in direct profit. And, happily, there are remedies available.

Notes

The most recent automation and machine learning solutions in digital marketing enable you to transform massive amounts of data into useful real-time action at scale. Here's how three businesses utilised these technologies to make sense of their data and improve their advertisements for profit.

How to use customer signals to find the most profitable prospects

Begin by analysing data signals using tools to better understand your most profitable consumers. That's how ZEE5, an Indian video streaming business, increased the number of members on its site.

ZEE5 switched their ad campaigns to Google Analytics for Firebase (GA4F), which allows for limitless event tracking and centralises data collection. This allowed it to analyse different data points from current subscribers, such as location, device, and browsing behaviour, and construct a model that determined its two most important subscriber segments: reinstalling subscribers and people obtained via seeing a show's trailer.

Because GA4F automatically links to Google Ads, ZEE5's campaigns began bidding higher on prospects who had the characteristics of persons who were more likely to subscribe and had greater lifetime values. It also started crediting campaigns for any reinstalls after 30 days of removing rather than the prior six months in order to tailor its efforts for the high-value category of customers reinstalling.

Its model was also developed further by machine learning, utilising performance data sent back into Firebase, allowing ZEE5 to better optimise Google Ads. It observed a 125 percent increase in transaction value and 2X more total subscriptions by concentrating on the most profitable audiences.

Brands that utilise GA4F have more indications about their users, allowing them to locate and bid on the greatest prospects. According to internal Google statistics, ads that move to GA4F attract more active users, with a 13% rise in installations and a 20% increase in in-app events, with no notable change in cost per conversion.

How to identify and prioritize customers who are likely to churn

Another profit-driven marketing strategy is to utilise data to identify and nurture consumers who are likely to leave. Voot, an Indian video-on-demand provider, teamed with Google to better understand which users were likely to quit their membership.

Ads Data Hub was utilised by the team to efficiently aggregate data from many platforms, including first-party Google Ads campaign data and third-party sources such as Mixpanel. Voot then utilised BigQuery to examine the unified data, which included more than 10 billion data points from throughout the customer funnel, and create prediction models for audiences who were likely to abandon their subscription. It looked at everything from material viewed to viewing behaviour and traffic sources, and it utilised machine learning to identify segments and critical qualities to look out for, such as new users who only watched one genre in their first month of subscribing.

Voot's remarketing funds are now focused on individuals who need an extra nudge to stay, thanks to data driving its user attrition prediction algorithm.

The audience that was most likely to churn was then divided into smaller groups

based on similar behaviours. It chose specific shows to recommend to each subscriber who was likely to cancel the service based on patterns of content that people in related groups had watched. In one study, the group that received personal recommendations had 35% fewer cancellations than the control group.

Voot's marketing efforts are now data-driven, with its unique churn prediction model, and its remarketing funds are only spent on customers who need an extra push to remain, not on those who are likely to return on their own.

How to optimize Search campaign bidding on higher-value customers

Analysing data from many customer journeys can assist you in identifying potential customers who are worth your marketing spend even before they become a lead. Oyo, a hotel booking website, had traditionally judged campaign performance in terms of cost-per-lead. However, when the pandemic increased pressure on the hospitality business, it noticed that many leads were not converting, so it chose to focus on high-intent audiences who were more likely to complete their transactions.

To optimise revenues rather than leads, Oyo changed its Search advertising to value-based bidding, which allows for larger bids on consumers who are more likely to provide lucrative results. It also kept feeding first-party conversion data like hotel stays and frequency, as well as transaction amounts, back into the machine learning-powered ads.

Oyo was able to identify which of its clients were the most valuable by utilizing machine learning in marketing tools to make sense of the numerous data signals. As a result, it allocated more of its money to targeting high-potential clients and increased its return on ad spend on Search ads by 75%.

Brands are losing money by failing to link advertising expenditure to data throughout the whole consumer experience. However, with the most recent marketing solutions, such as GA4F and Ads Data Hub, you can simply integrate and analyze numerous consumer data signals using machine learning and put your marketing spend to better use.

4.2.20 Case Study CorrMatrix Regression Analysis

Multiple Regression Analysis -- A Case Study

Case Study Method

The first stage in a case study analysis is to do research on the subject property and identify the important elements that influence it. Then, in order to assess any influence on value, case studies from comparable properties that are similarly located in relation to the subject property and its value drivers are generated. Case studies must make "apples to apples" comparisons in order for the study to be credible. This implies that the case studies being utilised must have similar property, market, and value drivers (highest and best usage) as the subject property.

The case-study method is typically seen as a subset of the Sales Comparisons approach to value. Statistical studies, including simple and multiple regression analysis, are seen to be a refined form of sales comparison analysis since they employ more

Notes

data and allow for statistical testing of the conclusions. The goal of this MRA case-study analysis is to show how this statistical approach may be used to isolate the influence of distinct value drivers, as well as to show how MRA results can be used to “test” the reasonableness of the factors that constitute the basis of market value.

The physical and functional aspects of the case-study subject property are described briefly below.

Current Use: The site has a one-story, 8,967± SqFt concrete tilt-up (“CTU”) warehouse structure with parking that was built in 1974.

Construction Type: Class “C” – Concrete Tilt-up.

Year Built: 1974. The actual age is 40 years.

Effective Age: 20 years

Total Economic Life: 45 years.

Remaining Economic Life: 25 years.

Multiple Regression Analysis

Description of Improvements: This is a concrete tilt-up warehouse/retail building with office space and these general specifications:

Vertical Clearance: 20’ Min. / 24’ Max.

Office Space: 796± SqFt (22.39% of GBA)

Parking: 12 Spaces

Parking Ratio: 1.509/1,000 SqFt

Sprinklers: The property is fully sprinklered.

Electrical Power

- o Main: 2000 Amps, 480 VAC, 3 Phase, 3 Wire
- o Subpanel: 400 Amps, 120/240 VAC, 3 Phase, 3 Wire

Loading:

- 0 “Dock High” Spots
- 3 Ground Level Doors
- 0 Rail Spurs

Functionality: According to interviews with local brokers and market participants, a lack of dock high loading locations, which are frequent in distribution warehouses, might be a practical negative to an industrial property. Manufacturing buildings developed in previous decades, such as the one under consideration, frequently have just ground level entrances. While our market leasing study revealed an overall market preference for dock high doors, the topic looked to be leased at market rates.

Multiple regression analysis (MRA) is a statistical approach that connects the behaviour or fluctuation of several elements, or independent variables, in order to

determine their individual and combined influence on a single component, known as the dependent variable. “The multiple regression equation represents the average connection between several variables, which is then utilised to predict or control the dependent variable.” In statistics, this is referred to as a hedonic pricing model. The main disadvantage of multiple regression analysis is that it requires a lot of data. For the study to be credible, large data sets must be provided. Smaller data sets carry the danger of a few observations having a large impact on the regression model’s conclusion. This might result in a lack of multivariate normality, one of the basic assumptions of a multiple regression study. According to some studies, “a decent rule of thumb is to utilise at least 10 comparable sales per independent variable included in the MRA [Multiple Regression Analysis] model.” Typically, appraisers do not have the sufficient amount of data in a specific market to establish a meaningful assessment. In the instance of the given case-study property, it is historically located in a particularly dynamic submarket. Historically, vacancy rates have been fairly low, and the market has remained reasonably active throughout the years. As a result, we were able to gather enough data to overcome this obstacle. Another disadvantage is that multiple regression analysis is statistical in nature, which means it relies on hard data features (such as building size, year built, clear height, and so on) and so cannot properly take into account all of a property’s “soft” attributes (i.e., individual location, effective age, etc.). As a result, adequate data on any comparative must not only be known, but also be quantitatively measurable. In our analysis of the case-study data set, we found seven significant factors, all of which are statistically measurable.

These variables included:

- Rented SqFt (X1)
- Percent Office Space (X2)
- Year Built (X3)
- Clear Height (X4)
- Truck-High Doors (X5)
- Ground Level Doors (X6)
- Date of Lease (X7)

After identifying the variables and the data set, we conducted the regression in Microsoft Excel using the Data Analysis ToolPack. “The Regression analysis tool performs linear regression analysis by employing the ‘least squares’ approach to fit a line across a set of observations,” according to the application. One of the most frequent methods for obtaining the regression equation is the “least squares” approach, which involves “solving a system of simultaneous linear equations in which the unknowns are the regression equation’s constants.” A first regression was performed using data from 167 leases. The remaining 20 were excluded because their residuals (the gap between anticipated and actual values) were more than one standard error. The remaining 147 leases were subjected to a second investigation. The results are shown below.

Notes

Regression Statistics						
Multiple R	0.963238794					
R Square	0.927828975					
Adjusted R Square	0.924194463					
Standard Error	845.2831017					
Observations	147					

ANOVA					
	df	SS	MS	F	Significance F
Regression	7	1276803988	182400569.7	255.282954	4.85766E-76
Residual	139	99315989.57	714503.5221		
Total	146	1376119978			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	-25199.92159	9041.39028	-2.787173301	0.006061244	-43076.35789	-7323.485294
Rented SqFt	0.299058402	0.010727598	27.87747948	8.86251E-59	0.277848032	0.320268771
% Office	1922.329272	859.5535812	2.236427506	0.026916907	222.8389989	3621.819545
Year Blt	6.205948575	4.584375236	1.353717411	0.178023625	-2.858176447	15.2700736
Clr Ht	46.36805005	25.80616359	1.796781993	0.074541676	-4.655324137	97.39142424
TD	280.7750856	76.40493815	3.674829041	0.000338913	129.7089368	431.8412344
GL	-140.4947672	72.23526439	-1.944960933	0.053798454	-283.3167296	2.327195116
Lease Date	0.354482196	0.090385688	3.921884132	0.000137489	0.175773619	0.533190774

The resulting regression equation is:

$$\begin{aligned}
 \text{Monthly Rent} = & \\
 & - 25199.92159 \\
 & + 0.299058402 \times \text{SqFt in Premises} \\
 & + 1922.329272 \times \text{Percent Office} \\
 & + 6.205948575 \times \text{Year Built} \\
 & + 46.36805005 \times \text{Feet of Clear Height} \\
 & + 280.7750856 \times \text{Number of Truck-High Doors} \\
 & - 140.4947672 \times \text{Number of Ground Level Doors} \\
 & + 0.354482196 \times \text{Date of Lease}
 \end{aligned}$$

Using the case-study property's attributes in the calculation provides a rent indication of \$11,432 per month, or \$0.367/SqFt/Mo., unadjusted for any improvements.

Reasonableness:

The majority of the other statistics in the regression summary above are tests that determine whether the model and variables are statistically significant. The "Adjusted R-Square" value is the most important. The "R-Square" statistic reveals how much of the change in the dependent variable (in this case, total rent) can be explained by changes in the independent variables (the property characteristics).

The R-Square must be between zero and one, with a greater score suggesting that the independent variables have more explanatory power. The Adjusted R-Square simply compensates for the degrees of freedom in the model (the number of observations minus the number of variables). The two numbers in this situation are quite close to one (R-Square of 0.9278 and Adjusted RSquare of 0.9242).

These results show that changes in the seven factors may explain more than 92 percent of the change in total rent. Furthermore, the model is not over-specified with irrelevant variables that might bias the findings. The "Standard Error" statistic calculates the average error in the dependent variable. This is simply a measure of the accuracy of the regression model's estimate. Estimates of the dependent variable based on

this model will be unreliable if the standard error is considerable in comparison to the forecast. One standard error is about comparable to a 66 percent confidence level. As a result, we can forecast that the total rent for the case-study property has a 66 percent chance of reaching \$11,432 \$845. We need a higher level of relevance for our aims. By increasing the standard error by 1.65, you may get a 90 percent confidence interval. Based on the original data set, we estimate a 90% likelihood that the subject case-study property will rent for anywhere between:

$$\$11,432 - (845 \times 1.65) = \$10,037/\text{Mo.}$$

And

$$\$11,432 + (845 \times 1.65) = \$12,826/\text{Mo.}$$

The “F” Statistic indicates whether or not we can reject the assumption (the “null hypothesis”) that the independent variables are negligible when considered together. In other words, how significant is the overall regression analysis? The value of our F Statistic is 255.282954. The crucial F statistic is 2.164 at a 95% confidence level with 147 degrees of freedom. The fact that the F Statistic is so much greater than the crucial value, a result of a vast and consistent data set, emphasises the model's relevance and dependability. The F probability for our model is 4.857×10^{-76} , which is nearly nil. As a result, we may infer that the coefficients revealed by the regression model did not happen by coincidence. For this regression model, we also computed the Durbin-Watson statistic. This is a test used to detect the presence of autocorrelation in the residuals, which, if present, might render the standard error measurements inaccurate. The test yields a result between 0 and 4. A score of two shows that there is no autocorrelation at all. A number of zero denotes extremely high positive autocorrelation, whereas a value of four denotes extremely high negative autocorrelation. The regression model's statistic is 2.375, which is substantially within acceptable limits. Individual “t-stats” for each independent variable are also included in the table. These are merely the coefficients' ratios to their respective standard errors. A higher t-stat implies that the variable is statistically more important in the model. A low t-statistic suggests the inverse. A t-stat greater than two indicates a significant variable, according to general rule. The majority of the factors in our model were very significant. The t-stat for the variable “Rented SqFt” was 27.8775, demonstrating an extraordinarily strong relationship between size and lease rate. This corresponds to our intuitive intuition that larger spaces command higher rents. The comparable lease date likewise has an unusually high t-stat of 3.9219. Again, this makes basic sense, because the date the lease was signed has a significant impact on the rental rate. The year the structure was erected appears to be the least important variable. This is explained by the fact that, while the year created is supposed to be a proxy for condition (i.e., newer structures are often in better condition than older buildings), it does not account for renovations or the level of care. An older, well-maintained structure might be in just as excellent, if not better, shape than a modern building with poorer quality construction and less upkeep.

The most intriguing association that can be drawn from the t-stats is that between truck-high and ground level doors. With a t-stat of 3.6748, truck high doors must have a significant effect in determining a rental pricing. Again, this makes logical sense since, everything else being equal, a building with more truck-high doors should rent for more than a structure without.

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The ground level doors variable, on the other hand, has a negative coefficient, suggesting that for every ground level door added to a structure, the rental rate decreases. This does not make sense at first. Why would adding anything to a structure that theoretically enhances functionality reduce the amount that the market will pay for it? The solution can be found in the market's average user. The outsourcing boom of the 1990s, as well as the import boom that coincided with the growth of China and India, resulted in the typical market user migrating from manufacturing to warehousing. According to market players, the average user is currently importing and exporting commodities to and from the Ports of Los Angeles and Long Beach. They require warehouse space to store their products prior to shipment. Because many of those commodities are transported by truck, truck-high doors are quite handy for the average warehouse user. As a result, the market does not always favour ground level doors. They are effectively advocating for additional truck high doors. The reality remains that there are no truck high doors in the case study. It features three ground-level entrances, one facing a key artery and two facing a secondary feeder street in the back yard. Nonetheless, the subject case study indicates that the subject case study property appears to be renting at market rates, as revealed by the regression analysis. The present tenant makes good use of the case study facilities as they are and does not require any truck-high doors.

4.2.21 Case Study Conclusion

Regression Analysis – Retail Case Study Example

A good grasp of regression analysis and modelling lays the groundwork for analysts to get a better understanding of nearly any other modelling approach, such as neural networks, logistic regression, and so on. But, before we go into regression, let's try to put some fundamental concepts in statistics into context by utilising the most watched event of the summer Olympics.

100 Meters Sprint

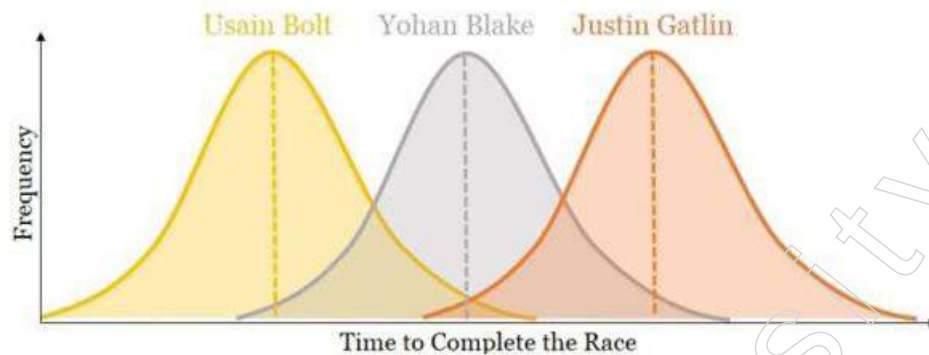
The first Olympic games I watched were held in Seoul, South Korea, in 1988. That was the same Olympics in which Ben Johnson set a new world mark for the 100-meter sprint by finishing in 9.79 seconds. Johnson later tested positive for the use of performance-enhancing substances. He was disqualified from the race and his gold medal was taken away from him. For a sporting event that lasts less than ten seconds, the 100 metres sprint is undoubtedly the most watched event of the summer Olympics. Usain Bolt set a new record in the 2012 Olympics after finishing the event in 9.63 seconds. The following is a list of medal winners from the 2012 Olympics.

Rank	Lane	Name	Nationality	Reaction	Result
	7	Usain Bolt	Jamaica	0.165	9.63
	5	Yohan Blake	Jamaica	0.179	9.75
	6	Justin Gatlin	USA	0.178	9.79

Usain Bolt is usually considered as the world's fastest man. However, I must state...

You Can Beat Usain Bolt in 100 Meters Sprint

Before I explain how, let us return to the medallists from the 2012 Olympics. For example, if we have Usain Bolt run the 100 metres a thousand times, he will finish each race with a different time, typically near to his Olympic record time. The same may be said for the two medallists, Yohan Blake and Justin Gatlin. For the purpose of argument, consider the following race finish time distributions for the three medal winners. The distributions shown below are all normal or Gaussian. For most natural phenomena, such as human running speed, a normal distribution is a suitable assumption.



Using the aforementioned distributions, Usain Bolt remains the most likely winner of the gold medal. There are still scenarios in which any sprinter might win the gold medal. This, in my opinion, is the cornerstone of statistical reasoning. Returning to our section title, if you compete with Usain Bolt Googolplex a number of times, there is still a chance that you will win at least one race against the fastest man in the world. Assume you are the Chief Analytics Officer & Business Strategy Head of Dress Mart Inc. in this case study. Set the following two objectives:

Objective 1: Increase the conversion rate of campaigns, i.e., the number of clients purchasing things from the marketing product catalogue.

Objective 2: Increase the profit earned by converted clients.

In the preceding sections of this case study example, you accomplished the first goal. Customers' propensities to respond to campaigns were estimated using categorization algorithms. This leaves you with the second goal of estimating the projected earnings from each consumer if he or she reacts to the ad. This is a standard regression problem. To create a regression model, you will utilise data from 4200 consumers who reacted to earlier campaigns out of hundreds of thousands of solicited clients. All 4200 clients reside in various places that may be classified into the three groups listed below:

1. Large Cities
2. Mid-Sized Cities
3. Small Towns

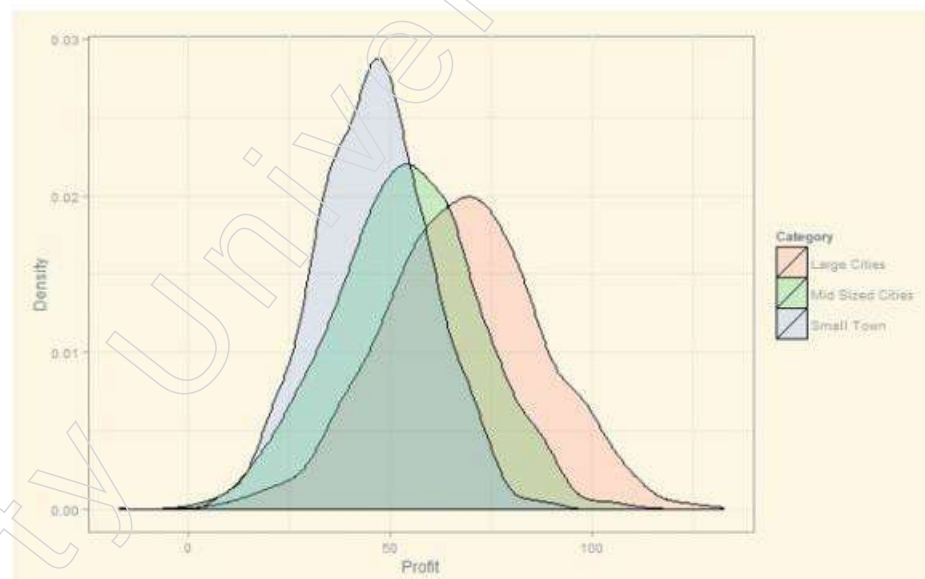
In addition, these clients are evenly divided into these three groups, with 1400 customers in each. The first item you looked at was the average profit created by these three types of cities. As seen in the graph below, the average profit values for various categories varies. Keep these average values in mind; they'll come in helpful when we're building our regression model.

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The second question is whether or not these average profit figures differ considerably. This question is solved using the distributions of all 4200 customers by location category. The graphic above depicts a depiction of these distributions (towards right). For our original data, the following are the density distributions by location category for all 4200 consumers. Because of consumer returns and other losses, profit is negative in some circumstances in this distribution.



The charts above provide a few of obvious insights:

- 1) Because of their larger earning capacity and disposable money, people of large metropolitan cities have a higher average value for profits than the rest.
- 2) Because of their higher socioeconomic variety, major metropolitan centres have a larger distribution of profit than the other two groups.

Keeping the above observations in mind, let's build a basic regression model with these categories as predictor variables. The following are the outcomes of our regression model:

Coefficients:	Estimate	Std. Error	t value	Pr(> t)
Intercept	46	0.4691	98.06	<2e-16
Mid Sized Cities	8	0.6635	12.06	<2e-16
Large Cities	22	0.6635	33.16	<2e-16
Multiple R-squared:	0.2069			
Adjusted R-squared:	0.2065			
F-statistic (P Value)	2.20E-16			

The following is the linear equation for this regression model

$$\text{Profit} = 46 + 8 \times \text{Mid sized cities} + 22 \times \text{Large cities}$$

It's worth noting that the model only uses mid-sized and bigger cities as predictor variables. The data regarding tiny towns is absorbed in the intercept section. Furthermore, because these predictor variables are dummy variables, the only possible values are 0 or 1. For example, if the location is a small town, mid-sized cities equal zero, and major cities equal zero, thus the profit is:

$$\text{Profit} = 46 + 8 \times 0 + 22 \times 0 = 46$$

Remember the values from before; this is the same average value for small towns. If the place is a medium-sized city, then

$$\text{Profit} = 46 + 8 \times 1 + 22 \times 0 = 54$$

This is, once again, the average value for mid-sized cities. Finally, the anticipated profit from a major city's resident customers is:

$$\text{Profit} = 46 + 8 \times 0 + 22 \times 1 = 68$$

Now the next question is: how good is this model? For this we will have to scroll up to the regression model results and look at the following three things:

1. P values for individual coefficients: Look at the coefficients in the rightmost column — the value is quite modest <2e-16. This suggests that the model is nearly guaranteed that the coefficients will not be zero. This is comparable to your odds of defeating Usain Bolt, which are extremely low but not zero.
2. Adjusted R-squared value: 0.2065 for our model. This suggests that merely the location category explains around 20% of the difference in earnings. This is not terrible for a single category variable, and if more important variables are added to the aforementioned model, the value of Adjusted R -squared will rise.
3. F-Statistics: Once again, the p-value is extremely low, 2.20E-16. This suggests that the model has a very low probability of being random, equivalent to your chances of defeating Usain Bolt at random.

Sign-off Note

The following sentences outline the key concepts behind the Olympic games. The most essential aspect of the Olympic Games is not to win, but to participate. The important thing is not to have won, but to have battled well.

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So, get out there, play well, and most importantly, have fun, even if your opponent is the world's fastest man.

4.2.22 Use Cases of Regression in Research (Case study Corr Matrix Regression Analysis)

Make the Investment or Not? A Case Study Using Regression-Based Driver Analysis

We frequently see businesses who are unsure on how to leverage their data to make better decisions.

Regression-based driver analysis is a tool that we commonly propose. This method aids in determining which underlying operational actions (drivers) will have the most influence on improving a measure (outcome) that is important to you. In other words, it demonstrates which of several causes is most predictive of a certain result.

We'll walk through how a store utilised it to prioritise its customer experience initiatives to further illustrate the strategy. Prior to COVID, a retail customer of ours employed a transactional feedback survey on a daily basis to measure five essential functional in-store drivers: cleanliness, merchandise availability, speed of checkout, competent staff, and friendliness of employees.

Retailer Example: To Invest in Cleanliness or Not?

The cleanliness driver's satisfaction dropped by six percentage points in June, which was statistically different from May's result. During the same time period, the client's outcome measure – "Likelihood to Return" – decreased by one point, although the decrease was not statistically significant.

The client went into full repair mode: Everyone must pitch in! Examine the cleaning crew! Should we hire fresh cleaning services? Should we replace the retail fixtures? Should we adjust the cleaning schedule for the bathrooms? Keep an eye out for garbage cans in shop atriums!

These were all commendable and potentially beneficial changes, but the change in cleaning contractors and renovation of store fixtures would cost the firm thousands of dollars and substantial employee time for retraining. Before making such investments, our customer requested that we review their data. The primary question to address before spending money: are these activities worthwhile? And, perhaps more fundamentally, will the existing condition of cleanliness deter customers from returning to the store?

To determine the answer, we examined how the five essential functional in-store drivers influenced the Likelihood to Return using existing data. This is what we did:

- To begin, we performed a linear regression using all five functional drivers as independent factors and the outcome measure – Likelihood to Return – as the dependent variable. We were startled by the results: even after adjusting for all the independent factors, speed of checkout, merchandise availability, and friendliness of personnel were predictive of Likelihood to Return, but cleanliness and competent staff were not (given the current timeframe and current state of all five drivers).

- Second, we looked at how important each of the driving factors was in terms of Likelihood to Return. Relative Importance is a regression extension that compensates for circumstances in which the input factors (our driving variables) are associated with one another, which is a typical problem in survey research. It enables us to see which driver variables are most important in predicting our outcome variable in relation to one another. The findings are displayed as a sequence of percentages for each variable that sum up to 100 percent. Cleanliness was shown to have a relatively low relative impact on Likelihood to Return.

The analysis was rather straightforward: the customer had a directional issue with cleanliness that needed to be addressed, but it was not an emergency and certainly not one that required thousands of dollars in new fixtures and other costly changes.

The analysis was pretty clear, the client had a directional issue to fix with cleanliness, but not an emergency and definitely not an issue that required thousands of dollars for new fixtures and other costly improvements.

Check your understanding

- 1) _____ is a collection of statistical methods for estimating connections between one or more independent variables and a dependent variable.
 - a) Regression analysis
 - b) Linear Analysis
 - c) Model Analysis
 - d) Descriptive Analysis
 - e) None of the above
- 2) Linear regression, also known as _____ and linear least squares, is the workhorse of regression.
 - a) Simple Linear Regression
 - b) Multiple Linear Regression
 - c) Ordinary least squares
 - d) Regression Line
 - e) None of the above
- 3) _____ is a statistical approach for predicting the result of a variable based on the values of two or more variables.
 - a) Simple Linear Regression
 - b) Multiple linear regression
 - c) Single Linear Regression
 - d) Regression Analysis
 - e) None of the above
- 4) The Regression Coefficient is also known as a _____.
 - a) Slope coefficient

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- b) Multivariate normality
 - c) Residual Analysis
 - d) Regression Model
 - e) None of the above
- 5) _____ is a type of regression analysis in which the connection between the independent and dependent variables is represented by an nth degree polynomial.
- a) Least Square Regression
 - b) Multiple Regression
 - c) Polynomial regression
 - d) Coefficient Regression
 - e) None of the above
- 6) What is the full form of RMSE?
- a) Root Mean Squared Error
 - b) Region Mean Squared Error
 - c) Root Mean Squared Experiment
 - d) Root Moderate Squared Error
 - e) None of the above
- 7) What is the full form of RSE?
- a) Regression Standard Error
 - b) Regional Standard Error
 - c) Residual Standard Error
 - d) Reserve Standard Error
 - e) None of the above
- 8) ROC is an abbreviation for_____.
- a) Receiver Operating Characteristic Curve
 - b) Receiver Optimization Characteristic Curve
 - c) Regression Operating Characteristic Curve
 - d) Receiver Orientational Characteristic Curve
 - e) None of the above

Summary

- Linear regression is used to predict the value of a dependent variable (y) based on a given independent variable (x).
- Linear regression models are frequently fitted using the least squares method, but they can also be fitted in other ways, such as by minimising the “lack of fit” in some other norm.

Notes

- The results of regression analysis might be either linear or nonlinear graphs.
- Linear regression is a technique for simulating the connection between two variables.
- Multicollinearity, which arises when the independent variables (explanatory variables) are highly interrelated, should not be present in the data.
- Linear regression is based on the assumption that the amount of error in the residuals is comparable at each point in the linear model. This is referred to as homoscedasticity.
- Regression analysis is often based on the sum of the squares of the observed values' departures from the lines of best fit.
- The least-squares approach is a type of mathematical regression analysis that is used to identify the best fit line for a set of data points, offering a visual representation of the connection between the data points.
- As additional variables are introduced to the model, the value of r-square always grows or remains constant, without recognising the relevance of the newly added variable.
- Linear regression finds the equation with the least difference between all of the observed and fitted values.

Activity

1. Write a case study on Linear and Regression Analysis.

Questions & exercise

1. Explain the model evaluation for MAPR RMSE and how model evaluation multicollinearity effect VIF.
2. What causes Multicollinearity problem. What causes Multicollinearity problem and how to fix the same?
3. Explain the concept of Correlation and Regression.
4. What is Polynomial Regression and Regression Coefficients.
5. Explain the concept of Linear Regression.

Glossary

- Regression is a statistical approach used in finance, investment, and other fields to identify the strength and type of a connection between one dependent variable and a sequence of other variables.
- There are various types of regression analysis, including linear, multiple linear, and nonlinear.
- The Regression Line is the line that best fits the data, with the least overall distance between the line and the dots (variable values) depicted on a graph.
- Multiple linear regression analysis is identical to basic linear regression analysis, with the distinction that the model includes multiple independent variables.

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- According to the least-square approach, the curve that best fits a given set of observations is one with the smallest sum of the squared residuals (or deviations or errors) from the provided data points.
- The K-S chart, also known as the Kolmogorov-Smirnov chart, is used to assess the effectiveness of classification models.
- The most significant advantage of adopting a ROC curve is that it is unaffected by changes in the percentage of dependant variables.

Further readings

1. <https://www.mygreatlearning.com/blog/model-evaluation-metrics-for-machine-learning/>
2. <https://www.knowledgehut.com/blog/data-science/interpret-r-squared-and-goodness-fit-regression-analysis>
3. <https://www.analyticsvidhya.com/blog/2021/10/evaluation-metric-for-regression-models/>
4. <https://www.investopedia.com/terms/v/variance-inflation-factor.asp>

Check your understanding – Answers

- 1) a)
- 2) c)
- 3) b)
- 4) a)
- 5) c)
- 6) a)
- 7) c)
- 8) a)

Module - 5: Field Project and Report Writing

Notes

Structure:

- 5.1 Pre-Writing Considerations
 - 5.1.1 Topic
 - 5.1.2 Audience
 - 5.1.3 Purpose
- 5.2 Literature Review Writing
 - 5.2.1 What is Literature Review?
 - 5.2.2 Purpose of Literature Review
 - 5.2.3 Types of Literature Reviews
 - 5.2.4 Structure & Writing Style
 - 5.2.5 Stages of Literature Review Development
 - 5.2.6 Ways to Organize Literature Review
 - 5.2.7 Writing Literature Review
- 5.3 Report Writing
 - 5.3.1 Meaning of Research Report
 - 5.3.2 Types of Report
 - 5.3.3 Components of a Research Report
 - 5.3.4 Steps in Report Writing
 - 5.3.5 APA Style Essentials
 - 5.3.6 Citing & Referencing Sources
 - 5.3.7 Footnotes
 - 5.3.8 Key Considerations/Factors

Notes

Unit - 5.1: Pre-Writing Considerations

Objectives

At the end of this unit, you will be able to:

- Understand the topic for writing.
- Learn about the audience.
- Comprehend the purpose.

Introduction



Ground reports are most often allocated in disciplines of the applied social sciences [e.g., social work, anthropology, gerontology, illegal justice, education, law, and the healthiness care professions] where it is imperative to build a bridge of relevancy flanked by the theoretical concepts educated in the classroom and the drill of actually doing the work you are being taught to do. Field information is also common in convinced science disciplines [e.g., geology] but these reports are organized otherwise and serve a different purpose than what is described below.

Professors will allocate a field report with the intention of enlightening your understanding of key theoretical ideas by applying methods of careful and well-thought-out observation of, and reflection about, people, places, or phenomena prevailing in their natural surroundings. Field reports enable the development of data collection methods and observation skills, and they help you to apprehend how theory applies to real world situations. Field reports are correspondingly an opportunity to obtain evidence over and done with methods of detecting professional practice that back or challenge prevailing theories.

We are all spectators of people, their relations, places, and events; however, your responsibility when composing a field report is to comportment research based on data generated by the act of scheming a specific study, deliberate reflection, synthesis of key findings, and interpretation of their denotation.

When you're creating a field report, you'll need to:

- Observe and describe as precisely as possible the many aspects of a scenario. At all times approach your field study with an exhaustive protocol about what you will observe, where you should conduct your observations, and the technique by which you will collect and record your data.
- Uninterruptedly analyze your observations. Continuously look for the significance underlying the movements you observe. What's working on here? What does this pragmatic activity mean? What else does this narrate to? Write down that this is an on-going process of replication and analysis captivating place for the duration of your field research.
- Stick to the report's goals in your mind even though you are observing. Recording what you detect should not be done arbitrarily or haphazardly; you must be focused and pay attention to details. Cross the threshold the observation site [i.e., "field"] with a clear plan, assess what you are intending to detect and record in relative to the research problem while, at the same time, being able to adapt to changing circumstances as they possibly will arise.
- Unintentionally observe, record, and investigate what you hear and see in the context of a theoretical framework. This is whatever separates data crowds from reporting. The theoretical framework controlling your field research should regulate what, when, and how you spot and act as the foundation from which you construe your findings in relation to the original assumptions embedded in the theoretic framework.

Specimens of Things to Manuscript While Observing

- Bodily setting. The features of an occupied space and the human use of the place where the observation(s) are being accompanied.
- Substances and material culture. This refers to the presence, placement, and arrangement of substances that impact the performance or actions of those being observed. If applicable, define the cultural artifacts on behalf of the beliefs [i.e., the values, ideas, attitudes, and expectations] of the persons you are observing [e.g., the choice of particular kinds of clothing in the observation of family congregations during culturally specific holidays].
- Usage of language. Don't just detect but listen to what is being said, how is it being said, and the tone of conversations among the subjects.
- Behavior phases. This refers to documenting when and who achieves what behavior or task and how frequently they occur. Best ever at which stage this behavior is happening within the setting.
- The instruction in which actions unfold. Note consecutive patterns of performance or the moment as soon as actions or events take place and their meaning. Also,

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be prepared to note moments that deviate from these sequential patterns of behavior or actions.

- Somatic characteristics of subjects. If pertinent, note personal characteristics of individuals being observed. Footnote that, unless this data can be situated verified in interviews or as of documentary evidence, you should only focus on features that can be clearly detected [e.g., clothing, physical look, body language].
- Easy-to-read body movements. This would consist of things like body posture or facial expressions. Note that it may be applicable to also assess whether easy-to-read body movements maintenance or contradict the language hand-me-down in conversation [e.g., detecting sarcasm].

The writing procedure consists of different stages: prewriting, drafting, revising, and editing. Prewriting is the most significant of these steps. Prewriting is the “generating ideas” share of the writing procedure when the student works to control the topic and the position or point-of-view for an objective audience. Pre-writing should be accessible with the time required for a student to create a plan or progress an outline to organize ingredients for the final product.

Why Prewrite?

The pre-writing phase could also be labeled the “talking stage” of writing. Researchers have discovered that strong-mindedness plays a significant effect in literacy. Andrew Wilkinson (1965) coined the term oracy, which he defined as “the ability to express oneself logically and to communicate easily with others by word of mouth.” Wilkinson clarified how oracy leads to augmented skill in reading and writing. In other arguments, talking about a topic will recover the writing. This connection amid talk and writing is best spoken by the author James Britton (1970) who specified: “talk is the depths upon which all else floats.”

- Good writers prewrite or plan, write the first draft, revise, edit, and proofread.
- The writing process is not as simple and as straightforward as this.
- Writing is personal; no two people go about it exactly the same way.
- In prewriting, you select your subject and topic, ideas and information, and determine the thesis and organizational patterns available to you.

Prewriting Methods

At hand are a number of ways that students can challenge the prewriting stage of the writing process. Subsequent are a few of the most shared methods and plans that students can use.

- o Brainstorming - Brainstorming is the method of coming up with as many ideas as possible about a topic without being worried about the viability or whether an idea is realistic or not. A tilt format is often the calmest to organize. This can be done separately and then shared with the class or complete as a group. Admission to this list during the writing process can aid students make influences they may want to use later in their writing.
- o Freewriting - The free writing strategy is when students write whatsoever comes into their mind related to the topic at hand for a specific quantity of time, approximately

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- to 10 or 15 minutes. In a free write, students ought to not worry about grammar, punctuation, or meaning. As an alternative, they should try and come up with as many ideas as they can to help them as soon as they get to the writing process.
- o Mind Maps - Mind maps or mind-mapping are countless strategies to use throughout the pre-writing stage. In cooperation are visual ways to framework information. There are many diversities of mind maps that can be quite valuable as students work in the prewriting stage. Lattice is a great tool that has scholars write a word in the central of a sheet of paper. Interconnected words or phrases are then associated by lines to this initial word in the center. They build on the idea so that, in the end, the student has a number of notions that are connected to this central idea. For instance, if the topic for a paper is about the role of the US President, the scholar would write this in the middle of the paper. By the side of that term, they might jot down each role that the president does in a circle connected by a line to this central idea. From these footings, the student could then add subsidiary details. In the end, they would have a nice roadmap for a essay on this topic.
 - o Drawing/Doodling - More or less students reply well to the idea of being able to connect words with drawings as they contemplate about what they need to write in the prewriting stage. This can reveal creative lines of understanding.
 - o Requesting Questions - Students often come up with extra creative thinking through the use of questioning. For example, if the student has to write about Heathcliff's title role in Wuthering Heights, they influence themselves more or less by asking questions about him and the reasons for his hatred. They might ask how a 'normal' person would react to better understand the depths of Heathcliff's malevolence. The opinion is that these questions can benefit the student uncover a profounder understanding of the topic before they begin lettering the essay.
 - o Outlining - Scholars can employ modern outlines to help them organize their opinions in a logical manner. The student would jolt with the overall topic in addition then list out their ideas with supportive details. It's helpful to remind students that the more complete their outline is at the start or near the end, the easier it will be for them to compose their paper.
 - o Libitum Sampling -- this method is not that dissimilar from what people do at the zoo; they detect whatever seems interesting at the moment. There is no organized system to noting the observations; you just note whatsoever seems relevant at the time. The benefit of this method is that you are often able to detect relatively rare or unusual behaviors that might be missed by more deliberately intended sampling methods. This method is also valuable for obtaining preliminary comments that can be used to develop your final field study. Problems using this method comprise the possibility of inherent bias in the direction of conspicuous behaviors or individuals, in that way missing mundane or repeated designs of behavior, and that you may miss brief connections in social settings.
 - o Behavior Sampling -- this includes watching the all-inclusive group of subjects and recording each incidence of a specific behavior of interest and by means of reference to which individuals were complicated. The method is valuable in recording rare behaviors missed by additional sampling methods and is often used in conjunction with focal or scan methods. However, sampling can be biased in the direction of specific conspicuous behaviors.

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- o Continuous Recording -- is in charge of a consistent pattern of activity, including occurrences, durations, and latencies [the period between a stimulus and a reaction]. This is a similarly challenging strategy because you're trying to record everything within the setting and, as a result, measurement consistency may be jeopardised. In addition, durations and latencies are only stable if subjects are present during the data gathering process. Nonetheless, this strategy facilitates the analysis of behavioural orders and enables the collection of a plethora of data about the observation site and the people that are present. With this type of sampling, the use of audio or video recording is quite beneficial.
- o Focal Sampling -- this includes observing one individual for an enumerated amount of time and recording all circumstance in point of that individual's behavior. Often you have a set of predetermined groups or types of compartments that you are attentive in observing [e.g., as soon as a teacher walks on or after place to place the classroom] in addition you hang onto track of the period of individuals behaviors. This approach doesn't tend to prefer one performance over another and arrange for significant detail about an individual's behavior. However, with this strategy, you should anticipate running a large number of focus samples before you have a good picture of how group members interact. It can also be difficult to maintain a discrete presence in certain environments for the duration of the observation period without becoming invasive.
- o Instantaneous Sampling -- This is when observation sessions are divided into short intervals and sample points are used to separate them. The observer records if predetermined deeds of interest occur at the selected sample point. This method is ineffective for recording discrete short-duration measures, because observers may commonly desire to capture novel events that occur just before or after the sampling point, resulting in a sampling mistake. Despite the fact that it is not exact, this method ensures that you have a notion of durations and is reasonably simple to implement. The preceding is also valid for film behaviour patterns that occur at a specific time, such as movement or body positions.
- o One-Zero Specimen -- This is quite similar to instantaneous sampling, except that the observer reports if the curious behaviours occurred at any point during the interval rather than at the sampling point. The method is useful for capturing data on behaviour patterns that start and stop frequently and abruptly, but only endure for a short length of time. The drawback with this strategy is that you only obtain one data point for each recording session because you get a dimensionless notch for the entire filming session.
- o Scan Sampling -- this method encompasses taking a census of the whole observed group at predetermined time ages and recording whatever each individual is doing at that moment. This is convenient for obtaining group interactive data and allows for data that are evenly representative across individuals and periods of time. On the other hand, this method may be biased towards more conspicuous behaviors and you may miss a lot of what is going on sandwiched between observations, exclusively rare or unusual behaviors. It is also problematic to record more than a few people in a group setting without missing what apiece individual is doing at each predetermined instant in time [e.g., children sitting at a table throughout lunch at school]. The usage of audio or video recording is useful with this classification of

sampling.

Writing takes place in stages. Writing is a process that produces a sentence, a paragraph, an essay, and other items.

Prewriting is the initial stage of the writing process, and it requires the writer to think about three primary factors: topic, audience, and goal.



Westend61/Getty Images

5.1.1 Topic

A student may be faced with one of two types of themes: assigned topics or topics chosen by the student. If the topic has been assigned, the assignment instructions will limit and select the technique to take. Instructions must be carefully studied and instructions must be followed to the letter. If the student is given free reign over the topic, it is critical that they consider the worth and significance of the final result. A writer should choose something he is passionate about and knows a lot about, but he should also think about the impression he wants to produce and the reaction he wants from the reader. Any issue can spark a lively debate if the following options are considered: choosing an unusual topic or taking a fresh and unique perspective to an old one.

"You get ideas from daydreaming. You get ideas from being bored. You get ideas all the time. The only difference between writers and other people is we notice when we're doing it."

— Neil Gaiman

Choosing a topic and then reducing it down to a length that can be addressed

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in the time allotted is an important part of prewriting. Oral storytelling is an excellent approach to come up with a solid personal narrative topic. Writers can quickly tell a narrative and gauge whether it will be an engaging topic to write about based on the emotions of the audience.

Freewriting, a practice popularized by Peter Elbow, is another way to come up with a theme. When you free write, you write down any and all thoughts that occur to mind. This could also be a written investigation of your current understanding of a large topic with the goal of narrowing down a writing topic. Freewriting is frequently timed. The writer is told to keep writing until the time limit expires, which motivates him or her to push past preconceived notions and, ideally, find a more intriguing topic.

Several alternative approaches of selecting a topic intersect with another broad aspect of prewriting, namely, research or information collecting. Reading can help you choose and focus down a topic as well as obtain knowledge for your writing. When a writer reads other works, it extends ideas, opens doors, points to potential topics, and narrates specific content for future writing. Creating annotated note cards with one chunk of information per card is a typical method of tracking the content read. To avoid plagiarism, writers must also document music, photographs, online sites, interviews, and any other source they used.

Aside from reading what others have written, make original insights about a topic. This may necessitate on-site visits, testing, or the acquisition of authentic or primary historical materials. Writers interact with the environment or materials and create notes on their observations. Sensory nuances should be given special consideration while creating a powerful piece (what the writer hears, tastes, touches, smells, and feels). When obtaining information, many writers pay close attention to the word used when describing the subject. This would include slang, specific terminology, term translations, and often used phrases. The writer frequently seeks definitions, synonyms, and examples of how other individuals utilize terms. Other approaches for acquiring information include lists, journals, and teacher-student conferences, creating pictures, employing imagination, restating an issue in numerous ways, viewing films, and inventorying interests.

Encourage pupils to keep a three-ring binder in which they can collect ideas for future writing projects to assist them in coming up with subjects. They can also include images from newspapers or periodicals, as well as their own personal photographs, to help them think. An intriguing headline, a movie, a short story by a well-known author, or a description of a book they read could all be sources of inspiration for their next writing project.

Choosing a Research Subject

The first stage in the procedure is to research and choose a topic. As you learn more from the literature, you may revise the topic/scope of your research. Make sure you choose a topic that you will be able to devote a significant amount of time to. It is critical to consider the following when considering a topic:

Is this a topic that you're interested in learning more about?

Working on a project that bores you will make the process laborious. Because the substance of your study should reflect your enthusiasm for learning, it is critical to

conduct research in your area of interest rather than selecting a topic that fascinates someone else. Expand your thoughts and imagination as you build your study topic to see what works best for you. Consider a topic that is extremely important to your field, or look for a research gap. It may take some time to narrow down a topic and get started, but it will be well worth your efforts.

Is the subject matter relevant?

Make sure your topic is appropriate for the assignment/research. When in doubt, go over the instructions again and ask your lecturer for clarification.

What is the scope and purpose of this project?

It's possible that your chosen topic is too broad. To get started, narrow the scope and aim of your research by choosing the themes you want to investigate. Once you've done that, you may fine-tune your topic by experimenting with keyword searches in our A-Z Databases until you're happy with the results.

Is your research supported by enough resources?

You may not be able to deliver the depth of results required if the topic is too limited. When choosing a topic, be sure you have enough resources to support your research. Investigate a wide range of materials, including journals, books, and online information.

5.1.2 Audience

For communication to be effective, the audience's experience and knowledge of the issue must be considered: too technical and specialist information may be beyond the reader's comprehension; a too basic or simple approach will weary the reader. The question to consider is: What will the reader benefit by reading this essay? The goal will be to educate, entertain, or persuade. These goals are frequently integrated in a paper, with each goal serving as a function of the others.

Motivation and audience awareness are frequently the first steps in prewriting: what is the student or writer trying to say, why it is necessary to deliver it properly, and who is the intended audience. It assists you in putting your thoughts on paper regarding what you want to write about. Writers normally start with a clear sense of their audience, topic, and the importance of their communication; nevertheless, for the best communication, one of these may need to be clarified. Because they are writing for a teacher or a grade rather than a genuine audience, student writers struggle to stay motivated. Teachers frequently invite students to read to younger classes or parents, post writing for others to read, write a blog, or write about real themes, such as a letter to the editor of a local newspaper, in order to give them a genuine audience.

The intended or probable reader or readers of a technical report—or any piece of writing for that matter—are the target audience. This is the most significant aspect for most technical writers when planning, drafting, and reviewing a paper. Your writing is “adapted” to match the requirements, interests, and backgrounds of the readers who will be reading it. In actuality, one of the core causes of most of the difficulties you'll discover in professional, technical documents—particularly instructions—is a lack of audience analysis and adaptation.

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Types of Audiences

When analysing an audience, one of the first steps is to determine its kind (or types—rarely it's simply one). The following is a popular categorization of audiences:

- **Experts:** These are persons who have a thorough understanding of the business or organisation (as well as the theory and product). They created it, tested it, and know everything there is to know about it. They frequently hold graduate degrees and work in academic environments or in government and technological research and development.
- **Technicians** are the individuals who construct, operate, maintain, and repair the products that specialists develop and speculate about. They have a high level of technical knowledge as well, but it is more practical.
- **Executives:** These are the persons who make judgments concerning the products of experts and technicians in terms of business, economics, and administration, legal, governmental, and political issues. Executives are likely to have the same level of technical understanding as non-specialists. For many of you, this will be your report's primary audience.
- **Non-specialists:** These are the readers who know the least about technology. They want to know how to use the new product to do their tasks, and they want to know enough about the new power technology to vote for or against it in the approaching bond election. Or they may simply be fascinated about a certain technological topic and wish to learn more about it for no particular reason. Most likely, these people will be your secondary audience.

5.1.3 Purpose

The fundamental goal of prewriting activities is to determine the paper's focus. The point of focus is where all of your energy is focused. The paper will be unclear, shallow, and potentially chaotic if the topic is too wide. Consider the audience to see if the issue is narrow enough. If your audience lacks detailed understanding of the subject, you may want to take a more general approach. Our own expertise of the subject also limits you. You can't be precise about something you don't understand. Of course, research will provide you with the necessary facts on a subject.

After you've decided on a strategy, you may start gathering ideas. Remember that you can always change your paper's focus if you give yourself enough time to make the necessary changes. If you're having difficulties narrowing down your topic, a prewriting activity might assist.

Often, your audience will assist you in determining your goal. Their beliefs will reveal whether or not they agree with what you're saying. Let's say you're writing to persuade readers to oppose Internet restriction. The goal of your work will vary based on the audience who will read it.

- Internet users are the first audience.
- Parents are the second audience.

The reason you're writing has a purpose. To help you remember what you need to buy, make a shopping list. You can write a laboratory report to describe a chemistry experiment in detail. You may create an argumentative essay to persuade someone on

campus to amend the parking laws. You may write a letter to a friend to congratulate her on her new job. You'll see that one of your options for achieving your goal is to choose the type of your writing (list, report, essay, letter). You can also choose the style, arrangement, and types of evidence that will assist you reach your goal.

Focusing on your purpose before you start writing will help you determine what form to use, how to focus and organize your writing, what kind of evidence to use, how professional or informal your style should be, and how much you should write.

Check your Understanding

1. _____ is the first stage of the writing process, typically followed by drafting, revision, editing and publishing. Prewriting can consist of a combination of outlining, diagramming, storyboarding, and clustering.
2. _____ is whatever you do to help you figure out what your key idea is or what information, examples, explanations, or content you'll include in your paper. Prewriting techniques include freewriting, brainstorming, and grouping.
3. For communication to be effective, the audience's experience and knowledge of the issue must be considered.
4. Motivation and audience awareness are frequently the first steps in prewriting.
5. The fundamental goal of prewriting activities is to determine the paper's focus. The point of focus is where all of your energy is focused.

Summary

- "The 'getting ready to write' stage is known as prewriting. The concept that authors have a topic totally planned out and ready to pour onto the page is absurd. Writers begin by conversing, reading, and brainstorming to see what they already know and where they want to go."
- "Prewriting is whatever you do to help you figure out what your key idea is or what information, examples, explanations, or content you'll include in your paper. Prewriting techniques include freewriting, brainstorming, and grouping. Prewriting includes things like thinking, talking to others, reading related material, and outlining or arranging ideas. You may obviously prewrite at any point during the writing process. Stop what you're doing and start using one of [these] strategies whenever you want to come up with new material..."
- "Prewriting tasks usually aid in the discovery of a good topic, the narrowing of overly broad ideas, and the consideration of aim. At the very least, you should have a statement and a list at the end of the prewriting exercises. Alternatively, you may have something more formal, such as a three-part thesis sentence and a well-developed outline. You'll have set the foundation in any case."
- Prewriting plans are only methods for creating and organizing ideas; they are not set in stone. Writers regularly alter their ideas while writing, removing some aspects and adding or changing others. That's why some authors argue that the term "prewriting" is misleading; they return to their plans often throughout the writing process, often altering and adjusting them as they go.
- Choosing a topic and then reducing it down to a length that can be addressed in

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the time allotted is an important part of prewriting. Oral storytelling is an excellent approach to come up with a solid personal narrative topic. Writers can quickly tell a narrative and gauge whether it will be an engaging topic to write about based on the emotions of the audience.

- For communication to be effective, the audience's experience and knowledge of the issue must be considered: too technical and specialist information may be beyond the reader's comprehension; a too basic or simple approach will weary the reader. The question to consider is: What will the reader benefit by reading this essay? The goal will be to educate, entertain, or persuade. These goals are frequently integrated in a paper, with each goal serving as a function of the others.
- Motivation and audience awareness are frequently the first steps in prewriting: what is the student or writer trying to say, why it is necessary to deliver it properly, and who is the intended audience. It assists you in putting your thoughts on paper regarding what you want to write about.
- The fundamental goal of prewriting activities is to determine the paper's focus. The point of focus is where all of your energy is focused. The paper will be unclear, shallow, and potentially chaotic if the topic is too wide. Consider the audience to see if the issue is narrow enough. If your audience lacks detailed understanding of the subject, you may want to take a more general approach. Our own expertise of the subject also limits you. You can't be precise about something you don't understand. Of course, research will provide you with the necessary facts on a subject.

Activity

1. Prepare a presentation on the topic "Pre-Writing Considerations".

Questions & Exercises

1. What is pre-writing?
2. State the importance of topic in pre writing.
3. What is the role of audience in pre writing?
4. State the significance of purpose in writing.
5. What is the role of purpose, audience and topic altogether in making the writing great?

Glossary

- Prewriting: it is the first stage of the writing process, typically followed by drafting, revision, editing and publishing. Prewriting can consist of a combination of outlining, diagramming, storyboarding, and clustering.

Further Readings

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Check your Understanding – Answers

1. Prewriting
2. Prewriting
3. True
4. True
5. True

Notes**Unit - 5.2: Literature Review Writing****Objectives:**

At the end of this unit, you will be able to:

- Understand what literature review is.
- Learn about the purpose of the literature review.
- Comprehend the types of literature reviews.
- Learn about structure & writing style.
- Understand the stages of literature review development.
- Evaluate ways to organize literature review.
- Analyze writing literature reviews.

Introduction

A portion of your proposal dedicated to a more deliberate analysis and synthesis of prior studies relevant to the research issue under investigation is connected to the context and importance of your research. The goal is to situate your project within the broader context of current research while also demonstrating to your readers that your work is unique and creative. Consider what questions other researchers have raised, the approaches they used, and how you interpret their results and, if applicable, their recommendations.

Since a literature review is dense with information, it's critical that this section is well-structured so that a reader can understand the main points that support your proposed study in comparison to those of other researchers.

Rather than systematically or chronologically presenting groups of materials one at a time, it is a reasonable technique to split the literature into "conceptual categories" [themes]. It's worth noting that conceptual categories usually emerge after you've read the majority of the relevant literature on your topic, so introducing new categories is a continuous process of exploration as you read more studies. How do you know you've covered all of the research literature's core conceptual categories? When you start to see repetition in the assumptions or suggestions that are being made, you can be certain that all of the important conceptual categories have been found.

A literature review examines published information in a certain subject area, as well as information published within a specific time period. As the topic develops, a literature review of a mature issue tackles the need for a critique of, and potential reconceptualization of, the topic's larger and more diversified knowledge base. The second type of literature review is for issues that are new or emerging and would benefit from a comprehensive conceptualization and synthesis of the literature. The review is more likely to lead to an initial or preliminary conceptualization of the topic, such as a new model or framework, because these topics are relatively new and have not yet undergone a complete evaluation of the literature.

A literature review can be done for a variety of reasons, including gathering information for formulating policies and providing evidence-based care, as part of the

research process, or as part of an academic assessment. Many eligible students find conducting a literature review to be a daunting endeavour. Questions range from where to begin, how to choose a subject, and how many articles to include, to what a review of the literature entails. A literature review can be as simple as a list of sources, but it usually follows a structure and includes both summary and synthesis. A summary is a re-organization or reshuffling of the source's significant information, whereas a synthesis is a re-organization or reshuffling of that material. It could provide a fresh perspective on existing content or merge new and old perspectives. It could also chart the field's intellectual evolution, including important controversies. The literature review may also evaluate the sources and advise the reader on the most pertinent or relevant ones, depending on the situation.

5.2.1 What is Literature Review?

A literature review is, in essence, a detailed overview of previous research on a certain topic. The overview informs the reader what is known about a topic as well as what is unknown, establishing the rationale or necessity for a new inquiry, which is what the actual study to which the literature review is related aims to do. Creswell (1994, pp. 20, 21) explains it a little differently:

In a research project, the literature serves numerous purposes: (a) It informs the reader about the findings of other research that are relevant to the one being reported (Fraenkel & Wallen, 1990). (b) It connects a study to a larger, ongoing conversation in the literature about a topic, filling in gaps and building on previous research (Marshall & Rossman, 1989). (c) It establishes a framework for determining the study's significance.

In general, a well-written literature review incorporates all of the major themes and subthemes found within the study's overall topic.

These themes and subthemes are frequently intertwined with the previous research's techniques or findings. A literature review also serves as a prelude to and justification for the purpose and techniques of the original study published in a manuscript. "The literature review is where you identify the theories and past studies that have impacted your choice of research topic and methods," is another way of putting it.

Articles from academic journals and books are the most usual and relevant sources to rely on and utilise as evidence in a topic review. However, depending on the research topic chosen, the availability of academic journal articles can vary greatly. Government documents and newspaper articles, to mention a few, are other frequently approved materials to consult. All of the important points or findings of a certain topic must be identified and discussed/explained in the literature review. In order to demonstrate an in-depth understanding of the topic at hand, both classic (if available) and modern studies must be included.

A literature review examines published information in a certain subject area, as well as information published within a specific time period. As the topic develops, a literature review of a mature issue tackles the need for a critique of, and potential reconceptualization of, the topic's larger and more diversified knowledge base. The second type of literature review is for issues that are new or emerging and would benefit from a comprehensive conceptualization and synthesis of the literature. The

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review is more likely to lead to an initial or preliminary conceptualization of the topic, such as a new model or framework, because these topics are relatively new and have not yet undergone a complete evaluation of the literature.

A literature review is a survey of scholarly articles, books, and other materials related to a specific subject, research area, or theory, including a description, summary, and critical evaluation of these works. Literature reviews are intended to give readers an overview of the sources you used when studying a certain issue and to show them how your research fits into the greater field of study. A literature review is a summary of relevant literature in a specific field or topic. It provides an overview of what has been said, who the important writers are, what ideas and hypotheses are prevalent, what questions are being posed, and what approaches and methodologies are appropriate and useful. As a result, it is not primary research in and of itself, but rather a report on other findings.

Although some of the initial reports in the literature are vocal, the great majority of reports are written documentation. Empirical, theoretical, critical analytic, and methodological scholarship are examples of different forms of scholarship. A literature review, on the other hand, aims to summarize, assess, clarify, and/or integrate the content of primary reports.

A literature review is an objective, comprehensive synthesis and critical study of all relevant research and non-research literature on the issue under consideration. Its purpose is to keep the reader up to date on current literature on a topic and to serve as the foundation for another goal, such as justifying future research in the field. A competent literature review collects information about a topic from a variety of sources. It's well-written and free of personal biases, assuming any exist. A precise search and selection process should be included. To improve the flow and readability of the review, proper structuring is required (Colling, 2003).

A literature review differs from an academic research paper in several ways. The fundamental goal of an academic research paper is to establish a fresh argument, and one of the aspects of a research paper is a literature review. You use the literature as a foundation and support for a new insight that you give in a research paper. A literature review, on the other hand, is designed to summarize and synthesize the arguments and ideas of others without providing new information.

The great majority of literature reviews are part of a primary research article that lays the theoretical groundwork for the main study that is the article's focus. Fink (2005) offers numerous uses for literature reviews in this role. The rest of a scholarly essay is supported by a literature review. It describes the content and quality of existing knowledge and makes the reader aware of the importance of earlier work (Okoli & Schabram, 2010). The review, as an academic document, cannot merely regurgitate the subject matter; rather, it must contribute to the work by synthesizing the available material and delivering a scholarly critique of theory.

Literature reviews are common in scholarly publications in the social sciences and "hard" sciences (biology, chemistry, etc.) and assess materials pertinent to a certain subject or field of inquiry. In this usage, the term literature refers to a body of study literature (as opposed to literary works covered by terms like American literature).

Literature reviews summarize and synthesize the knowledge that has already been published on a topic. A summary is a condensed version of a source's key points. A synthesis is the combining of numerous sources into a single text with the goal of gaining a better understanding of the information in those sources or providing an original viewpoint on that information, as well as identifying patterns, themes, conflicts, and gaps in the sources.

An argumentative research paper is not the same as a literature review. Without adding fresh contributions, literature reviews summarize and synthesize the knowledge already written about a certain sector or topic. While literature evaluations should include a critical analysis of the arguments offered in the sources, they should not go beyond that, that is, they should not incorporate the writer's own arguments or ideas.

Writing a literature review is frequently included as part of the argumentative paper writing process or as part of the argumentative paper itself. The purpose of the literature review is to place the current paper in perspective with the rest of the relevant literature.

5.2.2 Purpose of Literature Review

A literature review serves a number of purposes. For starters, it allows you to demonstrate what research has already been done on a certain topic. As part of this overview, you may be able to identify areas that are related to the topic at hand but haven't had as much research done on them. This can assist you in justifying your research project. You may need to update your study proposal if a lot of research on a subject has already been published. If you're doing research, you'll need to identify a topic where more research might be beneficial. A literature review can assist you in identifying such a topic.

The theories and concepts that underpin your research should be discussed in your literature review. This is not the same as the methodology you will use in your research; it is a description of the research area and the major challenges surrounding it. You should also define any terminology that you will use in your study topic and explain any terms that aren't evident. You must critically analyze the content you are looking at as part of your literature study - this is part of the review process.

For a variety of reasons, literature evaluations are essential. Literature evaluations, for the most part, force a writer to educate himself or herself on as much information as possible about the topic chosen. This will aid in both the learning process and the writing process by allowing you to see what has and has not been explored and established as knowledge in the previous study.

Second, literary evaluations show readers that the author is knowledgeable about the subject. This lends trust to the author and maintains the work's overall argument's integrity. In addition, by reviewing and commenting on all earlier literature, previous literature's flaws and failings will become more evident. This will not only help you locate or argue for the need for a specific research question to investigate, but it will also help you develop a better argument for why more research is required. In this approach, a research report's literature evaluation "prefigures" the researcher's own study.

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It's crucial to remember that expecting readers to be familiar with all of the relevant history and prior knowledge about a topic is unrealistic. Scientific information (on all topics) builds quickly, and staying current on any subject can be difficult. This is not a new concept; Cooper (1984, p. 9) argued three decades ago:

"Given the cumulative nature of science, trustworthy accounts of past research form a necessary condition for orderly knowledge building. Yet, research methods textbooks in the social sciences show a remarkable lack of attention to how an inquirer finds, evaluates, and integrates past research. This inattention is especially troubling today because the social sciences have recently undergone a huge increase in the amount of research being conducted ... The need for trustworthy accounts of past research is also strengthened by growing specialization with the social sciences."

When it comes to the professional value of a solid literature review for a manuscript that reflects on the findings of an original research study, it's crucial to remember that literature reviews are frequently focused on components of manuscripts that are being reviewed for publication. More than three-quarters (76.9%) of manuscript reviewers indicate the quality of a manuscript's literature review has a major influence on their assessment, according to Mustaine and Tewksbury (2008) in a survey of criminal justice and criminology journal manuscript reviewers. Likewise, the literature review of a manuscript is a highly crucial part of a manuscript, according to surveyed criminal justice and sociology journal editors (Mustaine & Tewksbury, in press). Problems with literature reviews are the fourth (of nine) most commonly criticized aspects of reviewed manuscripts in the top tier publication *Justice Quarterly*, with 57 percent of papers identified for problems in the literature review (Tewksbury & Mustaine, 2012). Surprisingly, it is graduate students and assistant professors who, as manuscript reviewers, are most likely to point out flaws in the literature reviews of reviewed manuscripts.

5.2.3 Types of Literature Reviews

There are many different forms of literature reviews, but the following are the most common in business studies:

Narrative literature

The body of a narrative literature review examines and summarizes the literature. A narrative review can also be used to establish conclusions about a topic and identify gaps or contradictions in a body of knowledge. To do a narrative literature review, you must have a suitably defined research issue.

Systematic Literature Review

In comparison to most other types of literature reviews, systematic literature review necessitates a more rigorous and well-defined strategy. A systematic literature review is thorough and includes information on the timeframe in which the material was chosen. Meta-analysis and meta-synthesis are the two types of systematic literature reviews.

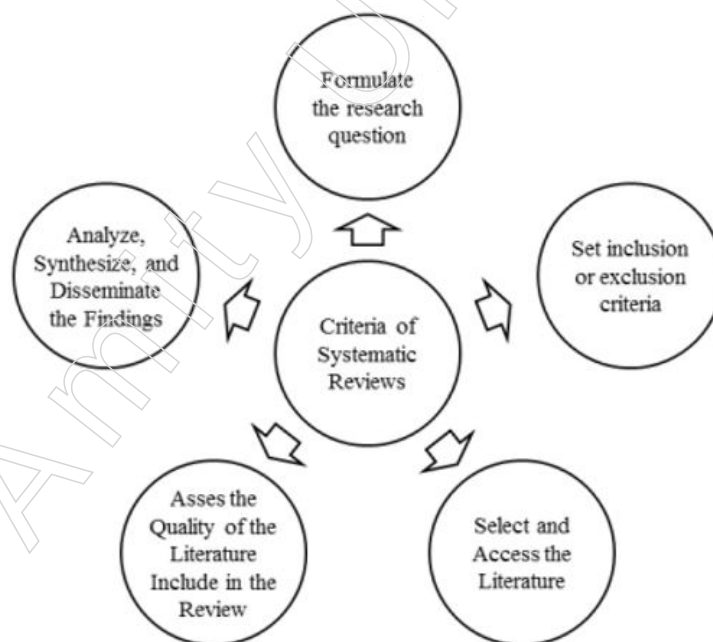
When you do a meta-analysis, you combine the results of multiple research on the same topic and analyse them using defined statistical processes. Patterns and links are identified, and conclusions are reached, in meta-analysis. Meta-analysis is linked to a

logical research strategy. Meta-synthesis, on the other hand, is based on procedures that aren't statistical. This method combines, analyses, and interprets the results of several qualitative research projects. When using an inductive research approach, a meta-synthesis literature review is frequently performed.

A systematic literature review's goal is to compile as comprehensive a list as possible of all published and unpublished works in a given subject area. Unlike traditional reviews, which attempt to summarise the findings of multiple research, systematic reviews utilise defined and rigorous criteria to discover, critically analyse, and synthesise all of the literature on a given issue (Cronin, et al, 2008). The primary goal of this essay is to give the reader a thorough framework for comprehending current knowledge and emphasising the need of fresh research. It can spark research ideas by pointing out gaps or contradictions in a body of information, assisting the researcher in defining research questions and hypotheses.

Before doing a literature evaluation, Beecroft et al. (2006) contend that a suitably focused research question is required. It can also be used to refine or focus a large research issue, and it can be used for both subject selection and topic refinement. It can also aid in the creation of conceptual or theoretical frameworks (Coughlan, et al, 2007; Cronnin, et al, 2008).

A systematic review, according to Parahoo (2006), should include the time span in which the literature was chosen, as well as the methodologies utilized to analyze and synthesize the findings of the research in question. The reviewer must present the precise criteria used to: (a) Formulate the research question; (b) Set inclusion or exclusion criteria; (c) Select and access the literature; (d) Assess the quality of the literature included in the review; and (e) Analyze, synthesise, and disseminate the findings in order for the reader to assess the review's reliability and validity (see figure below).



Criteria used to systematic review

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Argumentative literature review

As the name implies, an argumentative literature review selects books to support or contradict an argument, deeply ingrained assumption, or philosophical dilemma already established in the literature. It should be mentioned that one of the key drawbacks of an argumentative literature evaluation is the possibility of prejudice.

Integrative literature review

An integrative literature review examines, evaluates, and synthesizes secondary sources about a study topic in order to produce new frameworks and perspectives on the subject. Integrative literature review will be your only alternative if your research does not entail primary data collecting and analysis.

Theoretical literature review

Theoretical literature review is concerned with a body of knowledge that has accumulated in relation to a topic, concept, theory, or phenomenon. Theoretical literature reviews are useful for determining what ideas already exist, their relationships, and the extent to which existing theories have been studied, as well as for developing new hypotheses to test.

Methods used (SALSA)					
Label	Description	Search	Appraisal	Synthesis	Analysis
Rapid Review	Assessment of what is already known about a policy or practice issue, by using systematic review methods to search and critically appraise existing research	Completeness of searching determined by time constraints	Time-limited formal quality assessment	Typically narrative and tabular	Quantities of literature and overall quality/direction of effect of literature
Scoping review	Preliminary assessment of potential size and scope of available research literature. Aims to identify nature and extent of research evidence (usually including ongoing research)	Completeness of searching determined by time/scope constraints. May include research in progress	No formal quality assessment	Typically tabular with some narrative commentary	Characterizes quantity and quality of literature, perhaps by study design and other key features. Attempts to specify a viable review
State-of-the-art review	Tend to address more current matters in contrast to other combined retrospective and current approaches. May offer new perspectives on issue or point out area for further research	Aims for comprehensive searching of current literature	No formal quality assessment	Typically narrative, may have tabular accompaniment	Current state of knowledge and priorities for future investigation and research
Systematic review	Seeks to systematically search for, appraise and synthesis research evidence, often adhering to guidelines on the conduct of a review	Aims for exhaustive, comprehensive searching	Quality assessment may determine inclusion/exclusion	Minimal narrative, tabular summary of studies	What is known; recommendations for practice. What remains unknown; uncertainty around findings, recommendations for future research
Systematic search and review	Combines strengths of critical review with a comprehensive search process. Typically addresses broad questions to produce best evidence synthesis	Aims for exhaustive, comprehensive searching	May or may not include quality assessment	Typically narrative with tabular accompaniment	What is known; recommendations for practice. Limitations

System- atized review	Attempt to include elements of systematic review process while stopping short of systematic review. Typically conducted as postgraduate student assignment	May or may not include comprehensive searching	May or may not include quality assessment	Typically narrative with tabular accompaniment	What is known; uncertainty around findings, limitations of methodology
Umbrella review	Specifically refers to review compiling evidence from multiple reviews into one accessible and usable document. Focuses on broad condition or problem for which there are competing interventions and highlights reviews that address these interventions and their results	Identification of component reviews, but no search for primary studies	Quality assessment of studies within component reviews and/or of reviews themselves	Graphical and tabular with narrative commentary	What is known; recommendations for practice. What remains unknown; recommendations for future research

14 Review Types (Grant & Booth, 2009)

5.2.4 Structure & Writing Style

Structure

A literature review should follow the same format as any other essay, including an introduction, a main body, and a conclusion.

Introduction

The introduction should include the following information:

- Give an adequate framework for examining the literature by defining your topic and providing an appropriate context for reviewing the literature;
- Explain the organisation – i.e. sequence – of the review; clarify your reasons – i.e. point of view – for examining the literature;
- Declare the review's scope — that is, what is included and what isn't. For example, if you were evaluating the literature on childhood obesity, you might state, "There are several studies documenting obesity trends in the general population." However, because the focus of this study is on childhood obesity, these will not be discussed in depth and will only be mentioned when necessary.

The main body

The major body should:

- Organize the literature by common themes; provide insight into the relationship between your chosen issue and the larger subject area, such as the relationship between childhood obesity and obesity in general;
- From a broad, overarching view of the literature being studied, narrow your emphasis to the precise target of your research.

The major body of the essay is where you develop your argument in depth and make references to secondary sources that support your position or have influenced your thinking. You should offer both sides of an argument before stating why you

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believe one is more persuasive than the other. This should take up the majority of the text and include several author and article references. You should also strive to include one or more case studies (real-world examples) to support your point. Don't be hesitant to use diagrams and drawings if necessary, but always remember to reference them in the text. Graphics should be large and sharp enough to create an eye-catching visual effect.

Conclusion

The conclusion should include the following:

- Summarize the key elements of the present body of knowledge;
- Assess the current state of the evaluated literature;
- Find errors or holes in existing knowledge that are significant;
- A list of areas that should be researched in the future;
- Make a connection between your study and current knowledge.

Writing Style

A literature review is an examination of scholarly sources on a particular subject. It gives you a broad perspective of current knowledge, helping you to spot pertinent ideas, methodologies, and research gaps.

Finding relevant publications (such as books and journal articles), critically examining them, and summarizing what you discovered are all part of writing a literature review. There are five important stages to follow:

- Look up related material.
- Examine the sources
- Determine the themes, arguments, and gaps.
- Create a blueprint for the structure.
- Prepare a literature review.

A competent literature review does more than summarize sources; it also analyses, synthesizes, and critically evaluates them to provide a comprehensive picture of the current state of knowledge on the topic.

5.2.5 Stages of Literature Review Development

The remainder of this article focuses on the stages needed in doing a traditional or narrative review of the literature, due to the specific processes involved in systematic reviews, meta-analysis, and meta-synthesis. According to Cronin et al. (2008), the first stage is to determine the topic of the literature review. This may have already been decided by the researcher conducting a quantitative investigation.

This will, however, be the initial step for someone conducting a non-research-based literature review.

Choosing a review topic

The first and frequently most difficult aspect in producing a review of literature is deciding on a topic (Timmins and McCabe, 2005; Cronin, et al, 2008). The process

is frequently made more difficult due to a lack of content expertise. Here are some pointers to help you choose a topic more quickly.

First, go through your textbook to discover general areas of interest in the discipline. Second, read the chapters related to the themes you've chosen to gain a better understanding of the language (important words), primary investigators, and issues or debates in the field. Third, speaking with others, such as experts, or reading about a topic can assist identify which aspects of the subject the reviewer is interested in, as well as how much information is available on the issue (Timmins and McCabe, 2005). After you've decided on a topic, head to the library and look for journal articles published in the area. Find article titles for relevant themes using key phrases; abstracts are sometimes supplied for the reader's convenience. Abstracts can be time-saving tools since they help to separate relevant, related literature from unconnected, peripheral pieces.

According to Cronin (2008), having enough literature is also vital, especially when the review is an academic project. Because these academic activities frequently have tight deadlines, having enough literature is essential for completing the review and submitting it on time.

The word limit for literature evaluations that are part of academic homework is usually carefully enforced, and it is critical to stick to it. Topics that are too broad will result in either excessively long or shallow evaluations. As a general guideline, it's best to start with a restricted and concentrated topic and then expand the scope of the review as needed. Cutting information successfully is significantly more challenging, especially when time is limited.

Searching and selecting appropriate articles

Following the selection of a topic, the next stage is to organise the necessary and related material. A methodical approach is thought to be the most likely to produce a review that will be useful in practise (Hek and Langton, 2000; Cronin et al, 2008). Reviewers should examine comprehensiveness and relevancy, according to Newell and Burnard (2006), and the more specific the topic or query being searched, the more focused the result will be (Cronin et al, 2008). Theoretical presentations, review articles, and empirical research papers are the types of articles that are chosen for a good survey of literature. One technique for commencing a literature review is to select the work of a single researcher. If you provide contradictory theoretical viewpoints and data alongside the position or prediction that you support in your research, your presentation will be more compelling. You should select several researchers' works that have contributed to the body of knowledge in a certain field. Attempt to delete (or explain away) articles that employ flawed methodology or reasoning to justify their conclusions.

Computers and electronic databases are now the most frequent tools for doing literature searches. Computer databases provide access to massive amounts of data that can be retrieved much more readily and quickly than a human search (Younger, 2004). There are a plethora of electronic databases available, many of which specialise in specialised disciplines of knowledge. As a result, it's critical to figure out which databases are relevant to the subject (Cronin et al, 2008). Existing literature reviews and systematic reviews can be valuable data sources. They can provide a solid

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summary of previous research so that the relevance to the current project may be evaluated (Cronin et al, 2008). When conducting a literature search, Cronin et al. (2008) suggest that specifying the type of source is critical in evaluating whether a publication should be included in your review. Table below lists the four most common types of sources. It is critical to keep track of the keywords and procedures used in conducting the literature search because these will need to be recognized later when reporting how the search was completed (Timmins and McCabe, 2005)

When possible, it is always preferable to consult primary sources. In science, primary sources are usually articles published in respectable journals. Journals are generally thought to be more up-to-date than books as sources of knowledge (Cronin, et al, 2008). Textbooks and review articles are examples of secondary sources, as are descriptions or summaries written by anyone other than the original researcher. Secondary sources, like your literature review, do not contain fresh information. When beginning a literature study, it's a good idea to look at secondary sources first, but you should never rely only on secondary sources and should always examine primary sources for probable inaccuracies.

Source	Definitions
Primary source	Usually a report by the original researchers of a study
Secondary source	Description or summary by somebody other than the original researcher, e.g. a review article
Conceptual/ theoretical	Papers concerned with description or analysis of theories or concepts associated with the topic
Anecdotal/ opinion	Views or opinions about the subject that are not research, review or theoretical in nature

Analyzing and synthesizing the literature

You're ready to analyse each article (break it down and identify the important information in it) and then synthesise the collection of articles (integrate them and identify the conclusions that can be drawn from the articles as a whole) after you've collected the articles you want to use in your literature review.

To begin, it's a good idea to browse over the articles that have been gathered to get a sense of what they're about. The majority of published articles provide a summary or abstract at the beginning of the publication, which will aid in this process and allow you to decide whether it is worth reading further or including. It's also a good idea to do an initial classification and grouping of the articles by source type at this point.

Primary sources	Secondary sources -review	Non-research literature
Title:	Title:	Title:
Author and year:	Author and year:	Author and year:
Journal (full reference):	Journal (full reference):	Journal (full reference):
Purpose of study:	Review questions/ purpose:	Purpose of paper:
Type of study:	Key definitions:	Credibility:
Setting:	Review boundaries:	Quality:
Data collection method:	Appraisal criteria:	Content:
Major findings:	Synthesis of studies:	Coherence:
Recommendations:	Summary/ conclusions:	Recommendations:
Key thoughts/ comments, e.g. strengths/ weakness:	Key thoughts/ comments, e.g. strengths/ weakness:	Key thoughts/ comments, e.g. strengths/ weakness:

Organization of writing the review

The major goal of arranging your literature review is to help your reader comprehend why the type of literature review or research paper that you propose or have done is necessary. The ability to present the findings in such a way that they illustrate your knowledge in a clear and consistent manner is the key to a good literature review or research report (Cronin, et al, 2008).

The introduction and conclusion of your review of the literature should demonstrate how your research project will contribute to the ongoing discussion: define essential words and concepts, and explain how your research will address unanswered problems raised by others' work. You can also predict the direction of the next section/ chapter by outlining the format of the review itself — by previewing in the introduction or reviewing in the conclusion.

5.2.6 Ways to Organize Literature Review



The following are some examples of different approaches to organize a Literature Review:

- **Chronological (by date):** This is a popular method, especially for topics that have been discussed for a long period and have evolved over time. Organize it in stages to show how the topic has evolved through time: the original definitions, then key time periods of change as researchers discussed it, and finally how it is considered today.
- **Broad-to-specific:** Another strategy is to start with a section on the organization road type of issue you're looking at, then narrow down to increasingly specific issues in the literature until you find the articles that are most closely related to your research question, thesis statement, hypothesis, or proposal. When there isn't much directly on your issue but you're linking together several connected, bigger topics, this might be an excellent method to introduce a lot of background and related sides of your topic.

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- Major models or major theories: When there are numerous models or major ideas, it is a good idea to outline the theories or models that are used the most in your articles. That way, you may organize the articles you read according to the theoretical frameworks that each author likes, giving you a good overview of the most popular approaches to your subject.
- Prominent authors: If a particular researcher established an area, but numerous well-known persons further developed it, collecting the well-known authors/researchers and what each is known to have stated about the issue is a smart method. Then you can sort additional authors into groups based on which prominent authors' concepts they're following. Looking at the citations in your publications to see if there is one author that appears frequently can help with this organization.

Organizing Principles

The materials gathered for the literature review can be organized and presented in a variety of ways.

The following is a list of several organizational frameworks that can be utilized to display the results of your research.

Chronological: The chronological framework classifies the material according to when it was first published. If you're writing about a particular teaching approach, for example, you'd start with the resources that originally introduced the method. After that, you'd do case studies using that strategy. You'd wrap up your assessment with recent studies that might even provide a historical perspective on the method, from its conception to its current application.

- By publication: this framework is handy if you see a succession of pieces produced in reaction to one another that are all contained within the same publication. You keep the chronological sequence but break it up so that items that respond to each other are grouped together.
- By trend: This framework examines various trends and groups them according to their timeliness. If you were looking at the history of assistive technology in helping students with disabilities, for example, you could group the reviews by the disability being treated and then present the history of utilising assistive technology to treat that condition in chronological order.

Thematic: A thematic framework is similar to organising by trend, with the exception that the reviews are not organized in the order in which they were written. This isn't to say you won't think about how a topic or issue evolved over time, but it does imply you won't organize your reviews chronologically. Rather, the focus will be on the themes that emerge from the topic or issue — such as commonalities — and then you'll fit your assessments into the many ideas that emerge from there. If your review topic was arts-based research, for example, you may concentrate on the various ways artistic inquiry was utilised to explore the creative process, focusing on the concepts rather than the progress.

Methodological: A literature review might be organized using the method or practice used in a case study. This framework focuses on how the author(s) or person(s) in charge of a study used methods that were comparable to those used in another study.

As a result, the categories of literature covered in a methodological framework-based literature review tend to be comparable. If you were reviewing ways for treating post-traumatic stress disorder, for example, the studies would be organized by the methods employed to treat the patients rather than the order in which they were published.

Seven Strategies for Organizing a Research Project

1. Collect and interact with printed sources

- Depending on your research issue, choose books, scholarly articles, government documents, and popular sources.
- Examine the sources to see if they are relevant to the issue.
- To find further references, look through the bibliographies of articles and books.
- Examine the quality of online sources.

2. Consider of additional data collection strategies.

- Examine various data-gathering options, such as interviews, surveys, field work, experiments, and questionnaires, in accordance with the discipline's traditions.
- Use source footnotes to find further sources and follow the scholarly conversation's trajectory.

3. Choose on a note-taking strategy.

- Consider using Excel spreadsheets, a synthesis matrix, or index cards as examples.
- To verify information in the source and properly reference it in your paper, use a citation manager or jot down author/title/call number/page number/url/etc. information.
- In your notes, differentiate between paraphrase, summary, and quotation.
- A paraphrase is a short piece of writing that restates the original subject in your own words and is roughly the same length as the original.
- A summary is a condensed version of the original that gives an overview of the main points.
- Quotations use quotation marks and contain the actual words of your source.

4. Brainstorm ideas using your sources.

- Write about what you've been reading.
- Identify important terms.
- As you read, make a list of questions. Look for links, discrepancies, and "silences" in the sources.
- Decide on your key point (s).

5. Make a list of your ideas.

- To draw down the bare bones of your argument, use an outline, chart, or other map.

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- Spend a few minutes freewriting any portions that are finding challenging.

6. Complete your paper.

- Create a thesis statement that you can work with.
- Write your paper, supporting your thesis with proof.
- Review and modify to ensure that your point is appropriately reflected.

7. Assess your argument.

- Is your thesis statement in your introduction clear?
- Do you have solid evidence to back up your claim?
- Do you have a glossary of crucial terms?
- Do you take into account counter-arguments?
- Have you gone back to your sources for additional information or help, if necessary?
- Have you provided an interpretation of the evidence or have you simply reported the facts regarding your subject?

5.2.7 Writing Literature Review



A literature review is a compilation of scholarly sources that gives a broad overview of a subject. Literature reviews are compilations of the most relevant and significant publications on a given topic in order to provide a comprehensive overview of what has been said and by whom on that topic. The following are the basic elements of a literature review:

- a description of the publication;
- a summary of the major points of the publication;
- a discussion of research gaps;
- And an assessment of the publication's contribution to the topic.

Seven Steps to Writing a Literature Review

1. Narrow your topic and select papers accordingly
2. Search for literature
3. Read the selected articles thoroughly and evaluate them

4. Organize the selected papers by looking for patterns and by developing subtopics
5. Develop a thesis or purpose statement
6. Write the paper
7. Review the paper
1. Choose papers that are relevant to your topic.
 - Consider your chosen field of study. Consider your own interests as well as the interests of other scholars in your field.
 - Talk to your professor, have a brainstorming session, and examine class notes and recent issues of scholarly journals in the field.
 - Limit your scope to a certain issue area (for example, focusing on France's role in WWII rather than WWII in general).
2. Conduct a literature search
 - Make a list of the criteria you'll use to choose your sources (ie. articles published between a specific date range, focusing on a specific geographic region, or using a specific methodology).
 - Search a library database using keywords.
 - Recent article and review reference lists can link to more important studies.
 - Include any studies that contradict your viewpoint.
3. Carefully read and assess the articles you've chosen.
4. Organize the papers you've chosen by looking for patterns and creating subtopics.
 - Keep the following in mind:
 - Findings that are both common and controversial
 - The most important research trends
 - The most popular theories
 - Tip: If your literature study is lengthy, use a big table surface to divide your findings into sections using post-it notes or filing cards.
 - If you decide that (a) they belong under other categories or (b) you need to create new topic headings, move them around.
 - Make headings and subheadings that represent the main topics and patterns you discovered.
5. Create a thesis statement or a goal statement.

Write a one- or two-sentence statement that summarises the conclusion you've formed regarding the significant trends and changes you've noticed in your subject's research.
6. Finish your paper
 - Follow the above-mentioned organisational structure, including the headings and subheadings you created.
 - Make sure each section is connected logically to the one before and after it.
 - Sections should be organised by themes or subtopics rather than by specific theories or scholars.

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- Tip: If each paragraph opens with a researcher's name, it's possible that you've just recounted what research has been done rather than analysing and comparing the research literature from an analytical standpoint.
 - Give analysis precedence than description.
7. Go over your work again.
- Examine each paragraph's topic sentences. Would you say that your work offered a clear, logically developed position from beginning to conclusion if you simply read these sentences? The essential ideas of your literature review should be indicated in the topic sentences of each paragraph.
 - Make an outline of each portion of the document to see if you need to add material, delete extraneous information, or restructure sections.
 - Read your work aloud to yourself. That way, you'll be able to spot where punctuation marks are needed to indicate pauses or divisions inside sentences, where grammatical problems have occurred, and where your sentences are confusing.
 - Check to ensure that you have covered all of the significant, up-to-date, and topical texts, as the objective of a literature review is to demonstrate that the writer is conversant with the important professional literature on the chosen subject. It is critical in the sciences and some social sciences that your literature be recent; however, this is less crucial in the humanities.
 - Make sure all of your citations and references are right, and that you're using the correct referencing style for your profession. Consult your lecturer if you're unsure about the style to employ.
 - Check to see whether you've plagiarized by forgetting to cite a source of information or by utilising words directly quoted from a source. (If you use three or more words directly from another source, you should put them in quotation marks and cite the source page.)
 - Text should be written in an academic style that is clear and succinct; it should not be descriptive or utilize everyday language.
 - There should be no mistakes in grammar or spelling.
 - Sentences should be easy to read and understand.

Check your Understanding

1. _____ review is concerned with a body of knowledge that has accumulated in relation to a topic, concept, theory, or phenomenon.
2. An _____ review examines, evaluates, and synthesizes secondary sources about a study topic in order to produce new frameworks and perspectives on the subject.
3. A _____ review's goal is to compile as comprehensive a list as possible of all published and unpublished works in a given subject area.
4. A literature review examines published information in a certain subject area, as well as information published within a specific time period.

5. A literature review is a compilation of scholarly sources that gives a broad overview of a subject.

Summary

- A literature review examines published information in a certain subject area, as well as information published within a specific time period. As the topic develops, a literature review of a mature issue tackles the need for a critique of, and potential reconceptualization of, the topic's larger and more diversified knowledge base.
- Since a literature review is dense with information, it's critical that this section is well-structured so that a reader can understand the main points that support your proposed study in comparison to those of other researchers.
- A literature review can be done for a variety of reasons, including gathering information for formulating policies and providing evidence-based care, as part of the research process, or as part of an academic assessment. Many eligible students find conducting a literature review to be a daunting endeavour. In general, a well-written literature review incorporates all of the major themes and subthemes found within the study's overall topic.
- These themes and subthemes are frequently intertwined with the previous research's techniques or findings. A literature review also serves as a prelude to and justification for the purpose and techniques of the original study published in a manuscript. "The literature review is where you identify the theories and past studies that have impacted your choice of research topic and methods," is another way of putting it.
- A literature review examines published information in a certain subject area, as well as information published within a specific time period. As the topic develops, a literature review of a mature issue tackles the need for a critique of, and potential reconceptualization of, the topic's larger and more diversified knowledge base. The second type of literature review is for issues that are new or emerging and would benefit from a comprehensive conceptualization and synthesis of the literature. The review is more likely to lead to an initial or preliminary conceptualization of the topic, such as a new model or framework, because these topics are relatively new and have not yet undergone a complete evaluation of the literature.
- A literature review is a survey of scholarly articles, books, and other materials related to a specific subject, research area, or theory, including a description, summary, and critical evaluation of these works.
- A literature review is an objective, comprehensive synthesis and critical study of all relevant research and non-research literature on the issue under consideration. Its purpose is to keep the reader up to date on current literature on a topic and to serve as the foundation for another goal, such as justifying future research in the field. A competent literature review collects information about a topic from a variety of sources. It's well-written and free of personal biases, assuming any exist. A precise search and selection process should be included. To improve the flow and readability of the review, proper structuring is required (Colling, 2003).

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- An argumentative research paper is not the same as a literature review. Without adding fresh contributions, literature reviews summarize and synthesize the knowledge already written about a certain sector or topic. While literature evaluations should include a critical analysis of the arguments offered in the sources, they should not go beyond that, that is, they should not incorporate the writer's own arguments or ideas.
- A literature review serves a number of purposes. For starters, it allows you to demonstrate what research has already been done on a certain topic. As part of this overview, you may be able to identify areas that are related to the topic at hand but haven't had as much research done on them. This can assist you in justifying your research project. You may need to update your study proposal if a lot of research on a subject has already been published. If you're doing research, you'll need to identify a topic where more research might be beneficial. A literature review can assist you in identifying such a topic.
- The body of a narrative literature review examines and summarizes the literature. A narrative review can also be used to establish conclusions about a topic and identify gaps or contradictions in a body of knowledge. To do a narrative literature review, you must have a suitably defined research issue.
- In comparison to most other types of literature reviews, systematic literature review necessitates a more rigorous and well-defined strategy. A systematic literature review is thorough and includes information on the timeframe in which the material was chosen. Meta-analysis and meta-synthesis are the two types of systematic literature reviews.
- As the name implies, an argumentative literature review selects books to support or contradict an argument, deeply ingrained assumption, or philosophical dilemma already established in the literature. It should be mentioned that one of the key drawbacks of an argumentative literature evaluation is the possibility of prejudice.
- An integrative literature review examines, evaluates, and synthesizes secondary sources about a study topic in order to produce new frameworks and perspectives on the subject. Integrative literature review will be your only alternative if your research does not entail primary data collecting and analysis.
- Theoretical literature review is concerned with a body of knowledge that has accumulated in relation to a topic, concept, theory, or phenomenon. Theoretical literature reviews are useful for determining what ideas already exist, their relationships, and the extent to which existing theories have been studied, as well as for developing new hypotheses to test.
- The structure of literature review involves Introduction, main body and conclusion.
- The remainder of this article focuses on the stages needed in doing a traditional or narrative review of the literature, due to the specific processes involved in systematic reviews, meta-analysis, and meta-synthesis. According to Cronin et al. (2008), the first stage is to determine the topic of the literature review. This may have already been decided by the researcher conducting a quantitative investigation. This will, however, be the initial step for someone conducting a non-research-based literature review.

- A literature review is a compilation of scholarly sources that gives a broad overview of a subject. Literature reviews are compilations of the most relevant and significant publications on a given topic in order to provide a comprehensive overview of what has been said and by whom on that topic.

Activity

1. Formulate a case study describing all the aspects of Literature Review Writing.

Questions & Exercises

1. Briefly discuss most popular types of literature review mentioned above in the unit.
2. Define literature review.
3. What is the purpose of literature review?
4. Briefly explain the different types of literature reviews.
5. Elaborate on the structure and writing style in literature review.
6. What are the various stages of literature review development?
7. Explain the ways to organize literature review.
8. How to write a literature review.
9. Write an essay describing all the aspects of literature review.

Glossary

- Literature review: A literature review examines books, scholarly papers, and any other sources related to a specific issue, field of study, or theory, and provides a description, summary, and critical evaluation of these works in connection to the research problem under consideration. Literature reviews are intended to give readers an overview of the sources you used when researching a certain issue and to show them how your research fits into a larger field of study.

Further Readings

- 1) Bocken, N. M. P., Short, S. W., Rana, P., & Evans, S. (2014). A Literature and Practice Review to Develop Sustainable Business Model Archetypes. *Journal of Cleaner Production*, 65: 42-56.
- 2) Carnwell, R., & Daly, W. (2001) Strategies for the Construction of a Critical Review of the Literature. *Nurse Educ Pract* 1: 57-63
- 3) Colling, J (2003). Demystifying The Clinical Nursing Research Process: The Literature Review. *Urol Nurs* 23 (4): 297–9.
- 4) Cooper, H. M. (1988). The Structure of Knowledge Synthesis. *Knowledge in Society*, 1: 104-126.
- 5) Cronin, P., Ryan, F., & Coughlan, M. (2008). Undertaking a literature review: a step-by-step approach. *British Journal of Nursing*, 17(1): 38-43.
- 6) Galvan, J. L. (2006). Writing literature reviews. Glendale, CA: Pyczak Publishing.

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- 7) Hall, T., Beecham, S., Bowes, D., Gray, D., & Counsell, S. (n.d.). A Systematic Literature Review on Fault Prediction Performance in Software Engineering.
- 8) Hart, C. (1998). Doing a Literature Review. London: Sage Publications.
- 9) Lawrence, C. N. (2011). Writing a Literature Review in the Social Sciences.
- 10) Okoli, C., & Schabram, K. (2010). A Guide to Conducting a Systematic Literature Review of Information Systems Research. Sprouts: Working Papers on Information Systems, 10(26).

Check your Understanding – Answers

1. Theoretical literature
2. integrative literature
3. systematic literature
4. true
5. true

Unit - 5.3: Report Writing

Notes

Objectives:

At the end of this unit, you will be able to:

- Understand the meaning of the research report.
- Learn about the types of reports.
- Comprehend the components of a research report.
- Learn the steps in report writing.
- Understand APA style essentials.
- Evaluate citing and referencing sources.
- Comprehend footnotes.
- Analyze the key considerations/factors.

Introduction

A research report is an important part of any research project because the research isn't complete until the report is presented or prepared. The findings of a research study are of little use unless they are successfully documented and disseminated to others, regardless of how good the research study is or how meticulously it was conducted. The findings of study must invariably be added to the overall body of knowledge. Writing a report is the final phase of a research project, and it necessitates a distinct set of abilities than those required to do the research.

Since the research isn't complete until the report is presented or created, a research report is a crucial aspect of every research effort. Regardless of how thorough the research study is or how meticulously it was done, the findings of a research study are of little benefit unless they are successfully documented and disseminated to others. The results of a study must always be added to the body of knowledge. The last part of a research project, writing a report, demands a different set of skills than those required to conduct the research.

5.3.1 Meaning of Research Report

The methodical analysis of a natural phenomenon, materials, sources, or the current state of society in order to find facts, obtain further knowledge, and draw new conclusions is known as research. It is a production process that requires a lot of inputs in order to develop new knowledge and the application of new and existing knowledge to produce technology that, in turn, can lead to a nation's economic prosperity. Simply said, a research paper/report is a methodical report on the study's findings, including techniques, discussion, and conclusions, all written in a specific style. The study's findings in order to produce a high-quality report, writers should recall the adage, "Try to express, not to impress." A Research Report is a report or systematic write-up on the findings of a research study that includes an abstract/executive summary/summary, introduction (Background with literature review, justification, objectives, etc.), methodology/materials and methods (including statistical design, if any), results and

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discussion, conclusions and recommendations, references, and so on, and follows a specific style or format.

The research assignment is not complete until the report has been presented and/or written, hence it is regarded a major component of the research project. Even the most brilliant idea, well-designed and performed research study, and most startling generalizations and conclusions are useless unless they are successfully communicated to others. The objective of study isn't served adequately until the results are disseminated.

The findings of study must invariably be added to the overall body of knowledge. All of this explains why producing a research report is so important. Some people do not consider report writing to be an important aspect of the research process. However, the common consensus is that the presenting of research findings or the preparation of a report should be considered part of the research endeavor. The writing of a report is the final step of a research project, and it necessitates a distinct set of skills than those required in the other stages. This activity should be completed with extreme caution by the researcher; he may seek aid and instruction from professionals for this reason.

Reports are typically broad in scope, yet they are focused on providing information about a certain issue and a very specific target market. The major goal of research reports is to deliver important information about a study to marketers who are developing new tactics. Certain occurrences, facts, and other information based on incidents must be transmitted to those in control, and the most effective communication instrument is research reports. The best research reports are exceptionally accurate in the facts they provide and have a clear goal and conclusion. For these reports to be useful in relaying information, they should have a clean and ordered format.

Effective Research Report Characteristics

The following are the four characteristics of a good research report:

- Focus: A good report emphasises the most important details.
- Accuracy: a good report does not lead the reader astray.
- Clarity: A good report does not lead the reader astray.
- Conciseness: a good report does not squander the time of the reader.

5.3.2 Types of Report

Depending on the subject, research reports come in a variety of lengths and formats. For their annual reports, banks and other financial institutions, for example, prefer brief balance sheet tabulations. A scientist's report may be in the form of symbols and formulae, whereas a mathematician's report may be in the form of various algebraic notations. Literature students typically produce a lengthy paper critical of a writer or work.

Newspaper articles are also a type of report writing. Book reviews, government reports, PhD theses, and other types of reports are examples of reports. Any research study can be presented in a variety of ways, including a technical report, a public report, an article, a monograph, or even an oral presentation. The technical report is written for experts who want to learn more about the study project's technical procedures and

vocabulary. The report will be written in technical terms. The main focus of the technical report is on: (i) the methodologies used; (ii) the assumptions established during the investigation; and (iii) the full presentation of the findings, including their limitations and supporting data.

Popular data is designed for people who aren't interested in the technical components of research methods or results. The audience will consist of laypeople as well as top executives who require summary reports. One of the most popular reports is one that emphasizes simplicity and aesthetics. Clear language, the elimination of technical, particularly mathematical details, and the extensive use of charts and diagrams are all ways to simplify. Another quality of a successful report is attractive layouts, huge types, and several subheadings. Practical factors and policy consequences are emphasized in such a study.

Technical Report

The length and format of research papers might vary. Reports in the form of letters are preferred by most businesses. In their yearly reports to clients and shareholders, banks, insurance companies, and financial institutions must include a brief balance-sheet kind of tabulation. A research investigation's findings can be published in a variety of formats, including a technical report, a public report, an article, a monograph, or even an oral presentation. When a detailed written report of the study is necessary, whether for record-keeping or public dissemination, a technical report is employed. If the research findings have policy consequences, a popular report is employed. Since a technical report is written for other researchers, it should be organized in a distinct way. The researcher is expected to provide a detailed analysis of the technical features of the sampling procedures and the topic matter in such a report. Professional colleagues are more worried about the procedures used. In fact, the significance of the findings is determined by the methods used. The sample design's conceptual and analytical framework should be thoroughly defined.

The following elements make up a technical report.

1. Major results and contents: A technical report will only include two or three pages devoted to the major findings.
2. The nature of the research project: This section explains the study's overall goals, the problem's operationalization, the working hypothesis, the type of analysis, the data needed, and so on.
3. Research methodology: This section covers the study's numerous approaches and their limitations.

For example, sample size, sample selection, and so forth.

4. Data analysis: The report looks at the data and its sources, as well as its qualities and limits. If secondary data is employed, its applicability to the situation is thoroughly evaluated. The manner in which data was obtained should be explicitly documented in the case of a survey.
5. Findings presentation: The researcher delivers the study's key findings along with supporting data in the form of tables and figures. This section is the report's primary body, and it normally spans several chapters.

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6. Primary conclusion: This section contains the research's main results as well as the report's main body, which normally spans many chapters.
7. Bibliography: This section provides the primary secondary data sources.
8. Technical appendices: These comprise all technical aspects relating to questionnaires, mathematical derivations, elaboration on specific analysis methodologies, and so on. The above structure gives a broad sense of what a technical report is like; however, the order of presentation in all technical reports may not be the same. As a result, the presentation may vary; the various components indicated above may not always be the same, and not all of these sections will appear in every report.

Popular Report

This emphasizes the importance of simplicity and appeal. Its wording is simple, with little statistical facts and a lot of graphs and illustrations. It features a pleasing layout, a large print size, numerous subheadings, and possibly some cartoons. It also emphasizes the practical issues as well as policy ramifications.

A popular report's broad outline is as follows:

1. Key results and conclusions: The report will include practical findings as well as their consequences.
2. Follow-up action: Based on the findings of the study in this part, it will advise a course of action.
3. Study objectives: The problem is given here, along with the study's precise objectives.
4. Methodology: This section includes a discussion of the methodologies and procedures used, as well as a brief evaluation of the data used in the study.
5. Results: This is the report's major body, which is written in plain, non-technical language and liberally illustrated with charts, graphs, and other visual aids.
6. Appendices: This section contains detailed information on the methodologies utilized, as well as forms and other documents. If the report is intended for the general public, appendices are usually not included.

From an operational standpoint, a popular report emphasizes simplicity and policy consequences while avoiding technical details.

General types of reports are -

1. Informational
 - Inform or instruct – present information.
 - Reader sees the details of events, activities or conditions.
 - No analysis of the situation, no conclusion, no recommendations.
2. Analytical
 - Written to solve problems.
 - Information is analyzed.

- Conclusions are drawn and recommendations are made.

3. Persuasive

- An extension of analytical reports - main focus is to sell an idea, a service, or product.
- Proposals are the most common type.

Reports typically target a broader audience, serve many purposes, and provide more extensive information.

Other types of reports include:

- Incident Report: A report that details how near you are to completing a project.
- Accident Report: A document that details how many goods or services were sold, as well as the reasons for any deviations from the original plan.
- Sales Report: A report on what has occurred in a specific location and how close your company is to completing construction.
- Progress Report: A progress report is a formal academic report that explains how and why something has evolved through time.
- Feasibility Study/Report: A feasibility study/report is a document that describes what transpired.
- Recommendation Report: A report on the feasibility of a plan.
- Site: A report on what your company should be doing.
- Case Study: A case study is a report that details how someone was injured or something was harmed.
- Periodic Operating Reports: To track and oversee manufacturing, sales, shipping, and customer service, among other things.
- Situational Report: This type of report is used to discuss one-time activities like travels, conferences, and seminars.
- Investigative/Informative: To look into issues and provide information - with little analysis.
- Compliance entails adhering to government regulations and rules.
- Justification/Recommendation: To provide management with recommendations and to serve as instruments for solving problems and making decisions.
- Yardstick: Establishing criteria and evaluating alternatives by comparing them to the 'yardstick' criteria.
- Research Studies: Interpreting an issue, formulating hypotheses, gathering data, analyzing data, and making conclusions are all examples of scientific studies.

5.3.3 Components of a Research Report

Scientists use scientific research publications to communicate with other scientists about the findings of their research. These articles follow a standard structure in which the author presents the research in a logical and organized manner. This may or may

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not reflect the sequence in which you completed or considered the job. The following is a sample research report outline.

Beginning Material:	i.e. title page, abstract, key word list, table of contents, list of figures and tables, acknowledgements
Chapter 1:	Introduction - statement of the problem, hypotheses, why it is important, objectives of the work, scope of the work
Chapter 2:	Background and Literature Review - discuss related work and indicate how it relates to report
Chapter 3:	Procedure - describe the procedure used in project, data used, and how it was obtained
Chapter 4:	Results - indicate what happened and interpret what it means
Chapter 5:	Conclusions and Recommendations - summarize conclusions and what they mean (i.e., answer the question, "So what?"). What changes and further work do you recommend?

Title

- Only significant words are capitalized, and the title is placed at the top of the page. It offers a brief description of the study's major variables (independent and dependent).
- Make the title descriptive enough to describe the paper's contents, but not so technical that only experts will comprehend it. The title should be suitable for the target market.
- The article's topic matter is usually described in the title. For instance, consider the impact of smoking on academic performance.
- A title that describes the results is sometimes more effective. Students who smoke, for example, receive lower grades.
- Colon-based titles are popular right now, but they're not always as adorable as you might imagine.
- Notice how the title provides the reader with the most concise explanation of the experiment's primary 'players' and the type of their relationships with one another.

Authors

- The author's name, as well as the name of the university or research institution, is centered beneath the title.
- The first author of a research article is usually the person who did the work and wrote the paper.
- Other people who made significant contributions to the work are named as authors in published articles. Before naming your mentor as a co-author, get his or her consent.

Abstract

- A research paper is accompanied by an abstract or summary that gives the reader a "taste" of what's to come. Potential readers will use this to decide whether or not the paper is intriguing enough to read, thus it should be clear, brief, and comprehensive.

- The abstract should be a single paragraph of 100-250 words that explains the paper's goal, methodology, results, and conclusions, as well as a description of the independent variable manipulations and a summary of the research findings. Leave the review of the literature and theoretical background to the 'Introduction and Discussion' section.
- It's easiest to write the abstract after you've finished writing the paper. Writing an introductory sentence, one phrase every chapter of your report, and a few sentences summarizing your most important conclusions/recommendations is a good rule of thumb.
- In the abstract, don't utilize abbreviations or citations. It should be able to stand on its own without the need for any footnotes.

The following is a summary of ABSTRACT's content and presentation order.

- a. The subject population must be identified,
- b. The research design must be specified,
- c. The apparatus and data collection procedures must be specified,
- d. The results must be summarized, including statistical significance levels, and
- e. A report on the inferences or comparisons drawn from the results must be written.

Introduction

- The term 'INTRODUCTION' should be in the center of the page. It starts on a fresh, numbered page. Rewrite the title as it appeared on the front page, but do not duplicate the researcher's name.
- The following questions should be answered by a reader of the introduction, but not in great depth.
 - a) What is the purpose of the study?
 - b) What makes it relevant or significant?
 - d) What are the concerns or issues?
 - d) What is the recommended method or solution?
 - e) What can we expect from the remainder of the study?
- Explain the problem, the question you'll try to answer, and why it's essential to the reader. It could be crucial for practical or theoretical (or methodological) reasons related to the development of a scholarly subject. Don't dismiss either form of rationale.
- The introduction should build up to and close with a declaration of how you want to approach your question, as well as why your approach is better than previous attempts (or why it is worth doing even if it isn't). This is ultimately what makes your approach unique, your unique contribution.
- The introduction can be thought of as I a description of the psychological issues you'll be looking into; (ii) a discussion of the research question(s) or hypothesis you'll be looking into; and (iii) a reference to other studies (in the same field)

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that have findings that are relevant to your research project, whether they're in the same direction as your hypothesis or in the opposite direction. The final paragraph(s) should include a more detailed characterization of your variables (independent and dependent) as well as a clear statement of your predictions based on the background data you've provided.

- The related work section (also known as a literature review) is exactly that: a review of work that is relevant to the problem you are trying to solve. It should look back at previous approaches to the problem and evaluate them. It should also find similar solutions to yours that have been used to solve other problems that aren't necessarily connected to the one you're working on. Reviewing the successes and limitations of your proposed solution in other contexts provides valuable insight that can help you avoid past mistakes, capitalize on previous successes, and, most importantly, potentially improve your solution or technique in general when used in your and other contexts.
- In addition to the obvious objective, the related work section can be used to – explain the existence of the problem through examples and arguments, generate interest in your work by demonstrating relevance and importance, identify the key challenges, and provide context for your solution.
- By the end of the related work part, any residual concerns about the existence, justification, motivation, or relevance of your study topic or problem from the introduction should be dispelled.

Note that a literature review is just that, a review. It is not a list of papers and a description of their contents. A literature review should critique, categorize, evaluate, and summarize work related to your research. Related work is also not a brain dump of everything you know in the field. You are not writing a textbook: only include information directly related to your topic, problem, or solution. You should review only those points that are relevant to the arguments you will make. Do not say that 'X found Y' or 'demonstrated' if X's conclusions don't follow from X's results. You can use words like 'X claimed to show that Y' or 'suggested that when you are not sure. If you see a flaw, you can add, 'However.... Try to avoid expressions like 'Unfortunately, Smith and Jones neglected to examine [precisely what you are examining]. It might have been unfortunate for them or for the field, but it is fortunate for you, and everyone knows it.

- Editorializing, expressing one's personal viewpoint, and making judgmental judgments are all bad ideas. Maintain a close relationship with the data, theory, design, and hypotheses. Your hypotheses should never appear to emerge from nowhere or to be based on personal preference or opinion.
- In most cases, hypotheses are given formally in the introduction's final paragraph. If you have more than one hypothesis, number them and state them in a logical order. Although your experiment can have multiple independent and dependent variables, each hypothesis can only include one of each type of variable. Instead of stating them in terms of the exact procedures (operational definitions) utilized in your experiment, you explain them in conceptual terms.
- Concepts and terminology should be explicitly defined as soon as they are presented, and then utilized consistently after that. Make sure the phrase you're defining isn't included in the definition's body (tautology).

Method

- This part can be considered of as the paper's "cookbook," and by reading it, you can recreate (or, in research jargon, replicate) your study. As a result, it should be comprehensive, detailed, and easy to grasp so that another researcher can replicate your findings.
- The term 'METHOD' should be in the centre of the page. The past tense is used in this section.
- The techniques section is usually straightforward to write: simply state what you did, how you did it, and when you did it.

For example: Describe your participants how many? of what sex? mean (average) age? where did they come from? Response rate should be reported here. Reproduce the exact questions that you asked and don't forget to tell the reader the scale you used for participants' answers. For any scale that you construct, include the mean, standard deviation and Cronbach's alpha. If you 'threw out that participant's data (e.g., you didn't include it in your study, then you mention this fact 'one participant was discarded from the sample because....) in the Participant section,

- The strategy is broken down into a number of subsection titles that you should utilize to organize your paper.

Subjects

- (a) The 'Subjects' section details who took part in the research. The subjects are classified based on their age, gender, and other pertinent social and demographic factors.
- (b) What is the total number of subjects? List the total number of participants as well as the number of people that were assigned to each experimental condition. Give the number of subjects who did not complete the study and the explanation for their absence.
- (c) How are the subjects chosen? Report on how the subjects for the experiment were chosen, and how they were divided into groups. Was a randomization strategy employed, for example, or was another way required? Payments or promises made to subjects should be reported.

Apparatus

If one is required, this section provides a brief explanation of the study's equipment or materials. Stop watches, for example, do not need to be detailed in depth. Remember that the description must be precise enough for the reader to be able to repeat the experiment.

Procedure

These directions to the subjects must be written verbatim because they are a kind of recipe for each stage in the implementation of the experiment. The establishment of groups, as well as the exact experimental manipulations carried out on each group, are all covered. Randomization, counterbalancing, and other control mechanisms are also described in detail. Keep in mind that one of the aims of this part is to allow another

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experimenter to repeat exactly what you've done. This section should not contain any results. Preliminary results that were used to develop the primary experiment that you are reporting on are acceptable. Mention any ethical concerns that come to mind. Did the people you used give their approval to participate if you used human subjects? What pain-reduction techniques did you apply if you used animals?

Results

- The term 'RESULTS' should be in the center of the page. If space constraints necessitate it, start on a new page. This is (obviously) where you present the reader with the outcomes of your experiment.
- In the same way, the outcomes section is written in the past tense.
- The data and statistical treatment of the data are summarized in the results section. If the information is straightforward, it can be presented fully in text without the need of TABLES or FIGURES.
- Summarize your findings' major points and report whether or not your hypothesis were confirmed. Present your findings in the same sequence as your predictions (hypotheses) in the introduction, using simple words.
- In the RESULTS section, do not explore the ramifications, interpretations, or theoretical relevance of your findings. Except for nitpickers, most teachers favor tables and figures that are near to where they are needed.

Tables

- The most essential data directly linked to the experiment is retained for tables in the findings portion of the publication. Tables save space by compressing data and allowing the reader to see linkages that would otherwise be invisible at a glance.
- A good table should not repeat the text of the RESULTS, but rather highlight the data by referring to the table; it should also be self-explanatory and relevant to the content.
- Tables should be numbered and referred to in the text. If tables are placed in an APPENDIX, they should be recognized with capital letters (e.g., Table A).
- Tables are always displayed in the same order as they are stated in the text.
- Every table has a telegraphic-style title with a brief explanation. It's above the table and below the table number.
- There is enough white space to make the table readable. Abbreviations, parenthesis, and units of measurement are explained in the notes of subheadings.

Figures

- What has been discussed about Tables also applies to the figures in the Results section.
- Graphs, charts, and illustrations are examples of figures.
- Instead of being placed above the figure, the caption is placed below it. The caption comes first, followed by the word 'Figure 1'. The caption's initial word is capitalized alone.

- If there is enough room, you can put it on the same page as the text, but if there isn't, a separate page should be utilized, especially for publishing.
- Remember to put the dependent variable on the vertical axis and follow the requirements for proper data calibration. Both axes, as well as the graph lines when applicable, should be clearly labelled.

Discussion

- The word 'DISCUSSION' is in the center of the page. Use a new page only when absolutely essential.
- It's a good idea to start the discussion with a summary of the findings, for the benefit of those who wish to avoid the results section (and to remind those who didn't miss it but were interrupted by a phone call and forgot).
- The discussion section summarizes the primary findings (what you learned) and explains what you think they mean to the reader. There's no need to restate the data in this area; it's already in the results section.
- The studies you mentioned in your introduction should be referenced in your discussion section.
- Discuss how your findings (results) compare to the findings (results) of these research, or how they differ (don't panic) (and your ideas as to why they differ). Make an attempt to resolve and deal with these disparities by speculating about possible causes.
- Another smart strategy is to provide ideas for future research trials in this field, ones that follow up on the study you did, improve on it, and so on. For example, you could advise how to dig deeper into a finding you made, or how to figure out why your results differed from those of other researchers. If your discussion part is lengthy, include a brief summary paragraph (of the conclusion, interpretations, etc.) at the end to help the reader recall your main points.
- This is also a fantastic way to get ready to write the abstract. Remember that your paper's discussion part will make or break it, so give it your all and try to draw intelligent (and accurate) inferences from your facts.
- Your understanding of the inductive and deductive thought procedures involved in experimental work will be demonstrated in the discussion section.

Here is the summary of what is expected in the DISCUSSION section.

- a. Discuss the results in the context of the research and theory you already brought forward in the introduction. This will unify your work. Avoid personal opinion and irrelevant or undisciplined speculation.
- b. Show awareness of the shortcomings and uncontrolled variables in your work and qualify your results accordingly. showing your ability to identify any other explanations for your data that may suggest themselves.
- c. Specify what variables you would control or change in future research to correct for the problems in your present study.
- d. Taking your results at face value, suggest other research avenues for the

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future. An experiment may answer questions, but it generally raises other questions that may not have been considered before.

- e. Briefly draw out any practical implications of the study, if any.

References

- This is the final portion, and it should follow the APA format.
- At the top of the page, center the word 'REFERENCES,' but do not underline it or put it in quotation marks.
- A reference list cites books that are freely available on the internet.
- This section is always on its own page, and the page number is never included.
- The reference list must include all works cited in the text of your experiment, and each entry in the reference list must be cited in the text.
- Reference lists must be accurate and full because they are designed for the reader's use.
- The following general subsections make up a reference: AUTHOR, DATE OF PUBLICATION, TITLE, and PUBLICATION DATA.

Footnotes

In psychology, footnotes are rarely utilized. They can be used to thank others for their help in preparing a study or to acknowledge a research grant. Footnotes should be avoided. You may feel compelled to say something that isn't quite essential. This is when a footnote comes in handy. If you can avoid employing them, the reader's eyes will be saved. However, it's sometimes difficult to refrain from making lengthy but tangential statements. These go in the footnotes, not the body of the text. They will be read by the truly eager reader. Others, though, will not.

Writing a Research Report

Some points to consider:

- Define the Report's Context

As with writing an essay, determining the context for your research report will assist you in creating a detailed yet concise paper. This is why, before you begin writing, you should construct an outline to ensure that you do not overlook anything.

- Identify your target audience.

The tone of the report is determined by writing with your audience in mind. If you're writing for a general audience, you should offer the material in a straightforward and understandable manner. You would need to employ technical and field-specific words for a specialised readership.

- Include Important Findings

The purpose of a research report is to give a condensed version of your systematic inquiry. You should exclude irrelevant information from your report while emphasising just key data and findings.

- Use Illustrations

Illustrations and other visual representations of your data should be included in your study report. Graphs, pie charts and relevant images add credibility to your thorough investigation.

- Select the Appropriate Title

An excellent research report title is succinct, accurate and includes terms from your research. It should provide readers a clear concept of your systematic inquiry so that they may understand the complete emphasis of your research from the title.

- Review the Report

Before you publish the paper, go over it again to double-check the information. If possible, have someone else go over the report with you and you can also run it through proofreading and editing software.

To collect research data for the Report, we must:

- Recognise the Issue

Every research project attempts to solve a specific problem or group of problems and this should be kept in mind while you write your research report. Understanding the issue will allow you to filter the information you have and include only relevant facts in your report.

- Understand the purpose of your report.

This is related to the previous point in that the goal of your research report is in some ways connected with the goals of your systematic inquiry. Identifying the primary aim of producing a research report can assist you in identifying and presenting the necessary facts.

- Determine your target audience.

Knowing your target audience is critical when collecting data for a research report. If you are writing a research report for a company, you should include industry-specific information or demonstrate how the research findings are relevant to the company's work.

- Design Surveys/Questionnaires

A survey is a research approach that uses a series of questions to collect data from a specified group of people. It can be quantitative or qualitative in nature.

A survey is typically composed of structured questions and can be administered either online or offline. An online survey, on the other hand, is a more effective means of collecting research data because it saves time and makes data collection easier.

The following research report exemplifies report style and provides samples of the kind of information included in the various sections of an experimental document. It should be noted that the report has been abridged.

Different Components of a Research Proposal

Your research proposal should generally include the following information, regardless of whether you are submitting for the MJur, MPhil, or PhD programmes:

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1. Title

This is merely a working title for your prospective study. If you are accepted for admission, you will be free to change your title during your research.

2. Synopsis

A succinct explanation of your desired research, no more than 100 words long, should be included in the proposal. This could be a few phrases stating the problem you want to investigate or the fundamental question you want to solve.

3. Context of Research

You should describe the broad context in which your research will be conducted. Include a summary of the current state of knowledge and recent debates on the topic, as well as a brief overview of the general area of study in which your proposed research falls. This will help you to exhibit your knowledge of the field as well as your ability to communicate effectively and simply.

4. Research Issues

The essential goals and questions that will guide your research should be stated in the proposal. Take some time to think about the main questions you want to answer before you start writing your proposal. Many research ideas are too broad, so thinking about your main research topics can help you ensure that your project is focused and realistic (i.e., one that is likely to be completed with the normal period for a MJur, MPhil or PhD degree).

Prioritising one or two major questions from which you can derive a number of secondary research questions may be beneficial. The proposal should also state how you plan to respond to the questions: will you use an empirical, doctrinal, or theoretical approach, for example?

5. Research Techniques

The proposal should include a description of your research methods, including how you plan to conduct your study. Visits to specific libraries or archives, field work, or interviews could all be part of your strategy.

The majority of my research is done in libraries. If your research will be conducted at a library, you should describe where you will get your main resources (law reports, journal articles, etc.) (in the Law School's library, Westlaw, etc.). If you intend to perform field research or collect empirical data, you should include specifics (for example, if you intend to conduct interviews, who will you interview? How many interviews are you planning to conduct? Will there be any access issues?). This part should also include information on how you plan to analyse the results of your research.

6. The Importance of Research

The proposal should highlight the uniqueness of the research you intend to conduct. As a result, you should explain why your research is significant (for example, by explaining how your research builds on and adds to the current state of knowledge in the field or by setting out reasons why it is timely to research your proposed topic).

7. List of references

A brief bibliography identifying the most relevant works for your topic should be included in the proposal.

Research Proposal and Aspects

A research proposal is a document that proposes a research project, usually in the sciences or academia and usually serves as a request for funding. Proposals are assessed based on the proposed research's cost and possible impact, as well as the feasibility of the suggested plan for carrying it out. In general, research proposals cover the following topics:

- What research question(s) will be tackled and how will they be tackled?
- How much time and money will the research require?
- Have you done any prior research on the subject?
- How will the research findings be assessed?
- How will the research assist the sponsoring organisation and other stakeholders?

Unsolicited research proposals are those that are presented in response to a request with specific requirements, such as a request for proposal. Solicited research proposals are those that are submitted without prior request. Other types of proposals include "preproposals," in which a letter of intent or a brief abstract is submitted for review prior to submitting a full proposal; continuation proposals, in which an original proposal and its funding requirements are re-iterated in order to ensure continued funding; and renewal proposals, in which a project that would otherwise be terminated is sought to continue sponsorship.

Academic research proposals are usually written as part of the preliminary steps in the writing of a thesis, research paper, or dissertation. They usually have an introduction, a literature review, a discussion of research methodology and goals and a conclusion, just like a research paper. This basic structure may differ between projects and fields, with each having its own set of requirements.

What is a research proposal's purpose?

A research proposal's goal can be summed up as follows:

- To suggest a research project that will add to the body of knowledge in a substantial way.
- Develop a detailed project plan that includes a methodological approach and a theoretical framework.
- To make sure that the project research can be completed in the time allotted and with the resources available.
- To show that you have the necessary knowledge and experience to complete the assignment.

Even if writing a research proposal is not required as part of your candidacy, it is a good idea to do so. Writing a study proposal can help you to establish your goals and main points. It will allow you to consider each stage of the research process in order to create a clear and precise plan. It will also assist you in anticipating challenges that

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may develop throughout your candidature and urge you to consider how you will handle them if they do. What abilities do you need to write a research proposal?

Writing a research proposal necessitates a variety of abilities. These abilities can be divided into three categories:

- Subject expertise and research abilities

The proposal allows you to demonstrate your command of the topic matter as well as your familiarity with current research trends.

- Skills in critical thinking

A strong research proposal demonstrates sophisticated analytical, assessment and synthesis abilities, as well as inventiveness and the capacity to think vertically and laterally.

- Ability to communicate

The proposal demonstrates your ability to communicate clearly and concisely.

5.3.4 Steps in Report Writing

The aids of active project reporting

Reporting isn't just something you should do for the sake of responsibility it. Effective reporting has a variability of benefits. When you appropriately report on project status, you efficiently:

Cling to track of project fitness

The worst thing for a project is when you come to at the end of the timeline and realize you were off course the whole time. No one likes being blindsided—and as the project boss, you're empowered to make sure your team is conscious of your project progression at all times.

Development reports are a way to accomplish this without having to perform a lot of manual labor. Because this information combines high-level details with key measurements, everyone can assess the plan's health.

Abridge project progress

Project position reports are not real-time reports. These reports are summaries of what ensued during the past week, two weeks, or month of project effort. They're a chance for your stakeholders to stay knowledgeable on how well you're sticking to the scheme plan.

Decrease manual work

As the scheme manager, you already have enough on your plate. You don't need to also devote hours every week or month collecting data from different places. Project reportage tools make it easy to find all of this info in one place and create a project status report by means of the click of a button.

Segment next steps and accomplishment items

Project status reports must go out to your project team, project sponsor, significant stakeholders, and in addition cross-functional team members. For the reason that these are high-level reports, they're suitable for anybody who wants to stay informed about mission progress.

This is the ideal way to let everyone important know what's happening without getting into the particulars. If there are important development next steps or action items, share them at this time so everyone be acquainted with what to expect.

Proactively recognize blockers

If your project isn't progressing on track, your status description lets others know what the interruption is and what you're doing to resolve any blockers, allowing you to show off your hands-on approach to in get things back in order. Similar to the project risk organization process, proactive status fourth estate helps you identify and overcome issues beforehand they impact your project timeline.

Keep an eye on this report to understand what to consist of in your project status report and take care as we put each step hooked on exercise with an example of an Employee Satisfaction scheme.

1. Shape your report wherever work lives

Before you build your report, make sure you're already tracking your work data in a project management tool. That way, you don't have to physically grab data from a host of sources—instead, you can decrease manual work and create an account with a few clicks.

First off with a project management tool styles it easy to capture dependencies and note imminent tasks so you're not ever blindsided approximately your project health.

2. Term your report

A boundless option is to merely use the project name for clarity. If you're reporting on this project regularly, you ought to also consist of a date or timestamp.

Sample project report title: February 2020 - Worker satisfaction initiative

3. Designate project health

The project health is the current position of the project. Project health may vary from report to report, specifically if you run hooked on blockers or unblock big project risks. Use a project management tool that permits you to communicate the project's status and whether or not it's on track. One way to do this is to use a color coding system (green = on track, yellow = at risk, red = off track). Instance project health update: Project rank is on track.

4. Rapidly summarize the rank report

Your project status report summary should be short and sweet—about 2-3 sentences. The goal here is to inform readers who may not have time to recite the entire report a quick TL; DR of the furthestmost important facts.

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This is the primary section of your report, so it's the superlative place to:

Consist of highlights

Flag most important blockers

Note unforeseen project risks

Sample status report summary: Our survey consequences are in and being reviewed. At first glance, we're sighted 80% employee gratification, up 3 points from the last survey. The Engagement Committee is at work with the Executive team happening what new engagement creativities to implement in our key bull's eye areas, which include career development and transparency.

5. Complement a high-level general idea of each key area

Dependent on your project, your key extents may vary from report to report, or they may stay consistent. For instance, in an Agile scheme that's uninterruptedly improving, you'd likely use self-motivated key areas that concealment the belongings your team worked on all through the last sprint. Alternatively, for an occasion planning project, there are a set quantity of key areas that you at all times want to touch on, similar to promotion, signups, and utterers.

For each important area in the status report, enhance a few bullet points that give and bring up-to-date on progress, happenings, and forthcoming work.

Example high-level overview of a key area: Survey results

70% of staffs took the satisfaction survey.

Our complete satisfaction rating is 80%.

Simply 57% of employees report having an indistinct path towards career advancement, down 5% subsequently the last survey.

41% of employees enumerated transparency as the number one enhancement they'd like to see.

6. Enhance links to extra documents or resources

Despite the fact that you shouldn't include every little aspect about how your project is going, generally people will want to know more. Intended for stakeholders who are observing for more in-depth information, arrange for links to documents or resources. This can include more specific project data, like links to specific project milestones, or the wide-ranging impacts of the project, comparable a reference to the business goals the project be situated contributing to.

Case in point: Take account of a link to the employee gratification survey, as well as to the bigger company OKR around snowballing employee engagement over the course of the economic year.

7. Flag in the least blockers the project partakes run into

Altogether projects run into roadblocks. These can originate in the form of project risks, unforeseen increases to the budget, or interruptions that influence the project

timeline. Keep stakeholders in the loop when unforeseen obstacles arise will help everyone adjust consequently to stay on track.

Illustration roadblock: The decision-making team wants to aspect at results before the rendezvous committee meets again, nevertheless won't be able to do so for another three weeks. This will postponement our complete project timeline.

8. Icing on the cake next steps

These might be a list of next steps, compliments you want to give someone, or whatever else you want to emphasize.

Example: Thank you Sarah A. for conveyance out multiple communications to employees heartening them to take part in the survey!

5.3.5 APA Style Essentials

The APA Style web site (<http://www.apastyle.org/>) and the Publication Manual of the American Psychological Association (6th ed., 2014) provide a thorough reference guide on writing utilizing APA style, organisation, and content. The goal of this paper is to establish a common core of APA style features that all members of an academic department can use as minimum standards for any APA-style assignment.

I. General Document Guidelines

- A. Margins: One inch on all sides (top, bottom, left, right).
- B. Font Size and Type: 12-pt. Times New Roman font.
- C. Line Spacing: Double-space throughout the paper, including the title page, abstract, body of the document, references, appendixes, footnotes, tables, and figures.
- D. Spacing after Punctuation: Space once after commas, colons, and semicolons within sentences. Insert two spaces after punctuation marks that end sentences.
- E. Alignment: Flush left (creating uneven right margin).
- F. Paragraph Indentation: 5-7 spaces.
- G. Pagination: The page number appears one inch from the right edge of the paper on the first line of every page.
- H. Running Head: The running head is a short title that appears at the top of the pages of a paper or published article. The running head is typed flush left at the top of all pages. The running head should not exceed 50 characters, including punctuation and spacing. Using most word processors, the running head and page number can be inserted into a header, which then automatically appears on all pages.
- I. Active voice: As a general rule, use the active voice rather than the passive voice. For example, use 'We predicted that ...' rather than 'It was predicted that ...' Order of Pages: Title Page, Abstract, Body, References, Footnotes, Tables, Figures, Appendixes.

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II. Title Page

- A. Pagination: The Title Page is page 1.
- B. Running Head: The running head is typed flush left (all uppercase) following 'Running head'.
- C. Key Elements: Paper title, author(s), institutional affiliation(s), author note.
- D. Paper Title: Uppercase and lowercase letters, centered on the page.
- E. Author(s): Uppercase and lowercase letters, centered on the line following the title.
- F. Institutional affiliation: Uppercase and lowercase letters, centered on the line following the author(s).
- G. Author Note: Provide information about the author's departmental affiliation, acknowledgments of assistance or financial support, and a mailing address for correspondence.

III. Abstract The abstract is a one-paragraph, self-contained summary of the most important elements of the paper.

- A. Pagination: The abstract begins on a new page (page 2).
- B. Heading: 'Abstract' (centered on the first line below the running head).
- C. Format: The abstract (in block format) begins on the line following the Abstract heading. The abstract word limit is set by individual journals. Typically, the word limit is between 150 and 250 words. All numbers in the abstract (except those beginning a sentence) should be typed as digits rather than words.

IV. Body

- A. Pagination: The body of the paper begins on a new page (page 3). Subsections of the body of the paper do not begin on new pages.
- B. Title: The title of the paper (in uppercase and lowercase letters) is centered on the first line below the running head.
- C. Introduction: The introduction (which is not labeled) begins on the line following the paper title.
- D. Headings: Five levels of headings are available to be used to organize the paper and reflect the relative importance of sections. For example, many empirical research articles utilize two levels of headings: Main headings (such as Method, Results, Discussion, References) would use Level 1 (centered, boldface, uppercase and lowercase letters), and subheadings (such as Participants, Apparatus, and Procedure as subsections of the Method section) would use Level 2 (flush left, boldface, uppercase and lowercase letters)."

V. Text citations

In the body of the article, source material must be recorded by citing the author(s) and date(s) of the sources. The essential premise is that other people's ideas and words must be acknowledged in writing. The complete source citation may be found in the list of references that follows the body of the paper.

Notes

- A. When the names of a source's authors are part of the formal structure of the phrase, the year of publication appears in parentheses after the authors' names.

Consider the following situation: Although there was a two-week reduction in insulin dosage in the treatment condition compared to the control condition, Wirth and Mitchell (1994) concluded that the difference was not statistically significant. [Note: and is used when many authors are mentioned in the sentence's formal structure.]

- B. When the authors of a source aren't part of the sentence's formal structure, both the authors and the year of publication are placed in parenthesis. Consider the following illustration. According to reviews of studies on religion and health, at least some types of religious actions are linked to improved physical and mental health. [Note: When many writers are mentioned in parenthetical text, the & symbol is used. When multiple sources are cited in parentheses, they are arranged alphabetically by first authors' surnames and separated by semicolons.]
- C. When citing a source with two authors, both authors are listed each time the source is cited.
- D. When citing a source with three, four, or five authors, the first time the source is referenced, all authors are listed. The initial author's surname and 'et al.' are used when citing that source again. Consider the following illustration: According to reviews of studies on religion and health, at least some types of religious actions are linked to improved physical and mental health (Payne, Bergin, Bielema, & Jenkins, 1991). According to Payne et al. (1991),
- E. When citing a source with six or more authors, the first author's surname and the phrase 'et al.' are used each time the source is cited (including the first time)
- F. You should make every attempt to cite only sources that you have read. Use the following structure for the text citation and identify only the source you have read in the References list if you need to reference a source you have not read ('Grayson' in the following example) that is cited in a source you have read ('Murzynski & Degelman' in the following example). Grayson (as referenced in Murzynski & Degelman, 1996) identified four body language components that were linked to vulnerability judgments.
- G. Include initials, surname, and as precise a date as feasible when citing a personal communication (including letters, emails, and telephone interviews). Personal communications are not included in the 'References' section since they are not 'recoverable' information. Use the following format for the text citation: According to B. F. Skinner (personal communication, February 12, 1978),
- H. Use the author-date format to cite a Web document. Use the first few words of the title in place of the author if no author is provided. If no date is given, substitute 'n.d.' for the date. Take a look at the following examples: Degelman (2009) compiles a list of APA writing style rules. It has been documented that Americans' perceptions of gender status distinctions have shifted.

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- I. Provide the Bible's book, chapter, and verse when citing it. Identify the version of the Bible used the first time it is mentioned in the text. "You are forgiving and good, O Lord, abounding in affection to everyone who call to you," for example (Psalm 86:5, New International Version). [Note: The Bible does not require a reference entry in the References list.]

VI. Quotations

When using a direct quotation, make sure to include the author's name, year, and page number in the citation.

- A. A quote of less than 40 words should be included in double quotation marks and incorporated into the sentence's formal structure. Patients who received prayer, for example, had "reduced congestive heart failure, required less diuretic and antibiotic medication, experienced fewer episodes of pneumonia, had fewer cardiac arrests, and were less commonly incubated and ventilated" (Byrd, 1988, p. 829).
- B. A longer quotation of 40 or more words should be set off from the surrounding text in block style, with each line indented five spaces from the left margin (without quotation marks).

VII. References

All sources listed in the References section must also be mentioned in the body of the paper (and all sources cited in the body of the paper must also be included in the References section).

- A. Pagination: The References section is split into two pages.
- B. 'References' is the second heading (centered on the first line below the running head).
- C. Format: The references begin on the line after the 'References' heading (with a dangling indent).

The entries are arranged alphabetically by the initial authors' surnames. The following elements are found in most reference entries:

- ii. Authors: Authors are listed using surnames and initials in the same order as in the source. All authors are separated by commas. If there are eight or more authors, list the first six, then three ellipses (...), and finally the last author. If no author is given, the reference begins with the document's title.
- iii. Year of Publication: After the authors, in parentheses, with a period after the closing parenthesis. If no publishing date is given, put 'n.d.' in parenthesis after the writers' names.
- iv. References: Title, journal, volume, pages (for journal articles) or title, city of publishing, publisher (for a book). Italicize the titles of books, magazines, and the volume numbers of periodicals.
- v. Electronic Information Retrieval: Digital object identifiers (DOIs) or universal resource locators are examples of electronic retrieval information (URLs). DOIs are alphanumeric identifiers that point to digital source materials.

Examples of sources

1. "Journal article with DOI Murzynski, J., & Degelman, D. (1996). Body language of women and judgments of vulnerability to sexual assault. *Journal of Applied Social Psychology*, 26, 1617- 1626.
2. Journal article without DOI, print version Koenig, H. G. (1990). Research on religion and mental health in later life: A review and commentary. *Journal of Geriatric Psychiatry*, 23, 23-53.
3. Journal article without DOI, retrieved online [Note: For articles retrieved from databases, include the URL of the journal home page. Database information is not needed. Do not include the date of retrieval.] Aldridge, D. (1991). Spirituality, healing and medicine. *British Journal of General Practice*, 41, 425-427. Retrieved from <http://www.rcgp.org.uk/publications/bjgp.aspx>
4. Book Paloutzian, R. F. (1996). *Invitation to the psychology of religion* (2nd ed.). Boston, MA: Allyn and Bacon.
5. Informally published Web document Degelman, D. (2009). APA style essentials."

VIII. Footnotes

Content footnotes are used to supplement substantive information in the text on occasion. A content footnote can be found at the bottom of the page on which it is discussed, or it can be found after the References on a separate page.

- A. Pagination: Footnotes begin on a separate page from the rest of the text.
- B. Heading: 'Footnotes' is centred below the running head on the first line.
- C. Format: The first line of each footnote should be indented 5-7 spaces, and the footnotes should be numbered (slightly above the line) as they are identified in the text.

IX. Tables

The presentation of quantitative data or the results of statistical analysis (such as ANOVA) is a typical application of tables. In the text, tables must be specified.

- A. Pagination: Each Table is presented on its own page.
 'Table 1' (or 2 or 3, or whatever) is typed flush left on the first line below the running head.
 Double-space the table title and type it flush left (italicized in uppercase and lowercase letters).

X. Figures

Figures are frequently used to present graphs, pictures, or other illustrations (not tables).

- A. Pagination: Figures are placed on their own page.
- B. Figure Caption: On the first line below the figure, type 'Figure 1.' (or 2 or 3, etc.) flush left and italicised, followed by the caption (which should be a brief descriptive phrase) on the same line.

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XI. Appendixes

Appendices are frequently used to give previously unreported tests or to detail complicated equipment or stimulus materials.

- A. Pagination: Each Appendix starts on its own page.
- B. Heading: If there is only one appendix, the word 'Appendix' is centred below the text page header on the first line. Use Appendix A if there are multiple appendices (or B or C, etc.).

Double-space and type the title of the appendix (centered in uppercase and lowercase letters).

- C. Format: The first line should be indented 5-7 spaces.

5.3.6 Citing & Referencing Sources

Authors cite previously published material to back up their claims. Readers may choose to consult those publications in order to obtain information for their own purposes or to verify the author's use of the referenced sources. Citations and references must be clear and correct in order to help the reader. They are also easier to utilize if the style is constant. Most journals and editors have distinct preferences for how references are cited in the text and listed at the end of articles.

Citations

In today's scientific literature, there are three basic citation methods. There may be numerous variants within these systems. Name and year (sometimes known as author-date) number, with references in alphabetical order are the primary systems.

Name and Year: The name(s) of the author(s) and the year of publication are inserted into the text in the first system. Depending on the sentence construction, both or only the date may be enclosed in parenthesis. Examples - According to a recent study, the refereeing system is effective (Lock and Smith 1986). The refereeing system, according to Lock and Smith (1986), is effective. In the years leading up to 1984, manuscripts published in the BMJ received much more citations than either of the groups of rejected papers (Lock and Smith 1986, p. 312).

A lowercase letter is added to the date if there are multiple references with the same author(s) and year of publication. Lock and Smith (Lock and Smith 1986a), Lock and Smith (Lock and Smith 1986b), and so on. All cited references are listed in their entirety at the end of the paper. They are arranged alphabetically by the author's surname. If there are multiple authors, the name of the first author in the citation determines the alphabetical order. If two or more pieces by the same author(s) are found, they are ordered chronologically.

Number: References are numbered in the order they are mentioned in the text in the second system. The name(s) of the author(s) may also be given in some journals. Examples Scientific papers are not meant to be read; rather, they are meant to convey information. Any significant departure from the norm is likely to make it difficult for the reader to comprehend the material (2). Even if it is quoted numerous times at widely spaced points, the cited work retains its number each time it is cited throughout the text. All of the cited sources are provided in numerical order at the end of the text.

Number with Alphabetical References: The third system, Number with Alphabetical References, combines the benefits of the two preceding systems. All references are listed in alphabetical order by author first. That is the sequence in which they are numbered. These numbers are used in the text for citations.

References

At the end of the paper, each reference must be described in depth. 'Literature Cited,' 'References Cited,' or simply 'References' are common headings for the list. Any references that aren't cited should be removed.

Articles: Reference to an article should contain -

- names and initials of all authors (although, if there are many authors, some journals will list only the senior author et al)
- title and subtitle
- name of the journal (usually abbreviated)
- volume number
- first and last page number
- year of publication
- months or number of issue, if pages are not numbered consecutively through a volume.
- Example: Lock, S., and J. Smith. 1986. Peer review at work. Scholarly Publishing 17: 303- 16.

Books: A reference to a book should contain -

- names and initials of all authors
- title and subtitle
- number of edition, if there is more than one
- name and initials of editor or translator, if any
- place of publication
- name of publisher
- year of publication
- volume number, if more than one
- page numbers, if any need to be cited specifically.
- Example: Stainton, E.M. 1982. Author and editor at work. Toronto: University of Toronto Press.p.17

Proceedings: For the proceedings of symposia, conferences, and workshops, references should include-

- names and initials of authors
- title of paper
- names and initials of the volume editors
- title of symposium or conference
- date and place of meeting
- place of publication

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- name of publisher
- year of publication
- numbers of specific pages.
 - o The titles of books and periodicals are sometimes printed in italics in some publications. In some cases, such as the first two examples above, the date is placed immediately after the author(s).
 - o If there are only two or three authors, many publications mention all of their names. They only list the senior author et al. if there are more than those.
 - o Punctuation has been reduced to a bare minimum in some journals, particularly in the health sciences. They've deleted the space between initials and dropped the periods after initials in author names (thus Smith, J.L. becomes Smith JL).
 - o The first and last page numbers of articles are given in some magazines. This enables the reader to request copies from a library. It also informs the reader of the article's length.

5.3.7 Footnotes

In general, footnotes (notes at the bottom of the page) should be avoided. They disrupt the reader's flow of thought and raise publishing costs. References can be found in the footnotes of some periodicals.

However, in the sciences, this practice has almost vanished, and it appears to be declining in other fields. When citing references, it's usually more cost-effective to group them towards the end of an article or book rather than inserting them one or two at a time at the bottom of each page. Footnotes are typically used to give information that the author believes is significant but would otherwise disrupt the flow of the discussion. An editor can usually find a way to include such observations into the text's body, usually by putting them in parenthesis. A statement is usually worth stating in the text if it is worth saying. A superior number (1,2), a superior letter (a,b), or a symbol (*,+) are used to denote footnotes in the text. Not at the bottom of the text pages, but on separate sheets of paper, should be typed footnotes. They'll probably be typeset separately from the main text, and grouping them together in this way allows for more cost-effective typesetting. They should be double-spaced to make editing and setting them easier. Footnotes can be used to provide important information such as the author's address(es).

5.3.8 Key Considerations/Factors

Important Reminder

- Both the proposal and the final report must meet English requirements, including the use of technically accurate English and suitable citations to the reference sources used.
- The earlier you begin writing the final research/project report, the easier it will be. A common mistake is to put off writing the report until the very last minute, by which time you will have forgotten many of the details of your work and will have effectively completed the project twice.

General Advice

- Academic writing might come across as arrogant and complicated. Much of it is, but not all of it is. Do not employ words simply because they sound scholarly (especially if you have no idea what they mean).
- The most important rule of syntax is to write in such a way that a reader can parse sentences without comprehending what they imply - that is, figure out what modifies what, what is the object of what, and so on.
- The reader should not need the meaning to interpret the syntax; the reader should be able to figure out the syntax without it.
- Pay attention to the various ways in which people demonstrate the relationship of their work to the truth as you read. In academic speech, words like indicate, demonstrate, prove (not used outside of mathematics), speculate, suggest, assert, question, claim, conclude, debate, discover, define, and suppose have extremely precise meanings.

Good Scientific Writing

1. Vacant Lead Sentences: The most significant sentences are the opening sentences of each section, as well as the first sentences of each paragraph. They should state your major points in straightforward English. Then the specifics can be added.

Right: the end result. Cognitive treatment was more effective than pharmacological therapy in preventing recurrence. Drug therapy outperformed no treatment at all. Covariance analysis...

Results are incorrect. We did four covariance studies on our data after first converting them to z scores. We then used a Bonferroni correction to perform paired comparisons...

2. Qualifiers and Limitations Don't waste your chance to write persuasively by starting with secondary ideas and caveats. They belong in the paragraph or section's body, not as an opener.
3. Recognize the difference between strong and weak statements. Qualifiers and caveats should be used sparingly in scientific writing. Qualifiers are used to classify marginal results, debatable statements, guesses, and possible artefacts. Strong results, well-confirmed statements, and fundamental theory are exempt. Verbs like seem, appear, indicate, may, and suggest have meaning. They should not be used on a regular basis.

Water deprivation may reduce appetite because the volume was hardly meaningful. Electric shock, on the other hand, doubled hunger.

Wrong: According to our studies, electric shock may boost hunger. It also indicates that lack of water reduces hunger.

4. Long phrases and big words: The majority of readers are preoccupied. The majority of readers are slackers. Many readers are only interested in scanning. Use short sentences and simple words to assist these readers. Look for a plain word if a huge word tempts you. Find a way to break up a long statement if it tempts you. To compensate for the hindering reading, the huge word and the long sentence must greatly boost accuracy.

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Right: People who are wealthier are less depressed. As a result, we skewed our results by placing more of them in the wait-list control group.

Wrong: Assigning this group to the wait-list condition would prevent treatment effects from being unduly inflated by enrolling the higher-income group with a better prognosis in the early treatment phase.

5. Excessive rewriting. Remove words and concepts that the reader is already familiar with. Overwriting slows down the reader and has no effect on accuracy.

Correct: Psychotherapy and medications outperformed attention alone and were far superior to no treatment.

Wrong: When compared to the attention control group, the drug treatment group, and the psychotherapy treatment group, the wait list control group performed worse than the attention control group, the experimental drug treatment group, and the psychotherapy treatment group.

6. The Passive Voice and the Royal 'We' To avoid expressing 'I did such and such,' poor writers use the awkward passive voice. When used sparingly, the first person is fine. Write with vigour and if possible, use the active voice.

Right: I believe that animals can learn about non-contingency and become helpless as a result.

Wrong: Animals, it is said, can learn about non-contingency. This is incorrect. When an animal learns non-contingency, it becomes helpless.

7. In the middle, there are citations. Citations should not be used to break up sentences. The reader's speed slows to a crawl as a result of this slight increase in accuracy. If possible, place all of your citations at the end of the paragraph.

8. Statistical effects are oriented in one direction or the other. Always state the direction as well as the significance of the direction.

Right: Small doses of the medicine put small rats to sleep very immediately, but high doses kept huge animals awake (F weight X dosage (2, 31) = 14.56, $p.001$).

Wrong: There was a highly significant interaction between medication and weight (F (2, 31) = 14.56, $p.001$).

A. Write accurately

- Scientific writing has to be precise. Make certain you say exactly what you mean.
- Rather than: The rats were given the medication. I administered the medication into the rat, as written.
- Use caution while using often misunderstood terms - There is less food (I am unable to count the amount of foods). There are fewer animals (can count numbers of animals). There's a lot of food (I can't count how many). Animals in huge numbers (can count them).

B. Write clearly

- Write at a level suited for the target audience.
- Dangling participles should be avoided.

C. Write succinctly

- Instead of abstract nouns, use verbs.
- Instead of think about it, think about it. Consider writing:
- Instead of 'to be,' use strong verbs.
- Instead of: The enzyme was discovered to be the catalytic agent.... Use simple words to write: The enzyme catalysed...
- Shorten your sentences. If a sentence contains more than 40 words, it should be split into two sentences.

➤ Use short words.

Instead of:	possess	sufficient	utilize	demonstrate	assistance	terminate
Write:	have	enough	use	show	help	end

➤ Use concise terms.

Instead of:	prior to	due to the fact that	in a considerable number of cases	the vast majority of	During the time that	in close proximity to
Write:	before	because	often	most	when	near

D. Check your grammar, spelling and punctuation

- Use a spellchecker, but keep in mind that it will not capture all errors.
- It's possible that Spellchecker won't recognize scientific terminology. Try one of the technical dictionaries on the reference shelf in the Biology or Health Sciences libraries for the right spelling.
- Use commas sparingly, and proofread thoroughly to ensure that no words are missed.

The following concerns must be considered when writing a research report:

- The Goals
- Type of problem/subject
- Nature and scope of study
- Target audience or users of research
- Report size
- Handwritten, typed, or electronic is the type of writing.
- Cost and time
- Language
- Report Contents
- Number of copies
- Order of contents
- Format – paper type and size; report lengths, widths, and depths; and writing style, including paragraph, indent, numbering, font size and type, coloration, and so on.
- Binding (both soft and hard copy) – kind, quality of material, color, and other relevant factors.

Notes

Check your Understanding

1. Content footnotes are used to supplement substantive information in the text on occasion. A content footnote can be found at the bottom of the page on which it is discussed, or it can be found after the References on a separate page.
2. The methodical analysis of a natural phenomenon, materials, sources, or the current state of society in order to find facts, obtain further knowledge, and draw new conclusions is known as research.
3. A _____ is a reference, explanation, or comment¹ placed below the main text on a printed page. Footnotes are identified in the text by a numeral or a symbol. In research papers and reports, footnotes commonly acknowledge the sources of facts and quotations that appear in the text.
4. A _____ is a publication that reports on the findings of a research project or alternatively scientific observations on or about a subject.

Summary

- A research report is an important part of any research project because the research isn't complete until the report is presented or prepared. The findings of a research study are of little use unless they are successfully documented and disseminated to others, regardless of how good the research study is or how meticulously it was conducted. The findings of study must invariably be added to the overall body of knowledge. Writing a report is the final phase of a research project, and it necessitates a distinct set of abilities than those required to do the research.
- The methodical analysis of a natural phenomenon, materials, sources, or the current state of society in order to find facts, obtain further knowledge, and draw new conclusions is known as research. It is a production process that requires a lot of inputs in order to develop new knowledge and the application of new and existing knowledge to produce technology that, in turn, can lead to a nation's economic prosperity.
- The findings of study must invariably be added to the overall body of knowledge. All of this explains why producing a research report is so important. Some people do not consider report writing to be an important aspect of the research process. However, the common consensus is that the presenting of research findings or the preparation of a report should be considered part of the research endeavor. The writing of a report is the final step of a research project, and it necessitates a distinct set of skills than those required in the other stages. This activity should be completed with extreme caution by the researcher; he may seek aid and instruction from professionals for this reason.
- Depending on the subject, research reports come in a variety of lengths and formats. For their annual reports, banks and other financial institutions, for example, prefer brief balance sheet tabulations. A scientist's report may be in the form of symbols and formulae, whereas a mathematician's report may be in the form of various algebraic notations. Literature students typically produce a lengthy paper critical of a writer or work.
- Newspaper articles are also a type of report writing. Book reviews, government

reports, PhD theses, and other types of reports are examples of reports. Any research study can be presented in a variety of ways, including a technical report, a public report, an article, a monograph, or even an oral presentation. The technical report is written for experts who want to learn more about the study project's technical procedures and vocabulary. The report will be written in technical terms. The main focus of the technical report is on: (i) the methodologies used; (ii) the assumptions established during the investigation; and (iii) the full presentation of the findings, including their limitations and supporting data.

- Scientists use scientific research publications to communicate with other scientists about the findings of their research.
- Content footnotes are used to supplement substantive information in the text on occasion. A content footnote can be found at the bottom of the page on which it is discussed, or it can be found after the References on a separate page.
- Authors cite previously published material to back up their claims. Readers may choose to consult those publications in order to obtain information for their own purposes or to verify the author's use of the referenced sources. Citations and references must be clear and correct in order to help the reader.

Activity

1. Prepare a report covering all the aspects of the topic 'Report Writing'.

Questions & Exercises

1. Define a research report and explain its purpose.
2. What are the characteristics of a research report? What functions does it perform?
3. What are the format requirements of a research article to be published in a professional journal?
4. Describe the considerations and steps involved in planning report writing work.
5. What are the various kinds of target audience for research reports? What may be their requirements?
6. Describe the layout or format of a research report.
7. Why is a 'review of literature' included in a research report? What is its purpose?
8. Describe briefly the various elements included in a research report.
9. Compare and contrast the findings and the conclusions of a research study.
10. What precautions should a researcher take while interpreting his findings?
11. What is a bibliography? What is its purpose?
12. What are the principles for the organisation of a research report?
13. What style characteristics are desirable while writing a research report?
14. What are the different types of report?
15. What are the components of a research report?
16. Define footnotes.

Notes

17. What are citing and referencing sources?
18. State the key considerations in research.
19. What are the essential steps in report writing?
20. What is general advice to be followed in research report?

Glossary

- Footnotes: A footnote is a reference, explanation, or comment¹ placed below the main text on a printed page. Footnotes are identified in the text by a numeral or a symbol. In research papers and reports, footnotes commonly acknowledge the sources of facts and quotations that appear in the text.
- Research Report: A research report is a publication that reports on the findings of a research project or alternatively scientific observations on or about a subject.

Further Readings

1. Bocken, N. M. P., Short, S. W., Rana, P., & Evans, S. (2014). A Literature and Practice Review to Develop Sustainable Business Model Archetypes. *Journal of Cleaner Production*, 65: 42-56.
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3. Colling, J (2003). Demystifying The Clinical Nursing Research Process: The Literature Review. *Urol Nurs* 23 (4): 297–9.
4. Cooper, H. M. (1988). The Structure of Knowledge Synthesis. *Knowledge in Society*, 1: 104-126.
5. Cronin, P., Ryan, F., & Coughlan, M. (2008). Undertaking a literature review: a step-by-step approach. *British Journal of Nursing*, 17(1): 38-43.
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8. Hart, C. (1998). *Doing a Literature Review*. London: Sage Publications.
9. Lawrence, C. N. (2011). *Writing a Literature Review in the Social Sciences*.
10. Okoli, C., & Schabram, K. (2010). A Guide to Conducting a Systematic Literature Review of Information Systems Research. *Sprouts: Working Papers on Information Systems*, 10(26).

Check your Understanding – Answers

1. True
2. True
3. Footnote
4. research report